SYNOPSYS®

DesignWare® Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC

Reference Manual

dwc_32g_phy_tsmc12ffc_x2ns – Product Code: H976-0

Version 3.01a
April 4, 2023

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Revision History

The following table lists the revision history of the PHY for each release. For a detailed description of PHY component updates, refer to the PHY release notes.

Note

- In some instances, documentation-only updates occur. The DesignWare IP product (https://www.synopsys.com/designware-ip/interface-ip.html) contains the latest information.
- Documentation-only updates are designated using the following numbering structure: <phy_version>_d<x> (for example, 1.00a_d1).

PHY Version	Date	Description
3.01a	April 4, 2023	For a list of functional changes, refer to the Revision History in the PHY databook.

Preface

This Reference Manual describes the signals and registers of Synopsys DesignWare® Cores Multi-Protocol 32G PHY x2 for TSMC 12FFC, which is a high-performance, low-power transceiver for high-bandwidth low latency applications.

This chapter contains the following sections:

- "Reference Manual Organization" on page 7
- "Web Resources" on page 7
- "STARs on the Web (SotW)" on page 7
- "Synopsys Statement on Inclusivity and Diversity" on page 8
- "Customer Support" on page 8

Reference Manual Organization

This Reference Manual is organized as follows:

- Chapter 1, "Signal Descriptions", describes the PHY top-level signals.
- Chapter 2, "Register Descriptions", describes the PHY control registers.

Web Resources

- DesignWare IP product information: https://www.synopsys.com/designware-ip.html
- Your custom DesignWare IP page: https://www.synopsys.com/dw/mydesignware.php
- Documentation through SolvNetPlus: https://solvnetplus.synopsys.com(Synopsys password required)
- Synopsys Common Licensing (SCL): https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html

STARs on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the DesignWare IP product information:

https://www.synopsys.com/designware-ip.html

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Customer Support

To obtain support for your product, contact Support Center using one of the following methods:

- For the fastest response, enter a case through SolvNetPlus:
  a. https://solvnetplus.synopsys.com

  Note SolvNetPlus does not support Internet Explorer.

  b. Click the Cases menu and then click Create a New Case (below the list of cases).
  c. Complete the mandatory fields that are marked with an asterisk and click Save.

  Make sure to include the following:
  - Product L1: DesignWare Cores
  - Product L2: SERDES_32G_PHY

  For more information about general usage information, refer to the following article in SolvNetPlus:

  https://solvnetplus.synopsys.com/s/article/SolvNetPlus-Usage-Help-Resources
- Or, send an e-mail message to support_center@synopsys.com(your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - Include the Product L1 and Product L2 names, and Version number in your e-mail so it can be routed correctly.
  - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - Attach any debug files you created.
- Or, telephone your local support center:
  - North America:

    Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
  - All other countries:

    https://www.synopsys.com/support/global-support-centers.html

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Signal Descriptions

This chapter details all I/O signals in the IP. Inputs are on the left of the signal diagrams; outputs are on the right. In addition to describing the function of each signal, the signal descriptions in this chapter might include the following information:

- **Active State:** Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).
- **Registered:** Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of *No* does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the IP. A value of N/A indicates that this information is not provided for this IP title.
- **Synchronous to:** Indicates which clocks in the IP sample this input (drive for an output). This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook.

The I/O signals are grouped as follows:

- "Resistor Tune (Common Block Group) Signals" on page 11
- "Power Supply Signals" on page 14
- "Analog I/O Signals" on page 17
- "Boundary Scan Port Signals" on page 20
- "Scan Interface Signals" on page 23
- "Parallel Control Register Port Signals" on page 29
- "Reset and Test Signals" on page 31
- "Firmware Clock Signals" on page 33
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- "Context Restore Interface Signals" on page 37
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- "Per Transceiver Signals" on page 40
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- "Transmitter Equalization (Per-Lane Transmit Interface Group) Signals" on page 133
- "Transmitter Test Signals" on page 135

1.1 Resistor Tune (Common Block Group) Signals

Table 1-1 Resistor Tune (Common Block Group) Signals

Port Name	I/O	Description
txdn_term_offset[8:0]	I	Offset for TX down termination Specifies an additional fixed offset to calibrated TX down termination value. This is a signed input with 2's complement encoding. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txup_term_offset[8:0]	I	Offset for TX up termination Specifies an additional fixed offset to calibrated TX up termination value. This is a signed input with 2's complement encoding. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
res_ack_in	I	Arbitration lines for sharing single reference resistor (res_ack_in) Arbitration lines used for sharing a single-reference resistor among multiple cores **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
res_ack_out	O	Arbitration lines for sharing single reference resistor (res_ack_out) Arbitration lines used for sharing a single reference resistor among multiple cores **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-1 Resistor Tune (Common Block Group) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** High
res_req_in	I	Arbitration lines for sharing single reference resistor (res_req_in) Arbitration lines used for sharing a single-reference resistor among multiple cores **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
res_req_out	O	Arbitration lines for sharing single reference resistor (res_req_out) Arbitration lines used for sharing a single reference resistor among multiple cores **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rtune_ack	O	Resistor tune acknowledge Indicates that a resistor tune has completed **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rtune_req	I	Resistor tune request Assertion triggers a resistor tune request (if one is not already in progress) Resistor tuning is done automatically by PHY during POR and results are stored in Raw PCS AON registers. So external rtune_req is not needed for normal operation. In normal operation, if rtune_req is triggered, then customer will need to copy the results in Raw PCS AON registers. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rx_term_offset[4:0]	I	Offset for RX termination Specifies an additional fixed offset to calibrated RX termination value. This is a signed input with 2's complement encoding. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-1 Resistor Tune (Common Block Group) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A

1.2 Power Supply Signals

Table 1-2 Power Supply Signals

Port Name	I/O	Description
vddcore	IO	SoC core voltage for level shifters on the ASIC boundary inside the PMA hard macro. For more information, refer to the "Power Gating Connectivity for External PMA Switch" in Chapter 5, "PHY Usage and Configuration". **Voltage Range:** For voltage range information, see "Power Supply Limits" in the "Physical Implementation" chapter. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
vp	IO	Low voltage supply Off-chip analog power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
vpdig	IO	Digital power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
vph	IO	I/O Voltage supply An off-chip analog high voltage power supply **Voltage Range:** For more information about the voltage range, see the "Product Overview" chapter. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
vptxX (for X = 0; X <= 1)	IO	Analog transmitter termination voltage supply Off-chip analog power supply for TX lane X

Table 1-2 Power Supply Signals (continued)

Port Name	I/O	Description
		Voltage Range: For more information about the voltage range, see the "Product Overview" chapter. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
vsscore_X (for X = 0; X <= 1)	IO	Ground; used by internal core logic Ground used by the core; this input is used for ESD protection. **Voltage Range:** 0 V **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
gd	IO	Ground This signal can be tied to chip ground plane in package, but must not be connected to ground distribution network on-die. **Voltage Range:** 0 V **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
nominal_vp_sel[1:0]	I	VP nominal selection All settings of this pin are reserved for future use. For this version, tie the signal low to 2'b00. This signal indicates the supplied VP voltage level: ■ 2'b00: Reserved for future use ■ 2'b01: Reserved for future use ■ 2'b10: Reserved for future use ■ 2'b11: Reserved for future use **Note:**This input can be changed only when phy_reset is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
nominal_vph_sel[1:0]	I	VPH nominal selection All settings of this pin are reserved for future use. For this version, tie the signal low to 2'b00. This signal indicates the supplied VP voltage level: ■ 2'b00: Reserved for future use ■ 2'b01: Reserved for future use ■ 2'b10: Reserved for future use ■ 2'b11: Reserved for future use **Note:**This input can be changed only when phy_reset is asserted. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-2 Power Supply Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A

1.3 Analog I/O Signals

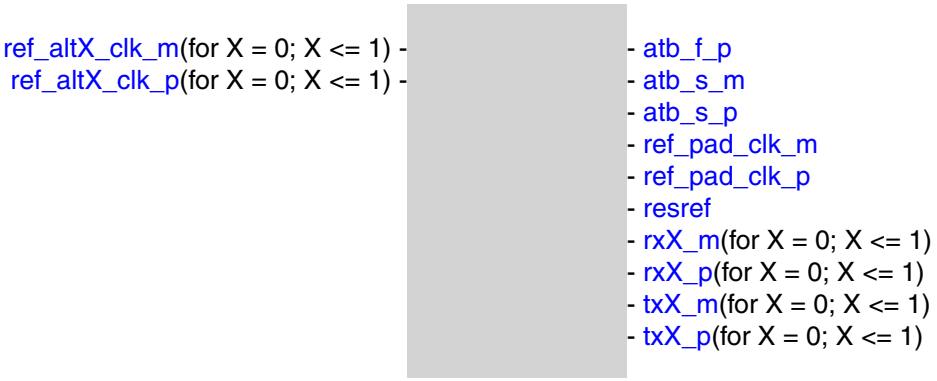


Table 1-3 Analog I/O Signals

Port Name	I/O	Description
atb_f_p	IO	Analog test bus force Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
atb_s_m	IO	Analog test bus sense (atb_s_m) Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
atb_s_p	IO	Analog test bus sense (atb_s_p) Reserved; leave unconnected when integrating the PHY **Voltage Range:** 0-vph **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
ref_altX_clk_m (for X = 0; X <= 1)	I	Low-swing differential input clock pair internal pin (ref_alt0_clk_m) The selection is based on whether the reference clock is provided to the core via an on-die distribution network or directly to the core from off- die. The pair that is selected depends on the value of the refa/b/repeat_clk_sel input. **Voltage Range:** Refer to the "Reference Clock Amplitude Requirements" table in the databook **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
ref_altX_clk_p	I	Low-swing differential input clock pair internal pin (ref_alt0_clk_p)

Table 1-3 Analog I/O Signals (continued)

Port Name	I/O	Description
(for X = 0; X <= 1)		The selection is based on whether the reference clock is provided to the core via an on-die distribution network or directly to the core from off- die. The pair that is selected depends on the value of the refa/b/repeat_clk_sel input. **Voltage Range:** Refer to the "Reference Clock Amplitude Requirements" table in the databook **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
ref_pad_clk_m	IO	Low-Swing differential input clock pair with pad (ref_pad_clk_m) The selection is based on whether the reference clock is provided to the core through an on-die distribution network or directly to the core from off-die. The pair that is selected depends on the value of the refa/b_clk_sel inputs. **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
ref_pad_clk_p	IO	Low-Swing differential input clock pair with pad (ref_pad_clk_p) The selection is based on whether the reference clock is provided to the core through an on-die distribution network or directly to the core from off-die. The pair that is selected depends on the value of the refa/b_clk_sel inputs. **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
resref	IO	Reference resistor connection Attach a 200 ohm 1% 100-ppm/C precision resistor-to- ground on the board. For information about resref, see the "Physical-Level Implementation" and "Board- and Package-Level Implementation" chapters. **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_m (for X = 0; X <= 1)	IO	High-Speed differential receive pair (rxX_m) Receive differential pair for lane X **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-3 Analog I/O Signals (continued)

Port Name	I/O	Description
rxX_p (for X = 0; X <= 1)	IO	High-Speed differential receive pair (rxX_p) Receive differential pair for lane X **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_m (for X = 0; X <= 1)	IO	High-Speed differential transmit pair (txX_m) Transmit differential pair for lane X **Voltage Range:** 0-vptxX **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_p (for X = 0; X <= 1)	IO	High-Speed differential transmit pair (txX_p) Transmit differential pair for lane X **Voltage Range:** 0-vptxX **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

1.4 Boundary Scan Port Signals

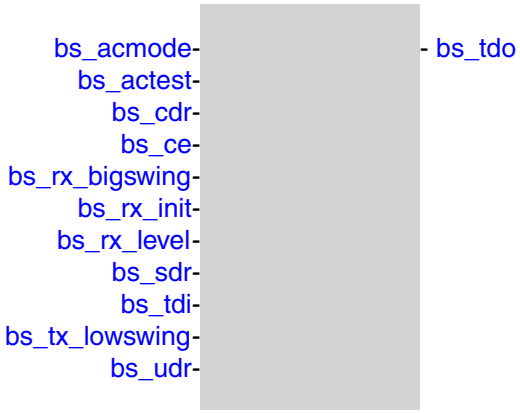


Table 1-4 Boundary Scan Port Signals

Port Name	I/O	Description
bs_acmode	I	Boundary Scan AC mode (Figure 51, IEEE 1149.6) Enables Boundary Scan AC mode **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
bs_actest	I	Boundary Scan AC test signal (Figure 51, IEEE 1149.6) The AC Test Signal is generated by the user logic when AC EXTEST instruction is decoded. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
bs_cdr	I	Boundary Scan Clock-DR (IEEE 1149.1) Clocks the Boundary Scan capture registers, in either Scan or Capture modes **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
bs_ce	I	Boundary scan compliance enable signal Enables the Boundary Scan Transmitter and Receiver register cells on all lanes. This input must be driven from a sequential element and not from any combinatorial logic. Any glitch on this input causes the PHY to go into reset state. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-4 Boundary Scan Port Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** High
bs_rx_bigswing	I	RX Boundary Scan big swing Enables RX ACJTAG big-swing mode **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
bs_rx_init	I	RX Boundary Scan initialize signal This signal is similar to the 'Init Clk' referenced in test receiver diagrams in the ACJTAG specification (Figure 48, IEEE 1149.6). While this signal is high, the hysteresis in the receiver comparator is reset. The falling edge of this signal must occur at appropriate times, as specified in the ACJTAG specification. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
bs_rx_level[4:0]	I	ACJTAG Receiver sensitivity level control Sets the internal thresholds for the ACJTAG receiver. The ACJTAG threshold level must be set to trigger for the expected amplitude for the given protocol. For threshold levels, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
bs_sdr	I	Boundary Scan Shift-DR (IEEE 1149.1) Enables shifting of the Boundary Scan capture registers **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** bs_cdr **Active State:** High
bs_tdi	I	Boundary Scan test data in Boundary scan shift register input signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** bs_cdr **Active State:** High
bs_tdo	O	Boundary Scan test data out Boundary scan shift register output signal

Table 1-4 Boundary Scan Port Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** bs_cdr **Active State:** High
bs_tx_lowswing	I	TX Boundary Scan low swing Enables TX JTAG low-swing mode (otherwise, it is in full- swing mode) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
bs_udr	I	Boundary Scan Update-DR (IEEE 1149.1) Clocks the Boundary Scan update registers **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

1.5 Scan Interface Signals

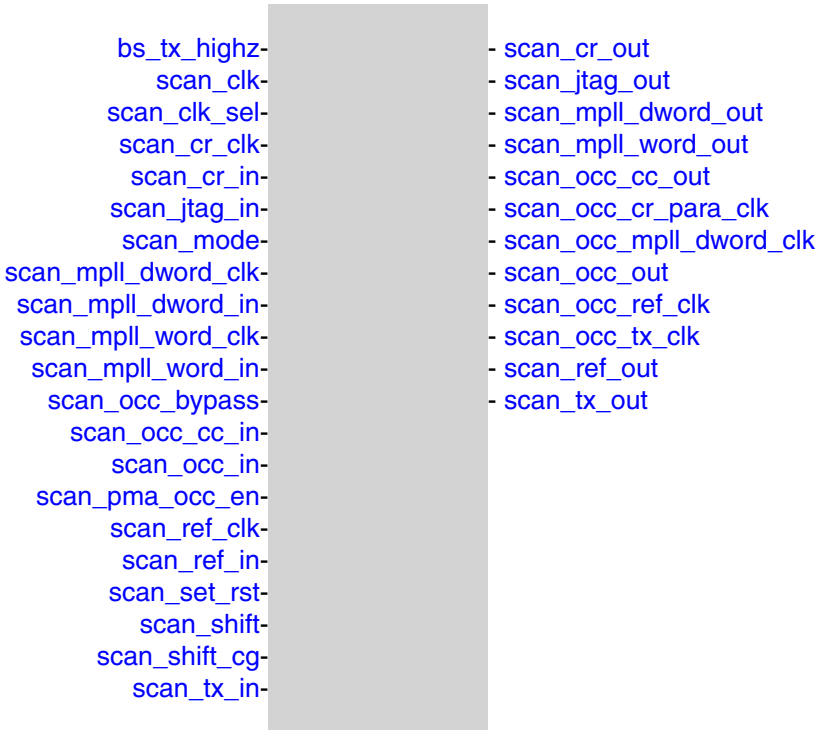


Table 1-5 Scan Interface Signals

Port Name	I/O	Description
bs_tx_highz	I	Support HighZ boundary scan All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Exists:** Always **Synchronous To:** N/A **Active State:** High
scan_clk	I	Stuck-at scan clock This input is selected when scan_clk_sel input is 1. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_clk_sel	I	This input selects the scan clock mode of the PHY ■ 0: At speed scan clock ■ 1: Stuck-at scan clock (scan_clk input) **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-5 Scan Interface Signals (continued)

Port Name	I/O	Description
		Synchronous To: N/A **Active State:** N/A
scan_cr_clk	I	Scan clock per clock domain (scan_cr_clk) Test Clock muxed into the clock pin of registers during scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_cr_in[299:0]	I	Scan chain inputs per clock domain (scan_cr_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_cr_out[299:0]	O	Scan chain outputs per clock domain (scan_cr_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_jtag_in[2:0]	I	Scan chain inputs per clock domain (scan_jtag_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_jtag_out[2:0]	O	Scan chain outputs per clock domain (scan_jtag_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_mode	I	Scan mode enable **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
scan_mpll_dword_clk	I	Scan clock per clock domain (scan_mpll_dword_clk)

Table 1-5 Scan Interface Signals (continued)

Port Name	I/O	Description
		Test Clock muxed into the clock pin of registers during scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_mpll_dword_in[83:0]	I	Scan chain inputs per clock domain (scan_mpll_dword_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_mpll_dword_out[83:0]	O	Scan chain outputs per clock domain (scan_mpll_dword_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_mpll_word_clk	I	Scan clock per clock domain (scan_mpll_word_clk) Test Clock muxed into the clock pin of registers during scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_mpll_word_in[6:0]	I	Scan chain inputs per clock domain (scan_mpll_word_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_mpll_word_out[6:0]	O	Scan chain outputs per clock domain (scan_mpll_word_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High

Table 1-5 Scan Interface Signals (continued)

Port Name	I/O	Description
scan_occ_bypass	I	This pin enables dynamic selection between shift and capture; when scan_shift is high the OCC controller selects scan_clk for shift, and when scan_shift is low, it selects the MPLL fast clocks for capture at speed. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_cc_in	I	OCC clock chain scan input, clock domain (scan_clk) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_cc_out	O	OCC clock chain scan output, clock domain (scan_clk) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_cr_para_clk	O	OCC controller cr_para clock output; when scan_shift is high, this clock is free running at the same frequency of scan_clk, and when scan_shift is low, this clock toggles twice at cr_para_clk frequency, one for launch and another for capture. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_in[2:0]	I	Scan chain inputs per clock domain (scan_occ_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_occ_mpll_dword_clk	O	OCC controller mpll dword clock output; when scan_shift is high, this clock is free running at the same frequency of scan_clk, and when scan_shift is low, this clock toggles twice at mpll_dword_clk frequency, one for launch and another for capture. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_out[2:0]	O	Scan chain outputs per clock domain (scan_occ_out)

Table 1-5 Scan Interface Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_occ_ref_clk	O	OCC controller reference clock output; when scan_shift is high, this clock is free running at the same frequency of scan_clk, and when scan_shift is low, this clock toggles twice at ref{a,b}_clk frequency, one for launch and another for capture. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_occ_tx_clk	O	OCC controller TX clock output; when scan_shift is high, this clock is free running at the same frequency of scan_clk, and when scan_shift is low, this clock toggles twice at tx_clk frequency, one for launch and another for capture. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_pma_occ_en	I	Scan OCC enable This signal enables the on-chip clocking (OCC) latch and hold circuit to latch the control signals going to the prescaler and MPLLs. **Note:**Assert this signal only if scan_mode is 1'b0 and the prescaler and MPLLs are powered up. For more information about OCC, see the OCC section. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_ref_clk	I	Scan clock per clock domain (scan_ref_clk) Test Clock muxed into the clock pin of registers during scan operation. All scan clock inputs are assumed to be asynchronous to each other. All scan clock inputs can be active at the same time during scan shift operation; however, only one scan clock can be active at any time during the scan capture cycle. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
scan_ref_in[53:0]	I	Scan chain inputs per clock domain (scan_ref_in)

Table 1-5 Scan Interface Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_ref_out[53:0]	O	Scan chain outputs per clock domain (scan_ref_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_set_rst	I	Scan set reset **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_shift	I	Scan shift **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_shift_cg	I	Scan shift for clock gates **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_tx_in[15:0]	I	Scan chain inputs per clock domain (scan_tx_in) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High
scan_tx_out[15:0]	O	Scan chain outputs per clock domain (scan_tx_out) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** scan_CLKDOMAIN_clk **Active State:** High

1.6 Parallel Control Register Port Signals

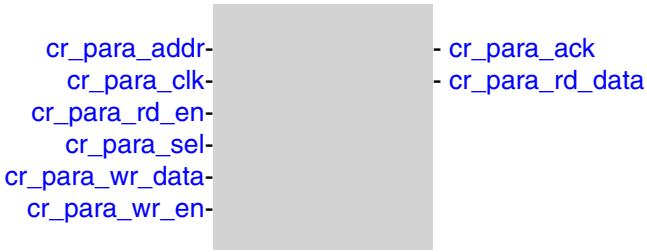


Table 1-6 Parallel Control Register Port Signals

Port Name	I/O	Description
cr_para_ack	O	Control Register (CR) read/write acknowledgment Acknowledge signal indicating completion of the CR parallel read/write access. For more information about operation during SRAM update, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** High
cr_para_addr[15:0]	I	Control Register address **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** N/A
cr_para_clk	I	Control Register clock **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
cr_para_rd_data[15:0]	O	Control Register read data **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** N/A
cr_para_rd_en	I	Control Register read enable Reads from the referenced address on cr_para_addr and provides the data on cr_para_rd_data **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** High

Table 1-6 Parallel Control Register Port Signals (continued)

Port Name	I/O	Description
cr_para_sel	I	Control Register parallel interface select Controls selection between JTAG and CR interfaces: ■ 0: JTAG ■ 1: Control Register This input can only be changed when the cr_para_clk and jtag_tck clock inputs are disabled. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
cr_para_wr_data[15:0]	I	Control Register write data **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** N/A
cr_para_wr_en	I	Control Register write enable Writes the data on cr_para_wr_data to the referenced address on cr_para_addr **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** cr_para_clk **Active State:** High

1.7 Reset and Test Signals

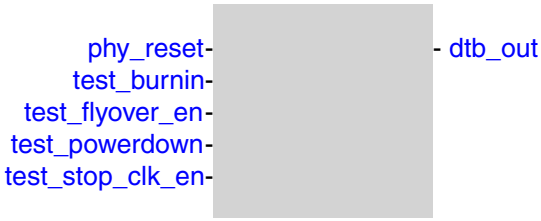


Table 1-7 Reset and Test Signals

Port Name	I/O	Description
dtb_out[1:0]	O	Digital test bus output Used in debug mode for probing the values of various internal signals in the PHY **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
phy_reset	I	PHY reset Asynchronously resets the core and all state machines. Asserting phy_reset resets all sequential elements in the design, including control registers. As a result, any register programming needs to be redone after phy_reset is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
test_burnin	I	All circuits activator Activates all circuitry in the PHY for burn-in testing. This signal must not be asserted while test_powerdown signal is set to 1'b1. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
test_flyover_en	I	Enable flyover test mode Flyover test mode enables direct access to the TX and RX bump pads, from the txX_flyover_data_p/m and rxX_flyover_data_p/m I/Os This mode can be enabled in either normal mode or scan mode. Assertion or de-assertion of this signal may result in a short pulse (glitch) on the outputs that are enabled by this signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
test_powerdown	I	All circuits power-down control

Table 1-7 Reset and Test Signals (continued)

Port Name	I/O	Description
		Powers down all circuitry in the PHY for IDDQ testing **Note:**The PHY is not functional in this mode and must be reset after this signal is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
test_stop_clk_en	I	Stop clock test mode enable Enables scan dump functionally. For additional information, see the "Scan Dump" section. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.8 Firmware Clock Signals

Table 1-8 Firmware Clock Signals

Port Name	I/O	Description
fw_clk	I	Input clock used by PHY firmware. **Note:**This clock must be running as long as fw_clk_ack is high. The fw_clk frequency must be: ■ x1, x2, x4: >= 100 MHz and <= 175 MHz ■ x6: >= 125 MHz and <= 175 MHz ■ x8: >= 150 MHz and <= 175 MHz Drive low if unused. This input should not exceed 175 MHz regardless of fw_clk_ack=0 or 1. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
fw_clk_ack	I	Firmware clock acknowledge fw_clk_ack must be asserted when fw_clk is running. **Note:**Drive low if fw_clk is unused. Once asserted, this input must stay high until fw_clk_req is de-asserted. Once fw_clk_req is de-asserted, this input must be de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
fw_clk_range[3:0]	I	Input firmware clock frequency range Specifies the frequency range of the input firmware clock. The code mapping of fw_clk_range is as follows: ■ 4'b0000: 13.5 - 26 MHz (not supported) ■ 4'b0001: 26.1 - 52 MHz (not supported) ■ 4'b0010: 52.1 - 78 MHz (not supported) ■ 4'b0011: 78.1 - 104 MHz (only >=100 is supported) ■ 4'b0100: 104.1 - 130 MHz ■ 4'b0101: 130.1 - 156 MHz ■ 4'b0110: 156.1 - 175 MHz ■ 4'b0111 and above: NA Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-8 Firmware Clock Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A
fw_clk_req	O	Firmware clock request, asserted when the PHY can use fw_clk instead of the internal reference clock **Note:**if multi_perst_en input is set to 1, then this output may glitch (because it is combinatorially generated from refa/b_clk_en/ack signals which may assert/de-assert asynchronously for different links) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.9 JTAG Port Signals

Table 1-9 JTAG Port Signals

Port Name	I/O	Description
jtag_tck	I	JTAG clock JTAG state machine clock signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
jtag_tdi	I	JTAG input port JTAG test data input signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** jtag_tck **Active State:** N/A
jtag_tdo	O	JTAG output port JTAG test data output signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** jtag_tck **Active State:** N/A
jtag_tdo_en	O	JTAG output enable JTAG test data output enable signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** jtag_tck **Active State:** High
jtag_tms	I	JTAG state machine control JTAG test mode select signal **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** jtag_tck **Active State:** High
jtag_trst_n	I	JTAG reset JTAG state machine reset signal. While this signal is held low, the state machine is held in test logic reset state.

Table 1-9 JTAG Port Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** Low

1.10 Context Restore Interface (for X = 0; X <= 1) Signals

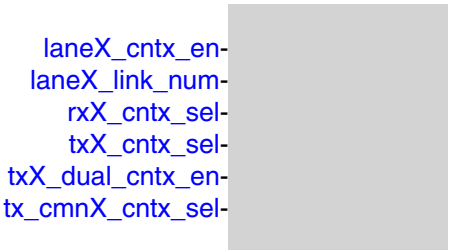


Table 1-10 Context Restore Interface (for X = 0; X <= 1) Signals

Port Name	I/O	Description
laneX_cntx_en	I	Per-lane context enable input Indicates if the context restore feature is enabled for this lane or not. After initial exit from the PHY reset, if laneX_cntx_en[0] is asserted, the context restore is enabled for this specific lane. Change this signal only during phy_reset. Detailed description of this feature is in Chapter 5. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
laneX_link_num[3:0]	I	Per-lane context link number input (when laneX_cntx_en=1) This input determines which link the current lane belongs to. This signal ensure that all lanes with the same laneX_link_num apply new context (configuration settings) simultaneously, thus reducing rate change latency. **Note:**The PHY waits for txX_req and rxX_req from all lanes in the link before executing context restore in a rate change scenario. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_cntx_sel[7:0]	I	Per-lane RX context select input (when laneX_cntx_en=1) This input specifies the ID of the context (PHY RX configuration stored in SRAM/ROM) that needs to be restored for this lane RX. A change in rxX_cntx_sel along with rxX_req triggers context restore of configurations related to RX for the lane. Otherwise, everything remains the same. The rxX_cntx_sel for all the lanes in the same link needs to be stable and the same, before rx_reset of any lane in the link gets de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_cntx_sel[7:0]	I	Per-lane TX context select input (when laneX_cntx_en=1) This input specifies the ID of the context (PHY TX configuration stored in SRAM/ROM) that needs to be restored for this lane TX.

Table 1-10 Context Restore Interface (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		A change in txX_cntx_sel along with txX_req triggers context restore of configurations related to TX for the lane. Otherwise, everything remains the same. The txX_cntx_sel for all the lanes in the same link needs to be stable and the same, before tx_reset of any lane in the link gets de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_dual_cntx_en	I	Per-lane TX dual context enable input (when laneX_cntx_en=1) When this signal is asserted, both MPLLA and MPLLB configurations are restored. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
tx_cmnX_cntx_sel[7:0]	I	Per-lane TX cmn context select input (when laneX_cntx_en=1) This input specifies the ID of the context (PHY cmn configuration stored in SRAM/ROM) that needs to be restored for this lane TX cmn. A change in tx_cmnX_cntx_sel along with txX_req triggers context restore of configurations related to SUP/CMN blocks of the PHY, such as MPLLA, MPLLB, and RTUNE. So, to avoid conflicting settings of these shared blocks among different lanes, the shared MPLL and external RTUNE trigger must be in disabled state when this input is changed. The tx_cmnX_cntx_sel for all the lanes in the same link needs to be stable and the same, before tx_reset of any lane in the link gets de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.11 PERST control inputs Signals

laneX_perst_n(for X = 0; X <= 1) -
multi_perst_en-

Table 1-11 PERST control inputs Signals

Port Name	I/O	Description
laneX_perst_n (for X = 0; X <= 1)	I	Per lane PERST input for PCIe/CXL applications. For non-PCIe/CXL applications, keep this input set to inactive value **Exists:** Always **Synchronous To:** Asynchronous **Active State:** Low
multi_perst_en	I	This input should be tied low and please contact synopsys if it needs to be set to 1. This input should be set to 1 if PERST can happen on different links independently. Please contact synopsys at this case. Note: Any change in this input must be followed by phy_reset assertion. **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.12 Per Transceiver (for X = 0; X <= 1) Signals

laneX_ref_sel-

Table 1-12 Per Transceiver (for X = 0; X <= 1) Signals

Port Name	I/O	Description
laneX_ref_sel	I	Lane reference clock selection It selects the refa or refb reference clock for digital logic. This input can be changed during phy_reset only. ■ 0: Select refa clock ■ 1: Select refb clock This input can be driven to 0 or 1 if PHY inputs refa_clk_sel=refb_clk_sel and refa_clk_en=refb_clk_en. Otherwise, drive this input based on the configuration settings in appendix A. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

1.13 Loopback Control (for X = 0; X <= 1) Signals

laneX_rx2tx_par_lb_en-
laneX_tx2rx_ser_lb_en-

Table 1-13 Loopback Control (for X = 0; X <= 1) Signals

Port Name	I/O	Description
laneX_rx2tx_par_lb_en	I	Parallel (RX to TX) loopback enable When this signal is asserted, recovered parallel data from the receiver is looped back to the transmit serializer. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
laneX_tx2rx_ser_lb_en	I	Analog serial (TX to RX) loopback enable When this signal is asserted, data from the TX pre-driver is looped back to the RX analog front end (AFE); signal detection (SIGDET) is bypassed and based on the txX_data_en input so that rxX_sigdet_lf = txX_data_en. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.14 MPLL Control Signals

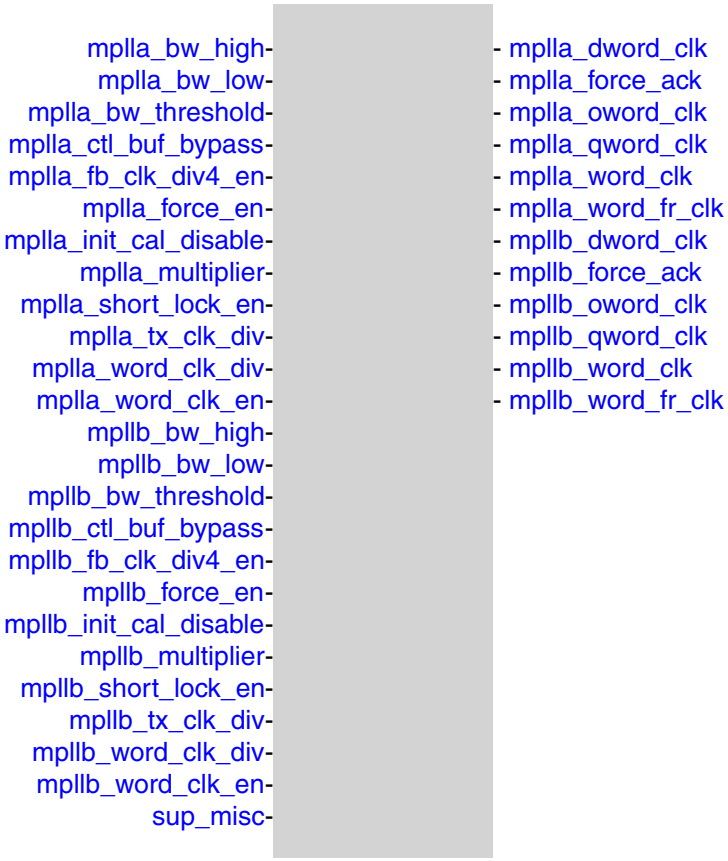


Table 1-14 MPLL Control Signals

Port Name	I/O	Description
mplla_bw_high[15:0]	I	MPLLA bandwidth control (mplla_bw_high) ■ 15:13: Reserved ■ 12: Enables additional cap on loop filter proportional net. Can be set for additional jitter filtering. Do not set for high bandwidth PCIe2 cases. ■ 11: Reserved ■ 10:4: Charge-pump current setting ❑ 10:9 = 00 + 8:4 = 00001 to 11111 gives 25 uA to 750 uA current setting range ❑ 10:9 = 01 + 8:4 = 00001 to 11111 gives 12.5 uA to 380 uA current setting range ❑ 10:9 = 10 + 8:4 = 00001 to 11111 gives 8.6 uA to 265 uA current setting range ❑ 10:9 = 11 + 8:4 = 00001 to 11111 gives 6.4 uA to 200 uA current setting range ■ 3:0: Loop filter resistor value ❑ 0001: 4 kohm to -- 1111 : 60 kohm**Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_bw_low[15:0]	I	MPLLA bandwidth control (mplla_bw_low) ■ 15:13: Reserved ■ 12: Enables additional cap on loop filter proportional net. Can be set for additional jitter filtering. Do not set for high bandwidth PCIe2 cases. ■ 11: Reserved ■ 10:4: Charge-pump current setting ❑ 10:9 = 00 + 8:4 = 00001 to 11111 gives 25 uA to 750 uA current setting range ❑ 10:9 = 01 + 8:4 = 00001 to 11111 gives 12.5 uA to 380 uA current setting range ❑ 10:9 = 10 + 8:4 = 00001 to 11111 gives 8.6 uA to 265 uA current setting range ❑ 10:9 = 11 + 8:4 = 00001 to 11111 gives 6.4 uA to 200 uA current setting range ■ 3:0: Loop filter resistor value ❑ 0001: 4 kohm to -- 1111: 60 kohm**Voltage Range:** 0-vddcore **Exists:** Always

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A
mplla_bw_threshold[7:0]	I	MPLLA bandwidth threshold (mplla_bw_threshold) Bandwidth threshold control for MPLLA. This value controls the threshold that the PHY uses to select between the mplla_bw_low and mplla_bw_high values. Set this input as per the tables provided in the "PHY Usage and Configuration" for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_ctl_buf_bypass	I	MPLLA control buffer bypass (mplla_clt_buf_bypass) Control buffer bypass enable for MPLLA. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_dword_clk	O	Double-Rate word clock (mplla_dword_clk) Double-symbol rate (x2) clock derived from MPLLA **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mplla_fb_clk_div4_en	I	MPLLA DIV clock output frequency divider control Additional frequency division factor used to generate mpll(a,b)_div_clk clock output from the MPLL(A,B) VCO. This signal controls whether the MPLL VCO is divided by 2 (when mpll(a,b)_fb_clk_div4_en is de-asserted) or 4 (when mpll(a,b)_fb_clk_div4_en is asserted) before being divided down by mpll(a,b)_div_multiplier. This input can only be changed when the txX_pll_en inputs for all lanes are de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_force_ack	O	MPLLA Force Acknowledge

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Acknowledge output for mpll(a,b)_force_en. This output asserts when the MPLL is powered up in response to mpll(a,b)_force_en assertion. This output de-asserts as a response to mpll(a,b)_force_en de-assertion, and at that point the MPLL is powered down if txX_pll_en of other lanes is de-asserted. **Note:**A four-way level handshake must be followed between mpll(a,b)_force_en and mpll(a,b)_force_ack. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_force_en	I	MPLLA force enable When asserted, the corresponding MPLL is forced to be powered up, irrespective of the txX_pll_en input. This input is used for applications where a free-running MPLL clock output is required. The mpll(a,b)_force_en signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. If MPLL is not powered up, Synopsys recommends that you follow the txX_pll_en controls as described in the "PHY Usage and Configuration" chapter. **Note:** ■ A four-way level handshake must be followed between mpll(a,b)_force_en and mpll(a,b)_force_ack. ■ This signal must be de-asserted before entering L1 substates. It must also be asserted no earlier than refa_clk_en when exiting L1 substates. ■ This signal must be asserted at the same time as, or after asserting txX_req when asserting it as part of an MPLL recalibration event (either mpll(a,b)_recal_bank_sel or mpll(a,b)_recal_force_en assertion). ■ This input must be de-asserted during PCIe PERST event. ■ Disclaimer: When Context Restore feature is enabled, the assertion of this signal must be delayed after the context restore engine has finished its operation (i.e. Tx req/ack four-way handshake initiated by the context restore engine fully terminated). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
mplla_init_cal_disable	I	MPLLA calibration disable control At power on, the internal PHY firmware will first calibrate both PLLs before powering down the unused one, unless mplla_force_en is asserted to force it to remain powered up. If context restore is enabled, only one of the PLLs will be calibrated based on the protocol configuration, unless txX_dual_cntx_en is asserted, where in this case, both PLLs are calibrated. The PLL configuration must be valid for calibration to complete in the specified time. **Note:**This signal will only disable initial power up calibration of the specified MPLL. The MPLL can still be enabled at a late time if required. ■ This is an asynchronous input tied high when the PLL is not used. ■ mplla_init_cal_disable = 1 disables mplla power up calibration ■ mpllb_init_cal_disable = 1 disables mpllb power up calibration ■ Asserting mplla_init_cal_disable allows invalid configurations if the PLL is unused. **Voltage Range:** 0-vpdig **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_multiplier[11:0]	I	MPLLA frequency multiplier control Multiplies the reference clock to a frequency suitable for intended operating speed. The reference clocks used by the MPLLs are generated after applying ref_clk_mpll(a,b)_div divisions on the ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p/ref_alt0/1_clk_m. **Note:** ■ These inputs can be changed only when the MPLL is powered down. ■ These inputs have 2 fractional bits and 10 integer bits, so the intended multiplication factor must be multiplied by 4. For example, for a multiplier value of 125, the bus must be set to 12'd500. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_oword_clk	O	Octuple-Rate word clock (mplla_oword_clk) Octuple-symbol rate (x8) clock derived from MPLLA **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
mplla_qword_clk	O	Quadruple-Rate word clock (mplla_qword_clk) Quadruple-symbol rate (x4) clock derived from MPLLA **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mplla_short_lock_en	I	MPLLA short lock enable This input indicates that a shared reference clock configuration is being used. When asserted, the MPLL(A,B) locking time will be reduced. **Note:**This signal can be changed only when the txX_pll_en input for all lanes is de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_tx_clk_div[3:0]	I	MPLLA TX Clock Divider Sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 4'b0000: div16 ■ 4'b0001: div1 ■ 4'b0010: div2 ■ 4'b0011: div3 ■ 4'b0100: div4 ■ 4'b0101: div5 ■ 4'b0110: div6 ■ 4'b0111: Reserved ■ 4'b1000: div8 ■ 4'b1001: div9 ■ 4'b1010: div10 ■ 4'b1011: Reserved ■ 4'b1100: Reserved ■ 4'b1101: Reserved ■ 4'b1110: Reserved ■ 4'b1111: div15 **Note:**Change this signal only when the txX_pll_en input for all lanes is de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_word_clk	O	Word clock (mplla_word_clk)

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Symbol rate clock derived from MPLLA **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mplla_word_clk_div[1:0]	I	MPLLA word clock divider control (mplla_word_clk_div) Controls the MPLLA word clock dividers for generating the mplla_word_clk output. These inputs can be changed only when the txX_pll_en input is de-asserted. ■ 2'b00: Divide by 4 ■ 2'b01: Divide by 5 ■ 2'b10: Divide by 8 ■ 2'b11: Divide by 10 This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_word_clk_en	I	MPLLA word clock output enable This input can be set to 0 in functional mode. In scan mode, set it to 1, if PMA clock output is required for at-speed scan. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
mplla_word_fr_clk	O	Free running mplla_word_clk This clock is a free-running version of the mplla_word_clk and is intended for use during at-speed scan capture when operating in on-chip clocking (OCC) mode. **Note:**This signal may glitch when MPLLA is disabled. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
mpllb_bw_high[15:0]	I	MPLLB bandwidth control (mpllb_bw_high) ■ 15:13: Reserved ■ 12: Enables additional cap on loop filter proportional net. Can be set for additional jitter filtering. Do not set for high BW PCIE2 cases. ■ 11: Reserved ■ 10:4: Charge-pump current setting ❑ 10:9 = 00 + 8:4 = 00001 to 11111 gives 25 uA to 750 uA current setting range ❑ 10:9 = 01 + 8:4 = 00001 to 11111 gives 12.5 uA to 380 uA current setting range ❑ 10:9 = 10 + 8:4 = 00001 to 11111 gives 8.6 uA to 265 uA current setting range ❑ 10:9 = 11 + 8:4 = 00001 to 11111 gives 6.4 uA to 200 uA current setting range ■ 3:0 : loop filter resistor value ❑ 0001: 4 kohm to -- 1111: 60 kohm**Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_bw_low[15:0]	I	MPLLB bandwidth control (mpllb_bw_low) ■ 15:13: Reserved ■ 12: Enables additional cap on loop filter proportional net. Can be set for additional jitter filtering. Do not set for high BW PCIE2 cases. ■ 11: Reserved ■ 10:4: Charge-pump current setting ❑ 10:9 = 00 + 8:4 = 00001 to 11111 gives 25 uA to 750 uA current setting range ❑ 10:9 = 01 + 8:4 = 00001 to 11111 gives 12.5 uA to 380 uA current setting range ❑ 10:9 = 10 + 8:4 = 00001 to 11111 gives 8.6 uA to 265 uA current setting range ❑ 10:9 = 11 + 8:4 = 00001 to 11111 gives 6.4 uA to 200 uA current setting range ■ 3:0: Loop filter resistor value ❑ 0001: 4 kohm to -- 1111: 60 kohm**Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_bw_threshold[7:0]	I	MPLLB bandwidth threshold (mpllb_bw_threshold)

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Bandwidth threshold control for MPLLB. This value controls the threshold that the PHY uses to select between the mpllb_bw_low and mpllb_bw_high values. Set this input as per the tables provided in the "PHY Usage and Configuration" for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_ctl_buf_bypass	I	MPLLB control buffer bypass (mpllb_clt_buf_bypass) Control buffer bypass enable for MPLLB. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_dword_clk	O	Double-Rate word clock (mpllb_dword_clk) Double-symbol rate (x2) clock derived from MPLLB **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mpllb_fb_clk_div4_en	I	MPLLB DIV clock output frequency divider control Additional frequency division factor used to generate mpll(a,b)_div_clk clock output from the MPLL(A,B) VCO. This signal controls whether the MPLL VCO is divided by 2 (when mpll(a,b)_fb_clk_div4_en is deasserted) or 4 (when mpll(a,b)_fb_clk_div4_en is asserted) before being divided down by mpll(a,b)_div_multiplier. This input can only be changed when the txX_pll_en inputs for all lanes are de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_force_ack	O	MPLLB Force Acknowledge Acknowledge output for mpll(a,b)_force_en.

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		This output asserts when the MPLL is powered up in response to mpll(a,b)_force_en assertion. This output de-asserts as a response to mpll(a,b)_force_en de-assertion, and at that point the MPLL is powered down if txX_pll_en of other lanes is de-asserted. **Note:**A four-way level handshake must be followed between mpll(a,b)_force_en and mpll(a,b)_force_ack. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_force_en	I	MPLLB force enable When asserted, the corresponding MPLL is forced to be powered up, irrespective of the txX_pll_en input. This input is used for applications where a free-running MPLL clock output is required. The mpll(a,b)_force_en signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. If MPLL is not powered up, Synopsys recommends that you follow the txX_pll_en controls as described in the "PHY Usage and Configuration" chapter. **Note:** ■ This signal must be de-asserted before entering L1 substates. It must also be asserted no earlier than refb_clk_en when exiting L1 substates. ■ This signal must be asserted at the same time as, or after asserting txX_req when asserting it as part of an MPLL recalibration event (either mpll(a,b)_recal_bank_sel or mpll(a,b)_recal_force_en assertion). ■ This input must be de-asserted during PCIe PERST event. ■ Disclaimer: When Context Restore feature is enabled, the assertion of this signal must be delayed after the context restore engine has finished its operation (i.e. Tx req/ack four-way handshake initiated by the context restore engine fully terminated). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
mpllb_init_cal_disable	I	MPLLB calibration disable control At power on, the internal PHY firmware will first calibrate both PLLs before powering down the unused one, unless mpllb_force_en is asserted to force it to remain powered up. If context restore is enabled, only one of the PLLs will be calibrated based on the protocol configuration, unless txX_dual_cntx_en is asserted, where in this case, both PLLs are calibrated. The PLL configuration must be valid for calibration to complete in the specified time. **Note:**This signal will only disable initial power up calibration of the specified MPLL. The MPLL can still be enabled at a late time if required. ■ This is an asynchronous input tied high when the PLL is not used. ■ mplla_init_cal_disable = 1 disables mplla power up calibration ■ mpllb_init_cal_disable = 1 disables mpllb power up calibration ■ Asserting mpllb_init_cal_disable allows invalid configurations if the PLL is unused. **Voltage Range:** 0-vpdig **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_multiplier[11:0]	I	MPLLB frequency multiplier control Multiplies the reference clock to a frequency suitable for intended operating speed. The reference clocks used by the MPLLs are generated after applying ref_clk_mpll(a,b)_div divisions on the ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p/ref_alt0/1_clk_m. **Note:** ■ These inputs can be changed only when the MPLL is powered down. ■ These inputs have 2 fractional bits and 10 integer bits, so the intended multiplication factor must be multiplied by 4. For example, for a multiplier value of 125, the bus must be set to 12'd500. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_oword_clk	O	Octuple-Rate word clock (mpllb_oword_clk) Octuple-symbol rate (x8) clock derived from MPLLBs **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
mpllb_qword_clk	O	Quadruple-Rate word clock (mpllb_qword_clk) Quadruple-symbol rate (x4) clock derived from MPLLB **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mpllb_short_lock_en	I	MPLLB short lock enable This input indicates that a shared reference clock configuration is being used. When asserted, the MPLL(A,B) locking time will be reduced. **Note:**This signal can be changed only when the txX_pll_en input for all lanes is de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_tx_clk_div[3:0]	I	MPLLB TX clock divider Sets the divide ratio on the MPLL clock sent to the TX as follows: ■ 4'b0000: div16 ■ 4'b0001: div1 ■ 4'b0010: div2 ■ 4'b0011: div3 ■ 4'b0100: div4 ■ 4'b0101: div5 ■ 4'b0110: div6 ■ 4'b0111: Reserved ■ 4'b1000: div8 ■ 4'b1001: div9 ■ 4'b1010: div10 ■ 4'b1011: Reserved ■ 4'b1100: Reserved ■ 4'b1101: Reserved ■ 4'b1110: Reserved ■ 4'b1111: div15 **Note:**Change this signal only when the txX_pll_en input for all lanes is de-asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_word_clk	O	Word clock (mpllb_word_clk)

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Symbol rate clock derived from MPLLB **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
mpllb_word_clk_div[1:0]	I	MPLLB word clock divider control (mpllb_word_clk_div) Controls the MPLLB word clock dividers for generating the mpllb_word_clk output. These inputs can be changed only when the txX_pll_en input is de-asserted. ■ 2'b00: Divide by 4 ■ 2'b01: Divide by 5 ■ 2'b10: Divide by 8 ■ 2'b11: Divide by 10 This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_word_clk_en	I	MPLLB word clock output enable This input can be set to 0 in functional mode. In scan mode, set it to 1, if PMA clock output is required for at-speed scan. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
mpllb_word_fr_clk	O	Free running mpllb_word_clk This clock is a free-running version of the mpllb_word_clk and is intended for use during at-speed scan capture when operating in on-chip clocking (OCC) mode. **Note:**This signal may glitch when MPLLB is disabled. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
sup_misc[7:0]	I	Support miscellaneous controls This input controls miscellaneous settings in the Support block. This input must be set to the values provided in the "PHY Usage and Configuration" chapter.

Table 1-14 MPLL Control Signals (continued)

Port Name	I/O	Description
		Note: Unlike txX/rxX_misc, this PHY input is NOT removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro because it not included as part of context restore (described in Chapter 5) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.15 MPLL(A,B) Fractional Division and SSC Control Signals

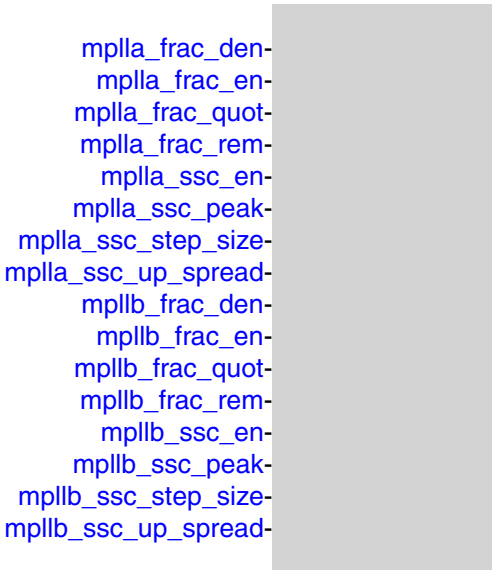


Table 1-15 MPLL(A,B) Fractional Division and SSC Control Signals

Port Name	I/O	Description
mplla_frac_den[15:0]	I	MPLLA fractional feedback denominator. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_frac_en	I	MPLLA fractional clocking enable Enable fractional clocking for MPLLA. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_frac_quot[15:0]	I	MPLLA fractional feedback quotient. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

Table 1-15 MPLL(A,B) Fractional Division and SSC Control Signals (continued)

Port Name	I/O	Description
mplla_frac_rem[15:0]	I	MPLLA fractional feedback remainder. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_ssc_en	I	Spread spectrum enable (mplla_ssc_en) Enables spread-spectrum clock (SSC) generation on the MPLL clocks outputs (mpll(a,b)_word_clk, mpll(a,b)_div_clk). If the reference clock already has spread spectrum applied, mpll(a,b)_ssc_en must be de-asserted. These inputs can be changed only when the txX_pll_en input for all lanes is de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_ssc_peak[19:0]	I	Spread Spectrum peak value (mplla_ssc_peak) Controls the ppm shift amplitude of the spread-spectrum profile. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_ssc_step_size[20:0]	I	Spread Spectrum step size value (mplla_ssc_step_size) Controls the spread-spectrum profile step. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mplla_ssc_up_spread	I	MPLLA SSC Up Spread Enable When asserted, the SSC is applied in upwards direction (positive ppm) and when de-asserted the SSC is applied downwards (negative ppm). **Note:**Change this input only when the MPLL is powered down.

Table 1-15 MPLL(A,B) Fractional Division and SSC Control Signals (continued)

Port Name	I/O	Description
		This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_frac_den[15:0]	I	MPLLB fractional feedback denominator. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_frac_en	I	MPLLB fractional clocking enable Enable fractional clocking for MPLLB. Set this input as per the tables provided in the "PHY Usage and Configuration", for all protocols. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_frac_quot[15:0]	I	MPLLB fractional feedback quotient. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_frac_rem[15:0]	I	MPLLB fractional feedback remainder. Used for fractional feedback divider programming. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

Table 1-15 MPLL(A,B) Fractional Division and SSC Control Signals (continued)

Port Name	I/O	Description
mpllb_ssc_en	I	Spread spectrum enable (mpllb_ssc_en) Enables spread-spectrum clock (SSC) generation on the MPLL clocks outputs (mpll(a,b)_word_clk, mpll(a,b)_div_clk). If the reference clock already has spread spectrum applied, mpll(a,b)_ssc_en must be deasserted. These inputs can be changed only when the txX_pll_en input for all lanes is de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_ssc_peak[19:0]	I	Spread Spectrum peak value (mpllb_ssc_peak) Controls the ppm shift amplitude of the spread-spectrum profile. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_ssc_step_size[20:0]	I	Spread Spectrum step size value (mpllb_ssc_step_size) Controls the spread-spectrum profile step. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_ssc_up_spread	I	MPLLB SSC Up spread enable When asserted, the SSC is applied in upwards direction (positive ppm) and when de-asserted the SSC is applied downwards (negative ppm). **Note:**This input can be changed only when the MPLL is powered down. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.16 Receiver Request (Per-Lane Receive Interface Group) Signals

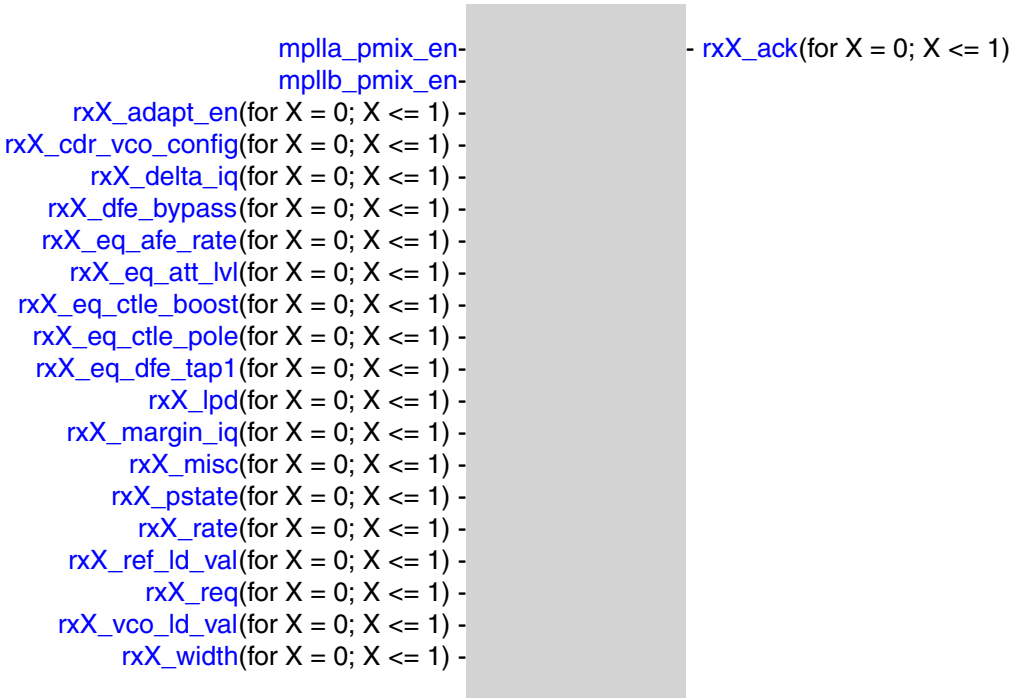


Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals

Port Name	I/O	Description
mplla_pmix_en	I	MPLLA phase mixer enable This signal enables the phase mixer of the MPLL, which is used for SSC generation on clocks (mpll(a,b)_word_clk, mpll(a,b)_div_clk). These inputs can be changed only when the txX_pll_en input for all lanes is de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Exists:** Always **Synchronous To:** Asynchronous
mpllb_pmix_en	I	MPLLB phase mixer enable This signal enables the phase mixer of the MPLL, which is used for SSC generation on clocks (mpll(a,b)_word_clk, mpll(a,b)_div_clk). These inputs can be changed only when the txX_pll_en input for all lanes is de-asserted. **Notes:** ■ Use of built-in SSC generation results in increased deterministic jitter (DJ). ■ The SSC profile is not synchronized across aggregated PHYs. **Exists:** Always **Synchronous To:** Asynchronous

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
rxX_ack (for X = 0; X <= 1)	O	Receiver acknowledge The assertion of this signal indicates the requested receiver setting is complete. This signal stays asserted until the rxX_req input is de-asserted. Once asserted, rxX_ack is de-asserted when rxX_req is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_adapt_en (for X = 0; X <= 1)	I	RX DFE adaptation enable This signal indicates adaptation of the feedback equalization (DFE) circuitry. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. **Exists:** Always **Synchronous To:** Asynchronous

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
rxX_cdr_vco_config[11:0] (for X = 0; X <= 1)	I	RX CDR VCO configuration bus ■ 11: ❑ 0: Selects either the dfe clock (dfe mode) or bypass clock (bypass mode) to align at 90deg with regard to phase clock b ❑ 1: Toggles XOR inputs between data and bypass clocks to align at 90deg with regard to phase clk ■ 10: Range mentioned depends on rx0_cdr_vco_config ❑ 0: Frequency band (14.5 - 16 GHz) ❑ 1: Frequency band (12 - 14.5 GHz) ❑ 0: Frequency band (9.5 - 12 GHz) ❑ 1: Frequency band (8 - 9.5 GHz) ■ 9: ❑ 0: Lower fcdac TC gain ❑ 1: fcdac higher gain selected (usually when ptcomp_en=1) ■ 8: ❑ 0: Disable oscillators process and temperature compensation circuit ❑ 1: Enable compensation ■ 7: ❑ 0: Enable ❑ 1: Disable DAC PMOS load regulator circuit ■ 6: ❑ 0: Disable ❑ 1: Enable and recenter low bias current intg path operation ■ 5: 0: Reserved ■ 4: ❑ 0: Normal operation ❑ 1: Bypass DAC filter big MOS resistors ■ 3: ❑ 0: Select rx_vref as bias reference ❑ 1: Select rx_iref as bias reference, both are sent from bandgap. ■ 2: ❑ 0: Regular bias setting ❑ 1: Low bias setting ■ 1: ❑ 0: Regular bias current gain ❑ 1: Low gain ■ 0: ❑ 0: High frequency band (12 - 16 GHz) ❑ 1: low (8.0 - 12 GHz)
		Voltage Range: 0-vddcore

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
		Exists: Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_delta_iq[3:0] (for X = 0; X <= 1)	I	IQ offset value Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This is an unsigned offset value and covers a range of 0UI to 0.2UI. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_dfe_bypass (for X = 0; X <= 1)	I	RX DFE bypass enable Indicates bypassing of the decision feedback equalization (DFE) circuitry. The bypass path is meant to save power for low data rates where the DFE is not required. The DFE is powered-off when in the bypass mode. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_eq_afe_rate[2:0] (for X = 0; X <= 1)	I	RX Equalization AFE rate Controls the amount of parasitic boost provided by the negative-C circuit in the Equalizer and VGA stages. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_att_lvl[2:0] (for X = 0; X <= 1)	I	RX equalization attenuation level Controls the AFE attenuation level from -2dB when set to 3'b000, to -6dB when set to 3'b111; binary encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A
rxX_eq_ctle_boost[4:0] (for X = 0; X <= 1)	I	RX equalization CTLE boost Controls the CTLE boost level; binary encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_ctle_pole[1:0] (for X = 0; X <= 1)	I	RX equalization CTLE pole Controls the continuous time linear equalizer (CTLE) boost pole location; binary encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_dfe_tap1[8:0] (for X = 0; X <= 1)	I	RX equalization DFE Tap1 Controls the value of DFE data Tap1. This is a signed input; two's complement encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_lpd (for X = 0; X <= 1)	I	Receiver lane power down Powers down the receiver into a power state equivalent to P1. This signal is useful for selectively powering down lanes in a link while maintaining the power state of the link as a whole (for example, link in P0). **Note:**If the rxX_pstate(1:0) input is set to P2, then that takes priority and the receiver is placed in P2. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_margin_iq[6:0] (for X = 0; X <= 1)	I	Value for RX IQ margining This is a signed value for skewing the IQ separation for the purpose of RX margining.

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
		It covers a range from -0.5UI to 0.5UI. Each margin step corresponds to approximately 1/35UI phase shift in magnitude. For PCIe margining the recommended limit for RX margining is +/-7 codes corresponding to +/-0.2UI. This input requires an rxX_req/rxX_ack handshake, and is only valid when the RX DFE is enabled (rxX_dfe_bypass=0). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_misc[7:0] (for X = 0; X <= 1)	I	RX Miscellaneous Controls This input controls miscellaneous settings in the RX. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. misc[7:4]: reserved Controls the division ratio of the RX VCO clock routed to rxX_hs_clk_p/m. misc[3]: divider enable misc[2:0]: division ratio ■ 000 div8 ■ 001 div10 ■ 010 div16 ■ 011 div20 ■ 100 div 32 ■ 101 div40 ■ 110 reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_pstate[1:0] (for X = 0; X <= 1)	I	Receiver power state Sets the power state of the receiver. The state mappings are as follows: ■ 2'b00: P0- Receiver is fully powered up and output receive clocks are active ■ 2'b01: P0S- RX voltage-controlled oscillator (VCO) is in continuous calibration mode, output receive clocks are not available. ■ 2'b10: P1- RX analog front-end (AFE) and voltage regulators are powered up, but RX VCO is in reset ■ 2'b11: P2- RX signal detector is powered up and the rest of RX is powered down **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A
rxX_rate[2:0] (for X = 0; X <= 1)	I	RX data rate Select the baud rate for the receive path ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b100: baud/16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_ref_ld_val[6:0] (for X = 0; X <= 1)	I	RX VCO calibration reference load value This input is used to load internal calibration registers used to perform RX VCO calibration. This input must be set to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
rxX_req (for X = 0; X <= 1)	I	Receiver request This signal indicates a new receiver setting request. This signal should stay high until the rxX_ack output is asserted. After rxX_ack is asserted, rxX_req must be de-asserted. **Notes:** The PHY captures the following input signals when rxX_req is asserted: ■ rxX_pstate ■ rxX_lpd ■ rxX_rate ■ rxX_width ■ rxX_dfe_bypass ■ rxX_eq_att_lvl ■ rxX_eq_afe_rate ■ rxX_eq_vga1_gain ■ rxX_eq_vga2_gain ■ rxX_eq_ctle_pole ■ rxX_eq_ctle_boost ■ rxX_eq_dfe_tap1 ■ rxX_ref_ld_val ■ rxX_vco_ld_val ■ rxX_adapt_sel ■ rxX_cdr_vco_config ■ rxX_delta_iq ■ rxX_margin_iq ■ rxX_misc ■ rxX_dcc_ctrl_range ■ rxX_recal_force_en ■ rxX_recal_skip_en ■ rxX_recal_bank_sel ■ rxX_*div* These signals must be stable before assertion of rxX_req and remain stable until rxX_req is de-asserted. Any combination of D300 changes in these signals is valid for a single request, and the completion of all changes are indicated by the assertion of rxX_ack. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
rxX_vco_ld_val[12:0] (for X = 0; X <= 1)	I	RX VCO calibration load value This input is used to load internal calibration registers used to perform RX VCO calibration. This input must be set to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro.

Table 1-16 Receiver Request (Per-Lane Receive Interface Group) Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_width[2:0] (for X = 0; X <= 1)	I	RX data width Select the width of the output receive data: ■ 3'b000: 8-bit ■ 3'b001: 10-bit ■ 3'b010: 16-bit ■ 3'b011: 20-bit ■ 3'b100: 32-bit ■ 3'b101: 40-bit ■ 3'b110: 64-bit **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.17 MPLL(A,B) Recalibration Signals

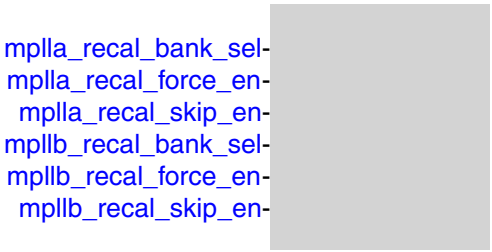


Table 1-17 MPLL(A,B) Recalibration Signals

Port Name	I/O	Description
mplla_recal_bank_sel[1:0]	I	MPLLA re-calibration bank select Selects the bank to store the MPLL and TX duty cycle correction (DCC) calibration values. **Note:**This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_recal_force_en	I	MPLLA force re-Calibration Forces a recalibration of the MPLL and TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:** ■ This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). This input has priority over the mpll(a,b)_recal_skip_en signal. ■ Only MPLL(s) selected by txX_ropll_refsel is calibrated. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mplla_recal_skip_en	I	MPLLA skip re-calibration Skips re-calibration of the MPLL and TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:**This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). This input has a lower priority than the mpll(a,b)_recal_force_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_recal_bank_sel[1:0]	I	MPLLB re-calibration bank select Selects the bank to store the MPLL and TX duty cycle correction (DCC) calibration values.

Table 1-17 MPLL(A,B) Recalibration Signals (continued)

Port Name	I/O	Description
		Note: This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_recal_force_en	I	MPLLB force re-calibration Forces a recalibration of the MPLL and TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:** ■ This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). This input has priority over the mpll(a,b)_recal_skip_en signal. ■ Only MPLL(s) selected by txX_ropll_refsel is calibrated. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
mpllb_recal_skip_en	I	MPLLB skip re-calibration Skips re-calibration of the MPLL and TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:**This input can be changed only when the MPLL is powered down (txX_pll_en = 0 and mpll(a,b)_force_en = 0). This input has a lower priority than the mpll(a,b)_recal_force_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.18 Transmitter Control Signals

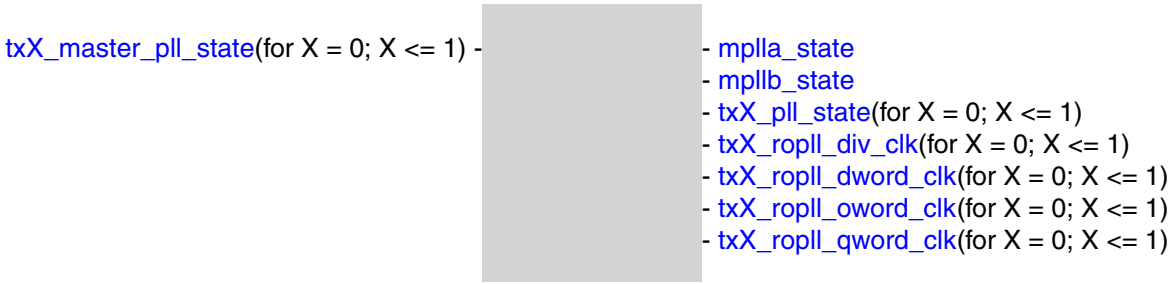


Table 1-18 Transmitter Control Signals

Port Name	I/O	Description
mplla_state	O	MPLLA state indicator This signal indicates the state of the MPLL. It is asserted when the MPLL is powered up and phase-locked. **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
mpllb_state	O	MPLLB state indicator This signal indicates the state of the MPLL. It is asserted when the MPLL is powered up and phase-locked. **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_master_pll_state (for X = 0; X <= 1)	I	Master PLL state per lane Indicates the state of PLL from the master lane of the PHY. For multi-lane link, connect to master lane's txX_pll_state output. For single lane link, connect to current lane's txX_pll_state output. For details, see the "Aggregation and Bifurcation Support" section in the PHY databook. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_pll_state (for X = 0; X <= 1)	O	TX PLL State Indicator Indicates the state of PLL for the lane. This signal is asserted when lane PLL (combination of MPLLA/B, ROPLL as determined by txX_ropll_refsel (1:0)) is powered up and phase-locked. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_ropll_div_clk (for X = 0; X <= 1)	O	TX ROPLL division clock output Frequency is TX PLL Freq/(4*(txX_ropll_out_div +4))

Table 1-18 Transmitter Control Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_dword_clk (for X = 0; X <= 1)	O	TX ROPLL dword clock output Note: This clock stops and restarts during rate change **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_oword_clk (for X = 0; X <= 1)	O	TX ROPLL oword clock output Note: This clock stops and restarts during rate change **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_qword_clk (for X = 0; X <= 1)	O	TX ROPLL qword clock output Note: This clock stops and restarts during rate change **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

1.19 Power Gating Signals

Table 1-19 Power Gating Signals

Port Name	I/O	Description
pcs_pwr_en	O	Power enable for Raw PCS power switches Enable signal for external switches to supply power to the power gated logic in Raw PCS. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
pcs_pwr_stable	I	Power stable for Raw PCS Status signal indicating that the power for the Raw PCS is stable. The pcs_pwr_stable signal should only be asserted if the supply is 90% of nominal or higher. For power gating sequencing, see the "PHY Usage and Configuration" chapter. If power gating mode is not used (pg_mode_en=0), then tie this input high. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
pg_mode_en	I	Power gating support enable Control input to enable the power gating support. When de-asserted, the control inputs related to power gating are ignored. For power gating sequencing, see the "PHY Usage and Configuration" chapter. This input is meant to be a tie-off and can be transitioned only during the PHY reset sequence. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
pg_reset	I	Power gated reset Reset signal for the power gating domain, which controls the generation of the pcs_pwr_en and pma_pwr_en outputs. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-19 Power Gating Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** High
pma_pwr_en	O	Power enable for PMA power switch Enable signal for PMA power switch (external) to supply power to the PMA. For power gating sequencing, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
pma_pwr_stable	I	Power stable for PMA Status signal indicating that the power for the PMA is stable. The pma_pwr_stable signal should only be asserted if the supply is 90% of nominal or higher. For power gating sequencing, see the "PHY Usage and Configuration" chapter. If power gating mode is not used (pg_mode_en=0), then tie this input high. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.20 Reference Clock Control Signals

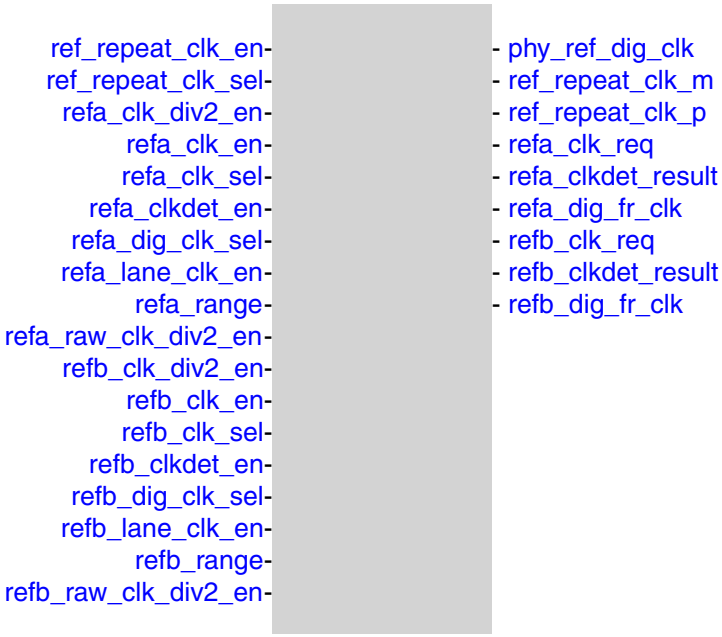


Table 1-20 Reference Clock Control Signals

Port Name	I/O	Description
phy_ref_dig_clk	O	Buffered version of input reference clock This is a buffered, single-ended version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt0_clk_p/ref_alt0_clk_m or ref_alt1_clk_p/ref_alt1_clk_m, depending on the settings of refa/b_clk_sel and refa/b_clk_en. If refa/b_raw_clk_div2_en and/or refa/b_clk_div2_en is asserted, then phy_ref_dig_clk is the divided-by-2 or divided-by-4 version of the reference clock. This is to ensure phy_ref_dig_clk has a maximum frequency of 175 MHz. **Notes:** ■ Do not use this signal as a precision clock source for other PHY instantiations. It is not appropriate for such use. ■ This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. ■ This signal toggles only when ref_clk_en is asserted and phy_reset is de-asserted. The phy_ref_dig_clk is selected as follows: **refa_clk_en, refb_clk_en = phy_ref_dig_clk** ■ 0,0 = N/A ■ 0,1 = REFB ■ 1,0 = REFA ■ 1,1 = REFB **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
		Synchronous To: N/A **Active State:** N/A
ref_repeat_clk_en	I	Repeat reference clock enable Enables the CML output clocks ref_repeat_clk_(p,m). This pair of clocks can be used as reference clocks for other on-chip PHYs. **Notes:** ■ The phy_reset must be de-asserted and (refa_clk_en OR refb_clk_en) must be asserted to enable accurate bandgap voltage reference for ref_repeat_clk_p/m output. Prior to enabling bandgap, the ref_repeat_clk_p/m output uses vp voltage reference, which may result in higher jitter clock. ■ This input can be changed only when phy_reset is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
ref_repeat_clk_m	O	Current-mode logic (CML) buffered version of input reference clock (ref_repeat_clk_m) This is a CML-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p/ref_alt0/1_clk_m, depending on the settings of refa/b_clk_sel. This pair of clocks can be used as reference clocks for other PHYs on chip. **Note:**This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
ref_repeat_clk_p	O	Current-mode logic (CML) buffered version of input reference clock (ref_repeat_clk_p) This is a CML-buffered version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p/ref_alt0/1_clk_m, depending on the settings of refa/b_clk_sel. This pair of clocks can be used as reference clocks for other PHYs on chip. **Note:**This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
ref_repeat_clk_sel[1:0]	I	Repeat reference clock select Selects the input reference clock for the CML output clocks ref_repeat_clk_(p,m). ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
refa_clk_div2_en	I	Input reference clock divider control for refa The reference clock is divided by 2 when asserted. Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refa_clk_en	I	Reference clock enable for refa_clk This signal must remain de-asserted until the reference clock is running at the appropriate frequency, at which point this signal can be asserted. For lower power states where the reference clock to the PHY is disabled, this signal must be de-asserted. **Note:**External CR/JTAG access is not allowed during refa/a_clk_en to refa/b_clk_req handshake **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refa_clk_req	O	Reference clock request for refa Request for output logic to supply reference clock to the PHY. When this input is asserted, reference clock to the PHY cannot be disabled. **Note:**This output may glitch (because it is combinatorially generated from refa_clk_en/ack signals). **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
refa_clk_sel[1:0]	I	Select reference clock source from ref_pad_clk_p/m or ref_alt0/1_clk_p/m **Note:**A transition on this input must be followed by the assertion of phy_reset. ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refa_clkdet_en	I	Enable Reference Clock Detection for refa Enables detection of the reference clock on either the pads input or the alt0/1 input (depending on refa_clk_sel) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
refa_clkdet_result	O	Reference clock detection result for refa_clk Indicates the presence of the reference clock on either the pads input or the alt0/1 input (depending on refa_clk_sel). The result is only valid when refa_clkdet_en and refa_clk_en are asserted. It is a one-time only capture of reference clock presence on refa_clk_en assertion. Output asserts after more than 36 cycles of reference clock on the prescaler input. This signal retains the state of previous detection until a reset occurs with de-assertion of refa_clk_en. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
refa_dig_clk_sel[1:0]	I	Allow selection of different reference clock for PLL and digital logic. Required for retimer or U2/U3 applications otherwise connect to 0 **Note:**A transition on this input must be followed by the assertion of phy_reset. ■ 2'b00 : Use ref A prescaler (phy_reset is required if refa_clk_sel changes) ■ 2'b01 : Use ref B prescaler (phy_reset is required if refb_clk_sel changes) ■ 2'b10 : Use ref repeat prescaler (phy_reset is required if ref_repeat_clk_sel changes. ref_repeat_clk_en should assert prior to refa_clk_en input assertion. ref_repeat_clk_en should de-assert after refa_clk_req output de-assertion) ■ 2'b11 : Reserved

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
		Exists: Always **Synchronous To:** N/A **Active State:** N/A
refa_dig_fr_clk	O	Free-running buffered version of input reference clock refa This is a buffered, single-ended version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p/ref_alt0/1_clk_m, depending on the settings of refa_clk_sel. The frequency of this clock matches that of the input reference clock and is not affected by the setting of refa_clk_div2_en. **Notes:** ■ Do not use this signal as a precision clock source for other PHY instantiations. It is not appropriate for such use. ■ This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. ■ This signal toggles only when refa_clk_en is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
refa_lane_clk_en	I	Ungates refa_clk from prescaler to lane blocks in analog Must be set to 1 when txX_ropll_refsel=2'b10. Any change in this input must be followed by phy_reset assertion. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
refa_range[3:0]	I	Input reference clock frequency range for refa_clk Specifies the frequency range of the input reference clock (post refa_clk_div2_en division, if any). The code mapping of ref_range is as follows: ■ 4'b0000: 13.5 - 26 MHz ■ 4'b0001: 26.1 - 52 MHz ■ 4'b0010: 52.1 - 78 MHz ■ 4'b0011: 78.1 - 104 MHz ■ 4'b0100: 104.1 - 130 MHz ■ 4'b0101: 130.1 - 156 MHz ■ 4'b0110: 156.1 - 182 MHz ■ 4'b0111: 182.1 - 200 MHz ■ 4'b1000: 200.1 - 226 MHz ■ 4'b1001: 226.1 - 252 MHz ■ 4'b1010 and above: N/A Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
refa_raw_clk_div2_en	I	Input reference clock divider control for refa clock in RAW PCS. When asserted, the refa clock is divided by 2. Make sure to assert phy_reset after any change in this input. Set this input to value defined in the configuration tables in the "Support Configuration Settings" and "Lane Configuration Settings" appendices. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refb_clk_div2_en	I	Input reference clock divider control for refb The reference clock is divided by 2 when asserted. Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refb_clk_en	I	Reference clock enable for refb_clk This signal must remain de-asserted until the reference clock is running at the appropriate frequency, at which point this signal can be asserted. For lower power states where the reference clock to the PHY is disabled, this signal must be de-asserted.

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
		Note: External CR/JTAG access is not allowed during refa/a_clk_en to refa/b_clk_req handshake. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refb_clk_req	O	Reference clock request for refb Request for output logic to supply reference clock to the PHY. When this input is asserted, reference clock to the PHY cannot be disabled. **Note:**This output may glitch (because it is combinatorially generated from refb_clk_en/ack signals) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refb_clk_sel[1:0]	I	Select reference clock source from ref_pad_clk_p/m, ref_alt0/1_clk_p/m **Note:**Assert phy_reset after a transition on this input. ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
refb_clkdet_en	I	Enable reference clock detection for refb Enables detection of the reference clock on either the pads input or the alt0/1 input (depending on refb_clk_sel) **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
refb_clkdet_result	O	Reference clock detection result for refb_clk Indicates the presence of the reference clock on either the pads input or the alt0/1 input (depending on refb_clk_sel). The result is only valid when refb_clkdet_en and refb_clk_en are asserted. It is a one-time only capture of reference clock presence on refa_clk_en assertion. Output asserts after more than 36 cycles of reference clock on the prescaler input. This signal retains the state of previous detection until a reset occurs with de-assertion of refb_clk_en. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** High
refb_dig_clk_sel[1:0]	I	Allow selection of different reference clock for PLL and digital logic. Required for retimer or U2/U3 applications, otherwise, connect to 0. **Note:**A transition on this input must be followed by the assertion of phy_reset. ■ 2'b00 : Use ref B prescaler (phy_reset is required if refb_clk_sel changes) ■ 2'b01 : Use ref A prescaler (phy_reset is required if refa_clk_sel changes) ■ 2'b10 : Use ref repeat prescaler (phy_reset is required if ref_repeat_clk_sel changes. ref_repeat_clk_en should assert prior to refb_clk_en input assertion. ref_repeat_clk_en should de-assert after refb_clk_req output de-assertion) ■ 2'b11 : Reserved **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
refb_dig_fr_clk	O	Free-running buffered version of input reference clock refb This is a buffered, single-ended version of either ref_pad_clk_p/ref_pad_clk_m or ref_alt0/1_clk_p, depending on the settings of refb_clk_sel. The frequency of this clock matches that of the input reference clock and is not affected by the setting of ref_clkb_div2_en. **Notes:** ■ Do not use this signal as a precision clock source for other PHY instantiations. It is not appropriate for such use. ■ This signal may be undriven when power gating is enabled. For more information on power gating, see the "PHY Usage and Configuration" chapter. ■ This signal toggles only when refb_clk_en is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
refb_lane_clk_en	I	Ungates refb_clk from prescaler to lane blocks in analog Must be set to 1 when txX_ropll_refsel=2'b11; Any change in this input must be followed by phy_reset assertion. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-20 Reference Clock Control Signals (continued)

Port Name	I/O	Description
refb_range[3:0]	I	Input reference clock frequency range for refb_clk Specifies the frequency range of the input reference clock (post refb_clk_div2_en division, if any). The code mapping of ref_range is as follows: ■ 4'b0000: 13.5 - 26 MHz ■ 4'b0001: 26.1 - 52 MHz ■ 4'b0010: 52.1 - 78 MHz ■ 4'b0011: 78.1 - 104 MHz ■ 4'b0100: 104.1 - 130 MHz ■ 4'b0101: 130.1 - 156 MHz ■ 4'b0110: 156.1 - 182 MHz ■ 4'b0111: 182.1 - 200 MHz ■ 4'b1000: 200.1 - 226 MHz ■ 4'b1001: 226.1 - 252 MHz ■ 4'b1010 and above: N/A Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
refb_raw_clk_div2_en	I	Input reference clock divider control for refb clock in RAW PCS. When asserted, the refa clock is divided by 2. Make sure to assert phy_reset after any change in this input. Set this input to value defined in the configuration tables in the "Support Configuration Settings" and "Lane Configuration Settings" appendices. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.21 Reference Clock Control (Common Block Group) Signals

ref_clk_mplla_div-
ref_clk_mpllb_div-

Table 1-21 Reference Clock Control (Common Block Group) Signals

Port Name	I/O	Description
ref_clk_mplla_div[2:0]	I	MPLLA reference clock divider control The reference clock used for MPLLA calibration and locking can be divided by 2/4/8/16. Note that this division is applied independently of ref_clk_div2_en. ■ 3'b000: Divide by 1 ■ 3'b001: Divide by 2 ■ 3'b010: Divide by 4 ■ 3'b011: Divide by 8 ■ 3'b100: Divide by 16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
ref_clk_mpllb_div[2:0]	I	MPLLB reference clock divider control The reference clock used for MPLLB calibration and locking can be divided by 2/4/8/16. Note that this division is applied independently of ref_clk_div2_en. ■ 3'b000: Divide by 1 ■ 3'b001: Divide by 2 ■ 3'b010: Divide by 4 ■ 3'b011: Divide by 8 ■ 3'b100: Divide by 16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.22 DCO Control Signals

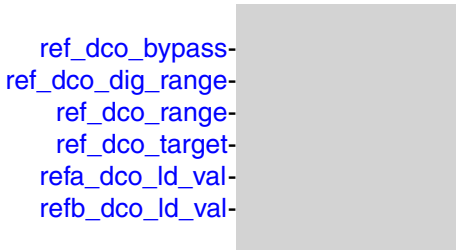


Table 1-22 DCO Control Signals

Port Name	I/O	Description
ref_dco_bypass	I	DCO (digitally controlled oscillator) is an internal free running clock for Raw PCS, firmware, and some PMA logic. Set this static input to 0 for any of the following conditions otherwise set to 1: ■ PCIe/CXL application and reference clock is < 100MHz ■ Single PMA with multiple bifurcated links and independent refa_clk and refb_clk sources ■ Any change in this input must be followed by phy_reset assertion. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous
ref_dco_dig_range[3:0]	I	Configuration controls for DCO **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous
ref_dco_range[1:0]	I	Configuration controls for DCO **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous
ref_dco_target[7:0]	I	Configuration controls for DCO **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous
refa_dco_ld_val[7:0]	I	Configuration controls for DCO **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous
refb_dco_ld_val[7:0]	I	Configuration controls for DCO **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous

1.23 External ROM Control Signals

Table 1-23 External ROM Control Signals

Port Name	I/O	Description
rom_addr[14:0]	O	ROM address bus Address bus for the external ROM **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** rom_clk **Active State:** N/A
rom_clk	O	ROM clock Clock used to drive external ROM interface **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rom_rd_data[15:0]	I	ROM read data bus Read data bus for the external ROM **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** rom_clk **Active State:** N/A

1.24 Ethernet Clock (for X = 0; X <= 1) Signals

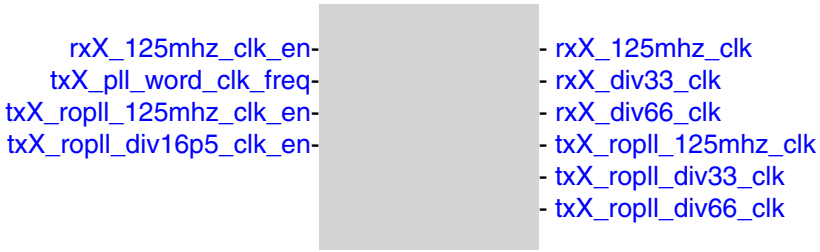


Table 1-24 Ethernet Clock (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_125mhz_clk	O	Per-lane RX 125 MHz clock output for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating rxX_clk. As a result, duty cycle of this clock will not be 50/50. Also, there will be period jitter, so synthesis and timing closure frequency will be greater than 125 MHz. For details, refer to Synopsys synthesis constraints in the release package. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_125mhz_clk_en	I	Per-lane RX 125 MHz clock enable signal. Typically used for some Ethernet protocols (used by Synopsys xPCS IP). This signal is only valid when RX is in P0 power state. rxX_req/ack handshake is required when changing this input. For details, see rxX_125mhz_clk signal description. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. In the appendix, if this input is 0 for a given protocol/data-rate/width, then this clock is not supported. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_div33_clk	O	Per-lane RX div33 clock output for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating rxX_clk. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than div33 of rxX_clk. For details, refer to Synopsys synthesis constraints in the release package. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-24 Ethernet Clock (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: N/A **Active State:** High
rxX_div66_clk	O	Per-lane RX div66 clock output for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating rxX_clk. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than div66 of rxX_clk. For details, refer to Synopsys synthesis constraints in the release package. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_pll_word_clk_freq[3:0]	I	Per-lane TX input for generating txX_ropll_div33/66/125mhz_clk outputs for some Ethernet protocols (used by Synopsys xPCS IP) and other pulse gated digital clocks. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_125mhz_clk	O	Per-lane TX 125 MHz clock enable signal for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating txX_ropll_dword_clk output. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than 125MHz. For details, refer to Synopsys synthesis constraints in the release package. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_125mhz_clk_en	I	Per-lane TX 125mhz clock enable signal for some Ethernet protocols (used by Synopsys xPCS IP) This signal is only valid when txX_pll_en=1 and txX_ropll_word_clk_en=1. txX_req/ack handshake is required when changing this input. For details, see txX_ropll_125mhz_clk signal description. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. In the appendix, if this input is 0 for a given protocol/data-rate/width, then this clock is not supported.

Table 1-24 Ethernet Clock (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_div16p5_clk_en	I	Per-lane TX div33 and div66 clock enable signal for some Ethernet protocols (used by Synopsys xPCS IP) This signal is only valid when txX_pll_en=1 and txX_ropll_word_clk_en=1. txX_req/ack handshake is required when changing this input. For details, see txX_ropll_div33_clk and txX_ropll_div66_clk signal description. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. In the appendix, if this input is 0 for a given protocol/data-rate/width, then this clock is not supported. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_div33_clk	O	Per-lane TX div33 clock output for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating txX_ropll_dword_clk output. As a result, duty cycle of this clock will not be 50/50. Also, there will be period jitter, so synthesis and timing closure frequency will be greater than div33. For details, refer to Synopsys synthesis constraints in the release package. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_div66_clk	O	Per-lane TX div66 clock output for some Ethernet protocols (used by Synopsys xPCS IP). Generated by pulse-gating txX_ropll_dword_clk output. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than div66. For details, refer to Synopsys synthesis constraints in the release package. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix.

Table 1-24 Ethernet Clock (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.25 Receiver Adaptation (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

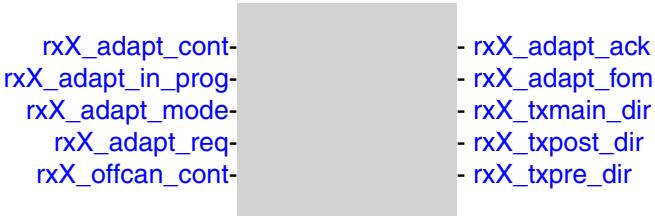


Table 1-25 Receiver Adaptation (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_adapt_ack	O	Receiver adaptation acknowledge The assertion of this output indicates completion of the receiver adaptation sequence and that values on rxX_adapt_fom, rxX_txmain_dir, rxX_txpre_dir, and rxX_txpost_dir are valid. **Note:**This signal remains asserted until rxX_adapt_req is de-asserted, after which rxX_adapt_ack de-asserts. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_adapt_cont	I	Receiver adaptation continuous operation This input must be asserted if continuous receiver adaptation is required. If this signal is de-asserted, the receiver adaptation stops when the adaptation acknowledge (rxX_adapt_ack) is asserted; otherwise, the receiver continues to adapt. **Note:**This signal can be kept asserted. Continuous adaptation will only be performed in rates which support adaptation (that is, rxX_dfe_bypass - 1'b0) and that have completed startup adaptation. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_adapt_fom[7:0]	O	Receiver figure of merit Output indicating the quality of the received data eye. A higher value of rxX_adapt_fom indicates better link equalization, with 8'd0 indicating worst equalization setting and 8'd255 indicating the best equalization setting. This output is valid only when rxX_adapt_ack is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

Table 1-25 Receiver Adaptation (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
rxX_adapt_in_prog	I	Receiver adaptation in progress This input indicates that an adaptation is in progress. This signal must be asserted prior to the rxX_reset assertion of the first adaptation iteration. It must be de-asserted after the rxX_adapt_ack of the final adaptation iteration. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_adapt_mode[2:0]	I	RX Adaptation and Coarse Adaptation mode select Selects an adaptation and coarse adaptation mode. rxX_adapt_mode mode defines an algorithm with varying degrees of length and accuracy, which defines the length of the capture window, number of iterations, and update gains. ■ For rxX_adapt_mode=0, setting=SHORT ■ For rxX_adapt_mode=1, setting=PCIe3 ■ For rxX_adapt_mode=2, setting=PCIe4 ■ For rxX_adapt_mode=3, setting=PCIe5 ■ For rxX_adapt_mode=4, setting=LONG Set this input according to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_adapt_req	I	Receiver adaptation request This input indicates a request for the receiver to adapt to incoming data. This signal must remain asserted until rxX_adapt_ack is asserted, after which rxX_adapt_req can be de- asserted. **Note:**Stopping rxX_adapt_req by negating before rxX_adapt_ack is asserted or by asserting rxX_reset may leave the receiver with a less than optimal settings. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_offcan_cont	I	Receiver offset cancellation continuous operation If continuous receiver offset cancellation is required, then assert this signal. If this signal is de-asserted, then offset cancellation runs only when the receiver exits the P2 power state. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro.

Table 1-25 Receiver Adaptation (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_txmain_dir[1:0]	O	Direction change for far-end TX coefficients (rxX_txmain_dir) Indicates the recommended direction change for the far-end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir(1:0). The INC/DEC definition for rxX_txpre_dir(1:0) and rxX_txpost_dir(1:0) may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_txpost_dir[1:0]	O	Direction change for far-end TX coefficients (rxX_txpost_dir) Indicates the recommended direction change for the far-end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir(1:0). The INC/DEC definition for rxX_txpre_dir(1:0) and rxX_txpost_dir(1:0) may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

Table 1-25 Receiver Adaptation (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
rxX_txpre_dir[1:0]	O	Direction change for far-end TX coefficients (rxX_txpre_dir) Indicates the recommended direction change for the far-end TX equalization coefficients (main, pre, and post). The encoding for each signal is as follows: ■ 2'b00: HOLD ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: Reserved These outputs are valid only when rxX_adapt_ack is asserted. The PHY does not generate INC/DEC commands for the rxX_txmain_dir[(1:0). The INC/DEC definition for rxX_txpre_dir(1:0) and rxX_txpost_dir(1:0) may be opposite to the definition provided in certain protocol standards. For definition of INC/DEC as implemented in the PHY, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.26 Receiver Request (for X = 0; X <= 1) Signals

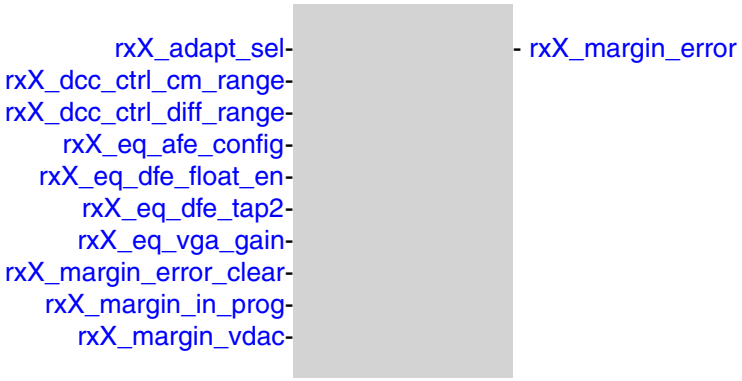


Table 1-26 Receiver Request (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_adapt_sel[1:0]	I	Select storage bank for RX Adaptation and Coarse Adaptation The PHY supports three internal register storage banks for saving adaptation and coarse adaptation results, and this input specifies which one is selected. The storage banks are used for restoring the adapted and coarse adapted values when needed (for example, when returning from another rate). This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. This input must not change when rxX_recal_force_en is high. After de-asserting PHY input rxX_recal_force_en, rxX_req/ack handshake is required to de-assert PHY's internal rxX_recal_force_en state. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_dcc_ctrl_cm_range[3:0]	I	Reserved **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_dcc_ctrl_diff_range[3:0]	I	RX internal differential clock duty cycle control range Sets the range of internal clock duty cycle correction circuitry. Set this input according to the tables provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

Table 1-26 Receiver Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
rxX_eq_afe_config[11:0]	I	These bits are used to configure the AFE ■ 11:10: ATT output common mode adjustment ■ 9:8: rtrim adjustment ■ 7:5: AFE Blocks negc current adjustment ■ 4:3: AFE bias current adjustment through resistors ■ 2: VGA2 Current adjustment ■ 1:0: AFE Bias current adjustment through PMOS switches (1:0)==2'b00 Highest current ■ When bit 0 = '1' then reduction of four units of current ■ When bit 1 = '1' then reduction of eight units of current **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_dfe_float_en	I	When asserted, this signal enables floating taps in RX_DFE Assert this signal only when RX_DFE is in dfe mode, that is it must be off in bypass mode. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_dfe_tap2[7:0]	I	RX Equalization DFE Tap2 Controls the value of DFE data Tap2. This is a signed input; two's complement encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_eq_vga_gain[4:0]	I	RX equalization VGA gain Controls the AFE first stage variable gain amplifier gain; binary encoded. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_margin_error[5:0]	O	Error counter to indicate the number of bit-errors during RX margining

Table 1-26 Receiver Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		This output is continuously updated when rxX_margin_in_prog is high except when rxX_ack is high. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_margin_error_clear	I	Clear the rxX_margin_error counter **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_margin_in_prog	I	Receiver margining in progress This input indicates that RX margining is in progress. Assert this signal at the same time or prior to any change in rxX_margin_vdac or rxX_margin_iq. It should stay asserted during the entire RX margining sequence and must be de-asserted to complete the RX margining sequence. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_margin_vdac[8:0]	I	Value for RX voltage margining This is a signed value. It covers a range from -50 mV to +50 mV. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

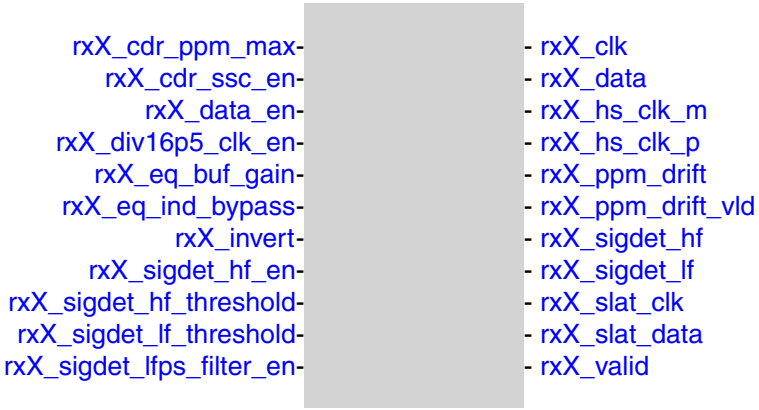


Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_cdr_ppm_max[4:0]	I	Maximum allowed PPM on the RX CDR clock This input specifies the maximum PPM drift to be allowed on the RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. The recovery circuit forces the CDR to exit the data tracking mode even if rxX_data_en is asserted. Instead, the CDR enters the reference tracking mode and is gradually brought back to the ideal frequency before allowing the data tracking again (if rxX_data_en is still asserted). This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_cdr_ssc_en	I	Receiver CDR SSC-Mode enable Controls the CDR tracking gains and duration. It must be asserted when the receive data has a spread-spectrum clock (SSC), and de-asserted when the receive data does not have SSC. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_clk	O	Receiver output clock Received data clock **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: N/A **Active State:** N/A
rxX_data[63:0]	O	Receiver output data MSB is received first. This output must be qualified with rxX_valid. The valid range of the data is determined by the rxX_width input as follows: ■ For rxX_width=3'b000, Valid Data Range=rxX_data[7:0] ■ For rxX_width=3'b001, Valid Data Range=rxX_data[9:0] ■ For rxX_width=3'b010, Valid Data Range=rxX_data[15:0] ■ For rxX_width=3'b011, Valid Data Range=rxX_data[19:0] ■ For rxX_width=3'b100, Valid Data Range=rxX_data[31:0] ■ For rxX_width=3'b101, Valid Data Range=rxX_data[39:0] ■ For rxX_width=3'b110, Valid Data Range=rxX_data[63:0] **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** rxX_clk **Active State:** N/A
rxX_data_en	I	RX data enable Enables the receiver data output. When asserted in P0, the RX datapath is enabled and the CDR begins tracking the received data. This signal is only valid when the RX is in the P0 power state and must be set to 0 in the other power states. This signal should also be set to 0 when rxX_reset is asserted. Note that when rxX_data_en is de-asserted in P0, then the CDR tracks the reference clock in order to stay close to the desired frequency of operation. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_div16p5_clk_en	I	Per-lane RX div33, div66 clock enable signal. Typically used for Ethernet protocols (used by Synopsys xPCS IP) This signal is only valid when RX is in P0 power state. rxX_req/ack handshake is required when changing this input. See rxX_div33_clk and rxX_div66_clk signal description for details. Set this input as per the tables referenced in the "Support and Lane Configuration Settings" appendix. In the appendix, if this input is 0 for a given protocol/data-rate/width, then this clock is not supported. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_eq_buf_gain[2:0]	I	Rx equalization buffer gain

Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		This signal controls the buffer gain; binary encoded. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_eq_ind_bypass	I	Rx equalization inductor bypass This signal controls the buffer gain; binary encoded. **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_hs_clk_m	O	Clock output to monitor internal clocks. **Notes:** ■ This circuit is currently available as a customization, If customization is required, contact Synopsys. ■ To gate this clock, rx_misc[3] need to be set to 0 ■ **This signal may not have a level shifter inside the PMA.** **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_hs_clk_p	O	Clock output to monitor internal clocks. **Notes:** ■ This circuit is currently available as a customization. If customization is required, contact Synopsys. ■ If tx_misc[0] == 0, it is connected to divided down RX VCO clock. ■ if tx_misc[0] == 1, it is connected to the divided down TX PLL clock. ■ To gate this clock, both tx_misc[0] and rx_misc[3] need to be set to 0 ■ **This signal may not have a level shifter inside the PMA.** **Voltage Range:** 0-vp **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_invert	I	Invert logical sense of rxX_p and rxX_m inputs This signal has the effect of inverting the received data stream. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_ppm_drift[5:0]	O	RX CDR PPM drift This value represents the amount of ppm on the rxX_clk with respect to the ideal desired frequency. For more information, see the "PHY Usage and Configuration" chapter.

Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		This output is valid when rxX_ppm_drift_vld is asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_ppm_drift_vld	O	RX CDR PPM drift valid Indicates when the rxX_ppm_drift(5:0) output is valid and can be sampled. For more information, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_sigdet_hf	O	Reserved **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_sigdet_hf_en	I	Reserved Reserved. Tie to 1'b0. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_sigdet_hf_threshold[2:0]	I	Reserved see the "Support Configuration Settings" and "Lane Configuration Settings" This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_sigdet_lf	O	Receive low-frequency signal detection output When asserted, indicates that the receiver is receiving low-frequency signals. This signal is valid only for low-frequency data on rxX_p/rxX_m. For more information about this signal's behavior, see the "PHY Usage and Configuration" chapter. Note: Changing the termination while Sigdet is active can cause this signal to glitch.

Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_sigdet_lf_threshold[2:0]	I	Receiver low-frequency signal detection threshold Sets the low-frequency signal detection threshold level trip point. For the correct setting for a specific protocol, see the "Support Configuration Settings" and "Lane Configuration Settings" appendices. ■ For rxX_sigdet_lf_threshold=3'b000, SigDet Threshold=Reserved ■ For rxX_sigdet_lf_threshold=3'b001, SigDet Threshold=60 mVpp (approximately) ■ For rxX_sigdet_lf_threshold=3'b010, SigDet Threshold=80 mVpp (approximately) ■ For rxX_sigdet_lf_threshold=3'b011, SigDet Threshold=100 mVpp (approximately) ■ Continue in a similar manner for other threshold values ■ For rxX_sigdet_lf_threshold=3'b110, SigDet Threshold=160 mVpp (approximately) ■ For rxX_sigdet_lf_threshold=3'b111, SigDet Threshold=180 mVpp (approximately) This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_sigdet_lfps_filter_en	I	Receiver LFPS signal detection filter enable Enables the LFPS filter on the low-frequency signal detection circuit by reducing the cutoff frequency from 500 MHz to 50 MHz. Assert this signal only if rx_sigdet_lf_en is also asserted. For per-protocol settings, see the "Support Configuration Settings" and "Lane Configuration Settings" appendices. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_slat_clk	O	Receiver's short latency mode output clock. This clock can enabled by writing to CREG (LANEN_DIG_RX_PWRCTL_RX_CTL/RX_SLAT_CLK_EN=1) rxX_slat_data is synchronous to rxX_slat_clk output of the same lane. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-27 Receiver Datapath (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: N/A **Active State:** N/A
rxX_slat_data[19:0]	O	Receiver's short latency mode output data. Synchronous to rxX_slat_clk. MSB is received first. This output must be qualified with rxX_valid. The valid range of the data is determined by the rxX_width input as follows: ■ For rxX_width=3'b010, Valid Data Range=rxX_data[15:0] ■ For rxX_width=3'b011, Valid Data Range=rxX_data[19:0] **Notes:** ■ rxX_slat_clk is high frequency due to 16 or 20-bit RX data width, as such timing closure may be difficult for data rates greater than 20 Gbps. ■ Latency reduction in RX short latency mode is 40 UI compared to normal 32-bit RX latency mode. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** rxX_slat_clk **Active State:** N/A
rxX_valid	O	Data on rxX_data valid indicator This signal is asserted when the RX clock and data recovery (CDR) is bit-locked. The rxX_valid is de-asserted if CDR drift detector is enabled (determined by RAWLANEN_DIG_AON_RX_CDR_DETECTOR_CTL.RX_CDR_DETECTOR_EN) and CDR has drifted beyond the specified maximum PPM (determined by rxX_cdr_ppm_max(4:0)). For more information, see the "PHY Usage and Configuration" chapter. **Note:**The rxX_sigdet_lf signal has no effect on the rxX_valid output. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** rxX_clk **Active State:** High

1.28 Receiver Datapath (for X = 0; X <= 1) Signals

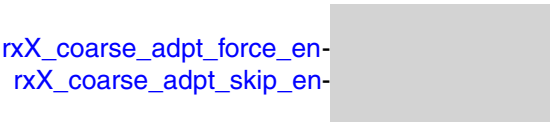


Table 1-28 Receiver Datapath (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_coarse_adpt_force_en	I	RX force coarse adaption Forces a coarse adaptation in the selected bank (rxX_adapt_sel) and mode (rxX_adapt_mode) regardless of the state of the corresponding "*done" signal. **Note:**PHY samples this input with the assertion of rxX_data_en. This input has a higher priority than the rxX_coarse_adpt_skip_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
rxX_coarse_adpt_skip_en	I	RX skip coarse adaption Skips coarse adaptation in the selected bank (rxX_adapt_sel) and mode (rxX_adapt_mode) regardless of the state of the corresponding "*done" signal. **Note:**PHY samples this input with the assertion of rxX_data_en. This input has lower priority than the rxX_coarse_adpt_force_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.29 Multi-protocol Clock (for X = 0; X <= 1) Signals

Table 1-29 Multi-protocol Clock (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_dig_div_clk	O	Per-lane RX divided clock output for non-PCIe protocols. Supported when rxX_width=2 or 4 (16-bit or 32-bit). Can be enabled using rxX_dig_div_clk_sel input along with rxX_req/ack handshake. Generated by pulse-gating rxX_clk. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than selected division ratio of rxX_clk. Refer to Synopsys synthesis constraints in the release package for details. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_dig_div_clk_sel[3:0]	I	Per-lane setting for the division ratio of the rxX_dig_div_clk output for non-PCIe protocols This input may be set to non-zero when rxX_width=2 or 4. rxX_req/ack handshake is required when changing this input. Division ratios: ■ 4'd0: Disabled ■ 4'd1: Div34 ■ 4'd2: Div65 ■ 4'd3: Div67 ■ 4'd4: Div68 ■ 4'd5: Div130 ■ 4'd6: Div132 ■ 4'd7: Div257 This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_div_clk	O	Per-lane RX divided clock output for non-PCIe protocols. **Note:**This circuit is currently available as a customization, If customization is required, contact Synopsys. Enabled when rxX_div_clk_en=1 Frequency is RX VCO Freq/(4*rxX_div_clk_sel)

Table 1-29 Multi-protocol Clock (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		rxX_req/ack handshake is required when changing rxX_div_clk_sel or rxX_div_clk_en Duty cycle of this clock output will not be 50/50 **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
rxX_div_clk_en	I	Per-lane setting to enable rxX_div_clk output for non-PCIe protocols. **Note:**This circuit is currently available as a customization, If customization is required, contact Synopsys. rxX_req/ack handshake is required when changing this input. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_div_clk_sel[7:0]	I	Per-lane setting for the division ratio of the rxX_div_clk output for non-PCIe protocols. Valid settings are 24 to 257. **Note:**This circuit is currently available as a customization, If customization is required, contact Synopsys. rxX_req/ack handshake is required when changing this input. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_dig_ropll_div_clk	O	Per-lane TX divided clock output for non-PCIe protocols Generated by pulse-gating txX_ropll_dword_clk. As a result, duty cycle of this clock will not be 50/50. Also there will be period jitter, so synthesis and timing closure frequency will be greater than selected division ratio of txX_ropll_dword_clk. Refer to Synopsys synthesis constraints in the release package for details. Can be enabled using txX_dig_ropll_div_clk_sel input along with txX_req/ack handshake. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-29 Multi-protocol Clock (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_dig_ropll_div_clk_sel[3:0]	I	Per-lane setting for the division ratio of the txX_dig_ropll_div_clk output. This input can be set to non-zero if txX_pll_word_clk_freq is not "NA" in tables provided in the "Lane Configuration Settings" appendix (txX_pll_word_clk_freq=0,4,6,8 are supported) txX_req/ack handshake is required when changing this input. Division ratios: ■ 4'd0: Disabled ■ 4'd1: Div34 ■ 4'd2: Div65 ■ 4'd3: Div67 ■ 4'd4: Div68 ■ 4'd5: Div130 ■ 4'd6: Div132 ■ 4'd7: Div257 **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.30 Receiver Control (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

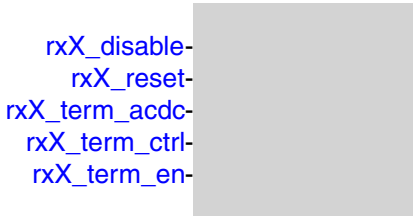


Table 1-30 Receiver Control (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_disable	I	Receiver disable Puts the receiver in a low-power mode where all the receiver blocks, including the signal detector (SigDet) circuit, are turned off. This signal must be asserted when in PCIe L1 Substates L1.1 and L1.2. The receiver must be in the P1 power state when using this signal. This input asynchronously disables RX terminations and does not require any clock to be supplied to the PHY. **Note:**The rxX_disable signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
rxX_reset	I	Receiver reset This signal internally resets the RX datapath, all the receiver settings and state machines. **Notes:** ■ If any rxX_*clk are active at the time rxX_reset is asserted, then these output clocks might experience a temporary glitch. ■ This input must be asserted for minimum 10 reference clock cycles. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
rxX_term_acdc	I	Receiver termination control Controls the terminations of the RX. Selector for floating versus grounded RX termination: ■ 0: Floating RX termination (DC coupled PHY) ■ 1: Grounded RX termination (AC coupled PHY) This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-30 Receiver Control (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** High
rxX_term_ctrl[3:0]	I	Receiver termination control This signal sets the target transmitter termination impedance. ■ For rxX_term_ctrl=3'b000, impedance=54 ohms ■ For rxX_term_ctrl=3'b001, impedance=52 ohms ■ For rxX_term_ctrl=3'b010, impedance=50 ohms (default) ■ For rxX_term_ctrl=3'b011, impedance=48 ohms ■ For rxX_term_ctrl=3'b100, impedance=46 ohms ■ For rxX_term_ctrl=3'b101, impedance=44 ohms ■ For rxX_term_ctrl=3'b110, impedance=42 ohms ■ For rxX_term_ctrl=3'b111, impedance=40 ohms **Note:**This input can be changed only when phy_reset or rxX_reset or rxX_req is asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_term_en	I	Receiver termination enable When asserted HIGH, the RX is terminated with nominal 50 ohm resistance. When asserted LOW, the termination is in high impedance. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.31 Receiver Test (for X = 0; X <= 1) Signals

- rxX_flyover_data_m
- rxX_flyover_data_p

Table 1-31 Receiver Test (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_flyover_data_m	O	Receive Flyover Data Outputs (rxX_flyover_data_m) Outputs are directly connected to RX bump pads when test_flyover_en is enabled. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
rxX_flyover_data_p	O	Receive Flyover Data Outputs (rxX_flyover_data_p) Outputs are directly connected to RX bump pads when test_flyover_en is enabled. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.32 Receiver Re-Calibration (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

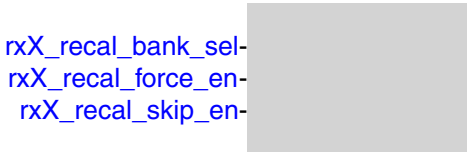


Table 1-32 Receiver Re-Calibration (Per-Lane Receive Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
rxX_recal_bank_sel[1:0]	I	RX re-calibration bank select Selects the bank to store the RX duty cycle correction (DCC) and IQ correction (IQC) calibration values. **Note:**This input is sampled by the PHY with the assertion of rxX_req. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_recal_force_en	I	RX force re-calibration Forces a re-calibration of the RX DCC and IQC in the selected bank regardless of the state of the corresponding done signal. **Note:**This input is sampled by the PHY with the assertion of rxX_req. This input has a higher priority than the rxX_recal_skip_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
rxX_recal_skip_en	I	RX skip re-calibration Skips re-calibration of the RX DCC and IQC in the selected bank regardless of the state of the corresponding done signal. **Note:**This input is sampled by the PHY with the assertion of rxX_req. This input has a lower priority than the rxX_recal_force_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.33 RX Biasing Current (Common Block Group) Signals

Table 1-33 RX Biasing Current (Common Block Group) Signals

Port Name	I/O	Description
rx_vref_ctrl[4:0]	I	RX biasing current control Sets the RX biasing current for RX analog front end. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.34 SRAM Control Signals

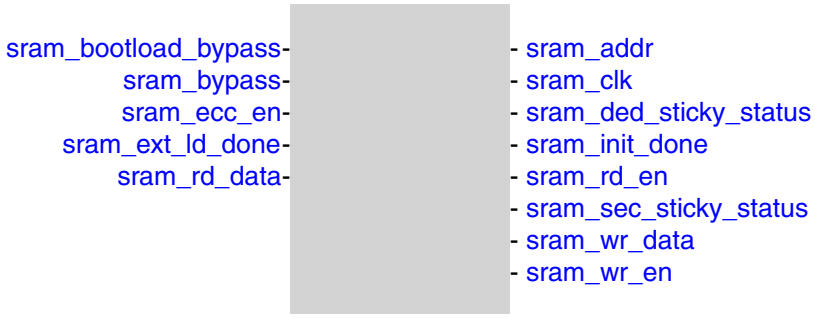


Table 1-34 SRAM Control Signals

Port Name	I/O	Description
sram_addr[14:0]	O	SRAM address bus Address port for the SRAM **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** sram_clk **Active State:** N/A
sram_bootload_bypass	I	SRAM bootload bypass Control signal when asserted, bypasses the SRAM bootloading function. This signal can be used to skip bootloading when coming out of phy_reset. For more information, see the "PHY Usage and Configuration" chapter. **Note:**This signal should only be changed during phy_reset and must not change after phy_reset is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
sram_bypass	I	SRAM bypass (Reserved - connect to 0) Control signal when asserted, bypasses the SRAM interface. In this case, the adaptation and calibration algorithms are executed from the hard wired values within the Raw PCS. If SRAM is not bypassed, the internal algorithms are first loaded by Raw PCS into the SRAM at which point you can change the contents of the SRAM. The updated SRAM contents are used for the adaptation and calibration routines. **Note:**This signal should only be changed during phy_reset and must not change after phy_reset is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
sram_clk	O	SRAM Clock Clock used to drive SRAM interface

Table 1-34 SRAM Control Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
sram_ded_sticky_status	O	Status output for optional ECC feature **Exists:** Always **Synchronous To:** Asynchronous
sram_ecc_en	I	SRAM ECC enable Control signal when asserted, enable SRAM ECC inside RAW PCS. **Note:**This signal should only be changed during phy_reset and must not change after phy_reset is de-asserted. SECDED Hamming code is used for ECC - so single bit error on sram_rd_data bus is corrected and double (two) bit errors are detected (but not corrected). In both cases, a CREG register status bit is set as described in register description chapter. There is no impact on firmware performance when enabling ECC and no timing closure issues are expected. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
sram_ext_ld_done	I	SRAM external load done Assert this signal after any updates to the SRAM have been loaded. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
sram_init_done	O	SRAM initialization done Signal indicating that the SRAM has been initialized by the boot loader in the Raw PCS This signal does not assert if (sram_bypass = 1 and sram_bootload_bypass = 0) which is an invalid signalling. External register access should not be done prior to assertion of sram_init_done. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
sram_rd_data[21:0]	I	SRAM read bus (sram_rd_data). Connects to read data port of the SRAM 16-bits wide when ECC is not enabled. Tie low upper 6 bits. 22-bits wide when ECC is enabled

Table 1-34 SRAM Control Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** sram_clk **Active State:** N/A
sram_rd_en	O	SRAM read enable Read enable control to read data from SRAM **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** sram_clk **Active State:** High
sram_sec_sticky_status	O	Status output for optional ECC feature **Exists:** Always **Synchronous To:** Asynchronous
sram_wr_data[21:0]	O	SRAM write bus (sram_wr_data). Connects to write data port of the SRAM 16-bits wide when ECC is not enabled. Upper 6-bits can be ignored. 22-bits wide when ECC is enabled **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** sram_clk **Active State:** N/A
sram_wr_en	O	SRAM write enable Write enable control to write data to SRAM **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** sram_clk **Active State:** High

1.35 Transmitter Request (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

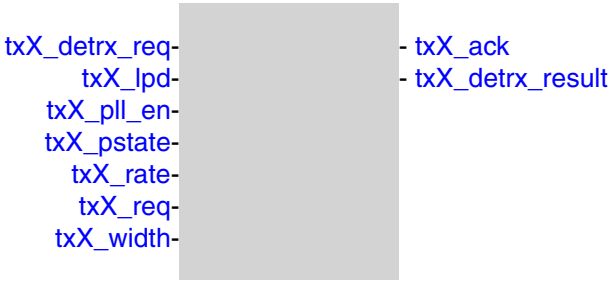


Table 1-35 Transmitter Request (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_ack	O	Transmitter acknowledge The assertion of this signal indicates the requested transmitter setting (txX_req) is complete or the requested RX detection operation is complete. This signal stays asserted until the request input (txX_req or txX_detrx_req) is de-asserted, after which txX_ack is de-asserted. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_detrx_req	I	Transmitter RX-detection request This signal indicates an RX-detection request. When txX_ack is asserted then the detection result on txX_detrx_result is valid. **Notes:** ■ This signal should stay high until the txX_ack output is asserted, then txX_detrx_req must be de-asserted. ■ RX detection is only valid when the TX is in the P1 power state. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
txX_detrx_result	O	Transmitter RX-detection result This signal is valid when txX_ack is asserted following a txX_detrx_req request. ■ 1'b0: Receiver not detected ■ 1'b1: Receiver detected **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_lpd	I	Transmitter lane power down

Table 1-35 Transmitter Request (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Powers down the transmitter into a power state equivalent to P1. This signal is useful for selectively powering down lanes in a link while maintaining the power state of the link as a whole (for example, link in P0). **Note:** If the txX_pstate(1:0) input is set to P2, it takes priority over txX_lpd and the transmitter is placed in P2 state. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_pll_en	I	TX MPLL enable Powers up or powers down the ROPLL, MPLLA or MPLLB (determined using txX_ropll_refsel and txX_ropll_bypass signals). For normal operation, this input must be set to 1 when txX_pstate (1:0) is in P0 or P0s state. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_pstate[1:0]	I	Transmitter power state Sets the power state of the transmitter. For example, the settings for PCIe are: ■ 2'b00: P0-Transmitter is fully powered up. ■ 2'b01: P0S-Transmitter common mode is held, TX analog clocks are active, but TX serializer is off. ■ 2'b10: P1-Transmitter common mode is held, but TX analog clocks and TX serializer are off. ■ 2'b11: P2-Transmitter is powered down. For settings for other protocols, see the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

Table 1-35 Transmitter Request (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_rate[2:0]	I	TX data rate Select the baud rate for the transmit path. ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b100: baud/16 ■ 3'b101: Reserved ■ 3'b110: Reserved ■ 3'b111: Reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_req	I	Transmitter request This signal indicates a new transmitter setting request. This signal must be asserted only when txX_ack is low and should stay high until the txX_ack output is asserted. After txX_ack is asserted, txX_req must be de-asserted. **Notes:** The following input signals are captured by the PHY when txX_req is asserted: ■ txX_pstate ■ txX_lpd ■ txX_rate ■ txX_width ■ txX_ropll_refsel ■ txX_ropll_bypass ■ txX_pll_en ■ txX_misc ■ txX_dcc_ctrl_range ■ mpll(a,b)_recal_force_en ■ mpll(a,b)_recal_skip_en ■ mpll(a,b)_recal_bank_sel These signals must be stable before txX_req is asserted, and remain stable until txX_req is de-asserted. Any combination of changes in these signals is valid for a single request, and the completion of all changes is indicated by the assertion of txX_ack. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High

Table 1-35 Transmitter Request (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_width[2:0]	I	TX data width Select the width of the input transmit data. ■ 3'b000: 8-bit ■ 3'b001: 10-bit ■ 3'b010: 16-bit ■ 3'b011: 20-bit ■ 3'b100: 32-bit ■ 3'b101: 40-bit ■ 3'b110: 64-bit **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A

1.36 Transmitter Datapath (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

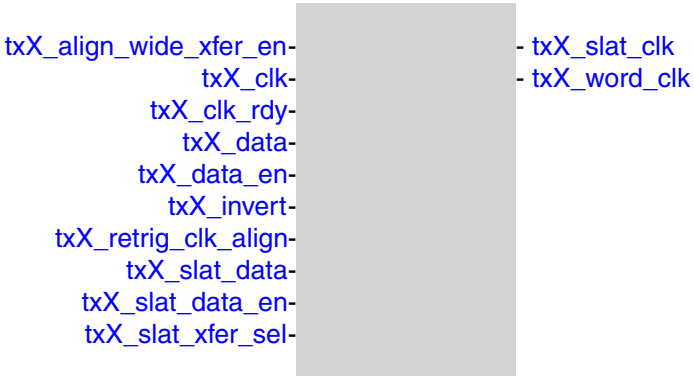


Table 1-36 Transmitter Datapath (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_align_wide_xfer_en	I	Transmitter alignment wide transfer enable When asserted, the internal data transfer from the txX_clk domain to the internal clock domain is performed using a width of 32 bits (instead of 16 bits). Assert this signal only when the data rate is > 16 Gbps. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_clk	I	Transmitter input clock Input data must be synchronous with this clock. If the lane is part of a multi-lane link and the link remains active (for example, in P0) then this clock should remain active (even if the lane is powered down or in reset). **Note:**The minimum high and low pulse width of this clock must be more than 2UI plus 100ps. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_clk_rdy	I	Transmitter input clock ready This signal is used for synchronizing TX lanes in a multi- lane link. When the txX_clk is active and stable, txX_clk_rdy must be asserted at the same time for all TX lanes and remain asserted as long as the link is active (for example, P0) even if some of the lanes in the link are powered down or in reset. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-36 Transmitter Datapath (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: txX_clk **Active State:** High
txX_data[63:0]	I	Transmitter input parallel data MSB is transmitted first. The valid range of the data is determined by the txX_width input as follows: ■ For txX_width=3'b000, Valid Data Range=txX_data(7:0) ■ For txX_width=3'b001, Valid Data Range=txX_data(9:0) ■ For txX_width=3'b010, Valid Data Range=txX_data(15:0) ■ For txX_width=3'b011, Valid Data Range=txX_data(19:0) ■ For txX_width=3'b100, Valid Data Range=txX_data(31:0) ■ For txX_width=3'b101, Valid Data Range=txX_data(39:0) ■ For txX_width=3'b110, Valid Data Range=txX_data(63:0) **Note:**The invalid bits of txX_data are ignored by the PHY. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** N/A
txX_data_en[1:0]	I	Transmitter data enable When asserted and the TX lane is in P0, then the TX output driver is enabled. If de-asserted or the TX lane is not in P0, then the TX output driver is disabled. In 64-bit mode: ■ tx_data_en[0] is used for tx_data[31:0] ■ tx_data_en[1] is used for tx_data[63:32] In all other modes, only tx_data_en[0] is used and tx_data_en[1] is not used and must be tied low. **Note:**For TX data rates less than 2 Gbps, txX_data_en must not be asserted within 15 txX_clk cycles after completing the txX_req and txX_ack handshake. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** High
txX_invert	I	Invert logical sense of txX_p and txX_m outputs This signal has the effect of inverting the output data stream. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** High
txX_retrig_clk_align	I	Pulse this PHY input to dynamically re-trigger clock alignment for retimer applications in txX_pstate=P0. Connect to 0 for non-retimer applications.

Table 1-36 Transmitter Datapath (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Note: Bit errors are expected on tx_p/m if this signal is pulsed when txX_data_en=1. **Exists:** Always **Synchronous To:** txX_clk **Active State:** High
txX_slat_clk	O	Transmitter's short latency mode output clock txX_slat_data and txX_slat_data_en must be synchronous to txX_slat_clk output of the same lane. For data rates less than 1 Gbps, this clock may glitch during a power state or rate change. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_slat_data[19:0]	I	Transmitter input parallel data for short latency mode which can be enabled by txX_slat_xfer_sel input. MSB is transmitted first. The valid range of the data is determined by the txX_width input as follows: ■ For txX_width=3'b010, Valid Data Range=txX_data(15:0) ■ For txX_width=3'b011, Valid Data Range=txX_data(19:0) **Notes:** ■ The invalid bits of txX_data are ignored by the PHY. ■ The txX_slat_data and txX_slat_data_en signals must be synchronous to txX_slat_clk output of the same lane. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_slat_clk **Active State:** N/A
txX_slat_data_en	I	Transmitter data enable for short latency mode When asserted and the TX lane is in P0, then the TX output driver is enabled. If de-asserted or the TX lane is not in P0, then the TX output driver is disabled. txX_slat_data and txX_slat_data_en must be synchronous to txX_slat_clk output of the same lane. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_slat_clk **Active State:** High

Table 1-36 Transmitter Datapath (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_slat_xfer_sel	I	Static input signal that selects between short latency TX mode and normal latency TX mode. txX_reset or phy_reset is required after changing this input. Tie low if unused. **Notes:** ■ txX_slat_data and txX_slat_data_en should to be synchronous to txX_slat_clk output of the same lane. ■ Due to lane-to-lane skew requirements, this TX short latency mode is not suitable for multi-lane link applications (like PCIe). ■ txX_slat_clk is high frequency due to 16 or 20-bit TX data width, as such timing closure may be difficult for data rates greater than 20 Gbps. ■ Latency reduction in TX short latency mode is up to 48 UI compared to normal 32-bit TX latency mode. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_word_clk	O	Transmitter word clock output The TX word clock from the TX lane. This output is meant for debugging purposes only. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

1.37 Transmitter Control (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

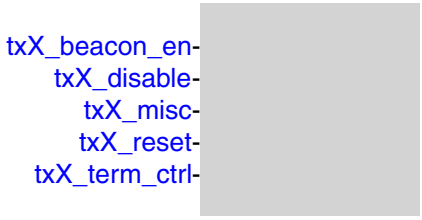


Table 1-37 Transmitter Control (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_beacon_en	I	Transmitter beaconing enable This signal enables or disables transmitter beaconing. The period for transmit pulses is between 20-50 ns. **Note:**Beacon signaling is not supported in P2.PG state. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_disable	I	Transmitter disable Puts the transmitter in a low-power mode where all the transmitter blocks, including TX common mode, are turned off. This signal must be asserted when in PCIe L1 Substate L1.2. The transmitter must be in the P1 power state when using this signal. This input asynchronously disables TX terminations and does not require any clock to be supplied to the PHY. **Note:**The txX_disable signal must be kept de-asserted from the negation of phy_reset until the txX_ack/rxX_ack signals de-assert. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
txX_misc[7:0]	I	TX miscellaneous controls This input controls miscellaneous settings in the TX analog block. This input must be set according to the values provided in the "PHY Usage and Configuration" chapter. tx_misc(7:1): reserved tx_misc(0): controls source of rxX_hs_clk_p/m clock ■ If tx_misc(0) == 0, rxX_hs_clk_p/m is connected to divided down RX VCO clock. ■ If tx_misc[0] == 1, rxX_hs_clk_p/m is connected to the divided down TX PLL clock. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always

Table 1-37 Transmitter Control (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Synchronous To: Asynchronous **Active State:** N/A
txX_reset	I	Transmitter reset This signal internally resets the TX datapath, all the transmitter settings and state machines, including common-mode adjustment and receiver detection state machines.**Note:**This input must be asserted for minimum 10 reference clock cycles. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous (must not glitch) **Active State:** High
txX_term_ctrl[3:0]	I	Transmit termination control This signal sets the target transmitter termination impedance. ■ For txX_term_ctrl=3'b000, impedance=54 ohms ■ For txX_term_ctrl=3'b001, impedance=52 ohms ■ For txX_term_ctrl=3'b010, impedance=50 ohms (default) ■ For txX_term_ctrl=3'b011, impedance=48 ohms ■ For txX_term_ctrl=3'b100, impedance=46 ohms ■ For txX_term_ctrl=3'b101, impedance=44 ohms ■ For txX_term_ctrl=3'b110, impedance=42 ohms ■ For txX_term_ctrl=3'b111, impedance=40 ohms **Note:**This input can be changed only when phy_reset or txX_reset or txX_req is asserted. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

1.38 Transmitter Request (for X = 0; X <= 1) Signals

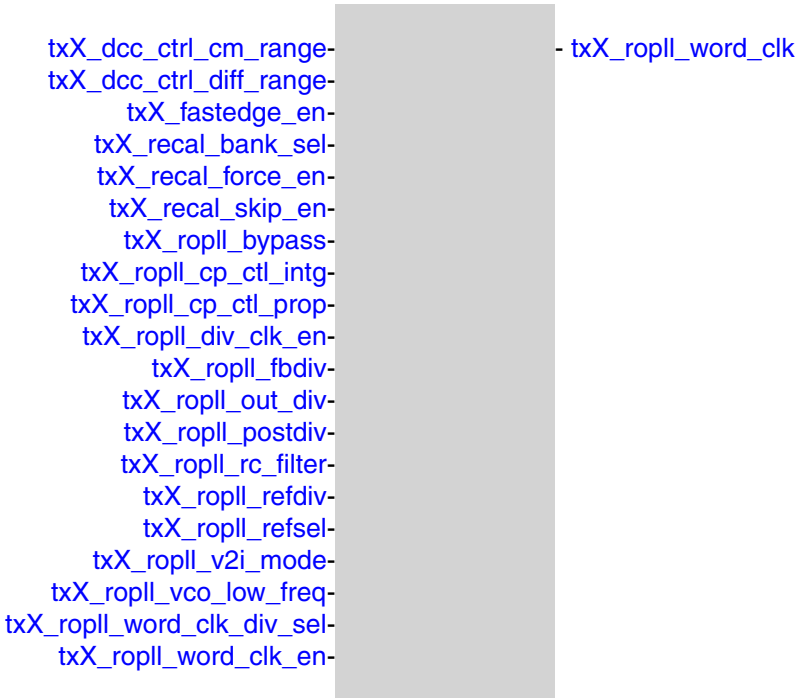


Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_dcc_ctrl_cm_range[3:0]	I	TX internal common mode clock duty cycle control range Sets the range of internal clock duty cycle correction circuitry. Set this input according to the tables provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_dcc_ctrl_diff_range[3:0]	I	TX Internal differential clock duty cycle control range Sets the range of internal clock duty cycle correction circuitry. Set this input according to the tables provided in the "PHY Usage and Configuration" chapter. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_fastedge_en	I	Enables "Fast Edge" mode in TX_DRIVER/Serializer Reduces rise and fall time of TX output when transitioning-allowing for wider TX output eye openings. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro.

Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Voltage Range: 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_recal_bank_sel[1:0]	I	TX re-calibration bank select Selects the bank to store the TX duty cycle correction (DCC) values **Note:**PHY samples this input with the assertion of txX_req. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_recal_force_en	I	TX Force re-calibration Forces a re-calibration of TX DCC in the selected bank regardless of the state of the corresponding "*done" signal. **Note:**PHY samples this input with the assertion of txX_req. This input has a higher priority than the txX_recal_skip_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_recal_skip_en	I	TX Skip re-calibration Skips re-calibration of TX DCC in the selected bank regardless of the state of the corresponding "*done" signal. **Note:**PHY samples this input with the assertion of txX_req. This input has a lower priority than the txX_recal_force_en signal. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_ropll_bypass	I	TX ROPLL bypass Function: Bypasses the ROPLL in the TX clock path and propagates the MPLL clock directly to the TX ■ 0: Cascaded mode (TX clock is provided by ROPLL) ■ 1: Direct mode (TX clock is provided by MPLL) This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_cp_ctl_intg[6:0]	I	TX ROPLL integral charge-pump controls

Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
		Set the integral charge-pump current to configure the PLL loop bandwidth. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_cp_ctl_prop[6:0]	I	TX ROPLL proportional charge-pump controls Set the proportional charge-pump current to configure the PLL loop bandwidth This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_div_clk_en	I	Per-lane enable for txX_ropll_div_clk output txX_req/ack handshake is required when changing this input This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High
txX_ropll_fbdiv[6:0]	I	TX ROPLL feedback clock division This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_out_div[7:0]	I	Per-lane setting for the division ratio of the txX_ropll_div_clk output for non-PCIe protocols. Valid settings are 4 to 257. txX_req/ack handshake is required when changing this input. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_ropll_postdiv[2:0]	I	TX ROPLL post divider Divides ROPLL VCO to ropll word clock divider clock paths ■ 2'b000: div1 ■ 2'b001: div2 ■ 2'b010: div4 ■ 2'b011: div8 ■ 2'b100: Reserved ■ 2'b101: Reserved ■ 2'b110: Reserved ■ 2'b111: Reserved This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_rc_filter[2:0]	I	TX ROPLL RC filter programming and feedback divider resampling bypass ■ bit (2) - bypasses resampling flop on feedback divider output ■ bit (1:0) - RC filter programming This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_ropll_refdiv[3:0]	I	TX ROPLL reference clock division Sets the divide ratio on the reference clock to the ROPLL as follows: ■ 4'b0000: div16 ■ 4'b0001: div1 ■ 4'b0010: div2 ■ 4'b0011: div3 ■ 4'b0100: div4 ■ 4'b0101: div5 ■ 4'b0110: div6 ■ 4'b0111: Reserved ■ 4'b1000: div8 ■ 4'b1001: div9 ■ 4'b1010: div10 ■ 4'b1011: Reserved ■ 4'b1100: Reserved ■ 4'b1101: Reserved ■ 4'b1110: Reserved ■ 4'b1111: div15 This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** N/A
txX_ropll_refsel[1:0]	I	TX ROPLL ref clock selection ■ 2'b00: MPLLA ■ 2'b01: MPLLB ■ 2'b10: REFA_CLK ■ 2'b11: REFB_CLK This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_v2i_mode[1:0]	I	TX ROPLL v2i bias programming This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A

Table 1-38 Transmitter Request (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_ropll_vco_low_freq[1:0]	I	TX ROPLL VCO frequency range ■ 2'b00: 13 GHz-16 GHz ■ 2'b01: Reserved ■ 2'b10: 10 GHz-13 GHz ■ 2'b11: 8 GHz-10 GHz This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_word_clk	O	TX ROPLL word clock output Note: This clock stops and restarts during rate change **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_word_clk_div_sel[1:0]	I	TX ROPLL word clock division selection This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** N/A
txX_ropll_word_clk_en	I	TX ROPLL word clock output enable **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.39 Per Lane Transmit Interface (for X = 0; X <= 1) Signals

txX_dly_cal_en-

Table 1-39 Per Lane Transmit Interface (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_dly_cal_en	I	Per-lane TX delay calibration enable input For per protocol settings for this control input, see the "PHY Usage and Configuration" chapter. This PHY input is removed if you define the {PREFIX}_REMOVE_CNTX_RSTR_INPUTS macro. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** N/A **Active State:** High

1.40 Transmitter Equalization (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

Table 1-40 Transmitter Equalization (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_eq_main[5:0]	I	Transmitter amplitude adjustment control Control for setting the transmitter driver output amplitude (main coefficient) txX_eq_main(5:0): Integer value (0 to 24) **Notes:** ■ If the fractional post is non-zero (in other words, txX_eq_post(1:0) != 2'b00) or the fractional pre is non-zero (in other words, txX_eq_pre(1:0) != 2'b00), then the txX_eq_main(5:0) input must be increased by 1. This is to complete the fractional portion given a fixed number of transmitter legs. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main(5:0) values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** N/A
txX_eq_post[5:0]	I	Transmitter post-emphasis level adjustment control Control for setting the transmitter driver output post- emphasis (post coefficient) ■ txX_eq_post(5:2): Integer value (0 to 8) ■ txX_eq_post(1:0): Fraction value (0, 0.25, 0.5, 0.75) **Notes:** ■ The maximum total post value is 8; therefore, txX_eq_post(5:2) = 8 is permitted only if the fractional part (txX_eq_post(1:0)) is zero. Otherwise, txX_eq_post(5:2) must be < 7. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main(5:0) values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** N/A

Table 1-40 Transmitter Equalization (Per-Lane Transmit Interface Group) (for X = 0; X <= 1) Signals (continued)

Port Name	I/O	Description
txX_eq_pre[5:0]	I	Transmitter pre-emphasis level adjustment control Controls transmitter driver output pre-emphasis (preshoot coefficient). ■ txX_eq_pre(5:2): Integer value (0 to 6) ■ txX_eq_pre(1:0): Fraction value (0, 0.25, 0.5, 0.75) **Notes:** ■ The maximum total pre value is 6; therefore, txX_eq_pre(5:2) = 6 is permitted only if the fractional part (txX_eq_pre(1:0)) is zero; otherwise, txX_eq_pre(5:2) must < 5. ■ This signal can be updated in a single txX_clk cycle to a known valid value. Alternately, the txX_eq_main(5:0) values are to be sequenced to free up resources as described in the "PHY Usage and Configuration" chapter. An invalid value causes TX serial data corruption in simulation. **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** txX_clk **Active State:** N/A

1.41 Transmitter Test (for X = 0; X <= 1) Signals

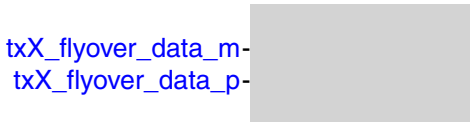


Table 1-41 Transmitter Test (for X = 0; X <= 1) Signals

Port Name	I/O	Description
txX_flyover_data_m	I	Transmit Flyover Data inputs (txX_flyover_data_m) Inputs are directly connected to TX bump pads when test_flyover_en is enabled **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High
txX_flyover_data_p	I	Transmit Flyover Data inputs (txX_flyover_data_p) Inputs are directly connected to TX bump pads when test_flyover_en is enabled **Voltage Range:** 0-vddcore **Exists:** Always **Synchronous To:** Asynchronous **Active State:** High

2

Register Descriptions

This chapter details all possible registers in the IP. They are arranged hierarchically into maps and blocks (banks).

Exists Expression

The Exist expressions (if present) indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the Exists expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address*.

Memory Access Attributes

The Memory Access attribute is defined as `<ReadBehavior>/<WriteBehavior>` which are defined in the following table.

Table 2-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field in a manner not described by any of the above.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.

Table 2-1 Possible Read and Write Behaviors (continued)

Read (or Write) Behavior	Description
WS	Any write sets this register field.
WM	A write modifies this register field in a manner not described by any of the above.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 2-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 2-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the IP updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

Register definitions for each component memory map.

Table 2-4 Registers for the E32 Memory Map

Register	Offset	Description
Describes all the registers in this PHY. E32_mem_map Exists: Always		
"SUP_DIG_IDCODE_LO" on page 217	0x0	Low 16 Bits of IDCODE.
"SUP_DIG_IDCODE_HI" on page 218	0x1	High 16 Bits of IDCODE.
"SUP_DIG_REFCLKA_OVRD_IN_0" on page 219	0x2	Override Values for Incoming REFCLKA and RESET Controls from ASIC.
"SUP_DIG_REFCLKB_OVRD_IN_0" on page 221	0x3	Override Values for Incoming REFCLKA and RESET Controls from ASIC.
"SUP_DIG_REFCLK_OVRD_IN_1" on page 223	0x4	Override Values for Incoming REFCLK and RESET Controls from ASIC.
"SUP_DIG_MPLLA_DIV_CLK_OVRD_IN" on page 225	0x5	Override Values for Incoming/Outgoing Control Signals.
"SUP_DIG_MPLLB_DIV_CLK_OVRD_IN" on page 227	0x6	Override Values for Incoming/Outgoing Control Signals.
"SUP_DIG_MPLLA_OVRD_IN_0" on page 229	0x7	Override Values for Incoming MPLLA Controls from ASIC, Register 0.
"SUP_DIG_MPLLA_OVRD_IN_1" on page 231	0x8	Override Values for Incoming MPLLA Controls from ASIC, Register 1.
"SUP_DIG_MPLLA_OVRD_IN_2" on page 232	0x9	Override Values for Incoming MPLLA Controls from ASIC, Register 2.
"SUP_DIG_MPLLA_BW_LOW_OVRD_IN" on page 233	0xa	Override Values for Incoming MPLLA Bandwidth Low Controls from ASIC.
"SUP_DIG_MPLLA_BW_HIGH_OVRD_IN" on page 234	0xb	Override Values for Incoming MPLLA Bandwidth High Controls from ASIC.
"SUP_DIG_MPLLA_SSC_OVRD_IN_0" on page 235	0xc	Override Values for Incoming MPLLA SSC Controls from ASIC, Register 0.
"SUP_DIG_MPLLA_SSC_OVRD_IN_1" on page 237	0xd	Override Values for Incoming MPLLA SSC Controls from ASIC, Register 1.
"SUP_DIG_MPLLA_SSC_OVRD_IN_2" on page 238	0xe	Override Values for Incoming MPLLA Controls from ASIC, Register 2.
"SUP_DIG_MPLLA_SSC_OVRD_IN_3" on page 239	0xf	Override Values for Incoming MPLLA Controls from ASIC, Register 3.
"SUP_DIG_MPLLA_SSC_OVRD_IN_4" on page 240	0x10	Override Values for Incoming MPLLA Controls from ASIC, Register 4.
"SUP_DIG_MPLLA_SSC_OVRD_IN_5" on page 241	0x11	Override Values for Incoming MPLLA Controls from ASIC, Register 5.
"SUP_DIG_MPLLA_SSC_OVRD_IN_6" on page 242	0x12	Override Values for Incoming MPLLA Controls from ASIC, Register 6.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_MPLLB_OVRD_IN_0" on page 243	0x13	Override Values for Incoming MPLLB Controls from ASIC, Register 0.
"SUP_DIG_MPLLB_OVRD_IN_1" on page 245	0x14	Override Values for Incoming MPLLB Controls from ASIC, Register 1.
"SUP_DIG_MPLLB_OVRD_IN_2" on page 246	0x15	Override Values for Incoming MPLLB Controls from ASIC, Register 2.
"SUP_DIG_MPLLB_BW_LOW_OVRD_IN" on page 247	0x16	Override Values for Incoming MPLLB Bandwidth Low Controls from ASIC.
"SUP_DIG_MPLLB_BW_HIGH_OVRD_IN" on page 248	0x17	Override Values for Incoming MPLLB Bandwidth High Controls from ASIC.
"SUP_DIG_MPLLB_SSC_OVRD_IN_0" on page 249	0x18	Override Values for Incoming MPLLB SSC Controls from ASIC, Register 0.
"SUP_DIG_MPLLB_SSC_OVRD_IN_1" on page 251	0x19	Override Values for Incoming MPLLB SSC Controls from ASIC, Register 1.
"SUP_DIG_MPLLB_SSC_OVRD_IN_2" on page 252	0x1a	Override Values for Incoming MPLLB Controls from ASIC, Register 2.
"SUP_DIG_MPLLB_SSC_OVRD_IN_3" on page 253	0x1b	Override Values for Incoming MPLLB Controls from ASIC, Register 3.
"SUP_DIG_MPLLB_SSC_OVRD_IN_4" on page 254	0x1c	Override Values for Incoming MPLLB Controls from ASIC, Register 4.
"SUP_DIG_MPLLB_SSC_OVRD_IN_5" on page 255	0x1d	Override Values for Incoming MPLLB Controls from ASIC, Register 5.
"SUP_DIG_MPLLB_SSC_OVRD_IN_6" on page 256	0x1e	Override Values for Incoming MPLLB Controls from ASIC, Register 6.
"SUP_DIG_SUP_OVRD_IN_0" on page 257	0x1f	Override Values for Support Block ASIC Inputs, Register 0.
"SUP_DIG_SUP_OVRD_IN_1" on page 259	0x20	Override Values for Support Block ASIC Inputs, Register 1.
"SUP_DIG_SUP_OVRD_IN_2" on page 260	0x21	Override Values for Support Block ASIC Inputs, Register 2.
"SUP_DIG_SUP_OVRD_OUT" on page 261	0x22	Override Values for Support Block ASIC Outputs.
"SUP_DIG_LVL_OVRD_IN" on page 264	0x23	Override Values for Level Settings.
"SUP_DIG_DEBUG" on page 266	0x24	Debug Controls.
"SUP_DIG_MPLLA_ASIC_IN_0" on page 267	0x25	Current Values for Incoming MPLLA Controls from ASIC, Register 0.
"SUP_DIG_MPLLA_ASIC_IN_1" on page 270	0x26	Current Values for Incoming MPLLA Controls from ASIC, Register 1.
"SUP_DIG_MPLLA_BW_LOW_ASIC_IN" on page 271	0x27	Current Values for Incoming MPLLA Bandwidth Low Controls from ASIC.
"SUP_DIG_MPLLA_BW_HIGH_ASIC_IN" on page 272	0x28	Current Values for Incoming MPLLA Bandwidth High Controls from ASIC.
"SUP_DIG_MPLLA_SSC_ASIC_IN_0" on page 273	0x29	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 0.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_MPLLA_SSC_ASIC_IN_1" on page 275	0x2a	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 1.
"SUP_DIG_MPLLA_SSC_ASIC_IN_2" on page 276	0x2b	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 2.
"SUP_DIG_MPLLA_SSC_ASIC_IN_3" on page 277	0x2c	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 3.
"SUP_DIG_MPLLA_SSC_ASIC_IN_4" on page 278	0x2d	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 4.
"SUP_DIG_MPLLA_SSC_ASIC_IN_5" on page 279	0x2e	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 5.
"SUP_DIG_MPLLA_SSC_ASIC_IN_6" on page 280	0x2f	Current Values for Incoming MPLLA SSC Controls from ASIC, Register 6.
"SUP_DIG_MPLLB_ASIC_IN_0" on page 281	0x30	Current Values for Incoming MPLLB Controls from ASIC, Register 0.
"SUP_DIG_MPLLB_ASIC_IN_1" on page 284	0x31	Current Values for Incoming MPLLB Controls from ASIC, Register 1.
"SUP_DIG_MPLLB_BW_LOW_ASIC_IN" on page 285	0x32	Current Values for Incoming MPLLB Bandwidth Low Controls from ASIC.
"SUP_DIG_MPLLB_BW_HIGH_ASIC_IN" on page 286	0x33	Current Values for Incoming MPLLB Bandwidth High Controls from ASIC.
"SUP_DIG_MPLLB_SSC_ASIC_IN_0" on page 287	0x34	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 0.
"SUP_DIG_MPLLB_SSC_ASIC_IN_1" on page 289	0x35	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 1.
"SUP_DIG_MPLLB_SSC_ASIC_IN_2" on page 290	0x36	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 2.
"SUP_DIG_MPLLB_SSC_ASIC_IN_3" on page 291	0x37	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 3.
"SUP_DIG_MPLLB_SSC_ASIC_IN_4" on page 292	0x38	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 4.
"SUP_DIG_MPLLB_SSC_ASIC_IN_5" on page 293	0x39	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 5.
"SUP_DIG_MPLLB_SSC_ASIC_IN_6" on page 294	0x3a	Current Values for Incoming MPLLB SSC Controls from ASIC, Register 6.
"SUP_DIG_ASIC_IN_0" on page 295	0x3b	Current Values for Incoming SUP Control Signals from ASIC, Register 0.
"SUP_DIG_REFCLKA_ASIC_IN" on page 297	0x3c	Current Values for Incoming SUP Control Signals from ASIC, Register 1.
"SUP_DIG_REFCLKB_ASIC_IN" on page 300	0x3d	Current Values for Incoming SUP Control Signals from ASIC, Register 0.
"SUP_DIG_ASIC_IN_1" on page 303	0x3e	Current Values for Incoming SUP Control Signals from ASIC, Register 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_ASIC_IN_2" on page 305	0x3f	Current Values for Incoming SUP Control Signals from ASIC, Register 2.
"SUP_DIG_LVL_ASIC_IN" on page 307	0x40	Current Values for Incoming Level Controls from ASIC.
"SUP_DIG_ASIC_IN_3" on page 309	0x41	Current Values for Incoming Controls from ASIC, Register 3.
"SUP_DIG_SUP_OVRD_MISC" on page 310	0x42	Override Values for Incoming SUP MISC BUS from ASIC.
"SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN" on page 311	0x43	Current Values for Incoming Level Controls from ASIC.
"SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN" on page 312	0x44	Current Values for Incoming Level Controls from ASIC.
"SUP_DIG_NBR_OF_LANES_IN" on page 313	0x45	Current Values for Incoming Level Controls from ASIC.
"SUP_DIG_MPLLA_AON_OVRD_IN" on page 314	0x46	Override Values for MPLLA Incoming AON tune code from ASIC.
"SUP_DIG_MPLLB_AON_OVRD_IN" on page 316	0x47	Override Values for MPLLB Incoming AON tune code from ASIC.
"SUP_DIG_REFCLK_OVRD_IN_2" on page 318	0x48	Override Values for Incoming REFCLK Controls from ASIC
"SUP_DIG_ASIC_IN_4" on page 320	0x49	Current Values for Incoming Controls from ASIC, Register 4.
"SUP_DIG_ASIC_IN_5" on page 321	0x4a	Current Values for Incoming Controls from ASIC, Register 5.
"SUP_DIG_ASIC_OUT_0" on page 322	0x4b	Current Values for Incoming Controls from ASIC, Register 0.
"SUP_DIG_ASIC_OUT_1" on page 324	0x4c	Current Values for Incoming Controls from ASIC, Register 1.
"SUP_DIG_RTUNE_DEBUG" on page 326	0x60	Resistor Tuning Debug Controls.
"SUP_DIG_RTUNE_CONFIG" on page 328	0x61	Configure Rtune Operation.
"SUP_DIG_RTUNE_STAT" on page 330	0x62	Resistor Tuning Register Status.
"SUP_DIG_RTUNE_RX_SET_VAL" on page 331	0x63	Set Value of RX Resistor.
"SUP_DIG_RTUNE_TXDN_SET_VAL" on page 332	0x64	Set Value of TX-DN Resistor.
"SUP_DIG_RTUNE_TXUP_SET_VAL" on page 333	0x65	Set Value of TX-UP Resistor.
"SUP_DIG_RTUNE_RX_STAT" on page 334	0x66	RX Resistor Tuning Register Status.
"SUP_DIG_RTUNE_TXDN_STAT" on page 335	0x67	TX-DN Resistor Tuning Register Status.
"SUP_DIG_RTUNE_TXUP_STAT" on page 336	0x68	TX-UP Resistor Tuning Register Status.
"SUP_DIG_RTUNE_CONFIG2" on page 337	0x69	Configure Rtune Operation.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_BG_CTL_BG_PWRUP_TIME_0" on page 338	0x70	Bandgap Power-Up Time Register 0.
"SUP_DIG_BG_CTL_BG_PWRUP_TIME_1" on page 339	0x71	Bandgap Power-Up Time Register 1.
"SUP_DIG_BG_CTL_BG_PWRUP_TIME_2" on page 340	0x72	Bandgap Power-Up Time Register 2.
"SUP_DIG_BG_CTL_BG_PWRUP_TIME_3" on page 341	0x73	Bandgap Power-Up Time Register 3.
"SUP_DIG_BG_CTL_REF_PWRUP_TIME_0" on page 342	0x74	Prescaler Power-Up Time Register.
"SUP_DIG_BG_CTL_BG_STATE_STATUS " on page 343	0x75	Current Values for Bandgap State Machine State Signals.
"SUP_DIG_ANA_MPLLA_OVRD_OUT0" on page 344	0x80	Override Value for MPLLA Signals Going to ANA.
"SUP_DIG_ANA_MPLLA_OVRD_OUT1" on page 347	0x81	Override Value for MPLLA Signals Going to ANA.
"SUP_DIG_ANA_MPLLB_OVRD_OUT0" on page 350	0x82	Override Value for MPLLB Signals Going to ANA.
"SUP_DIG_ANA_MPLLB_OVRD_OUT1" on page 353	0x83	Override Value for MPLLB Signals Going to ANA.
"SUP_DIG_ANA_RTUNE_OVRD_OUT" on page 356	0x84	Override Value for RTUNE Signals Going to ANA.
"SUP_DIG_ANA_STAT" on page 358	0x85	SUP Input Status Register for SUP ANA Outputs.
"SUP_DIG_ANA_ANA_OVRD_OUT" on page 360	0x86	Override Values for ana_async_rst and bandgap Signals Going to ANA.
"SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_0" on page 362	0x87	Override Value for MPLLA PMIX Signals Going to ANA, Register 0.
"SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_1" on page 364	0x88	Override Value for MPLLA PMIX Signals Going to ANA, Register 1.
"SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_0" on page 365	0x89	Override Value for MPLLB PMIX Signals Going to ANA, Register 0.
"SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_1" on page 367	0x8a	Override Value for MPLLB PMIX Signals Going to ANA, Register 1.
"SUP_DIG_ANA_REF_DCO_OVRD_OUT" on page 368	0x8b	Control Settings to Analog for Prescaler Digitally Controlled Oscillator (DCO).
"SUP_DIG_ANA_SUP_MISC_OVRD_OUT" on page 370	0x8c	Override Value for sup_misc Bus Going to ANA.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_0" on page 371	0xa0	MPLL Calibration Controls, Register 0.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1" on page 373	0xa1	MPLL Calibration Controls, Register 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_2" on page 375	0xa2	MPLL Calibration Controls, Register 2.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT" on page 377	0xa3	MPLL Status Register.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0" on page 380	0xa4	MPLL Power Up Time Register 0.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1" on page 382	0xa5	MPLL Power Up Time Register 1.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2" on page 383	0xa6	MPLL Power Up Time Register 2.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3" on page 384	0xa7	MPLL Power Up Time Register 3.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_TUNE_VAL" on page 385	0xa8	MPLL Coarse and Fine Tune Value Registers.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE" on page 387	0xa9	Value for MPLL Coarse and Fine Tune when Skipping Calibration.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0" on page 389	0xaa	MPLL Power Up FSM Config, Register 0.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1" on page 390	0xab	MPLL Power Up FSM Config, Register 1.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2" on page 391	0xac	MPLL Power Up FSM Config, Register 2.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3" on page 392	0xad	MPLL Power Up FSM Config, Register 3.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4" on page 393	0xae	Thresholds for MPLL CAL Update Timer and MPLL VCO Stabilization Short Timer.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5" on page 394	0xaf	Thresholds for PCLK Enable and MPLL VCO Clock Stabilization Short Timer.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6" on page 395	0xb0	MPLL Power Up Time Register 6.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7" on page 396	0xb1	MPLL Power Up Time Register 7.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL" on page 397	0xb2	MPLL VCO Gain Calibration FB Count1 Registers.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL" on page 398	0xb3	MPLL VCO Gain Calibration FB Count2 Registers.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0" on page 399	0xb4	MPLL VCO Gain Calibration Controls, Register 0.
"SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL" on page 400	0xb6	MPLL VCO Drift Count Register.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_3" on page 401	0xb7	MPLL Calibration Controls, Register 3
"SUP_DIG_MPLLA_SSC_FRAC_OUT" on page 402	0xb8	frac_out Override for Fractional Control.
"SUP_DIG_MPLLA_SSC_SSC_RAMP" on page 403	0xb9	ssc_ramp Override for Spread Spectrum Control.
"SUP_DIG_MPLLA_SSC_CONFIG" on page 404	0xba	Configuration Register for SSC and Fractional Control.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_0" on page 406	0xc0	MPLL Calibration Controls, Register 0.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1" on page 408	0xc1	MPLL Calibration Controls, Register 1.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_2" on page 410	0xc2	MPLL Calibration Controls, Register 2.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT" on page 412	0xc3	MPLL Status Register.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0" on page 415	0xc4	MPLL Power Up Time Register 0.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1" on page 417	0xc5	MPLL Power Up Time Register 1.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2" on page 418	0xc6	MPLL Power Up Time Register 2.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3" on page 419	0xc7	MPLL Power Up Time Register 3.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_TUNE_VAL" on page 420	0xc8	MPLL Coarse and Fine Tune Value Registers.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE" on page 422	0xc9	Value for MPLL Coarse and Fine Tune when Skipping Calibration.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0" on page 424	0xca	MPLL Power Up FSM Config, Register 0.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1" on page 425	0xcb	MPLL Power Up FSM Config, Register 1.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2" on page 426	0xcc	MPLL Power Up FSM Config, Register 2.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3" on page 427	0xcd	MPLL Power Up FSM Config, Register 3.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4" on page 428	0xce	Thresholds for MPLL CAL Update Timer and MPLL VCO Stabilization Short Timer.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5" on page 429	0xcf	Thresholds for PCLK Enable and MPLL VCO Clock Stabilization Short Timer.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6" on page 430	0xd0	MPLL Power Up Time Register 6.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7" on page 431	0xd1	MPLL Power Up Time Register 7.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL" on page 432	0xd2	MPLL VCO Gain Calibration FB Count1 Registers.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL" on page 433	0xd3	MPLL VCO Gain Calibration FB Count2 Registers.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0" on page 434	0xd4	MPLL VCO Gain Calibration Controls, Register 0.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL" on page 435	0xd6	MPLL VCO Drift Count Register.
"SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_3" on page 436	0xd7	MPLL Calibration Controls, Register 3
"SUP_DIG_MPLLB_SSC_FRAC_OUT" on page 437	0xd8	frac_out Override for Fractional Control.
"SUP_DIG_MPLLB_SSC_SSC_RAMP" on page 438	0xd9	ssc_ramp Override for Spread Spectrum Control.
"SUP_DIG_MPLLB_SSC_CONFIG" on page 439	0xda	Configuration Register for SSC and Fractional Control.
"SUP_ANA_MPLLA_LOOP_CTRL" on page 441	0xe0	MPLLA_LOOP_CTRL
"SUP_ANA_MPLLA_OVRD" on page 443	0xe1	MPLLA_OVRD
"SUP_ANA_MPLLA_ATB1" on page 445	0xe2	MPLLA_ATB1
"SUP_ANA_MPLLA_ATB2" on page 447	0xe3	MPLLA_ATB2
"SUP_ANA_MPLLA_ATB3" on page 449	0xe4	MPLLA_ATB3
"SUP_ANA_MPLLB_LOOP_CTRL" on page 451	0xe5	MPLLB_LOOP_CTRL
"SUP_ANA_MPLLB_OVRD" on page 453	0xe6	MPLLB_OVRD
"SUP_ANA_MPLLB_ATB1" on page 455	0xe7	MPLLB_ATB1
"SUP_ANA_MPLLB_ATB2" on page 457	0xe8	MPLLB_ATB2
"SUP_ANA_MPLLB_ATB3" on page 459	0xe9	MPLLB_ATB3
"SUP_ANA_RTUNE_CTRL" on page 461	0xea	RTUNE_CTRL
"SUP_ANA_SWITCH_PWR_MEAS" on page 463	0xeb	SWITCH_PWR_MEAS
"SUP_ANA_SWITCH_MISC_MEAS" on page 465	0xec	SWITCH_MISC_MEAS
"SUP_ANA_BG" on page 467	0xed	BG
"SUP_ANA_MPLLA" on page 469	0xee	MPLLA
"SUP_ANA_MPLLB" on page 471	0xef	MPLLB
"SUP_ANA_PREA" on page 473	0xf0	PREA

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"SUP_ANA_PREB" on page 475	0xf1	PREB
"SUP_ANA_PREC" on page 477	0xf2	PREC
"SUP_ANA_PRED" on page 479	0xf3	PRED
"SUP_ANA_MPLLA_VCO_GAIN_CAL" on page 481	0xf4	MPLLA_VCO_GAIN_CAL
"SUP_ANA_MPLLA_RESERVED_2" on page 482	0xf5	MPLLA_RESERVED_2
"SUP_ANA_MPLLB_VCO_GAIN_CAL" on page 483	0xf6	MPLLB_VCO_GAIN_CAL
"SUP_ANA_MPLLB_RESERVED_2" on page 484	0xf7	MPLLB_RESERVED_2
"RAWCMN_DIG_CMN_CTL" on page 485	0x100	Common Control Register.
"RAWCMN_DIG_LANE_FSM_OP_XTND" on page 486	0x101	Lane FSM OP XTND Control Register.
"RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0" on page 487	0x102	Override Values for Incoming TX Term Offset.
"RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1" on page 488	0x103	Override Values for Incoming TX/RX Term Offset.
"RAWCMN_DIG_CMN_CTL_1" on page 489	0x104	Common Control Register 1.
"RAWCMN_DIG_MPLL_CONFIG" on page 490	0x105	MPLL Configuration Parameters.
"RAWCMN_DIG_ATE_ALU_CTRL" on page 492	0x106	ATE ALU Module Control Register.
"RAWCMN_DIG_ATE_ALU_ADDR" on page 493	0x107	Address for Read and Write Command for ATE ALU Module.
"RAWCMN_DIG_ATE_ALU_DATA " on page 494	0x108	Data for Write Command for ATE ALU Module.
"RAWCMN_DIG_ATE_ALU_FLAGS" on page 495	0x109	Flags from ATE ALU Module.
"RAWCMN_DIG_ATE_ALU_ACCUM" on page 496	0x10a	Stores ATE ALU Accumulator Result.
"RAWCMN_DIG_MPLL_IN" on page 497	0x10b	Status of mplla/b_init_cal_disable Inputs.
"RAWCMN_DIG_FW_PWRUP_DONE_A" on page 498	0x10c	Firmware Power-up Done_A Status.
"RAWCMN_DIG_MPLL_CLK_ASYNC_OVRD" on page 499	0x10d	Controls MPLL Output Clocks Asynchronously.
"RAWCMN_DIG_MPLL_RECAL_BANK_OVRD" on page 500	0x10e	Override Incoming mplla/b_recal Inputs.
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_1" on page 503	0x10f	ATE Override Input to Control Top-Level Inputs Register 1. RO: stands for re-ordered

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_2" on page 504	0x110	ATE Override Input to Control Top-Level Inputs Register 2. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_3" on page 505	0x111	ATE Override Input to Control Top-Level Inputs Register 3. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_4" on page 506	0x112	ATE Override Input to Control Top-Level Inputs Register 4. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_5" on page 508	0x113	ATE Override Input to Control Top-Level Inputs Register 5. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_6" on page 509	0x114	ATE Override Input to Control Top-Level Inputs Register 6. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_7" on page 510	0x115	ATE Override Input to Control Top-Level Inputs Register 7. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_8" on page 511	0x116	ATE Override Input to Control Top-Level Inputs Register 8. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_9" on page 512	0x117	ATE Override Input to Control Top-Level Inputs Register 9. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_10" on page 513	0x118	ATE Override Input to Control Top-Level Inputs Register 10. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_11" on page 514	0x119	ATE Override Input to Control Top-Level Inputs Register 11. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_12" on page 515	0x11a	ATE Override Input to Control Top-Level Inputs Register 12. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_13" on page 516	0x11b	ATE Override Input to Control Top-Level Inputs Register 13. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_14" on page 518	0x11c	ATE Override Input to Control Top-Level Inputs Register 14. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_15" on page 519	0x11d	ATE Override Input to Control Top-Level Inputs Register 15. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_16" on page 521	0x11e	ATE Override Input to Control Top-Level Inputs Register 16. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_17" on page 522	0x11f	ATE Override Input to Control Top-Level Inputs Register 17. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_18" on page 523	0x120	ATE Override Input to Control Top-Level Inputs Register 18. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_19" on page 524	0x121	ATE Override Input to Control Top-Level Inputs Register 19. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_20" on page 525	0x122	ATE Override Input to Control Top-Level Inputs Register 20. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_21" on page 526	0x123	ATE Override Input to Control Top-Level Inputs Register 21. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_22" on page 527	0x124	ATE Override Input to Control Top-Level Inputs Register 22. RO: stands for re-ordered

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_23" on page 528	0x125	ATE Override Input to Control Top-Level Inputs Register 23. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_24" on page 529	0x126	ATE Override Input to Control Top-Level Inputs Register 24. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_25" on page 531	0x127	ATE Override Input to Control Top-Level Inputs Register 25. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_26" on page 532	0x128	ATE Override Input to Control Top-Level Inputs Register 26. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_27" on page 535	0x129	ATE Override Input to Control Top-Level Inputs Register 27. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_28" on page 537	0x12a	ATE Override Input to Control Top-Level Inputs Register 28. RO: stands for re-ordered
"RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_29" on page 539	0x12b	ATE Override Input to Control Top-Level Inputs Register 29. RO: stands for re-ordered
"RAWCMN_DIG_NUMBER_OF_LANES" on page 542	0x12c	Stores Number of Lanes in PHY.
"RAWCMN_DIG_CONFIG_MASTER_VERSION" on page 543	0x12d	The current config master version used.
"RAWCMN_DIG_METADATA_LOCATION " on page 544	0x12e	Override register for location of metadata in memory
"RAWCMN_DIG_CNTX_RSTR_CLIENTS_MASK" on page 545	0x12f	Override register for client masks used by context restore module
"RAWCMN_DIG_MEM_POWER_SAVING" on page 546	0x130	Enables disabling the SRAM clk when SRAM isn't used to save power.
"RAWCMN_DIG_FW_PWRUP_DONE_B" on page 547	0x131	Firmware Power-up Done_B Status.
"RAWCMN_DIG_TOKEN_DLY" on page 548	0x132	Time Spent By a Token In a Lane.
"RAWCMN_DIG_RTUNE_REQ" on page 550	0x133	The read register for rtune_req.
"RAWCMN_DIG_PWR_EN" on page 551	0x134	power_en new register to sequence isolation
"RAWCMN_DIG_MPLL_FORCE_ACK" on page 552	0x135	The read register for mplla/b_force_ack.
"RAWCMN_DIG_MPLL_SSC_EN" on page 553	0x136	overwrite register for mplla/b_ssc_en before sending to pma.
"RAWCMN_DIG_MPLLA_FORCE_CTL_STATUS" on page 554	0x137	The read register for mplla_force_ctl_status
"RAWCMN_DIG_MPLLB_FORCE_CTL_STATUS" on page 555	0x138	The read register for mpllb_force_ctl_status
"RAWCMN_DIG_CNTX_CTL_STATUS_I " on page 556	0x139	The read register for monitoring maing cntx rstr signals.(for debug purposes)
"RAWCMN_DIG_CNTX_CTL_STATUS_II " on page 558	0x13a	The read register for monitoring maing cntx rstr signals.(for debug purposes)

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_MPLL_STATE_OVRD " on page 561	0x13b	Mpll_state ovrd
"RAWCMN_DIG_DCOCTRL_STATUS " on page 562	0x13c	The read register for monitoring main DCO signals.
"RAWCMN_DIG_DCOCTRL_STATUS_2 " on page 563	0x13d	The read register for monitoring main DCO signals.
"RAWCMN_DIG_MPLL_PMIX_EN" on page 564	0x13f	overwrite register for mplla/b_pmix_en before sending to pma.
"RAWCMN_DIG_AON_SRAM_PGATE_BL_EN" on page 565	0x160	Enable SRAM bootloader on power-gated exit.
"RAWCMN_DIG_AON_MPLLA_TUNE_BANK_0" on page 566	0x161	MPLLA Tune Value for Bank 0.
"RAWCMN_DIG_AON_MPLLA_TUNE_BANK_1" on page 567	0x162	MPLLA Tune Value for Bank 1.
"RAWCMN_DIG_AON_MPLLA_TUNE_BANK_2" on page 568	0x163	MPLLA Tune Value for Bank 2.
"RAWCMN_DIG_AON_MPLLA_TUNE_BANK_3" on page 569	0x164	MPLLA Tune Value for Bank 3.
"RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL" on page 570	0x165	Value for MPLLA Bank Select.
"RAWCMN_DIG_AON_MPLLA_TUNE_DONE" on page 571	0x166	MPLLA Calibration Bank Valid Calibrated Value.
"RAWCMN_DIG_AON_MPLLB_TUNE_BANK_0" on page 572	0x167	MPLLB Tune Value for Bank 0.
"RAWCMN_DIG_AON_MPLLB_TUNE_BANK_1" on page 573	0x168	MPLLB Tune Value for Bank 1.
"RAWCMN_DIG_AON_MPLLB_TUNE_BANK_2" on page 574	0x169	MPLLB Tune Value for Bank 2.
"RAWCMN_DIG_AON_MPLLB_TUNE_BANK_3" on page 575	0x16a	MPLLB Tune Value for Bank 3.
"RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL" on page 576	0x16b	MPLLB Bank Select Value.
"RAWCMN_DIG_AON_MPLLB_TUNE_DONE" on page 577	0x16c	MPLLB Calibration Bank Valid Calibrated Value.
"RAWCMN_DIG_AON_MPLLA_TUNE" on page 578	0x16e	MPLLA Selected TUNE Value.
"RAWCMN_DIG_AON_MPLLB_TUNE" on page 579	0x16f	MPLLB Selected TUNE Value.
"RAWCMN_DIG_AON_MPLLA_IN_RECAL" on page 580	0x170	MPLLA Recalibration.
"RAWCMN_DIG_AON_MPLLB_IN_RECAL" on page 581	0x171	MPLLB Recalibration.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_AON_PG_OVRD_IN" on page 582	0x172	Override Values for Incoming Power-gating Signals.
"RAWCMN_DIG_AON_PG_OVRD_OUT" on page 584	0x173	Override Values for Outgoing Power-gating Signals.
"RAWCMN_DIG_AON_SUP_OVRD_IN" on page 585	0x174	Override Values for Incoming Support Block Signals.
"RAWCMN_DIG_AON_SUP_OVRD_OUT" on page 587	0x175	Override Values for Outgoing Support Block Signals.
"RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL" on page 589	0x176	Current Bank Selected for MPLLA Coarse Tune in PMA.
"RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL" on page 590	0x177	Current Bank Selected for MPLLB Coarse Tune in PMA.
"RAWCMN_DIG_AON_RTUNE_RX_VAL" on page 591	0x178	Resistor Tune RX Value.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL" on page 592	0x179	Resistor Tune TX Down Value.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL" on page 593	0x17a	Resistor Tune TX Up Value.
"RAWCMN_DIG_AON_CMNCAL_MPLL_STATUS" on page 594	0x17b	MPLL Common Calibration Status.
"RAWCMN_DIG_AON_CMNCAL_RTUNE_STATUS" on page 596	0x17c	RTUNE Common Calibration Status.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_0" on page 597	0x17d	Resistor Tune RX Value 0.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_0" on page 598	0x17e	Resistor Tune TX Down Value 0.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_0" on page 599	0x17f	Resistor Tune TX Up Value 0.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_1" on page 600	0x180	Resistor Tune RX Value 1.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_1" on page 601	0x181	Resistor Tune TX Down Value 1.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_1" on page 602	0x182	Resistor Tune TX Up Value 1.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_2" on page 603	0x183	Resistor Tune RX Value 2.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_2" on page 604	0x184	Resistor Tune TX Down Value 2.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_2" on page 605	0x185	Resistor Tune TX Up Value 2.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_3" on page 606	0x186	Resistor Tune RX Value 3.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_3" on page 607	0x187	Resistor Tune TX Down Value 3.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_3" on page 608	0x188	Resistor Tune TX Up Value 3.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_4" on page 609	0x189	Resistor Tune RX Value 4.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_4" on page 610	0x18a	Resistor Tune TX Down Value 4.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_4" on page 611	0x18b	Resistor Tune TX Up Value 4.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_5" on page 612	0x18c	Resistor Tune RX Value 5.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_5" on page 613	0x18d	Resistor Tune TX Down Value 5.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_5" on page 614	0x18e	Resistor Tune TX Up Value 5.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_6" on page 615	0x18f	Resistor Tune RX Value 6.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_6" on page 616	0x190	Resistor Tune TX Down Value 6.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_6" on page 617	0x191	Resistor Tune TX Up Value 6.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_7" on page 618	0x192	Resistor Tune RX Value 7.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_7" on page 619	0x193	Resistor Tune TX Down Value 7.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_7" on page 620	0x194	Resistor Tune TX Up Value 7.
"RAWCMN_DIG_AON_SRAM_OVRD_IN" on page 621	0x195	Override for Incoming SRAM Inputs.
"RAWCMN_DIG_AON_SRAM_IN" on page 623	0x196	Monitor for SRAM Inputs.
"RAWCMN_DIG_AON_SRAM_OUT" on page 625	0x197	Monitor for SRAM Outputs and ECC Status.
"RAWCMN_DIG_AON_FW_VERSION_0" on page 627	0x198	Firmware Version Register 0.
"RAWCMN_DIG_AON_FW_VERSION_1" on page 628	0x199	Firmware Version Register 1.
"RAWCMN_DIG_AON_BG_SEL" on page 629	0x19a	bg_sel.
"RAWCMN_DIG_AON_SRAM_EOF_ADDR" on page 632	0x19b	Value for SRAM End Of FW address.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_AON_SRAM_BOC_ADDR" on page 633	0x19c	Value for SRAM Begin Of Context Restore address.
"RAWCMN_DIG_AON_ATE_MODE_EN" on page 634	0x19d	Determines if we are in the ATE mode
"RAWCMN_DIG_AON_SRAM_ECC_ADDR" on page 635	0x19e	SRAM ECC Address Register
"RAWCMN_DIG_AON_SRAM_ECC_STATUS_CLR" on page 636	0x19f	SRAM ECC error status and recorded SRAM address clear.
"RAWCMN_DIG_AON_RAW_VERSION" on page 637	0x1a0	Determines the RAW version
"RAWCMN_DIG_AON_RO_REG_AONCMN_1" on page 638	0x1a1	Read register for few signals mentioned below
"RAWCMN_DIG_AON_RW_REG_AONCMN_1" on page 640	0x1a2	Read/write register for few signals mentioned below
"RAWCMN_DIG_AON_RTUNE_RX_VAL_8" on page 642	0x1a3	Resistor Tune RX Value 8.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_8" on page 643	0x1a4	Resistor Tune TX Down Value 8.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_8" on page 644	0x1a5	Resistor Tune TX Up Value 8.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_9" on page 645	0x1a6	Resistor Tune RX Value 9.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_9" on page 646	0x1a7	Resistor Tune TX Down Value 9.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_9" on page 647	0x1a8	Resistor Tune TX Up Value 9.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_10" on page 648	0x1a9	Resistor Tune RX Value 10.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_10" on page 649	0x1aa	Resistor Tune TX Down Value 10.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_10" on page 650	0x1ab	Resistor Tune TX Up Value 10.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_11" on page 651	0x1ac	Resistor Tune RX Value 11.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_11" on page 652	0x1ad	Resistor Tune TX Down Value 11.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_11" on page 653	0x1ae	Resistor Tune TX Up Value 11.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_12" on page 654	0x1af	Resistor Tune RX Value 12.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_12" on page 655	0x1b0	Resistor Tune TX Down Value 12.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_12" on page 656	0x1b1	Resistor Tune TX Up Value 12.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_13" on page 657	0x1b2	Resistor Tune RX Value 13.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_13" on page 658	0x1b3	Resistor Tune TX Down Value 13.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_13" on page 659	0x1b4	Resistor Tune TX Up Value 13.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_14" on page 660	0x1b5	Resistor Tune RX Value 14.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_14" on page 661	0x1b6	Resistor Tune TX Down Value 14.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_14" on page 662	0x1b7	Resistor Tune TX Up Value 14.
"RAWCMN_DIG_AON_RTUNE_RX_VAL_15" on page 663	0x1b8	Resistor Tune RX Value 15.
"RAWCMN_DIG_AON_RTUNE_TXDN_VAL_15" on page 664	0x1b9	Resistor Tune TX Down Value 15.
"RAWCMN_DIG_AON_RTUNE_TXUP_VAL_15" on page 665	0x1ba	Resistor Tune TX Up Value 15.
"RAWCMN_DIG_AON_BYP_RES_CHAIN" on page 666	0x1bb	register for chicken bit for PMA res tunning chain.
"RAWCMN_DIG_AON_RW_FW_CLK_CTL" on page 667	0x1bc	Read/write register for few signals mentioned below
"RAWCMN_DIG_AON_FW_CLK_STATUS " on page 670	0x1bd	Values FW CLK related signals.
"RAWCMN_DIG_AON_DCO_CTL_1" on page 672	0x1be	DCO Control Signals
"RAWCMN_DIG_AON_DCO_CTL_2" on page 675	0x1bf	Override incoming refa_dco_ld_val input.
"RAWCMN_DIG_AON_DCO_CTL_3" on page 676	0x1c0	DCO Control Signals (For debug only)
"RAWCMN_DIG_AON_DCO_CAL_STATUS " on page 678	0x1c1	Values DCO CAL related signals.
"RAWCMN_DIG_AON_CMN_PERST_CTL" on page 680	0x1c2	PERST common control
"RAWCMN_DIG_AON_DCO_FW_ACCESS" on page 682	0x1c3	FW access status for DCO overrides
"RAWCMN_DIG_AON_REF_CLK_MUX_CTL" on page 683	0x1c4	REF CLK MUX Control Signals
"RAWCMN_DIG_AON_DCO_CTL_4" on page 685	0x1c5	Override incoming refb_dco_ld_val input. (For debug only)

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWCMN_DIG_AON_DCO_CTL_5" on page 687	0x1c6	Override incoming ref_dco_target input. (For debug only)
"RAWCMN_DIG_AON_DCO_CAL_STATUS_2" on page 689	0x1c7	Values DCO CAL related signals.
"RAWCMN_DIG_AON_DCO_CAL_STATUS_3" on page 690	0x1c8	Values DCO CAL related signals.
"RAWCMN_DIG_AON_CMN_PERST_CTL_2" on page 691	0x1c9	PERST common control 2
"RAWCMN_DIG_AON_LANE_PERST_STATUS" on page 692	0x1ca	Values of lane_perst after invert polarity.
"RAWCMN_DIG_AON_CMN_PERST_STATUS" on page 693	0x1cb	Values of cmn_perst_status.
"RAWCMN_DIG_AON_LANE_REFA_CLK_DIS_STATUS " on page 694	0x1cc	Values of lane_refa_clk_dis.
"RAWCMN_DIG_AON_LANE_REFB_CLK_DIS_STATUS " on page 695	0x1cd	Values of lane_refb_clk_dis.
"RAWCMN_DIG_AON_DCO_CTL_6" on page 696	0x1ce	Control DCO restoration timers.
"RAWCMN_DIG_AON_DCO_CAL_STATUS_4" on page 697	0x1cf	Values DCO CAL related signals.
"RAWCMN_DIG_AON_RSVD_CTL_1" on page 698	0x1d0	Reserved control signals.
"RAWCMN_DIG_AON_DCO_CTL_7" on page 699	0x1d1	Override DCO finetune
"RAWCMN_DIG_AON_RO_REG_AONCMN_2" on page 701	0x1d2	Read register for few signals mentioned below
"RAWCMN_DIG_AON_RO_REG_AONCMN_3" on page 702	0x1d3	Read register for few signals mentioned below
"RAWCMN_DIG_AON_RSVD_CTL_2" on page 703	0x1d4	Reserved control signals.
"RAWCMN_DIG_AON_CMN_PERST_CTL_3" on page 704	0x1d5	PERST common control
"RAWCMN_DIG_AON_SUP_MISC" on page 705	0x1d6	E32_RAWCMN_DIG_AON_SUP_MISC
"LANEN_DIG_ASIC_LANE_OVRD_IN(for N = 0; N <= NLANES-1)" on page 706	0x1(N*2+0)00	Override Values for Incoming LANE Controls from ASIC.
"LANEN_DIG_ASIC_TX_OVRD_IN_0(for N = 0; N <= NLANES-1)" on page 708	0x1(N*2+0)01	Override Values for Incoming TX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_TX_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 711	0x1(N*2+0)02	Override Values for Incoming TX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_TX_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 713	0x1(N*2+0)03	Override Values for Incoming TX Drive Controls from ASIC, Register 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ASIC_TX_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 715	0x1(N*2+0)04	Override Values for Incoming TX Drive Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_TX_OVRD_IN_4(for N = 0; N <= NLANES-1)" on page 716	0x1(N*2+0)05	Override Values for Incoming TX Drive Controls from ASIC, Register 4.
"LANEN_DIG_ASIC_TX_OVRD_IN_5(for N = 0; N <= NLANES-1)" on page 718	0x1(N*2+0)06	Override Values for Incoming TX Drive Controls from ASIC, Register 5.
"LANEN_DIG_ASIC_TX_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 720	0x1(N*2+0)07	Override Values for Outgoing TX Controls to ASIC.
"LANEN_DIG_ASIC_ROPLL_OVRD_IN_0(for N = 0; N <= NLANES-1)" on page 722	0x1(N*2+0)08	Current Values for Incoming RX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_ROPLL_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 724	0x1(N*2+0)09	Current Values for Incoming RX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_ROPLL_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 725	0x1(N*2+0)0a	Current Values for Incoming RX Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_ROPLL_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 727	0x1(N*2+0)0b	Current Values for Incoming RX Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_TX_OVRD_MISC(for N = 0; N <= NLANES-1)" on page 729	0x1(N*2+0)0c	Override Values for Incoming TX MISC BUS from ASIC.
"LANEN_DIG_ASIC_TX_ASIC_IN_0(for N = 0; N <= NLANES-1)" on page 731	0x1(N*2+0)0d	Current Values for Incoming TX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_TX_ASIC_IN_1(for N = 0; N <= NLANES-1)" on page 734	0x1(N*2+0)0e	Current Values for Incoming TX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_TX_ASIC_IN_2(for N = 0; N <= NLANES-1)" on page 736	0x1(N*2+0)0f	Current Values for Incoming TX Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_TX_ASIC_IN_3(for N = 0; N <= NLANES-1)" on page 737	0x1(N*2+0)10	Override Values for Incoming TX Drive Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_TX_ASIC_IN_4(for N = 0; N <= NLANES-1)" on page 738	0x1(N*2+0)11	Override Values for Incoming TX Drive Controls from ASIC, Register 4.
"LANEN_DIG_ASIC_TX_ASIC_OUT(for N = 0; N <= NLANES-1)" on page 739	0x1(N*2+0)12	Current Values for Outgoing TX Status Controls from PHY.
"LANEN_DIG_ASIC_ROPLL_ASIC_IN_0(for N = 0; N <= NLANES-1)" on page 740	0x1(N*2+0)13	Current Values for Incoming RX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_ROPLL_ASIC_IN_1(for N = 0; N <= NLANES-1)" on page 742	0x1(N*2+0)14	Current Values for Incoming RX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_ROPLL_ASIC_IN_2(for N = 0; N <= NLANES-1)" on page 744	0x1(N*2+0)15	Current Values for Incoming RX Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_ROPLL_ASIC_IN_3(for N = 0; N <= NLANES-1)" on page 745	0x1(N*2+0)16	Current Values for Incoming RX Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_TX_DLY_CAL_OVRD_IN (for N = 0; N <= NLANES-1)" on page 746	0x1(N*2+0)17	Override Values for Incoming TX Delay Calibration Signals from ASIC.
"LANEN_DIG_ASIC_TX_DLY_CAL_ASIC_IN (for N = 0; N <= NLANES-1)" on page 748	0x1(N*2+0)18	Current Values for Incoming TX Delay Calibration Controls from ASIC Register.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_UP(for N = 0; N <= NLANES-1)" on page 750	0x1(N*2+0)19	Override Values for Incoming TXUP termination code from ASIC.
"LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_DN(for N = 0; N <= NLANES-1)" on page 751	0x1(N*2+0)1a	Override Values for Incoming TXDN termination code from ASIC.
"LANEN_DIG_ASIC_TX_TERM_CODE_UP_ASIC_IN(for N = 0; N <= NLANES-1)" on page 752	0x1(N*2+0)1b	Current Values for Incoming TXUP termination code from ASIC.
"LANEN_DIG_ASIC_TX_TERM_CODE_DN_ASIC_IN(for N = 0; N <= NLANES-1)" on page 753	0x1(N*2+0)1c	Current Values for Incoming TXDN termination code from ASIC.
"LANEN_DIG_ASIC_RX_OVRD_IN_0(for N = 0; N <= NLANES-1)" on page 754	0x1(N*2+0)20	Override Values for Incoming RX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_RX_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 757	0x1(N*2+0)21	Override Values for Incoming RX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_RX_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 759	0x1(N*2+0)22	Override Values for Incoming RX Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_RX_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 760	0x1(N*2+0)23	Override Values for Incoming RX Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_RX_OVRD_IN_4(for N = 0; N <= NLANES-1)" on page 762	0x1(N*2+0)24	Override Values for Incoming RX Drive Controls from ASIC, Register 4.
"LANEN_DIG_ASIC_RX_OVRD_IN_5(for N = 0; N <= NLANES-1)" on page 764	0x1(N*2+0)25	Override Values for Incoming RX Drive Controls from ASIC, Register 5.
"LANEN_DIG_ASIC_RX_OVRD_IN_6(for N = 0; N <= NLANES-1)" on page 765	0x1(N*2+0)26	Override Values for Incoming RX Drive Controls from ASIC, Register 6.
"LANEN_DIG_ASIC_RX_OVRD_IN_7(for N = 0; N <= NLANES-1)" on page 766	0x1(N*2+0)27	Override Values for Incoming RX Drive Controls from ASIC, Register 7.
"LANEN_DIG_ASIC_RX_OVRD_SIGDET_IN(for N = 0; N <= NLANES-1)" on page 767	0x1(N*2+0)28	Override Values for Incoming RX SIGDET Controls from ASIC.
"LANEN_DIG_ASIC_RX_OVRD_VCO_IN(for N = 0; N <= NLANES-1)" on page 769	0x1(N*2+0)29	Override Values for Incoming RX VCO Controls from ASIC.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0(for N = 0; N <= NLANES-1)" on page 770	0x1(N*2+0)2a	Override Values for Incoming RX EQ Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1(for N = 0; N <= NLANES-1)" on page 771	0x1(N*2+0)2b	Override Values for Incoming RX EQ Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2(for N = 0; N <= NLANES-1)" on page 772	0x1(N*2+0)2c	Override Values for Incoming RX EQ Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3(for N = 0; N <= NLANES-1)" on page 773	0x1(N*2+0)2d	Override Values for Incoming RX EQ Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_4(for N = 0; N <= NLANES-1)" on page 774	0x1(N*2+0)2e	Override Values for Incoming RX EQ Controls from ASIC, Register 4.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ASIC_RX_OVRD_OUT_0(for N = 0; N <= NLANES-1)" on page 775	0x1(N*2+0)2f	Override Values for Outgoing RX Controls to ASIC, Register 0.
"LANEN_DIG_ASIC_LANE_ASIC_IN(for N = 0; N <= NLANES-1)" on page 777	0x1(N*2+0)30	Current Values for Incoming LANE Controls from ASIC.
"LANEN_DIG_ASIC_RX_ASIC_IN_0(for N = 0; N <= NLANES-1)" on page 779	0x1(N*2+0)31	Current Values for Incoming RX Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_RX_ASIC_IN_1(for N = 0; N <= NLANES-1)" on page 782	0x1(N*2+0)32	Current Values for Incoming RX Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_RX_ASIC_IN_2(for N = 0; N <= NLANES-1)" on page 784	0x1(N*2+0)33	Current Values for Incoming RX Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_RX_ASIC_IN_3(for N = 0; N <= NLANES-1)" on page 786	0x1(N*2+0)34	Current Values for Incoming RX Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_RX_ASIC_IN_4(for N = 0; N <= NLANES-1)" on page 787	0x1(N*2+0)35	Current Values for Incoming RX Controls from ASIC, Register 4.
"LANEN_DIG_ASIC_RX_ASIC_IN_5(for N = 0; N <= NLANES-1)" on page 788	0x1(N*2+0)36	Current Values for Incoming RX Controls from ASIC, Register 5.
"LANEN_DIG_ASIC_RX_ASIC_IN_6(for N = 0; N <= NLANES-1)" on page 789	0x1(N*2+0)37	Current Values for Incoming RX Controls from ASIC, Register 6.
"LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN (for N = 0; N <= NLANES-1)" on page 790	0x1(N*2+0)38	Current Values for Incoming RX CDR VCO Controls from ASIC.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0(for N = 0; N <= NLANES-1)" on page 791	0x1(N*2+0)39	Current Values for Incoming RX EQ Controls from ASIC, Register 0.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1(for N = 0; N <= NLANES-1)" on page 793	0x1(N*2+0)3a	Current Values for Incoming RX EQ Controls from ASIC, Register 1.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_2(for N = 0; N <= NLANES-1)" on page 795	0x1(N*2+0)3b	Current Values for Incoming RX EQ Controls from ASIC, Register 2.
"LANEN_DIG_ASIC_RX_ASIC_OUT_0(for N = 0; N <= NLANES-1)" on page 796	0x1(N*2+0)3c	Current Values for Outgoing RX Status Controls from PHY, Register 0.
"LANEN_DIG_ASIC_RX_OVRD_MISC(for N = 0; N <= NLANES-1)" on page 798	0x1(N*2+0)3d	Override Values for Incoming RX MISC BUS from ASIC.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_5(for N = 0; N <= NLANES-1)" on page 799	0x1(N*2+0)3e	Override Values for Incoming RX EQ Controls from ASIC, Register 5.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_6(for N = 0; N <= NLANES-1)" on page 800	0x1(N*2+0)3f	Override Values for Incoming RX EQ Controls from ASIC, Register 6.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_7(for N = 0; N <= NLANES-1)" on page 801	0x1(N*2+0)40	Override Values for Incoming RX EQ Controls from ASIC, Register 7.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_8(for N = 0; N <= NLANES-1)" on page 802	0x1(N*2+0)41	Override Values for Incoming RX EQ Controls from ASIC, Register 8.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_9(for N = 0; N <= NLANES-1)" on page 803	0x1(N*2+0)42	Override Values for Incoming RX EQ Controls from ASIC, Register 9.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_10(for N = 0; N <= NLANES-1)" on page 804	0x1(N*2+0)43	Override Values for Incoming RX EQ Controls from ASIC, Register 10.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_11(for N = 0; N <= NLANES-1)" on page 805	0x1(N*2+ 0)44	Override Values for Incoming RX EQ Controls from ASIC, Register 11.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_12(for N = 0; N <= NLANES-1)" on page 806	0x1(N*2+ 0)45	Override Values for Incoming RX EQ Controls from ASIC, Register 12.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_13(for N = 0; N <= NLANES-1)" on page 807	0x1(N*2+ 0)46	Override Values for Incoming RX EQ Controls from ASIC, Register 13.
"LANEN_DIG_ASIC_RX_OVRD_EQ_IN_14(for N = 0; N <= NLANES-1)" on page 808	0x1(N*2+ 0)47	Override Values for Incoming RX EQ Controls from ASIC, Register 14.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_3(for N = 0; N <= NLANES-1)" on page 809	0x1(N*2+ 0)48	Current Values for Incoming RX Controls from ASIC, Register 3.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_4(for N = 0; N <= NLANES-1)" on page 810	0x1(N*2+ 0)49	Current Values for Incoming RX Controls from ASIC, Register 4.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_5(for N = 0; N <= NLANES-1)" on page 811	0x1(N*2+ 0)4a	Current Values for Incoming RX Controls from ASIC, Register 5.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_6(for N = 0; N <= NLANES-1)" on page 812	0x1(N*2+ 0)4b	Current Values for Incoming RX Controls from ASIC, Register 6.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_7(for N = 0; N <= NLANES-1)" on page 813	0x1(N*2+ 0)4c	Current Values for Incoming RX Controls from ASIC, Register 7.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_8(for N = 0; N <= NLANES-1)" on page 815	0x1(N*2+ 0)4d	Current Values for Incoming RX Controls from ASIC, Register 8.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_9(for N = 0; N <= NLANES-1)" on page 816	0x1(N*2+ 0)4e	Current Values for Incoming RX Controls from ASIC, Register 9.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_10(for N = 0; N <= NLANES-1)" on page 817	0x1(N*2+ 0)4f	Current Values for Incoming RX Controls from ASIC, Register 10.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_11(for N = 0; N <= NLANES-1)" on page 818	0x1(N*2+ 0)50	Current Values for Incoming RX Controls from ASIC, Register 11.
"LANEN_DIG_ASIC_RX_EQ_ASIC_IN_12(for N = 0; N <= NLANES-1)" on page 819	0x1(N*2+ 0)51	Current Values for Incoming RX Controls from ASIC, Register 12.
"LANEN_DIG_ASIC_RX_OVRD_IN_8(for N = 0; N <= NLANES-1)" on page 820	0x1(N*2+ 0)52	Override Values for Incoming RX Controls from ASIC, Register 8
"LANEN_DIG_ASIC_RX_ASIC_IN_13(for N = 0; N <= NLANES-1)" on page 822	0x1(N*2+ 0)53	Current Values for Incoming RX Controls from ASIC, Register 13.
"LANEN_DIG_ASIC_RX_ASIC_IN_14(for N = 0; N <= NLANES-1)" on page 823	0x1(N*2+ 0)54	Current Values for Incoming RX Controls from ASIC, Register 14.
"LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 (for N = 0; N <= NLANES-1)" on page 824	0x1(N*2+ 0)55	TX Power State Control Register for P0.
"LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 S(for N = 0; N <= NLANES-1)" on page 826	0x1(N*2+ 0)56	TX Power State Control Register for P0S.
"LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 (for N = 0; N <= NLANES-1)" on page 828	0x1(N*2+ 0)57	TX Power State Control Register for P1.
"LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 (for N = 0; N <= NLANES-1)" on page 830	0x1(N*2+ 0)58	TX Power State Control Register for P2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0(for N = 0; N <= NLANES-1)" on page 832	0x1(N*2+0)59	TX Power UP Time Register 0.
"LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1(for N = 0; N <= NLANES-1)" on page 833	0x1(N*2+0)5a	TX Power UP Time Register 1.
"LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2(for N = 0; N <= NLANES-1)" on page 834	0x1(N*2+0)5b	TX Power UP Time Register 2.
"LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3(for N = 0; N <= NLANES-1)" on page 835	0x1(N*2+0)5c	TX Power UP Time Register 3.
"LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4(for N = 0; N <= NLANES-1)" on page 836	0x1(N*2+0)5d	TX Power UP Time Register 4.
"LANEN_DIG_TX_PWRCTL_TX_CTL(for N = 0; N <= NLANES-1)" on page 837	0x1(N*2+0)5e	TX Power UP Control Register.
"LANEN_DIG_TX_PWRCTL_TX_CTL_2(for N = 0; N <= NLANES-1)" on page 840	0x1(N*2+0)5f	TX Power UP Control Register 2.
"LANEN_DIG_TX_PWRCTL_TX_STATUS (for N = 0; N <= NLANES-1)" on page 842	0x1(N*2+0)60	TX Power Up Control FSM StatusLane Input Status Register 1.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_OVRD(for N = 0; N <= NLANES-1)" on page 843	0x1(N*2+0)61	ROPLL Override Controls.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_0(for N = 0; N <= NLANES-1)" on page 845	0x1(N*2+0)62	ROPLL Timers Register 0.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_1(for N = 0; N <= NLANES-1)" on page 846	0x1(N*2+0)63	ROPLL Timers Register 1.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_2(for N = 0; N <= NLANES-1)" on page 847	0x1(N*2+0)64	ROPLL Timers Register 2.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_3(for N = 0; N <= NLANES-1)" on page 848	0x1(N*2+0)65	ROPLL Timers Register 3.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_4(for N = 0; N <= NLANES-1)" on page 849	0x1(N*2+0)66	ROPLL Timers Register 4.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_5(for N = 0; N <= NLANES-1)" on page 850	0x1(N*2+0)67	ROPLL Timers Register 5.
"LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_6(for N = 0; N <= NLANES-1)" on page 851	0x1(N*2+0)68	ROPLL Timers Register 6.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_TX_ROPLL_CTL_STAT (for N = 0; N <= NLANES-1)" on page 853	0x1(N*2+0)69	ROPLL status register.
"LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_FR(for N = 0; N <= NLANES-1)" on page 856	0x1(N*2+0)6a	TX DCC Differential IDAC OFFSET for Full Rate.
"LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_HR(for N = 0; N <= NLANES-1)" on page 858	0x1(N*2+0)6b	TX DCC Differential IDAC OFFSET for Half Rate.
"LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_FR(for N = 0; N <= NLANES-1)" on page 860	0x1(N*2+0)6c	TX DCC Common-Mode IDAC OFFSET for Full Rate.
"LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_HR(for N = 0; N <= NLANES-1)" on page 861	0x1(N*2+0)6d	TX DCC Common-Mode IDAC OFFSET for Half Rate.
"LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0(for N = 0; N <= NLANES-1)" on page 862	0x1(N*2+0)6e	TX Clock Alignment Control Register 0.
"LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1(for N = 0; N <= NLANES-1)" on page 864	0x1(N*2+0)6f	TX Clock Alignment Control Register 1.
"LANEN_DIG_TX_CLK_ALIGN_CLK_ALIGN_STATUS (for N = 0; N <= NLANES-1)" on page 865	0x1(N*2+0)70	Tx Clock Align FSM Status.
"LANEN_DIG_TX_LBERT_CTL(for N = 0; N <= NLANES-1)" on page 866	0x1(N*2+0)71	Pattern Generator controls
"LANEN_DIG_TX_LBERT_PAT1_0 (for N = 0; N <= NLANES-1)" on page 868	0x1(N*2+0)72	Extended Pattern register 0.
"LANEN_DIG_TX_LBERT_PAT1_1 (for N = 0; N <= NLANES-1)" on page 869	0x1(N*2+0)73	Extended Pattern register 1.
"LANEN_DIG_TX_LBERT_PAT1_2 (for N = 0; N <= NLANES-1)" on page 870	0x1(N*2+0)74	Extended Pattern register 2.
"LANEN_DIG_TX_LBERT_PAT1_3 (for N = 0; N <= NLANES-1)" on page 871	0x1(N*2+0)75	Extended Pattern register 3.
"LANEN_DIG_TX_FIFO_CTL(for N = 0; N <= NLANES-1)" on page 872	0x1(N*2+0)76	TX Digital Control Register.
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_0 (for N = 0; N <= NLANES-1)" on page 874	0x1(N*2+0)77	TX Delay Configuration Register 0.
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_1 (for N = 0; N <= NLANES-1)" on page 875	0x1(N*2+0)78	TX Delay Configuration Register 1.
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_2 (for N = 0; N <= NLANES-1)" on page 876	0x1(N*2+0)79	TX Delay Configuration Register 2.
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_3 (for N = 0; N <= NLANES-1)" on page 877	0x1(N*2+0)7a	TX Delay Configuration Register 3.
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_4 (for N = 0; N <= NLANES-1)" on page 878	0x1(N*2+0)7b	TX Delay Configuration Register 4.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_TX_DLY_CAL_DLY_CNFG_5 (for N = 0; N <= NLANES-1)" on page 880	0x1(N*2+ 0)7c	TX Delay Configuration Register 5.
"LANEN_DIG_TX_DLY_CAL_DLY_STAT_0 (for N = 0; N <= NLANES-1)" on page 882	0x1(N*2+ 0)7d	Pre-Driver DLY Status Register 0.
"LANEN_DIG_TX_DLY_CAL_DLY_STAT_1 (for N = 0; N <= NLANES-1)" on page 883	0x1(N*2+ 0)7e	Pre-driver DLY Status Register 1.
"LANEN_DIG_RX_PWRCTL_RX_PSTATE_P 0(for N = 0; N <= NLANES-1)" on page 884	0x1(N*2+ 0)80	RX Power State Control Register for P0.
"LANEN_DIG_RX_PWRCTL_RX_PSTATE_P 0S(for N = 0; N <= NLANES-1)" on page 887	0x1(N*2+ 0)81	RX Power State Control Register for P0S.
"LANEN_DIG_RX_PWRCTL_RX_PSTATE_P 1(for N = 0; N <= NLANES-1)" on page 890	0x1(N*2+ 0)82	RX Power State Control Register for P1.
"LANEN_DIG_RX_PWRCTL_RX_PSTATE_P 2(for N = 0; N <= NLANES-1)" on page 893	0x1(N*2+ 0)83	RX Power State Control Register for P2.
"LANEN_DIG_RX_PWRCTL_RX_PWRUP_TI ME_0(for N = 0; N <= NLANES-1)" on page 896	0x1(N*2+ 0)84	RX Power UP Time Register 0.
"LANEN_DIG_RX_PWRCTL_RX_PWRUP_TI ME_1(for N = 0; N <= NLANES-1)" on page 897	0x1(N*2+ 0)85	RX Power UP Time Register 1.
"LANEN_DIG_RX_PWRCTL_RX_PWRUP_TI ME_2(for N = 0; N <= NLANES-1)" on page 899	0x1(N*2+ 0)86	RX Power UP Time Register 2.
"LANEN_DIG_RX_PWRCTL_RX_CTL(for N = 0; N <= NLANES-1)" on page 901	0x1(N*2+ 0)87	RX Power UP Control Register.
"LANEN_DIG_RX_PWRCTL_RX_CTL_2(for N = 0; N <= NLANES-1)" on page 904	0x1(N*2+ 0)88	RX Power UP Control Register 2.
"LANEN_DIG_RX_PWRCTL_RX_STATUS (for N = 0; N <= NLANES-1)" on page 906	0x1(N*2+ 0)89	RX Power Up Control FSM StatusLane Input Status Register 1.
"LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_ CTRL_0(for N = 0; N <= NLANES-1)" on page 907	0x1(N*2+ 0)8a	RX VCO Calibration Controls Register 0.
"LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_ CTRL_1(for N = 0; N <= NLANES-1)" on page 909	0x1(N*2+ 0)8b	RX VCO Calibration Controls Register 1.
"LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_ CTRL_2(for N = 0; N <= NLANES-1)" on page 911	0x1(N*2+ 0)8c	RX VCO Calibration Controls Register 2.
"LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_ TIME_0(for N = 0; N <= NLANES-1)" on page 912	0x1(N*2+ 0)8d	RX Power UP Time Register 0.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1(for N = 0; N <= NLANES-1)" on page 913	0x1(N*2+0)8e	RX Power UP Time Register 1.
"LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0(for N = 0; N <= NLANES-1)" on page 915	0x1(N*2+0)8f	RX VCO Status Register 0.
"LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1(for N = 0; N <= NLANES-1)" on page 917	0x1(N*2+0)90	RX VCO Status Register 1.
"LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2(for N = 0; N <= NLANES-1)" on page 919	0x1(N*2+0)91	RX VCO Status Register 2.
"LANEN_DIG_RX_VCOCAL_RX_DPLL_FREQ_SUM_RST(for N = 0; N <= NLANES-1)" on page 920	0x1(N*2+0)92	RX DPLL frequency integrator reset value.
"LANEN_DIG_RX_LBERT_CTL(for N = 0; N <= NLANES-1)" on page 921	0x1(N*2+0)93	Pattern Matcher controls
"LANEN_DIG_RX_LBERT_ERR(for N = 0; N <= NLANES-1)" on page 923	0x1(N*2+0)94	Pattern Match Error Counter.
"LANEN_DIG_RX_LBERT_LBERT_PM_STAT0(for N = 0; N <= NLANES-1)" on page 924	0x1(N*2+0)95	LBERT PM Status Register #0
"LANEN_DIG_RX_LBERT_LBERT_PM_STAT1(for N = 0; N <= NLANES-1)" on page 925	0x1(N*2+0)96	E32_LANE0_DIG_RX_LBERT_LBERT_PM_STAT1
"LANEN_DIG_RX_CDR_CDR_CTL_0(for N = 0; N <= NLANES-1)" on page 926	0x1(N*2+0)97	Control Bits for Receiver in Recovered Domain.
"LANEN_DIG_RX_CDR_CDR_CTL_1(for N = 0; N <= NLANES-1)" on page 928	0x1(N*2+0)98	CDR Control Register 1.
"LANEN_DIG_RX_CDR_CDR_CTL_2(for N = 0; N <= NLANES-1)" on page 929	0x1(N*2+0)99	CDR Control Register 2.
"LANEN_DIG_RX_CDR_CDR_CTL_3(for N = 0; N <= NLANES-1)" on page 930	0x1(N*2+0)9a	CDR Control Register 3.
"LANEN_DIG_RX_CDR_CDR_CTL_4(for N = 0; N <= NLANES-1)" on page 932	0x1(N*2+0)9b	CDR Control Register 4.
"LANEN_DIG_RX_CDR_STAT (for N = 0; N <= NLANES-1)" on page 934	0x1(N*2+0)9c	Current Output Values to Dpll (phug, frug).
"LANEN_DIG_RX_DPLL_FREQ(for N = 0; N <= NLANES-1)" on page 935	0x1(N*2+0)9d	Current Frequency Integrator Value.
"LANEN_DIG_RX_DPLL_FREQ_BOUND_0(for N = 0; N <= NLANES-1)" on page 936	0x1(N*2+0)9e	Frequency Bounds for Incoming Data Stream 0.
"LANEN_DIG_RX_DPLL_FREQ_BOUND_1(for N = 0; N <= NLANES-1)" on page 937	0x1(N*2+0)9f	Frequency Bounds for Incoming Data Stream 1.
"LANEN_DIG_RX_IQC_CTL_RESET_ADJUST(for N = 0; N <= NLANES-1)" on page 938	0x1(N*2+0)a0	RX IQC Reset iq_adjust Bus Register.
"LANEN_DIG_RX_IQC_CTL_CONFIG(for N = 0; N <= NLANES-1)" on page 939	0x1(N*2+0)a1	RX IQC Config Register.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_IQC_CTL_STATUS (for N = 0; N <= NLANES-1)" on page 941	0x1(N*2+0)a2	RX IQC Status Register.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0(for N = 0; N <= NLANES-1)" on page 942	0x1(N*2+0)a5	Adaptation Configuration Register 0.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1(for N = 0; N <= NLANES-1)" on page 943	0x1(N*2+0)a6	Adaptation Configuration Register 1.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2(for N = 0; N <= NLANES-1)" on page 945	0x1(N*2+0)a7	Adaptation Configuration Register 2.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3(for N = 0; N <= NLANES-1)" on page 946	0x1(N*2+0)a8	Adaptation Configuration Register 3.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4(for N = 0; N <= NLANES-1)" on page 948	0x1(N*2+0)a9	Adaptation Configuration Register 4.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5(for N = 0; N <= NLANES-1)" on page 949	0x1(N*2+0)aa	Adaptation Configuration Register 5.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6(for N = 0; N <= NLANES-1)" on page 950	0x1(N*2+0)ab	Adaptation Configuration Register 6.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7(for N = 0; N <= NLANES-1)" on page 951	0x1(N*2+0)ac	Adaptation Configuration Register 7.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8(for N = 0; N <= NLANES-1)" on page 953	0x1(N*2+0)ad	Adaptation Configuration Register 8.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9(for N = 0; N <= NLANES-1)" on page 954	0x1(N*2+0)ae	Adaptation Configuration Register 9.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10 (for N = 0; N <= NLANES-1)" on page 956	0x1(N*2+0)af	Adaptation Configuration Register 10.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_11 (for N = 0; N <= NLANES-1)" on page 957	0x1(N*2+0)b0	Adaptation Configuration Register 11.
"LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG(for N = 0; N <= NLANES-1)" on page 958	0x1(N*2+0)b1	Reset Adaptation Configuration Register.
"LANEN_DIG_RX_ADPTCTL_ATT_STATUS (for N = 0; N <= NLANES-1)" on page 960	0x1(N*2+0)b2	Value of ATT Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_VGA_STATUS (for N = 0; N <= NLANES-1)" on page 961	0x1(N*2+0)b3	Value of VGA Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_CTLE_STATUS (for N = 0; N <= NLANES-1)" on page 962	0x1(N*2+0)b4	Value of CTLE Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS(for N = 0; N <= NLANES-1)" on page 964	0x1(N*2+0)b5	Value of DFE TAP1 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS(for N = 0; N <= NLANES-1)" on page 965	0x1(N*2+0)b6	Value of DFE TAP2 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS(for N = 0; N <= NLANES-1)" on page 966	0x1(N*2+0)b7	Value of DFE TAP3 Adaptation Code.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS(for N = 0; N <= NLANES-1)" on page 967	0x1(N*2+0)b8	Value of DFE TAP4 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS(for N = 0; N <= NLANES-1)" on page 968	0x1(N*2+0)b9	Value of DFE TAP5 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 969	0x1(N*2+0)ba	Offset Values for RX DFE Data0 Even High vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 970	0x1(N*2+0)bb	Offset Values for RX DFE Data0 Even Low vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 971	0x1(N*2+0)bc	Offset Values for RX DFE Data0 Odd High vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 972	0x1(N*2+0)bd	Offset Values for RX DFE Data0 Odd Low vDAC.
"LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL(for N = 0; N <= NLANES-1)" on page 973	0x1(N*2+0)be	Sets Values for RX SLICER CTRL Signals Going to ANA.
"LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_MAP(for N = 0; N <= NLANES-1)" on page 974	0x1(N*2+0)bf	Sets Mapping Values for RX SLICER CTRL.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 976	0x1(N*2+0)c0	Offset Values for RX DFE Data1 Even High vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 977	0x1(N*2+0)c1	Offset Values for RX DFE Data1 Odd High vDAC.
"LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_ODD_LEVEL(for N = 0; N <= NLANES-1)" on page 978	0x1(N*2+0)c2	Value of Odd Error Slicer Level.
"LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_EVEN_LEVEL(for N = 0; N <= NLANES-1)" on page 979	0x1(N*2+0)c3	Value of Even Error Slicer Level.
"LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 980	0x1(N*2+0)c4	Offset Values for RX DFE By-Pass Even vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 981	0x1(N*2+0)c5	Offset values for RX DFE Bypass Odd vDAC.
"LANEN_DIG_RX_ADPTCTL_ADPT_RESET(for N = 0; N <= NLANES-1)" on page 982	0x1(N*2+0)c6	Adaptation Reset Register.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_12 (for N = 0; N <= NLANES-1)" on page 983	0x1(N*2+0)c7	Adaptation Configuration Register 12.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_IDAC_OFST_OVRD_EN(for N = 0; N <= NLANES-1)" on page 984	0x1(N*2+0)c8	Offset Override Enables for RX DCC iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_DIFF_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 986	0x1(N*2+0)c9	Offset Values for RX DCC Phase Differential iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_CM_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 987	0x1(N*2+0)ca	Offset Values for RX DCC Phase Common Mode iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_DIFF_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 988	0x1(N*2+0)cb	Offset Values for RX DCC Data Differential iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_CM_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 989	0x1(N*2+0)cc	Offset Values for RX DCC Data Common Mode iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_DIFF_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 990	0x1(N*2+0)cd	Offset Values for RX DCC Bypass Differential iDAC.
"LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_CM_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 991	0x1(N*2+0)ce	Offset Values for RX DCC Bypass Common Mode iDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 992	0x1(N*2+0)cf	Offset Values for RX DFE Data1 Even Low vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 993	0x1(N*2+0)d0	Offset Values for RX DFE Data1 Odd Low vDAC
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_13 (for N = 0; N <= NLANES-1)" on page 994	0x1(N*2+0)d1	Adaptation Configuration Register 13.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_14 (for N = 0; N <= NLANES-1)" on page 995	0x1(N*2+0)d2	Adaptation Configuration Register 14.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_15 (for N = 0; N <= NLANES-1)" on page 996	0x1(N*2+0)d3	Adaptation Configuration Register 15.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_16 (for N = 0; N <= NLANES-1)" on page 997	0x1(N*2+0)d4	Adaptation Configuration Register 16.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP6_STATUS(for N = 0; N <= NLANES-1)" on page 998	0x1(N*2+0)d5	Value of DFE TAP6 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_TAP7_STATUS(for N = 0; N <= NLANES-1)" on page 999	0x1(N*2+0)d6	Value of DFE TAP7 Adaptation Code.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_ADPTCTL_DFE_TAP8_STATUS (for N = 0; N <= NLANES-1)" on page 1000	0x1(N*2+0)d7	Value of DFE TAP8 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_FLOAT0_STATUS (for N = 0; N <= NLANES-1)" on page 1001	0x1(N*2+0)d8	Value of DFE Float0 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_FLOAT1_STATUS (for N = 0; N <= NLANES-1)" on page 1002	0x1(N*2+0)d9	Value of DFE Float1 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_FLOAT2_STATUS (for N = 0; N <= NLANES-1)" on page 1003	0x1(N*2+0)da	Value of DFE Float2 Adaptation Code.
"LANEN_DIG_RX_ADPTCTL_DFE_FLOAT3_STATUS (for N = 0; N <= NLANES-1)" on page 1004	0x1(N*2+0)db	Value of DFE Float3 Adaptation Code
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG0(for N = 0; N <= NLANES-1)" on page 1005	0x1(N*2+0)dc	DAC Address Config 0.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG1(for N = 0; N <= NLANES-1)" on page 1006	0x1(N*2+0)dd	DAC Address Config 1.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG2(for N = 0; N <= NLANES-1)" on page 1007	0x1(N*2+0)de	DAC Address Config 2.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG3(for N = 0; N <= NLANES-1)" on page 1008	0x1(N*2+0)df	DAC Address Config 3.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG4(for N = 0; N <= NLANES-1)" on page 1009	0x1(N*2+0)e0	DAC Address Config 4.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG5(for N = 0; N <= NLANES-1)" on page 1010	0x1(N*2+0)e1	DAC Address Config 5.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG6(for N = 0; N <= NLANES-1)" on page 1011	0x1(N*2+0)e2	DAC Address Config 6 .
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG7(for N = 0; N <= NLANES-1)" on page 1012	0x1(N*2+0)e3	DAC Address Config 7.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG8(for N = 0; N <= NLANES-1)" on page 1013	0x1(N*2+0)e4	DAC Address Config 8.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG9(for N = 0; N <= NLANES-1)" on page 1014	0x1(N*2+0)e5	DAC Address Config 9.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG10(for N = 0; N <= NLANES-1)" on page 1015	0x1(N*2+0)e6	DAC Address Config 10.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG11(for N = 0; N <= NLANES-1)" on page 1016	0x1(N*2+0)e7	DAC Address Config 11.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG12(for N = 0; N <= NLANES-1)" on page 1017	0x1(N*2+0)e8	DAC Address Config 12.
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG13(for N = 0; N <= NLANES-1)" on page 1018	0x1(N*2+0)e9	DAC Address Config 13.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_17(for N = 0; N <= NLANES-1)" on page 1019	0x1(N*2+0)ea	Adaptation Configuration Register 17.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_18(for N = 0; N <= NLANES-1)" on page 1020	0x1(N*2+0)eb	Adaptation Configuration Register 18.
"LANEN_DIG_RX_ADPTCTL_ADPT_CFG_19(for N = 0; N <= NLANES-1)" on page 1021	0x1(N*2+0)ec	ADPT and SSM Fast Flags
"LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG14(for N = 0; N <= NLANES-1)" on page 1022	0x1(N*2+0)ed	DAC Address Config 14.
"LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_EVEN_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1023	0x1(N*2+0)ee	Offset Values for RX DFE Error Bypass Even vDAC.
"LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_ODD_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1024	0x1(N*2+0)ef	Offset values for RX DFE Error Bypass Odd vDAC.
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_0(for N = 0; N <= NLANES-1)" on page 1025	0x1(N*2+0)f1	Search State Machine Configuration Register 0.
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_1(for N = 0; N <= NLANES-1)" on page 1026	0x1(N*2+0)f2	Search State Machine Configuration Register 1.
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_2(for N = 0; N <= NLANES-1)" on page 1027	0x1(N*2+0)f3	Search State Machine Configuration Register 2.
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_3(for N = 0; N <= NLANES-1)" on page 1028	0x1(N*2+0)f4	Search State Machine Configuration Register 3.
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_4(for N = 0; N <= NLANES-1)" on page 1029	0x1(N*2+0)f5	Search State Machine Configuration Register 4.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_5(for N = 0; N <= NLANES-1)" on page 1030	0x1(N*2+0)f6	Search State Machine Configuration Register 0.
"LANEN_DIG_RX_ADPTCTL_SSM_FINAL_CODE(for N = 0; N <= NLANES-1)" on page 1031	0x1(N*2+0)f7	Value of Final DAC Code.
"LANEN_DIG_RX_STAT_LD_VAL_1 (for N = 0; N <= NLANES-1)" on page 1033	0x1(N*2+1)00	Stat Load Value for the Sample Counter 1.
"LANEN_DIG_RX_STAT_DATA_MSK (for N = 0; N <= NLANES-1)" on page 1034	0x1(N*2+1)01	Stat Data Mask Bits [15:0]
"LANEN_DIG_RX_STAT_MATCH_CTL0 (for N = 0; N <= NLANES-1)" on page 1035	0x1(N*2+1)02	Stat Match Controls Register 0.
"LANEN_DIG_RX_STAT_MATCH_CTL1 (for N = 0; N <= NLANES-1)" on page 1036	0x1(N*2+1)03	Stat Match Controls Register 1.
"LANEN_DIG_RX_STAT_STAT_CTL0 (for N = 0; N <= NLANES-1)" on page 1038	0x1(N*2+1)04	Stat Controls Register 0.
"LANEN_DIG_RX_STAT_STAT_CTL1 (for N = 0; N <= NLANES-1)" on page 1041	0x1(N*2+1)05	Stat Controls Register 1.
"LANEN_DIG_RX_STAT_SMPL_CNT1 (for N = 0; N <= NLANES-1)" on page 1044	0x1(N*2+1)06	Bits[14:0] of Sample Counter 1 Status.
"LANEN_DIG_RX_STAT_SMPL_CNT1_29DN15(for N = 0; N <= NLANES-1)" on page 1045	0x1(N*2+1)07	Bits[29:15] of Sample Counter 1 Status.
"LANEN_DIG_RX_STAT_SMPL_CNT1_44DN30(for N = 0; N <= NLANES-1)" on page 1046	0x1(N*2+1)08	Bits[44:30] of Sample Counter 1 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_0 (for N = 0; N <= NLANES-1)" on page 1047	0x1(N*2+1)09	Stat Counter 0 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_1 (for N = 0; N <= NLANES-1)" on page 1048	0x1(N*2+1)0a	Stat Counter 1 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_2 (for N = 0; N <= NLANES-1)" on page 1049	0x1(N*2+1)0b	Stat Counter 2 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_3 (for N = 0; N <= NLANES-1)" on page 1050	0x1(N*2+1)0c	Stat Counter 3 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_4 (for N = 0; N <= NLANES-1)" on page 1051	0x1(N*2+1)0d	Stat Counter 4 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_5 (for N = 0; N <= NLANES-1)" on page 1052	0x1(N*2+1)0e	Stat Counter 5 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_6 (for N = 0; N <= NLANES-1)" on page 1053	0x1(N*2+1)0f	Stat Counter 6 Status.
"LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL(for N = 0; N <= NLANES-1)" on page 1054	0x1(N*2+1)10	Calibration Comparator Control.
"LANEN_DIG_RX_STAT_MATCH_CTL2 (for N = 0; N <= NLANES-1)" on page 1055	0x1(N*2+1)11	Stat Match Controls Register 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_RX_STAT_MATCH_CTL3 (for N = 0; N <= NLANES-1)" on page 1056	0x1(N*2+1)12	Stat Match Controls Register 3.
"LANEN_DIG_RX_STAT_MATCH_CTL4 (for N = 0; N <= NLANES-1)" on page 1057	0x1(N*2+1)13	Stat Match Controls Register 4.
"LANEN_DIG_RX_STAT_MATCH_CTL5 (for N = 0; N <= NLANES-1)" on page 1058	0x1(N*2+1)14	Stat Match Controls Register 5.
"LANEN_DIG_RX_STAT_STAT_CTL2 (for N = 0; N <= NLANES-1)" on page 1059	0x1(N*2+1)15	Stat Controls Register 2.
"LANEN_DIG_RX_STAT_STAT_STOP (for N = 0; N <= NLANES-1)" on page 1061	0x1(N*2+1)16	Stat Stop Register.
"LANEN_DIG_RX_STAT_STAT_CNT_7 (for N = 0; N <= NLANES-1)" on page 1062	0x1(N*2+1)17	Stat Counter 7 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_8 (for N = 0; N <= NLANES-1)" on page 1063	0x1(N*2+1)18	Stat Counter 8 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_9 (for N = 0; N <= NLANES-1)" on page 1064	0x1(N*2+1)19	Stat Counter 9 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_10 (for N = 0; N <= NLANES-1)" on page 1065	0x1(N*2+1)1a	Stat Counter 10 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_11 (for N = 0; N <= NLANES-1)" on page 1066	0x1(N*2+1)1b	Stat Counter 11 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_12 (for N = 0; N <= NLANES-1)" on page 1067	0x1(N*2+1)1c	Stat Counter 12 Status.
"LANEN_DIG_RX_STAT_STAT_CNT_13 (for N = 0; N <= NLANES-1)" on page 1068	0x1(N*2+1)1d	Stat Counter 13 Status.
"LANEN_DIG_RX_STAT_STAT_CTL3 (for N = 0; N <= NLANES-1)" on page 1069	0x1(N*2+1)1e	Stat Controls Register 3.
"LANEN_DIG_RX_STAT_STAT_CTL4 (for N = 0; N <= NLANES-1)" on page 1070	0x1(N*2+1)1f	Stat Controls Register 4.
"LANEN_DIG_RX_STAT_MATCH_CTL6 (for N = 0; N <= NLANES-1)" on page 1072	0x1(N*2+1)20	Stat Match Controls Register 6.
"LANEN_DIG_RX_STAT_STAT_CNT_N_SHD (for N = 0; N <= NLANES-1)" on page 1073	0x1(N*2+1)21	Stat Counter N Shadow Status.
"LANEN_DIG_RX_STAT_SMPL_CNT2 (for N = 0; N <= NLANES-1)" on page 1074	0x1(N*2+1)22	Sample Counter 2 Status.
"LANEN_DIG_RX_STAT_LD_VAL_EXT_1 (for N = 0; N <= NLANES-1)" on page 1075	0x1(N*2+1)23	Bits[14:0] of Stat Extended Load Value for the Sample Counter 1.
"LANEN_DIG_RX_STAT_LD_VAL_EXT_2 (for N = 0; N <= NLANES-1)" on page 1076	0x1(N*2+1)24	Bits[29:15] of Stat Extended Load Value for the Sample Counter 1.
"LANEN_DIG_ANA_TX_PWR_OVRD_OUT_0 (for N = 0; N <= NLANES-1)" on page 1077	0x1(N*2+1)40	Override Values for TX PWR UP/DN Signals Going to ANA, Register 0.
"LANEN_DIG_ANA_TX_PWR_OVRD_OUT_1 (for N = 0; N <= NLANES-1)" on page 1080	0x1(N*2+1)41	Override Values for TX PWR UP/DN Signals Going to ANA, Register 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ANA_TX_PWR_OVRD_OUT_2 (for N = 0; N <= NLANES-1)" on page 1083	0x1(N*2+ 1)42	Override Values for TX PWR UP/DN Signals Going to ANA, Register 2.
"LANEN_DIG_ANA_TX_TERM_CODE_OVRD _OUT(for N = 0; N <= NLANES-1)" on page 1085	0x1(N*2+ 1)43	Override Value for TX Termination Code Going to ANA.
"LANEN_DIG_ANA_TX_TERM_CODE_CLK_ OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1086	0x1(N*2+ 1)44	Override Value for TX Termination Code Clocks Going to ANA.
"LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0(for N = 0; N <= NLANES-1)" on page 1088	0x1(N*2+ 1)45	Override Values for TX EQ Signals Going to ANA, Register 0.
"LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1(for N = 0; N <= NLANES-1)" on page 1089	0x1(N*2+ 1)46	Override Values for TX EQ Signals Going to ANA, Register 1.
"LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2(for N = 0; N <= NLANES-1)" on page 1091	0x1(N*2+ 1)47	Override Values for TX EQ Signals Going to ANA, Register 2.
"LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3(for N = 0; N <= NLANES-1)" on page 1093	0x1(N*2+ 1)48	Override Values for TX EQ Signals Going to ANA, Register 3.
"LANEN_DIG_ANA_TX_DCC_EN(for N = 0; N <= NLANES-1)" on page 1095	0x1(N*2+ 1)49	TX DCC Control for tx_ana_dcc_en.
"LANEN_DIG_ANA_TX_DCC_DSQ_EN(for N = 0; N <= NLANES-1)" on page 1096	0x1(N*2+ 1)4a	TX DCC Control for tx_ana_dcc_dsq_en.
"LANEN_DIG_ANA_TX_DCC_CAL_OVRD_E N(for N = 0; N <= NLANES-1)" on page 1097	0x1(N*2+ 1)4b	Overrides TX DCC Control BUS (en/val/sel) Going to ANA.
"LANEN_DIG_ANA_TX_DCC_CAL_COMP_E N(for N = 0; N <= NLANES-1)" on page 1098	0x1(N*2+ 1)4c	TX DCC Control for tx_ana_dcc_cal_comp_en.
"LANEN_DIG_ANA_TX_DCC_CAL_CTRL_EN (for N = 0; N <= NLANES-1)" on page 1099	0x1(N*2+ 1)4d	TX DCC Control for tx_ana_dcc_cal_ctrl_en.
"LANEN_DIG_ANA_TX_DCC_CAL_CTRL_SE L(for N = 0; N <= NLANES-1)" on page 1100	0x1(N*2+ 1)4e	TX DCC Control for tx_ana_dcc_cal_ctrl_sel.
"LANEN_DIG_ANA_TX_DCC_CAL_DATA (for N = 0; N <= NLANES-1)" on page 1102	0x1(N*2+ 1)4f	TX DCC Control for tx_ana_dcc_cal_data.
"LANEN_DIG_ANA_ROPLL_ANA_OUT_0(for N = 0; N <= NLANES-1)" on page 1103	0x1(N*2+ 1)50	Override Values for Outgoing ROPLL Controls to ANA, Register 0.
"LANEN_DIG_ANA_ROPLL_ANA_OUT_1(for N = 0; N <= NLANES-1)" on page 1105	0x1(N*2+ 1)51	Override Values for Outgoing ROPLL Controls to ANA, Register 1.
"LANEN_DIG_ANA_ROPLL_ANA_OUT_2(for N = 0; N <= NLANES-1)" on page 1106	0x1(N*2+ 1)52	Override Values for Outgoing ROPLL Controls to ANA, Register 2.
"LANEN_DIG_ANA_ROPLL_ANA_OUT_3(for N = 0; N <= NLANES-1)" on page 1108	0x1(N*2+ 1)53	Override Values for Outgoing ROPLL Controls to ANA, Register 3.
"LANEN_DIG_ANA_ROPLL_ANA_OUT_4(for N = 0; N <= NLANES-1)" on page 1110	0x1(N*2+ 1)54	Override Values for Outgoing ROPLL Controls to ANA, Register 4.
"LANEN_DIG_ANA_RX_CTL_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1113	0x1(N*2+ 1)55	Override Values for RX Control Signals Going to ANA.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ANA_RX_PWR_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1116	0x1(N*2+ 1)56	Override Values for RX PWR UP/DN Signals Going to ANA.
"LANEN_DIG_ANA_RX_SIGDET_CAL_EN(for N = 0; N <= NLANES-1)" on page 1119	0x1(N*2+ 1)57	Value for RX Sigdet Calibration Enable Signal Going to ANA.
"LANEN_DIG_ANA_RX_SIGDET_HF_CAL(for N = 0; N <= NLANES-1)" on page 1121	0x1(N*2+ 1)58	Value for RX Sigdet High Frequency Calibration Signal Going to ANA.
"LANEN_DIG_ANA_RX_SIGDET_LF_CAL(for N = 0; N <= NLANES-1)" on page 1122	0x1(N*2+ 1)59	Value for RX Sigdet Low Frequency Calibration Signal Going to ANA.
"LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0(for N = 0; N <= NLANES-1)" on page 1123	0x1(N*2+ 1)5a	Override Values for RX VCO Signals Going to ANA 0.
"LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1(for N = 0; N <= NLANES-1)" on page 1125	0x1(N*2+ 1)5b	Override Values for RX VCO Signals Going to ANA 1.
"LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2(for N = 0; N <= NLANES-1)" on page 1126	0x1(N*2+ 1)5c	Override Values for RX VCO Signals Going to ANA 2.
"LANEN_DIG_ANA_RX_CAL_0(for N = 0; N <= NLANES-1)" on page 1128	0x1(N*2+ 1)5d	Sets Values for RX CAL Signals Going to ANA, Register 0.
"LANEN_DIG_ANA_RX_CAL_1(for N = 0; N <= NLANES-1)" on page 1130	0x1(N*2+ 1)5e	Sets Values for RX CAL Signals Going to ANA, Register 1.
"LANEN_DIG_ANA_RX_VDAC_RANGE_SEL(for N = 0; N <= NLANES-1)" on page 1131	0x1(N*2+ 1)5f	Sets Values for RX DAC CTRL Value Going to ANA.
"LANEN_DIG_ANA_RX_DAC_CTRL(for N = 0; N <= NLANES-1)" on page 1132	0x1(N*2+ 1)60	Sets Values for RX DAC CTRL Value Going to ANA.
"LANEN_DIG_ANA_RX_ANA_RTRIM(for N = 0; N <= NLANES-1)" on page 1133	0x1(N*2+ 1)61	Set Value for RX RTRIM Going to ANA.
"LANEN_DIG_ANA_RX_DAC_CTRL_OVRD(for N = 0; N <= NLANES-1)" on page 1135	0x1(N*2+ 1)62	Overrides RX DAC CTRL Bus (en/val/sel) Going to ANA.
"LANEN_DIG_ANA_RX_DAC_CTRL_SEL(for N = 0; N <= NLANES-1)" on page 1136	0x1(N*2+ 1)63	Sets Values for RX DAC CTRL Select Signal Going to ANA.
"LANEN_DIG_ANA_RX_AFE_ATT_VGA (for N = 0; N <= NLANES-1)" on page 1137	0x1(N*2+ 1)64	Value for RX AFE ATT & VGA Signals Going to ANA.
"LANEN_DIG_ANA_RX_AFE_CTLE(for N = 0; N <= NLANES-1)" on page 1139	0x1(N*2+ 1)65	Values for RX AFE CTLE Signals Going to ANA.
"LANEN_DIG_ANA_RX_ANA_AFE_CONFIG(for N = 0; N <= NLANES-1)" on page 1141	0x1(N*2+ 1)66	RX AFE Config Control.
"LANEN_DIG_ANA_RX_SCOPE(for N = 0; N <= NLANES-1)" on page 1142	0x1(N*2+ 1)67	Values for RX SCOPE Signals Going to ANA.
"LANEN_DIG_ANA_RX_SLICER_CTRL(for N = 0; N <= NLANES-1)" on page 1144	0x1(N*2+ 1)68	Sets Values for RX Slicer Ctrl Signals Going to ANA Register.
"LANEN_DIG_ANA_RX_ANA_IQ(for N = 0; N <= NLANES-1)" on page 1145	0x1(N*2+ 1)69	Sets Values for RX ANA IQ Signals.
"LANEN_DIG_ANA_RX_ANA_IQC_BYP_OVRD(for N = 0; N <= NLANES-1)" on page 1147	0x1(N*2+ 1)6a	Sets Value for RX ANA IQC BYPASS Adjust Signal Going to ANA Register.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ANA_RX_ANA_IQC_BYPASS_ADJUST_CLK(for N = 0; N <= NLANES-1)" on page 1148	0x1(N*2+1)6b	IQ Calibration Bypass Adjust Clock Register.
"LANEN_DIG_ANA_RX_ANA_IQC_DATA_OVRD(for N = 0; N <= NLANES-1)" on page 1149	0x1(N*2+1)6c	Sets Value for RX ANA IQC DATA Adjust Signal Going to ANA Register.
"LANEN_DIG_ANA_RX_ANA_IQC_DATA_ADJUST_CLK(for N = 0; N <= NLANES-1)" on page 1150	0x1(N*2+1)6d	IQ Calibration Data Adjust Clock Register.
"LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN(for N = 0; N <= NLANES-1)" on page 1151	0x1(N*2+1)6e	DAC CTRL Enable Signal.
"LANEN_DIG_ANA_RX_ANA_AFE_UPDATE_EN(for N = 0; N <= NLANES-1)" on page 1152	0x1(N*2+1)6f	AFE Update Enable Clock.
"LANEN_DIG_ANA_RX_ANA_DFE_SAMP_SEL(for N = 0; N <= NLANES-1)" on page 1154	0x1(N*2+1)70	RX DCC Control for rx_ana_dfe_samp_sel.
"LANEN_DIG_ANA_RX_ANA_BYP_SAMP_SEL(for N = 0; N <= NLANES-1)" on page 1155	0x1(N*2+1)71	RX DCC Control for rx_ana_byp_samp_sel.
"LANEN_DIG_ANA_RX_ANA_PHS_SAMP_SEL(for N = 0; N <= NLANES-1)" on page 1156	0x1(N*2+1)72	RX DCC Control for rx_ana_phs_samp_sel.
"LANEN_DIG_ANA_RX_TERM_CODE_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1157	0x1(N*2+1)73	Override Value for RX Termination Code Going to ANA.
"LANEN_DIG_ANA_RX_TERM_CODE_CLK_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1158	0x1(N*2+1)74	Override Value for RX Termination Code Clock Going to ANA.
"LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_ENABLE(for N = 0; N <= NLANES-1)" on page 1159	0x1(N*2+1)75	RX DFE Control for Floating TAP Enables.
"LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_POS(for N = 0; N <= NLANES-1)" on page 1161	0x1(N*2+1)76	RX DFE Control for Floating Positions.
"LANEN_DIG_ANA_RX_DCC_CTRL_RANGE(for N = 0; N <= NLANES-1)" on page 1163	0x1(N*2+1)77	RX DCC Control for .rx_ana_dcc_cal_diff_range and rx_ana_dcc_cal_cm_range.
"LANEN_DIG_ANA_STATUS_0 (for N = 0; N <= NLANES-1)" on page 1165	0x1(N*2+1)78	Lane Input Status Register 0.
"LANEN_DIG_ANA_STATUS_1 (for N = 0; N <= NLANES-1)" on page 1168	0x1(N*2+1)79	Lane Input Status Register 1.
"LANEN_DIG_ANA_RX_MISC_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1169	0x1(N*2+1)7a	Override Values for RX MISC Signals Going to ANA register.
"LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_0(for N = 0; N <= NLANES-1)" on page 1170	0x1(N*2+1)7b	TX Data Delay Enable Clock Register 0.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_1(for N = 0; N <= NLANES-1)" on page 1172	0x1(N*2+1)7c	TX Data Delay Enable Clock Register 1.
"LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_2(for N = 0; N <= NLANES-1)" on page 1174	0x1(N*2+1)7d	TX Data Delay Enable Clock Register 2.
"LANEN_DIG_ANA_RX_DIV_CLK_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1176	0x1(N*2+1)7e	Override Values for RX DIV CLK Signals Going to ANA register.
"LANEN_ANA_TX_OVRD_MEAS(for N = 0; N <= NLANES-1)" on page 1178	0x1(N*2+1)c0	TX_OVRD_MEAS
"LANEN_ANA_TX_PWR_OVRD(for N = 0; N <= NLANES-1)" on page 1180	0x1(N*2+1)c1	TX_PWR_OVRD
"LANEN_ANA_TX_ALT_BUS(for N = 0; N <= NLANES-1)" on page 1182	0x1(N*2+1)c2	TX_ALT_BUS
"LANEN_ANA_TX_ATB1(for N = 0; N <= NLANES-1)" on page 1184	0x1(N*2+1)c3	TX_ATB1
"LANEN_ANA_TX_ATB2(for N = 0; N <= NLANES-1)" on page 1186	0x1(N*2+1)c4	TX_ATB2
"LANEN_ANA_TX_ATB3(for N = 0; N <= NLANES-1)" on page 1188	0x1(N*2+1)c5	TX_ATB3
"LANEN_ANA_TX_TERM_CODE_DN(for N = 0; N <= NLANES-1)" on page 1190	0x1(N*2+1)c6	TX_TERM_CODE_DN
"LANEN_ANA_TX_TERM_CODE_UP(for N = 0; N <= NLANES-1)" on page 1192	0x1(N*2+1)c7	TX_TERM_CODE_UP
"LANEN_ANA_TX_IBOOST_CODE(for N = 0; N <= NLANES-1)" on page 1194	0x1(N*2+1)c8	TX_IBOOST_CODE
"LANEN_ANA_TX_OVRD_CLK(for N = 0; N <= NLANES-1)" on page 1196	0x1(N*2+1)c9	TX_OVRD_CLK
"LANEN_ANA_TX_MISC(for N = 0; N <= NLANES-1)" on page 1198	0x1(N*2+1)ca	TX_MISC
"LANEN_ANA_RX_ATB_IQSKEW(for N = 0; N <= NLANES-1)" on page 1200	0x1(N*2+1)cb	RX_ATB_IQSKEW
"LANEN_ANA_RX_DCC_OVRD(for N = 0; N <= NLANES-1)" on page 1202	0x1(N*2+1)cc	RX_DCC_OVRD
"LANEN_ANA_RX_PWR_CTRL1(for N = 0; N <= NLANES-1)" on page 1204	0x1(N*2+1)cd	RX_PWR_CTRL1
"LANEN_ANA_RX_ATB_REGREF(for N = 0; N <= NLANES-1)" on page 1206	0x1(N*2+1)ce	RX_ATB_REGREF
"LANEN_ANA_RX_CDR_AFE(for N = 0; N <= NLANES-1)" on page 1208	0x1(N*2+1)cf	RX_CDR_AFE
"LANEN_ANA_RX_PWR_CTRL2(for N = 0; N <= NLANES-1)" on page 1210	0x1(N*2+1)d0	RX_PWR_CTRL2

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_ANA_RX_MISC_OVRD(for N = 0; N <= NLANES-1)" on page 1212	0x1(N*2+1)d1	RX_MISC_OVRD
"LANEN_ANA_RX_CAL_MUXA(for N = 0; N <= NLANES-1)" on page 1214	0x1(N*2+1)d2	RX_CAL_MUXA
"LANEN_ANA_RX_ATB_MEAS1(for N = 0; N <= NLANES-1)" on page 1215	0x1(N*2+1)d3	RX_ATB_MEAS1
"LANEN_ANA_RX_ATB_MEAS2(for N = 0; N <= NLANES-1)" on page 1217	0x1(N*2+1)d4	RX_ATB_MEAS2
"LANEN_ANA_RX_CAL_MUXB(for N = 0; N <= NLANES-1)" on page 1219	0x1(N*2+1)d5	RX_CAL_MUXB
"LANEN_ANA_RX_TERM(for N = 0; N <= NLANES-1)" on page 1221	0x1(N*2+1)d6	RX_TERM
"LANEN_ANA_RX_SLC_CTRL(for N = 0; N <= NLANES-1)" on page 1223	0x1(N*2+1)d7	RX_SLC_CTRL
"LANEN_ANA_RX_ATB_VREG(for N = 0; N <= NLANES-1)" on page 1225	0x1(N*2+1)d8	RX_ATB_VREG
"LANEN_ANA_RX_CDR_VCO_CTRL(for N = 0; N <= NLANES-1)" on page 1227	0x1(N*2+1)d9	RX_CDR_VCO_CTRL
"LANEN_ANA_RX_VREG_CTRL2(for N = 0; N <= NLANES-1)" on page 1229	0x1(N*2+1)da	RX_VREG_CTRL2
"LANEN_ANA_RX_MISC_OVRD_2(for N = 0; N <= NLANES-1)" on page 1231	0x1(N*2+1)db	RX_MISC_OVRD_2
"LANEN_ANA_RX_MISC_OVRD_3(for N = 0; N <= NLANES-1)" on page 1232	0x1(N*2+1)dc	RX_MISC_OVRD_3
"LANEN_ANA_RX_AFE_OVRD_1(for N = 0; N <= NLANES-1)" on page 1234	0x1(N*2+1)dd	RX_AFE_OVRD_1
"LANEN_ANA_RX_AFE_OVRD_2(for N = 0; N <= NLANES-1)" on page 1235	0x1(N*2+1)de	RX_AFE_OVRD_2
"LANEN_ANA_RX_AFE_OVRD_3(for N = 0; N <= NLANES-1)" on page 1237	0x1(N*2+1)df	RX_AFE_OVRD_3
"LANEN_ANA_ROPLL_0(for N = 0; N <= NLANES-1)" on page 1238	0x1(N*2+1)e0	ROPLL_0
"LANEN_ANA_ROPLL_1(for N = 0; N <= NLANES-1)" on page 1239	0x1(N*2+1)e1	ROPLL_1
"LANEN_ANA_ROPLL_2(for N = 0; N <= NLANES-1)" on page 1240	0x1(N*2+1)e2	ROPLL_2
"LANEN_ANA_ROPLL_3(for N = 0; N <= NLANES-1)" on page 1242	0x1(N*2+1)e3	ROPLL_3
"LANEN_ANA_ROPLL_4(for N = 0; N <= NLANES-1)" on page 1243	0x1(N*2+1)e4	ROPLL_4
"LANEN_ANA_ROPLL_5(for N = 0; N <= NLANES-1)" on page 1244	0x1(N*2+1)e5	ROPLL_5

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"LANEN_ANA_ROPLL_6(for N = 0; N <= NLANES-1)" on page 1245	0x1(N*2+1)e6	ROPLL_6
"LANEN_ANA_ROPLL_7(for N = 0; N <= NLANES-1)" on page 1246	0x1(N*2+1)e7	ROPLL_7
"LANEN_ANA_ROPLL_8(for N = 0; N <= NLANES-1)" on page 1247	0x1(N*2+1)e8	ROPLL_8
"LANEN_ANA_ROPLL_9(for N = 0; N <= NLANES-1)" on page 1248	0x1(N*2+1)e9	ROPLL_9
"LANEN_ANA_ROPLL_10(for N = 0; N <= NLANES-1)" on page 1249	0x1(N*2+1)ea	ROPLL_10
"LANEN_ANA_ROPLL_11(for N = 0; N <= NLANES-1)" on page 1250	0x1(N*2+1)eb	ROPLL_11
"LANEN_ANA_RX_AFE_OVRD_4(for N = 0; N <= NLANES-1)" on page 1251	0x1(N*2+1)ec	RX_AFE_OVRD_4
"LANEN_ANA_RX_AFE_OVRD_5(for N = 0; N <= NLANES-1)" on page 1252	0x1(N*2+1)ed	RX_AFE_OVRD_5
"LANEN_ANA_RX_AFE_OVRD_6(for N = 0; N <= NLANES-1)" on page 1253	0x1(N*2+1)ee	RX_AFE_OVRD_6
"RAWLANEN_DIG_PCS_XF_TX_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1254	0x2(N*2+0)00	Override Values for Incoming TX Controls from PCS.
"RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1)" on page 1257	0x2(N*2+0)01	Override Values for Incoming TX Controls from PCS, Register 1.
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN(for N = 0; N <= NLANES-1)" on page 1259	0x2(N*2+0)02	Current Values for Incoming TX ontrols from PCS (Pre-Override Input Monitor).
"RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1)" on page 1262	0x2(N*2+0)03	Override Values for Outgoing TX Controls to PCS.
"RAWLANEN_DIG_PCS_XF_TX_PCS_OUT(for N = 0; N <= NLANES-1)" on page 1264	0x2(N*2+0)04	Current Values for Outgoing TX Status Controls from Raw PCS (Pre-Override Output Monitor).
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1265	0x2(N*2+0)05	Override Values for Incoming RX Controls from PCS, Register 0.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1)" on page 1268	0x2(N*2+0)06	Override Values for Incoming RX Controls from PCS, Register 1.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1)" on page 1270	0x2(N*2+0)07	Override Values for Incoming RX Controls from PCS, Register 2.
"RAWLANEN_DIG_PCS_XF_CNTX_RES_IN_ PROG(for N = 0; N <= NLANES-1)" on page 1272	0x2(N*2+0)08	TX/RX cntx res in prog .
"RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR _FW_1(for N = 0; N <= NLANES-1)" on page 1273	0x2(N*2+0)09	E32_RAWLANE0_DIG_PCS_XF_RX_OVRD_FOR_FW_1
"RAWLANEN_DIG_PCS_XF_ADAPT_CONT_ OVRD_IN(for N = 0; N <= NLANES-1)" on page 1274	0x2(N*2+0)0a	Override Values for Incoming Adaptation Controls from PCS.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1)" on page 1276	0x2(N*2+ 0)0b	RX Adaptation Mode Override Register.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1)" on page 1278	0x2(N*2+ 0)0c	Override Register for RX IQ Setting.
"RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_ IN_1(for N = 0; N <= NLANES-1)" on page 1280	0x2(N*2+ 0)0d	Override Values for Incoming RX EQ Controls from PCS, Register 1.
"RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_ IN_2(for N = 0; N <= NLANES-1)" on page 1282	0x2(N*2+ 0)0e	Override values for incoming RX EQ controls from PCS, register #2
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN(for N = 0; N <= NLANES-1)" on page 1283	0x2(N*2+ 0)0f	Current Values for Incoming RX Controls from PCS (Monitor on Post ATE Override).
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1(for N = 0; N <= NLANES-1)" on page 1286	0x2(N*2+ 0)10	Current Values for Incoming RX Controls from PCS, Register 1 (Monitor on Pre-Override...
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2(for N = 0; N <= NLANES-1)" on page 1287	0x2(N*2+ 0)11	Current Values for Incoming RX Controls from PCS, Register 2 (Monitor on Pre-Override...
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3(for N = 0; N <= NLANES-1)" on page 1288	0x2(N*2+ 0)12	Current Values for Incoming RX Controls from PCS, Register 3 (Monitor on Pre-Override...
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4(for N = 0; N <= NLANES-1)" on page 1290	0x2(N*2+ 0)13	Current Values for Incoming RX Controls from PCS, Register 4 (Monitor on Pre-Override...
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5(for N = 0; N <= NLANES-1)" on page 1291	0x2(N*2+ 0)14	Current Values for Incoming RX Controls from PCS, Register 5.
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6(for N = 0; N <= NLANES-1)" on page 1293	0x2(N*2+ 0)15	Current Values for Incoming RX Controls from PCS, Register 6.
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_7(for N = 0; N <= NLANES-1)" on page 1295	0x2(N*2+ 0)16	Current Values for Incoming RX Controls from PCS, Register 7.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1)" on page 1296	0x2(N*2+ 0)17	Override Values for Outgoing RX Controls to PCS.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT _1(for N = 0; N <= NLANES-1)" on page 1297	0x2(N*2+ 0)18	Override Values for Outgoing RX Controls to PCS, Register 1.
"RAWLANEN_DIG_PCS_XF_RX_PCS_OUT(for N = 0; N <= NLANES-1)" on page 1299	0x2(N*2+ 0)19	Current Values for Outgoing RX Status Controls from Raw PCS.
"RAWLANEN_DIG_PCS_XF_RX_ADAPT_AC K(for N = 0; N <= NLANES-1)" on page 1300	0x2(N*2+ 0)1a	RX Adaptation Acknowledge.
"RAWLANEN_DIG_PCS_XF_RX_ADAPT_FO M(for N = 0; N <= NLANES-1)" on page 1301	0x2(N*2+ 0)1b	RX Adaptation Figure of Merit.
"RAWLANEN_DIG_PCS_XF_RX_TXPRE_DI R(for N = 0; N <= NLANES-1)" on page 1302	0x2(N*2+ 0)1c	RX Calculated Direction for TX-pre.
"RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DI R(for N = 0; N <= NLANES-1)" on page 1303	0x2(N*2+ 0)1d	RX Calculated Direction for TX-Main.
"RAWLANEN_DIG_PCS_XF_RX_TXPOST_DI R(for N = 0; N <= NLANES-1)" on page 1304	0x2(N*2+ 0)1e	RX Calculated Direction for TX-Post.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_LANE_NUMBER (for N = 0; N <= NLANES-1)" on page 1305	0x2(N*2+0)1f	Current Lane Number.
"RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (for N = 0; N <= NLANES-1)" on page 1306	0x2(N*2+0)20	ATE Override Input to Control Top-Level Inputs.
"RAWLANEN_DIG_PCS_XF_RX_DELTA_IQ (for N = 0; N <= NLANES-1)" on page 1309	0x2(N*2+0)21	RX Delta IQ after Override. This CREG is in RX_PCS_IN category
"RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RX_RO_OUT(for N = 0; N <= NLANES-1)" on page 1310	0x2(N*2+0)22	read only registers for TX/RX related output signals
"RAWLANEN_DIG_PCS_XF_TXRX_TERM_CTRL_IN(for N = 0; N <= NLANES-1)" on page 1312	0x2(N*2+0)23	tx/rx_term_ctrl status. This CREG is in RX_PCS_IN category
"RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT_1(for N = 0; N <= NLANES-1)" on page 1313	0x2(N*2+0)24	Override Values for Outgoing TX Controls to PCS, Register 1.
"RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 1315	0x2(N*2+0)25	ATE Override input to control top-level inputs, Register 2.
"RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 1317	0x2(N*2+0)26	ATE Override Input to Control Top-Level Inputs, Register 3.
"RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 1319	0x2(N*2+0)27	ATE Override Input to Control Top-Level Inputs, Register 3.
"RAWLANEN_DIG_PCS_XF_LANE_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1321	0x2(N*2+0)28	Override Values for Incoming Lane Controls from PCS.
"RAWLANEN_DIG_PCS_XF_SIGDET_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1323	0x2(N*2+0)29	Override Values for Outgoing SIGDET Signals from PCS.
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_1(for N = 0; N <= NLANES-1)" on page 1325	0x2(N*2+0)2a	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_RX_VALID_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1327	0x2(N*2+0)2b	Override Value for RX VALID Signal from PCS.
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_2(for N = 0; N <= NLANES-1)" on page 1329	0x2(N*2+0)2c	read only registers for context restore related signals
"RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RO_1(for N = 0; N <= NLANES-1)" on page 1330	0x2(N*2+0)2d	read only registers for TX related signals
"RAWLANEN_DIG_PCS_XF_LANE_PCS_RX_RO_1(for N = 0; N <= NLANES-1)" on page 1332	0x2(N*2+0)2e	read only registers for RX related signals
"RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1)" on page 1335	0x2(N*2+0)2f	Override Values for Incoming TX Controls from PCS, Register 3.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_RX_RECAL_BANK_OVRD(for N = 0; N <= NLANES-1)" on page 1337	0x2(N*2+0)30	Override the Incoming rx_recal_inputs.
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_2(for N = 0; N <= NLANES-1)" on page 1339	0x2(N*2+0)31	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_3(for N = 0; N <= NLANES-1)" on page 1340	0x2(N*2+0)32	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_4(for N = 0; N <= NLANES-1)" on page 1341	0x2(N*2+0)33	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_CUST_TX_CNTX_RSTR_DONE(for N = 0; N <= NLANES-1)" on page 1343	0x2(N*2+0)34	Current Values for TX_CNTX_RSTR_DONE.
"RAWLANEN_DIG_PCS_XF_CUST_RX_CNTX_RSTR_DONE(for N = 0; N <= NLANES-1)" on page 1344	0x2(N*2+0)35	read values for rx_eq_dfe_tap2 and RX_CNTX_RSTR_DONE.
"RAWLANEN_DIG_PCS_XF_PAUSE_AFTER_CNTX_RSTR(for N = 0; N <= NLANES-1)" on page 1345	0x2(N*2+0)36	A register for customer to pause the FW from proceeding after the context restore is done
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_1(for N = 0; N <= NLANES-1)" on page 1346	0x2(N*2+0)37	read only registers for context restore related signals
"RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_8(for N = 0; N <= NLANES-1)" on page 1347	0x2(N*2+0)38	Override Values for Incoming TX Controls from PCS, Register 8.
"RAWLANEN_DIG_PCS_XF_TX_RECAL_BANK_OVRD(for N = 0; N <= NLANES-1)" on page 1349	0x2(N*2+0)39	Override the Incoming tx_recal_inputs.
"RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_9(for N = 0; N <= NLANES-1)" on page 1351	0x2(N*2+0)3a	Override Register for RX IQ Setting.
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_8(for N = 0; N <= NLANES-1)" on page 1353	0x2(N*2+0)3b	Current Values for Incoming RX Controls from PCS, Register 8.
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_5(for N = 0; N <= NLANES-1)" on page 1355	0x2(N*2+0)3c	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_TX_PCS_IN_6(for N = 0; N <= NLANES-1)" on page 1356	0x2(N*2+0)3d	Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
"RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_2(for N = 0; N <= NLANES-1)" on page 1358	0x2(N*2+0)3e	E32_RAWLANE0_DIG_PCS_XF_RX_OVRD_FOR_FW_2
"RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_10(for N = 0; N <= NLANES-1)" on page 1360	0x2(N*2+0)3f	Override Values for Incoming TX Controls from PCS, Register 10.
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_9(for N = 0; N <= NLANES-1)" on page 1362	0x2(N*2+0)40	Current Values for Incoming RX Controls from PCS, Register 9.
"RAWLANEN_DIG_PCS_XF_RX_PCS_IN_10(for N = 0; N <= NLANES-1)" on page 1364	0x2(N*2+0)41	Current Values for Incoming RX Controls from PCS, Register 10.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_1(for N = 0; N <= NLANES-1)" on page 1365	0x2(N*2+0)42	Override register for CNTX_RSTR primary input signals
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_2(for N = 0; N <= NLANES-1)" on page 1366	0x2(N*2+0)43	Override register for CNTX_RSTR primary input signals
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_3(for N = 0; N <= NLANES-1)" on page 1367	0x2(N*2+0)44	Override register for CNTX_RSTR primary input signals
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 1368	0x2(N*2+0)45	ATE Override Input to Control Top-Level Inputs, Register 1. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 1370	0x2(N*2+0)46	Override Values for Incoming TX Controls from PCS, Register 2. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 1372	0x2(N*2+0)47	Override Values for Incoming TX Controls from PCS, Register 3. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_4(for N = 0; N <= NLANES-1)" on page 1373	0x2(N*2+0)48	Override Values for Incoming TX Controls from PCS, Register 4. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_5(for N = 0; N <= NLANES-1)" on page 1375	0x2(N*2+0)49	Override Values for Incoming TX Controls from PCS, Register 5. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_6(for N = 0; N <= NLANES-1)" on page 1377	0x2(N*2+0)4a	Override Incoming Values for tx/rx_term_ctrl, Register 6. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_7(for N = 0; N <= NLANES-1)" on page 1379	0x2(N*2+0)4b	Override Values for Incoming TX Controls from PCS, Register 7. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_8(for N = 0; N <= NLANES-1)" on page 1380	0x2(N*2+0)4c	Override Register for RX PCS Input, Register 8. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_9(for N = 0; N <= NLANES-1)" on page 1383	0x2(N*2+0)4d	Override Register for RX PCS Input, Register 9. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_PWR_EN_LANE(for N = 0; N <= NLANES-1)" on page 1386	0x2(N*2+0)4e	power_en new register to sequence isolation
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_1(for N = 0; N <= NLANES-1)" on page 1387	0x2(N*2+0)4f	Override Values for Incoming RX EQ Controls from PCS, Register 1. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_2(for N = 0; N <= NLANES-1)" on page 1389	0x2(N*2+0)50	Override Register for RX PCS Input. RO: stands for re-ordered

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_3(for N = 0; N <= NLANES-1)" on page 1390	0x2(N*2+0)51	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_4(for N = 0; N <= NLANES-1)" on page 1391	0x2(N*2+0)52	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_5(for N = 0; N <= NLANES-1)" on page 1392	0x2(N*2+0)53	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_6(for N = 0; N <= NLANES-1)" on page 1394	0x2(N*2+0)54	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_7(for N = 0; N <= NLANES-1)" on page 1395	0x2(N*2+0)55	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_8(for N = 0; N <= NLANES-1)" on page 1396	0x2(N*2+0)56	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_9(for N = 0; N <= NLANES-1)" on page 1398	0x2(N*2+0)57	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_10(for N = 0; N <= NLANES-1)" on page 1400	0x2(N*2+0)58	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_11(for N = 0; N <= NLANES-1)" on page 1402	0x2(N*2+0)59	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_12(for N = 0; N <= NLANES-1)" on page 1403	0x2(N*2+0)5a	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_13(for N = 0; N <= NLANES-1)" on page 1406	0x2(N*2+0)5b	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_14(for N = 0; N <= NLANES-1)" on page 1409	0x2(N*2+0)5c	Override Register for RX PCS Input. RO: stands for re-ordered
"RAWLANEN_DIG_PCS_XF_TX_CNTX_RSTR_DONE(for N = 0; N <= NLANES-1)" on page 1411	0x2(N*2+0)5d	TX Context Restore done.
"RAWLANEN_DIG_PCS_XF_RX_CNTX_RSTR_DONE(for N = 0; N <= NLANES-1)" on page 1412	0x2(N*2+0)5e	RX Context Restore done.
"RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_EN(for N = 0; N <= NLANES-1)" on page 1413	0x2(N*2+0)5f	Lane context restore enable

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_PCS_XF_DEBUG_MODE(for N = 0; N <= NLANES-1)" on page 1414	0x2(N*2+0)60	A register added for FW to know if we are in the debug mode or not
"RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1415	0x2(N*2+0)b0	Override Values for Incoming LANE Controls.
"RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1417	0x2(N*2+0)b1	Override Values for Outgoing LANE Controls.
"RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1419	0x2(N*2+0)b2	Override Values for Incoming SUP Controls from PMA.
"RAWLANEN_DIG_PMA_XF_SUP_PMA_IN(for N = 0; N <= NLANES-1)" on page 1421	0x2(N*2+0)b3	Current Values for Coming SUP Status Controls from PMA.
"RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1422	0x2(N*2+0)b4	Override Values for Outgoing TX Controls to PMA.
"RAWLANEN_DIG_PMA_XF_TX_PMA_IN(for N = 0; N <= NLANES-1)" on page 1424	0x2(N*2+0)b5	Current Values for Coming TX Status Controls from PMA.
"RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT(for N = 0; N <= NLANES-1)" on page 1425	0x2(N*2+0)b6	Override Values for Outgoing RX Controls to PMA.
"RAWLANEN_DIG_PMA_XF_RX_PMA_IN(for N = 0; N <= NLANES-1)" on page 1427	0x2(N*2+0)b7	Current Values for Coming RX Status Controls from PMA.
"RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL(for N = 0; N <= NLANES-1)" on page 1428	0x2(N*2+0)b8	Lane Rtune Controls.
"RAWLANEN_DIG_PMA_XF_SUP_PMA_IN_1(for N = 0; N <= NLANES-1)" on page 1429	0x2(N*2+0)b9	Current Values for Coming SUP Status Controls from PMA.
"RAWLANEN_DIG_PMA_XF_RX_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1430	0x2(N*2+0)ba	Override Values for Incoming RX Controls from PMA.
"RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ(for N = 0; N <= NLANES-1)" on page 1431	0x2(N*2+0)c0	Reset Routine Request.
"RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ(for N = 0; N <= NLANES-1)" on page 1432	0x2(N*2+0)c1	RX Reset Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ(for N = 0; N <= NLANES-1)" on page 1433	0x2(N*2+0)c2	RX Request Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ(for N = 0; N <= NLANES-1)" on page 1434	0x2(N*2+0)c3	RX Rate Change Interrupt Request.
"RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ(for N = 0; N <= NLANES-1)" on page 1435	0x2(N*2+0)c4	RX PSTATE Change Interrupt Request.
"RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ(for N = 0; N <= NLANES-1)" on page 1436	0x2(N*2+0)c5	RX adaptation request interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ(for N = 0; N <= NLANES-1)" on page 1437	0x2(N*2+0)c6	RX adaptation disable interrupt.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1438	0x2(N*2+0)c7	RX reset interrupt clear.
"RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1439	0x2(N*2+0)c8	RX request interrupt clear.
"RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1440	0x2(N*2+0)c9	RX rate change interrupt clear.
"RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1441	0x2(N*2+0)ca	RX pstate change interrupt clear.
"RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1442	0x2(N*2+0)cb	RX Adaptation Request Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1443	0x2(N*2+0)cc	RX Adaptation Disable Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_IRQ_MASK(for N = 0; N <= NLANES-1)" on page 1444	0x2(N*2+0)cd	Interrupt Mask.
"RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ (for N = 0; N <= NLANES-1)" on page 1448	0x2(N*2+0)ce	TX Rate Interrupt.
"RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1449	0x2(N*2+0)cf	TX Rate Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ(for N = 0; N <= NLANES-1)" on page 1450	0x2(N*2+0)d2	TX Reset Interrupt.
"RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1451	0x2(N*2+0)d3	TX Reset Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ(for N = 0; N <= NLANES-1)" on page 1452	0x2(N*2+0)d4	TX REQ Interrupt.
"RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1453	0x2(N*2+0)d5	TX req Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ(for N = 0; N <= NLANES-1)" on page 1454	0x2(N*2+0)d6	RX2TX Parallel Loopback Enable Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1455	0x2(N*2+0)d7	RX2TX Parallel Loopback Enable Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ(for N = 0; N <= NLANES-1)" on page 1456	0x2(N*2+0)d8	RX2TX Parallel Loopback Disable Interrupt.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1457	0x2(N*2+0)d9	RX2TX Parallel Loopback Disable Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_IRQ_EN_FLAGS(for N = 0; N <= NLANES-1)" on page 1458	0x2(N*2+0)da	Interrupt Enable Flags.
"RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ(for N = 0; N <= NLANES-1)" on page 1461	0x2(N*2+0)dd	TX Termination Control Interrupt.
"RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1462	0x2(N*2+0)de	TX Termination Control Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ(for N = 0; N <= NLANES-1)" on page 1463	0x2(N*2+0)df	RX Termination Control Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1464	0x2(N*2+0)e0	RX Termination Control Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_IRQ_MASK_2(for N = 0; N <= NLANES-1)" on page 1465	0x2(N*2+0)e1	Interrupt Mask 2.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ (for N = 0; N <= NLANES-1)" on page 1467	0x2(N*2+0)e2	RX MARGIN iq_start Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ_CLR (for N = 0; N <= NLANES-1)" on page 1468	0x2(N*2+0)e3	RX MARGIN iq_start Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ (for N = 0; N <= NLANES-1)" on page 1469	0x2(N*2+0)e4	RX MARGIN vdac_start Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ_CLR (for N = 0; N <= NLANES-1)" on page 1470	0x2(N*2+0)e5	RX MARGIN vdac_start Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ(for N = 0; N <= NLANES-1)" on page 1471	0x2(N*2+0)e6	RX MARGIN Error Clear Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1472	0x2(N*2+0)e7	RX MARGIN Error Clear Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ(for N = 0; N <= NLANES-1)" on page 1473	0x2(N*2+0)e8	RX MARGIN init Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1474	0x2(N*2+0)e9	RX MARGIN init Interrupt Clear.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ(for N = 0; N <= NLANES-1)" on page 1475	0x2(N*2+0)ea	RX MARGIN Finish Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1476	0x2(N*2+0)eb	RX MARGIN Finish Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ(for N = 0; N <= NLANES-1)" on page 1477	0x2(N*2+0)ec	RX MARGIN glb Interrupt.
"RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ_CLR(for N = 0; N <= NLANES-1)" on page 1478	0x2(N*2+0)ed	RX MARGIN glb Interrupt Clear.
"RAWLANEN_DIG_IRQ_CTL_LANE_PERST_CTL(for N = 0; N <= NLANES-1)" on page 1480	0x2(N*2+0)ee	PERST related control.
"RAWLANEN_DIG_TX_CTL_TX_FSM_CTL(for N = 0; N <= NLANES-1)" on page 1482	0x2(N*2+0)f8	TX FSM Control.
"RAWLANEN_DIG_TX_CTL_TX_CLK_CTL(for N = 0; N <= NLANES-1)" on page 1484	0x2(N*2+0)f9	Select Clock to Act as TX Input Clock.
"RAWLANEN_DIG_TX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)" on page 1486	0x2(N*2+0)fa	TX DCC Continuous Calibration Status.
"RAWLANEN_DIG_TX_CTL_TX_RATE_IRQ_ACK(for N = 0; N <= NLANES-1)" on page 1487	0x2(N*2+0)fb	TX FSM Rate IRQ ACK Register.
"RAWLANEN_DIG_TX_CTL_REQ_TX_PLL_EN_CTL(for N = 0; N <= NLANES-1)" on page 1488	0x2(N*2+0)fc	TX PLL Enable override register to be used by FW
"RAWLANEN_DIG_TX_CTL_TX_CTL_STATUS(for N = 0; N <= NLANES-1)" on page 1489	0x2(N*2+0)fd	Tx_ctl status
"RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_LEFT(for N = 0; N <= NLANES-1)" on page 1492	0x2(N*2+0)ff	RX Adapted Value of Left Code.
"RAWLANEN_DIG_RX_CTL_RX_SIGDET_MASK_CTL(for N = 0; N <= NLANES-1)" on page 1493	0x2(N*2+1)00	RX SIGDET Mask Control.
"RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_RIGHT(for N = 0; N <= NLANES-1)" on page 1494	0x2(N*2+1)01	RX Adapted Value of Right Code.
"RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)" on page 1495	0x2(N*2+1)02	RX continuous offset cancellation status.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS (for N = 0; N <= NLANES-1)" on page 1496	0x2(N*2+1)03	RX Continuous Adaptation Status.
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE(for N = 0; N <= NLANES-1)" on page 1497	0x2(N*2+1)04	RX Adaptation Mode.
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL(for N = 0; N <= NLANES-1)" on page 1498	0x2(N*2+1)05	Select Between Two Banks Adaptation Settings for RX Adaptation.
"RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT(for N = 0; N <= NLANES-1)" on page 1499	0x2(N*2+1)06	RX CDR PPM Drift on RX Clock.
"RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS (for N = 0; N <= NLANES-1)" on page 1500	0x2(N*2+1)07	RX Misc Status Register.
"RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL(for N = 0; N <= NLANES-1)" on page 1502	0x2(N*2+1)08	Values for Outgoing RX Controls to PMA.
"RAWLANEN_DIG_RX_CTL_RX_CTL_STATUS(for N = 0; N <= NLANES-1)" on page 1504	0x2(N*2+1)09	Rx_ctl status
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN(for N = 0; N <= NLANES-1)" on page 1507	0x2(N*2+1)0a	Enable RX Adapt Mode Override (using Custom Setting for RX Adaptation).
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN(for N = 0; N <= NLANES-1)" on page 1509	0x2(N*2+1)0b	Enable Using Four Different RX Adaptation Modes.
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM(for N = 0; N <= NLANES-1)" on page 1510	0x2(N*2+1)0c	RX Adaptation Mission-Mode Figure of Merit.
"RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM(for N = 0; N <= NLANES-1)" on page 1511	0x2(N*2+1)0d	RX Adaptation Start-Up Figure of Merit (Base Mode or Extended Mode).
"RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN(for N = 0; N <= NLANES-1)" on page 1512	0x2(N*2+1)0e	RX Adapted Value of Reference Level for Error Even Slicer.
"RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD(for N = 0; N <= NLANES-1)" on page 1513	0x2(N*2+1)0f	RX Adapted Value of Reference Level for Error Odd Slicer.
"RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN(for N = 0; N <= NLANES-1)" on page 1514	0x2(N*2+1)10	RX Phase Adjust Linear Value.
"RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP(for N = 0; N <= NLANES-1)" on page 1515	0x2(N*2+1)11	RX Phase Adjust Mapped Value.
"RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN_UPDATE (for N = 0; N <= NLANES-1)" on page 1516	0x2(N*2+1)12	RX Phase Adjust Linear Update Value.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP_UPDATE (for N = 0; N <= NLANES-1)" on page 1517	0x2(N*2+1)13	RX Phase Adjust Mapped Update Value.
"RAWLANEN_DIG_RX_CTL_RX_MARGIN_IQ_DELTA (for N = 0; N <= NLANES-1)" on page 1518	0x2(N*2+1)14	RX Margin IQ Delta Value.
"RAWLANEN_DIG_RX_CTL_RX_FSM_CTL(for N = 0; N <= NLANES-1)" on page 1519	0x2(N*2+1)15	RX FSM Control.
"RAWLANEN_DIG_RX_CTL_RX_RATE_IRQ_ACK(for N = 0; N <= NLANES-1)" on page 1520	0x2(N*2+1)16	RX FSM Rate IRQ ACK Register.
"RAWLANEN_DIG_RX_CTL_DFE_EEH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1521	0x2(N*2+1)17	Value for DFE Error Even High Slicer in Mission Mode.
"RAWLANEN_DIG_RX_CTL_DFE_EOH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1522	0x2(N*2+1)18	Value for DFE Error Odd High Slicer in Mission Mode.
"RAWLANEN_DIG_RX_CTL_DFE_EEL_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1523	0x2(N*2+1)19	Value for DFE Error Even Low Slicer in Mission Mode.
"RAWLANEN_DIG_RX_CTL_DFE_EOL_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1524	0x2(N*2+1)1a	Value for DFE Error Odd Low Slicer in Mission Mode.
"RAWLANEN_DIG_RX_CTL_SETUP_REF_SCO_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1525	0x2(N*2+1)1b	Offset Value for AFE SCO IDAC During Reference-Level Calibration.
"RAWLANEN_DIG_RX_CTL_SETUP_REF_VGA2_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1526	0x2(N*2+1)1c	Offset Value for AFE VGA2 IDAC During Reference-Level Calibration.
"RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA1_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1527	0x2(N*2+1)1d	Offset Value for AFE VGA1 for Slicer Set-Up Calibration.
"RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA2_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1528	0x2(N*2+1)1e	Offset Value for AFE VGA2 for Slicer Set-Up Calibration.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP0_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1529	0x2(N*2+1)20	RX DFE Floating TAP Correlation Magnitude Sorting Input 0.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP1_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1530	0x2(N*2+1)21	RX DFE Floating TAP Correlation Magnitude Sorting Input 1.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP2_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1531	0x2(N*2+1)22	RX DFE Floating TAP Correlation Magnitude Sorting Input 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP3_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1532	0x2(N*2+1)23	RX DFE Floating TAP Correlation Magnitude Sorting Input 3.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP4_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1533	0x2(N*2+1)24	RX DFE Floating TAP Correlation Magnitude Sorting Input 4.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP5_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1534	0x2(N*2+1)25	RX DFE Floating TAP Correlation Magnitude Sorting Input 5.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP6_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1535	0x2(N*2+1)26	RX DFE Floating TAP Correlation Magnitude Sorting Input 6.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP7_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1536	0x2(N*2+1)27	RX DFE Floating TAP Correlation Magnitude Sorting Input 7.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP8_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1537	0x2(N*2+1)28	RX DFE Floating TAP Correlation Magnitude Sorting Input 8.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP9_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1538	0x2(N*2+1)29	RX DFE Floating TAP Correlation Magnitude Sorting Input 9.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP10_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1539	0x2(N*2+1)2a	RX DFE Floating TAP Correlation Magnitude Sorting Input 10.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP11_CORR_MAG(for N = 0; N <= NLANES-1)" on page 1540	0x2(N*2+1)2b	RX DFE Floating TAP Correlation Magnitude Sorting Input 11.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_CTL(for N = 0; N <= NLANES-1)" on page 1541	0x2(N*2+1)2c	RX DFE Floating TAP Sorting Control.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_STATE (for N = 0; N <= NLANES-1)" on page 1542	0x2(N*2+1)2d	RX DFE Floating TAP Sorting ACK.
"RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_RESULT (for N = 0; N <= NLANES-1)" on page 1543	0x2(N*2+1)2e	RX DFE Floating TAP Sorting Result.
"RAWLANEN_DIG_RX_CTL_RX_MARGIN_VDAC_DELTA (for N = 0; N <= NLANES-1)" on page 1545	0x2(N*2+1)2f	RX Margin VDAC Delta Value.
"RAWLANEN_DIG_RX_CTL_RX_MARGIN_ERROR(for N = 0; N <= NLANES-1)" on page 1546	0x2(N*2+1)30	RX Margin Error.
"RAWLANEN_DIG_RX_CTL_RX_VALID_COARSE_ADAPT(for N = 0; N <= NLANES-1)" on page 1547	0x2(N*2+1)31	RX Valid Coarse Adapt.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_WR(for N = 0; N <= NLANES-1)" on page 1548	0x2(N*2+1)32	RX IQ Linear Code Input Register for Conversion.
"RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_RD(for N = 0; N <= NLANES-1)" on page 1549	0x2(N*2+1)33	RX IQ Step Code Readback Value.
"RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_WR(for N = 0; N <= NLANES-1)" on page 1550	0x2(N*2+1)34	RX IQ Step Code Input Register for Conversion.
"RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_RD(for N = 0; N <= NLANES-1)" on page 1551	0x2(N*2+1)35	RX IQ Linear Code Readback Value.
"RAWLANEN_DIG_RX_CTL_FOM_CTL(for N = 0; N <= NLANES-1)" on page 1552	0x2(N*2+1)36	FOM CTL Setting.
"RAWLANEN_DIG_RX_CTL_RX_MARGIN_CTL(for N = 0; N <= NLANES-1)" on page 1553	0x2(N*2+1)37	RX Margin CTL.
"RAWLANEN_DIG_RX_CTL_PRE_EE_DAC_CODE(for N = 0; N <= NLANES-1)" on page 1554	0x2(N*2+1)38	EE Precursor Measurement Code.
"RAWLANEN_DIG_RX_CTL_PRE_EO_DAC_CODE(for N = 0; N <= NLANES-1)" on page 1555	0x2(N*2+1)39	EO Precursor Measurement Code.
"RAWLANEN_DIG_RX_CTL_SETUP_SLC_CTLE_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1556	0x2(N*2+1)3a	Offset Value for AFE CTLE for Slicer Set-Up Calibration.
"RAWLANEN_DIG_RX_CTL_RX_GAPPED_CLK_EN_TIME(for N = 0; N <= NLANES-1)" on page 1557	0x2(N*2+1)3b	Eth RX Gapped Clock Enable Timer.
"RAWLANEN_DIG_RX_CTL_RX_SIGDET_FILT_CTL(for N = 0; N <= NLANES-1)" on page 1558	0x2(N*2+1)3c	RX SIGDET LF/HF Filtering Control
"RAWLANEN_DIG_FSM_FSM_OVRD_CTL(for N = 0; N <= NLANES-1)" on page 1560	0x2(N*2+1)80	FSM Override Control Register.
"RAWLANEN_DIG_FSM_FSM_JMP_BANK(for N = 0; N <= NLANES-1)" on page 1562	0x2(N*2+1)81	FSM Jump Bank Register.
"RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0(for N = 0; N <= NLANES-1)" on page 1563	0x2(N*2+1)82	FSM Breakpoint 0 on SRAM Address for Debugging.
"RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1(for N = 0; N <= NLANES-1)" on page 1564	0x2(N*2+1)83	FSM Breakpoint 1 on SRAM Address for Debugging.
"RAWLANEN_DIG_FSM_SKIP_RX_MARGINING(for N = 0; N <= NLANES-1)" on page 1565	0x2(N*2+1)84	Status of Skip RX Margining.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_FSM_MEM_ADDR_MON(for N = 0; N <= NLANES-1)" on page 1566	0x2(N*2+ 1)85	Memory Address Monitor.
"RAWLANEN_DIG_FSM_STATUS_MON (for N = 0; N <= NLANES-1)" on page 1567	0x2(N*2+ 1)86	FSM Status Monitor.
"RAWLANEN_DIG_FSM_FAST_RX_STARTU P_CAL(for N = 0; N <= NLANES-1)" on page 1569	0x2(N*2+ 1)87	Status of Fast RX Start-Up Calibration.
"RAWLANEN_DIG_FSM_FAST_RX_ADAPT (for N = 0; N <= NLANES-1)" on page 1570	0x2(N*2+ 1)88	Status of Fast RX Adaptation.
"RAWLANEN_DIG_FSM_FAST_SUP (for N = 0; N <= NLANES-1)" on page 1571	0x2(N*2+ 1)89	Status of Fast Support Block.
"RAWLANEN_DIG_FSM_FAST_TX_CMN_M ODE(for N = 0; N <= NLANES-1)" on page 1572	0x2(N*2+ 1)8a	Status of Fast TX Common-Mode Charge Up.
"RAWLANEN_DIG_FSM_FAST_TX_RXDET (for N = 0; N <= NLANES-1)" on page 1573	0x2(N*2+ 1)8b	Status of Fast TX Detect RX.
"RAWLANEN_DIG_FSM_FAST_RX_PWRUP (for N = 0; N <= NLANES-1)" on page 1574	0x2(N*2+ 1)8c	Status of Fast RX Power Up.
"RAWLANEN_DIG_FSM_FAST_RX_VCO_W AIT(for N = 0; N <= NLANES-1)" on page 1575	0x2(N*2+ 1)8d	Status of Fast RX VCO Wait Times.
"RAWLANEN_DIG_FSM_FAST_RX_VCO_CA L(for N = 0; N <= NLANES-1)" on page 1576	0x2(N*2+ 1)8e	Status of Fast RX VCO Calibration.
"RAWLANEN_DIG_FSM_FAST_RX_CONT_C AL_ADAPT(for N = 0; N <= NLANES-1)" on page 1577	0x2(N*2+ 1)8f	Status of Fast RX Continuous Calibration/Adaptation.
"RAWLANEN_DIG_FSM_FAST_RX_CONT_A DAPT(for N = 0; N <= NLANES-1)" on page 1578	0x2(N*2+ 1)90	Status of Fast RX Continuous Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_AFE_STA RTUP_CAL(for N = 0; N <= NLANES-1)" on page 1579	0x2(N*2+ 1)91	Status of RX AFE Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DFE_STA RTUP_CAL(for N = 0; N <= NLANES-1)" on page 1580	0x2(N*2+ 1)92	Status of RX DFE Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT _STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1581	0x2(N*2+ 1)93	Status of DFE Extended Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELT A_STARTUP (for N = 0; N <= NLANES-1)" on page 1582	0x2(N*2+ 1)94	Status of RX IQ Fixed Offset.
"RAWLANEN_DIG_FSM_SKIP_RX_IQ_STAR TUP_CAL(for N = 0; N <= NLANES-1)" on page 1583	0x2(N*2+ 1)95	Status of RX IQ Start-Up Calibration.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT(for N = 0; N <= NLANES-1)" on page 1584	0x2(N*2+1)96	Status of RX AFE Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT(for N = 0; N <= NLANES-1)" on page 1585	0x2(N*2+1)97	Status of RX DFE Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT(for N = 0; N <= NLANES-1)" on page 1586	0x2(N*2+1)98	Status of RX IO Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL(for N = 0; N <= NLANES-1)" on page 1587	0x2(N*2+1)99	Status of RX Continuous Phase Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL(for N = 0; N <= NLANES-1)" on page 1588	0x2(N*2+1)9a	Status of RX AFE Continuous Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT(for N = 0; N <= NLANES-1)" on page 1589	0x2(N*2+1)9b	Status of RX Reference Level Continuous Adaptation .
"RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT(for N = 0; N <= NLANES-1)" on page 1590	0x2(N*2+1)9c	Status of RX VGA Continuous Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1591	0x2(N*2+1)9d	Status of RX Phase Start-up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1592	0x2(N*2+1)9e	Status of RX Phase Extended Start-up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1593	0x2(N*2+1)9f	Status of RX DCC Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_CONT_CAL(for N = 0; N <= NLANES-1)" on page 1594	0x2(N*2+1)a0	Status of RX DCC Continuous Calibration.
"RAWLANEN_DIG_FSM_SKIP_TX_DCC_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1595	0x2(N*2+1)a1	Status of TX DCC Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_TX_DCC_CONT_CAL(for N = 0; N <= NLANES-1)" on page 1596	0x2(N*2+1)a2	Status of TX DCC Continuous Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_RATE_CAL(for N = 0; N <= NLANES-1)" on page 1597	0x2(N*2+1)a3	Status of RX DCC Rate-Change Calibration.
"RAWLANEN_DIG_FSM_SKIP_TX_DCC_RATE_CAL(for N = 0; N <= NLANES-1)" on page 1598	0x2(N*2+1)a4	Status of TX DCC Rate-Change Calibration.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_FSM_SKIP_SIGDET_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1599	0x2(N*2+1)a5	Status of SIGDET Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_VGEN_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1600	0x2(N*2+1)a6	Status of VGEN Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_IQC_RATE_CAL(for N = 0; N <= NLANES-1)" on page 1601	0x2(N*2+1)a7	Status of RX IQC Rate-hange Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_DATA_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1602	0x2(N*2+1)a8	Status of RX DCC DATA Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1603	0x2(N*2+1)a9	Status of RX DCC BYPASS Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_DCC_PHASE_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1604	0x2(N*2+1)aa	Status of RX DCC PHASE Start-up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_HALF_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1605	0x2(N*2+1)ab	Status of RX Half-Rate Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_FULL_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1606	0x2(N*2+1)ac	Status of RX Full-Rate Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_ERROR_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1607	0x2(N*2+1)ad	Status of RX Error Slicer Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1608	0x2(N*2+1)ae	Status of RX Bypass Slicer Start-Up Calibration.
"RAWLANEN_DIG_FSM_SKIP_RX_VGA_SLICER_STARTUP_CAL (for N = 0; N <= NLANES-1)" on page 1609	0x2(N*2+1)af	Status of RX VGA Slicer Start-Up Calibration.
"RAWLANEN_DIG_FSM_FW_SCRATCH_0 (for N = 0; N <= NLANES-1)" on page 1610	0x2(N*2+1)b0	Firmware Scratch Register 0.
"RAWLANEN_DIG_FSM_FW_SCRATCH_1 (for N = 0; N <= NLANES-1)" on page 1611	0x2(N*2+1)b1	Firmware Scratch Register 1.
"RAWLANEN_DIG_FSM_FW_SCRATCH_2 (for N = 0; N <= NLANES-1)" on page 1612	0x2(N*2+1)b2	Firmware Scratch Register 2.
"RAWLANEN_DIG_FSM_FW_SCRATCH_3 (for N = 0; N <= NLANES-1)" on page 1613	0x2(N*2+1)b3	Firmware Scratch Register 3.
"RAWLANEN_DIG_FSM_FAST_TX_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1614	0x2(N*2+1)b4	Status of Fast TX Start-Up Calibration.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_FSM_SKIP_RX_BUF_STARTUP_CAL(for N = 0; N <= NLANES-1)" on page 1615	0x2(N*2+1)b5	Status of RX Buffer Start-Up Calibration.
"RAWLANEN_DIG_FSM_FAST_TX_PWRUP (for N = 0; N <= NLANES-1)" on page 1616	0x2(N*2+1)b6	Status of Fast TX Power Up.
"RAWLANEN_DIG_FSM_SKIP_RX_ADAPT_RELOAD(for N = 0; N <= NLANES-1)" on page 1617	0x2(N*2+1)b7	Status of RX Adaptation Initialization.
"RAWLANEN_DIG_FSM_FW_SCRATCH_4 (for N = 0; N <= NLANES-1)" on page 1618	0x2(N*2+1)b8	Firmware Scratch Register 4.
"RAWLANEN_DIG_FSM_FW_SCRATCH_5 (for N = 0; N <= NLANES-1)" on page 1619	0x2(N*2+1)b9	Firmware Scratch Register 5.
"RAWLANEN_DIG_FSM_FW_SCRATCH_6 (for N = 0; N <= NLANES-1)" on page 1620	0x2(N*2+1)ba	Firmware Scratch Register 6.
"RAWLANEN_DIG_FSM_FW_SCRATCH_7 (for N = 0; N <= NLANES-1)" on page 1621	0x2(N*2+1)bb	Firmware Scratch Register 7.
"RAWLANEN_DIG_FSM_FW_SCRATCH_8 (for N = 0; N <= NLANES-1)" on page 1622	0x2(N*2+1)bc	Firmware Scratch Register 8.
"RAWLANEN_DIG_FSM_FW_SCRATCH_9 (for N = 0; N <= NLANES-1)" on page 1623	0x2(N*2+1)bd	Firmware Scratch Register 9.
"RAWLANEN_DIG_FSM_FW_SCRATCH_10 (for N = 0; N <= NLANES-1)" on page 1624	0x2(N*2+1)be	Firmware Scratch Register 10.
"RAWLANEN_DIG_FSM_FW_SCRATCH_11 (for N = 0; N <= NLANES-1)" on page 1625	0x2(N*2+1)bf	Firmware Scratch Register 11.
"RAWLANEN_DIG_FSM_FW_SCRATCH_12 (for N = 0; N <= NLANES-1)" on page 1626	0x2(N*2+1)c0	Firmware Scratch Register 12.
"RAWLANEN_DIG_FSM_FW_SCRATCH_13 (for N = 0; N <= NLANES-1)" on page 1627	0x2(N*2+1)c1	Firmware Scratch Register 13.
"RAWLANEN_DIG_FSM_FW_SCRATCH_14 (for N = 0; N <= NLANES-1)" on page 1628	0x2(N*2+1)c2	Firmware Scratch Register 14.
"RAWLANEN_DIG_FSM_FW_SCRATCH_15 (for N = 0; N <= NLANES-1)" on page 1629	0x2(N*2+1)c3	Firmware Scratch Register 15.
"RAWLANEN_DIG_FSM_SKIP_RX_ATT_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1630	0x2(N*2+1)c4	Status of RX ATT Coarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_VGA_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1631	0x2(N*2+1)c5	Status of RX VGA Ccoarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_CTLE_POLE_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1632	0x2(N*2+1)c6	Status of RX CTLE_POLE Coarse Start-Up Adaptation.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEN_DIG_FSM_SKIP_RX_CTLE_BOOST_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1633	0x2(N*2+1)c7	Status of RX CTLE_BOOST Coarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP1_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1634	0x2(N*2+1)c8	Status of RX DFE_TAP1 Coarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP2_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1635	0x2(N*2+1)c9	Status of RX DFE_TAP2 Coarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_SKIP_RX_IQ_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)" on page 1636	0x2(N*2+1)ca	Status of RX IQ Coarse Start-Up Adaptation.
"RAWLANEN_DIG_FSM_FAST_RX_COARSE_ADAPT(for N = 0; N <= NLANES-1)" on page 1637	0x2(N*2+1)cb	Status of Fast RX Coarse Adapt.
"RAWLANEN_DIG_FSM_SKIP_INIT_PWRUP_END_RX_RESET_TOGGLE(for N = 0; N <= NLANES-1)" on page 1638	0x2(N*2+1)cc	Status of RX Init Power-Up End rx_reset Toggle.
"RAWLANEN_DIG_FSM_SKIP_RATE_CHANGE_SETTING(for N = 0; N <= NLANES-1)" on page 1639	0x2(N*2+1)cd	Status of Skip Rate Change Setting.
"RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_STARTUP (for N = 0; N <= NLANES-1)" on page 1640	0x2(N*2+1)ce	Status of Skip DFE FLT Adapt Start Up.
"RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_CONT(for N = 0; N <= NLANES-1)" on page 1641	0x2(N*2+1)cf	Status of Skip DFE FLT Adapt Continuous.
"RAWLANEN_DIG_FSM_FW_SCRATCH_16 (for N = 0; N <= NLANES-1)" on page 1642	0x2(N*2+1)d0	Firmware Scratch Register 16.
"RAWLANEN_DIG_FSM_FW_SCRATCH_17 (for N = 0; N <= NLANES-1)" on page 1643	0x2(N*2+1)d1	Firmware Scratch Register 17.
"RAWLANEN_DIG_FSM_FW_SCRATCH_18 (for N = 0; N <= NLANES-1)" on page 1644	0x2(N*2+1)d2	Firmware Scratch Register 18.
"RAWLANEN_DIG_FSM_FW_SCRATCH_19 (for N = 0; N <= NLANES-1)" on page 1645	0x2(N*2+1)d3	Firmware Scratch Register 19.
"RAWLANEN_DIG_FSM_FW_SCRATCH_20 (for N = 0; N <= NLANES-1)" on page 1646	0x2(N*2+1)d4	Firmware Scratch Register 20.
"RAWLANEN_DIG_FSM_FW_SCRATCH_21 (for N = 0; N <= NLANES-1)" on page 1647	0x2(N*2+1)d5	Firmware Scratch Register 21.
"RAWLANEN_DIG_FSM_FW_SCRATCH_22 (for N = 0; N <= NLANES-1)" on page 1648	0x2(N*2+1)d6	Firmware Scratch Register 22.
"RAWLANEN_DIG_FSM_FW_SCRATCH_23 (for N = 0; N <= NLANES-1)" on page 1649	0x2(N*2+1)d7	Firmware Scratch Register 23.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_AFE_ATT_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1650	0x3(N*2+0)00	Offset Value for RX AFE ATT iDAC.
"RAWLANEAONN_DIG_AFE_CTLE_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1651	0x3(N*2+0)01	Offset Value for RX AFE CTLE iDAC.
"RAWLANEAONN_DIG_AFE_VGA1_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1652	0x3(N*2+0)02	Offset Values for RX AFE VGA1 iDAC.
"RAWLANEAONN_DIG_DFE_PHASE_EVEN_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1653	0x3(N*2+0)03	Offset Values for RX DFE Phase Even vDAC.
"RAWLANEAONN_DIG_DFE_PHASE_ODD_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1654	0x3(N*2+0)04	Offset Values for RX DFE Phase Odd vDAC.
"RAWLANEAONN_DIG_DFE_PHASE_EVEN_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1655	0x3(N*2+0)05	Offset Values for RX DFE Phase Even Low vDAC.
"RAWLANEAONN_DIG_DFE_PHASE_ODD_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1656	0x3(N*2+0)06	Offset Values for RX DFE Phase Odd Low vDAC.
"RAWLANEAONN_DIG_AFE_BUF_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1657	0x3(N*2+0)07	Offset Value for RX AFE Buffer iDAC.
"RAWLANEAONN_DIG_DFE_DATA0_EVEN_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1658	0x3(N*2+0)09	Offset Values for RX DFE Data0 Even High vDAC.
"RAWLANEAONN_DIG_DFE_DATA0_EVEN_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1659	0x3(N*2+0)0a	Offset Values for RX DFE Data0 Even Low vDAC.
"RAWLANEAONN_DIG_DFE_DATA0_ODD_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1660	0x3(N*2+0)0b	Offset Values for RX DFE Data0 Odd High vDAC.
"RAWLANEAONN_DIG_DFE_DATA0_ODD_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1661	0x3(N*2+0)0c	Offset Values for RX DFE Data0 Odd Low vDAC.
"RAWLANEAONN_DIG_DFE_BYPASS_EVEN_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1662	0x3(N*2+0)0d	Offset Values for RX DFE By-Pass Even vDAC.
"RAWLANEAONN_DIG_DFE_BYPASS_ODD_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1663	0x3(N*2+0)0e	Offset Values for RX DFE By-Pass Odd vDAC.
"RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL(for N = 0; N <= NLANES-1)" on page 1664	0x3(N*2+0)12	Set to Skip Firmware Start-up Algorithms.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_CONT_ALGO_CTL(for N = 0; N <= NLANES-1)" on page 1667	0x3(N*2+0)13	Set to Skip Firmware Mission-mode Algorithms.
"RAWLANEAONN_DIG_INIT_PWRUP_DONE (for N = 0; N <= NLANES-1)" on page 1670	0x3(N*2+0)14	Initial Power-Up Done Status.
"RAWLANEAONN_DIG_RX_ADPT_ATT (for N = 0; N <= NLANES-1)" on page 1671	0x3(N*2+0)15	RX Adapted Value of ATT.
"RAWLANEAONN_DIG_RX_ADPT_VGA (for N = 0; N <= NLANES-1)" on page 1672	0x3(N*2+0)16	RX Adapted Value of VGA.
"RAWLANEAONN_DIG_RX_ADPT_CTLE (for N = 0; N <= NLANES-1)" on page 1673	0x3(N*2+0)17	RX Adapted Value of CTLE.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1(for N = 0; N <= NLANES-1)" on page 1674	0x3(N*2+0)18	RX Adapted Value of DFE TAP1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2(for N = 0; N <= NLANES-1)" on page 1675	0x3(N*2+0)19	RX Adapted Value of DFE TAP2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3(for N = 0; N <= NLANES-1)" on page 1676	0x3(N*2+0)1a	RX Adapted Value of DFE TAP3.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4(for N = 0; N <= NLANES-1)" on page 1677	0x3(N*2+0)1b	RX Adapted Value of DFE TAP4.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5(for N = 0; N <= NLANES-1)" on page 1678	0x3(N*2+0)1c	RX Adapted Value of DFE TAP5.
"RAWLANEAONN_DIG_RX_ADPT_IQ(for N = 0; N <= NLANES-1)" on page 1679	0x3(N*2+0)1d	RX Adapted Value of IQ for Bank 0.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN(for N = 0; N <= NLANES-1)" on page 1680	0x3(N*2+0)1e	RX Adapted Value of Reference Level for Error Even Slicer for Bank 0.
"RAWLANEAONN_DIG_RX_ADAPT_DONE(for N = 0; N <= NLANES-1)" on page 1681	0x3(N*2+0)1f	RX Adaptation Done Status.
"RAWLANEAONN_DIG_RX_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)" on page 1682	0x3(N*2+0)20	RX Adapted Value of ATT for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_VGA_B1 (for N = 0; N <= NLANES-1)" on page 1683	0x3(N*2+0)21	RX Adapted Value of VGA for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_CTLE_B1 (for N = 0; N <= NLANES-1)" on page 1684	0x3(N*2+0)22	RX Adapted Value of CTLE for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B1(for N = 0; N <= NLANES-1)" on page 1685	0x3(N*2+0)23	RX Adapted Value of DFE TAP1 for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B1(for N = 0; N <= NLANES-1)" on page 1686	0x3(N*2+0)24	RX Adapted Value of DFE TAP2 for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B1(for N = 0; N <= NLANES-1)" on page 1687	0x3(N*2+0)25	RX Adapted value of DFE TAP3 for bank 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B1(for N = 0; N <= NLANES-1)" on page 1688	0x3(N*2+0)26	RX Adapted Value of DFE TAP4 for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B1(for N = 0; N <= NLANES-1)" on page 1689	0x3(N*2+0)27	RX Adapted Value of DFE TAP5 for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_IQ_B1(for N = 0; N <= NLANES-1)" on page 1690	0x3(N*2+0)28	RX Adapted Value of IQ for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B1(for N = 0; N <= NLANES-1)" on page 1691	0x3(N*2+0)29	RX Adapted Value of Reference Level for Error Even Slicer for Bank 1.
"RAWLANEAONN_DIG_RX_ADAPT_DONE_B1(for N = 0; N <= NLANES-1)" on page 1692	0x3(N*2+0)2a	RX Adaptation Done Status for Bank 1.
"RAWLANEAONN_DIG_FAST_FLAGS (for N = 0; N <= NLANES-1)" on page 1693	0x3(N*2+0)2b	Fast Flags for Simulation Only.
"RAWLANEAONN_DIG_FW_STATES_0 (for N = 0; N <= NLANES-1)" on page 1696	0x3(N*2+0)2c	Stores Various States for Firmware - State 1.
"RAWLANEAONN_DIG_FW_STATES_1 (for N = 0; N <= NLANES-1)" on page 1698	0x3(N*2+0)2d	Stores Various States for Firmware - State 2.
"RAWLANEAONN_DIG_TXRX_OVRD_IN(for N = 0; N <= NLANES-1)" on page 1699	0x3(N*2+0)2e	Override Values for Incoming AON TX/RX Controls from PCS.
"RAWLANEAONN_DIG_TX_EQ_DIR_POLARITY_CTL(for N = 0; N <= NLANES-1)" on page 1700	0x3(N*2+0)2f	Controls Equalization Direction Polarity Setting in PCS RAW.
"RAWLANEAONN_DIG_TX_PRE_DIV(for N = 0; N <= NLANES-1)" on page 1701	0x3(N*2+0)30	TX Pre-Threshold Div.
"RAWLANEAONN_DIG_TX_MAIN_ATT_THRESHOLD(for N = 0; N <= NLANES-1)" on page 1702	0x3(N*2+0)31	TX Main ATT High and Low Threshold.
"RAWLANEAONN_DIG_TX_MAIN_VGA_THRESHOLD(for N = 0; N <= NLANES-1)" on page 1703	0x3(N*2+0)32	TX Main VGA High and Low Threshold.
"RAWLANEAONN_DIG_TX_POST_BOOST_THRESHOLD(for N = 0; N <= NLANES-1)" on page 1704	0x3(N*2+0)33	TX Post Boost High and Low Threshold.
"RAWLANEAONN_DIG_TX_POST_TAP1_LOW_THRESHOLD(for N = 0; N <= NLANES-1)" on page 1705	0x3(N*2+0)34	TX Post TAP1 Low Threshold.
"RAWLANEAONN_DIG_RX_CAL_REF_EE_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1706	0x3(N*2+0)35	Value for Reference Level Error Even Slicer.
"RAWLANEAONN_DIG_RX_CAL_REF_EO_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1707	0x3(N*2+0)36	Value for Reference Level Error Odd Slicer.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_ERR_SLC_MODE(for N = 0; N <= NLANES-1)" on page 1708	0x3(N*2+0)39	RX Adaptation Error Slicer Mode for Reference-Level Calibration.
"RAWLANEAONN_DIG_ADPT_CTL_0(for N = 0; N <= NLANES-1)" on page 1709	0x3(N*2+0)40	Adaptation Control Register 0.
"RAWLANEAONN_DIG_ADPT_CTL_1(for N = 0; N <= NLANES-1)" on page 1710	0x3(N*2+0)41	Adaptation Control Register 1.
"RAWLANEAONN_DIG_ADPT_CTL_2(for N = 0; N <= NLANES-1)" on page 1711	0x3(N*2+0)42	Adaptation Control Register 2.
"RAWLANEAONN_DIG_ADPT_CTL_3(for N = 0; N <= NLANES-1)" on page 1712	0x3(N*2+0)43	Adaptation Control Register 3.
"RAWLANEAONN_DIG_ADPT_CTL_4(for N = 0; N <= NLANES-1)" on page 1713	0x3(N*2+0)44	Adaptation Control Register 4.
"RAWLANEAONN_DIG_ADPT_CTL_5(for N = 0; N <= NLANES-1)" on page 1714	0x3(N*2+0)45	Adaptation Control Register 5.
"RAWLANEAONN_DIG_ADPT_CTL_6(for N = 0; N <= NLANES-1)" on page 1715	0x3(N*2+0)46	Adaptation Control Register 6.
"RAWLANEAONN_DIG_ADPT_CTL_7(for N = 0; N <= NLANES-1)" on page 1716	0x3(N*2+0)47	Adaptation Control Register 7.
"RAWLANEAONN_DIG_ADPT_CTL_8(for N = 0; N <= NLANES-1)" on page 1717	0x3(N*2+0)48	Adaptation Control Register 8.
"RAWLANEAONN_DIG_ADPT_CTL_9(for N = 0; N <= NLANES-1)" on page 1718	0x3(N*2+0)49	Adaptation Control Register 9.
"RAWLANEAONN_DIG_ADPT_CTL_10(for N = 0; N <= NLANES-1)" on page 1719	0x3(N*2+0)4a	Adaptation Control Register 10.
"RAWLANEAONN_DIG_ADPT_CTL_11(for N = 0; N <= NLANES-1)" on page 1720	0x3(N*2+0)4b	Adaptation Control Register 11.
"RAWLANEAONN_DIG_ADPT_CTL_12(for N = 0; N <= NLANES-1)" on page 1721	0x3(N*2+0)4c	Adaptation Control Register 12.
"RAWLANEAONN_DIG_ADPT_CTL_13(for N = 0; N <= NLANES-1)" on page 1722	0x3(N*2+0)4d	Adaptation Control Register 13.
"RAWLANEAONN_DIG_ADPT_CTL_14(for N = 0; N <= NLANES-1)" on page 1723	0x3(N*2+0)4e	Adaptation Control Register 14.
"RAWLANEAONN_DIG_ADPT_CTL_15(for N = 0; N <= NLANES-1)" on page 1724	0x3(N*2+0)4f	Adaptation Control Register 15.
"RAWLANEAONN_DIG_ADPT_CTL_16(for N = 0; N <= NLANES-1)" on page 1725	0x3(N*2+0)50	Adaptation Control Register 16.
"RAWLANEAONN_DIG_ADPT_CTL_17(for N = 0; N <= NLANES-1)" on page 1726	0x3(N*2+0)51	Adaptation Control Register 17.
"RAWLANEAONN_DIG_ADPT_CTL_18(for N = 0; N <= NLANES-1)" on page 1727	0x3(N*2+0)52	Adaptation Control Register 18.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_ADPT_CTL_19(for N = 0; N <= NLANES-1)" on page 1728	0x3(N*2+0)53	Adaptation Control Register 19.
"RAWLANEAONN_DIG_ADPT_CTL_20(for N = 0; N <= NLANES-1)" on page 1729	0x3(N*2+0)54	Adaptation Control Register 20.
"RAWLANEAONN_DIG_ADPT_CTL_21(for N = 0; N <= NLANES-1)" on page 1730	0x3(N*2+0)55	Adaptation ControlRregister 21.
"RAWLANEAONN_DIG_ADPT_CTL_22(for N = 0; N <= NLANES-1)" on page 1731	0x3(N*2+0)56	Adaptation Control Register 22.
"RAWLANEAONN_DIG_ADPT_CTL_23(for N = 0; N <= NLANES-1)" on page 1732	0x3(N*2+0)57	Adaptation Control Register 23.
"RAWLANEAONN_DIG_SRAM_REC_EN(for N = 0; N <= NLANES-1)" on page 1733	0x3(N*2+0)58	SRAM Record Enable Status.
"RAWLANEAONN_DIG_SRAM_REC_CTRL(for N = 0; N <= NLANES-1)" on page 1734	0x3(N*2+0)59	SRAM Record Control.
"RAWLANEAONN_DIG_SRAM_REC_ADDR(for N = 0; N <= NLANES-1)" on page 1735	0x3(N*2+0)5a	Current SRAM Recording Address.
"RAWLANEAONN_DIG_SRAM_REC_ITER(for N = 0; N <= NLANES-1)" on page 1736	0x3(N*2+0)5b	Current Iteration Count for SRAM Recording.
"RAWLANEAONN_DIG_IQ_MARGIN_RANGE (for N = 0; N <= NLANES-1)" on page 1737	0x3(N*2+0)5c	IQ Calibration Margin Range.
"RAWLANEAONN_DIG_RX_CAL_IQ_MAX(for N = 0; N <= NLANES-1)" on page 1738	0x3(N*2+0)5d	IQ Calibration Max Limit.
"RAWLANEAONN_DIG_RX_CAL_IQ_MIN(for N = 0; N <= NLANES-1)" on page 1739	0x3(N*2+0)5e	IQ Calibration Min Limit.
"RAWLANEAONN_DIG_RX_CAL_IQ_RESET(for N = 0; N <= NLANES-1)" on page 1740	0x3(N*2+0)5f	IQ Calibration Reset Code.
"RAWLANEAONN_DIG_RX_CAL_IQ_ADJUS T(for N = 0; N <= NLANES-1)" on page 1741	0x3(N*2+0)60	IQ Calibration Adjust Values.
"RAWLANEAONN_DIG_RX_SIGDET_CAL(for N = 0; N <= NLANES-1)" on page 1742	0x3(N*2+0)61	SigDet Calibration Values.
"RAWLANEAONN_DIG_RX_STARTUP_ALG O_CTL_1(for N = 0; N <= NLANES-1)" on page 1743	0x3(N*2+0)62	Set to Skip Firmware Start-up/Rate-change Algorithms.
"RAWLANEAONN_DIG_CCA_START_LOOP _CNT(for N = 0; N <= NLANES-1)" on page 1746	0x3(N*2+0)63	Number of Back-to-Back Iterations of CCA to Run when Starting CCA.
"RAWLANEAONN_DIG_CCA_WAIT_CNT(for N = 0; N <= NLANES-1)" on page 1747	0x3(N*2+0)64	Wait Time Between CCA Iterations.
"RAWLANEAONN_DIG_VGEN_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1748	0x3(N*2+0)65	VGEN VDAC Offset Register.
"RAWLANEAONN_DIG_RX_VDAC_RANGE_ SEL(for N = 0; N <= NLANES-1)" on page 1749	0x3(N*2+0)66	DAC Calibration Extension Mode.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL(for N = 0; N <= NLANES-1)" on page 1750	0x3(N*2+0)67	RX CDR Detector Control.
"RAWLANEAONN_DIG_RX_CDR_RECOVERY_TIME(for N = 0; N <= NLANES-1)" on page 1752	0x3(N*2+0)68	RX CDR Recovery Time in Reference Clock for Intended PPM.
"RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_0(for N = 0; N <= NLANES-1)" on page 1753	0x3(N*2+0)69	TX DCC Full-Rate Code Value Bank 0.
"RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_0(for N = 0; N <= NLANES-1)" on page 1754	0x3(N*2+0)6a	TX DCC Half-Rate CM Code Value for Bank 0.
"RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_1(for N = 0; N <= NLANES-1)" on page 1755	0x3(N*2+0)6b	TX DCC Full-Rate CM Code Value for Bank 1.
"RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_1(for N = 0; N <= NLANES-1)" on page 1756	0x3(N*2+0)6c	TX DCC Half-Rate CM Code Value for Bank 1.
"RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_2(for N = 0; N <= NLANES-1)" on page 1757	0x3(N*2+0)6d	TX DCC Full-Rate CM Code Value for Bank 2.
"RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_2(for N = 0; N <= NLANES-1)" on page 1758	0x3(N*2+0)6e	TX DCC Half-Rate CM Code Value for Bank 2.
"RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_3(for N = 0; N <= NLANES-1)" on page 1759	0x3(N*2+0)6f	TX DCC Full-Rate CM Code Value for Bank 3.
"RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_3(for N = 0; N <= NLANES-1)" on page 1760	0x3(N*2+0)70	TX DCC Half-Rate CM Code Value for Bank 3.
"RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_0(for N = 0; N <= NLANES-1)" on page 1761	0x3(N*2+0)71	TX DCC Full-Rate Differential Code Value for Bank 0.
"RAWLANEAONN_DIG_TX_DCC_HALF_DIF F_BANK_0(for N = 0; N <= NLANES-1)" on page 1762	0x3(N*2+0)72	TX DCC Half-Rate Differential Code Value for Bank 0.
"RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_1(for N = 0; N <= NLANES-1)" on page 1763	0x3(N*2+0)73	TX DCC Full-Rate Differential Code Value for Bank 1.
"RAWLANEAONN_DIG_TX_DCC_HALF_DIF F_BANK_1(for N = 0; N <= NLANES-1)" on page 1764	0x3(N*2+0)74	TX DCC Half-Rate Differential Code Value for Bank 1.
"RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_2(for N = 0; N <= NLANES-1)" on page 1765	0x3(N*2+0)75	TX DCC Full-Rate Differential Code Value for Bank 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_2(for N = 0; N <= NLANES-1)" on page 1766	0x3(N*2+0)76	TX DCC Half-Rate Differential Code Value for Bank 2.
"RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_3(for N = 0; N <= NLANES-1)" on page 1767	0x3(N*2+0)77	TX DCC Full-Rate Differential Code Value for Bank 3.
"RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_3(for N = 0; N <= NLANES-1)" on page 1768	0x3(N*2+0)78	TX DCC Half-Rate Differential Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_FULL_BANK_0(for N = 0; N <= NLANES-1)" on page 1769	0x3(N*2+0)79	RX DCC Full-Rate Data/Bypass Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_0(for N = 0; N <= NLANES-1)" on page 1770	0x3(N*2+0)7a	RX DCC Full-Rate Phase Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_HALF_BANK_0(for N = 0; N <= NLANES-1)" on page 1771	0x3(N*2+0)7b	RX DCC Half-Rate Data/Bypass Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_0(for N = 0; N <= NLANES-1)" on page 1772	0x3(N*2+0)7c	RX DCC Half-Rate Phase Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_IQ_CAL_BANK_0(for N = 0; N <= NLANES-1)" on page 1773	0x3(N*2+0)7d	RX IQ Calibration Codes Value for Bank 0.
"RAWLANEAONN_DIG_RX_CAL_DONE_BANK_0(for N = 0; N <= NLANES-1)" on page 1774	0x3(N*2+0)7e	RX Calibration Done Flag for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_FULL_BANK_1(for N = 0; N <= NLANES-1)" on page 1775	0x3(N*2+0)7f	RX DCC Full-Rate Data/Bypass Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_1(for N = 0; N <= NLANES-1)" on page 1776	0x3(N*2+0)80	RX DCC Full-Rate Phase Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_DCC_HALF_BANK_1(for N = 0; N <= NLANES-1)" on page 1777	0x3(N*2+0)81	RX DCC Half-Rate Data/Bypass Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_1(for N = 0; N <= NLANES-1)" on page 1778	0x3(N*2+0)82	RX DCC Half-Rate Phase Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_IQ_CAL_BANK_1(for N = 0; N <= NLANES-1)" on page 1779	0x3(N*2+0)83	RX IQ Calibration Codes Value for Bank 1.
"RAWLANEAONN_DIG_RX_CAL_DONE_BANK_1(for N = 0; N <= NLANES-1)" on page 1780	0x3(N*2+0)84	RX Calibration Done Flag for Bank 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_DCC_FULL_BANK_2(for N = 0; N <= NLANES-1)" on page 1781	0x3(N*2+0)85	RX DCC Full-Rate Data/Bypass Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_2(for N = 0; N <= NLANES-1)" on page 1782	0x3(N*2+0)86	RX DCC Full-Rate Phase Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_HALF_BANK_2(for N = 0; N <= NLANES-1)" on page 1783	0x3(N*2+0)87	RX DCC Half-Rate Data/Bypass Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_2(for N = 0; N <= NLANES-1)" on page 1784	0x3(N*2+0)88	RX DCC Half-Rate Phase Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_IQ_CAL_BANK_2(for N = 0; N <= NLANES-1)" on page 1785	0x3(N*2+0)89	RX IQ Calibration Codes Value for Bank 2.
"RAWLANEAONN_DIG_RX_CAL_DONE_BANK_2(for N = 0; N <= NLANES-1)" on page 1786	0x3(N*2+0)8a	RX Calibration Done Flag for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_FULL_BANK_3(for N = 0; N <= NLANES-1)" on page 1787	0x3(N*2+0)8b	RX DCC Full-Rate Data/Bypass Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_3(for N = 0; N <= NLANES-1)" on page 1788	0x3(N*2+0)8c	RX DCC Full-Rate Phase Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_HALF_BANK_3(for N = 0; N <= NLANES-1)" on page 1789	0x3(N*2+0)8d	RX DCC Half-Rate Data/Bypass Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_3(for N = 0; N <= NLANES-1)" on page 1790	0x3(N*2+0)8e	RX DCC Half-Rate Phase Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_IQ_CAL_BANK_3(for N = 0; N <= NLANES-1)" on page 1791	0x3(N*2+0)8f	RX IQ Calibration Codes Value for Bank 3.
"RAWLANEAONN_DIG_RX_CAL_DONE_BANK_3(for N = 0; N <= NLANES-1)" on page 1792	0x3(N*2+0)90	RX Calibration Done Flag for Bank 3.
"RAWLANEAONN_DIG_RX_CAL_BANK_SEL(for N = 0; N <= NLANES-1)" on page 1793	0x3(N*2+0)92	Value for RX Bank Select.
"RAWLANEAONN_DIG_TX_DCC_CM_CODE(for N = 0; N <= NLANES-1)" on page 1794	0x3(N*2+0)93	Selected Values for TX DCC Codes.
"RAWLANEAONN_DIG_RX_DCC_CODE(for N = 0; N <= NLANES-1)" on page 1795	0x3(N*2+0)94	Selected Values for RX DCC Codes.
"RAWLANEAONN_DIG_RX_DCC_PHASE_CODE(for N = 0; N <= NLANES-1)" on page 1796	0x3(N*2+0)95	Selected Values for RX DCC Phase Codes.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_IQ_CAL(for N = 0; N <= NLANES-1)" on page 1797	0x3(N*2+0)96	Selected Value for RX IQ Calibration Code.
"RAWLANEAONN_DIG_RX_CAL_DONE(for N = 0; N <= NLANES-1)" on page 1798	0x3(N*2+0)97	Selected RX Calibration Bank Valid Calibrated Value.
"RAWLANEAONN_DIG_TX_CAL_DONE_BANK_0(for N = 0; N <= NLANES-1)" on page 1799	0x3(N*2+0)98	TX DCC Calibration Done Flag for Bank 0.
"RAWLANEAONN_DIG_TX_CAL_DONE_BANK_1(for N = 0; N <= NLANES-1)" on page 1800	0x3(N*2+0)99	TX DCC Calibration Done Flag for Bank 1.
"RAWLANEAONN_DIG_TX_CAL_DONE_BANK_2(for N = 0; N <= NLANES-1)" on page 1801	0x3(N*2+0)9a	TX DCC Calibration Done Flag for Bank 2.
"RAWLANEAONN_DIG_TX_CAL_DONE_BANK_3(for N = 0; N <= NLANES-1)" on page 1802	0x3(N*2+0)9b	TX DCC Calibration Done Flag for Bank 3.
"RAWLANEAONN_DIG_TX_CAL_DONE(for N = 0; N <= NLANES-1)" on page 1803	0x3(N*2+0)a0	Selected TX DCC Calibration Bank Valid Calibrated Value.
"RAWLANEAONN_DIG_RX_IQ_CAL_DIVN(for N = 0; N <= NLANES-1)" on page 1804	0x3(N*2+0)a2	Value for RX IQ Calibrated Divide-by N Value.
"RAWLANEAONN_DIG_MEM_BREAKPOINT_2(for N = 0; N <= NLANES-1)" on page 1805	0x3(N*2+0)a3	FSM Breakpoint 2 on SRAM Address for Debugging.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST(for N = 0; N <= NLANES-1)" on page 1806	0x3(N*2+0)a4	Offset Values for RX DFE Data Even High TAP1.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST(for N = 0; N <= NLANES-1)" on page 1807	0x3(N*2+0)a5	Offset Values for RX DFE Data Even Low TAP1.
"RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST(for N = 0; N <= NLANES-1)" on page 1808	0x3(N*2+0)a6	Offset Values for RX DFE Data Odd High TAP1.
"RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD(for N = 0; N <= NLANES-1)" on page 1809	0x3(N*2+0)a7	Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6(for N = 0; N <= NLANES-1)" on page 1810	0x3(N*2+0)a8	RX Adapted value of DFE TAP6.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7(for N = 0; N <= NLANES-1)" on page 1811	0x3(N*2+0)a9	RX Adapted Value of DFE TAP7.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8(for N = 0; N <= NLANES-1)" on page 1812	0x3(N*2+0)aa	RX Adapted Value of DFE TAP8.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B1(for N = 0; N <= NLANES-1)" on page 1813	0x3(N*2+0)ab	RX Adapted value of DFE TAP6 for Bank 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B1(for N = 0; N <= NLANES-1)" on page 1814	0x3(N*2+0)ac	RX Adapted Value of DFE TAP7 for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B1(for N = 0; N <= NLANES-1)" on page 1815	0x3(N*2+0)ad	RX Adapted Value of DFE TAP8 for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS(for N = 0; N <= NLANES-1)" on page 1816	0x3(N*2+0)ae	Positions for RX DFE Floating TAPs 0-3.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP0 (for N = 0; N <= NLANES-1)" on page 1817	0x3(N*2+0)af	Adapted Value for RX DFE Floating TAP1 and Control.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP1 (for N = 0; N <= NLANES-1)" on page 1818	0x3(N*2+0)b0	Adapted Value for RX DFE Floating TAP2 and Control.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP2 (for N = 0; N <= NLANES-1)" on page 1819	0x3(N*2+0)b1	Adapted Value for RX DFE Floating TAP3 and Control.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP3 (for N = 0; N <= NLANES-1)" on page 1820	0x3(N*2+0)b2	Adapted Value for RX DFE Floating TAP4 and Control.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B1(for N = 0; N <= NLANES-1)" on page 1821	0x3(N*2+0)b4	Positions for RX DFE Floating TAPs 0-3 for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B1(for N = 0; N <= NLANES-1)" on page 1822	0x3(N*2+0)b5	Adapted Value for RX DFE Floating TAP1 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B1(for N = 0; N <= NLANES-1)" on page 1823	0x3(N*2+0)b6	Adapted Value for RX DFE Floating TAP2 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B1(for N = 0; N <= NLANES-1)" on page 1824	0x3(N*2+0)b7	Adapted Value for RX DFE Floating TAP3 and its Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B1(for N = 0; N <= NLANES-1)" on page 1825	0x3(N*2+0)b8	Adapted Value for RX DFE Floating TAP4 and Control for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)" on page 1826	0x3(N*2+0)ba	RX Adapted Value of ATT for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_VGA_B2(for N = 0; N <= NLANES-1)" on page 1827	0x3(N*2+0)bb	RX Adapted Value of VGA for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_CTLE_B2 (for N = 0; N <= NLANES-1)" on page 1828	0x3(N*2+0)bc	RX Adapted Value of CTLE for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B2(for N = 0; N <= NLANES-1)" on page 1829	0x3(N*2+0)bd	RX Adapted Value of DFE TAP1 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B2(for N = 0; N <= NLANES-1)" on page 1830	0x3(N*2+0)be	RX Adapted Value of DFE TAP2 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B2(for N = 0; N <= NLANES-1)" on page 1831	0x3(N*2+0)bf	RX Adapted Value of DFE TAP3 for Bank 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B2(for N = 0; N <= NLANES-1)" on page 1832	0x3(N*2+0)c0	RX Adapted Value of DFE TAP4 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B2(for N = 0; N <= NLANES-1)" on page 1833	0x3(N*2+0)c1	RX Adapted Value of DFE TAP5 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_IQ_B2(for N = 0; N <= NLANES-1)" on page 1834	0x3(N*2+0)c2	RX Adapted Value of IQ for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B2(for N = 0; N <= NLANES-1)" on page 1835	0x3(N*2+0)c3	RX Adapted Value of Reference Level for Error Even Slicer for Bank 2.
"RAWLANEAONN_DIG_RX_ADAPT_DONE_B2(for N = 0; N <= NLANES-1)" on page 1836	0x3(N*2+0)c4	RX Adaptation Done Status for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B2(for N = 0; N <= NLANES-1)" on page 1837	0x3(N*2+0)c5	RX Adapted Value of DFE TAP6 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B2(for N = 0; N <= NLANES-1)" on page 1838	0x3(N*2+0)c6	RX Adapted Value of DFE TAP7 for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B2(for N = 0; N <= NLANES-1)" on page 1839	0x3(N*2+0)c7	RX Adapted Value of DFE TAP8 for Bank 2.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B2(for N = 0; N <= NLANES-1)" on page 1840	0x3(N*2+0)c8	Positions for RX DFE Floating TAPs 0-3 for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B2(for N = 0; N <= NLANES-1)" on page 1841	0x3(N*2+0)c9	Adapted Value for RX DFE Floating TAP1 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B2(for N = 0; N <= NLANES-1)" on page 1842	0x3(N*2+0)ca	Adapted Value for RX DFE Floating TAP2 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B2(for N = 0; N <= NLANES-1)" on page 1843	0x3(N*2+0)cb	Adapted Value for RX DFE Floating TAP3 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B2(for N = 0; N <= NLANES-1)" on page 1844	0x3(N*2+0)cc	Adapted Value for RX DFE Floating TAP4 and Control for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1845	0x3(N*2+0)ce	Offset values for RX DFE Data1 Even High vDAC.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1846	0x3(N*2+0)cf	Offset Values for RX DFE Data1 Even Low vDAC.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1847	0x3(N*2+0)d0	Offset Values for RX DFE Data1 Odd High vDAC.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)" on page 1848	0x3(N*2+0)d1	Offset values for RX DFE Data1 Odd Low vDAC.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1849	0x3(N*2+0)d2	Offset Values for RX DFE Data Even High TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1850	0x3(N*2+0)d3	Offset values for RX DFE Data Even Low TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1851	0x3(N*2+0)d4	Offset Values for RX DFE Data Odd High TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)" on page 1852	0x3(N*2+0)d5	Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)" on page 1853	0x3(N*2+0)d6	Offset Values for RX DFE Data Even High TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)" on page 1854	0x3(N*2+0)d7	Offset Values for RX DFE Data Even Low TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)" on page 1855	0x3(N*2+0)d8	Offset Values for RX DFE Data Odd High TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B2 (for N = 0; N <= NLANES-1)" on page 1856	0x3(N*2+0)d9	Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid for Bank 2.
"RAWLANEAONN_DIG_AFE_VGA2_IDAC_OFST(for N = 0; N <= NLANES-1)" on page 1857	0x3(N*2+0)da	Offset Values for RX AFE VGA2 iDAC.
"RAWLANEAONN_DIG_ADPT_CTL_24 (for N = 0; N <= NLANES-1)" on page 1858	0x3(N*2+0)db	Adaptation Control Register 24.
"RAWLANEAONN_DIG_ADPT_CTL_25 (for N = 0; N <= NLANES-1)" on page 1859	0x3(N*2+0)dc	Adaptation Control Register 25.
"RAWLANEAONN_DIG_ADPT_CTL_26 (for N = 0; N <= NLANES-1)" on page 1860	0x3(N*2+0)dd	Adaptation Control Register 26.
"RAWLANEAONN_DIG_ADPT_CTL_27 (for N = 0; N <= NLANES-1)" on page 1861	0x3(N*2+0)de	Adaptation Control Register 27.
"RAWLANEAONN_DIG_ADPT_CTL_28 (for N = 0; N <= NLANES-1)" on page 1862	0x3(N*2+0)df	Adaptation Control Register 28.
"RAWLANEAONN_DIG_RX_IQ_CTL (for N = 0; N <= NLANES-1)" on page 1863	0x3(N*2+0)e0	RX IQ Adaptation Control Register 1.
"RAWLANEAONN_DIG_RX_IQ_CTL_1 (for N = 0; N <= NLANES-1)" on page 1864	0x3(N*2+0)e1	RX IQ Adaptation Control Register 2.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_IQ_LIMIT(for N = 0; N <= NLANES-1)" on page 1865	0x3(N*2+ 0)e2	RX IQ Adaptation Offset Limit.
"RAWLANEAONN_DIG_TX_DCC_HALF_DIF F_CODE(for N = 0; N <= NLANES-1)" on page 1866	0x3(N*2+ 0)e3	Selected Values for TX DCC Half Differential Codes.
"RAWLANEAONN_DIG_TX_DCC_HALF_CM_ CODE(for N = 0; N <= NLANES-1)" on page 1867	0x3(N*2+ 0)e4	Selected Values for TX DCC HALF Common-Mode Codes.
"RAWLANEAONN_DIG_RX_DCC_DIFF_COD E(for N = 0; N <= NLANES-1)" on page 1868	0x3(N*2+ 0)e5	Selected Values for RX DCC Differential Codes.
"RAWLANEAONN_DIG_RX_DCC_CM_CODE (for N = 0; N <= NLANES-1)" on page 1869	0x3(N*2+ 0)e6	Selected Values for RX DCC Common-Mode Codes.
"RAWLANEAONN_DIG_RX_DCC_PHASE_DI FF_CODE(for N = 0; N <= NLANES-1)" on page 1870	0x3(N*2+ 0)e7	Selected Values for RX DCC Phase Differential Codes.
"RAWLANEAONN_DIG_RX_DCC_PHASE_C M_CODE(for N = 0; N <= NLANES-1)" on page 1871	0x3(N*2+ 0)e8	Selected Values for RX DCC Phase Common-Mode Codes.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_ HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)" on page 1872	0x3(N*2+ 0)e9	Offset Values for RX DFE Data1 Even High TAP1.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_ LOW_TAP1_OFST (for N = 0; N <= NLANES-1)" on page 1873	0x3(N*2+ 0)ea	Offset Values for RX DFE Data1 Even Low TAP1.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_ HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)" on page 1874	0x3(N*2+ 0)eb	Offset Values for RX DFE Data1 Odd High TAP1.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_L OW_TAP1_OFST_W_VLD (for N = 0; N <= NLANES-1)" on page 1875	0x3(N*2+ 0)ec	Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_ HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1876	0x3(N*2+ 0)ed	Offset Values for RX DFE Data1 Even High TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_ LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1877	0x3(N*2+ 0)ee	Offset Values for RX DFE Data1 Even Low TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_ HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)" on page 1878	0x3(N*2+ 0)ef	Offset Values for RX DFE Data1 Odd High TAP1 for Bank 1.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_L OW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)" on page 1879	0x3(N*2+ 0)f0	Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid for Bank 1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST_B2(for N = 0; N <= NLANES-1)" on page 1880	0x3(N*2+0)f1	Offset Values for RX DFE Data1 Even High TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST_B2(for N = 0; N <= NLANES-1)" on page 1881	0x3(N*2+0)f2	Offset Values for RX DFE Data1 Even Low TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST_B2(for N = 0; N <= NLANES-1)" on page 1882	0x3(N*2+0)f3	Offset Values for RX DFE Data1 Odd High TAP1 for Bank 2.
"RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD_B2(for N = 0; N <= NLANES-1)" on page 1883	0x3(N*2+0)f4	Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_0(for N = 0; N <= NLANES-1)" on page 1884	0x3(N*2+0)f6	RX DCC Full-Rate Bypass Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_0(for N = 0; N <= NLANES-1)" on page 1885	0x3(N*2+0)f7	RX DCC Half-Rate Bypass Code Value for Bank 0.
"RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_1(for N = 0; N <= NLANES-1)" on page 1886	0x3(N*2+0)f8	RX DCC Full-Rate Bypass Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_1(for N = 0; N <= NLANES-1)" on page 1887	0x3(N*2+0)f9	RX DCC Half-Rate Bypass Code Value for Bank 1.
"RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_2(for N = 0; N <= NLANES-1)" on page 1888	0x3(N*2+0)fa	RX DCC Full-Rate Bypass Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_2(for N = 0; N <= NLANES-1)" on page 1889	0x3(N*2+0)fb	RX DCC Half-Rate Bypass Code Value for Bank 2.
"RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_3(for N = 0; N <= NLANES-1)" on page 1890	0x3(N*2+0)fc	RX DCC Full-Rate Bypass Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_3(for N = 0; N <= NLANES-1)" on page 1891	0x3(N*2+0)fd	RX DCC Half-Rate Bypass Code Value for Bank 3.
"RAWLANEAONN_DIG_RX_DCC_BYPASS_DIFF_CODE(for N = 0; N <= NLANES-1)" on page 1892	0x3(N*2+0)fe	Selected Values for RX DCC Bypass Differential Codes.
"RAWLANEAONN_DIG_RX_DCC_BYPASS_CM_CODE(for N = 0; N <= NLANES-1)" on page 1893	0x3(N*2+0)ff	Selected Values for RX DCC Bypass Common-Mode Codes.
"RAWLANEAONN_DIG_RX_DCC_BYPASS_CODE(for N = 0; N <= NLANES-1)" on page 1894	0x3(N*2+1)00	Selected Values for RX DCC Bypass Codes.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_TX_CAL_BANK_SEL (for N = 0; N <= NLANES-1)" on page 1895	0x3(N*2+ 1)01	Value for TX Bank Select.
"RAWLANEAONN_DIG_TX_DCC_DIFF_CODE(for N = 0; N <= NLANES-1)" on page 1896	0x3(N*2+ 1)02	Selected Values for TX DCC Codes.
"RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_CODE(for N = 0; N <= NLANES-1)" on page 1897	0x3(N*2+ 1)03	Selected Values for TX DCC Full Differential Codes.
"RAWLANEAONN_DIG_TX_DCC_FULL_CM_CODE(for N = 0; N <= NLANES-1)" on page 1898	0x3(N*2+ 1)04	Selected Values for TX DCC Full Common-Mode Codes.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD(for N = 0; N <= NLANES-1)" on page 1899	0x3(N*2+ 1)05	RX Adapted Value of Reference Level for Error Odd Slicer for Bank 0.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B1(for N = 0; N <= NLANES-1)" on page 1900	0x3(N*2+ 1)06	RX Adapted Value of Reference Level for Error Odd Slicer for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B2(for N = 0; N <= NLANES-1)" on page 1901	0x3(N*2+ 1)07	RX Adapted Value of Reference Level for Error Odd Slicer for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_0(for N = 0; N <= NLANES-1)" on page 1902	0x3(N*2+ 1)08	RX Coarse Adapt Done Flag for Bank 0.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_1(for N = 0; N <= NLANES-1)" on page 1903	0x3(N*2+ 1)09	RX Coarse Adapt Done Flag for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_2(for N = 0; N <= NLANES-1)" on page 1904	0x3(N*2+ 1)0a	RX Coarse Adapt Done Flag for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE(for N = 0; N <= NLANES-1)" on page 1905	0x3(N*2+ 1)0b	Selected RX Coarse Adapt Bank Valid Sdapt Value.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT (for N = 0; N <= NLANES-1)" on page 1906	0x3(N*2+ 1)0c	RX Coarse Adapted Value of ATT.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA(for N = 0; N <= NLANES-1)" on page 1907	0x3(N*2+ 1)0d	RX Coarse Adapted Value of VGA.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE(for N = 0; N <= NLANES-1)" on page 1908	0x3(N*2+ 1)0e	RX Coarse Adapted Value of CTLE.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1 (for N = 0; N <= NLANES-1)" on page 1909	0x3(N*2+ 1)0f	RX Coarse Adapted Value of DFE TAP1.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2 (for N = 0; N <= NLANES-1)" on page 1910	0x3(N*2+1)10	RX Coarse Adapted Value of DFE TAP2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ(for N = 0; N <= NLANES-1)" on page 1911	0x3(N*2+1)11	RX Coarse Adapted Value of IQ.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)" on page 1912	0x3(N*2+1)12	RX Coarse Adapted Value of ATT for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B1(for N = 0; N <= NLANES-1)" on page 1913	0x3(N*2+1)13	RX Coarse Adapted Value of VGA for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B1(for N = 0; N <= NLANES-1)" on page 1914	0x3(N*2+1)14	RX Coarse Adapted Value of CTLE for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B1 (for N = 0; N <= NLANES-1)" on page 1915	0x3(N*2+1)15	RX Coarse Adapted Value of DFE TAP1 for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B1 (for N = 0; N <= NLANES-1)" on page 1916	0x3(N*2+1)16	RX Coarse Adapted Value of DFE TAP2 for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B1(for N = 0; N <= NLANES-1)" on page 1917	0x3(N*2+1)17	RX Coarse Adapted Value of IQ for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)" on page 1918	0x3(N*2+1)18	RX Coarse Adapted Value of ATT for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B2(for N = 0; N <= NLANES-1)" on page 1919	0x3(N*2+1)19	RX Coarse Adapted Value of VGA for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B2(for N = 0; N <= NLANES-1)" on page 1920	0x3(N*2+1)1a	RX Coarse Adapted Value of CTLE for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B2 (for N = 0; N <= NLANES-1)" on page 1921	0x3(N*2+1)1b	RX Coarse Adapted Value of DFE TAP1 for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B2 (for N = 0; N <= NLANES-1)" on page 1922	0x3(N*2+1)1c	RX Coarse Adapted Value of DFE TAP2 for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B2(for N = 0; N <= NLANES-1)" on page 1923	0x3(N*2+1)1d	RX Coarse Adapted Value of IQ for Bank 2.
"RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_2(for N = 0; N <= NLANES-1)" on page 1924	0x3(N*2+1)1e	Set to Skip Firmware Start-up Coarse Adapt Algorithms.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_0(for N = 0; N <= NLANES-1)" on page 1926	0x3(N*2+1)1f	TX DCC ctl Range Value Bank 0.
"RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_1(for N = 0; N <= NLANES-1)" on page 1927	0x3(N*2+1)20	TX DCC ctl Range Value Bank 1.
"RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_2(for N = 0; N <= NLANES-1)" on page 1928	0x3(N*2+1)21	TX DCC ctl Range Value Bank 2.
"RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_3(for N = 0; N <= NLANES-1)" on page 1929	0x3(N*2+1)22	TX DCC ctl Range Value Bank 3.
"RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_0(for N = 0; N <= NLANES-1)" on page 1930	0x3(N*2+1)23	RX DCC ctl range value Bank 0.
"RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_1(for N = 0; N <= NLANES-1)" on page 1931	0x3(N*2+1)24	RX DCC ctl range value Bank 1.
"RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_2(for N = 0; N <= NLANES-1)" on page 1932	0x3(N*2+1)25	RX DCC ctl range value Bank 2.
"RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_3(for N = 0; N <= NLANES-1)" on page 1933	0x3(N*2+1)26	RX DCC ctl Range Value Bank 3.
"RAWLANEAONN_DIG_SRAM_REC_BASE(for N = 0; N <= NLANES-1)" on page 1934	0x3(N*2+1)27	Current SRAM Recording Base.
"RAWLANEAONN_DIG_CAL_INIT_WAIT(for N = 0; N <= NLANES-1)" on page 1935	0x3(N*2+1)28	Wait Cycles During Initialization.
"RAWLANEAONN_DIG_CAL_WAIT(for N = 0; N <= NLANES-1)" on page 1936	0x3(N*2+1)29	Wait Cycles After DAC Write.
"RAWLANEAONN_DIG_AFE_RTRIM(for N = 0; N <= NLANES-1)" on page 1937	0x3(N*2+1)2a	AFE Resistor Trim Code.
"RAWLANEAONN_DIG_PRE_PTTRN(for N = 0; N <= NLANES-1)" on page 1938	0x3(N*2+1)2b	Precursor Pattern.
"RAWLANEAONN_DIG_ADAPT_ATT_IQ_TH(for N = 0; N <= NLANES-1)" on page 1939	0x3(N*2+1)2c	Adapted IQ Offset Threshold.
"RAWLANEAONN_DIG_TX_MAIN_DFE2_TH(for N = 0; N <= NLANES-1)" on page 1940	0x3(N*2+1)2d	TX Main DFE2 threshold.
"RAWLANEAONN_DIG_SKIP_RX_DFE_FLT_ADAPT(for N = 0; N <= NLANES-1)" on page 1941	0x3(N*2+1)2e	Skip RX DFE FLT Adapt.
"RAWLANEAONN_DIG_VGA_ADAPT_TH(for N = 0; N <= NLANES-1)" on page 1943	0x3(N*2+1)2f	VGA Adapt Threshold.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_TX_POST_TAP1_HIGH_THRESHOLD(for N = 0; N <= NLANES-1)" on page 1944	0x3(N*2+1)30	TX Post TAP1 High Threshold.
"RAWLANEAONN_DIG_RX_CAL_REF_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1945	0x3(N*2+1)31	Reference Level Offset Value.
"RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_MUXED(for N = 0; N <= NLANES-1)" on page 1946	0x3(N*2+1)32	TX DCC CTL Range Multiplexed Value.
"RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_MUXED(for N = 0; N <= NLANES-1)" on page 1947	0x3(N*2+1)33	RX DCC CTL Range Multiplexed Value.
"RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_0(for N = 0; N <= NLANES-1)" on page 1948	0x3(N*2+1)34	Full Rate TX Delay Calibration for Bank 0.
"RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_1(for N = 0; N <= NLANES-1)" on page 1949	0x3(N*2+1)35	Full Rate TX Delay Calibration for Bank 1.
"RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_2(for N = 0; N <= NLANES-1)" on page 1950	0x3(N*2+1)36	Full Rate TX Delay Calibration for Bank 2.
"RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_3(for N = 0; N <= NLANES-1)" on page 1951	0x3(N*2+1)37	Full Rate TX Delay Calibration for Bank 3.
"RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_0(for N = 0; N <= NLANES-1)" on page 1952	0x3(N*2+1)38	Half Rate TX Delay Calibration for Bank 0.
"RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_1(for N = 0; N <= NLANES-1)" on page 1953	0x3(N*2+1)39	Half Rate TX Delay Calibration for Bank 1.
"RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_2(for N = 0; N <= NLANES-1)" on page 1954	0x3(N*2+1)3a	Half Rate TX Delay Calibration for Bank 2.
"RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_3(for N = 0; N <= NLANES-1)" on page 1955	0x3(N*2+1)3b	Half Rate TX Delay Calibration for Bank 3.
"RAWLANEAONN_DIG_TX_DLY_CAL_MUX (for N = 0; N <= NLANES-1)" on page 1956	0x3(N*2+1)3c	TX Delay Calibration Multiplexer Value.
"RAWLANEAONN_DIG_TX_DLY_CAL_GAIN (for N = 0; N <= NLANES-1)" on page 1957	0x3(N*2+1)3d	TX Delay Calibration Gain.
"RAWLANEAONN_DIG_RX_ADPT_AFE_RATE(for N = 0; N <= NLANES-1)" on page 1958	0x3(N*2+1)3e	RX Adapted AFE RATE.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG(for N = 0; N <= NLANES-1)" on page 1959	0x3(N*2+1)3f	RX Adapted AFE CONFIG.
"RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES(for N = 0; N <= NLANES-1)" on page 1960	0x3(N*2+1)40	RX Adapted DFE Enables.
"RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B1(for N = 0; N <= NLANES-1)" on page 1961	0x3(N*2+1)41	RX Adapted AFE RATE for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B1(for N = 0; N <= NLANES-1)" on page 1962	0x3(N*2+1)42	RX Adapted AFE CONFIG for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B1(for N = 0; N <= NLANES-1)" on page 1963	0x3(N*2+1)43	RX Adapted DFE Enables for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B2(for N = 0; N <= NLANES-1)" on page 1964	0x3(N*2+1)44	RX Adapted AFE RATE for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B2(for N = 0; N <= NLANES-1)" on page 1965	0x3(N*2+1)45	RX Adapted AFE CONFIG for Bank 2.
"RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B2(for N = 0; N <= NLANES-1)" on page 1966	0x3(N*2+1)46	RX Adapted DFE Enables for Bank 2.
"RAWLANEAONN_DIG_RX_ADAPT_IQ_CTRL(for N = 0; N <= NLANES-1)" on page 1967	0x3(N*2+1)47	Rx Adaption IQ Controls.
"RAWLANEAONN_DIG_RX_CTLE_ADAPT_CTRL(for N = 0; N <= NLANES-1)" on page 1968	0x3(N*2+1)48	Rx CTEL Adapt Controls.
"RAWLANEAONN_DIG_SRAM_REC_MAX_ITER(for N = 0; N <= NLANES-1)" on page 1969	0x3(N*2+1)49	The maximum number of Iterations for SRAM Recording.
"RAWLANEAONN_DIG_USE_RX_COARSE_ADPT_VAL(for N = 0; N <= NLANES-1)" on page 1970	0x3(N*2+1)4a	Use RX Coarse Adapt Value .
"RAWLANEAONN_DIG_RX_RO_AONLANE_1(for N = 0; N <= NLANES-1)" on page 1972	0x3(N*2+1)4b	Value for rx_disable.
"RAWLANEAONN_DIG_RX_DCC_IQ_SKIPS(for N = 0; N <= NLANES-1)" on page 1973	0x3(N*2+1)4c	E32_RAWLANEAON0_DIG_RX_DCC_IQ_SKIPS
"RAWLANEAONN_DIG_TX_DCC_SKIPS(for N = 0; N <= NLANES-1)" on page 1975	0x3(N*2+1)4d	E32_RAWLANEAON0_DIG_TX_DCC_SKIPS
"RAWLANEAONN_DIG_CAL_VGEN_DAC_CODE(for N = 0; N <= NLANES-1)" on page 1976	0x3(N*2+1)4e	VGEN calibration code

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_COARSE_ADPT_SKIP_BITS(for N = 0; N <= NLANES-1)" on page 1977	0x3(N*2+1)4f	RX Coarse Adapt Skip bits for all three banks
"RAWLANEAONN_DIG_DFE_BYPASS_EVEN_ERROR_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1978	0x3(N*2+1)50	Offset Values for RX DFE By-Pass Even Error vDAC.
"RAWLANEAONN_DIG_DFE_BYPASS_ODD_ERROR_VDAC_OFST(for N = 0; N <= NLANES-1)" on page 1979	0x3(N*2+1)51	Offset Values for RX DFE By-Pass Odd Error vDAC.
"RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS(for N = 0; N <= NLANES-1)" on page 1980	0x3(N*2+1)52	E32_RAWLANEAON0_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS
"RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B1(for N = 0; N <= NLANES-1)" on page 1981	0x3(N*2+1)53	RX Adapted BUF GAIN for Bank 1.
"RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B2(for N = 0; N <= NLANES-1)" on page 1982	0x3(N*2+1)54	RX Adapted BUF GAIN for Bank 2.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS(for N = 0; N <= NLANES-1)" on page 1983	0x3(N*2+1)55	E32_RAWLANEAON0_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS
"RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B1(for N = 0; N <= NLANES-1)" on page 1984	0x3(N*2+1)56	RX Adapted BUF GAIN for Bank 1.
"RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B2(for N = 0; N <= NLANES-1)" on page 1985	0x3(N*2+1)57	RX Adapted BUF GAIN for Bank 2.
"RAWLANEAONN_DIG_ADPT_CTL_29(for N = 0; N <= NLANES-1)" on page 1986	0x3(N*2+1)58	Adaptation Control Register 29.
"RAWLANEAONN_DIG_ADPT_CTL_30(for N = 0; N <= NLANES-1)" on page 1987	0x3(N*2+1)59	Adaptation Control Register 30.
"RAWLANEAONN_DIG_ADPT_CTL_31(for N = 0; N <= NLANES-1)" on page 1988	0x3(N*2+1)5a	Adaptation Control Register 31.
"RAWLANEAONN_DIG_ADPT_CTL_32(for N = 0; N <= NLANES-1)" on page 1989	0x3(N*2+1)5b	Adaptation Control Register 32.
"RAWLANEAONN_DIG_ADPT_CTL_33(for N = 0; N <= NLANES-1)" on page 1990	0x3(N*2+1)5c	Adaptation Control Register 33.
"RAWLANEAONN_DIG_ADPT_CTL_34(for N = 0; N <= NLANES-1)" on page 1991	0x3(N*2+1)5d	Adaptation Control Register 34.
"RAWLANEAONN_DIG_ADPT_CTL_35(for N = 0; N <= NLANES-1)" on page 1992	0x3(N*2+1)5e	Adaptation Control Register 35.
"RAWLANEAONN_DIG_ADPT_CTL_36(for N = 0; N <= NLANES-1)" on page 1993	0x3(N*2+1)5f	Adaptation Control Register 36.

Table 2-4 Registers for the E32 Memory Map (continued)

Register	Offset	Description
"RAWLANEAONN_DIG_RX_MISC (for N = 0; N <= NLANES-1)" on page 1994	0x3(N*2+1)60	OVRD register for RX_MISC.
"RAWLANEAONN_DIG_TX_MISC_OVRD (for N = 0; N <= NLANES-1)" on page 1995	0x3(N*2+1)61	OVRD register for TX_MISC.
"RAWMEM_DIG_ROM_CMN0_B0_R0" on page 1996	0x8000	First ROM address.
"RAWMEM_DIG_ROM_CMN31_B15_R31" on page 1997	0xbfff	Last ROM address.
"RAWMEM_DIG_RAM_CMN0_B0_R0" on page 1998	0xc000	First RAM address.
"RAWMEM_DIG_RAM_CMN31_B15_R31" on page 1999	0xffff	Last RAM address.

2.1 E32_mem_map Registers

2.1.1 SUP_DIG_IDCODE_LO

- **Description:** Low 16 Bits of IDCODE.
- **Size:** 16 bits
- **Offset:** 0x0
- **Exists:** Always

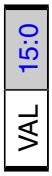


Table 2-5 Fields for Register: SUP_DIG_IDCODE_LO

Bits	Name	Memory Access	Description
15:0	VAL	R	**Value After Reset:** 0x74cd **Exists:** Always **Reset Mask:** 0xffff

2.1.2 SUP_DIG_IDCODE_HI

- **Description:** High 16 Bits of IDCODE.
- **Size:** 16 bits
- **Offset:** 0x1
- **Exists:** Always

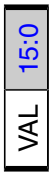


Table 2-6 Fields for Register: SUP_DIG_IDCODE_HI

Bits	Name	Memory Access	Description
15:0	VAL	R	**Value After Reset:** 0x7d38 **Exists:** Always **Reset Mask:** 0xffff

2.1.3 SUP_DIG_REFCLKA_OVRD_IN_0

- **Description:** Override Values for Incoming REFCLKA and RESET Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x2
- **Exists:** Always

15	14	13	12:9	8	7	6	5:4	3	2	1	0
REFA_LANE_CLK_EN_OVRD_EN	REFA_LANE_CLK_EN	REFA_CLK_RANGE_OVRD_EN	REFA_CLK_RANGE	REFA_CLKDET_EN_OVRD_EN	REFA_CLKDET_EN	REFA_CLK_SEL_OVRD_EN	REFA_CLK_SEL	REFA_CLK_DIV2_EN_OVRD_EN	REFA_CLK_DIV2_EN	REFA_CLK_EN_OVRD_EN	REFA_CLK_EN

Table 2-7 Fields for Register: SUP_DIG_REFCLKA_OVRD_IN_0

Bits	Name	Memory Access	Description
15	REFA_LANE_CLK_EN_OVRD_EN	R/W	This bit enables override for refa_lane_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	REFA_LANE_CLK_EN	R/W	This bit is the override value for refa_lane_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	REFA_CLK_RANGE_OVRD_EN	R/W	This bit enables override for refa_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:9	REFA_CLK_RANGE	R/W	This field is the override value for refa_range. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xf
8	REFA_CLKDET_EN_OVRD_EN	R/W	This bit enables override for refa_clkdet_en. **Value After Reset:** 0x0

Table 2-7 Fields for Register: SUP_DIG_REFCLKA_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	REFA_CLKDET_EN	R/W	This bit is the override value for refa_clkdet_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	REFA_CLK_SEL_OVRD_EN	R/W	This bit enables override for refa_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	REFA_CLK_SEL	R/W	This field is the override value for refa_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	REFA_CLK_DIV2_EN_OVRD_EN	R/W	This bit enables override for refa_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	REFA_CLK_DIV2_EN	R/W	This bit is the override value for refa_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	REFA_CLK_EN_OVRD_EN	R/W	This bit enables override for refa_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	REFA_CLK_EN	R/W	This bit is the override value for refa_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.4 SUP_DIG_REFCLKB_OVRD_IN_0

- **Description:** Override Values for Incoming REFCLKA and RESET Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x3
- **Exists:** Always

15	14	13	12:9	8	7	6	5:4	3	2	1	0
REFB_LANE_CLK_EN_OVRD_EN	REFB_LANE_CLK_EN	REFB_CLK_RANGE_OVRD_EN	REFB_CLK_RANGE	REFB_CLKDET_EN_OVRD_EN	REFB_CLKDET_EN	REFB_CLK_SEL_OVRD_EN	REFB_CLK_SEL	REFB_CLK_DIV2_EN_OVRD_EN	REFB_CLK_DIV2_EN	REFB_CLK_EN_OVRD_EN	REFB_CLK_EN

Table 2-8 Fields for Register: SUP_DIG_REFCLKB_OVRD_IN_0

Bits	Name	Memory Access	Description
15	REFB_LANE_CLK_EN_OVRD_EN	R/W	This bit enables override for refb_lane_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	REFB_LANE_CLK_EN	R/W	This bit is the override value for refb_lane_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	REFB_CLK_RANGE_OVRD_EN	R/W	This bit enables override for refb_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:9	REFB_CLK_RANGE	R/W	This field is the override value for refb_range. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xf
8	REFB_CLKDET_EN_OVRD_EN	R/W	This bit enables override for refb_clkdet_en. **Value After Reset:** 0x0

Table 2-8 Fields for Register: SUP_DIG_REFCLKB_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	REFB_CLKDET_EN	R/W	This bit is the override value for refb_clkdet_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	REFB_CLK_SEL_OVRD_EN	R/W	This bit enables override for refb_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	REFB_CLK_SEL	R/W	This field is the override value for refb_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	REFB_CLK_DIV2_EN_OVRD_EN	R/W	This bit enables override for refb_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	REFB_CLK_DIV2_EN	R/W	This bit is the override value for refb_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	REFB_CLK_EN_OVRD_EN	R/W	This bit enables override for refb_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	REFB_CLK_EN	R/W	This bit is the override value for refb_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.5 SUP_DIG_REFCLK_OVRD_IN_1

- **Description:** Override Values for Incoming REFCLK and RESET Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x4
- **Exists:** Always

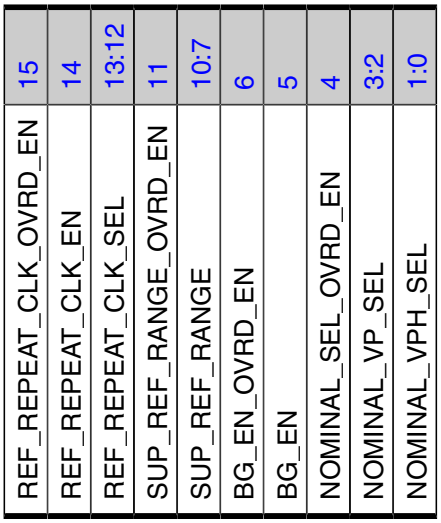


Table 2-9 Fields for Register: SUP_DIG_REFCLK_OVRD_IN_1

Bits	Name	Memory Access	Description
15	REF_REPEAT_CLK_OVRD_EN	R/W	This bit enables override for ref_repeat_clk_sel and ref_repeat_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	REF_REPEAT_CLK_EN	R/W	This bit is the override value for ref_repeat_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:12	REF_REPEAT_CLK_SEL	R/W	This field is the override value for ref_repeat_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
11	SUP_REF_RANGE_OVRD_EN	R/W	This bit enables override for sup_ref_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-9 Fields for Register: SUP_DIG_REFCLK_OVRD_IN_1 (continued)

Bits	Name	Memory Access	Description
10:7	SUP_REF_RANGE	R/W	This field is the override value for sup_ref_range[3:0]. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xf
6	BG_EN_OVRD_EN	R/W	This bit is the override enable for bg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	BG_EN	R/W	This bit is the override value for bg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	NOMINAL_SEL_OVRD_EN	R/W	This bit is the override enable for nominal_vp_sel and nominal_vph_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	NOMINAL_VP_SEL	R/W	This field is the override value for nominal_vp_sel[1:0]. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
1:0	NOMINAL_VPH_SEL	R/W	This field is the override value for nominal_vph_sel[1:0]. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3

2.1.6 SUP_DIG_MPLLA_DIV_CLK_OVRD_IN

- **Description:** Override Values for Incoming/Outgoing Control Signals.
- **Size:** 16 bits
- **Offset:** 0x5
- **Exists:** Always

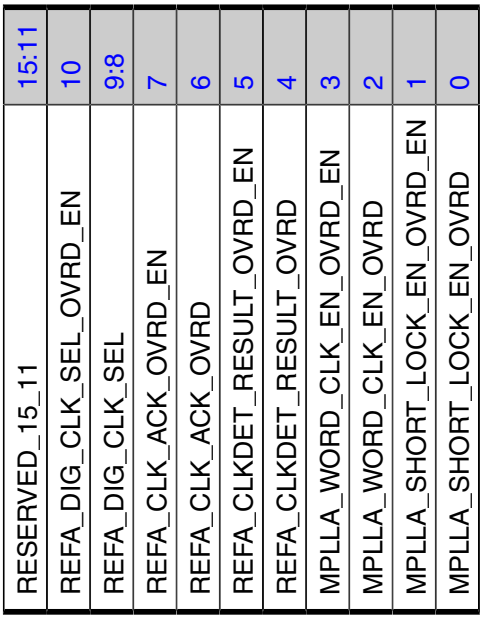


Table 2-10 Fields for Register: SUP_DIG_MPLLA_DIV_CLK_OVRD_IN

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	REFA_DIG_CLK_SEL_OVRD_EN	R/W	This bit enables override for refa_dig_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:8	REFA_DIG_CLK_SEL	R/W	This field is the override value for refa_dig_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
7	REFA_CLK_ACK_OVRD_EN	R/W	This bit enables override for refa_clk_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-10 Fields for Register: SUP_DIG_MPLLA_DIV_CLK_OVRD_IN (continued)

Bits	Name	Memory Access	Description
6	REFA_CLK_ACK_OVRD	R/W	This bit is the override value for refa_clk_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	REFA_CLKDET_RESULT_OVRD_EN	R/W	This bit enabled override for refa_clkdet_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	REFA_CLKDET_RESULT_OVRD	R/W	This bit is the override value for refa_clkdet_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLA_WORD_CLK_EN_OVRD_EN	R/W	This bit enables override for mplla_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_WORD_CLK_EN_OVRD	R/W	This bit is the override value for mplla_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_SHORT_LOCK_EN_OVRD_EN	R/W	This bit enables override for mplla_short_lock_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_SHORT_LOCK_EN_OVRD	R/W	This bit is the override value for mplla_short_lock_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.7 SUP_DIG_MPLLB_DIV_CLK_OVRD_IN

- **Description:** Override Values for Incoming/Outgoing Control Signals.
- **Size:** 16 bits
- **Offset:** 0x6
- **Exists:** Always

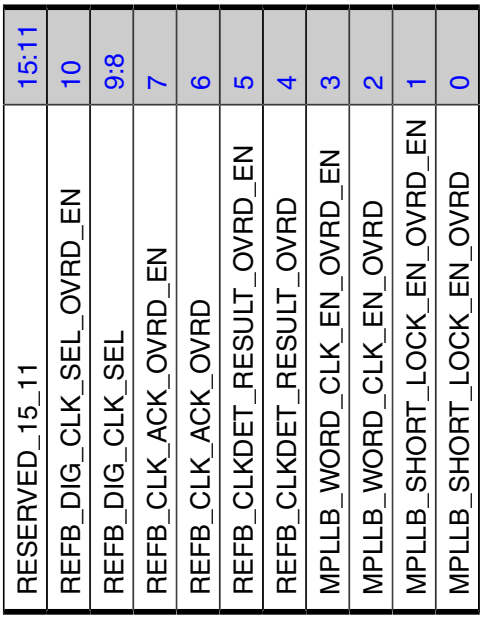


Table 2-11 Fields for Register: SUP_DIG_MPLLB_DIV_CLK_OVRD_IN

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	REFB_DIG_CLK_SEL_OVRD_EN	R/W	This bit enables override for refb_dig_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:8	REFB_DIG_CLK_SEL	R/W	This field is the override value for refb_dig_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
7	REFB_CLK_ACK_OVRD_EN	R/W	This bit enables override for refb_clk_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-11 Fields for Register: SUP_DIG_MPLLB_DIV_CLK_OVRD_IN (continued)

Bits	Name	Memory Access	Description
6	REFB_CLK_ACK_OVRD	R/W	This bit is the override value for refb_clk_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	REFB_CLKDET_RESULT_OVRD_EN	R/W	This bit enables override for refb_clkdet_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	REFB_CLKDET_RESULT_OVRD	R/W	This bit is the override value for refb_clkdet_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_WORD_CLK_EN_OVRD_EN	R/W	This bit enables override for mpllb_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_WORD_CLK_EN_OVRD	R/W	This bit is the override value for mpllb_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_SHORT_LOCK_EN_OVRD_EN	R/W	This bit enables override for mpllb_short_lock_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLB_SHORT_LOCK_EN_OVRD	R/W	This bit is the override value for mpllb_short_lock_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.8 SUP_DIG_MPLLA_OVRD_IN_0

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x7
- **Exists:** Always

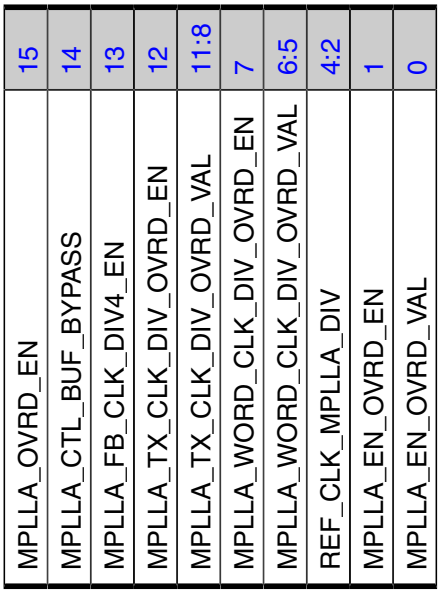


Table 2-12 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_0

Bits	Name	Memory Access	Description
15	MPLLA_OVRD_EN	R/W	This bit enable override values for all static MPLLA inputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLLA_CTL_BUF_BYPASS	R/W	This bit is the override value for mplla_ctl_buf_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLA_FB_CLK_DIV4_EN	R/W	This bit is the override value for mplla_fb_clk_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_TX_CLK_DIV_OVRD_EN	R/W	This bit is the override enable for mplla_tx_clk_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-12 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
11:8	MPLLA_TX_CLK_DIV_OVRD_VAL	R/W	This field is the override value for mplla_tx_clk_div. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf
7	MPLLA_WORD_CLK_DIV_OVRD_EN	R/W	This bit is the override enable for mplla_word_clk_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:5	MPLLA_WORD_CLK_DIV_OVRD_VAL	R/W	This field is the override value for mplla_word_clk_div. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
4:2	REF_CLK_MPLLA_DIV	R/W	This field is the override value for ref_clk_mplla_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
1	MPLLA_EN_OVRD_EN	R/W	This bit is the override enable value for mplla_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_EN_OVRD_VAL	R/W	This bit is the override value for mplla_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.9 SUP_DIG_MPLLA_OVRD_IN_1

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x8
- **Exists:** Always

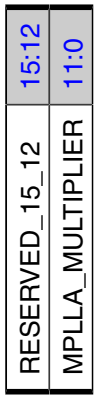


Table 2-13 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	MPLLA_MULTIPLIER	R/W	This field is the override value for mplla_multiplier. **Value After Reset:** 0x19 **Exists:** Always **Reset Mask:** 0xfff

2.1.10 SUP_DIG_MPLLA_OVRD_IN_2

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x9
- **Exists:** Always

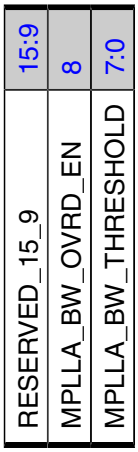


Table 2-14 Fields for Register: SUP_DIG_MPLLA_OVRD_IN_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLLA_BW_OVRD_EN	R/W	This bit is the override enable for mplla_bw_low, mplla_bw_high, and mplla_bw_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	MPLLA_BW_THRESHOLD	R/W	This field is the override value for mplla_bw_threshold. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xff

2.1.11 SUP_DIG_MPLLA_BW_LOW_OVRD_IN

- **Description:** Override Values for Incoming MPLLA Bandwidth Low Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0xa
- **Exists:** Always

Table 2-15 Fields for Register: SUP_DIG_MPLLA_BW_LOW_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_LOW	R/W	This field is the override value for mplla_bw_low[15:0]. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0xffff

2.1.12 SUP_DIG_MPLLA_BW_HIGH_OVRD_IN

- **Description:** Override Values for Incoming MPLLA Bandwidth High Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0xb
- **Exists:** Always

Table 2-16 Fields for Register: SUP_DIG_MPLLA_BW_HIGH_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_HIGH	R/W	This field is the override value for mplla_bw_high[15:0]. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0xffff

2.1.13 SUP_DIG_MPLLA_SSC_OVRD_IN_0

- **Description:** Override Values for Incoming MPLLA SSC Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0xc
- **Exists:** Always

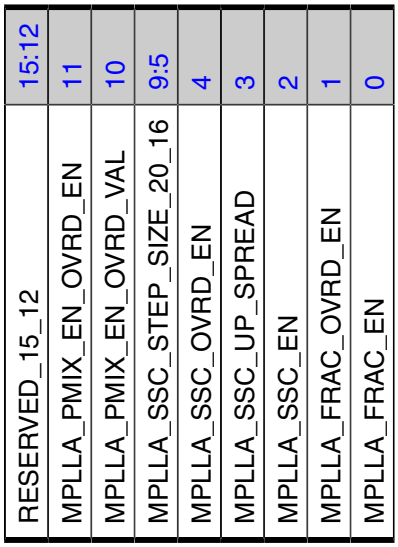


Table 2-17 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_0

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLA_PMIX_EN_OVRD_EN	R/W	This bit is the override enable for mplla_pmix_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	MPLLA_PMIX_EN_OVRD_VAL	R/W	This bit is the override value for mplla_pmix_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:5	MPLLA_SSC_STEP_SIZE_20_16	R/W	This field is the override value for mplla_ssc_step_size[20:16]. This signal is overridden by MPLLA_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

Table 2-17 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
4	MPLLA_SSC_OVRD_EN	R/W	This bit is the override enable for SSC control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLA_SSC_UP_SPREAD	R/W	OThis bit is the override value for mplla_ssc_up_spread. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_SSC_EN	R/W	This bit is the override value for mplla_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_FRAC_OVRD_EN	R/W	This bit is the override enable for Fractional ref_clk control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_FRAC_EN	R/W	This bit is the override value for mplla_frac_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.14 SUP_DIG_MPLLA_SSC_OVRD_IN_1

- **Description:** Override Values for Incoming MPLLA SSC Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0xd
- **Exists:** Always

Table 2-18 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_1

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_STEP_SIZE_15_0	R/W	This field is the override value for mplla_ssc_step_size[15:0]. This signal is overridden by MPLLA_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.15 SUP_DIG_MPLLA_SSC_OVRD_IN_2

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0xe
- **Exists:** Always

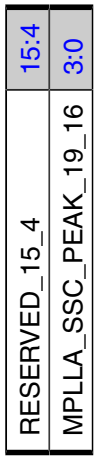


Table 2-19 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_2

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	MPLLA_SSC_PEAK_19_16	R/W	This field is the override value for mplla_ssc_peak[19:16]. This signal is overridden by MPLLA_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.16 SUP_DIG_MPLLA_SSC_OVRD_IN_3

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0xf
- **Exists:** Always

Table 2-20 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_3

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_PEAK_15_0	R/W	This field is the override value for mplla_ssc_peak[15:0]. This signal is overridden by MPLLA_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.17 SUP_DIG_MPLLA_SSC_OVRD_IN_4

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x10
- **Exists:** Always

Table 2-21 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_QUOT	R/W	This field is the override value for mplla_frac_quot. This signal is overridden by MPLLA_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.18 SUP_DIG_MPLLA_SSC_OVRD_IN_5

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x11
- **Exists:** Always

Table 2-22 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_5

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_DEN	R/W	This field is the override value for mplla_frac_den. This signal is overridden by MPLLA_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.19 SUP_DIG_MPLLA_SSC_OVRD_IN_6

- **Description:** Override Values for Incoming MPLLA Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x12
- **Exists:** Always

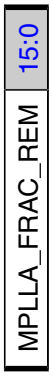


Table 2-23 Fields for Register: SUP_DIG_MPLLA_SSC_OVRD_IN_6

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_REM	R/W	This field is the override value for mplla_frac_rem. This signal is overridden by MPLLA_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.20 SUP_DIG_MPLLB_OVRD_IN_0

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x13
- **Exists:** Always

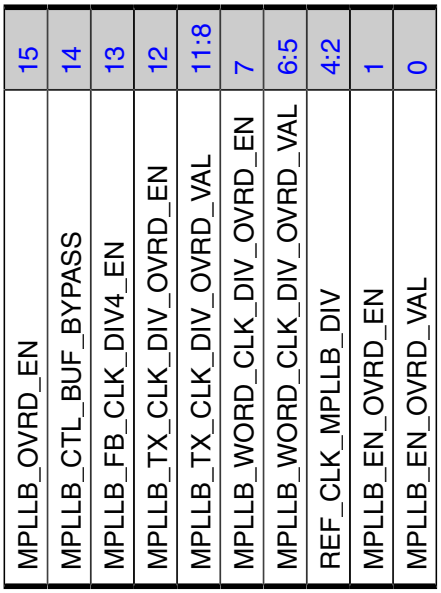


Table 2-24 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_0

Bits	Name	Memory Access	Description
15	MPLLB_OVRD_EN	R/W	This bit enables override values for all static MPLLB inputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLLB_CTL_BUF_BYPASS	R/W	This bit is the override value for mpllb_ctl_buf_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLB_FB_CLK_DIV4_EN	R/W	This bit is the override value for mpllb_fb_clk_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_TX_CLK_DIV_OVRD_EN	R/W	This bit is the override enable for mpllb_tx_clk_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-24 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
11:8	MPLLB_TX_CLK_DIV_OVRD_VAL	R/W	This field is the override value for mpllb_tx_clk_div. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf
7	MPLLB_WORD_CLK_DIV_OVRD_EN	R/W	This bit is the override enable for mpllb_word_clk_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:5	MPLLB_WORD_CLK_DIV_OVRD_VAL	R/W	This field is the override value for mpllb_word_clk_div. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
4:2	REF_CLK_MPLLB_DIV	R/W	This field is the override value for ref_clk_mpllb_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
1	MPLLB_EN_OVRD_EN	R/W	This bit is the override enable value for mpllb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLB_EN_OVRD_VAL	R/W	This bit is the override value for mpllb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.21 SUP_DIG_MPLLB_OVRD_IN_1

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x14
- **Exists:** Always

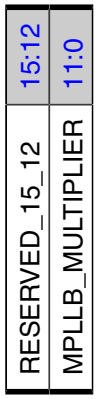


Table 2-25 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	MPLLB_MULTIPLIER	R/W	This field is the override value for mpllb_multiplier. **Value After Reset:** 0x19 **Exists:** Always **Reset Mask:** 0xfff

2.1.22 SUP_DIG_MPLLB_OVRD_IN_2

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x15
- **Exists:** Always

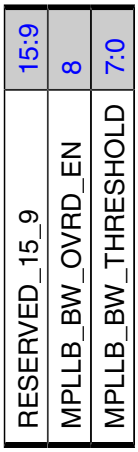


Table 2-26 Fields for Register: SUP_DIG_MPLLB_OVRD_IN_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLLB_BW_OVRD_EN	R/W	This bit is the override enable for mpllb_bw_low, mpllb_bw_high, and mpllb_bw_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	MPLLB_BW_THRESHOLD	R/W	This field is the override enable for mpllb_bw_threshold. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xff

2.1.23 SUP_DIG_MPLLB_BW_LOW_OVRD_IN

- **Description:** Override Values for Incoming MPLLB Bandwidth Low Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x16
- **Exists:** Always

Table 2-27 Fields for Register: SUP_DIG_MPLLB_BW_LOW_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_LOW	R/W	This field is the override value for mpllb_bw_low[15:0]. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0xffff

2.1.24 SUP_DIG_MPLLB_BW_HIGH_OVRD_IN

- **Description:** Override Values for Incoming MPLLB Bandwidth High Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x17
- **Exists:** Always

Table 2-28 Fields for Register: SUP_DIG_MPLLB_BW_HIGH_OVRD_IN

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_HIGH	R/W	This field is the override value for mpllb_bw_high[15:0]. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0xffff

2.1.25 SUP_DIG_MPLLB_SSC_OVRD_IN_0

- **Description:** Override Values for Incoming MPLLB SSC Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x18
- **Exists:** Always

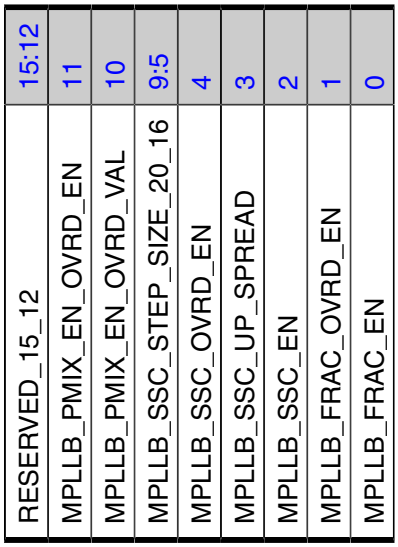


Table 2-29 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_0

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLB_PMIX_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_pmix_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	MPLLB_PMIX_EN_OVRD_VAL	R/W	This bit is the override value for mpllb_pmix_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:5	MPLLB_SSC_STEP_SIZE_20_16	R/W	This field is the override value for mpllb_ssc_step_size[20:16]. This signal is overridden by MPLLB_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

Table 2-29 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
4	MPLLB_SSC_OVRD_EN	R/W	This bit is the override enable for SSC control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_SSC_UP_SPREAD	R/W	This bit is the override value for mpllb_ssc_up_spread. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_SSC_EN	R/W	This bit is the override value for mpllb_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_FRAC_OVRD_EN	R/W	This bit is the override enable for fractional ref_clk control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLB_FRAC_EN	R/W	This bit is the override value for mpllb_frac_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.26 SUP_DIG_MPLLB_SSC_OVRD_IN_1

- **Description:** Override Values for Incoming MPLLB SSC Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x19
- **Exists:** Always

Table 2-30 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_1

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_STEP_SIZE_15_0	R/W	This field is the override value for mpllb_ssc_step_size[15:0]. This signal is overridden by MPLLB_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.27 SUP_DIG_MPLLB_SSC_OVRD_IN_2

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1a
- **Exists:** Always

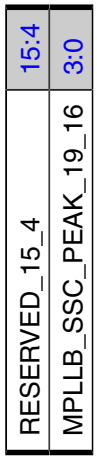


Table 2-31 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_2

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	MPLLB_SSC_PEAK_19_16	R/W	This field is the override value for mpllb_ssc_peak[19:16]. This signal is overridden by MPLLB_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.28 SUP_DIG_MPLLB_SSC_OVRD_IN_3

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1b
- **Exists:** Always

Table 2-32 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_3

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_PEAK_15_0	R/W	This field is the override value for mpllb_ssc_peak[15:0]. This signal is overridden by MPLLB_SSC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.29 SUP_DIG_MPLLB_SSC_OVRD_IN_4

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1c
- **Exists:** Always

Table 2-33 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_QUOT	R/W	This field is the override value for mpllb_frac_quot. This signal is overridden by MPLLB_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.30 SUP_DIG_MPLLB_SSC_OVRD_IN_5

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1d
- **Exists:** Always

Table 2-34 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_5

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_DEN	R/W	This field is the override value for mpllb_frac_den. This signal is overridden by MPLLB_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.31 SUP_DIG_MPLLB_SSC_OVRD_IN_6

- **Description:** Override Values for Incoming MPLLB Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x1e
- **Exists:** Always

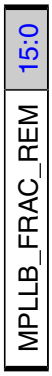


Table 2-35 Fields for Register: SUP_DIG_MPLLB_SSC_OVRD_IN_6

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_REM	R/W	This field is the override value for mpllb_frac_rem. This signal is overridden by MPLLB_FRAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.32 SUP_DIG_SUP_OVRD_IN_0

- **Description:** Override Values for Support Block ASIC Inputs, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1f
- **Exists:** Always

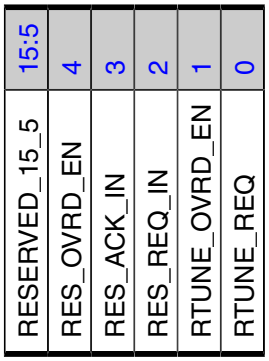


Table 2-36 Fields for Register: SUP_DIG_SUP_OVRD_IN_0

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	RES_OVRD_EN	R/W	This bit enables override of res_req_in and res_ack_in. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RES_ACK_IN	R/W	This bit is the override value for res_ack_in. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RES_REQ_IN	R/W	This bit is the override value for res_req_in. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RTUNE_OVRD_EN	R/W	This bit enables override of rtune_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RTUNE_REQ	R/W	This bit is the override value for rtune_req. **Value After Reset:** 0x0

Table 2-36 Fields for Register: SUP_DIG_SUP_OVRD_IN_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.33 SUP_DIG_SUP_OVRD_IN_1

- **Description:** Override Values for Support Block ASIC Inputs, Register 1.
- **Size:** 16 bits
- **Offset:** 0x20
- **Exists:** Always

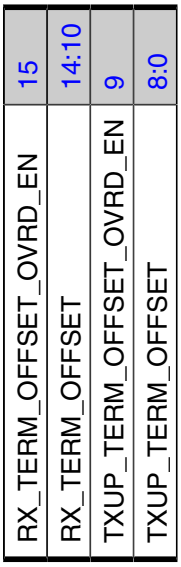


Table 2-37 Fields for Register: SUP_DIG_SUP_OVRD_IN_1

Bits	Name	Memory Access	Description
15	RX_TERM_OFFSET_OVRD_EN	R/W	This bit is the offset enable for RX termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:10	RX_TERM_OFFSET	R/W	This field is the offset value for RX termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
9	TXUP_TERM_OFFSET_OVRD_EN	R/W	This bit is the offset enable for TXUP termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	TXUP_TERM_OFFSET	R/W	This field is the offset value for TXUP termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.34 SUP_DIG_SUP_OVRD_IN_2

- **Description:** Override Values for Support Block ASIC Inputs, Register 2.
- **Size:** 16 bits
- **Offset:** 0x21
- **Exists:** Always

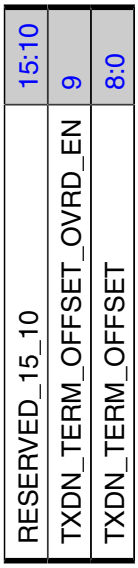


Table 2-38 Fields for Register: SUP_DIG_SUP_OVRD_IN_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TXDN_TERM_OFFSET_OVRD_EN	R/W	This bit is the offset enable for TXDN termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	TXDN_TERM_OFFSET	R/W	This field is the offset value for TXDN termination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.35 SUP_DIG_SUP_OVRD_OUT

- **Description:** Override Values for Support Block ASIC Outputs.
- **Size:** 16 bits
- **Offset:** 0x22
- **Exists:** Always

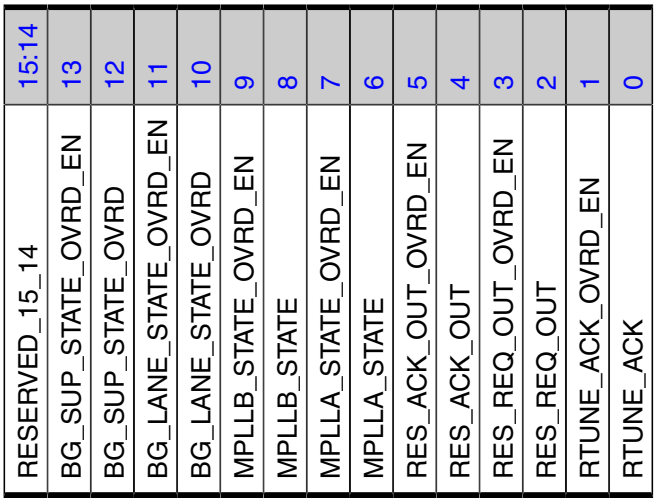


Table 2-39 Fields for Register: SUP_DIG_SUP_OVRD_OUT

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	BG_SUP_STATE_OVRD_EN	R/W	This bit enables override value for bg_sup_state signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	BG_SUP_STATE_OVRD	R/W	This bit is the override value for bg_sup_state signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	BG_LANE_STATE_OVRD_EN	R/W	This bit enables override value for bg_lane_state signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	BG_LANE_STATE_OVRD	R/W	This bit is the override value for bg_lane_state signal. **Value After Reset:** 0x0

Table 2-39 Fields for Register: SUP_DIG_SUP_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9	MPLLB_STATE_OVRD_EN	R/W	This bit enables override for mpllb_state output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	MPLLB_STATE	R/W	This bit is the override value for mpllb_state output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	MPLLA_STATE_OVRD_EN	R/W	This bit enables override for mplla_state output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	MPLLA_STATE	R/W	This bit is the override value for mplla_state output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RES_ACK_OUT_OVRD_EN	R/W	This bit enables override for res_ack_out output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RES_ACK_OUT	R/W	This bit is the override value for res_ack_out output. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	RES_REQ_OUT_OVRD_EN	R/W	This bit enables override for res_req_out output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RES_REQ_OUT	R/W	This bit is the override value for res_req_out output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-39 Fields for Register: SUP_DIG_SUP_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
1	RTUNE_ACK_OVRD_EN	R/W	This bit enables override for rtune_ack output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RTUNE_ACK	R/W	This bit is the override value for rtune_ack output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.36 SUP_DIG_LVL_OVRD_IN

- **Description:** Override Values for Level Settings.
- **Size:** 16 bits
- **Offset:** 0x23
- **Exists:** Always

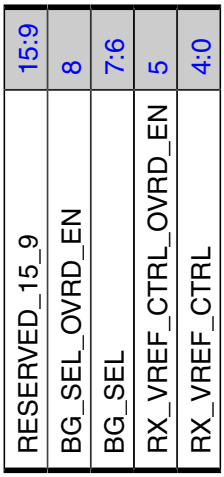


Table 2-40 Fields for Register: SUP_DIG_LVL_OVRD_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	BG_SEL_OVRD_EN	R/W	This bit enables override value for bg_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:6	BG_SEL	R/W	This field is the override value for bg_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5	RX_VREF_CTRL_OVRD_EN	R/W	This bit enables override value for rx_vref_ctrl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4:0	RX_VREF_CTRL	R/W	This field is the override value for rx_vref_ctrl. **Value After Reset:** 0x10

Table 2-40 Fields for Register: SUP_DIG_LVL_OVRD_IN (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1f

2.1.37 SUP_DIG_DEBUG

- **Description:** Debug Controls.
- **Size:** 16 bits
- **Offset:** 0x24
- **Exists:** Always

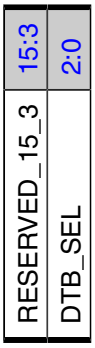


Table 2-41 Fields for Register: SUP_DIG_DEBUG

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	DTB_SEL	R/W	The lane DTB's are OR'd together with the support DTB signals selected with the below encodings. ■ 0 - none ■ 1 - mplla DTB output ■ 2 - mpllb DTB output ■ 3 - rtune DTB output **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.38 SUP_DIG_MPLLA_ASIC_IN_0

- **Description:** Current Values for Incoming MPLLA Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x25
- **Exists:** Always

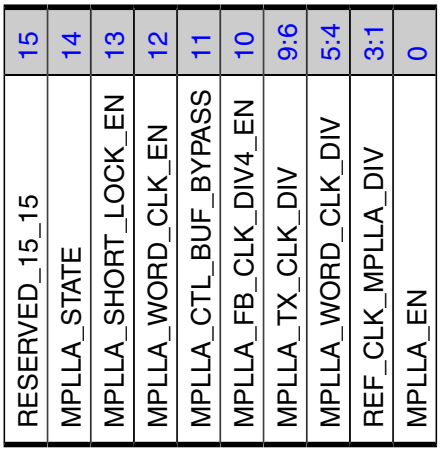


Table 2-42 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLA_STATE	R	This bit is the value to ASIC for mplla_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	MPLLA_SHORT_LOCK_EN	R	This bit is the value from ASIC for mplla_short_lock_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	MPLLA_WORD_CLK_EN	R	This bit is the value from ASIC for mplla_word_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-42 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11	MPLLA_CTL_BUF_BYPASS	R	This bit is the value from ASIC for mplla_ctl_buf_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	MPLLA_FB_CLK_DIV4_EN	R	This bit is the value from ASIC for mplla_fb_clk_div4_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:6	MPLLA_TX_CLK_DIV	R	This field is the value from ASIC for mplla_tx_clk_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	MPLLA_WORD_CLK_DIV	R	This field is the value from ASIC for mplla_word_clk_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:1	REF_CLK_MPLLA_DIV	R	This field is the value from ASIC for ref_clk_mplla_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLA_EN	R	This bit is the value from ASIC for mplla_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-42 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.39 SUP_DIG_MPLLA_ASIC_IN_1

- **Description:** Current Values for Incoming MPLLA Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x26
- **Exists:** Always

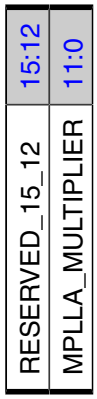


Table 2-43 Fields for Register: SUP_DIG_MPLLA_ASIC_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	MPLLA_MULTIPLIER	R	This field is the value from ASIC for mplla_multiplier. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.40 SUP_DIG_MPLLA_BW_LOW_ASIC_IN

- **Description:** Current Values for Incoming MPLLA Bandwidth Low Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x27
- **Exists:** Always

Table 2-44 Fields for Register: SUP_DIG_MPLLA_BW_LOW_ASIC_IN

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_LOW	R	This field is the value from ASIC for mplla_bw_low[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.41 SUP_DIG_MPLLA_BW_HIGH_ASIC_IN

- **Description:** Current Values for Incoming MPLLA Bandwidth High Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x28
- **Exists:** Always

Table 2-45 Fields for Register: SUP_DIG_MPLLA_BW_HIGH_ASIC_IN

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_HIGH	R	This field is the value from ASIC for mplla_bw_high[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.42 SUP_DIG_MPLLA_SSC_ASIC_IN_0

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x29
- **Exists:** Always

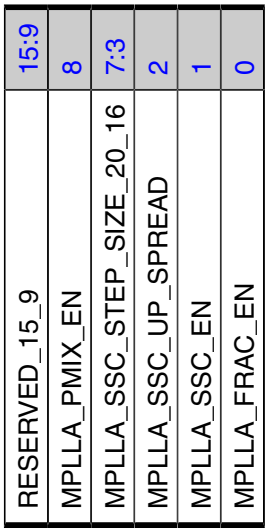


Table 2-46 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLLA_PMIX_EN	R	This bit is the value from mplla_pmix_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:3	MPLLA_SSC_STEP_SIZE_20_16	R	This field is the value from mplla_ssc_step_size[20:16]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	MPLLA_SSC_UP_SPREAD	R	This bit is the value from mplla_ssc_up_spread. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-46 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
1	MPLLA_SSC_EN	R	This bit is the value from mplla_ssc_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLA_FRAC_EN	R	This bit is the value from mplla_frac_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.43 SUP_DIG_MPLLA_SSC_ASIC_IN_1

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2a
- **Exists:** Always

Table 2-47 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_1

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_STEP_SIZE_15_0	R	This field is the value from mplla_ssc_step_size[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.44 SUP_DIG_MPLLA_SSC_ASIC_IN_2

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x2b
- **Exists:** Always

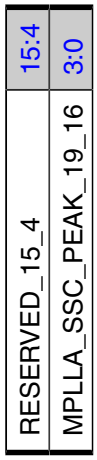


Table 2-48 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_2

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	MPLLA_SSC_PEAK_19_16	R	This field is the value from mplla_ssc_peak[19:16]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.45 SUP_DIG_MPLLA_SSC_ASIC_IN_3

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x2c
- **Exists:** Always

Table 2-49 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_3

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_PEAK_15_0	R	This field is the value from mplla_ssc_peak[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.46 SUP_DIG_MPLLA_SSC_ASIC_IN_4

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x2d
- **Exists:** Always

Table 2-50 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_QUOT	R	This field is the value from mplla_frac_quot. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.47 SUP_DIG_MPLLA_SSC_ASIC_IN_5

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x2e
- **Exists:** Always

Table 2-51 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_5

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_DEN	R	This field is the value from mplla_frac_den. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.48 SUP_DIG_MPLLA_SSC_ASIC_IN_6

- **Description:** Current Values for Incoming MPLLA SSC Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x2f
- **Exists:** Always

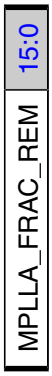


Table 2-52 Fields for Register: SUP_DIG_MPLLA_SSC_ASIC_IN_6

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_REM	R	This field is the value from mplla_frac_rem. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.49 SUP_DIG_MPLLB_ASIC_IN_0

- **Description:** Current Values for Incoming MPLLB Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x30
- **Exists:** Always

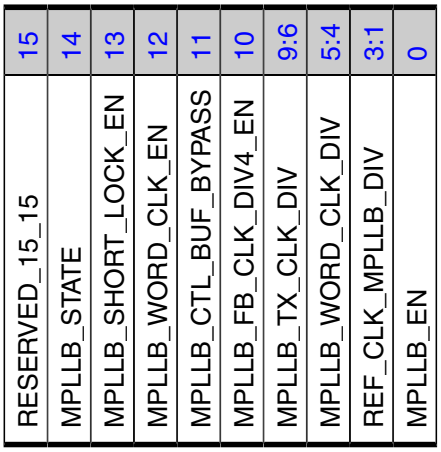


Table 2-53 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLB_STATE	R	This bit is the value to ASIC for mpllb_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	MPLLB_SHORT_LOCK_EN	R	This bit is the value from ASIC for mpllb_short_lock_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	MPLLB_WORD_CLK_EN	R	This bit is the value from ASIC for mpllb_word_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-53 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11	MPLLB_CTL_BUF_BYPASS	R	This bit is the value from ASIC for mpllb_ctl_buf_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	MPLLB_FB_CLK_DIV4_EN	R	This bit is the value from ASIC for mpllb_fb_clk_div4_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:6	MPLLB_TX_CLK_DIV	R	This field is the value from ASIC for mpllb_tx_clk_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	MPLLB_WORD_CLK_DIV	R	This field is the value from ASIC for mpllb_word_clk_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:1	REF_CLK_MPLLB_DIV	R	This field is the value from ASIC for ref_clk_mpllb_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLB_EN	R	This bit is the value from ASIC for mpllb_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-53 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.50 SUP_DIG_MPLLB_ASIC_IN_1

- **Description:** Current Values for Incoming MPLLB Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x31
- **Exists:** Always

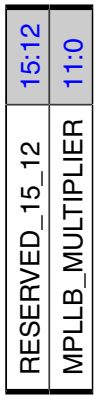


Table 2-54 Fields for Register: SUP_DIG_MPLLB_ASIC_IN_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	MPLLB_MULTIPLIER	R	This field is the value from ASIC for mpllb_multiplier. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.51 SUP_DIG_MPLLB_BW_LOW_ASIC_IN

- **Description:** Current Values for Incoming MPLLB Bandwidth Low Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x32
- **Exists:** Always

Table 2-55 Fields for Register: SUP_DIG_MPLLB_BW_LOW_ASIC_IN

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_LOW	R	This field is the value from ASIC for mpllb_bw_low[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.52 SUP_DIG_MPLLB_BW_HIGH_ASIC_IN

- **Description:** Current Values for Incoming MPLLB Bandwidth High Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x33
- **Exists:** Always

Table 2-56 Fields for Register: SUP_DIG_MPLLB_BW_HIGH_ASIC_IN

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_HIGH	R	This field is the value from ASIC for mpllb_bw_high[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.53 SUP_DIG_MPLLB_SSC_ASIC_IN_0

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x34
- **Exists:** Always

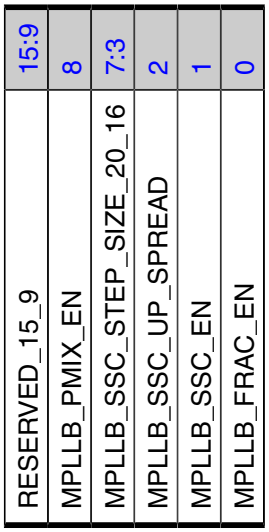


Table 2-57 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLLB_PMIX_EN	R	This bit is the value from mpllb_pmix_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:3	MPLLB_SSC_STEP_SIZE_20_16	R	This field is the value from mpllb_ssc_step_size[20:16]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	MPLLB_SSC_UP_SPREAD	R	This bit is the value from mpllb_ssc_up_spread. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-57 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
1	MPLLB_SSC_EN	R	This bit is the value from mpllb_ssc_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLB_FRAC_EN	R	This bit is the value from mpllb_frac_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.54 SUP_DIG_MPLLB_SSC_ASIC_IN_1

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x35
- **Exists:** Always

Table 2-58 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_1

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_STEP_SIZE_15_0	R	This field is the value from mpllb_ssc_step_size[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.55 SUP_DIG_MPLLB_SSC_ASIC_IN_2

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x36
- **Exists:** Always

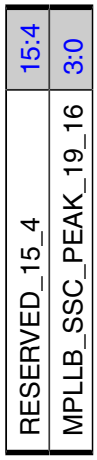


Table 2-59 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_2

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	MPLLB_SSC_PEAK_19_16	R	This field is the value from mpllb_ssc_peak[19:16]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.56 SUP_DIG_MPLLB_SSC_ASIC_IN_3

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x37
- **Exists:** Always

Table 2-60 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_3

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_PEAK_15_0	R	This field is the value from mpllb_ssc_peak[15:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.57 SUP_DIG_MPLLB_SSC_ASIC_IN_4

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x38
- **Exists:** Always

Table 2-61 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_4

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_QUOT	R	This field is the value from mpllb_frac_quot. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.58 SUP_DIG_MPLLB_SSC_ASIC_IN_5

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x39
- **Exists:** Always

Table 2-62 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_5

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_DEN	R	This field is the value from mpllb_frac_den. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.59 SUP_DIG_MPLLB_SSC_ASIC_IN_6

- **Description:** Current Values for Incoming MPLLB SSC Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x3a
- **Exists:** Always

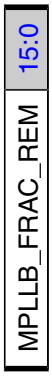
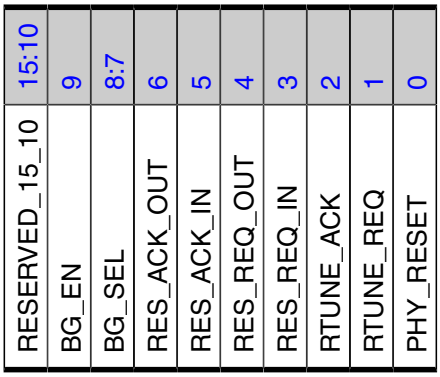


Table 2-63 Fields for Register: SUP_DIG_MPLLB_SSC_ASIC_IN_6

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_REM	R	This field is the value from mpllb_frac_rem. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.60 SUP_DIG_ASIC_IN_0

- **Description:** Current Values for Incoming SUP Control Signals from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x3b
- **Exists:** Always



15:10	9	8:7	6	5	4	3	2	1	0
RESERVED_15_10	BG_EN	BG_SEL	RES_ACK_OUT	RES_ACK_IN	RES_REQ_OUT	RES_REQ_IN	RTUNE_ACK	RTUNE_REQ	PHY_RESET

Table 2-64 Fields for Register: SUP_DIG_ASIC_IN_0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	BG_EN	R	This bit is the value from ASIC for bg_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:7	BG_SEL	R	This field is the value from ASIC for bg_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	RES_ACK_OUT	R	This bit is the value to ASIC for res_ack_out_i. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-64 Fields for Register: SUP_DIG_ASIC_IN_0 (continued)

Bits	Name	Memory Access	Description
5	RES_ACK_IN	R	This bit is the value from ASIC for res_req_in. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	RES_REQ_OUT	R	This bit is the value to ASIC for res_ack_out_i. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	RES_REQ_IN	R	This bit is the value from ASIC for res_req_in. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	RTUNE_ACK	R	This bit is the value to ASIC for rtune_ack_i. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	RTUNE_REQ	R	This bit is the value from ASIC for rtune_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	PHY_RESET	R	This bit is the value from ASIC for phy_reset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.61 SUP_DIG_REFCLKA_ASIC_IN

- **Description:** Current Values for Incoming SUP Control Signals from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x3c
- **Exists:** Always

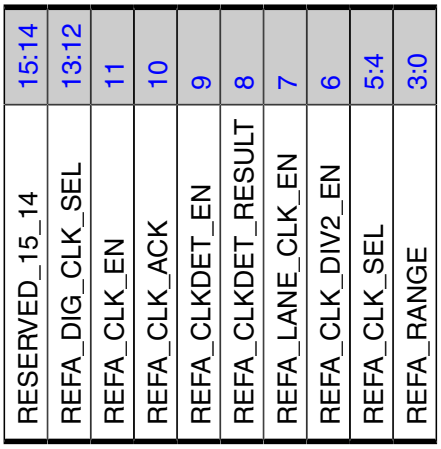


Table 2-65 Fields for Register: SUP_DIG_REFCLKA_ASIC_IN

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:12	REFA_DIG_CLK_SEL	R	This field is the value from ASIC for refa_dig_clk_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	REFA_CLK_EN	R	This bit is the value from ASIC for refa_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	REFA_CLK_ACK	R	This bit is the value to ASIC for refa_clk_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-65 Fields for Register: SUP_DIG_REFCLKA_ASIC_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
9	REFA_CLKDET_EN	R	This bit is the value from ASIC for refa_clkdet_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	REFA_CLKDET_RESULT	R	This bit is the value from ASIC for ref_clkdet_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	REFA_LANE_CLK_EN	R	This bit is the value from ASIC for refa_lane_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	REFA_CLK_DIV2_EN	R	This bit is the value from ASIC for refa_clk_div2_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	REFA_CLK_SEL	R	This field is the value from ASIC for refa_clk_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	REFA_RANGE	R	This field is the value from ASIC for refa_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-65 Fields for Register: SUP_DIG_REFCLKA_ASIC_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.62 SUP_DIG_REFCLKB_ASIC_IN

- **Description:** Current Values for Incoming SUP Control Signals from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x3d
- **Exists:** Always

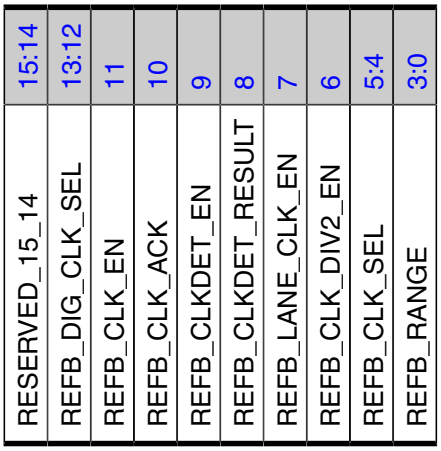


Table 2-66 Fields for Register: SUP_DIG_REFCLKB_ASIC_IN

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:12	REFB_DIG_CLK_SEL	R	This field is the value from ASIC for refb_dig_clk_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	REFB_CLK_EN	R	This bit is the value from ASIC for refb_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	REFB_CLK_ACK	R	This bit is the value to ASIC for refb_clk_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-66 Fields for Register: SUP_DIG_REFCLKB_ASIC_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
9	REFB_CLKDET_EN	R	This bit is the value from ASIC for refb_clkdet_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	REFB_CLKDET_RESULT	R	This bit is the value from ASIC for refb_clkdet_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	REFB_LANE_CLK_EN	R	This bit is the value from ASIC for refb_lane_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	REFB_CLK_DIV2_EN	R	This bit is the value from ASIC for reba_clk_div2_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	REFB_CLK_SEL	R	This field is the value from ASIC for refb_clk_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	REFB_RANGE	R	This field is the value from ASIC for refb_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-66 Fields for Register: SUP_DIG_REFCLKB_ASIC_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.63 SUP_DIG_ASIC_IN_1

- **Description:** Current Values for Incoming SUP Control Signals from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x3e
- **Exists:** Always

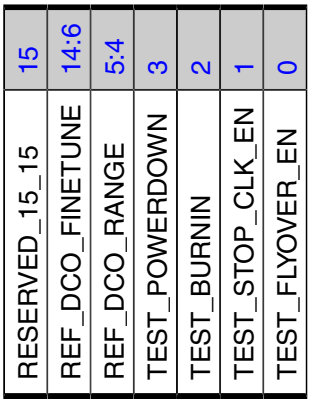


Table 2-67 Fields for Register: SUP_DIG_ASIC_IN_1

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:6	REF_DCO_FINETUNE	R	This bit is the value from ASIC for ref_dco_finetune. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	REF_DCO_RANGE	R	This bit is the value from ASIC for ref_dco_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	TEST_POWERDOWN	R	This bit is the value from ASIC for test_powerdown. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-67 Fields for Register: SUP_DIG_ASIC_IN_1 (continued)

Bits	Name	Memory Access	Description
2	TEST_BURNIN	R	This bit is the value from ASIC for test_burnin. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	TEST_STOP_CLK_EN	R	This bit is the value from ASIC for test_stop_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TEST_FLYOVER_EN	R	This bit is the value from ASIC for test_flyover_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.64 SUP_DIG_ASIC_IN_2

- **Description:** Current Values for Incoming SUP Control Signals from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x3f
- **Exists:** Always

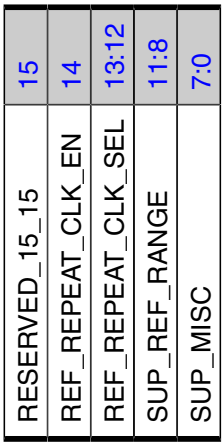


Table 2-68 Fields for Register: SUP_DIG_ASIC_IN_2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	REF_REPEAT_CLK_EN	R	This bit is the value from ASIC for ref_repeat_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:12	REF_REPEAT_CLK_SEL	R	This field is the value from ASIC for ref_repeat_clk_sel[1:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:8	SUP_REF_RANGE	R	This field is the value from ASIC for sup_ref_range[2:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-68 Fields for Register: SUP_DIG_ASIC_IN_2 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
7:0	SUP_MISC	R	This field is the value from ASIC for sup_misc. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.65 SUP_DIG_LVL_ASIC_IN

- **Description:** Current Values for Incoming Level Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x40
- **Exists:** Always

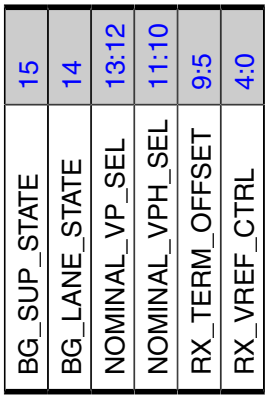


Table 2-69 Fields for Register: SUP_DIG_LVL_ASIC_IN

Bits	Name	Memory Access	Description
15	BG_SUP_STATE	R	This bit is the value to ASIC for bg_sup_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	BG_LANE_STATE	R	This bit is the value to ASIC for bg_lane_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:12	NOMINAL_VP_SEL	R	This field is the value from nominal_vp_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:10	NOMINAL_VPH_SEL	R	This field is the value from nominal_vph_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-69 Fields for Register: SUP_DIG_LVL_ASIC_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
9:5	RX_TERM_OFFSET	R	This field is the value from ASIC for rx_term_offset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4:0	RX_VREF_CTRL	R	This field is the value from ASIC for rx_vref_ctrl. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.66 SUP_DIG_ASIC_IN_3

- **Description:** Current Values for Incoming Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x41
- **Exists:** Always

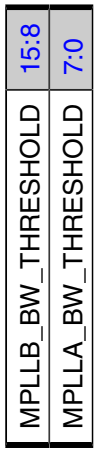


Table 2-70 Fields for Register: SUP_DIG_ASIC_IN_3

Bits	Name	Memory Access	Description
15:8	MPLLB_BW_THRESHOLD	R	This field is the value from ASIC for mpllb_bw_threshold. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	MPLLA_BW_THRESHOLD	R	This field is the value from ASIC for mplla_bw_threshold. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.67 SUP_DIG_SUP_OVRD_MISC

- **Description:** Override Values for Incoming SUP MISC BUS from ASIC.
- **Size:** 16 bits
- **Offset:** 0x42
- **Exists:** Always

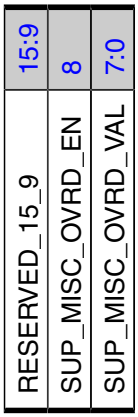


Table 2-71 Fields for Register: SUP_DIG_SUP_OVRD_MISC

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	SUP_MISC_OVRD_EN	R/W	This bit is the override enable for sup_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	SUP_MISC_OVRD_VAL	R/W	This field is the override value for sup_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.68 SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN

- **Description:** Current Values for Incoming Level Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x43
- **Exists:** Always

Table 2-72 Fields for Register: SUP_DIG_TXUP_TERM_OFFSET_ASIC_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	TXUP_TERM_OFFSET	R	This field is the value from ASIC for txup_term_offset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.69 SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN

- **Description:** Current Values for Incoming Level Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x44
- **Exists:** Always

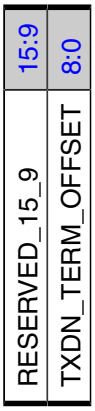


Table 2-73 Fields for Register: SUP_DIG_TXDN_TERM_OFFSET_ASIC_IN

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	TXDN_TERM_OFFSET	R	This field is the value from ASIC for txup_term_offset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.70 SUP_DIG_NBR_OF_LANES_IN

- **Description:** Current Values for Incoming Level Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x45
- **Exists:** Always

Table 2-74 Fields for Register: SUP_DIG_NBR_OF_LANES_IN

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4:0	NBR_OF_LANES	R	This field is the value indicating the number of lanes in the current configuration. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.71 SUP_DIG_MPLLA_AON_OVRD_IN

- **Description:** Override Values for MPLLA Incoming AON tune code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x46
- **Exists:** Always

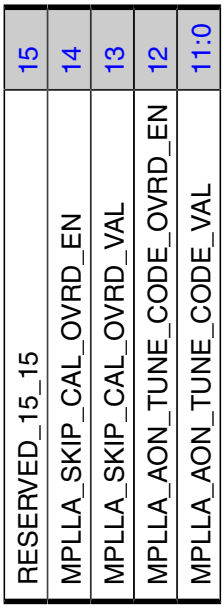


Table 2-75 Fields for Register: SUP_DIG_MPLLA_AON_OVRD_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLA_SKIP_CAL_OVRD_EN	R/W	This bit is the override enable for mplla_skip_cal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLA_SKIP_CAL_OVRD_VAL	R/W	This bit is the override enable for mplla_skip_cal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_AON_TUNE_CODE_OVRD_EN	R/W	This bit is the override enable for mplla_aon_tune_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-75 Fields for Register: SUP_DIG_MPLLA_AON_OVRD_IN (continued)

Bits	Name	Memory Access	Description
11:0	MPLLA_AON_TUNE_CODE_VAL	R/W	This field is the override value for mplla_aon_tune_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.72 SUP_DIG_MPLLB_AON_OVRD_IN

- **Description:** Override Values for MPLLB Incoming AON tune code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x47
- **Exists:** Always

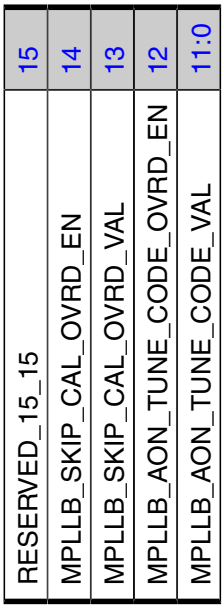


Table 2-76 Fields for Register: SUP_DIG_MPLLB_AON_OVRD_IN

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLB_SKIP_CAL_OVRD_EN	R/W	This bit is the override enable for mpllb_skip_cal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLB_SKIP_CAL_OVRD_VAL	R/W	This bit is the override enable for mpllb_skip_cal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_AON_TUNE_CODE_OVRD_EN	R/W	This bit is the override enable for mpllb_aon_tune_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-76 Fields for Register: SUP_DIG_MPLLB_AON_OVRD_IN (continued)

Bits	Name	Memory Access	Description
11:0	MPLLB_AON_TUNE_CODE_VAL	R/W	This field is the override value for mpllb_aon_tune_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.73 SUP_DIG_REFCLK_OVRD_IN_2

- **Description:** Override Values for Incoming REFCLK Controls from ASIC
- **Size:** 16 bits
- **Offset:** 0x48
- **Exists:** Always

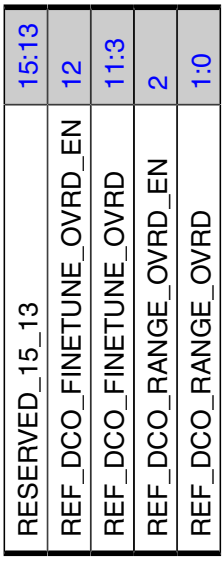


Table 2-77 Fields for Register: SUP_DIG_REFCLK_OVRD_IN_2

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	REF_DCO_FINETUNE_OVRD_EN	R/W	Enable override for ref_dco_finetune **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:3	REF_DCO_FINETUNE_OVRD	R/W	Override value for ref_dco_finetune **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff
2	REF_DCO_RANGE_OVRD_EN	R/W	Enable override for ref_dco_range **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-77 Fields for Register: SUP_DIG_REFCLK_OVRD_IN_2 (continued)

Bits	Name	Memory Access	Description
1:0	REF_DCO_RANGE_OVRD	R/W	Override value for ref_dco_range **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.74 SUP_DIG_ASIC_IN_4

- **Description:** Current Values for Incoming Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x49
- **Exists:** Always

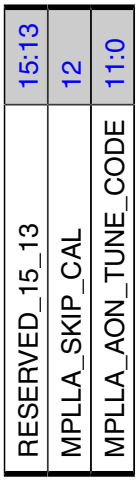


Table 2-78 Fields for Register: SUP_DIG_ASIC_IN_4

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLLA_SKIP_CAL	R	This field is the value from ASIC for mplla_skip_cal. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	MPLLA_AON_TUNE_CODE	R	This field is the value from ASIC for mplla_aon_tune_code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.75 SUP_DIG_ASIC_IN_5

- **Description:** Current Values for Incoming Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x4a
- **Exists:** Always

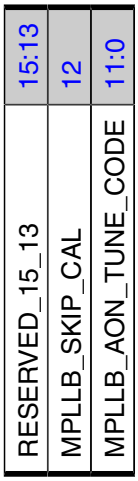


Table 2-79 Fields for Register: SUP_DIG_ASIC_IN_5

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLLB_SKIP_CAL	R	This field is the value from ASIC for mpllb_skip_cal. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	MPLLB_AON_TUNE_CODE	R	This field is the value from ASIC for mpllb_aon_tune_code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.76 SUP_DIG_ASIC_OUT_0

- **Description:** Current Values for Incoming Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x4b
- **Exists:** Always

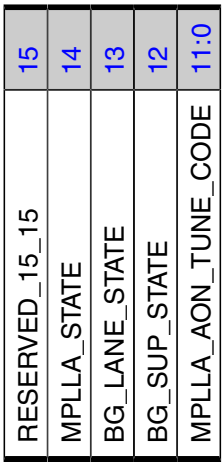


Table 2-80 Fields for Register: SUP_DIG_ASIC_OUT_0

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLA_STATE	R	This bit is the value to ASIC for mplla_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	BG_LANE_STATE	R	This bit is the value to ASIC for bg_lane_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	BG_SUP_STATE	R	This bit is the value to ASIC for bg_sup_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-80 Fields for Register: SUP_DIG_ASIC_OUT_0 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11:0	MPLLA_AON_TUNE_CODE	R	This field is the value from ASIC for mplla_aon_tune_code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.77 SUP_DIG_ASIC_OUT_1

- **Description:** Current Values for Incoming Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x4c
- **Exists:** Always

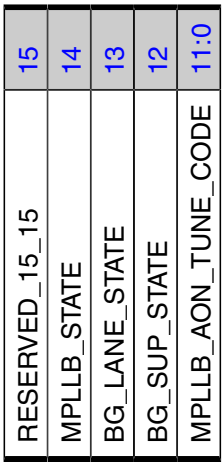


Table 2-81 Fields for Register: SUP_DIG_ASIC_OUT_1

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLLB_STATE	R	This bit is the value to ASIC for mplla_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	BG_LANE_STATE	R	This bit is the value to ASIC for bg_lane_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	BG_SUP_STATE	R	This bit is the value to ASIC for bg_sup_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-81 Fields for Register: SUP_DIG_ASIC_OUT_1 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11:0	MPLLB_AON_TUNE_CODE	R	This field is the value from ASIC for mpllb_aon_tune_code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.78 SUP_DIG_RTUNE_DEBUG

- **Description:** Resistor Tuning Debug Controls.
- **Size:** 16 bits
- **Offset:** 0x60
- **Exists:** Always

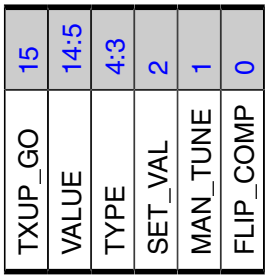


Table 2-82 Fields for Register: SUP_DIG_RTUNE_DEBUG

Bits	Name	Memory Access	Description
15	TXUP_GO	R/W	This bit enables TxUP tune to continue in manual tune mode, when TYPE is TxUP tune. When in non TxUP manual tune mode, this bit must be 0. When in TxUP manual tune mode and after TxUP manual tune is triggered, if the read only register SUP.RTUNE_STAT.DTB_RTUNE equals to 2'b01, it is required to first release the rtune related asynchronous override on analog/digital interface then turn this bit TXUP_GO from low to high to continue TxUP manual tune which also indicates that this bit needs to be low when triggering TxUP manual tune. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:5	VALUE	R/W	This field is the value to use when triggering SET_VAL field. Only the 6 LSB's are used when setting rx or tx cal values. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
4:3	TYPE	R/W	This field is the type of manual tuning or register read/write to execute. ■ 0 - ADC, or read/write rt_value. ■ 1 - Rx tune, or read/write rx_cal_val (only 6 bits). ■ 2 - TxDN tune, or read/write txdn_cal_val (10 bits). ■ 3 - TxUP tune, or read/write txup_cal_val (10 bits) or Resref detect (no affect when triggering SET_VAL field). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-82 Fields for Register: SUP_DIG_RTUNE_DEBUG (continued)

Bits	Name	Memory Access	Description
2	SET_VAL	R/W	This bit sets value. Write to a 1 to manually write the register specified by the TYPE field to the value in the VALUE field. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MAN_TUNE	R/W	Write to a 1 to do a manual tuning specified by TYPE field starting a manual tune while a tune is currently running can cause unpredictable results. For use only when you know what the part is doing (w.r.t. resistor tuning) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	FLIP_COMP	R/W	Invert analog comparator output **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.79 SUP_DIG_RTUNE_CONFIG

- **Description:** Configure Rtune Operation.
- **Size:** 16 bits
- **Offset:** 0x61
- **Exists:** Always

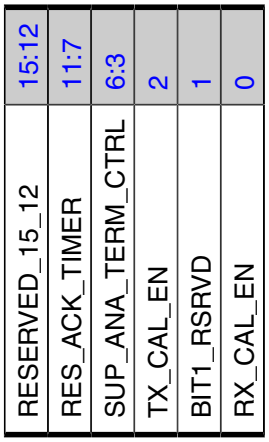


Table 2-83 Fields for Register: SUP_DIG_RTUNE_CONFIG

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:7	RES_ACK_TIMER	R/W	This field sets the Resistor ack timer. **Value After Reset:** 0x5 **Exists:** Always **Reset Mask:** 0x1f
6:3	SUP_ANA_TERM_CTRL	R/W	This field sets the reference resistor in the analog. Value Impedence (Ohms): ■ 0000 54 ■ 0001 52 ■ 0010 50 ■ 0011 48 ■ 0100 46 ■ 0101 44 ■ 0110 42 ■ 0111 40 ■ 1000 38 ■ 1001 36 ■ 1010 34 ■ 1011 32 ■ 1100 30

Table 2-83 Fields for Register: SUP_DIG_RTUNE_CONFIG (continued)

Bits	Name	Memory Access	Description
			■ 1101 30 ■ 1110 30 ■ 1111 30 **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
2	TX_CAL_EN	R/W	This bit enables calibration of TX resistor. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	BIT1_RSRVD	R/W	This bit is reserved **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_CAL_EN	R/W	This bit enables calibration of RX resistor. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.80 SUP_DIG_RTUNE_STAT

- **Description:** Resistor Tuning Register Status.
- **Size:** 16 bits
- **Offset:** 0x62
- **Exists:** Always

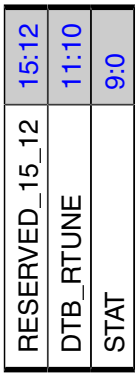


Table 2-84 Fields for Register: SUP_DIG_RTUNE_STAT

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	DTB_RTUNE	R	This field is the DTB sampling for rtune. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:0	STAT	R	This field is the current value of the register specified by the DEBUG.TYPE field. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.81 SUP_DIG_RTUNE_RX_SET_VAL

- **Description:** Set Value of RX Resistor.
- **Size:** 16 bits
- **Offset:** 0x63
- **Exists:** Always

Table 2-85 Fields for Register: SUP_DIG_RTUNE_RX_SET_VAL

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_SET_VAL	R/W	This field sets value of rx resistor. Writing a value to this register will set the rx resistor value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.82 SUP_DIG_RTUNE_TXDN_SET_VAL

- **Description:** Set Value of TX-DN Resistor.
- **Size:** 16 bits
- **Offset:** 0x64
- **Exists:** Always

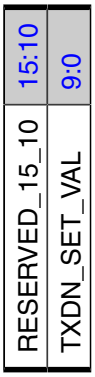


Table 2-86 Fields for Register: SUP_DIG_RTUNE_TXDN_SET_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TXDN_SET_VAL	R/W	This field sets value of tx-dn resistor. Writing a value to this register will set the tx-dn resistor value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.83 SUP_DIG_RTUNE_TXUP_SET_VAL

- **Description:** Set Value of TX-UP Resistor.
- **Size:** 16 bits
- **Offset:** 0x65
- **Exists:** Always

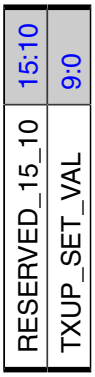


Table 2-87 Fields for Register: SUP_DIG_RTUNE_TXUP_SET_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TXUP_SET_VAL	R/W	This field sets value of tx-up resistor. Writing a value to this register will set the tx-up resistor value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.84 SUP_DIG_RTUNE_RX_STAT

- **Description:** RX Resistor Tuning Register Status.
- **Size:** 16 bits
- **Offset:** 0x66
- **Exists:** Always

Table 2-88 Fields for Register: SUP_DIG_RTUNE_RX_STAT

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_STAT	R	This field is the current value of the rx resistor tuning register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.85 SUP_DIG_RTUNE_TXDN_STAT

- **Description:** TX-DN Resistor Tuning Register Status.
- **Size:** 16 bits
- **Offset:** 0x67
- **Exists:** Always

Table 2-89 Fields for Register: SUP_DIG_RTUNE_TXDN_STAT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TXDN_STAT	R	This field is the current value of the tx-dn resistor tuning register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.86 SUP_DIG_RTUNE_TXUP_STAT

- **Description:** TX-UP Resistor Tuning Register Status.
- **Size:** 16 bits
- **Offset:** 0x68
- **Exists:** Always

Table 2-90 Fields for Register: SUP_DIG_RTUNE_TXUP_STAT

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TXUP_STAT	R	This field is the current value of the tx-up resistor tuning register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.87 SUP_DIG_RTUNE_CONFIG2

- **Description:** Configure Rtune Operation.
- **Size:** 16 bits
- **Offset:** 0x69
- **Exists:** Always

Table 2-91 Fields for Register: SUP_DIG_RTUNE_CONFIG2

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RTUNE	R/W	This bit enables fast resitor tuning (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.88 SUP_DIG_BG_CTL_BG_PWRUP_TIME_0

- **Description:** Bandgap Power-Up Time Register 0.
- **Size:** 16 bits
- **Offset:** 0x70
- **Exists:** Always

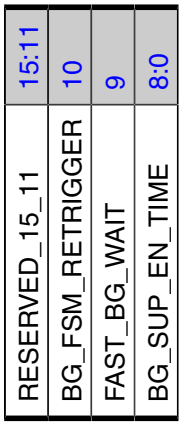


Table 2-92 Fields for Register: SUP_DIG_BG_CTL_BG_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	BG_FSM_RETRIGGER	R/W	This bit enables fast BG times (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	FAST_BG_WAIT	R/W	This bit enables fast BG times (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	BG_SUP_EN_TIME	R/W	This field is the power-up time (in ref_range cycles) for bandgap in SUP (spec >=5us). **Value After Reset:** 0x82 **Exists:** Always **Reset Mask:** 0x1ff

2.1.89 SUP_DIG_BG_CTL_BG_PWRUP_TIME_1

- **Description:** Bandgap Power-Up Time Register 1.
- **Size:** 16 bits
- **Offset:** 0x71
- **Exists:** Always

Table 2-93 Fields for Register: SUP_DIG_BG_CTL_BG_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	BG_LANE_EN_TIME	R/W	This field is the power-up time (in ref_range cycles) for bandgap in the lane (spec >=5us). **Value After Reset:** 0x82 **Exists:** Always **Reset Mask:** 0x1ff

2.1.90 SUP_DIG_BG_CTL_BG_PWRUP_TIME_2

- **Description:** Bandgap Power-Up Time Register 2.
- **Size:** 16 bits
- **Offset:** 0x72
- **Exists:** Always

Table 2-94 Fields for Register: SUP_DIG_BG_CTL_BG_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	BG_KICK_START_EN_TIME	R/W	This field is the assertion time (in ref_range cycles) for the bandgap kick-start signal (spec >=4us). Note that this time must be less than or equal to BG_SUP_EN_TIME. **Value After Reset:** 0x68 **Exists:** Always **Reset Mask:** 0x1ff

2.1.91 SUP_DIG_BG_CTL_BG_PWRUP_TIME_3

- **Description:** Bandgap Power-Up Time Register 3.
- **Size:** 16 bits
- **Offset:** 0x73
- **Exists:** Always

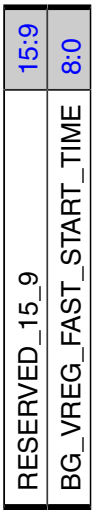


Table 2-95 Fields for Register: SUP_DIG_BG_CTL_BG_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	BG_VREG_FAST_START_TIME	R/W	This field is the assertion time (in ref_range cycles) for the bandgap kick-start signal (spec >=0us). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.92 SUP_DIG_BG_CTL_REF_PWRUP_TIME_0

- **Description:** Prescaler Power-Up Time Register.
- **Size:** 16 bits
- **Offset:** 0x74
- **Exists:** Always

Table 2-96 Fields for Register: SUP_DIG_BG_CTL_REF_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	FAST_REF_WAIT	R/W	This bit enables fast prescaler times (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4:0	REF_VREG_FAST_START_TIME	R/W	This field is the power-up time (in ref_range cycles) for the prescaler (spec >=400ns). **Value After Reset:** 0xb **Exists:** Always **Reset Mask:** 0x1f

2.1.93 SUP_DIG_BG_CTL_BG_STATE_STATUS

- **Description:** Current Values for Bandgap State Machine State Signals.
- **Size:** 16 bits
- **Offset:** 0x75
- **Exists:** Always

Table 2-97 Fields for Register: SUP_DIG_BG_CTL_BG_STATE_STATUS

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	BG_LANE_STATE	R	This bit is the value from the bandgap state machine for bg_lane_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	BG_SUP_STATE	R	This bit is the value from the bandgap state machine for bg_sup_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.94 SUP_DIG_ANA_MPLLA_OVRD_OUT0

- **Description:** Override Value for MPLLA Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x80
- **Exists:** Always

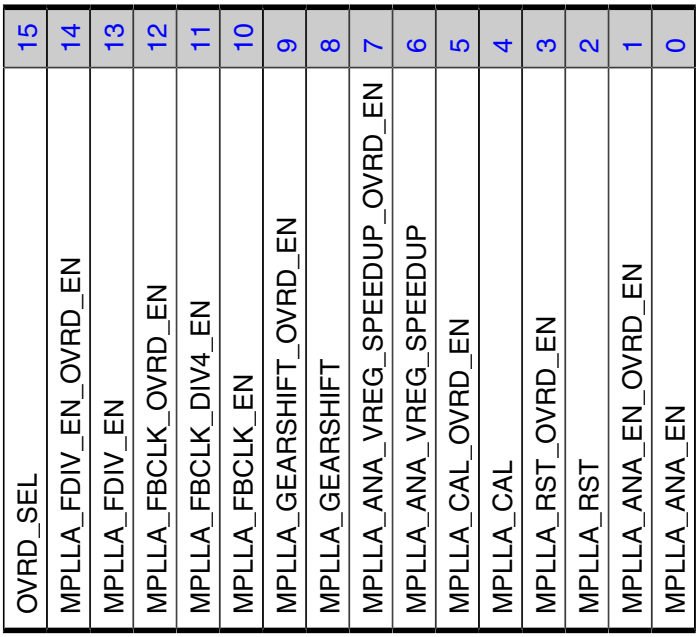


Table 2-98 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT0

Bits	Name	Memory Access	Description
15	OVRD_SEL	R/W	This bit is the override enable for signals listed below. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLLA_FDIV_EN_OVRD_EN	R/W	This bit is the override enable for mplla_ana_fdiv_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLA_FDIV_EN	R/W	This bit is the override value for mplla_ana_fdiv_en signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_FBCLK_OVRD_EN	R/W	This bit is the override enable for mplla_fb_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-98 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT0 (continued)

Bits	Name	Memory Access	Description
11	MPLLA_FBCLK_DIV4_EN	R/W	This bit is the override value for mplla_ana_fb_clk_div4_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	MPLLA_FBCLK_EN	R/W	This bit is the override value for mplla_fb_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLA_GEARSHIFT_OVRD_EN	R/W	This bit is the override enable for mplla_ana_gearshift signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	MPLLA_GEARSHIFT	R/W	This bit is the override value for mplla_ana_gearshift signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	MPLLA_ANA_VREG_SPEEDUP_OVRD_EN	R/W	This bit is the override enable for mplla_ana_vreg_speedup signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	MPLLA_ANA_VREG_SPEEDUP	R/W	This bit is the override value for mplla_ana_vreg_speedup signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	MPLLA_CAL_OVRD_EN	R/W	This bit is the override enable for mplla_ana_cal signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLLA_CAL	R/W	This bit is the override value for mplla_ana_cal signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-98 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT0 (continued)

Bits	Name	Memory Access	Description
3	MPLLA_RST_OVRD_EN	R/W	This bit is the override enable for mplla_ana_rst signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_RST	R/W	This bit is the override value for mplla_ana_rst signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_ANA_EN_OVRD_EN	R/W	This bit is the override enable for mplla_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_ANA_EN	R/W	This bit is the override value for mplla_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.95 SUP_DIG_ANA_MPLLA_OVRD_OUT1

- **Description:** Override Value for MPLLA Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x81
- **Exists:** Always

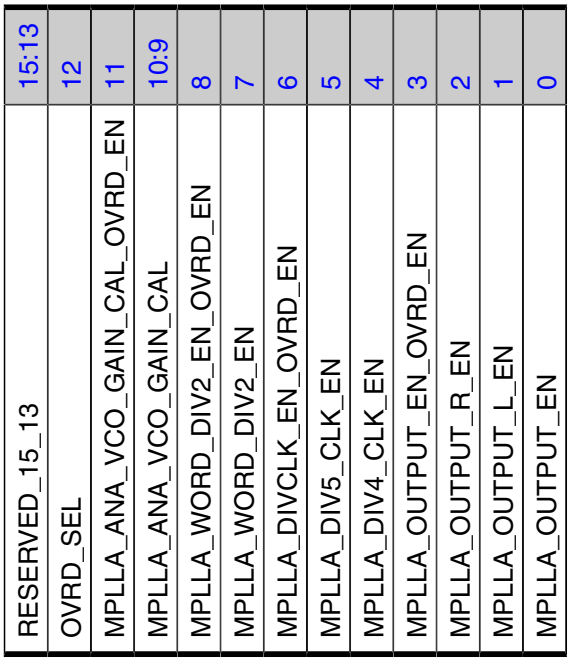


Table 2-99 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	OVRD_SEL	R/W	This bit is the global override enable for signals listed below. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MPLLA_ANA_VCO_GAIN_CAL_OVRD_EN	R/W	This bit is the override enable for mplla_ana_vco_gain_cal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:9	MPLLA_ANA_VCO_GAIN_CAL	R/W	This field is the override value for mplla_ana_vco_gain_cal. **Value After Reset:** 0x0

Table 2-99 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT1 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
8	MPLLA_WORD_DIV2_EN_OVRD_EN	R/W	This bit overrides the mplla_ana_word_div2_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	MPLLA_WORD_DIV2_EN	R/W	This bit overrides the mplla_ana_word_div2_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	MPLLA_DIVCLK_EN_OVRD_EN	R/W	This bit overrides the mplla_ana_div5_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLLA_DIV5_CLK_EN	R/W	This bit overrides the mplla_ana_div5_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLLA_DIV4_CLK_EN	R/W	This bit overrides the mplla_ana_div4_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLA_OUTPUT_EN_OVRD_EN	R/W	This bit overrides the mplla_ana_output_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_OUTPUT_R_EN	R/W	This bit overrides the mplla_ana_output_r_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_OUTPUT_L_EN	R/W	This bit overrides the mplla_ana_output_l_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-99 Fields for Register: SUP_DIG_ANA_MPLLA_OVRD_OUT1 (continued)

Bits	Name	Memory Access	Description
0	MPLLA_OUTPUT_EN	R/W	This bit overrides the mplla_ana_output_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.96 SUP_DIG_ANA_MPLLB_OVRD_OUT0

- **Description:** Override Value for MPLLB Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x82
- **Exists:** Always

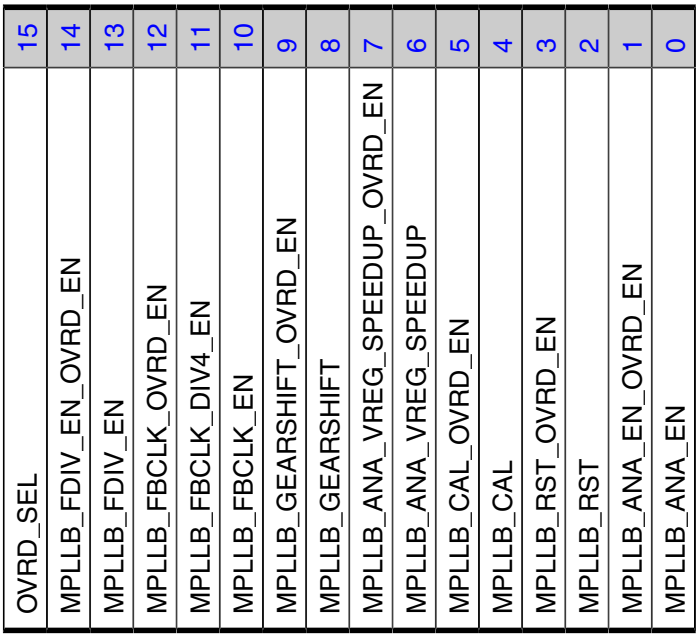


Table 2-100 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT0

Bits	Name	Memory Access	Description
15	OVRD_SEL	R/W	This is the override bit for mpllb_ana outputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLLB_FDIV_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_fdiv_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	MPLLB_FDIV_EN	R/W	This bit is the override value for mpllb_ana_fdiv_en signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_FBCLK_OVRD_EN	R/W	This bit is the override enable for mpllb_fb_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-100 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT0 (continued)

Bits	Name	Memory Access	Description
11	MPLLB_FBCLK_DIV4_EN	R/W	This bit is the override value for mpllb_ana_fb_clk_div4_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	MPLLB_FBCLK_EN	R/W	This bit is the override value for mpllb_fb_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLB_GEARSHIFT_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_gearshift signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	MPLLB_GEARSHIFT	R/W	This bit is the override value for mpllb_ana_gearshift signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	MPLLB_ANA_VREG_SPEEDUP_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_vreg_speedup signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	MPLLB_ANA_VREG_SPEEDUP	R/W	This bit is the override value for mpllb_ana_vreg_speedup signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	MPLLB_CAL_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_cal signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLLB_CAL	R/W	This bit is the override value for mpllb_ana_cal signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-100 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT0 (continued)

Bits	Name	Memory Access	Description
3	MPLLB_RST_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_rst signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_RST	R/W	This bit is the override value for mpllb_ana_rst signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_ANA_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLB_ANA_EN	R/W	This bit is the override value for mpllb_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.97 SUP_DIG_ANA_MPLLB_OVRD_OUT1

- **Description:** Override Value for MPLLB Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x83
- **Exists:** Always

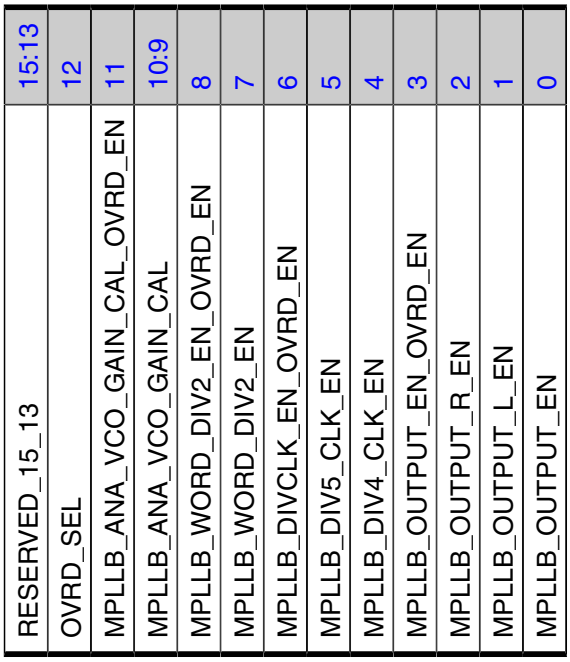


Table 2-101 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT1

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	OVRD_SEL	R/W	This is the override bit for all signals listed below. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MPLLB_ANA_VCO_GAIN_CAL_OVRD_EN	R/W	This bit is the override enable for mpllb_ana_vco_gain_cal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:9	MPLLB_ANA_VCO_GAIN_CAL	R/W	This field is the override enable for mpllb_ana_vco_gain_cal. **Value After Reset:** 0x0

Table 2-101 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT1 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
8	MPLLB_WORD_DIV2_EN_OVRD_EN	R/W	This bit overrides the mpllb_ana_word_div2_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	MPLLB_WORD_DIV2_EN	R/W	This bit overrides the mpllb_ana_word_div2_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	MPLLB_DIVCLK_EN_OVRD_EN	R/W	This bit overrides the mpllb_ana_div5_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLLB_DIV5_CLK_EN	R/W	This bit overrides the mpllb_ana_div5_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLLB_DIV4_CLK_EN	R/W	This bit overrides the mpllb_ana_div4_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_OUTPUT_EN_OVRD_EN	R/W	This bit overrides the mpllb_ana_output_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_OUTPUT_R_EN	R/W	This bit overrides the mpllb_ana_output_r_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_OUTPUT_L_EN	R/W	This bit overrides the mpllb_ana_output_l_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-101 Fields for Register: SUP_DIG_ANA_MPLLB_OVRD_OUT1 (continued)

Bits	Name	Memory Access	Description
0	MPLLB_OUTPUT_EN	R/W	This bit overrides the mpllb_ana_output_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.98 SUP_DIG_ANA_RTUNE_OVRD_OUT

- **Description:** Override Value for RTUNE Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x84
- **Exists:** Always

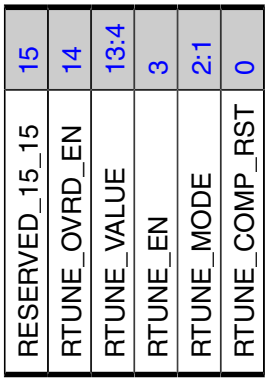


Table 2-102 Fields for Register: SUP_DIG_ANA_RTUNE_OVRD_OUT

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RTUNE_OVRD_EN	R/W	This is an override bit for rtune (rt_ana_* and term) outputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:4	RTUNE_VALUE	R/W	This field overrides the rt_ana_value[9:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
3	RTUNE_EN	R/W	This bit overrides the rt_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:1	RTUNE_MODE	R/W	This field overrides the rt_ana_mode[1:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-102 Fields for Register: SUP_DIG_ANA_RTUNE_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
0	RTUNE_COMP_RST	R/W	This bit overrides the rt_ana_comp_rst signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.99 SUP_DIG_ANA_STAT

- **Description:** SUP Input Status Register for SUP ANA Outputs.
- **Size:** 16 bits
- **Offset:** 0x85
- **Exists:** Always

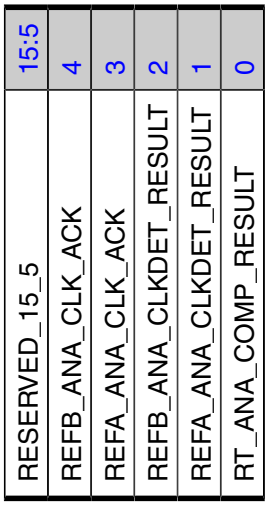


Table 2-103 Fields for Register: SUP_DIG_ANA_STAT

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	REFB_ANA_CLK_ACK	R	This bit is the value from ANA for refb_ana_clk_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	REFA_ANA_CLK_ACK	R	This bit is the value from ANA for refa_ana_clk_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	REFB_ANA_CLKDET_RESULT	R	This bit is the value from ANA for refb_ana_clkdet_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-103 Fields for Register: SUP_DIG_ANA_STAT (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
1	REFA_ANA_CLKDET_RESULT	R	This bit is the value from ANA for refa_ana_clkdet_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	RT_ANA_COMP_RESULT	R	This bit is the value from ANA for rt_ana_comp_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.100 SUP_DIG_ANA_ANA_OVRD_OUT

- **Description:** Override Values for ana_async_rst and bandgap Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x86
- **Exists:** Always

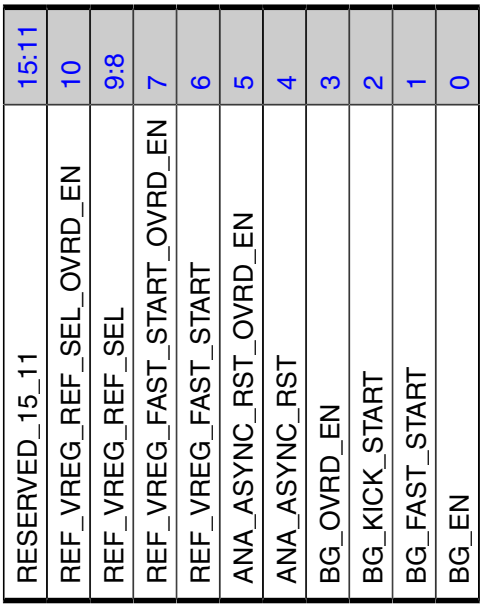


Table 2-104 Fields for Register: SUP_DIG_ANA_ANA_OVRD_OUT

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	REF_VREG_REF_SEL_OVRD_EN	R/W	This bit is the override enable for ref_ana_vreg_ref_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:8	REF_VREG_REF_SEL	R/W	This field is the override value for ref_ana_vreg_ref_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
7	REF_VREG_FAST_START_OVRD_EN	R/W	This bit is the override enable for ref_ana_vreg_fast_start. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-104 Fields for Register: SUP_DIG_ANA_ANA_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
6	REF_VREG_FAST_START	R/W	This bit is the override value for ref_ana_vreg_fast_start. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	ANA_ASYNC_RST_OVRD_EN	R/W	This bit is the override enable for ana_async_rst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ANA_ASYNC_RST	R/W	This bit is the override value for reset register for analog latches. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	BG_OVRD_EN	R/W	This is the override bit for bandgap outputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	BG_KICK_START	R/W	This bit overrides the bg_ana_kick_start_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	BG_FAST_START	R/W	This bit overrides the bg_ana_vref_fast_start signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	BG_EN	R/W	This bit overrides the bg_ana_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.101 SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_0

- **Description:** Override Value for MPLLA PMIX Signals Going to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x87
- **Exists:** Always

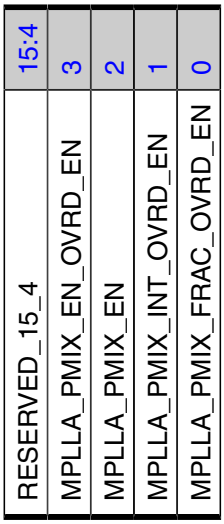


Table 2-105 Fields for Register: SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_0

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	MPLLA_PMIX_EN_OVRD_EN	R/W	This is the override bit for mplla_ana_pmix_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_PMIX_EN	R/W	This bit is the override value for mplla_ana_pmix_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_PMIX_INT_OVRD_EN	R/W	This is the override bit for mplla_ana_pmix_int signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_PMIX_FRAC_OVRD_EN	R/W	This is the override bit for mplla_ana_pmix_frac signal. **Value After Reset:** 0x0

Table 2-105 Fields for Register: SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.102 SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_1

- **Description:** Override Value for MPLLA PMIX Signals Going to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x88
- **Exists:** Always

Table 2-106 Fields for Register: SUP_DIG_ANA_MPLLA_PMIX_OVRD_OUT_1

Bits	Name	Memory Access	Description
15:6	MPLLA_PMIX_INT	R/W	This field is the override value for mplla_ana_pmix_int signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
5:0	MPLLA_PMIX_FRAC	R/W	This field is the override value for mplla_ana_pmix_frac signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.103 SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_0

- **Description:** Override Value for MPLLB PMIX Signals Going to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x89
- **Exists:** Always

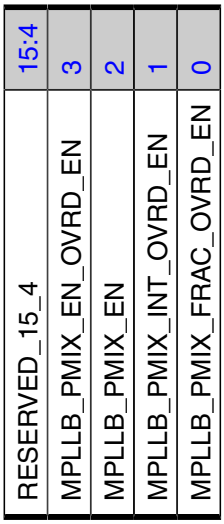


Table 2-107 Fields for Register: SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_0

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	MPLLB_PMIX_EN_OVRD_EN	R/W	This is the override bit for mpllb_ana_pmix_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_PMIX_EN	R/W	This bit is the override value for mpllb_ana_pmix_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_PMIX_INT_OVRD_EN	R/W	This is the override bit for mpllb_ana_pmix_int signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLB_PMIX_FRAC_OVRD_EN	R/W	This is the override bit for mpllb_ana_pmix_frac signal. **Value After Reset:** 0x0

Table 2-107 Fields for Register: SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.104 SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_1

- **Description:** Override Value for MPLLB PMIX Signals Going to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x8a
- **Exists:** Always

Table 2-108 Fields for Register: SUP_DIG_ANA_MPLLB_PMIX_OVRD_OUT_1

Bits	Name	Memory Access	Description
15:6	MPLLB_PMIX_INT	R/W	This field is the override value for mpllb_ana_pmix_int signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
5:0	MPLLB_PMIX_FRAC	R/W	This field is the override value for mpllb_ana_pmix_frac signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.105 SUP_DIG_ANA_REF_DCO_OVRD_OUT

- **Description:** Control Settings to Analog for Prescaler Digitally Controlled Oscillator (DCO).
- **Size:** 16 bits
- **Offset:** 0x8b
- **Exists:** Always

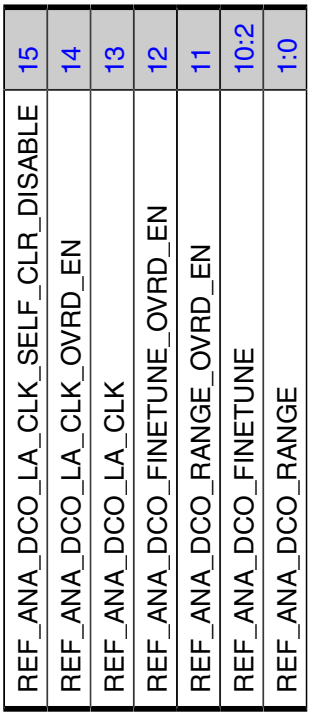


Table 2-109 Fields for Register: SUP_DIG_ANA_REF_DCO_OVRD_OUT

Bits	Name	Memory Access	Description
15	REF_ANA_DCO_LA_CLK_SELF_CLR_DISABLE	R/W	Disable self-clearing for the ref_ana_dco_la_clk register **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	REF_ANA_DCO_LA_CLK_OVRD_EN	R/W	This bit is the override enable for ref_ana_dco_la_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	REF_ANA_DCO_LA_CLK	R/W	This bit is the override value for ref_ana_dco_la_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	REF_ANA_DCO_FINETUNE_OVRD_EN	R/W	This field is the override enable for ref_ana_dco_finetune. **Value After Reset:** 0x0

Table 2-109 Fields for Register: SUP_DIG_ANA_REF_DCO_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11	REF_ANA_DCO_RANGE_OVRD_EN	R/W	This field is the override enable for ref_ana_dco_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:2	REF_ANA_DCO_FINETUNE	R/W	This field is the value for ref_ana_dco_finetune. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff
1:0	REF_ANA_DCO_RANGE	R/W	This field is the value for ref_ana_dco_range. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.106 SUP_DIG_ANA_SUP_MISC_OVRD_OUT

- **Description:** Override Value for sup_misc Bus Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x8c
- **Exists:** Always

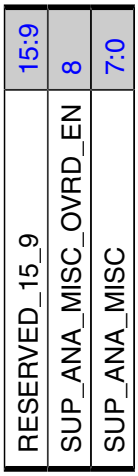


Table 2-110 Fields for Register: SUP_DIG_ANA_SUP_MISC_OVRD_OUT

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	SUP_ANA_MISC_OVRD_EN	R/W	This bit is the value for sup_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	SUP_ANA_MISC	R/W	This field is the override value for sup_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.107 SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_0

- **Description:** MPLL Calibration Controls, Register 0.
- **Size:** 16 bits
- **Offset:** 0xa0
- **Exists:** Always

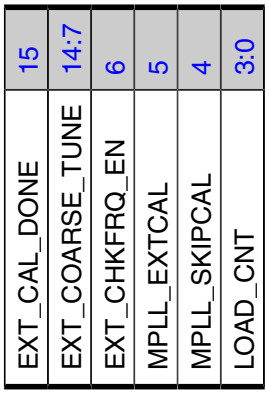


Table 2-111 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_0

Bits	Name	Memory Access	Description
15	EXT_CAL_DONE	R/W	This bit sets the external calibration status to done. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:7	EXT_COARSE_TUNE	R/W	This field is the value of mpll_ana_coarse_tune_i[7:0] in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
6	EXT_CHKFRQ_EN	R/W	This bit checks the frequency of the MPLL. This is only valid in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLL_EXTCAL	R/W	This bit enables external calibration of MPLL. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLL_SKIPCAL	R/W	This bit skips automatic (internal) calibration of MPLL (and also skips external calibration if it is enabled). If skipcal is enabled, then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset:** 0x1

Table 2-111 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3:0	LOAD_CNT	R/W	This field is the MSBs for the CHKFRQ FSM ld_val[10:0] load value. ■ 0x0 - gives a ld_val of 0, no PPM difference can be detected ■ 0x8 - gives a load value of 1024, 3000PPM resolution possible ■ 0xA - gives a load value of 1280, 2343PPM resolution possible ■ 0xB - gives a load value of 1408, 2130PPM resolution possible ■ 0xC - gives a load value of 1536, 1953PPM resolution possible ■ 0xF - gives a load value of 1920, 1600PPM resolution possible **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf

2.1.108 SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1

- **Description:** MPLL Calibration Controls, Register 1.
- **Size:** 16 bits
- **Offset:** 0xa1
- **Exists:** Always

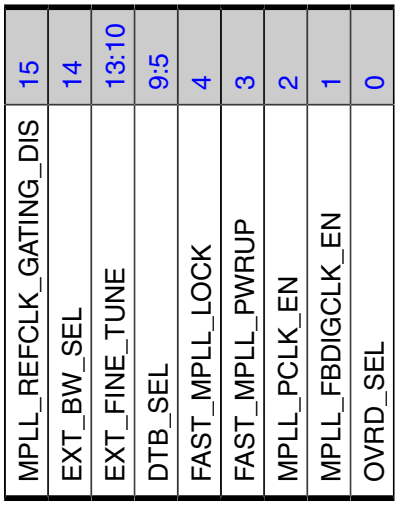


Table 2-112 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1

Bits	Name	Memory Access	Description
15	MPLL_REFCLK_GATING_DIS	R/W	This bit disables waiting for the IDLE state to deassert ref_ana_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	EXT_BW_SEL	R/W	This bit is the value of mpll_bw_sel in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	EXT_FINE_TUNE	R/W	This field is the value of mpll_ana_fine_tune_i[3:0] in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
9:5	DTB_SEL	R/W	This field is the DTB select for MPLL dtb signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

Table 2-112 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_1 (continued)

Bits	Name	Memory Access	Description
4	FAST_MPLL_LOCK	R/W	This bit enables fast MPLL locking. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	FAST_MPLL_PWRUP	R/W	This bit enables fast MPLL powerup. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLL_PCLK_EN	R/W	This bit overrides the PWR FSM mpll_pclk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLL_FBDIGCLK_EN	R/W	This bit overrides the PWR FSM mpll_fb_dig_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	OVRD_SEL	R/W	This bit is the override enable bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.109 SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_2

- **Description:** MPLL Calibration Controls, Register 2.
- **Size:** 16 bits
- **Offset:** 0xa2
- **Exists:** Always

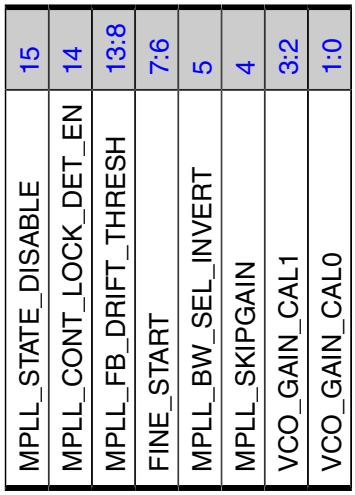


Table 2-113 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_2

Bits	Name	Memory Access	Description
15	MPLL_STATE_DISABLE	R/W	This bit enables de-asserting mpll_state when MPLL VCO drift away. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLL_CONT_LOCK_DET_EN	R/W	This bit enables continous MPLL lock detection **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:8	MPLL_FB_DRIFT_THRESH	R/W	This field is the FB clock drift threshold **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x3f
7:6	FINE_START	R/W	This field is the value for the starting point of the MPLL fine bank. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3

Table 2-113 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_2 (continued)

Bits	Name	Memory Access	Description
5	MPLL_BW_SEL_INVERT	R/W	This bit selects low bandwidth if **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLL_SKIPGAIN	R/W	This bit skips gain calibration of MPLL. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	VCO_GAIN_CAL1	R/W	This field is the value of vco_gain_cal[1:0] = 2'b10. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3
1:0	VCO_GAIN_CAL0	R/W	This field is the value of vco_gain_cal[1:0] = 2'b01. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.110 SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT

- **Description:** MPLL Status Register.
- **Size:** 16 bits
- **Offset:** 0xa3
- **Exists:** Always

15	14	13	12	11	10	9	8	7	6	5	4	3:0
MPLL_ANA_VREG_SPEEDUP	MPLL_ANA_EN	MPLL_RST	MPLL_CAL	MPLL_FBCLK_EN	MPLL_OUTPUT_EN	MPLL_PCLK_EN	MPLL_L_LANES	MPLL_R_LANES	MPLL_CAL_RDY	CHKFRQ_DONE	MPLL_TOOSLOW	FSM_STATE

Table 2-114 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT

Bits	Name	Memory Access	Description
15	MPLL_ANA_VREG_SPEEDUP	R	This bit is the current value of mpll_ana_vreg_speedup_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLL_ANA_EN	R	This bit is the current value of mpll_ana_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLL_RST	R	This bit is the current value of mpll_ana_rst_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_CAL	R	This bit is the current value of mpll_ana_cal_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-114 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT (continued)

Bits	Name	Memory Access	Description
11	MPLL_FBCLK_EN	R	This bit is the current value of mpll_ana_fb_clk_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	MPLL_OUTPUT_EN	R	This bit is the current value of mpll_ana_output_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	MPLL_PCLK_EN	R	This bit is the current value of mpll_pclk_en. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLL_L_LANES	R	This bit is the current value of lane_mpll_l. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	MPLL_R_LANES	R	This bit is the current value of lane_mpll_r. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	MPLL_CAL_RDY	R	This bit is the current value of mpll_cal_rdy. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	CHKFRQ_DONE	R	This bit is the current value of mpll_chkfrq_done. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-114 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_STAT (continued)

Bits	Name	Memory Access	Description
4	MPLL_TOOSLOW	R	This bit is the current value of mpll_tooslow. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	FSM_STATE	R	This field is the current value of the PWR FSM state register. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.111 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

- **Description:** MPLL Power Up Time Register 0.
- **Size:** 16 bits
- **Offset:** 0xa4
- **Exists:** Always

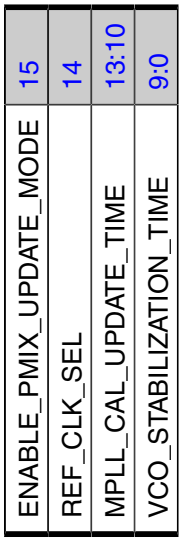


Table 2-115 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15	ENABLE_PMIX_UPDATE_MODE	R/W	Whether to check MPLL status when updating enable_pmix **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	REF_CLK_SEL	R/W	This bit enbale using the ref_dig_clk, ref_range_clk instead of sup_ref_dig_clk and sup_ref_range_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	MPLL_CAL_UPDATE_TIME	R/W	This field is the threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xf
9:0	VCO_STABILIZATION_TIME	R/W	This field is the threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles (default 12us). This timer is used when mpll_short_lock_en_i is set to 0. **Value After Reset:** 0x138

Table 2-115 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3ff

2.1.112 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

- **Description:** MPLL Power Up Time Register 1.
- **Size:** 16 bits
- **Offset:** 0xa5
- **Exists:** Always

Table 2-116 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:11	PCLK_EN_TIME	R/W	This field is the threshold for the PCLK enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
10:0	VCO_CLK_STABILIZATION_TIME	R/W	This field is the threshold for the VCO clock stabilization timer in terms of number of ref_range_clk cycles (default is 3us). This timer is used when mpll_short_lock_en_i is set to 0. **Value After Reset:** 0x4e **Exists:** Always **Reset Mask:** 0x7ff

2.1.113 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

- **Description:** MPLL Power Up Time Register 2.
- **Size:** 16 bits
- **Offset:** 0xa6
- **Exists:** Always

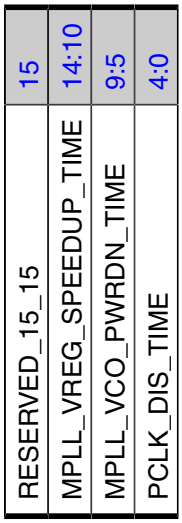


Table 2-117 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:10	MPLL_VREG_SPEEDUP_TIME	R/W	This field is the threshold for the MPLL vreg speedup timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xb **Exists:** Always **Reset Mask:** 0x1f
9:5	MPLL_VCO_PWRDN_TIME	R/W	This field is the threshold for the MPLL VCO power down timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
4:0	PCLK_DIS_TIME	R/W	This field is the threshold for the PCLK disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f

2.1.114 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

- **Description:** MPLL Power Up Time Register 3.
- **Size:** 16 bits
- **Offset:** 0xa7
- **Exists:** Always

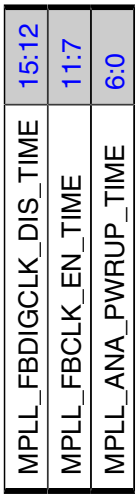


Table 2-118 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:12	MPLL_FBDIGCLK_DIS_TIME	R/W	This field is the threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xf
11:7	MPLL_FBCLK_EN_TIME	R/W	This field is the threshold for the MPLL feedback digital clock enable timer in terms of number of mpll_fb_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
6:0	MPLL_ANA_PWRUP_TIME	R/W	This field is the threshold for the MPLL analog power up timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0x7f

2.1.115 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_TUNE_VAL

- **Description:** MPLL Coarse and Fine Tune Value Registers.
- **Size:** 16 bits
- **Offset:** 0xa8
- **Exists:** Always

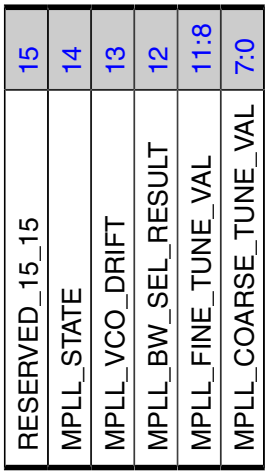


Table 2-119 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_TUNE_VAL

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLL_STATE	R	This bit is the current value of mpll_state. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLL_VCO_DRIFT	R	This bit indicates if the MPLL lost LOCK while in the ON state . (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_BW_SEL_RESULT	R	This bit is the current value of mpll_bw_sel_result. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-119 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_TUNE_VAL (continued)

Bits	Name	Memory Access	Description
11:8	MPLL_FINE_TUNE_VAL	R	This field is the current value of mpll_ana_fine_tune_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	MPLL_COARSE_TUNE_VAL	R	This field is the current value of mpll_ana_coarse_tune_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.116 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE

- **Description:** Value for MPLL Coarse and Fine Tune when Skipping Calibration.
- **Size:** 16 bits
- **Offset:** 0xa9
- **Exists:** Always

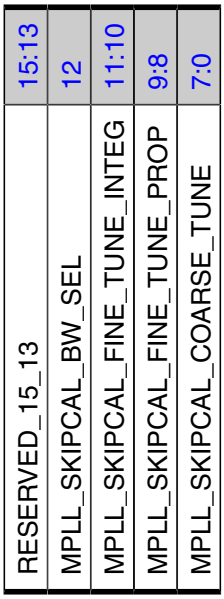


Table 2-120 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_SKIPCAL_BW_SEL	R/W	This bit is the value for MPLL bw_sel when calibration or gain calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:10	MPLL_SKIPCAL_FINE_TUNE_INTEG	R/W	This field is the value for MPLL fine_tune (integral portion) when calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:8	MPLL_SKIPCAL_FINE_TUNE_PROP	R/W	This field is the value for MPLL fine_tune (proportional portion) when calibration is skipped. **Value After Reset:** 0x0

Table 2-120 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
7:0	MPLL_SKIPCAL_COARSE_TUNE	R/W	This field is the value for MPLL coarse_tune when calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.117 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0

- **Description:** MPLL Power Up FSM Config, Register 0.
- **Size:** 16 bits
- **Offset:** 0xaa
- **Exists:** Always

Table 2-121 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_0	R/W	This field is the value for the ending point of the MPLL coarse range 0. **Value After Reset:** 0x28 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_0	R/W	This field is the value for the starting point of the MPLL coarse range 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.118 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1

- **Description:** MPLL Power Up FSM Config, Register 1.
- **Size:** 16 bits
- **Offset:** 0xab
- **Exists:** Always

Table 2-122 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_1	R/W	This field is the value for the ending point of the MPLL coarse range 1. **Value After Reset:** 0x50 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_1	R/W	This field is the value for the starting point of the MPLL coarse range 1. **Value After Reset:** 0x18 **Exists:** Always **Reset Mask:** 0xff

2.1.119 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2

- **Description:** MPLL Power Up FSM Config, Register 2.
- **Size:** 16 bits
- **Offset:** 0xac
- **Exists:** Always

Table 2-123 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_2	R/W	This field is the value for the ending point of the MPLL coarse range 2. **Value After Reset:** 0x78 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_2	R/W	This field is the value for the starting point of the MPLL coarse range 2. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0xff

2.1.120 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3

- **Description:** MPLL Power Up FSM Config, Register 3.
- **Size:** 16 bits
- **Offset:** 0xad
- **Exists:** Always

Table 2-124 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_3	R/W	This field is the value for the ending point of the MPLL coarse range 3. **Value After Reset:** 0x7f **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_3	R/W	This field is the value for the starting point of the MPLL coarse range 3. **Value After Reset:** 0x68 **Exists:** Always **Reset Mask:** 0xff

2.1.121 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

- **Description:** Thresholds for MPLL CAL Update Timer and MPLL VCO Stabilization Short Timer.
- **Size:** 16 bits
- **Offset:** 0xae
- **Exists:** Always

Table 2-125 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VCO_STABILIZATION_SHORT_TIME	R/W	This field is the threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles (default is 4us). This timer is used when mpll_short_lock_en_i is set to 1. **Value After Reset:** 0x68 **Exists:** Always **Reset Mask:** 0x3ff

2.1.122 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5

- **Description:** Thresholds for PCLK Enable and MPLL VCO Clock Stabilization Short Timer.
- **Size:** 16 bits
- **Offset:** 0xaf
- **Exists:** Always

Table 2-126 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:0	VCO_CLK_STABILIZATION_SHORT_TIME	R/W	This field is the threshold for the VCO clock stabilization short timer in terms of number of ref_range_clk cycles (default is 3.5us). This timer is used when mpll_short_lock_en_i is set to 1. . **Value After Reset:** 0x5b **Exists:** Always **Reset Mask:** 0x7ff

2.1.123 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6

- **Description:** MPLL Power Up Time Register 6.
- **Size:** 16 bits
- **Offset:** 0xb0
- **Exists:** Always

Table 2-127 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	MPLL_FDIV_EN_TIME	R/W	This field is the threshold for the MPLL fdiv enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff

2.1.124 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7

- **Description:** MPLL Power Up Time Register 7.
- **Size:** 16 bits
- **Offset:** 0xb1
- **Exists:** Always

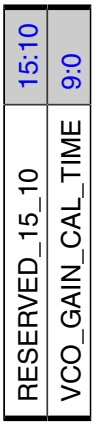


Table 2-128 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VCO_GAIN_CAL_TIME	R/W	This field is the threshold for the MPLL VCO control voltage setlling timer in terms of number of ref_range_clk cycles (default is 10us). **Value After Reset:** 0x104 **Exists:** Always **Reset Mask:** 0x3ff

2.1.125 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL

- **Description:** MPLL VCO Gain Calibration FB Count1 Registers.
- **Size:** 16 bits
- **Offset:** 0xb2
- **Exists:** Always

Table 2-129 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_ANA_VCO_FB_CNT1_VAL	R	This field is the current value of mpll_ana_vco_fb_cnt1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.126 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL

- **Description:** MPLL VCO Gain Calibration FB Count2 Registers.
- **Size:** 16 bits
- **Offset:** 0xb3
- **Exists:** Always

Table 2-130 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_ANA_VCO_FB_CNT2_VAL	R	This field is the current value of mpll_ana_vco_fb_cnt2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.127 SUP_DIG_MPLLA_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0

- **Description:** MPLL VCO Gain Calibration Controls, Register 0.
- **Size:** 16 bits
- **Offset:** 0xb4
- **Exists:** Always

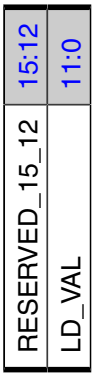


Table 2-131 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	LD_VAL	R/W	This field is the value of ld_val. **Value After Reset:** 0xc4e **Exists:** Always **Reset Mask:** 0xfff

2.1.128 SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL

- **Description:** MPLL VCO Drift Count Register.
- **Size:** 16 bits
- **Offset:** 0xb6
- **Exists:** Always

Table 2-132 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_FB_DRIFT_CNT_VAL	R	This field is the current value of mpll_fb_drift_cnt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.129 SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_3

- **Description:** MPLL Calibration Controls, Register 3
- **Size:** 16 bits
- **Offset:** 0xb7
- **Exists:** Always

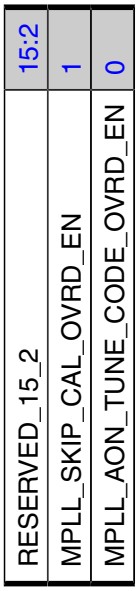


Table 2-133 Fields for Register: SUP_DIG_MPLLA_MPLL_PWR_CTL_CAL_CTRL_3

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLL_SKIP_CAL_OVRD_EN	R/W	Override the mpll_skip_cal_i input When enabled, the MPLL will use the skip flag value in the MPLL_SKIPCAL field **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	MPLL_AON_TUNE_CODE_OVRD_EN	R/W	Override the mpll_aon_tune_code_i input When enabled, the MPLL will use the coarse tune value in the MPLL_SKIPCAL_COARSE_TUNE field **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.130 SUP_DIG_MPLLA_SSC_FRAC_OUT

- **Description:** frac_out Override for Fractional Control.
- **Size:** 16 bits
- **Offset:** 0xb8
- **Exists:** Always

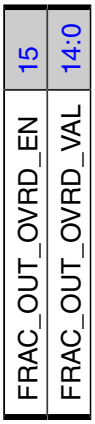


Table 2-134 Fields for Register: SUP_DIG_MPLLA_SSC_FRAC_OUT

Bits	Name	Memory Access	Description
15	FRAC_OUT_OVRD_EN	R/W	This bit is the frac_out override enable. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14:0	FRAC_OUT_OVRD_VAL	R/W	This field is the frac_out[19:5] override value. This register will override the MSB's <19:5> of the frac_out register. When overridden, the LSBs <4:0> will be set to 5'd0. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff **Volatile:** true

2.1.131 SUP_DIG_MPLLA_SSC_SSC_RAMP

- **Description:** ssc_ramp Override for Spread Spectrum Control.
- **Size:** 16 bits
- **Offset:** 0xb9
- **Exists:** Always

Table 2-135 Fields for Register: SUP_DIG_MPLLA_SSC_SSC_RAMP

Bits	Name	Memory Access	Description
15	SSC_RAMP_OVRD_EN	R/W	This bit is the ssc_ramp override enable. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14:0	SSC_RAMP_OVRD_VAL	R/W	This field is the ssc_ramp[20:6] override value. This register will override the MSB's <20:6> of the ssc_ramp register. When overridden, the LSBs <5:0> will be set to 6'd0. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff **Volatile:** true

2.1.132 SUP_DIG_MPLLA_SSC_CONFIG

- **Description:** Configuration Register for SSC and Fractional Control.
- **Size:** 16 bits
- **Offset:** 0xba
- **Exists:** Always

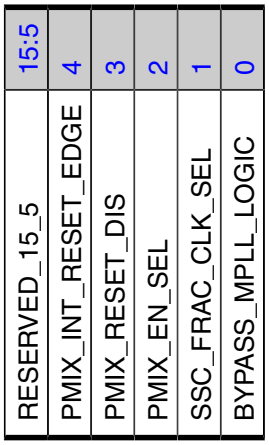


Table 2-136 Fields for Register: SUP_DIG_MPLLA_SSC_CONFIG

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	PMIX_INT_RESET_EDGE	R/W	This bit selects which ssc_en edge is used to reet pmix_int[1:0]. ■ 0x0 - ssc_en_fall_edge ■ 0x1 - ssc_en_rise_edge **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	PMIX_RESET_DIS	R/W	This bit disables reseting the SSC when ssc_en is asserting. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	PMIX_EN_SEL	R/W	This bit selects mpll_pmix_en or ssc_en as the enable for SSC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-136 Fields for Register: SUP_DIG_MPLLA_SSC_CONFIG (continued)

Bits	Name	Memory Access	Description
1	SSC_FRAC_CLK_SEL	R/W	This bit is the ssc_frac_clk select. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	BYPASS_MPLL_LOGIC	R/W	This bit is to bypass the mint_i[11:0] override logic. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.133 SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_0

- **Description:** MPLL Calibration Controls, Register 0.
- **Size:** 16 bits
- **Offset:** 0xc0
- **Exists:** Always

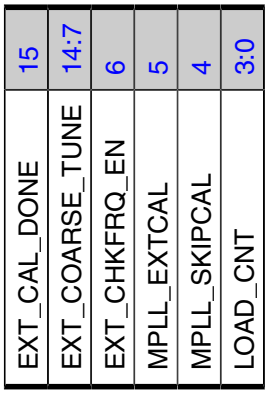


Table 2-137 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_0

Bits	Name	Memory Access	Description
15	EXT_CAL_DONE	R/W	This bit sets the external calibration status to done. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:7	EXT_COARSE_TUNE	R/W	This field is the value of mpll_ana_coarse_tune_i[7:0] in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
6	EXT_CHKFRQ_EN	R/W	This bit checks the frequency of the MPLL. This is only valid in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLL_EXTCAL	R/W	This bit enables external calibration of MPLL. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLL_SKIPCAL	R/W	This bit skips automatic (internal) calibration of MPLL (and also skips external calibration if it is enabled). If skipcal is enabled, then MPLL_SKIPCAL_COARSE_TUNE is used. **Value After Reset:** 0x1

Table 2-137 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3:0	LOAD_CNT	R/W	This field is the MSBs for the CHKFRQ FSM ld_val[10:0] load value. ■ 0x0 - gives a ld_val of 0, no PPM difference can be detected ■ 0x8 - gives a load value of 1024, 3000PPM resolution possible ■ 0xA - gives a load value of 1280, 2343PPM resolution possible ■ 0xB - gives a load value of 1408, 2130PPM resolution possible ■ 0xC - gives a load value of 1536, 1953PPM resolution possible ■ 0xF - gives a load value of 1920, 1600PPM resolution possible **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf

2.1.134 SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1

- **Description:** MPLL Calibration Controls, Register 1.
- **Size:** 16 bits
- **Offset:** 0xc1
- **Exists:** Always

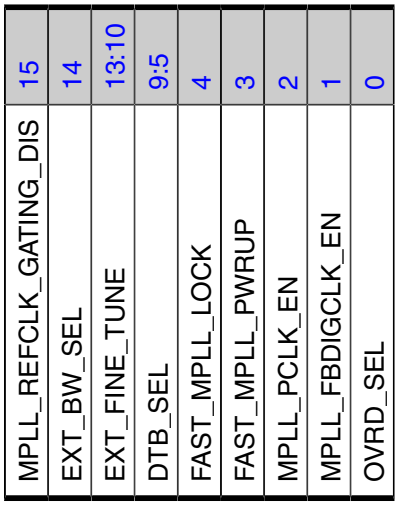


Table 2-138 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1

Bits	Name	Memory Access	Description
15	MPLL_REFCLK_GATING_DIS	R/W	This bit disables waiting for the IDLE state to deassert ref_ana_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	EXT_BW_SEL	R/W	This bit is the value of mpll_bw_sel in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	EXT_FINE_TUNE	R/W	This field is the value of mpll_ana_fine_tune_i[3:0] in external calibration mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
9:5	DTB_SEL	R/W	This field is the DTB select for MPLL dtb signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

Table 2-138 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_1 (continued)

Bits	Name	Memory Access	Description
4	FAST_MPLL_LOCK	R/W	This bit enables fast MPLL locking. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	FAST_MPLL_PWRUP	R/W	This bit enables fast MPLL powerup. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLL_PCLK_EN	R/W	This bit overrides the PWR FSM mpll_pclk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLL_FBDIGCLK_EN	R/W	This bit overrides the PWR FSM mpll_fb_dig_clk_en signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	OVRD_SEL	R/W	This bit is the override enable bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.135 SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_2

- **Description:** MPLL Calibration Controls, Register 2.
- **Size:** 16 bits
- **Offset:** 0xc2
- **Exists:** Always

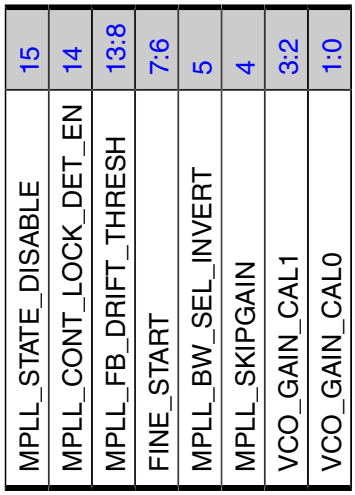


Table 2-139 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_2

Bits	Name	Memory Access	Description
15	MPLL_STATE_DISABLE	R/W	This bit enables de-asserting mpll_state when MPLL VCO drift away. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	MPLL_CONT_LOCK_DET_EN	R/W	This bit enables continous MPLL lock detection **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:8	MPLL_FB_DRIFT_THRESH	R/W	This field is the FB clock drift threshold **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x3f
7:6	FINE_START	R/W	This field is the value for the starting point of the MPLL fine bank. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3

Table 2-139 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_2 (continued)

Bits	Name	Memory Access	Description
5	MPLL_BW_SEL_INVERT	R/W	This bit selects low bandwidth if **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLL_SKIPGAIN	R/W	This bit skips gain calibration of MPLL. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	VCO_GAIN_CAL1	R/W	This field is the value of vco_gain_cal[1:0] = 2'b10. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3
1:0	VCO_GAIN_CAL0	R/W	This field is the value of vco_gain_cal[1:0] = 2'b01. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.136 SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT

- **Description:** MPLL Status Register.
- **Size:** 16 bits
- **Offset:** 0xc3
- **Exists:** Always

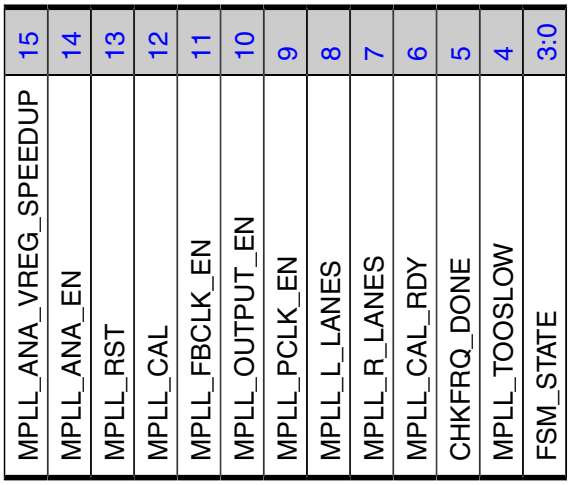


Table 2-140 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT

Bits	Name	Memory Access	Description
15	MPLL_ANA_VREG_SPEEDUP	R	This bit is the current value of mpll_ana_vreg_speedup_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLL_ANA_EN	R	This bit is the current value of mpll_ana_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLL_RST	R	This bit is the current value of mpll_ana_rst_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_CAL	R	This bit is the current value of mpll_ana_cal_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-140 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT (continued)

Bits	Name	Memory Access	Description
11	MPLL_FBCLK_EN	R	This bit is the current value of mpll_ana_fb_clk_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	MPLL_OUTPUT_EN	R	This bit is the current value of mpll_ana_output_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	MPLL_PCLK_EN	R	This bit is the current value of mpll_pclk_en. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	MPLL_L_LANES	R	This bit is the current value of lane_mpll_l. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	MPLL_R_LANES	R	This bit is the current value of lane_mpll_r. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	MPLL_CAL_RDY	R	This bit is the current value of mpll_cal_rdy. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	CHKFRQ_DONE	R	This bit is the current value of mpll_chkfrq_done. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-140 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_STAT (continued)

Bits	Name	Memory Access	Description
4	MPLL_TOOSLOW	R	This bit is the current value of mpll_tooslow. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	FSM_STATE	R	This field is the current value of the PWR FSM state register. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.137 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

- **Description:** MPLL Power Up Time Register 0.
- **Size:** 16 bits
- **Offset:** 0xc4
- **Exists:** Always

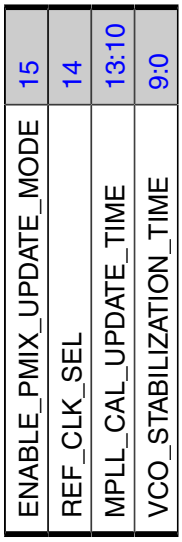


Table 2-141 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0

Bits	Name	Memory Access	Description
15	ENABLE_PMIX_UPDATE_MODE	R/W	Whether to check MPLL status when updating enable_pmix **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	REF_CLK_SEL	R/W	This bit enbale using the ref_dig_clk, ref_range_clk instead of sup_ref_dig_clk and sup_ref_range_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	MPLL_CAL_UPDATE_TIME	R/W	This field is the threshold for the MPLL calibration control word update timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xf
9:0	VCO_STABILIZATION_TIME	R/W	This field is the threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles (default 12us). This timer is used when mpll_short_lock_en_i is set to 0. **Value After Reset:** 0x138

Table 2-141 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_0 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3ff

2.1.138 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

- **Description:** MPLL Power Up Time Register 1.
- **Size:** 16 bits
- **Offset:** 0xc5
- **Exists:** Always

Table 2-142 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_1

Bits	Name	Memory Access	Description
15:11	PCLK_EN_TIME	R/W	This field is the threshold for the PCLK enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
10:0	VCO_CLK_STABILIZATION_TIME	R/W	This field is the threshold for the VCO clock stabilization timer in terms of number of ref_range_clk cycles (default is 3us). This timer is used when mpll_short_lock_en_i is set to 0. **Value After Reset:** 0x4e **Exists:** Always **Reset Mask:** 0x7ff

2.1.139 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

- **Description:** MPLL Power Up Time Register 2.
- **Size:** 16 bits
- **Offset:** 0xc6
- **Exists:** Always

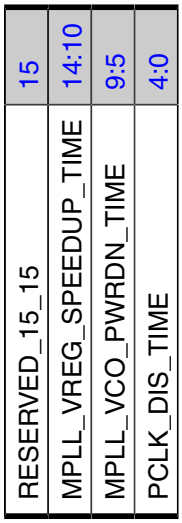


Table 2-143 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_2

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:10	MPLL_VREG_SPEEDUP_TIME	R/W	This field is the threshold for the MPLL vreg speedup timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xb **Exists:** Always **Reset Mask:** 0x1f
9:5	MPLL_VCO_PWRDN_TIME	R/W	This field is the threshold for the MPLL VCO power down timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
4:0	PCLK_DIS_TIME	R/W	This field is the threshold for the PCLK disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f

2.1.140 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

- **Description:** MPLL Power Up Time Register 3.
- **Size:** 16 bits
- **Offset:** 0xc7
- **Exists:** Always

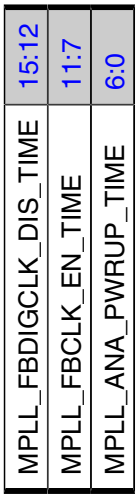


Table 2-144 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_3

Bits	Name	Memory Access	Description
15:12	MPLL_FBDIGCLK_DIS_TIME	R/W	This field is the threshold for the MPLL feedback digital clock disable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xf
11:7	MPLL_FBCLK_EN_TIME	R/W	This field is the threshold for the MPLL feedback digital clock enable timer in terms of number of mpll_fb_clk cycles. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
6:0	MPLL_ANA_PWRUP_TIME	R/W	This field is the threshold for the MPLL analog power up timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x43 **Exists:** Always **Reset Mask:** 0x7f

2.1.141 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_TUNE_VAL

- **Description:** MPLL Coarse and Fine Tune Value Registers.
- **Size:** 16 bits
- **Offset:** 0xc8
- **Exists:** Always

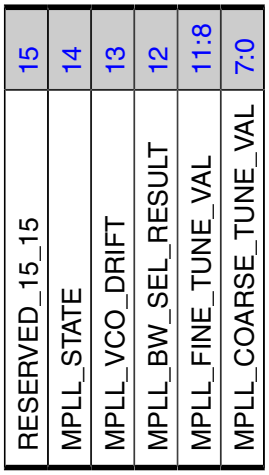


Table 2-145 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_TUNE_VAL

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MPLL_STATE	R	This bit is the current value of mpll_state. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLL_VCO_DRIFT	R	This bit indicates if the MPLL lost LOCK while in the ON state . (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_BW_SEL_RESULT	R	This bit is the current value of mpll_bw_sel_result. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-145 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_TUNE_VAL (continued)

Bits	Name	Memory Access	Description
11:8	MPLL_FINE_TUNE_VAL	R	This field is the current value of mpll_ana_fine_tune_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	MPLL_COARSE_TUNE_VAL	R	This field is the current value of mpll_ana_coarse_tune_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.142 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE

- **Description:** Value for MPLL Coarse and Fine Tune when Skipping Calibration.
- **Size:** 16 bits
- **Offset:** 0xc9
- **Exists:** Always

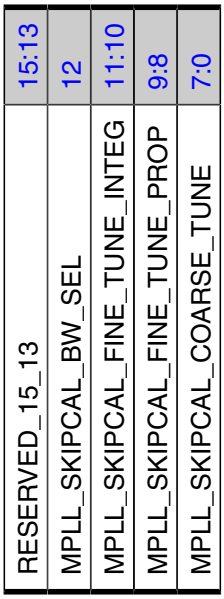


Table 2-146 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLL_SKIPCAL_BW_SEL	R/W	This bit is the value for MPLL bw_sel when calibration or gain calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:10	MPLL_SKIPCAL_FINE_TUNE_INTEG	R/W	This field is the value for MPLL fine_tune (integral portion) when calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:8	MPLL_SKIPCAL_FINE_TUNE_PROP	R/W	This field is the value for MPLL fine_tune (proportional portion) when calibration is skipped. **Value After Reset:** 0x0

Table 2-146 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_SKIPCAL_TUNE (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
7:0	MPLL_SKIPCAL_COARSE_TUNE	R/W	This field is the value for MPLL coarse_tune when calibration is skipped. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.143 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0

- **Description:** MPLL Power Up FSM Config, Register 0.
- **Size:** 16 bits
- **Offset:** 0xca
- **Exists:** Always

Table 2-147 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_0

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_0	R/W	This field is the value for the ending point of the MPLL coarse range 0. **Value After Reset:** 0x28 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_0	R/W	This field is the value for the starting point of the MPLL coarse range 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.144 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1

- **Description:** MPLL Power Up FSM Config, Register 1.
- **Size:** 16 bits
- **Offset:** 0xcb
- **Exists:** Always

Table 2-148 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_1

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_1	R/W	This field is the value for the ending point of the MPLL coarse range 1. **Value After Reset:** 0x50 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_1	R/W	This field is the value for the starting point of the MPLL coarse range 1. **Value After Reset:** 0x18 **Exists:** Always **Reset Mask:** 0xff

2.1.145 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2

- **Description:** MPLL Power Up FSM Config, Register 2.
- **Size:** 16 bits
- **Offset:** 0xcc
- **Exists:** Always

Table 2-149 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_2

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_2	R/W	This field is the value for the ending point of the MPLL coarse range 2. **Value After Reset:** 0x78 **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_2	R/W	This field is the value for the starting point of the MPLL coarse range 2. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0xff

2.1.146 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3

- **Description:** MPLL Power Up FSM Config, Register 3.
- **Size:** 16 bits
- **Offset:** 0xcd
- **Exists:** Always

Table 2-150 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWR_FSM_CFG_3

Bits	Name	Memory Access	Description
15:8	COARSE_LIMIT_3	R/W	This field is the value for the ending point of the MPLL coarse range 3. **Value After Reset:** 0x7f **Exists:** Always **Reset Mask:** 0xff
7:0	COARSE_START_3	R/W	This field is the value for the starting point of the MPLL coarse range 3. **Value After Reset:** 0x68 **Exists:** Always **Reset Mask:** 0xff

2.1.147 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

- **Description:** Thresholds for MPLL CAL Update Timer and MPLL VCO Stabilization Short Timer.
- **Size:** 16 bits
- **Offset:** 0xce
- **Exists:** Always

Table 2-151 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_4

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VCO_STABILIZATION_SHORT_TIME	R/W	This field is the threshold for the VCO stabilization timer in terms of number of ref_range_clk cycles (default is 4us). This timer is used when mpll_short_lock_en_i is set to 1. **Value After Reset:** 0x68 **Exists:** Always **Reset Mask:** 0x3ff

2.1.148 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5

- **Description:** Thresholds for PCLK Enable and MPLL VCO Clock Stabilization Short Timer.
- **Size:** 16 bits
- **Offset:** 0xcf
- **Exists:** Always

Table 2-152 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_5

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:0	VCO_CLK_STABILIZATION_SHORT_TIME	R/W	This field is the threshold for the VCO clock stabilization short timer in terms of number of ref_range_clk cycles (default is 3.5us). This timer is used when mpll_short_lock_en_i is set to 1. . **Value After Reset:** 0x5b **Exists:** Always **Reset Mask:** 0x7ff

2.1.149 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6

- **Description:** MPLL Power Up Time Register 6.
- **Size:** 16 bits
- **Offset:** 0xd0
- **Exists:** Always

Table 2-153 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_6

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	MPLL_FDIV_EN_TIME	R/W	This field is the threshold for the MPLL fdiv enable timer in terms of number of ref_range_clk cycles. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff

2.1.150 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7

- **Description:** MPLL Power Up Time Register 7.
- **Size:** 16 bits
- **Offset:** 0xd1
- **Exists:** Always

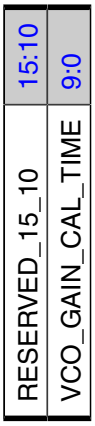


Table 2-154 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_PWRUP_TIME_7

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VCO_GAIN_CAL_TIME	R/W	This field is the threshold for the MPLL VCO control voltage setlling timer in terms of number of ref_range_clk cycles (default is 10us). **Value After Reset:** 0x104 **Exists:** Always **Reset Mask:** 0x3ff

2.1.151 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL

- **Description:** MPLL VCO Gain Calibration FB Count1 Registers.
- **Size:** 16 bits
- **Offset:** 0xd2
- **Exists:** Always

Table 2-155 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT1_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_ANA_VCO_FB_CNT1_VAL	R	This field is the current value of mpll_ana_vco_fb_cnt1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.152 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL

- **Description:** MPLL VCO Gain Calibration FB Count2 Registers.
- **Size:** 16 bits
- **Offset:** 0xd3
- **Exists:** Always

Table 2-156 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_VCO_GAIN_FB_CNT2_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_ANA_VCO_FB_CNT2_VAL	R	This field is the current value of mpll_ana_vco_fb_cnt2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.153 SUP_DIG_MPLLB_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0

- **Description:** MPLL VCO Gain Calibration Controls, Register 0.
- **Size:** 16 bits
- **Offset:** 0xd4
- **Exists:** Always

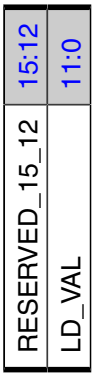


Table 2-157 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CHK_FRQ_MPLL_VCO_GAIN_CTL_0

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	LD_VAL	R/W	This field is the value of ld_val. **Value After Reset:** 0xc4e **Exists:** Always **Reset Mask:** 0xfff

2.1.154 SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL

- **Description:** MPLL VCO Drift Count Register.
- **Size:** 16 bits
- **Offset:** 0xd6
- **Exists:** Always

Table 2-158 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_MPLL_FB_DRIFT_CNT_VAL

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	MPLL_FB_DRIFT_CNT_VAL	R	This field is the current value of mpll_fb_drift_cnt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.155 SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_3

- **Description:** MPLL Calibration Controls, Register 3
- **Size:** 16 bits
- **Offset:** 0xd7
- **Exists:** Always

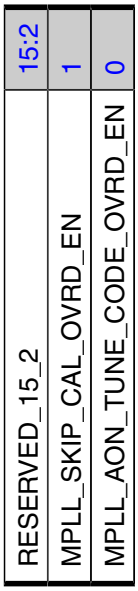


Table 2-159 Fields for Register: SUP_DIG_MPLLB_MPLL_PWR_CTL_CAL_CTRL_3

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLL_SKIP_CAL_OVRD_EN	R/W	Override the mpll_skip_cal_i input When enabled, the MPLL will use the skip flag value in the MPLL_SKIPCAL field **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	MPLL_AON_TUNE_CODE_OVRD_EN	R/W	Override the mpll_aon_tune_code_i input When enabled, the MPLL will use the coarse tune value in the MPLL_SKIPCAL_COARSE_TUNE field **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.156 SUP_DIG_MPLLB_SSC_FRAC_OUT

- **Description:** frac_out Override for Fractional Control.
- **Size:** 16 bits
- **Offset:** 0xd8
- **Exists:** Always

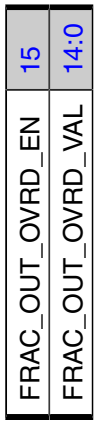


Table 2-160 Fields for Register: SUP_DIG_MPLLB_SSC_FRAC_OUT

Bits	Name	Memory Access	Description
15	FRAC_OUT_OVRD_EN	R/W	This bit is the frac_out override enable. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14:0	FRAC_OUT_OVRD_VAL	R/W	This field is the frac_out[19:5] override value. This register will override the MSB's <19:5> of the frac_out register. When overridden, the LSBs <4:0> will be set to 5'd0. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff **Volatile:** true

2.1.157 SUP_DIG_MPLLB_SSC_SSC_RAMP

- **Description:** ssc_ramp Override for Spread Spectrum Control.
- **Size:** 16 bits
- **Offset:** 0xd9
- **Exists:** Always

Table 2-161 Fields for Register: SUP_DIG_MPLLB_SSC_SSC_RAMP

Bits	Name	Memory Access	Description
15	SSC_RAMP_OVRD_EN	R/W	This bit is the ssc_ramp override enable. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14:0	SSC_RAMP_OVRD_VAL	R/W	This field is the ssc_ramp[20:6] override value. This register will override the MSB's <20:6> of the ssc_ramp register. When overridden, the LSBs <5:0> will be set to 6'd0. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff **Volatile:** true

2.1.158 SUP_DIG_MPLLB_SSC_CONFIG

- **Description:** Configuration Register for SSC and Fractional Control.
- **Size:** 16 bits
- **Offset:** 0xda
- **Exists:** Always

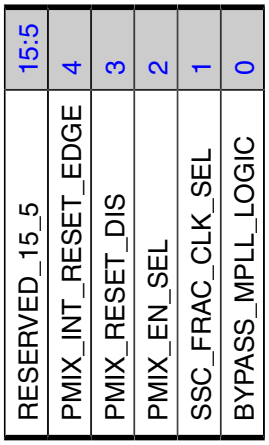


Table 2-162 Fields for Register: SUP_DIG_MPLLB_SSC_CONFIG

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	PMIX_INT_RESET_EDGE	R/W	This bit selects which ssc_en edge is used to reet pmix_int[1:0]. ■ 0x0 - ssc_en_fall_edge ■ 0x1 - ssc_en_rise_edge **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	PMIX_RESET_DIS	R/W	This bit disables reseting the SSC when ssc_en is asserting. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	PMIX_EN_SEL	R/W	This bit selects mpll_pmix_en or ssc_en as the enable for SSC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-162 Fields for Register: SUP_DIG_MPLLB_SSC_CONFIG (continued)

Bits	Name	Memory Access	Description
1	SSC_FRAC_CLK_SEL	R/W	This bit is the ssc_frac_clk select. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	BYPASS_MPLL_LOGIC	R/W	This bit is to bypass the mint_i[11:0] override logic. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.159 SUP_ANA_MPLLA_LOOP_CTRL

- **Description:** MPLLA_LOOP_CTRL
- **Size:** 16 bits
- **Offset:** 0xe0
- **Exists:** Always

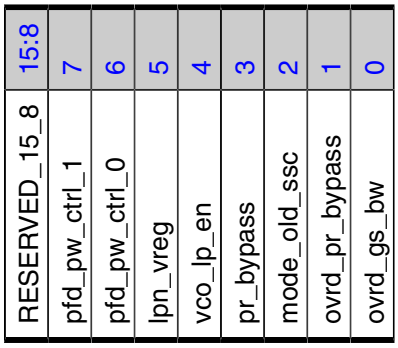


Table 2-163 Fields for Register: SUP_ANA_MPLLA_LOOP_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	pfd_pw_ctrl_1	R/W	PFD reset pulse width control (3=max pulse width, 0=min pulse width) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	pfd_pw_ctrl_0	R/W	PFD reset pulse width control (3=max pulse width, 0=min pulse width) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	lpn_vreg	R/W	Voltage Reference Selection for vreg_left/right. 1'b1 (default) = vbg_tx_vref, 1'b0 = vp/2 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	vco_lp_en	R/W	Not used **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-163 Fields for Register: SUP_ANA_MPLLA_LOOP_CTRL (continued)

Bits	Name	Memory Access	Description
3	pr_bypass	R/W	Set local pr bypass control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	mode_old_ssc	R/W	Switch back to 2-bit ssc (feature not available in esm25) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	ovrd_pr_bypass	R/W	Enable local control of PR bypass signal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_gs_bw	R/W	Enable local control of mpll_bandwidth 8-0 bus during gear shfit mode **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.160 SUP_ANA_MPLLA_OVRD

- **Description:** MPLLA_OVRD
- **Size:** 16 bits
- **Offset:** 0xe1
- **Exists:** Always

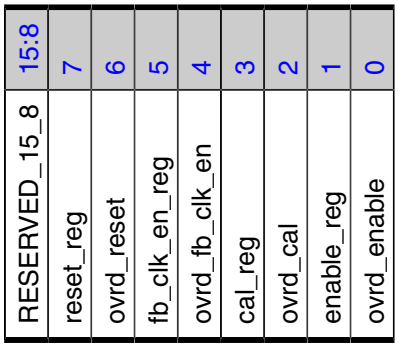


Table 2-164 Fields for Register: SUP_ANA_MPLLA_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	reset_reg	R/W	set local reset control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ovrd_reset	R/W	enable local control of reset signal (mpll_rst) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	fb_clk_en_reg	R/W	set local feedback clock control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_fb_clk_en	R/W	enable local control of feedback clock control signal (mpll_fb_clk_en) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-164 Fields for Register: SUP_ANA_MPLLA_OVRD (continued)

Bits	Name	Memory Access	Description
3	cal_reg	R/W	set local calibration control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_cal	R/W	enable local control of calibration signal (mpll_cal) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	enable_reg	R/W	set local enable control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_enable	R/W	enable local control of enable signal (mpll_en) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.161 SUP_ANA_MPLLA_ATB1

- **Description:** MPLLA_ATB1
- **Size:** 16 bits
- **Offset:** 0xe2
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_7	meas_iv_6	meas_iv_5	meas_iv_4	meas_iv_3	meas_iv_2	meas_iv_1	meas_iv_0

Table 2-165 Fields for Register: SUP_ANA_MPLLA_ATB1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_7	R/W	Measure vreg_left_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_6	R/W	Measure vreg_s_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_5	R/W	Measure vreg_vco_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_4	R/W	Override vreg_s_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	meas_iv_3	R/W	Override vreg_outclock_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0

Table 2-165 Fields for Register: SUP_ANA_MPLLA_ATB1 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	meas_iv_2	R/W	Override vreg_cp_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_1	R/W	Override vreg_left_bg and vreg_right_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_0	R/W	Override vreg_vco_bg to be 1X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.162 SUP_ANA_MPLLA_ATB2

- **Description:** MPLLA_ATB2
- **Size:** 16 bits
- **Offset:** 0xe3
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_15	meas_iv_14	meas_iv_13	meas_iv_12	meas_iv_11	meas_iv_10	meas_iv_9	meas_iv_8

Table 2-166 Fields for Register: SUP_ANA_MPLLA_ATB2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_15	R/W	Measure reference voltage for VCO magnitude control amp (atb_s_m) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_14	R/W	Measure VCO magntitude control signal vdrive (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_13	R/W	Measure buffered version of ctrl_fine_int (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_12	R/W	Measure GD in MPLL (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-166 Fields for Register: SUP_ANA_MPLLA_ATB2 (continued)

Bits	Name	Memory Access	Description
3	meas_iv_11	R/W	Measure VP in MPLL (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_10	R/W	Measure vreg_cp_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_9	R/W	Measure vreg_outclock_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_8	R/W	Measure vreg_right_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.163 SUP_ANA_MPLLA_ATB3

- **Description:** MPLLA_ATB3
- **Size:** 16 bits
- **Offset:** 0xe4
- **Exists:** Always

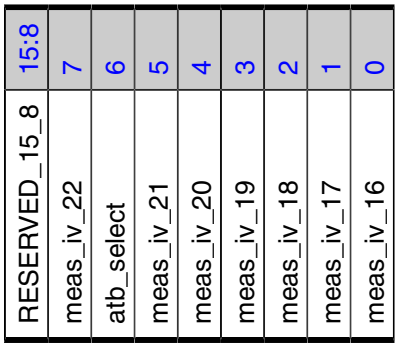


Table 2-167 Fields for Register: SUP_ANA_MPLLA_ATB3

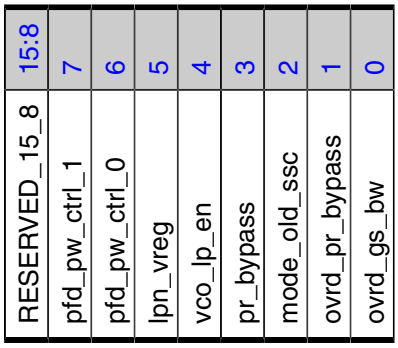
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_22	R/W	Measure CP vsense (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	atb_select	R/W	Connect internal ATB_S signals to external ATB_S signals **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_21	R/W	Measure DLL nbias (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_20	R/W	Override VCO amp reference to be smaller than the default setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-167 Fields for Register: SUP_ANA_MPLLA_ATB3 (continued)

Bits	Name	Memory Access	Description
3	meas_iv_19	R/W	Override VCO amp reference to be larger than the default setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_18	R/W	Override VCO amp reference to be the voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_17	R/W	Force fine control voltage to be ~45mV lower than mid value in cal mode (min value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_16	R/W	Force fine control voltage to be ~45mV higher than mid value in cal mode (max value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.164 SUP_ANA_MPLLB_LOOP_CTRL

- **Description:** MPLLB_LOOP_CTRL
- **Size:** 16 bits
- **Offset:** 0xe5
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	pfd_pw_ctrl_1	pfd_pw_ctrl_0	lpn_vreg	vco_lp_en	pr_bypass	mode_old_ssc	ovrd_pr_bypass	ovrd_gs_bw

Table 2-168 Fields for Register: SUP_ANA_MPLLB_LOOP_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	pfd_pw_ctrl_1	R/W	PFD reset pulse width control (3=max pulse width, 0=min pulse width) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	pfd_pw_ctrl_0	R/W	PFD reset pulse width control (3=max pulse width, 0=min pulse width) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	lpn_vreg	R/W	Voltage Reference Selection for vreg_left/right. 1'b1 (default) = vbg_tx_vref, 1'b0 = vp/2 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	vco_lp_en	R/W	Not used **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-168 Fields for Register: SUP_ANA_MPLLB_LOOP_CTRL (continued)

Bits	Name	Memory Access	Description
3	pr_bypass	R/W	Set local pr bypass control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	mode_old_ssc	R/W	Switch back to 2-bit ssc (feature not available in esm25) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	ovrd_pr_bypass	R/W	Enable local control of PR bypass signal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_gs_bw	R/W	Enable local control of mpll_bandwidth 8-0 bus during gear shfit mode **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.165 SUP_ANA_MPLLB_OVRD

- **Description:** MPLLB_OVRD
- **Size:** 16 bits
- **Offset:** 0xe6
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	reset_reg	ovrd_reset	fb_clk_en_reg	ovrd_fb_clk_en	cal_reg	ovrd_cal	enable_reg	ovrd_enable

Table 2-169 Fields for Register: SUP_ANA_MPLLB_OVRD

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	reset_reg	R/W	set local reset control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ovrd_reset	R/W	enable local control of reset signal (mpll_rst) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	fb_clk_en_reg	R/W	set local feedback clock control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_fb_clk_en	R/W	enable local control of feedback clock control signal (mpll_fb_clk_en) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-169 Fields for Register: SUP_ANA_MPLLB_OVRD (continued)

Bits	Name	Memory Access	Description
3	cal_reg	R/W	set local calibration control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_cal	R/W	enable local control of calibration signal (mpll_cal) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	enable_reg	R/W	set local enable control to ON **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_enable	R/W	enable local control of enable signal (mpll_en) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.166 SUP_ANA_MPLLB_ATB1

- **Description:** MPLLB_ATB1
- **Size:** 16 bits
- **Offset:** 0xe7
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_7	meas_iv_6	meas_iv_5	meas_iv_4	meas_iv_3	meas_iv_2	meas_iv_1	meas_iv_0

Table 2-170 Fields for Register: SUP_ANA_MPLLB_ATB1

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_7	R/W	Measure vreg_left_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_6	R/W	Measure vreg_s_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_5	R/W	Measure vreg_vco_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_4	R/W	Override vreg_s_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	meas_iv_3	R/W	Override vreg_outclock_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0

Table 2-170 Fields for Register: SUP_ANA_MPLLB_ATB1 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	meas_iv_2	R/W	Override vreg_cp_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_1	R/W	Override vreg_left_bg and vreg_right_bg to be 1/2X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_0	R/W	Override vreg_vco_bg to be 1X of voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.167 SUP_ANA_MPLLB_ATB2

- **Description:** MPLLB_ATB2
- **Size:** 16 bits
- **Offset:** 0xe8
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_15	meas_iv_14	meas_iv_13	meas_iv_12	meas_iv_11	meas_iv_10	meas_iv_9	meas_iv_8

Table 2-171 Fields for Register: SUP_ANA_MPLLB_ATB2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_15	R/W	Measure reference voltage for VCO magnitude control amp (atb_s_m) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_14	R/W	Measure VCO magntitude control signal vdrive (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_13	R/W	Measure buffered version of ctrl_fine_int (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_12	R/W	Measure GD in MPLL (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-171 Fields for Register: SUP_ANA_MPLLB_ATB2 (continued)

Bits	Name	Memory Access	Description
3	meas_iv_11	R/W	Measure VP in MPLL (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_10	R/W	Measure vreg_cp_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_9	R/W	Measure vreg_outclock_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_8	R/W	Measure vreg_right_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.168 SUP_ANA_MPLLB_ATB3

- **Description:** MPLLB_ATB3
- **Size:** 16 bits
- **Offset:** 0xe9
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_22	atb_select	meas_iv_21	meas_iv_20	meas_iv_19	meas_iv_18	meas_iv_17	meas_iv_16

Table 2-172 Fields for Register: SUP_ANA_MPLLB_ATB3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_22	R/W	Measure CP vsense (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	atb_select	R/W	Connect internal ATB_S signals to external ATB_S signals **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_21	R/W	Measure DLL nbias (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_20	R/W	Override VCO amp reference to be smaller than the default setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-172 Fields for Register: SUP_ANA_MPLLB_ATB3 (continued)

Bits	Name	Memory Access	Description
3	meas_iv_19	R/W	Override VCO amp reference to be larger than the default setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_18	R/W	Override VCO amp reference to be the voltage on atb_s_m. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_iv_17	R/W	Force fine control voltage to be ~45mV lower than mid value in cal mode (min value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_iv_16	R/W	Force fine control voltage to be ~45mV higher than mid value in cal mode (max value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.169 SUP_ANA_RTUNE_CTRL

- **Description:** RTUNE_CTRL
- **Size:** 16 bits
- **Offset:** 0xea
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	rt_en_frcon	temp_sensor_sel	vref_sel	rt_dac_mode	rt_dac_chop	rt_atb	rt_sel_atbp	rt_sel_atbf

Table 2-173 Fields for Register: SUP_ANA_RTUNE_CTRL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	rt_en_frcon	R/W	Local rtune block enable control, force rtune block on if asserted. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	temp_sensor_sel	R/W	Enables temperature sensor **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	vref_sel	R/W	Enables reference voltages during temperature sensor mode **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	rt_dac_mode	R/W	Margin DAC mode control bits **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-173 Fields for Register: SUP_ANA_RTUNE_CTRL (continued)

Bits	Name	Memory Access	Description
3	rt_dac_chop	R/W	ref to engineering doc **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	rt_atb	R/W	RTUNE ATB mode control. Combines with rt_ana_mode_1:0 to perform different function. See section 8.4.4 for details. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	rt_sel_atbp	R/W	RTUNE ATB input select **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	rt_sel_atbf	R/W	RTUNE ATB input select **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.170 SUP_ANA_SWITCH_PWR_MEAS

- **Description:** SWITCH_PWR_MEAS
- **Size:** 16 bits
- **Offset:** 0xeb
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	temp_meas	atb_switch_6	atb_switch_5	atb_switch_4	atb_switch_3	atb_switch_2	atb_switch_1	atb_switch_0

Table 2-174 Fields for Register: SUP_ANA_SWITCH_PWR_MEAS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	temp_meas	R/W	Enables temperature measurement if asserted: vbe is sent to atb_s_m and vbg to atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	atb_switch_6	R/W	atb_s_p to atb_s_m **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	atb_switch_5	R/W	atb_s_p to vph/2 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	atb_switch_4	R/W	atb_s_m to gd **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-174 Fields for Register: SUP_ANA_SWITCH_PWR_MEAS (continued)

Bits	Name	Memory Access	Description
3	atb_switch_3	R/W	atb_s_p to vp **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	atb_switch_2	R/W	atb_s_p to fastreg_vref (vbg_vref_rx) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	atb_switch_1	R/W	atb_s_p to vph **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	atb_switch_0	R/W	atb_s_p to slowreg_vref **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.171 SUP_ANA_SWITCH_MISC_MEAS

- **Description:** SWITCH_MISC_MEAS
- **Size:** 16 bits
- **Offset:** 0xec
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	sel_vpll_ref_1	sel_vpll_ref_0	sel_rx_vco_vref_1	sel_rx_vco_vref_0	sel_vbg_vref_rx_1	sel_vbg_vref_rx_0	ref_hyst_1	ref_hyst_0

Table 2-175 Fields for Register: SUP_ANA_SWITCH_MISC_MEAS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	sel_vpll_ref_1	R/W	Voltage level selection for MPLL core regulator. 25mV increments. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	sel_vpll_ref_0	R/W	Voltage level selection for MPLL core regulator. 25mV increments. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	sel_rx_vco_vref_1	R/W	Voltage level selection for RX lane VCO regulator. 25mV increments. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	sel_rx_vco_vref_0	R/W	Voltage level selection for RX lane VCO regulator. 25mV increments. **Value After Reset:** 0x0

Table 2-175 Fields for Register: SUP_ANA_SWITCH_MISC_MEAS (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	sel_vbg_vref_rx_1	R/W	Voltage level selection for RX lane regulator. 25mV increments. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	sel_vbg_vref_rx_0	R/W	Voltage level selection for RX lane regulator. 25mV increments. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ref_hyst_1	R/W	Sets prescaler input buffer hysteresis; 00 is no hysteresis; 01 is 18mVpp hysteresis; 10 is 35mVpp hyst; 11 is 50mVpp hyst **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	ref_hyst_0	R/W	Sets prescaler input buffer hysteresis; 00 is no hysteresis; 01 is 18mVpp hysteresis; 10 is 35mVpp hyst; 11 is 50mVpp hyst **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.172 SUP_ANA_BG

- **Description:** BG
- **Size:** 16 bits
- **Offset:** 0xed
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	sel_vbg_vref_tx_1	sel_vbg_vref_tx_0	chop_en_int	temp_sensor_chop_en	temp_sensor_en	atb_switch_7	sel_rx_vco_vref_2	bypass_bg

Table 2-176 Fields for Register: SUP_ANA_BG

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	sel_vbg_vref_tx_1	R/W	Voltage level selection for TX regulator and MPLL output path regulator. 25mV increments. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	sel_vbg_vref_tx_0	R/W	Voltage level selection for TX regulator and MPLL output path regulator. 25mV increments. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	chop_en_int	R/W	Enable chopper clock for bandgap **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	temp_sensor_chop_en	R/W	Chopper clock for temp sensor **Value After Reset:** 0x0

Table 2-176 Fields for Register: SUP_ANA_BG (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	temp_sensor_en	R/W	Enable Temperature Sensor block **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	atb_switch_7	R/W	atb_s_p to fastreg_vref (vb_vref_tx) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	sel_rx_vco_vref_2	R/W	Voltage level selection for RX lane VCO regulator. 25mV increments. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	bypass_bg	R/W	Bypass bandgap internal feedback loop with VP **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.173 SUP_ANA_MPLLA

- **Description:** MPLLA
- **Size:** 16 bits
- **Offset:** 0xee
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_28	meas_iv_27	meas_iv_26	meas_iv_25	meas_iv_24	meas_iv_23	gear_shift_en	ovrd_gear_shift_en

Table 2-177 Fields for Register: SUP_ANA_MPLLA

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_28	R/W	Measure vreg_pfd_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_27	R/W	Measure vdd_h in regulator with (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_26	R/W	Measure vreg_vdd_drv in regulator with (atb_s_m) currently probe gd (spare; not use) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_25	R/W	Measure vreg_pmix_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-177 Fields for Register: SUP_ANA_MPLLA (continued)

Bits	Name	Memory Access	Description
3	meas_iv_24	R/W	Measure vreg_pmixdiv_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_23	R/W	Change regulator doubler divide ratio from 2 to 2.1 (active high) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	gear_shift_en	R/W	override mpll_gear_shift_en when ovrd_gear_shift_en is asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_gear_shift_en	R/W	if asserted, override mpll_gear_shift_en with gear_shift_en signal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.174 SUP_ANA_MPLLB

- **Description:** MPLLB
- **Size:** 16 bits
- **Offset:** 0xef
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_iv_28	meas_iv_27	meas_iv_26	meas_iv_25	meas_iv_24	meas_iv_23	gear_shift_en	ovrd_gear_shift_en

Table 2-178 Fields for Register: SUP_ANA_MPLLB

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_iv_28	R/W	Measure vreg_pfd_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_iv_27	R/W	Measure vdd_h in regulator with (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_iv_26	R/W	Measure vreg_vdd_drv in regulator with (atb_s_m) currently probe gd (spare; not use) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_iv_25	R/W	Measure vreg_pmix_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-178 Fields for Register: SUP_ANA_MPLLB (continued)

Bits	Name	Memory Access	Description
3	meas_iv_24	R/W	Measure vreg_pmixdiv_bg (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_iv_23	R/W	Change regulator doubler divide ratio from 2 to 2.1 (active high) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	gear_shift_en	R/W	override mpll_gear_shift_en when ovrd_gear_shift_en is asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_gear_shift_en	R/W	if asserted, override mpll_gear_shift_en with gear_shift_en signal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.175 SUP_ANA_PREA

- **Description:** PREA
- **Size:** 16 bits
- **Offset:** 0xf0
- **Exists:** Always

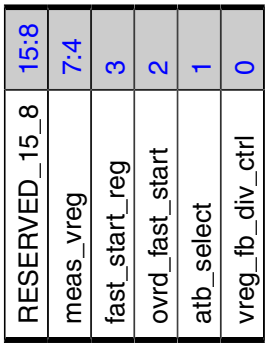


Table 2-179 Fields for Register: SUP_ANA_PREA

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	meas_vreg	R/W	ATB Output Selection: (must also set atb_select=1) bit0 -vreg_mplla(Default), bit4-vreg_mpllb, bit5-vreg_repeat, bit6-vreg_dco **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3	fast_start_reg	R/W	Fast-Start Overide Value, enabled by setting ovrd_fast_start=1'b1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_fast_start	R/W	Fast-Start Override Enable for Regulator Doubler **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	atb_select	R/W	Enable ATB probe capability by shorting atb_s_p/m_int - external atb_s_p/m **Value After Reset:** 0x0

Table 2-179 Fields for Register: SUP_ANA_PREA (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	vreg_fb_div_ctrl	R/W	Change VREG doubler reference level VR: 1'b0=2.09x(default), 1'b1=2.2x **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.176 SUP_ANA_PREB

- **Description:** PREB
- **Size:** 16 bits
- **Offset:** 0xf1
- **Exists:** Always

15:8	7	6	5	4	3:2	1	0
RESERVED_15_8	vreg_mplla_boost_1	vreg_mplla_boost_0	refa_hyst_1	refa_hyst_0	refa_gm_gain	ts_temp_meas	temp_meas_2

Table 2-180 Fields for Register: SUP_ANA_PREB

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	vreg_mplla_boost_1	R/W	Regulator Output Level Adjustement for vreg_mplla (Default: 2'b00 = Vref*2) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	vreg_mplla_boost_0	R/W	Regulator Output Level Adjustement for vreg_mplla (Default: 2'b00 = Vref*2) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	refa_hyst_1	R/W	RefA Receiver CML-to-CMOS Hysterisis Control (Default: 2'b10), 2'd0 - no hysterisis, 2'd1 - 18mVpp, 2'd2 - 35mVpp (Default), 2'd3 - 50mVpp **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-180 Fields for Register: SUP_ANA_PREB (continued)

Bits	Name	Memory Access	Description
4	refa_hyst_0	R/W	RefA Receiver CML-to-CMOS Hysterisis Control (Default: 2'b10), 2'd0 - no hysterisis, 2'd1 - 18mVpp, 2'd2 - 35mVpp (Default), 2'd3 - 50mVpp **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	refa_gm_gain	R/W	RefA CML Receiver Gain Control (Default: 2'b00) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1	ts_temp_meas	R/W	Enables temperature measurement for temperature sensor if asserted: vbe is sent to atb_s_m and vbg to atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	temp_meas_2	R/W	If enabled measures vip on atb_s_p and rsb_0 on atb_s_m **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.177 SUP_ANA_PREC

- **Description:** PREC
- **Size:** 16 bits
- **Offset:** 0xf2
- **Exists:** Always

15:8	7	6	5	4	3:2	1	0
RESERVED_15_8	vreg_mpllb_boost_1	vreg_mpllb_boost_0	refb_hyst_1	refb_hyst_0	refb_gm_gain	spare_7	spare_6

Table 2-181 Fields for Register: SUP_ANA_PREC

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	vreg_mpllb_boost_1	R/W	Regulator Output Level Adjustement for vreg_mpllb (Default: 2'b00 = Vref*2) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	vreg_mpllb_boost_0	R/W	Regulator Output Level Adjustement for vreg_mpllb (Default: 2'b00 = Vref*2) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	refb_hyst_1	R/W	RefB Receiver CML-to-CMOS Hysterisis Control (Default: 2'b10), 2'd0 - no hysterisis, 2'd1 - 18mVpp, 2'd2 - 35mVpp (Default), 2'd3 - 50mVpp **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-181 Fields for Register: SUP_ANA_PREC (continued)

Bits	Name	Memory Access	Description
4	refb_hyst_0	R/W	RefB Receiver CML-to-CMOS Hysterisis Control (Default: 2'b10), 2'd0 - no hysterisis, 2'd1 - 18mVpp, 2'd2 - 35mVpp (Default), 2'd3 - 50mVpp **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	refb_gm_gain	R/W	RefB CML Receiver Gain Control (Default: 2'b00) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1	spare_7	R/W	spare **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	spare_6	R/W	spare **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.178 SUP_ANA_PRED

- **Description:** PRED
- **Size:** 16 bits
- **Offset:** 0xf3
- **Exists:** Always

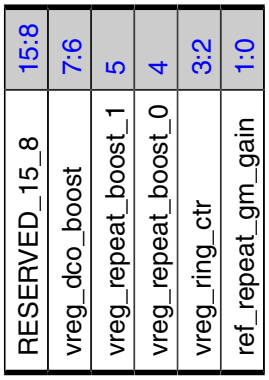


Table 2-182 Fields for Register: SUP_ANA_PRED

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:6	vreg_dco_boost	R/W	Regulator Output Level Adjustement for vreg_dco (Default: 2'b00 = Vref*2) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5	vreg_repeat_boost_1	R/W	Regulator Output Level Adjustement for vreg_repeat (Default: 2'b00 = Vref*2) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	vreg_repeat_boost_0	R/W	Regulator Output Level Adjustement for vreg_repeat (Default: 2'b00 = Vref*2) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3:2	vreg_ring_ctr	R/W	VREG Internal Doubler RO Adjustment (Default: 2'b00) **Value After Reset:** 0x0

Table 2-182 Fields for Register: SUP_ANA_PRED (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
1:0	ref_repeat_gm_gain	R/W	Repeater CML Receiver Gain Control (Default: 2'b00) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.179 SUP_ANA_MPLLA_VCO_GAIN_CAL

- **Description:** MPLLA_VCO_GAIN_CAL
- **Size:** 16 bits
- **Offset:** 0xf4
- **Exists:** Always

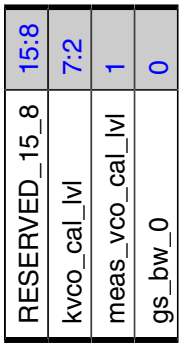


Table 2-183 Fields for Register: SUP_ANA_MPLLA_VCO_GAIN_CAL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:2	kvco_cal_lvl	R/W	Set vco gain calibration voltage level for phase 1 (default set to +200mV with 5:3=111) and phase 2 (default set to 0mV with 2:0=011). The voltage level can for phase 1 and 2 can be change from 200mV to -150mV in ~50mV step. **Value After Reset:** 0x3b **Exists:** Always **Reset Mask:** 0x3f
1	meas_vco_cal_lvl	R/W	Measure vco calibration control voltage (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	gs_bw_0	R/W	override bit 0 of mpll_bandwidth bus during gear shift mode when ovrd_gs_bw is asserted **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.180 SUP_ANA_MPLLA_RESERVED_2

- **Description:** MPLLA_RESERVED_2
- **Size:** 16 bits
- **Offset:** 0xf5
- **Exists:** Always

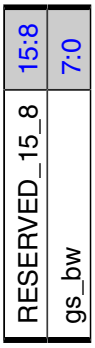


Table 2-184 Fields for Register: SUP_ANA_MPLLA_RESERVED_2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	gs_bw	R/W	override bit [8:1] of mpll_bandwidth bus during gear shift mode when ovrd_gs_bw is asserted **Value After Reset:** 0xff **Exists:** Always **Reset Mask:** 0xff

2.1.181 SUP_ANA_MPLLB_VCO_GAIN_CAL

- **Description:** MPLLB_VCO_GAIN_CAL
- **Size:** 16 bits
- **Offset:** 0xf6
- **Exists:** Always

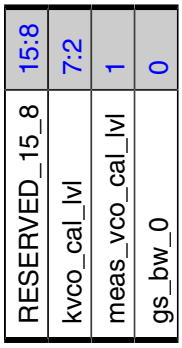


Table 2-185 Fields for Register: SUP_ANA_MPLLB_VCO_GAIN_CAL

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:2	kvco_cal_lvl	R/W	Set vco gain calibration voltage level for phase 1 (default set to +200mV with 5:3=111) and phase 2 (default set to 0mV with 2:0=011). The voltage level can for phase 1 and 2 can be change from 200mV to -150mV in ~50mV step. **Value After Reset:** 0x3b **Exists:** Always **Reset Mask:** 0x3f
1	meas_vco_cal_lvl	R/W	Measure vco calibration control voltage (atb_s_p) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	gs_bw_0	R/W	override bit 0 of mpll_bandwidth bus during gear shift mode when ovrd_gs_bw is asserted **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.182 SUP_ANA_MPLLB_RESERVED_2

- **Description:** MPLLB_RESERVED_2
- **Size:** 16 bits
- **Offset:** 0xf7
- **Exists:** Always

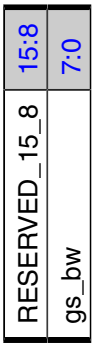


Table 2-186 Fields for Register: SUP_ANA_MPLLB_RESERVED_2

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	gs_bw	R/W	override bit [8:1] of mpll_bandwidth bus during gear shift mode when ovrd_gs_bw is asserted **Value After Reset:** 0xff **Exists:** Always **Reset Mask:** 0xff

2.1.183 RAWCMN_DIG_CMN_CTL

- **Description:** Common Control Register.
- **Size:** 16 bits
- **Offset:** 0x100
- **Exists:** Always

Table 2-187 Fields for Register: RAWCMN_DIG_CMN_CTL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	PHY_FUNC_RST	R/W	This bit resets the PHY except registers in the Raw PCS common and always-on registers. This bit is useful for resetting the PHY after reloading and without resetting the memory. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.184 RAWCMN_DIG_LANE_FSM_OP_XTND

- **Description:** Lane FSM OP XTND Control Register.
- **Size:** 16 bits
- **Offset:** 0x101
- **Exists:** Always

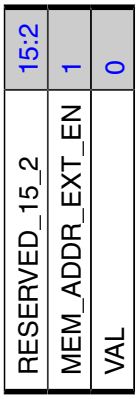


Table 2-188 Fields for Register: RAWCMN_DIG_LANE_FSM_OP_XTND

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MEM_ADDR_EXT_EN	R/W	This bit enables access to upper 16K **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit is required when accessing the analog CREGs in firmware to prevent timing violations across the analog-digital interface. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.185 RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0

- **Description:** Override Values for Incoming TX Term Offset.
- **Size:** 16 bits
- **Offset:** 0x102
- **Exists:** Always

Table 2-189 Fields for Register: RAWCMN_DIG_TERM_OFFSET_OVRD_IN_0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TXUP_TERM_OFFSET_OVRD_EN	R/W	This bit is the override enable for txup_term_offset[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	TXUP_TERM_OFFSET_OVRD_VAL	R/W	These bits are the override value for txup_term_offset[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.186 RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1

- **Description:** Override Values for Incoming TX/RX Term Offset.
- **Size:** 16 bits
- **Offset:** 0x103
- **Exists:** Always

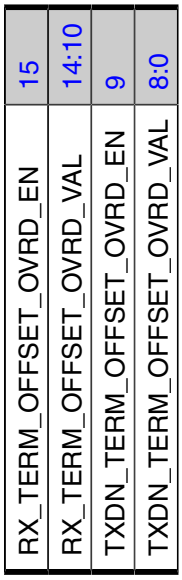


Table 2-190 Fields for Register: RAWCMN_DIG_TERM_OFFSET_OVRD_IN_1

Bits	Name	Memory Access	Description
15	RX_TERM_OFFSET_OVRD_EN	R/W	This bit is the override enable for rx_term_offset[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:10	RX_TERM_OFFSET_OVRD_VAL	R/W	These bits are the verride value for rx_term_offset[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
9	TXDN_TERM_OFFSET_OVRD_EN	R/W	This bit is the override enable for txdn_term_offset[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	TXDN_TERM_OFFSET_OVRD_VAL	R/W	These bits are the override value for txdn_term_offset[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.187 RAWCMN_DIG_CMN_CTL_1

- **Description:** Common Control Register 1.
- **Size:** 16 bits
- **Offset:** 0x104
- **Exists:** Always

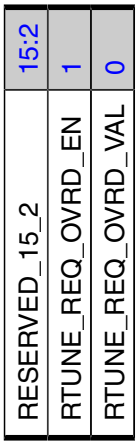


Table 2-191 Fields for Register: RAWCMN_DIG_CMN_CTL_1

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RTUNE_REQ_OVRD_EN	R/W	This bit is the override enable for rtune_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RTUNE_REQ_OVRD_VAL	R/W	This bit is the override value for rtune_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.188 RAWCMN_DIG_MPLL_CONFIG

- **Description:** MPLL Configuration Parameters.
- **Size:** 16 bits
- **Offset:** 0x105
- **Exists:** Always

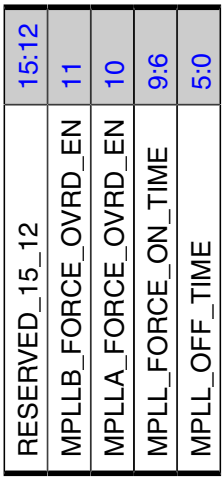


Table 2-192 Fields for Register: RAWCMN_DIG_MPLL_CONFIG

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLB_FORCE_OVRD_EN	R/W	This bita is used to bypass FW MPLLB calib in mpllb_force_ctrl **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	MPLLA_FORCE_OVRD_EN	R/W	This bita is used to bypass FW MPLLA calib in mplla_force_ctrl **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:6	MPLL_FORCE_ON_TIME	R/W	These bits are the wait counter for turning on MPLL using mplla/b_force_en. This is to ensure there is no collision between the force_en logic and the tx_req logic. **Value After Reset:** 0xc **Exists:** Always **Reset Mask:** 0xf

Table 2-192 Fields for Register: RAWCMN_DIG_MPLL_CONFIG (continued)

Bits	Name	Memory Access	Description
5:0	MPLL_OFF_TIME	R/W	These bits are the wait counter for turning off the MPLL. **Value After Reset:** 0x1e **Exists:** Always **Reset Mask:** 0x3f

2.1.189 RAWCMN_DIG_ATE_ALU_CTRL

- **Description:** ATE ALU Module Control Register.
- **Size:** 16 bits
- **Offset:** 0x106
- **Exists:** Always

Table 2-193 Fields for Register: RAWCMN_DIG_ATE_ALU_CTRL

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	ATE_ALU_OPCODE	R/W	These bit are the OPCode for ATE ALU module. Encoding: ■ 4'b0000: ZERO ■ 4'b0001: ADD_RD_DATA ■ 4'b0010: ADD_DATA ■ 4'b0011: SHL1 ■ 4'b0100: SHR1 ■ 4'b0101: ASHR1 ■ 4'b0110: INV ■ 4'b0111: NEG ■ 4'b1000: ABS ■ 4'b1110: READ ■ 4'b1111: WRITE **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.190 RAWCMN_DIG_ATE_ALU_ADDR

- **Description:** Address for Read and Write Command for ATE ALU Module.
- **Size:** 16 bits
- **Offset:** 0x107
- **Exists:** Always

Table 2-194 Fields for Register: RAWCMN_DIG_ATE_ALU_ADDR

Bits	Name	Memory Access	Description
15:0	ADDR	R/W	These bits are the address for read and write command for the ATE ALU module. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.191 RAWCMN_DIG_ATE_ALU_DATA

- **Description:** Data for Write Command for ATE ALU Module.
- **Size:** 16 bits
- **Offset:** 0x108
- **Exists:** Always

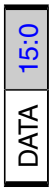


Table 2-195 Fields for Register: RAWCMN_DIG_ATE_ALU_DATA

Bits	Name	Memory Access	Description
15:0	DATA	R/W	These bits are the data for read and write command for the ATE ALU module. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.192 RAWCMN_DIG_ATE_ALU_FLAGS

- **Description:** Flags from ATE ALU Module.
- **Size:** 16 bits
- **Offset:** 0x109
- **Exists:** Always

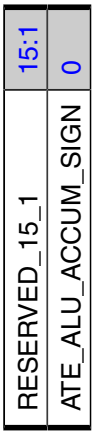


Table 2-196 Fields for Register: RAWCMN_DIG_ATE_ALU_FLAGS

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ATE_ALU_ACCUM_SIGN	R	This bit indicates the sign of the accumulator in ATE ALU. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.193 RAWCMN_DIG_ATE_ALU_ACCUM

- **Description:** Stores ATE ALU Accumulator Result.
- **Size:** 16 bits
- **Offset:** 0x10a
- **Exists:** Always

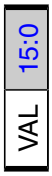


Table 2-197 Fields for Register: RAWCMN_DIG_ATE_ALU_ACCUM

Bits	Name	Memory Access	Description
15:0	VAL	R	These bits store the ATE ALU Accumulator result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.194 RAWCMN_DIG_MPLL_IN

- **Description:** Status of mplla/b_init_cal_disable Inputs.
- **Size:** 16 bits
- **Offset:** 0x10b
- **Exists:** Always

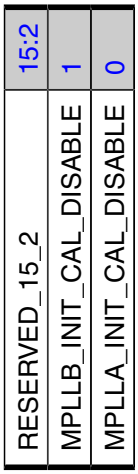


Table 2-198 Fields for Register: RAWCMN_DIG_MPLL_IN

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLLB_INIT_CAL_DISABLE	R	This bit is the value of the mpllb_init_cal_disable input. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLA_INIT_CAL_DISABLE	R	This bit is the value of the mplla_init_cal_disable input. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.195 RAWCMN_DIG_FW_PWRUP_DONE_A

- **Description:** Firmware Power-up Done_A Status.
- **Size:** 16 bits
- **Offset:** 0x10c
- **Exists:** Always

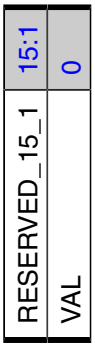


Table 2-199 Fields for Register: RAWCMN_DIG_FW_PWRUP_DONE_A

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	Partial FW Power-up Status for MPLLA after phy_reset and pg_reset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.196 RAWCMN_DIG_MPLL_CLK_ASYNC_OVRD

- **Description:** Controls MPLL Output Clocks Asynchronously.
- **Size:** 16 bits
- **Offset:** 0x10d
- **Exists:** Always

Table 2-200 Fields for Register: RAWCMN_DIG_MPLL_CLK_ASYNC_OVRD

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLLB_CLK_OVRD	R/W	This bit asynchronously enables the MPLLB output clocks. This can only be changed if the MPLLB are off to avoid the glitch on the MPLLB output clocks. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_CLK_OVRD	R/W	This bit asynchronously enables the MPLLA output clocks. This can only be changed if the MPLLA is off to avoid the glitch on the MPLLA output clocks. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.197 RAWCMN_DIG_MPLL_RECAL_BANK_OVRD

- **Description:** Override Incoming mplla/b_recal Inputs.
- **Size:** 16 bits
- **Offset:** 0x10e
- **Exists:** Always

15:14	13	12	11	10	9	8:7	6	5	4	3	2	1:0
RESERVED_15_14	MPLLB_RECAL_SKIP_EN_OVRD_EN	MPLLB_RECAL_SKIP_EN_OVRD_VAL	MPLLB_RECAL_FORCE_EN_OVRD_EN	MPLLB_RECAL_FORCE_EN_OVRD_VAL	MPLLB_RECAL_BANK_SEL_OVRD_EN	MPLLB_RECAL_BANK_SEL_OVRD_VAL	MPLLA_RECAL_SKIP_EN_OVRD_EN	MPLLA_RECAL_SKIP_EN_OVRD_VAL	MPLLA_RECAL_FORCE_EN_OVRD_EN	MPLLA_RECAL_FORCE_EN_OVRD_VAL	MPLLA_RECAL_BANK_SEL_OVRD_EN	MPLLA_RECAL_BANK_SEL_OVRD_VAL

Table 2-201 Fields for Register: RAWCMN_DIG_MPLL_RECAL_BANK_OVRD

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_RECAL_SKIP_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_RECAL_SKIP_EN_OVRD_VAL	R/W	This bit is the override value for incoming input mpllb_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MPLLB_RECAL_FORCE_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_recal_force_en. **Value After Reset:** 0x0

Table 2-201 Fields for Register: RAWCMN_DIG_MPLL_RECAL_BANK_OVRD (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
10	MPLLB_RECAL_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for incoming input mpllb_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLB_RECAL_BANK_SEL_OVRD_EN	R/W	This bit is the override enable for mpllb_cal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:7	MPLLB_RECAL_BANK_SEL_OVRD_VAL	R/W	This bit is the override value for incoming input mpllb_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6	MPLLA_RECAL_SKIP_EN_OVRD_EN	R/W	This bit is the override enable for mplla_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLLA_RECAL_SKIP_EN_OVRD_VAL	R/W	This bit is theverride value for incoming input mplla_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLLA_RECAL_FORCE_EN_OVRD_EN	R/W	This bit is the override enable for mplla_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLA_RECAL_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for incoming input mplla_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_RECAL_BANK_SEL_OVRD_EN	R/W	This bit is the override enable for mplla_recal_bank_sel. **Value After Reset:** 0x0

Table 2-201 Fields for Register: RAWCMN_DIG_MPLL_RECAL_BANK_OVRD (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
1:0	MPLLA_RECAL_BANK_SEL_OVRD_VAL	R/W	This bit is the override value for incoming input mplla_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.198 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_1

- **Description:** ATE Override Input to Control Top-Level Inputs Register 1. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x10f
- **Exists:** Always

Table 2-202 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:9	RX_TERM_OFFSET_OVRD_VAL	R/W	Override value for rx_term_offset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
8:0	TXUP_TERM_OFFSET_OVRD_VAL	R/W	Override value for txup_term_offset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.199 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_2

- **Description:** ATE Override Input to Control Top-Level Inputs Register 2. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x110
- **Exists:** Always

Table 2-203 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:9	RX_VREF_CTRL_R	R/W	Override value for rx_vref_ctl_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
8:0	TXDN_TERM_OFFSET_OVRD_VAL	R/W	Override value for txdn_term_offset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.200 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_3

- **Description:** ATE Override Input to Control Top-Level Inputs Register 3. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x111
- **Exists:** Always

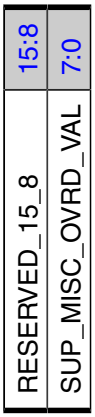


Table 2-204 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	SUP_MISC_OVRD_VAL	R/W	Override value for sup_misc **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.201 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_4

- **Description:** ATE Override Input to Control Top-Level Inputs Register 4. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x112
- **Exists:** Always

Table 2-205 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_4

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLA_INIT_CAL_DISABLE_OVRD_VAL	R/W	Override value for mplla_init_cal_disable **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10:3	MPLLA_BW_THRESHOLD_OVRD_VAL	R/W	Override value for mplla_bw_threshold **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
2:0	REF_CLK_MPLLA_DIV_OVRD_VAL	R/W	Override value for ref_clk_mplla_div **Value After Reset:** 0x0

Table 2-205 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_4 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7

2.1.202 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_5

- **Description:** ATE Override Input to Control Top-Level Inputs Register 5. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x113
- **Exists:** Always

Table 2-206 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_5

Bits	Name	Memory Access	Description
15:4	MPLLA_MULTIPLIER_OVRD_VAL	R/W	Override value for mplla_multiplier **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff
3:0	MPLLA_TX_CLK_DIV_OVRD_ VAL	R/W	Override value for mplla_tx_clk_div **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.203 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_6

- **Description:** ATE Override Input to Control Top-Level Inputs Register 6. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x114
- **Exists:** Always

Table 2-207 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_6

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_LOW_OVRD_VAL	R/W	Override value for mplla_bw_low **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.204 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_7

- **Description:** ATE Override Input to Control Top-Level Inputs Register 7. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x115
- **Exists:** Always

Table 2-208 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_7

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_HIGH_OVRD_VAL	R/W	Override value for mplla_bw_high **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.205 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_8

- **Description:** ATE Override Input to Control Top-Level Inputs Register 8. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x116
- **Exists:** Always

Table 2-209 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_8

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_QUOT_OVRD_ VAL	R/W	Override value for mplla_frac_quot **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.206 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_9

- **Description:** ATE Override Input to Control Top-Level Inputs Register 9. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x117
- **Exists:** Always

Table 2-210 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_9

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_DEN_OVRD_VAL	R/W	Override value for mplla_frac_den **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.207 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_10

- **Description:** ATE Override Input to Control Top-Level Inputs Register 10. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x118
- **Exists:** Always

Table 2-211 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_10

Bits	Name	Memory Access	Description
15:0	MPLLA_FRAC_REM_OVRD_VAL	R/W	Override value for mplla_frac_rem **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.208 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_11

- **Description:** ATE Override Input to Control Top-Level Inputs Register 11. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x119
- **Exists:** Always

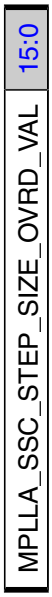


Table 2-212 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_11

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_STEP_SIZE_OVRD_VAL	R/W	Override value for mplla_ssc_step_size **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.209 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_12

- **Description:** ATE Override Input to Control Top-Level Inputs Register 12. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11a
- **Exists:** Always

Table 2-213 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_12

Bits	Name	Memory Access	Description
15:0	MPLLA_SSC_PEAK_OVRD_VAL	R/W	Override value for mplla_ssc_peak **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.210 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_13

- **Description:** ATE Override Input to Control Top-Level Inputs Register 13. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11b
- **Exists:** Always

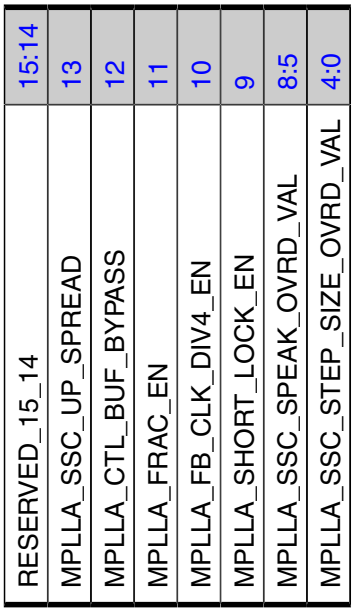


Table 2-214 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_13

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_SSC_UP_SPREAD	R/W	Override value for mplla_ssc_up_spread **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_CTL_BUF_BYPASS	R/W	Override value for mplla_ctl_buf_bypass **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MPLLA_FRAC_EN	R/W	Override value for mplla_frac_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-214 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_13 (continued)

Bits	Name	Memory Access	Description
10	MPLLA_FB_CLK_DIV4_EN	R/W	Override value for mplla_fb_clk_div4_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLA_SHORT_LOCK_EN	R/W	Override value for mplla_short_lock_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:5	MPLLA_SSC_SPEAK_OVRD_ VAL	R/W	Override value for mplla_ssc_peak (4 MSB bits) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
4:0	MPLLA_SSC_STEP_SIZE_OVRD _VAL	R/W	Override value for mplla_ssc_step_size (5 MSB bits) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.211 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_14

- **Description:** ATE Override Input to Control Top-Level Inputs Register 14. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11c
- **Exists:** Always

Table 2-215 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_14

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	REFA_LANE_CLK_EN_OVRD_VAL	R/W	Override value for refa_lane_clk_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	MPLLA_WORD_CLK_DIV_OVRD_VAL	R/W	Override value for mplla_word_clk_div **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.212 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_15

- **Description:** ATE Override Input to Control Top-Level Inputs Register 15. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11d
- **Exists:** Always

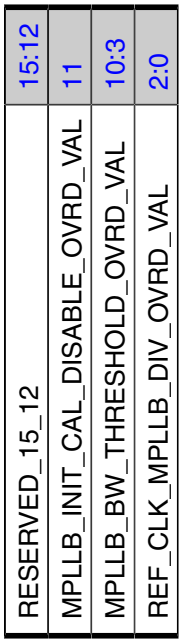


Table 2-216 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_15

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLB_INIT_CAL_DISABLE_OVRD_VAL	R/W	Override value for mpllb_init_cal_disable **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10:3	MPLLB_BW_THRESHOLD_OVRD_VAL	R/W	Override value for mpllb_bw_threshold **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
2:0	REF_CLK_MPLLB_DIV_OVRD_VAL	R/W	Override value for ref_clk_mpllb_div **Value After Reset:** 0x0

Table 2-216 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_15 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7

2.1.213 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_16

- **Description:** ATE Override Input to Control Top-Level Inputs Register 16. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11e
- **Exists:** Always

Table 2-217 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_16

Bits	Name	Memory Access	Description
15:4	MPLLB_MULTIPLIER_OVRD_VAL	R/W	Override value for mpllb_multiplier **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff
3:0	MPLLB_TX_CLK_DIV_OVRD_VAL	R/W	Override value for mpllb_tx_clk_div **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.214 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_17

- **Description:** ATE Override Input to Control Top-Level Inputs Register 17. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x11f
- **Exists:** Always

Table 2-218 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_17

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_LOW_OVRD_VAL	R/W	Override value for mpllb_bw_low **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.215 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_18

- **Description:** ATE Override Input to Control Top-Level Inputs Register 18. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x120
- **Exists:** Always

Table 2-219 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_18

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_HIGH_OVRD_VAL	R/W	Override value for mpllb_bw_high **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.216 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_19

- **Description:** ATE Override Input to Control Top-Level Inputs Register 19. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x121
- **Exists:** Always

Table 2-220 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_19

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_QUOT_OVRD_VAL	R/W	Override value for mpllb_frac_quot **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.217 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_20

- **Description:** ATE Override Input to Control Top-Level Inputs Register 20. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x122
- **Exists:** Always

Table 2-221 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_20

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_DEN_OVRD_VAL	R/W	Override value for mpllb_frac_den **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.218 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_21

- **Description:** ATE Override Input to Control Top-Level Inputs Register 21. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x123
- **Exists:** Always

Table 2-222 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_21

Bits	Name	Memory Access	Description
15:0	MPLLB_FRAC_REM_OVRD_VAL	R/W	Override value for mpllb_frac_rem **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.219 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_22

- **Description:** ATE Override Input to Control Top-Level Inputs Register 22. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x124
- **Exists:** Always

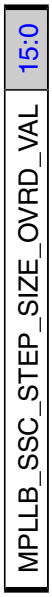


Table 2-223 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_22

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_STEP_SIZE_OVRD_VAL	R/W	Override value for mpllb_ssc_step_size **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.220 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_23

- **Description:** ATE Override Input to Control Top-Level Inputs Register 23. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x125
- **Exists:** Always

Table 2-224 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_23

Bits	Name	Memory Access	Description
15:0	MPLLB_SSC_PEAK_OVRD_VAL	R/W	Override value for mpllb_ssc_peak **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.221 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_24

- **Description:** ATE Override Input to Control Top-Level Inputs Register 24. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x126
- **Exists:** Always

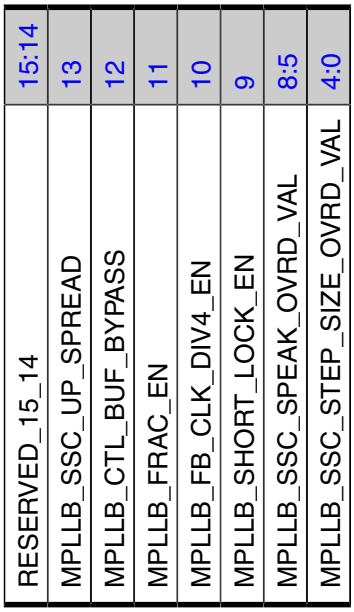


Table 2-225 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_24

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_SSC_UP_SPREAD	R/W	Override value for mpllb_ssc_up_spread **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_CTL_BUF_BYPASS	R/W	Override value for mpllb_ctl_buf_bypass **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MPLLB_FRAC_EN	R/W	Override value for mpllb_frac_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-225 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_24 (continued)

Bits	Name	Memory Access	Description
10	MPLLB_FB_CLK_DIV4_EN	R/W	Override value for mpllb_fb_clk_div4 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLB_SHORT_LOCK_EN	R/W	Override value for mpllb_short_lock_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:5	MPLLB_SSC_SPEAK_OVRD_ VAL	R/W	Override value for mpllb_ssc_peak **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
4:0	MPLLB_SSC_STEP_SIZE_OVRD _VAL	R/W	Override value for mpllb_ssc_step_size **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.222 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_25

- **Description:** ATE Override Input to Control Top-Level Inputs Register 25. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x127
- **Exists:** Always

Table 2-226 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_25

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	REFB_LANE_CLK_EN_OVRD_VAL	R/W	Override value for refb_lane_clk_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	MPLLB_WORD_CLK_DIV_OVRD_VAL	R/W	Override value for mpllb_word_clk_div **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.223 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_26

- **Description:** ATE Override Input to Control Top-Level Inputs Register 26. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x128
- **Exists:** Always

15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_14	MPLLA_INIT_CAL_DISABLE_OVRD_EN	MPLLA_FRAC_QUOT_EN_OVRD_EN	MPLLA_BW_HIGH_EN_OVRD_EN	MPLLA_BW_LOW_EN_OVRD_EN	MPLLA_MULTIPLIER_OVRD_EN	MPLLA_TX_CLK_DIV_EN_OVRD_EN	MPLLA_BW_THRESHOLD_OVRD_EN	REFA_LANE_CLK_EN_OVRD_EN	REF_CLK_MPLLA_DIV_OVRD_EN	SUP_MISC_OVRD_EN	RX_VREF_CTRL_OVRD_EN	TXDN_TERM_OFFSET_OVRD_EN	RX_TERM_OFFSET_OVRD_EN	TXUP_TERM_OFFSET_OVRD_EN

Table 2-227 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_26

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_INIT_CAL_DISABLE_OVRD_EN	R/W	Override value for mplla_init_cal_disable **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_FRAC_QUOT_EN_OVRD_EN	R/W	Override value for mplla_frac_quot_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
11	MPLLA_BW_HIGH_EN_OVRD_EN	R/W	Override value for mplla_bw_high_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-227 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_26 (continued)

Bits	Name	Memory Access	Description
10	MPLLA_BW_LOW_EN_OVRD_EN	R/W	Override value for mplla_bw_low_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	MPLLA_MULTIPLIER_OVRD_EN	R/W	Override value for mplla_multiplier_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	MPLLA_TX_CLK_DIV_EN_OVRD_EN	R/W	Override value for mplla_tx_clk_div_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	MPLLA_BW_THRESHOLD_OVRD_EN	R/W	Override value for mplla_bw_threshold_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	REFA_LANE_CLK_EN_OVRD_EN	R/W	Override value for refa_lane_clk_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	REF_CLK_MPLLA_DIV_OVRD_EN	R/W	Override value for ref_clk_mplla_div_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	SUP_MISC_OVRD_EN	R/W	Override value for sup_misc_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_VREF_CTRL_OVRD_EN	R/W	Override value for rx_vref_ctrl_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	TXDN_TERM_OFFSET_OVRD_EN	R/W	Override value for txdn_term_offset_en **Value After Reset:** 0x1

Table 2-227 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_26 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
1	RX_TERM_OFFSET_OVRD_EN	R/W	Override value for rx_term_offset_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TXUP_TERM_OFFSET_OVRD_EN	R/W	Override value for txup_term_offset_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.224 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_27

- **Description:** ATE Override Input to Control Top-Level Inputs Register 27. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x129
- **Exists:** Always

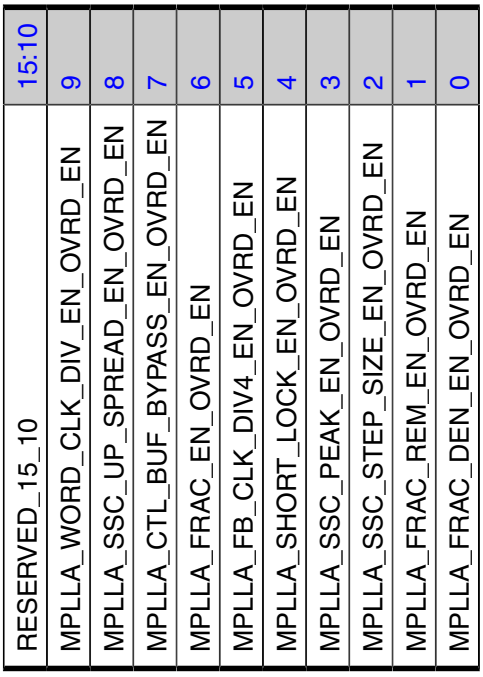


Table 2-228 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_27

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	MPLLA_WORD_CLK_DIV_EN_OVRD_EN	R/W	Override value for mplla_word_clk_div_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	MPLLA_SSC_UP_SPREAD_EN_OVRD_EN	R/W	Override value for mplla_ssc_up_spread_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	MPLLA_CTL_BUF_BYPASS_EN_OVRD_EN	R/W	Override value for mplla_ctl_buf_bypass_en **Value After Reset:** 0x1

Table 2-228 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_27 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
6	MPLLA_FRAC_EN_OVRD_EN	R/W	Override value for mplla_frac_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	MPLLA_FB_CLK_DIV4_EN_OVRD_EN	R/W	Override value for mplla_fb_clk_div4_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	MPLLA_SHORT_LOCK_EN_OVRD_EN	R/W	Override value for mplla_short_lock_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	MPLLA_SSC_PEAK_EN_OVRD_EN	R/W	Override value for mplla_ssc_peak_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	MPLLA_SSC_STEP_SIZE_EN_OVRD_EN	R/W	Override value for mplla_ssc_step_size_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_FRAC_REM_EN_OVRD_EN	R/W	Override value for mplla_frac_rem_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_FRAC_DEN_EN_OVRD_EN	R/W	Override value for mplla_frac_den_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.225 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_28

- **Description:** ATE Override Input to Control Top-Level Inputs Register 28. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x12a
- **Exists:** Always

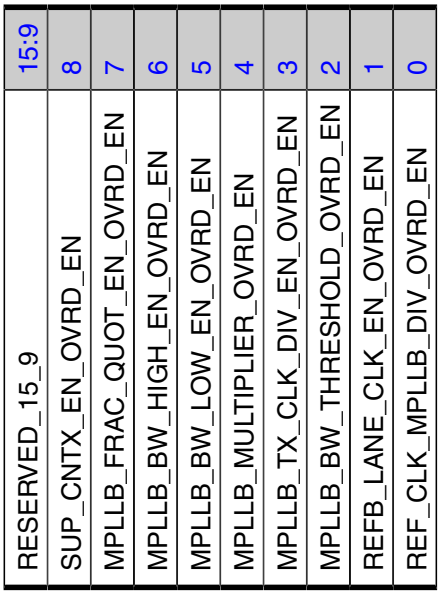


Table 2-229 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_28

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	SUP_CNTX_EN_OVRD_EN	R/W	Override value for sup_cntx_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	MPLLB_FRAC_QUOT_EN_OVRD_EN	R/W	Override value for mpllb_frac_quot_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	MPLLB_BW_HIGH_EN_OVRD_EN	R/W	Override value for mpllb_bw_high_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-229 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_28 (continued)

Bits	Name	Memory Access	Description
5	MPLLB_BW_LOW_EN_OVRD_EN	R/W	Override value for mpllb_bw_low_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	MPLLB_MULTIPLIER_OVRD_EN	R/W	Override value for mpllb_multiplier_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_TX_CLK_DIV_EN_OVRD_EN	R/W	Override value for mpllb_tx_clk_div_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_BW_THRESHOLD_OVRD_EN	R/W	Override value for mpllb_bw_threshold_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	REFB_LANE_CLK_EN_OVRD_EN	R/W	Override value for refb_lane_clk_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	REF_CLK_MPLLB_DIV_OVRD_EN	R/W	Override value for ref_clk_mpllb_div_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.226 RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_29

- **Description:** ATE Override Input to Control Top-Level Inputs Register 29. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x12b
- **Exists:** Always

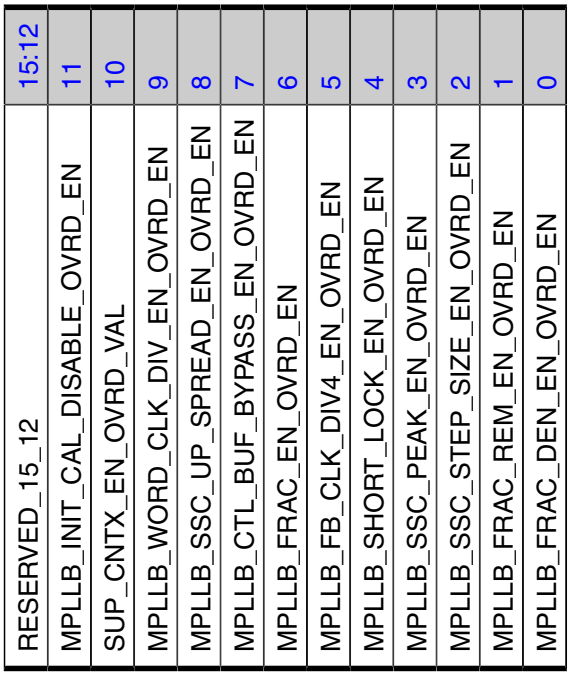


Table 2-230 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_29

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	MPLLB_INIT_CAL_DISABLE_OVRD_EN	R/W	Override value for mpllb_init_cal_disable **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	SUP_CNTX_EN_OVRD_VAL	R/W	Override value for sup_cntx_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	MPLLB_WORD_CLK_DIV_EN_OVRD_EN	R/W	Override value for mpllb_word_clk_div_en **Value After Reset:** 0x1

Table 2-230 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_29 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8	MPLLB_SSC_UP_SPREAD_EN_OVRD_EN	R/W	Override value for mpllb_ssc_up_spread_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	MPLLB_CTL_BUF_BYPASS_EN_OVRD_EN	R/W	Override value for mpllb_ctl_buf_bypass_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	MPLLB_FRAC_EN_OVRD_EN	R/W	Override value for mpllb_frac_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	MPLLB_FB_CLK_DIV4_EN_OVRD_EN	R/W	Override value for mpllb_fb_clk_div4_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	MPLLB_SHORT_LOCK_EN_OVRD_EN	R/W	Override value for mpllb_short_lock_en_ovrd_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_SSC_PEAK_EN_OVRD_EN	R/W	Override value for mpllb_ssc_peak_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_SSC_STEP_SIZE_EN_OVRD_EN	R/W	Override value for mpllb_ssc_step_size_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_FRAC_REM_EN_OVRD_EN	R/W	Override value for mpllb_frac_rem_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-230 Fields for Register: RAWCMN_DIG_CMN_PACKED_CNTX_SUP_RO_OVRD_IN_29 (continued)

Bits	Name	Memory Access	Description
0	MPLLB_FRAC_DEN_EN_OVRD_EN	R/W	Override value for mpllb_frac_den_en **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.227 RAWCMN_DIG_NUMBER_OF_LANES

- **Description:** Stores Number of Lanes in PHY.
- **Size:** 16 bits
- **Offset:** 0x12c
- **Exists:** Always

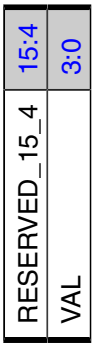


Table 2-231 Fields for Register: RAWCMN_DIG_NUMBER_OF_LANES

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	VAL	R	These bits store the number of lanes in the PHY. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.228 RAWCMN_DIG_CONFIG_MASTER_VERSION

- **Description:** The current config master version used.
- **Size:** 16 bits
- **Offset:** 0x12d
- **Exists:** Always

Table 2-232 Fields for Register: RAWCMN_DIG_CONFIG_MASTER_VERSION

Bits	Name	Memory Access	Description
15:0	CONFIG_VERSION	R	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.229 RAWCMN_DIG_METADATA_LOCATION

- **Description:** Override register for location of metadata in memory
- **Size:** 16 bits
- **Offset:** 0x12e
- **Exists:** Always

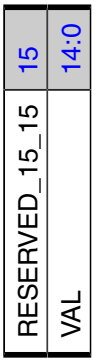


Table 2-233 Fields for Register: RAWCMN_DIG_METADATA_LOCATION

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	VAL	R/W	lower 15-bits. The upper two bits are determined by sram_bypass signal. **Value After Reset:** 0x7fff **Exists:** Always **Reset Mask:** 0x7fff

2.1.230 RAWCMN_DIG_CNTX_RSTR_CLIENTS_MASK

- **Description:** Override register for client masks used by context restore module
- **Size:** 16 bits
- **Offset:** 0x12f
- **Exists:** Always

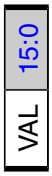


Table 2-234 Fields for Register: RAWCMN_DIG_CNTX_RSTR_CLIENTS_MASK

Bits	Name	Memory Access	Description
15:0	VAL	R/W	**Value After Reset:** 0x200 **Exists:** Always **Reset Mask:** 0xffff

2.1.231 RAWCMN_DIG_MEM_POWER_SAVING

- **Description:** Enables disabling the SRAM clk when SRAM isn't used to save power.
- **Size:** 16 bits
- **Offset:** 0x130
- **Exists:** Always

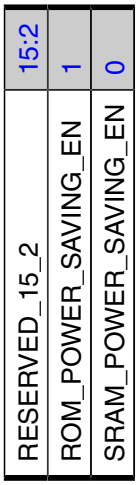


Table 2-235 Fields for Register: RAWCMN_DIG_MEM_POWER_SAVING

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	ROM_POWER_SAVING_EN	R/W	When this bit is set, ROM clk is disabled when it's not needed (no transactions) to save power. Clear this bit to keep the ROM clk always ON **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	SRAM_POWER_SAVING_EN	R/W	When this bit is set, SRAM clk is disabled when it's not needed (no transactions) to save power. Clear this bit to keep the SRAM clk always ON **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.232 RAWCMN_DIG_FW_PWRUP_DONE_B

- **Description:** Firmware Power-up Done_B Status.
- **Size:** 16 bits
- **Offset:** 0x131
- **Exists:** Always

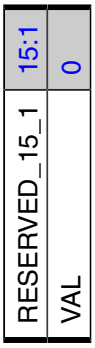


Table 2-236 Fields for Register: RAWCMN_DIG_FW_PWRUP_DONE_B

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	Partial FW Power-up Status for MPLLB after phy_reset and pg_reset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.233 RAWCMN_DIG_TOKEN_DLY

- **Description:** Time Spent By a Token In a Lane.
- **Size:** 16 bits
- **Offset:** 0x132
- **Exists:** Always

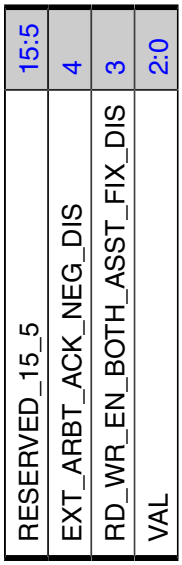


Table 2-237 Fields for Register: RAWCMN_DIG_TOKEN_DLY

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	EXT_ARBT_ACK_NEG_DIS	R/W	This bit is the checken bit to disable using negedge of rd/wr_ack in pcatch **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RD_WR_EN_BOTH_ASST_FIX_DIS	R/W	This bit is the checken bit to disable fix of rd/wr_en both asserted in ext_arbt **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	VAL	R/W	These bits are the time spent by the token in a lane. Value 001 is a limitation. It should be configured only once after reset. **Value After Reset:** 0x0

Table 2-237 Fields for Register: RAWCMN_DIG_TOKEN_DLY (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7

2.1.234 RAWCMN_DIG_RTUNE_REQ

- **Description:** The read register for rtune_req.
- **Size:** 16 bits
- **Offset:** 0x133
- **Exists:** Always

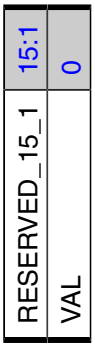


Table 2-238 Fields for Register: RAWCMN_DIG_RTUNE_REQ

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R	The read value for rtune_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.235 RAWCMN_DIG_PWR_EN

- **Description:** power_en new register to sequence isolation
- **Size:** 16 bits
- **Offset:** 0x134
- **Exists:** Always

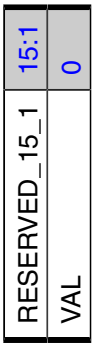


Table 2-239 Fields for Register: RAWCMN_DIG_PWR_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	**Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.236 RAWCMN_DIG_MPLL_FORCE_ACK

- **Description:** The read register for mplla/b_force_ack.
- **Size:** 16 bits
- **Offset:** 0x135
- **Exists:** Always

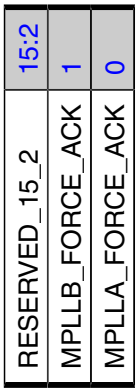


Table 2-240 Fields for Register: RAWCMN_DIG_MPLL_FORCE_ACK

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLLB_FORCE_ACK	R	The read value for mpllb_force_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLA_FORCE_ACK	R	The read value for mplla_force_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.237 RAWCMN_DIG_MPLL_SSC_EN

- **Description:** overwrite register for mplla/b_ssc_en before sending to pma.
- **Size:** 16 bits
- **Offset:** 0x136
- **Exists:** Always

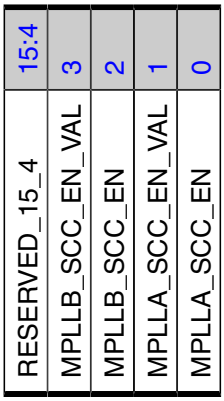


Table 2-241 Fields for Register: RAWCMN_DIG_MPLL_SSC_EN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	MPLLB_SCC_EN_VAL	R/W	override value for mpllb_ssc_en_val **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_SCC_EN	R/W	override value for mpllb_ssc_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_SCC_EN_VAL	R/W	override value for mplla_ssc_en_val **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_SCC_EN	R/W	override value for mplla_ssc_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.238 RAWCMN_DIG_MPLLA_FORCE_CTL_STATUS

- **Description:** The read register for mplla_force_ctl_status
- **Size:** 16 bits
- **Offset:** 0x137
- **Exists:** Always

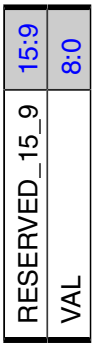


Table 2-242 Fields for Register: RAWCMN_DIG_MPLLA_FORCE_CTL_STATUS

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R	mplla_force_ctl status (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.239 RAWCMN_DIG_MPLLB_FORCE_CTL_STATUS

- **Description:** The read register for mpllb_force_ctl_status
- **Size:** 16 bits
- **Offset:** 0x138
- **Exists:** Always

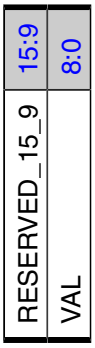


Table 2-243 Fields for Register: RAWCMN_DIG_MPLLB_FORCE_CTL_STATUS

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R	mpllb_force_ctl status (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.240 RAWCMN_DIG_CNTX_CTL_STATUS_I

- **Description:** The read register for monitoring maing cntx rstr signals.(for debug purposes)
- **Size:** 16 bits
- **Offset:** 0x139
- **Exists:** Always

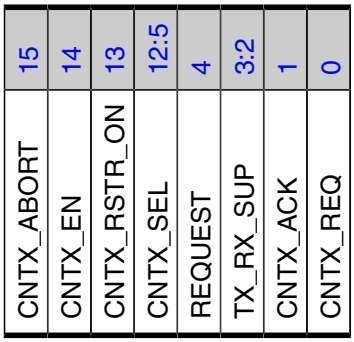


Table 2-244 Fields for Register: RAWCMN_DIG_CNTX_CTL_STATUS_I

Bits	Name	Memory Access	Description
15	CNTX_ABORT	R	shows an aobrt condition has happened so the context restore should stop (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	CNTX_EN	R	shows the context restore core is enabled (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	CNTX_RSTR_ON	R	when asserted shows the context restore is active (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:5	CNTX_SEL	R	the selected context number in the image (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-244 Fields for Register: RAWCMN_DIG_CNTX_CTL_STATUS_I (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
4	REQUEST	R	shows the restoration request has been sent to context restore core (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:2	TX_RX_SUP	R	shows the type of context being restored (00: tx, 01: rx, 10: sup) (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	CNTX_ACK	R	The read value for cntx_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	CNTX_REQ	R	The read value for cntx_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.241 RAWCMN_DIG_CNTX_CTL_STATUS_II

- **Description:** The read register for monitoring maing cntx rstr signals.(for debug purposes)
- **Size:** 16 bits
- **Offset:** 0x13a
- **Exists:** Always

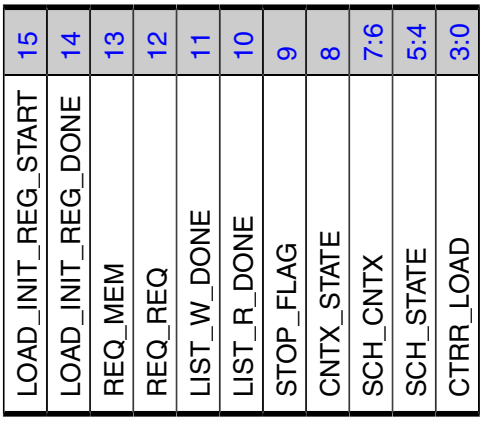


Table 2-245 Fields for Register: RAWCMN_DIG_CNTX_CTL_STATUS_II

Bits	Name	Memory Access	Description
15	LOAD_INIT_REG_START	R	The read value for load_init_reg_start (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	LOAD_INIT_REG_DONE	R	The read value for load_init_reg_done (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	REQ_MEM	R	The read value for req_mem (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	REQ_REQ	R	The read value for req_reg (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-245 Fields for Register: RAWCMN_DIG_CNTX_CTL_STATUS_II (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11	LIST_W_DONE	R	The read value for list_w_done (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	LIST_R_DONE	R	The read value for list_r_done (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	STOP_FLAG	R	The read value for STOP_flag (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	CNTX_STATE	R	The read value for cntx_state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:6	SCH_CNTX	R	The read value for sch_cntx (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:4	SCH_STATE	R	The read value for sch_state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-245 Fields for Register: RAWCMN_DIG_CNTX_CTL_STATUS_II (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
3:0	CTRR_LOAD	R	The read value for cntr_load (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.242 RAWCMN_DIG_MPLL_STATE_OVRD

- **Description:** Mpll_state ovrd
- **Size:** 16 bits
- **Offset:** 0x13b
- **Exists:** Always

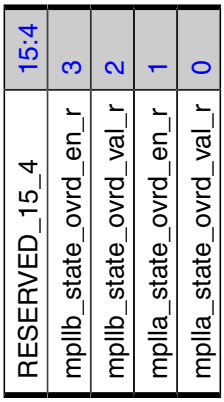


Table 2-246 Fields for Register: RAWCMN_DIG_MPLL_STATE_OVRD

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	mpllb_state_ovrd_en_r	R/W	mpllb_state_ovrd enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	mpllb_state_ovrd_val_r	R/W	mpllb_state_ovrd value **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	mplla_state_ovrd_en_r	R/W	mplla_state_ovrd enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	mplla_state_ovrd_val_r	R/W	mplla_state_ovrd value **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.243 RAWCMN_DIG_DCOCTRL_STATUS

- **Description:** The read register for monitoring main DCO signals.
- **Size:** 16 bits
- **Offset:** 0x13c
- **Exists:** Always

Table 2-247 Fields for Register: RAWCMN_DIG_DCOCTRL_STATUS

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	DCO_CALDONE	R	The read value for dco_caldone (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:0	DCO_FINETUNE	R	The read value for dco_finetune (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.244 RAWCMN_DIG_DCOCTRL_STATUS_2

- **Description:** The read register for monitoring main DCO signals.
- **Size:** 16 bits
- **Offset:** 0x13d
- **Exists:** Always

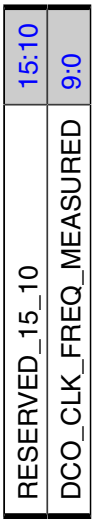


Table 2-248 Fields for Register: RAWCMN_DIG_DCOCTRL_STATUS_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	DCO_CLK_FREQ_MEASURED	R	The read value for the measured DCO clock frequency. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.245 RAWCMN_DIG_MPLL_PMIX_EN

- **Description:** overwrite register for mplla/b_pmix_en before sending to pma.
- **Size:** 16 bits
- **Offset:** 0x13f
- **Exists:** Always

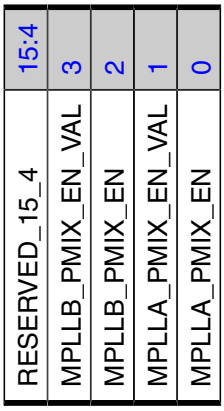


Table 2-249 Fields for Register: RAWCMN_DIG_MPLL_PMIX_EN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	MPLLB_PMIX_EN_VAL	R/W	override value for mpllb_pmix_en_val **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_PMIX_EN	R/W	override value for mpllb_pmix_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_PMIX_EN_VAL	R/W	override value for mplla_pmix_en_val **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_PMIX_EN	R/W	override value for mplla_pmix_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.246 RAWCMN_DIG_AON_SRAM_PGATE_BL_EN

- **Description:** Enable SRAM bootloader on power-gated exit.
- **Size:** 16 bits
- **Offset:** 0x160
- **Exists:** Always

Table 2-250 Fields for Register: RAWCMN_DIG_AON_SRAM_PGATE_BL_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SRAM_PGATE_BL_EN	R/W	This bit enables the SRAM bootloader on power-gated exit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.247 RAWCMN_DIG_AON_MPLLA_TUNE_BANK_0

- **Description:** MPLLA Tune Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x161
- **Exists:** Always

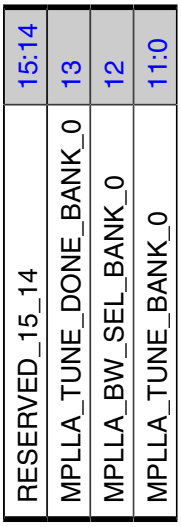


Table 2-251 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE_BANK_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_TUNE_DONE_BANK_0	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_BW_SEL_BANK_0	R/W	This bit is the stored Bank 0 bw_sel value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLA_TUNE_BANK_0	R/W	These bits are the stored Bank 0 tune value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.248 RAWCMN_DIG_AON_MPLLA_TUNE_BANK_1

- **Description:** MPLLA Tune Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x162
- **Exists:** Always

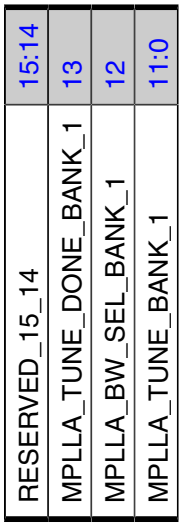


Table 2-252 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE_BANK_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_TUNE_DONE_BANK_1	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_BW_SEL_BANK_1	R/W	This bit is the stored Bank 1 bw_sel value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLA_TUNE_BANK_1	R/W	These bits are the stored Bank 1 tune value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.249 RAWCMN_DIG_AON_MPLLA_TUNE_BANK_2

- **Description:** MPLLA Tune Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x163
- **Exists:** Always

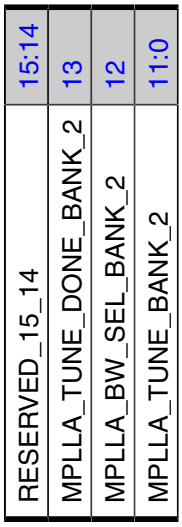


Table 2-253 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE_BANK_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_TUNE_DONE_BANK_2	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_BW_SEL_BANK_2	R/W	This bit is the stored Bank 2 bw_sel value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLA_TUNE_BANK_2	R/W	These bits are the stored Bank 2 tune value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.250 RAWCMN_DIG_AON_MPLLA_TUNE_BANK_3

- **Description:** MPLLA Tune Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x164
- **Exists:** Always

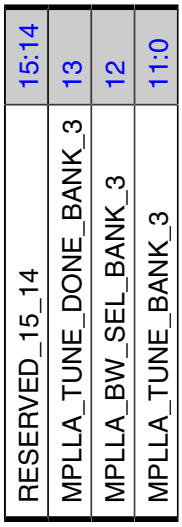


Table 2-254 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE_BANK_3

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLA_TUNE_DONE_BANK_3	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLA_BW_SEL_BANK_3	R/W	This bit is the stored Bank 2 bw_sel value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLA_TUNE_BANK_3	R/W	These bits are the stored Bank 3 tune value for the MPLLA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.251 RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL

- **Description:** Value for MPLLA Bank Select.
- **Size:** 16 bits
- **Offset:** 0x165
- **Exists:** Always

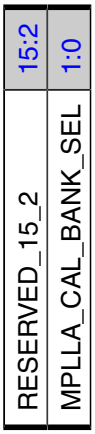


Table 2-255 Fields for Register: RAWCMN_DIG_AON_MPLLA_CAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	MPLLA_CAL_BANK_SEL	R	These bits are the value for the MPLLA Bank select. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.252 RAWCMN_DIG_AON_MPLLA_TUNE_DONE

- **Description:** MPLLA Calibration Bank Valid Calibrated Value.
- **Size:** 16 bits
- **Offset:** 0x166
- **Exists:** Always

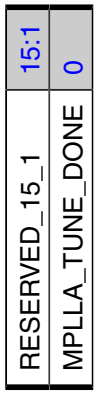


Table 2-256 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	MPLLA_TUNE_DONE	R	This bit indicates if the selected MPLLA calibration bank has the valid calibrated value. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.253 RAWCMN_DIG_AON_MPLLB_TUNE_BANK_0

- **Description:** MPLLB Tune Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x167
- **Exists:** Always

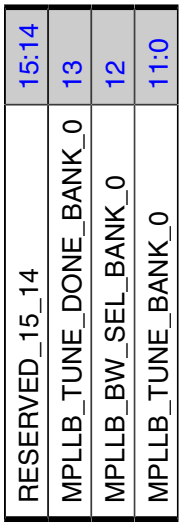


Table 2-257 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE_BANK_0

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_TUNE_DONE_BANK_0	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_BW_SEL_BANK_0	R/W	This bit stores the Bank 0 bw_sel value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLB_TUNE_BANK_0	R/W	This bit stores the Bank 0 tune value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.254 RAWCMN_DIG_AON_MPLLB_TUNE_BANK_1

- **Description:** MPLLB Tune Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x168
- **Exists:** Always

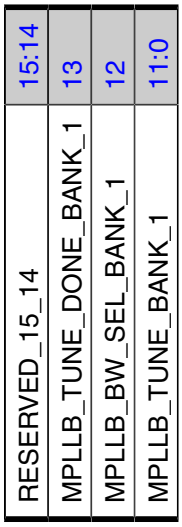


Table 2-258 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE_BANK_1

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_TUNE_DONE_BANK_1	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_BW_SEL_BANK_1	R/W	This bit stores the Bank 1 bw_sel value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLB_TUNE_BANK_1	R/W	These bits store the Bank 1 tune value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.255 RAWCMN_DIG_AON_MPLLB_TUNE_BANK_2

- **Description:** MPLLB Tune Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x169
- **Exists:** Always

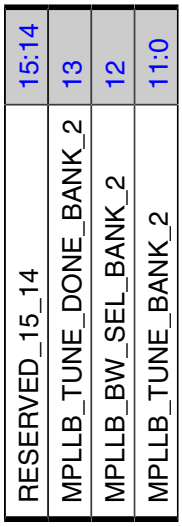


Table 2-259 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE_BANK_2

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_TUNE_DONE_BANK_2	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_BW_SEL_BANK_2	R/W	This bit stores the Bank 2 bw_sel value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLB_TUNE_BANK_2	R/W	These bits store the Bank 2 tune value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.256 RAWCMN_DIG_AON_MPLLB_TUNE_BANK_3

- **Description:** MPLLB Tune Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x16a
- **Exists:** Always

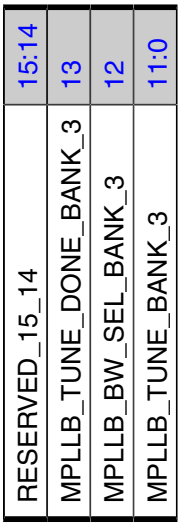


Table 2-260 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE_BANK_3

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MPLLB_TUNE_DONE_BANK_3	R/W	This bit indicates if the tune value is a calibrated value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MPLLB_BW_SEL_BANK_3	R/W	This bit stores the Bank 3 bw_sel value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	MPLLB_TUNE_BANK_3	R/W	These bits store the Bank 3 tune value for MPLLB. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.257 RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL

- **Description:** MPLLB Bank Select Value.
- **Size:** 16 bits
- **Offset:** 0x16b
- **Exists:** Always

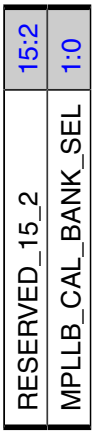


Table 2-261 Fields for Register: RAWCMN_DIG_AON_MPLLB_CAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	MPLLB_CAL_BANK_SEL	R	These bits are the value for the MPLLB Bank select. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.258 RAWCMN_DIG_AON_MPLLB_TUNE_DONE

- **Description:** MPLLB Calibration Bank Valid Calibrated Value.
- **Size:** 16 bits
- **Offset:** 0x16c
- **Exists:** Always

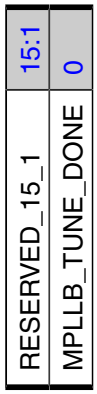


Table 2-262 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE_DONE

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	MPLLB_TUNE_DONE	R	This bit indicates if the selected MPLLB calibration bank has the valid calibrated value. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.259 RAWCMN_DIG_AON_MPLLA_TUNE

- **Description:** MPLLA Selected TUNE Value.
- **Size:** 16 bits
- **Offset:** 0x16e
- **Exists:** Always

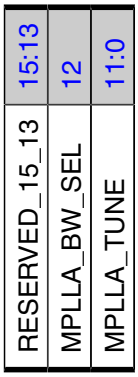


Table 2-263 Fields for Register: RAWCMN_DIG_AON_MPLLA_TUNE

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLLA_BW_SEL	R	This bit is the selected bw_sel value for MPLLA. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	MPLLA_TUNE	R	These bits are the selected tune value for MPLLA. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.260 RAWCMN_DIG_AON_MPLLB_TUNE

- **Description:** MPLLB Selected TUNE Value.
- **Size:** 16 bits
- **Offset:** 0x16f
- **Exists:** Always

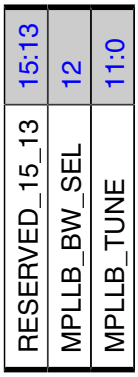


Table 2-264 Fields for Register: RAWCMN_DIG_AON_MPLLB_TUNE

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MPLLB_BW_SEL	R	This bit is the selected bw_sel value for MPLLB. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	MPLLB_TUNE	R	These bits are the selected tune value for MPLLB. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.261 RAWCMN_DIG_AON_MPLLA_IN_RECAL

- **Description:** MPLLA Recalibration.
- **Size:** 16 bits
- **Offset:** 0x170
- **Exists:** Always

Table 2-265 Fields for Register: RAWCMN_DIG_AON_MPLLA_IN_RECAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	MPLLA_IN_RECAL	R/W	This bit indicates if the MPLLA is in recalibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.262 RAWCMN_DIG_AON_MPLLB_IN_RECAL

- **Description:** MPLLB Recalibration.
- **Size:** 16 bits
- **Offset:** 0x171
- **Exists:** Always

Table 2-266 Fields for Register: RAWCMN_DIG_AON_MPLLB_IN_RECAL

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	MPLLB_IN_RECAL	R/W	This bit indicates if MPLLB is in recalibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.263 RAWCMN_DIG_AON_PG_OVRD_IN

- **Description:** Override Values for Incoming Power-gating Signals.
- **Size:** 16 bits
- **Offset:** 0x172
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	BG_EN_OVRD_EN	BG_EN_OVRD_VAL	PG_MODE_EN_OVRD_EN	PG_MODE_EN_OVRD_VAL	PG_RESET_OVRD_EN	PG_RESET_OVRD_VAL	PCS_PWR_STABLE_OVRD	PMA_PWR_STABLE_OVRD

Table 2-267 Fields for Register: RAWCMN_DIG_AON_PG_OVRD_IN

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	BG_EN_OVRD_EN	R/W	Override enable for bg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	BG_EN_OVRD_VAL	R/W	Override value for bg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	PG_MODE_EN_OVRD_EN	R/W	This bit is the override enable for pg_mode_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	PG_MODE_EN_OVRD_VAL	R/W	This bit is the override value for pg_mode_en. **Value After Reset:** 0x0

Table 2-267 Fields for Register: RAWCMN_DIG_AON_PG_OVRD_IN (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	PG_RESET_OVRD_EN	R/W	This bit is the override enable for pg_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	PG_RESET_OVRD_VAL	R/W	This bit is the override value for pg_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	PCS_PWR_STABLE_OVRD	R/W	This bit enables the overriding value for pcs_pwr_stable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	PMA_PWR_STABLE_OVRD	R/W	This bit enables the overriding value for pma_pwr_stable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.264 RAWCMN_DIG_AON_PG_OVRD_OUT

- **Description:** Override Values for Outgoing Power-gating Signals.
- **Size:** 16 bits
- **Offset:** 0x173
- **Exists:** Always

Table 2-268 Fields for Register: RAWCMN_DIG_AON_PG_OVRD_OUT

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	PCS_PWR_EN_OVRD	R/W	This bit enables the overriding value for pcs_pwr_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	PMA_PWR_EN_OVRD	R/W	This bit enables the overriding value for pma_pwr_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.265 RAWCMN_DIG_AON_SUP_OVRD_IN

- **Description:** Override Values for Incoming Support Block Signals.
- **Size:** 16 bits
- **Offset:** 0x174
- **Exists:** Always

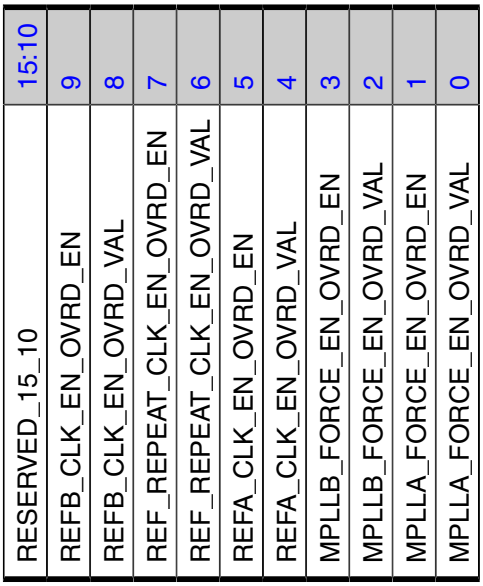


Table 2-269 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_IN

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	REFB_CLK_EN_OVRD_EN	R/W	This bit is the override enable for refb_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	REFB_CLK_EN_OVRD_VAL	R/W	This bit is the override value for refb_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	REF_REPEAT_CLK_EN_OVRD_EN	R/W	This bit is the override enable for ref_repeat_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-269 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_IN (continued)

Bits	Name	Memory Access	Description
6	REF_REPEAT_CLK_EN_OVRD_VAL	R/W	This bit is the override value for ref_repeat_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	REFA_CLK_EN_OVRD_EN	R/W	This bit is the override enable for refa_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	REFA_CLK_EN_OVRD_VAL	R/W	This bit is the override value for refa_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLLB_FORCE_EN_OVRD_EN	R/W	This bit is the override enable for mpllb_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for mpllb_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_FORCE_EN_OVRD_EN	R/W	This bit is the override enable for mplla_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	MPLLA_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for mplla_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.266 RAWCMN_DIG_AON_SUP_OVRD_OUT

- **Description:** Override Values for Outgoing Support Block Signals.
- **Size:** 16 bits
- **Offset:** 0x175
- **Exists:** Always

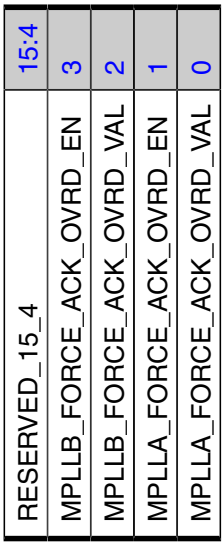


Table 2-270 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_OUT

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	MPLLB_FORCE_ACK_OVRD_EN	R/W	This bit is the override enable for the mpllb_force_ack output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLLB_FORCE_ACK_OVRD_ VAL	R/W	This bit is the override value for the mpllb_force_ack output. Its override signal is SUP_OVRD_OUT.MPLLB_FORCE_ACK_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLA_FORCE_ACK_OVRD_EN	R/W	This bit is the override enable for the mplla_force_ack output. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-270 Fields for Register: RAWCMN_DIG_AON_SUP_OVRD_OUT (continued)

Bits	Name	Memory Access	Description
0	MPLLA_FORCE_ACK_OVRD_VAL	R/W	This bit is the override value for the mplla_force_ack output. Its override signal is SUP_OVRD_OUT.MPLLA_FORCE_ACK_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.267 RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL

- **Description:** Current Bank Selected for MPLLA Coarse Tune in PMA.
- **Size:** 16 bits
- **Offset:** 0x176
- **Exists:** Always

Table 2-271 Fields for Register: RAWCMN_DIG_AON_PMA_MPLLA_RECAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	PMA_MPLLA_RECAL_BANK_SEL	R/W	These bits are the current bank selected for the MPLLA coarse tune in the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.268 RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL

- **Description:** Current Bank Selected for MPLLB Coarse Tune in PMA.
- **Size:** 16 bits
- **Offset:** 0x177
- **Exists:** Always

Table 2-272 Fields for Register: RAWCMN_DIG_AON_PMA_MPLLB_RECAL_BANK_SEL

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	PMA_MPLLB_RECAL_BANK_ SEL	R/W	These bits are the current bank selected for the MPLLB coarse tune in the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.269 RAWCMN_DIG_AON_RTUNE_RX_VAL

- **Description:** Resistor Tune RX Value.
- **Size:** 16 bits
- **Offset:** 0x178
- **Exists:** Always

Table 2-273 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL	R/W	These bits are the stored resistor tune RX value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.270 RAWCMN_DIG_AON_RTUNE_TXDN_VAL

- **Description:** Resistor Tune TX Down Value.
- **Size:** 16 bits
- **Offset:** 0x179
- **Exists:** Always

Table 2-274 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL	R/W	These bits are the stored resistor tune TX down value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.271 RAWCMN_DIG_AON_RTUNE_TXUP_VAL

- **Description:** Resistor Tune TX Up Value.
- **Size:** 16 bits
- **Offset:** 0x17a
- **Exists:** Always

Table 2-275 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL	R/W	These bits are the stored resistor tune TX up value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.272 RAWCMN_DIG_AON_CMNCAL_MPLL_STATUS

- **Description:** MPLL Common Calibration Status.
- **Size:** 16 bits
- **Offset:** 0x17b
- **Exists:** Always

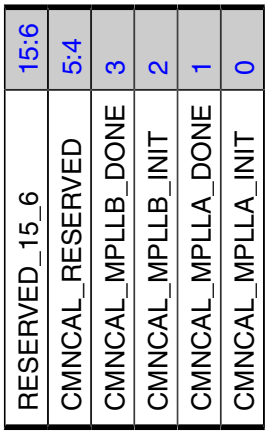


Table 2-276 Fields for Register: RAWCMN_DIG_AON_CMNCAL_MPLL_STATUS

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:4	CMNCAL_RESERVED	R/W	These bits are common calibration reserved bits by FW. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	CMNCAL_MPLLB_DONE	R/W	This bit indicates whether MPLLB common calibration has been completed by this lane. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	CMNCAL_MPLLB_INIT	R/W	This bit indicates whether MPLLB common calibration has been started by this lane. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	CMNCAL_MPLLA_DONE	R/W	This bit indicates whether MPLLA common calibration has been completed by this lane. **Value After Reset:** 0x0

Table 2-276 Fields for Register: RAWCMN_DIG_AON_CMNCAL_MPLL_STATUS (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	CMNCAL_MPLLA_INIT	R/W	This bit indicates whether MPLLA common calibration has been started by this lane. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.273 RAWCMN_DIG_AON_CMNCAL_RTUNE_STATUS

- **Description:** RTUNE Common Calibration Status.
- **Size:** 16 bits
- **Offset:** 0x17c
- **Exists:** Always

Table 2-277 Fields for Register: RAWCMN_DIG_AON_CMNCAL_RTUNE_STATUS

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	CMNCAL_RTUNE_DONE	R/W	This bit indicates whether RTUNE common calibration has been completed by this lane. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	CMNCAL_RTUNE_INIT	R/W	This bit indicates whether RTUNE common calibration has been started by this lane. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.274 RAWCMN_DIG_AON_RTUNE_RX_VAL_0

- **Description:** Resistor Tune RX Value 0.
- **Size:** 16 bits
- **Offset:** 0x17d
- **Exists:** Always

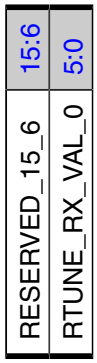


Table 2-278 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_0

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_0	R/W	These bits store the resistor tune RX value 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.275 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_0

- **Description:** Resistor Tune TX Down Value 0.
- **Size:** 16 bits
- **Offset:** 0x17e
- **Exists:** Always

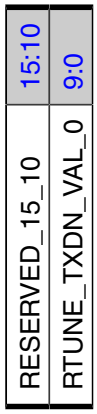


Table 2-279 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_0	R/W	These bits store the resistor tune TX down value 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.276 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_0

- **Description:** Resistor Tune TX Up Value 0.
- **Size:** 16 bits
- **Offset:** 0x17f
- **Exists:** Always

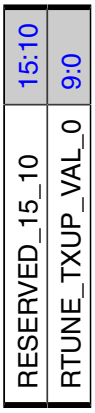


Table 2-280 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_0

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_0	R/W	These bits store the resistor tune TX up value 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.277 RAWCMN_DIG_AON_RTUNE_RX_VAL_1

- **Description:** Resistor Tune RX Value 1.
- **Size:** 16 bits
- **Offset:** 0x180
- **Exists:** Always

Table 2-281 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_1

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_1	R/W	These bits store the resistor tune RX value 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.278 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_1

- **Description:** Resistor Tune TX Down Value 1.
- **Size:** 16 bits
- **Offset:** 0x181
- **Exists:** Always

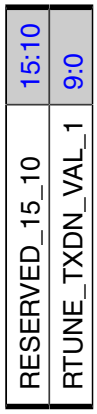


Table 2-282 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_1	R/W	These bits store the resistor tune TX down value 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.279 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_1

- **Description:** Resistor Tune TX Up Value 1.
- **Size:** 16 bits
- **Offset:** 0x182
- **Exists:** Always

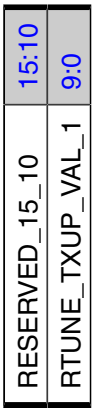


Table 2-283 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_1

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_1	R/W	These bits store the resistor tune TX up value 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.280 RAWCMN_DIG_AON_RTUNE_RX_VAL_2

- **Description:** Resistor Tune RX Value 2.
- **Size:** 16 bits
- **Offset:** 0x183
- **Exists:** Always

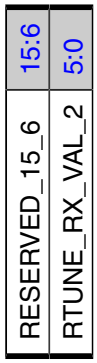


Table 2-284 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_2

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_2	R/W	These bits store the resistor tune RX value 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.281 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_2

- **Description:** Resistor Tune TX Down Value 2.
- **Size:** 16 bits
- **Offset:** 0x184
- **Exists:** Always

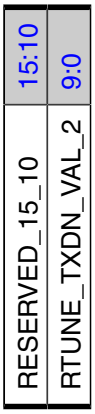


Table 2-285 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_2	R/W	These bits store the resistor tune TX down value 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.282 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_2

- **Description:** Resistor Tune TX Up Value 2.
- **Size:** 16 bits
- **Offset:** 0x185
- **Exists:** Always

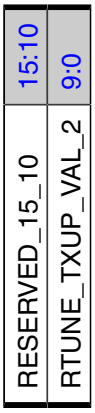


Table 2-286 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_2

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_2	R/W	These bits store the resistor tune TX up value 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.283 RAWCMN_DIG_AON_RTUNE_RX_VAL_3

- **Description:** Resistor Tune RX Value 3.
- **Size:** 16 bits
- **Offset:** 0x186
- **Exists:** Always

Table 2-287 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_3

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_3	R/W	These bits store the resistor resister tune RX value 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.284 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_3

- **Description:** Resistor Tune TX Down Value 3.
- **Size:** 16 bits
- **Offset:** 0x187
- **Exists:** Always

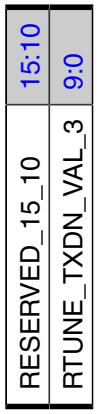


Table 2-288 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_3

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_3	R/W	These bits store the resistor tune TX down value 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.285 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_3

- **Description:** Resistor Tune TX Up Value 3.
- **Size:** 16 bits
- **Offset:** 0x188
- **Exists:** Always

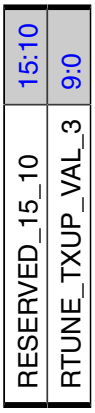


Table 2-289 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_3

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_3	R/W	These bits store the resistor tune TX up value 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.286 RAWCMN_DIG_AON_RTUNE_RX_VAL_4

- **Description:** Resistor Tune RX Value 4.
- **Size:** 16 bits
- **Offset:** 0x189
- **Exists:** Always

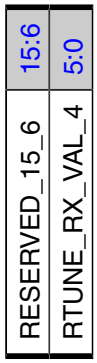


Table 2-290 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_4

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_4	R/W	These bits store the resistor tune RX value 4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.287 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_4

- **Description:** Resistor Tune TX Down Value 4.
- **Size:** 16 bits
- **Offset:** 0x18a
- **Exists:** Always

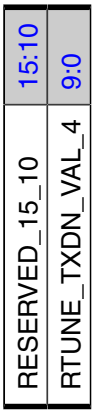


Table 2-291 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_4

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_4	R/W	These bits store the resistor tune TX down value 4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.288 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_4

- **Description:** Resistor Tune TX Up Value 4.
- **Size:** 16 bits
- **Offset:** 0x18b
- **Exists:** Always

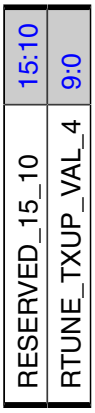


Table 2-292 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_4

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_4	R/W	These bits store the resistor tune TX up value 4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.289 RAWCMN_DIG_AON_RTUNE_RX_VAL_5

- **Description:** Resistor Tune RX Value 5.
- **Size:** 16 bits
- **Offset:** 0x18c
- **Exists:** Always

Table 2-293 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_5

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_5	R/W	These bits store the resistor tune RX value 5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.290 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_5

- **Description:** Resistor Tune TX Down Value 5.
- **Size:** 16 bits
- **Offset:** 0x18d
- **Exists:** Always

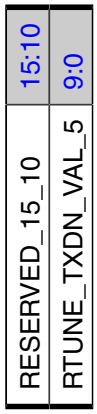


Table 2-294 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_5

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_5	R/W	These bits store the resistor tune TX down value 5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.291 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_5

- **Description:** Resistor Tune TX Up Value 5.
- **Size:** 16 bits
- **Offset:** 0x18e
- **Exists:** Always

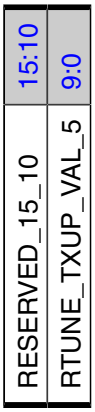


Table 2-295 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_5

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_5	R/W	These bits store the resistor tune TX up value 5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.292 RAWCMN_DIG_AON_RTUNE_RX_VAL_6

- **Description:** Resistor Tune RX Value 6.
- **Size:** 16 bits
- **Offset:** 0x18f
- **Exists:** Always

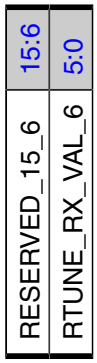


Table 2-296 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_6

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_6	R/W	These bits store the resistor tune RX value 6. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.293 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_6

- **Description:** Resistor Tune TX Down Value 6.
- **Size:** 16 bits
- **Offset:** 0x190
- **Exists:** Always

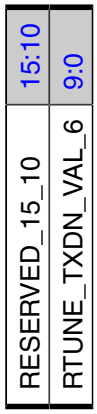


Table 2-297 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_6

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_6	R/W	These bits store the resistor tune TX down value 6. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.294 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_6

- **Description:** Resistor Tune TX Up Value 6.
- **Size:** 16 bits
- **Offset:** 0x191
- **Exists:** Always

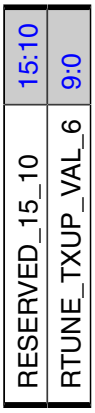


Table 2-298 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_6

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_6	R/W	These bits store the resistor tune TX up value 6. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.295 RAWCMN_DIG_AON_RTUNE_RX_VAL_7

- **Description:** Resistor Tune RX Value 7.
- **Size:** 16 bits
- **Offset:** 0x192
- **Exists:** Always

Table 2-299 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_7

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_7	R/W	These bits store the resistor tune RX value 7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.296 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_7

- **Description:** Resistor Tune TX Down Value 7.
- **Size:** 16 bits
- **Offset:** 0x193
- **Exists:** Always

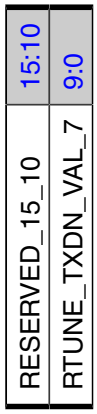


Table 2-300 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_7

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_7	R/W	These bits store the resistor tune TX down value 7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.297 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_7

- **Description:** Resistor Tune TX Up Value 7.
- **Size:** 16 bits
- **Offset:** 0x194
- **Exists:** Always

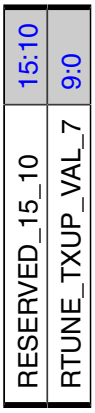


Table 2-301 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_7

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_7	R/W	These bits store the resistor tune TX up value 7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.298 RAWCMN_DIG_AON_SRAM_OVRD_IN

- **Description:** Override for Incoming SRAM Inputs.
- **Size:** 16 bits
- **Offset:** 0x195
- **Exists:** Always

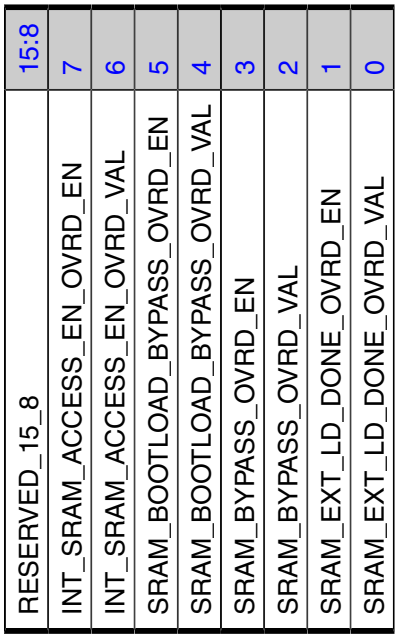


Table 2-302 Fields for Register: RAWCMN_DIG_AON_SRAM_OVRD_IN

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	INT_SRAM_ACCESS_EN_OVRD_EN	R/W	Override enable for sram_int_access_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	INT_SRAM_ACCESS_EN_OVRD_VAL	R/W	Override value for sram_int_access_en. Enable external access to update/load internal SRAM **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	SRAM_BOOTLOAD_BYPASS_OVRD_EN	R/W	This bit is the override enable for thesram_bootload_bypass input. **Value After Reset:** 0x0

Table 2-302 Fields for Register: RAWCMN_DIG_AON_SRAM_OVRD_IN (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
4	SRAM_BOOTLOAD_BYPASS_OVRD_VAL	R/W	This bit is the override value for the sram_bootload_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SRAM_BYPASS_OVRD_EN	R/W	This bit is the override enable for the sram_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SRAM_BYPASS_OVRD_VAL	R/W	This bit is the override value for the sram_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SRAM_EXT_LD_DONE_OVRD_EN	R/W	This bit is the override enable for the sram_ext_ld_done input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SRAM_EXT_LD_DONE_OVRD_VAL	R/W	This bit is the override value for the sram_ext_ld_done input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.299 RAWCMN_DIG_AON_SRAM_IN

- **Description:** Monitor for SRAM Inputs.
- **Size:** 16 bits
- **Offset:** 0x196
- **Exists:** Always

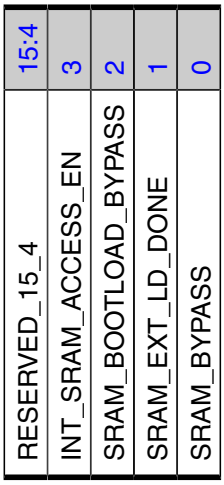


Table 2-303 Fields for Register: RAWCMN_DIG_AON_SRAM_IN

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	INT_SRAM_ACCESS_EN	R	This bit is the value from RAW PCS for sram_int_access_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	SRAM_BOOTLOAD_BYPASS	R	This bit is the value from RAW PCS for sram_bootload_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	SRAM_EXT_LD_DONE	R	This bit is the value from RAW PCS for sram_ext_ld_done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-303 Fields for Register: RAWCMN_DIG_AON_SRAM_IN (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
0	SRAM_BYPASS	R	This bit is the value from RAW PCS for sram_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.300 RAWCMN_DIG_AON_SRAM_OUT

- **Description:** Monitor for SRAM Outputs and ECC Status.
- **Size:** 16 bits
- **Offset:** 0x197
- **Exists:** Always

Table 2-304 Fields for Register: RAWCMN_DIG_AON_SRAM_OUT

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	SRAM_ECC_DED	R	Double error detection. Logic 1 on this bit indicates double (unrepairable) errors have been detected in corresponding User Segment. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	SRAM_ECC_SEC	R	Single error correction. Logic 1 on this bit indicates a single error has been detected and corrected in corresponding User Segment. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	SRAM_INIT_DONE	R	This bit is the value from RAW PCS for sram_init_done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-304 Fields for Register: RAWCMN_DIG_AON_SRAM_OUT (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.301 RAWCMN_DIG_AON_FW_VERSION_0

- **Description:** Firmware Version Register 0.
- **Size:** 16 bits
- **Offset:** 0x198
- **Exists:** Always

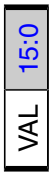


Table 2-305 Fields for Register: RAWCMN_DIG_AON_FW_VERSION_0

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware version. This register stores the firmware version as a 16-bit number in the format a.b.c, where ■ a = bits 15:12 ■ b = bits 11:4 ■ c = bits 3:0 (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff **Volatile:** true

2.1.302 RAWCMN_DIG_AON_FW_VERSION_1

- **Description:** Firmware Version Register 1.
- **Size:** 16 bits
- **Offset:** 0x199
- **Exists:** Always

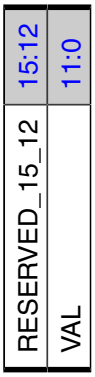


Table 2-306 Fields for Register: RAWCMN_DIG_AON_FW_VERSION_1

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	VAL	R/W	Firmware version. This register stores the firmware compile date as a 12-bit number in the format day/month/year. Where ■ Day = bits 11:7 ■ Month = bits 6:3 ■ Year = bits 2:0 (2018 = 0, 2019 = 1, and so on) (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff **Volatile:** true

2.1.303 RAWCMN_DIG_AON_BG_SEL

- **Description:** bg_sel.
- **Size:** 16 bits
- **Offset:** 0x19a
- **Exists:** Always

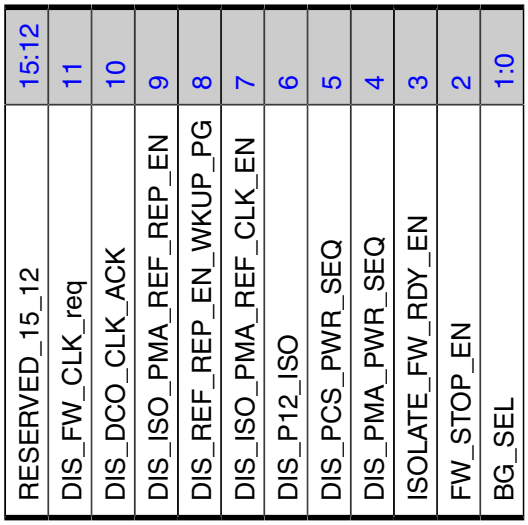


Table 2-307 Fields for Register: RAWCMN_DIG_AON_BG_SEL

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	DIS_FW_CLK_req	R/W	disable the wait for FW_CLK_REQ de-assertion before PG entry. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10	DIS_DCO_CLK_ACK	R/W	disable the wiat for DCO_CLK_ACK de-assertion before PG entry. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	DIS_ISO_PMA_REF_REP_EN	R/W	disable isolate pma_ref_repeat_en (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-307 Fields for Register: RAWCMN_DIG_AON_BG_SEL (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
8	DIS_REF_REP_EN_WKUP_PG	R/W	disable ref_repeat_clk_en wake up from PG (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
7	DIS_ISO_PMA_REF_CLK_EN	R/W	disable isolate pma_ref_clk_en (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
6	DIS_P12_ISO	R/W	disable p12_iso. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
5	DIS_PCS_PWR_SEQ	R/W	disable pcs_pwr_en sequencing. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4	DIS_PMA_PWR_SEQ	R/W	disable pma_pwr_en sequencing. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
3	ISOLATE_FW_RDY_EN	R/W	isolate_fw_rdy enable. (volatile) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	FW_STOP_EN	R/W	fw_stop enable. (volatile) **Value After Reset:** 0x1 **Exists:** Always

Table 2-307 Fields for Register: RAWCMN_DIG_AON_BG_SEL (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
1:0	BG_SEL	R/W	bg_sel. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3 **Volatile:** true

2.1.304 RAWCMN_DIG_AON_SRAM_EOF_ADDR

- **Description:** Value for SRAM End Of FW address.
- **Size:** 16 bits
- **Offset:** 0x19b
- **Exists:** Always

Table 2-308 Fields for Register: RAWCMN_DIG_AON_SRAM_EOF_ADDR

Bits	Name	Memory Access	Description
15:0	SRAM_EOF_ADDR	R	This field is the value for SRAM End Of FW address. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.305 RAWCMN_DIG_AON_SRAM_BOC_ADDR

- **Description:** Value for SRAM Begin Of Context Restore address.
- **Size:** 16 bits
- **Offset:** 0x19c
- **Exists:** Always

Table 2-309 Fields for Register: RAWCMN_DIG_AON_SRAM_BOC_ADDR

Bits	Name	Memory Access	Description
15:0	SRAM_BOC_ADDR	R	This field is the value for SRAM Begin Of Context Restore address. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.306 RAWCMN_DIG_AON_ATE_MODE_EN

- **Description:** Determines if we are in the ATE mode
- **Size:** 16 bits
- **Offset:** 0x19d
- **Exists:** Always

Table 2-310 Fields for Register: RAWCMN_DIG_AON_ATE_MODE_EN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ATE_MODE_EN	R/W	"1" if the ATE mode is enabled (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.307 RAWCMN_DIG_AON_SRAM_ECC_ADDR

- **Description:** SRAM ECC Address Register
- **Size:** 16 bits
- **Offset:** 0x19e
- **Exists:** Always

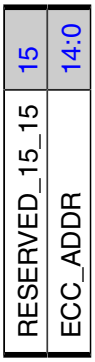


Table 2-311 Fields for Register: RAWCMN_DIG_AON_SRAM_ECC_ADDR

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	ECC_ADDR	R	Recorded SRAM address when double ECC errors have been detected. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.308 RAWCMN_DIG_AON_SRAM_ECC_STATUS_CLR

- **Description:** SRAM ECC error status and recorded SRAM address clear.
- **Size:** 16 bits
- **Offset:** 0x19f
- **Exists:** Always

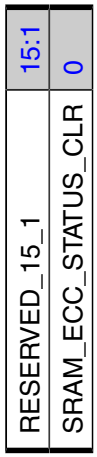


Table 2-312 Fields for Register: RAWCMN_DIG_AON_SRAM_ECC_STATUS_CLR

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SRAM_ECC_STATUS_CLR	R/W	This bit is used to clear SRAM ECC error status and recorded SRAM address (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.309 RAWCMN_DIG_AON_RAW_VERSION

- **Description:** Determines the RAW version
- **Size:** 16 bits
- **Offset:** 0x1a0
- **Exists:** Always

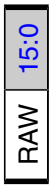


Table 2-313 Fields for Register: RAWCMN_DIG_AON_RAW_VERSION

Bits	Name	Memory Access	Description
15:0	RAW	R	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.310 RAWCMN_DIG_AON_RO_REG_AONCMN_1

- **Description:** Read register for few signals mentioned below
- **Size:** 16 bits
- **Offset:** 0x1a1
- **Exists:** Always

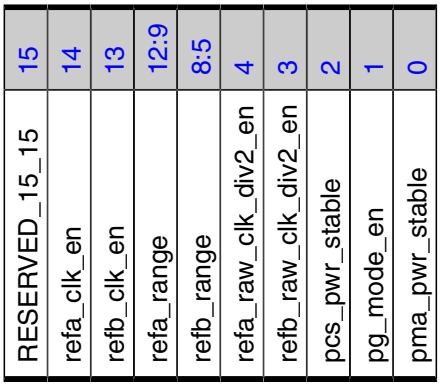


Table 2-314 Fields for Register: RAWCMN_DIG_AON_RO_REG_AONCMN_1

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	refa_clk_en	R	This field is to read refa_clk_en (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	refb_clk_en	R	This field is to read refb_clk_en (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:9	refa_range	R	This field is to read refa_range (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:5	refb_range	R	This field is to read refb_range (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-314 Fields for Register: RAWCMN_DIG_AON_RO_REG_AONCMN_1 (continued)

Bits	Name	Memory Access	Description
4	refa_raw_clk_div2_en	R	This field is to read refa_raw_clk_div2_en (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	refb_raw_clk_div2_en	R	This field is to read refb_raw_clk_div2_en (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	pcs_pwr_stable	R	This field is to read pcs_pwr_stable (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	pg_mode_en	R	This field is to read pg_mode_en (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	pma_pwr_stable	R	This field is to read pma_pwr_stable (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.311 RAWCMN_DIG_AON_RW_REG_AONCMN_1

- **Description:** Read/write register for few signals mentioned below
- **Size:** 16 bits
- **Offset:** 0x1a2
- **Exists:** Always

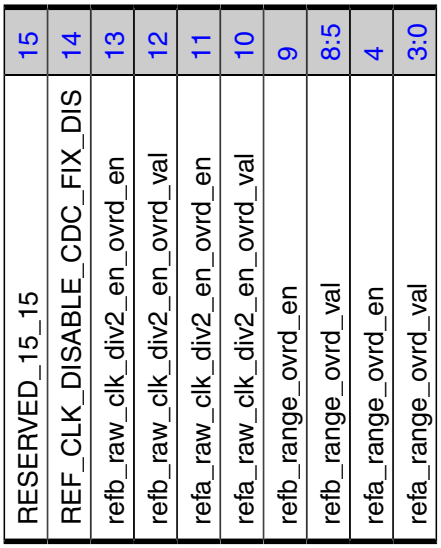


Table 2-315 Fields for Register: RAWCMN_DIG_AON_RW_REG_AONCMN_1

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	REF_CLK_DISABLE_CDC_FIX_DIS	R/W	This is the chicken bit to disable CDC fix of ref_clk_disable This register bit will be kept as 0 to have this CDC fix active (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
13	refb_raw_clk_div2_en_ovrd_en	R/W	This is the ovrd en for refb_raw_clk_div2_en (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
12	refb_raw_clk_div2_en_ovrd_val	R/W	This is the ovrd val for refb_raw_clk_div2_en (volatile) **Value After Reset:** 0x0

Table 2-315 Fields for Register: RAWCMN_DIG_AON_RW_REG_AONCMN_1 (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1 **Volatile:** true
11	refa_raw_clk_div2_en_ovrd_en	R/W	This is the ovrd en for refa_raw_clk_div2_en (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10	refa_raw_clk_div2_en_ovrd_val	R/W	This is the ovrd val for refa_raw_clk_div2_en (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	refb_range_ovrd_en	R/W	This is the ovrd en for refb_range (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
8:5	refb_range_ovrd_val	R/W	This is the ovrd val for refb_range (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf **Volatile:** true
4	refa_range_ovrd_en	R/W	This is the ovrd en for refa_range (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
3:0	refa_range_ovrd_val	R/W	This is the ovrd val for refa_range (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf **Volatile:** true

2.1.312 RAWCMN_DIG_AON_RTUNE_RX_VAL_8

- **Description:** Resistor Tune RX Value 8.
- **Size:** 16 bits
- **Offset:** 0x1a3
- **Exists:** Always

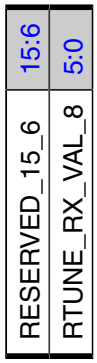


Table 2-316 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_8

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_8	R/W	These bits store the resistor tune RX value 8. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.313 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_8

- **Description:** Resistor Tune TX Down Value 8.
- **Size:** 16 bits
- **Offset:** 0x1a4
- **Exists:** Always

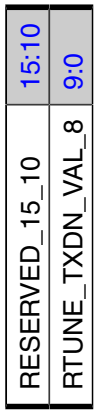


Table 2-317 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_8

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_8	R/W	These bits store the resistor tune TX down value 8. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.314 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_8

- **Description:** Resistor Tune TX Up Value 8.
- **Size:** 16 bits
- **Offset:** 0x1a5
- **Exists:** Always

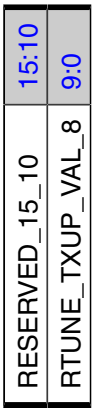


Table 2-318 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_8

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_8	R/W	These bits store the resistor tune TX up value 8. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.315 RAWCMN_DIG_AON_RTUNE_RX_VAL_9

- **Description:** Resistor Tune RX Value 9.
- **Size:** 16 bits
- **Offset:** 0x1a6
- **Exists:** Always

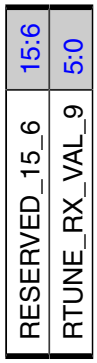


Table 2-319 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_9

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_9	R/W	These bits store the resistor tune RX value 9. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.316 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_9

- **Description:** Resistor Tune TX Down Value 9.
- **Size:** 16 bits
- **Offset:** 0x1a7
- **Exists:** Always

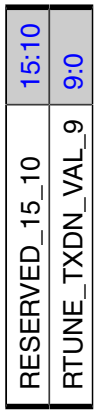


Table 2-320 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_9

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_9	R/W	These bits store the resistor tune TX down value 9. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.317 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_9

- **Description:** Resistor Tune TX Up Value 9.
- **Size:** 16 bits
- **Offset:** 0x1a8
- **Exists:** Always

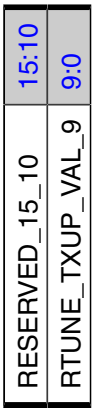


Table 2-321 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_9

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_9	R/W	These bits store the resistor tune TX up value 9. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.318 RAWCMN_DIG_AON_RTUNE_RX_VAL_10

- **Description:** Resistor Tune RX Value 10.
- **Size:** 16 bits
- **Offset:** 0x1a9
- **Exists:** Always

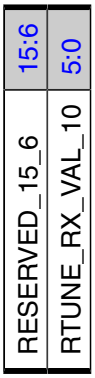


Table 2-322 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_10

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_10	R/W	These bits store the resistor tune RX value 10. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.319 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_10

- **Description:** Resistor Tune TX Down Value 10.
- **Size:** 16 bits
- **Offset:** 0x1aa
- **Exists:** Always

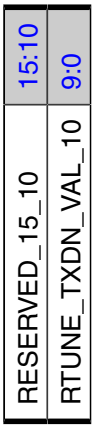


Table 2-323 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_10

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_10	R/W	These bits store the resistor tune TX down value 10. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.320 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_10

- **Description:** Resistor Tune TX Up Value 10.
- **Size:** 16 bits
- **Offset:** 0x1ab
- **Exists:** Always

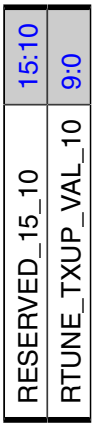


Table 2-324 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_10

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_10	R/W	These bits store the resistor tune TX up value 10. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.321 RAWCMN_DIG_AON_RTUNE_RX_VAL_11

- **Description:** Resistor Tune RX Value 11.
- **Size:** 16 bits
- **Offset:** 0x1ac
- **Exists:** Always

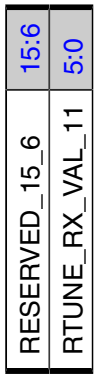


Table 2-325 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_11

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_11	R/W	These bits store the resistor tune RX value 11. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.322 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_11

- **Description:** Resistor Tune TX Down Value 11.
- **Size:** 16 bits
- **Offset:** 0x1ad
- **Exists:** Always

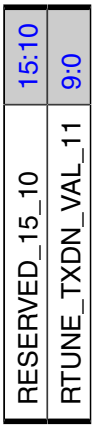


Table 2-326 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_11

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_11	R/W	These bits store the resistor tune TX down value 11. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.323 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_11

- **Description:** Resistor Tune TX Up Value 11.
- **Size:** 16 bits
- **Offset:** 0x1ae
- **Exists:** Always

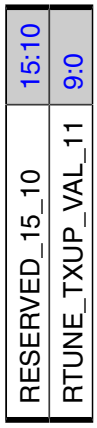


Table 2-327 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_11

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_11	R/W	These bits store the resistor tune TX up value 11. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.324 RAWCMN_DIG_AON_RTUNE_RX_VAL_12

- **Description:** Resistor Tune RX Value 12.
- **Size:** 16 bits
- **Offset:** 0x1af
- **Exists:** Always

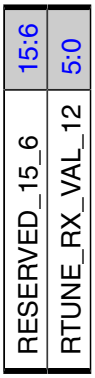


Table 2-328 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_12

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_12	R/W	These bits store the resistor tune RX value 12. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.325 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_12

- **Description:** Resistor Tune TX Down Value 12.
- **Size:** 16 bits
- **Offset:** 0x1b0
- **Exists:** Always

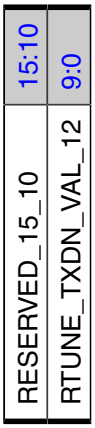


Table 2-329 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_12

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_12	R/W	These bits store the resistor tune TX down value 12. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.326 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_12

- **Description:** Resistor Tune TX Up Value 12.
- **Size:** 16 bits
- **Offset:** 0x1b1
- **Exists:** Always

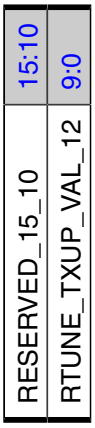


Table 2-330 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_12

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_12	R/W	These bits store the resistor tune TX up value 12. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.327 RAWCMN_DIG_AON_RTUNE_RX_VAL_13

- **Description:** Resistor Tune RX Value 13.
- **Size:** 16 bits
- **Offset:** 0x1b2
- **Exists:** Always

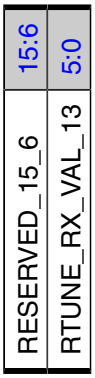


Table 2-331 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_13

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_13	R/W	These bits store the resistor tune RX value 13. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.328 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_13

- **Description:** Resistor Tune TX Down Value 13.
- **Size:** 16 bits
- **Offset:** 0x1b3
- **Exists:** Always

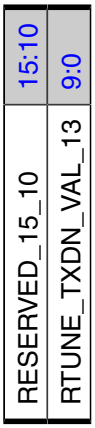


Table 2-332 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_13

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_13	R/W	These bits store the resistor tune TX down value 13. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.329 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_13

- **Description:** Resistor Tune TX Up Value 13.
- **Size:** 16 bits
- **Offset:** 0x1b4
- **Exists:** Always

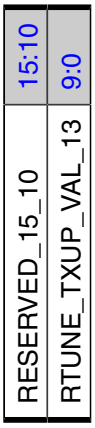


Table 2-333 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_13

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_13	R/W	These bits store the resistor tune TX up value 13. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.330 RAWCMN_DIG_AON_RTUNE_RX_VAL_14

- **Description:** Resistor Tune RX Value 14.
- **Size:** 16 bits
- **Offset:** 0x1b5
- **Exists:** Always

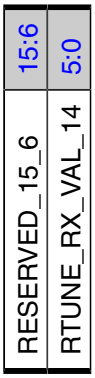


Table 2-334 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_14

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_14	R/W	These bits store the resistor tune RX value 14. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.331 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_14

- **Description:** Resistor Tune TX Down Value 14.
- **Size:** 16 bits
- **Offset:** 0x1b6
- **Exists:** Always

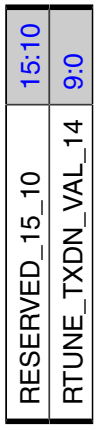


Table 2-335 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_14

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_14	R/W	These bits store the resistor tune TX down value 14. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.332 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_14

- **Description:** Resistor Tune TX Up Value 14.
- **Size:** 16 bits
- **Offset:** 0x1b7
- **Exists:** Always

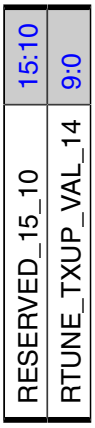


Table 2-336 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_14

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_14	R/W	These bits store the resistor tune TX up value 14. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.333 RAWCMN_DIG_AON_RTUNE_RX_VAL_15

- **Description:** Resistor Tune RX Value 15.
- **Size:** 16 bits
- **Offset:** 0x1b8
- **Exists:** Always

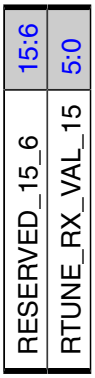


Table 2-337 Fields for Register: RAWCMN_DIG_AON_RTUNE_RX_VAL_15

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RTUNE_RX_VAL_15	R/W	These bits store the resistor tune RX value 15. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.334 RAWCMN_DIG_AON_RTUNE_TXDN_VAL_15

- **Description:** Resistor Tune TX Down Value 15.
- **Size:** 16 bits
- **Offset:** 0x1b9
- **Exists:** Always

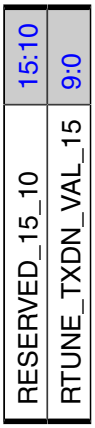


Table 2-338 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXDN_VAL_15

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXDN_VAL_15	R/W	These bits store the resistor tune TX down value 15. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.335 RAWCMN_DIG_AON_RTUNE_TXUP_VAL_15

- **Description:** Resistor Tune TX Up Value 15.
- **Size:** 16 bits
- **Offset:** 0x1ba
- **Exists:** Always

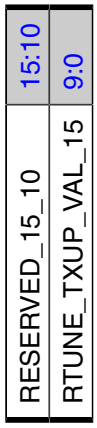


Table 2-339 Fields for Register: RAWCMN_DIG_AON_RTUNE_TXUP_VAL_15

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RTUNE_TXUP_VAL_15	R/W	These bits store the resistor tune TX up value 15. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.336 RAWCMN_DIG_AON_BYP_RES_CHAIN

- **Description:** register for chicken bit for PMA res tunning chain.
- **Size:** 16 bits
- **Offset:** 0x1bb
- **Exists:** Always

Table 2-340 Fields for Register: RAWCMN_DIG_AON_BYP_RES_CHAIN

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	BYP_RES_CHAIN	R/W	chicken bit for PMA res tunning chain **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.337 RAWCMN_DIG_AON_RW_FW_CLK_CTL

- **Description:** Read/write register for few signals mentioned below
- **Size:** 16 bits
- **Offset:** 0x1bc
- **Exists:** Always

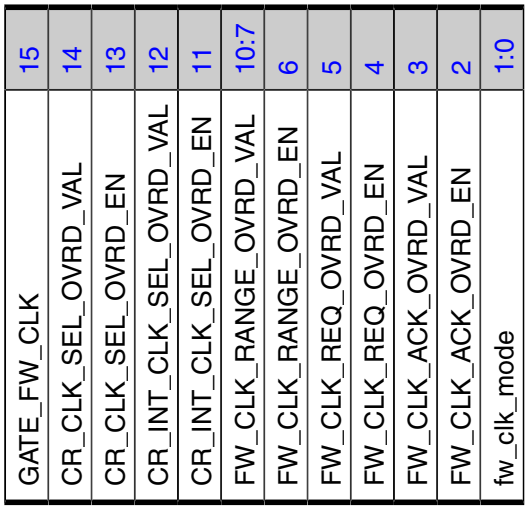


Table 2-341 Fields for Register: RAWCMN_DIG_AON_RW_FW_CLK_CTL

Bits	Name	Memory Access	Description
15	GATE_FW_CLK	R/W	Block FW CLK if set to 0 (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14	CR_CLK_SEL_OVRD_VAL	R/W	Override value for CR_CLK_SEL (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
13	CR_CLK_SEL_OVRD_EN	R/W	Override enable for CR_CLK_SEL (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
12	CR_INT_CLK_SEL_OVRD_VAL	R/W	Override value for CR_INT_CLK_SEL (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-341 Fields for Register: RAWCMN_DIG_AON_RW_FW_CLK_CTL (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
11	CR_INT_CLK_SEL_OVRD_EN	R/W	Override enable for CR_INT_CLK_SEL (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10:7	FW_CLK_RANGE_OVRD_VAL	R/W	Override value for FW_CLK_RANGE (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf **Volatile:** true
6	FW_CLK_RANGE_OVRD_EN	R/W	Override enable for FW_CLK_RANGE (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
5	FW_CLK_REQ_OVRD_VAL	R/W	Override value for FW_CLK_REQ (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4	FW_CLK_REQ_OVRD_EN	R/W	Override enable for FW_CLK_REQ (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
3	FW_CLK_ACK_OVRD_VAL	R/W	Override value for FW_CLK_ACK (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	FW_CLK_ACK_OVRD_EN	R/W	Override enable for FW_CLK_ACK (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-341 Fields for Register: RAWCMN_DIG_AON_RW_FW_CLK_CTL (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
1:0	fw_clk_mode	R/W	FW clk mode ■ 2'b00: IRQ MODE - FW clk requested whenever FW is serving a request or during POR/PG (fw_clk_ack is optional) ■ 2'b01: REF MODE - FW clk requested whenever FW is running. ■ 2'b10: Disable mode - FW clk is never requested. ■ 2'b11: RSRVD (volatile) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3 **Volatile:** true

2.1.338 RAWCMN_DIG_AON_FW_CLK_STATUS

- **Description:** Values FW CLK related signals.
- **Size:** 16 bits
- **Offset:** 0x1bd
- **Exists:** Always

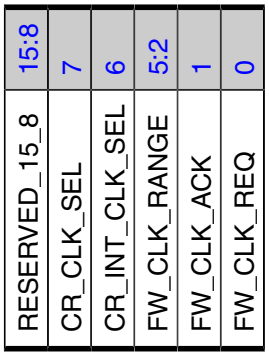


Table 2-342 Fields for Register: RAWCMN_DIG_AON_FW_CLK_STATUS

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	CR_CLK_SEL	R	This bit is the value for CR_CLK_SEL. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	CR_INT_CLK_SEL	R	This bit is the value for CR_INT_CLK_SEL. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:2	FW_CLK_RANGE	R	This bit is the value for FW_CLK_RANGE (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-342 Fields for Register: RAWCMN_DIG_AON_FW_CLK_STATUS (continued)

Bits	Name	Memory Access	Description
1	FW_CLK_ACK	R	This bit is the value for FW_CLK_ACK. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	FW_CLK_REQ	R	This bit is the value for FW_CLK_REQ. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.339 RAWCMN_DIG_AON_DCO_CTL_1

- **Description:** DCO Control Signals
- **Size:** 16 bits
- **Offset:** 0x1be
- **Exists:** Always

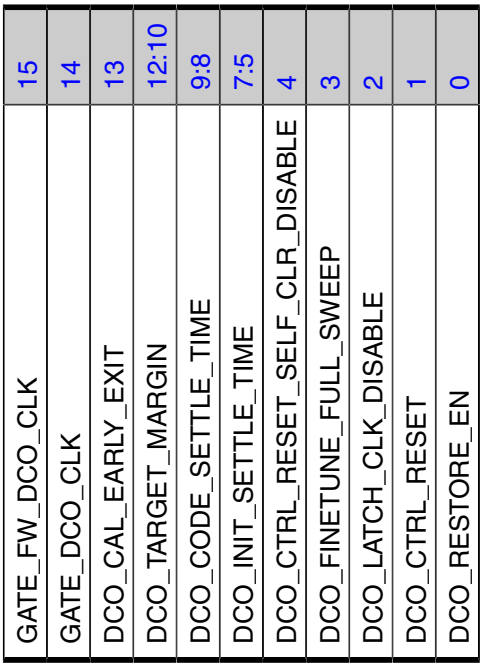


Table 2-343 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_1

Bits	Name	Memory Access	Description
15	GATE_FW_DCO_CLK	R/W	Block FW/DCO CLK if set to 1 (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14	GATE_DCO_CLK	R/W	Block DCO CLK if set to 1 (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
13	DCO_CAL_EARLY_EXIT	R/W	DCO calibration will exit once DCO target frequency +/- DCO_TARGET_MARGIN is achieved during calibration. (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-343 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_1 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
12:10	DCO_TARGET_MARGIN	R/W	DCO clock frequency tolerance in MHz, in case of DCO_CAL_EARLY_EXIT is activated. (volatile) **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7 **Volatile:** true
9:8	DCO_CODE_SETTLE_TIME	R/W	Time in micro seconds needed for dco_clk to be stable after changing the finetune code. It should not be assigned to 0, since it will violate the analog spec. (volatile) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3 **Volatile:** true
7:5	DCO_INIT_SETTLE_TIME	R/W	Time in micro seconds needed for dco_clk to be ready after asserting ref_clk_en. It should not be assigned to 0, since it will violate the analog spec. (volatile) **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7 **Volatile:** true
4	DCO_CTRL_RESET_SELF_CLR_DISABLE	R/W	This bit disables self clearing DCO_CTRL_RESET register when it is set to 1. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
3	DCO_FINETUNE_FULL_SWEEP	R/W	This bit controls using the whole 9 bits of finetune in DCO calibration or only 7 bits. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	DCO_LATCH_CLK_DISABLE	R/W	This bit activates DCO latch clock that is used inside analog to latch DCO finetune. (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-343 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_1 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
1	DCO_CTRL_RESET	R/W	This bit resets the DCO controller when it is set to 1. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	DCO_RESTORE_EN	R/W	This bit is the DCO restore enable. (volatile) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.340 RAWCMN_DIG_AON_DCO_CTL_2

- **Description:** Override incoming refa_dco_ld_val input.
- **Size:** 16 bits
- **Offset:** 0x1bf
- **Exists:** Always

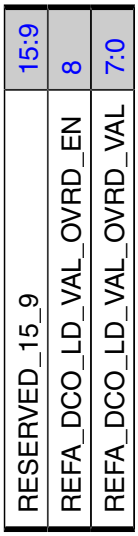


Table 2-344 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_2

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	REFA_DCO_LD_VAL_OVRD_EN	R/W	Override enable for refa_dco_ld_val input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
7:0	REFA_DCO_LD_VAL_OVRD_VAL	R/W	Override value for refa_dco_ld_val input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff **Volatile:** true

2.1.341 RAWCMN_DIG_AON_DCO_CTL_3

- **Description:** DCO Control Signals (For debug only)
- **Size:** 16 bits
- **Offset:** 0x1c0
- **Exists:** Always

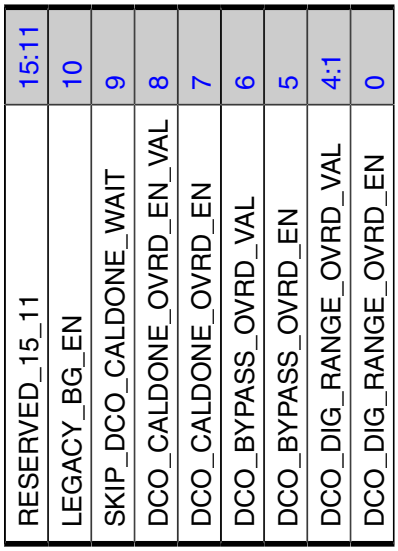


Table 2-345 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_3

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	LEGACY_BG_EN	R/W	Chicken bit to keep BG_EN behavior as RAW < 1.17. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	SKIP_DCO_CALDONE_WAIT	R/W	Bypass delaying BG enable till DCO is calibrated. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
8	DCO_CALDONE_OVRD_EN_VAL	R/W	Override value for DCO_CALDONE (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-345 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_3 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
7	DCO_CALDONE_OVRD_EN	R/W	Override value for DCO_CALDONE (volatile) **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
6	DCO_BYPASS_OVRD_VAL	R/W	Override value for DCO_BYPASS (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
5	DCO_BYPASS_OVRD_EN	R/W	Override enable for DCO_BYPASS (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4:1	DCO_DIG_RANGE_OVRD_VAL	R/W	Override value for DCO_DIG_RANGE (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf **Volatile:** true
0	DCO_DIG_RANGE_OVRD_EN	R/W	Override enable for DCO_DIG_RANGE (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.342 RAWCMN_DIG_AON_DCO_CAL_STATUS

- **Description:** Values DCO CAL related signals.
- **Size:** 16 bits
- **Offset:** 0x1c1
- **Exists:** Always

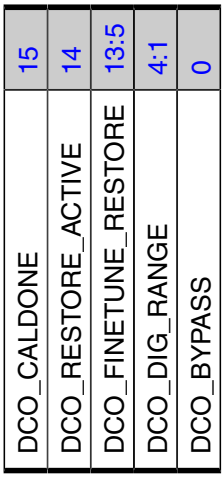


Table 2-346 Fields for Register: RAWCMN_DIG_AON_DCO_CAL_STATUS

Bits	Name	Memory Access	Description
15	DCO_CALDONE	R	This bit indicates DCO calibration is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	DCO_RESTORE_ACTIVE	R	This bit indicates DCO restoration will be activated after PG exit. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:5	DCO_FINETUNE_RESTORE	R	This field is DCO finetune value used in DCO restoration after PG exit. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-346 Fields for Register: RAWCMN_DIG_AON_DCO_CAL_STATUS (continued)

Bits	Name	Memory Access	Description
4:1	DCO_DIG_RANGE	R	This field is the value for DCO_DIG_RANGE. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	DCO_BYPASS	R	This bit is the value for DCO_BYPASS. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.343 RAWCMN_DIG_AON_CMN_PERST_CTL

- **Description:** PERST common control
- **Size:** 16 bits
- **Offset:** 0x1c2
- **Exists:** Always

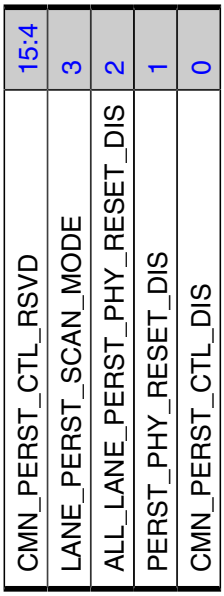


Table 2-347 Fields for Register: RAWCMN_DIG_AON_CMN_PERST_CTL

Bits	Name	Memory Access	Description
15:4	CMN_PERST_CTL_RSVD	R/W	Resved bit in cmn_perst_ctl ■ rest: RESERVED. ■ b5 : Disable perst_ref_clk_dis ■ b4 : Disable perst_ref_clk_watermark_dis (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff **Volatile:** true
3	LANE_PERST_SCAN_MODE	R/W	Set lane_perst value in scan mode (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	ALL_LANE_PERST_PHY_RESET_DIS	R/W	Disable all lane PERST trigger phy_reset (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-347 Fields for Register: RAWCMN_DIG_AON_CMN_PERST_CTL (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
1	PERST_PHY_RESET_DIS	R/W	Disable PERST trigger phy_reset (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	CMN_PERST_CTL_DIS	R/W	Disable PERST feature checiken bit (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.344 RAWCMN_DIG_AON_DCO_FW_ACCESS

- **Description:** FW access status for DCO overrides
- **Size:** 16 bits
- **Offset:** 0x1c3
- **Exists:** Always

Table 2-348 Fields for Register: RAWCMN_DIG_AON_DCO_FW_ACCESS

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DCO_CAL_FW_ACCESS_DONE	R/W	This bit indicates that FW is done with accessing DCO CAL overrides. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	DCO_CAL_FW_ACCESS_INIT	R/W	This bit indicates that FW is accessing DCO CAL overrides. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.345 RAWCMN_DIG_AON_REF_CLK_MUX_CTL

- **Description:** REF CLK MUX Control Signals
- **Size:** 16 bits
- **Offset:** 0x1c4
- **Exists:** Always

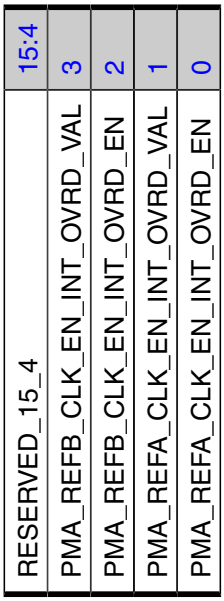


Table 2-349 Fields for Register: RAWCMN_DIG_AON_REF_CLK_MUX_CTL

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	PMA_REFB_CLK_EN_INT_OVRD_VAL	R/W	Override value for pma_refb_clk_en_int (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	PMA_REFB_CLK_EN_INT_OVRD_EN	R/W	Override enable for pma_refb_clk_en_int (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
1	PMA_REFA_CLK_EN_INT_OVRD_VAL	R/W	Override value for pma_refa_clk_en_int (volatile) **Value After Reset:** 0x0

Table 2-349 Fields for Register: RAWCMN_DIG_AON_REF_CLK_MUX_CTL (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1 **Volatile:** true
0	PMA_REFA_CLK_EN_INT_OVRD_EN	R/W	Override enable for pma_refa_clk_en_int (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.346 RAWCMN_DIG_AON_DCO_CTL_4

- **Description:** Override incoming refb_dco_ld_val input. (For debug only)
- **Size:** 16 bits
- **Offset:** 0x1c5
- **Exists:** Always

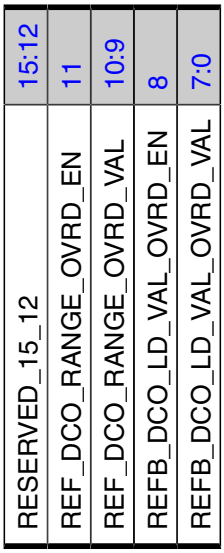


Table 2-350 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_4

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	REF_DCO_RANGE_OVRD_EN	R/W	Override enable for ref_dco_range input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10:9	REF_DCO_RANGE_OVRD_VAL	R/W	Override value for ref_dco_range input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3 **Volatile:** true
8	REFB_DCO_LD_VAL_OVRD_EN	R/W	Override enable for refb_dco_ld_val input. (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-350 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_4 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
7:0	REFB_DCO_LD_VAL_OVRD_VAL	R/W	Override value for refb_dco_ld_val input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff **Volatile:** true

2.1.347 RAWCMN_DIG_AON_DCO_CTL_5

- **Description:** Override incoming ref_dco_target input. (For debug only)
- **Size:** 16 bits
- **Offset:** 0x1c6
- **Exists:** Always

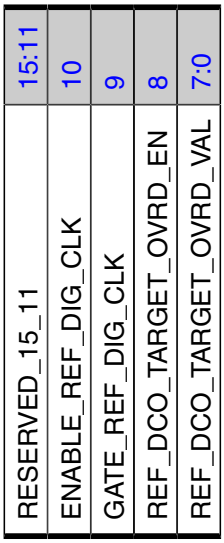


Table 2-351 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_5

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	ENABLE_REF_DIG_CLK	R/W	Ungates ref_dig_clk inputs. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	GATE_REF_DIG_CLK	R/W	Gates ref_dig_clk inputs. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
8	REF_DCO_TARGET_OVRD_EN	R/W	Override enable for ref_dco_target input. (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-351 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_5 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
7:0	REF_DCO_TARGET_OVRD_VAL	R/W	Override value for ref_dco_target input. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff **Volatile:** true

2.1.348 RAWCMN_DIG_AON_DCO_CAL_STATUS_2

- **Description:** Values DCO CAL related signals.
- **Size:** 16 bits
- **Offset:** 0x1c7
- **Exists:** Always

Table 2-352 Fields for Register: RAWCMN_DIG_AON_DCO_CAL_STATUS_2

Bits	Name	Memory Access	Description
15:8	REF_DCO_LD_VAL	R	The read value for ref_dco_ld_val used in DCO controller. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	REF_DCO_TARGET	R	The read value for ref_dco_target used in DCO controller. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.349 RAWCMN_DIG_AON_DCO_CAL_STATUS_3

- **Description:** Values DCO CAL related signals.
- **Size:** 16 bits
- **Offset:** 0x1c8
- **Exists:** Always

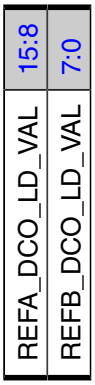


Table 2-353 Fields for Register: RAWCMN_DIG_AON_DCO_CAL_STATUS_3

Bits	Name	Memory Access	Description
15:8	REFA_DCO_LD_VAL	R	The read value for refa_dco_ld_val after overriding it in PHY. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	REFB_DCO_LD_VAL	R	The read value for refb_dco_ld_val after overriding it in PHY. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.350 RAWCMN_DIG_AON_CMN_PERST_CTL_2

- **Description:** PERST common control 2
- **Size:** 16 bits
- **Offset:** 0x1c9
- **Exists:** Always

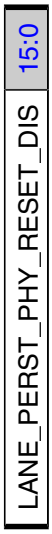


Table 2-354 Fields for Register: RAWCMN_DIG_AON_CMN_PERST_CTL_2

Bits	Name	Memory Access	Description
15:0	LANE_PERST_PHY_RESET_DIS	R/W	disable lane perst trigger phy_reset (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff **Volatile:** true

2.1.351 RAWCMN_DIG_AON_LANE_PERST_STATUS

- **Description:** Values of lane_perst after invert polarity.
- **Size:** 16 bits
- **Offset:** 0x1ca
- **Exists:** Always

Table 2-355 Fields for Register: RAWCMN_DIG_AON_LANE_PERST_STATUS

Bits	Name	Memory Access	Description
15:0	LANE_PERST	R	The readback value for lane_perst after invert polarity. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.352 RAWCMN_DIG_AON_CMN_PERST_STATUS

- **Description:** Values of cmn_perst_status.
- **Size:** 16 bits
- **Offset:** 0x1cb
- **Exists:** Always

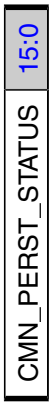


Table 2-356 Fields for Register: RAWCMN_DIG_AON_CMN_PERST_STATUS

Bits	Name	Memory Access	Description
15:0	CMN_PERST_STATUS	R	The readback value for cmn_perst_status. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.353 RAWCMN_DIG_AON_LANE_REFA_CLK_DIS_STATUS

- **Description:** Values of lane_refa_clk_dis.
- **Size:** 16 bits
- **Offset:** 0x1cc
- **Exists:** Always

Table 2-357 Fields for Register: RAWCMN_DIG_AON_LANE_REFA_CLK_DIS_STATUS

Bits	Name	Memory Access	Description
15:0	LANE_REFA_CLK_DIS	R	The readback value for lane_refa_clk_dis. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.354 RAWCMN_DIG_AON_LANE_REFB_CLK_DIS_STATUS

- **Description:** Values of lane_refb_clk_dis.
- **Size:** 16 bits
- **Offset:** 0x1cd
- **Exists:** Always

Table 2-358 Fields for Register: RAWCMN_DIG_AON_LANE_REFB_CLK_DIS_STATUS

Bits	Name	Memory Access	Description
15:0	LANE_REFB_CLK_DIS	R	The readback value for lane_refb_clk_dis. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.355 RAWCMN_DIG_AON_DCO_CTL_6

- **Description:** Control DCO restoration timers.
- **Size:** 16 bits
- **Offset:** 0x1ce
- **Exists:** Always

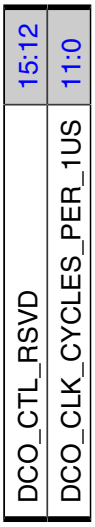


Table 2-359 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_6

Bits	Name	Memory Access	Description
15:12	DCO_CTL_RSVD	R/W	Reserved CREG space for DCO control. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf **Volatile:** true
11:0	DCO_CLK_CYCLES_PER_1US	R/W	Number of uncalibrated DCO clock cycles in 1 micro second duration. (volatile) **Value After Reset:** 0x1f4 **Exists:** Always **Reset Mask:** 0xfff **Volatile:** true

2.1.356 RAWCMN_DIG_AON_DCO_CAL_STATUS_4

- **Description:** Values DCO CAL related signals.
- **Size:** 16 bits
- **Offset:** 0x1cf
- **Exists:** Always

Table 2-360 Fields for Register: RAWCMN_DIG_AON_DCO_CAL_STATUS_4

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	REF_DCO_RANGE	R	The read value for ref_dco_range after overriding it in PHY. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.357 RAWCMN_DIG_AON_RSVD_CTL_1

- **Description:** Reserved control signals.
- **Size:** 16 bits
- **Offset:** 0x1d0
- **Exists:** Always

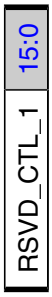


Table 2-361 Fields for Register: RAWCMN_DIG_AON_RSVD_CTL_1

Bits	Name	Memory Access	Description
15:0	RSVD_CTL_1	R/W	Reserved CREG space for future use. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff **Volatile:** true

2.1.358 RAWCMN_DIG_AON_DCO_CTL_7

- **Description:** Override DCO finetune
- **Size:** 16 bits
- **Offset:** 0x1d1
- **Exists:** Always

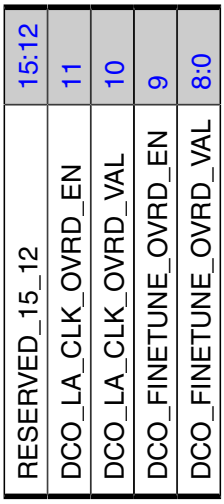


Table 2-362 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_7

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	DCO_LA_CLK_OVRD_EN	R/W	Override enable for dco_la_clk. It should change after changing DCO_LA_CLK_OVRD_VAL. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10	DCO_LA_CLK_OVRD_VAL	R/W	Override value for dco_la_clk. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	DCO_FINETUNE_OVRD_EN	R/W	Override enable for dco_finetune. It should change after changing DCO_FINETUNE_OVRD_VAL. (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-362 Fields for Register: RAWCMN_DIG_AON_DCO_CTL_7 (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
8:0	DCO_FINETUNE_OVRD_VAL	R/W	Override value for dco_finetune. (volatile) **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff **Volatile:** true

2.1.359 RAWCMN_DIG_AON_RO_REG_AONCMN_2

- **Description:** Read register for few signals mentioned below
- **Size:** 16 bits
- **Offset:** 0x1d2
- **Exists:** Always

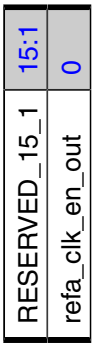


Table 2-363 Fields for Register: RAWCMN_DIG_AON_RO_REG_AONCMN_2

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	refa_clk_en_out	R	REF CLK MUX is using refa clk (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.360 RAWCMN_DIG_AON_RO_REG_AONCMN_3

- **Description:** Read register for few signals mentioned below
- **Size:** 16 bits
- **Offset:** 0x1d3
- **Exists:** Always

Table 2-364 Fields for Register: RAWCMN_DIG_AON_RO_REG_AONCMN_3

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	refb_clk_en_out	R	REF CLK MUX is using refb clk (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.361 RAWCMN_DIG_AON_RSVD_CTL_2

- **Description:** Reserved control signals.
- **Size:** 16 bits
- **Offset:** 0x1d4
- **Exists:** Always

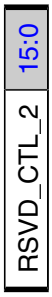


Table 2-365 Fields for Register: RAWCMN_DIG_AON_RSVD_CTL_2

Bits	Name	Memory Access	Description
15:0	RSVD_CTL_2	R/W	Reserved CREG space for future use. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff **Volatile:** true

2.1.362 RAWCMN_DIG_AON_CMN_PERST_CTL_3

- **Description:** PERST common control
- **Size:** 16 bits
- **Offset:** 0x1d5
- **Exists:** Always

Table 2-366 Fields for Register: RAWCMN_DIG_AON_CMN_PERST_CTL_3

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MULTI_PERST_VAL	R/W	multi_perst_en_val (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	MULTI_PERST_EN	R/W	multi_perst_en (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.363 RAWCMN_DIG_AON_SUP_MISC

- **Description:** E32_RAWCMN_DIG_AON_SUP_MISC
- **Size:** 16 bits
- **Offset:** 0x1d6
- **Exists:** Always

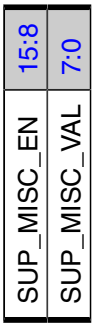


Table 2-367 Fields for Register: RAWCMN_DIG_AON_SUP_MISC

Bits	Name	Memory Access	Description
15:8	SUP_MISC_EN	R/W	sup_misc enable (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff **Volatile:** true
7:0	SUP_MISC_VAL	R/W	sup_misc val (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff **Volatile:** true

2.1.364 LANEN_DIG_ASIC_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming LANE Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)00
- **Exists:** Always

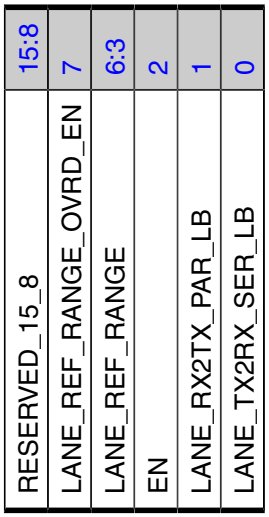


Table 2-368 Fields for Register: LANEN_DIG_ASIC_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	LANE_REF_RANGE_OVRD_EN	R/W	This bit enables override values for lane_ref_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:3	LANE_REF_RANGE	R/W	This field provides override values for lane_ref_range[3:0]. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xf
2	EN	R/W	This bit enables override values for lane_rx2tx_par_lb_en and lane_tx2rx_ser_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	LANE_RX2TX_PAR_LB	R/W	This bit is the override value for lane_rx2tx_par_lb_en_r. **Value After Reset:** 0x0

Table 2-368 Fields for Register: LANEN_DIG_ASIC_LANE_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	LANE_TX2RX_SER_LB	R/W	This bit is the override value for lane_tx2rx_ser_lb_en_r. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.365 LANEN_DIG_ASIC_TX_OVRD_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)01
- **Exists:** Always

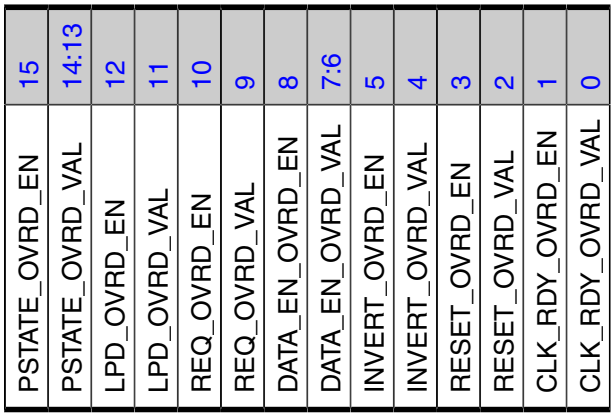


Table 2-369 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	PSTATE_OVRD_EN	R/W	This bit is the override enable for tx_pstate[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:13	PSTATE_OVRD_VAL	R/W	This field is the override value for tx_pstate[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
12	LPD_OVRD_EN	R/W	This bit is the override enable for tx_lpd. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	LPD_OVRD_VAL	R/W	This bit is the override value for tx_lpd. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	REQ_OVRD_EN	R/W	This bit is the override enable for tx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-369 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	REQ_OVRD_VAL	R/W	This bit is the override value for tx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	DATA_EN_OVRD_EN	R/W	This bit is the override enable for tx_data_en[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:6	DATA_EN_OVRD_VAL	R/W	This field is the override value for tx_data_en[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5	INVERT_OVRD_EN	R/W	This bit is the override enable for tx_invert. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	INVERT_OVRD_VAL	R/W	This bit is the override value for tx_invert. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RESET_OVRD_EN	R/W	This bit is the override enable for tx_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RESET_OVRD_VAL	R/W	This bit is the override value for tx_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	CLK_RDY_OVRD_EN	R/W	This bit is the override enable for tx_clk_rdy. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	CLK_RDY_OVRD_VAL	R/W	This bit is the override value for tx_clk_rdy. **Value After Reset:** 0x0

Table 2-369 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.366 LANEN_DIG_ASIC_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)02
- **Exists:** Always

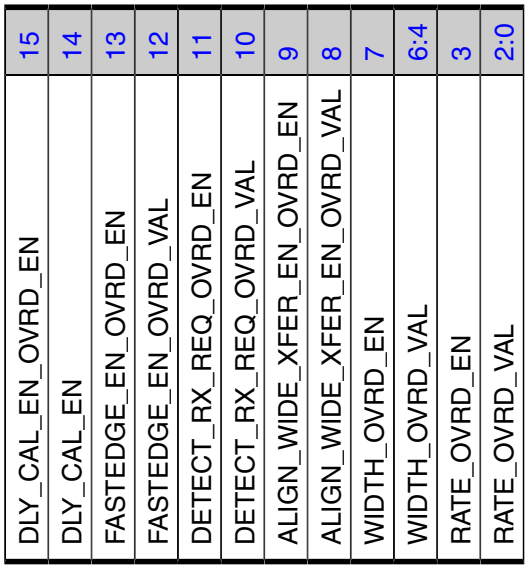


Table 2-370 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DLY_CAL_EN_OVRD_EN	R/W	This bit is the override enable for tx_dly_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	DLY_CAL_EN	R/W	This bit is the override value for tx_dly_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	FASTEDGE_EN_OVRD_EN	R/W	This bit is the override enable for tx_fastedge_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	FASTEDGE_EN_OVRD_VAL	R/W	This bit is the override value for tx_fastedge_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-370 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	DETECT_RX_REQ_OVRD_EN	R/W	This bit is the override enable for tx_detrx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	DETECT_RX_REQ_OVRD_VAL	R/W	This bit is the override value for tx_detrx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	ALIGN_WIDE_XFER_EN_OVRD_EN	R/W	This bit is the override enable for tx_align_wide_xfer_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	ALIGN_WIDE_XFER_EN_OVRD_VAL	R/W	This bit is the override value for tx_align_wide_xfer_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	WIDTH_OVRD_EN	R/W	This bit is the override enable for internal tx_width[2:0] bus. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:4	WIDTH_OVRD_VAL	R/W	This field is the override value for internal tx_width[2:0] bus. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3	RATE_OVRD_EN	R/W	This bit is the override enable for tx_rate[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	RATE_OVRD_VAL	R/W	This field is the override value for tx_rate[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.367 LANEN_DIG_ASIC_TX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)03
- **Exists:** Always

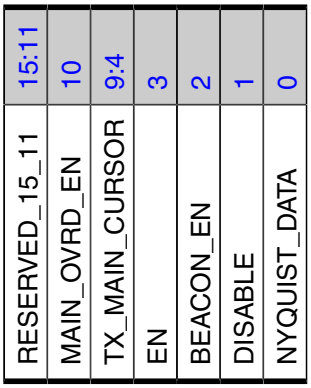


Table 2-371 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	MAIN_OVRD_EN	R/W	This bit enables override values for TX EQ main input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:4	TX_MAIN_CURSOR	R/W	This field is the override value for tx_eq_main. **Value After Reset:** 0x18 **Exists:** Always **Reset Mask:** 0x3f
3	EN	R/W	This bit enables override values for the following registers: - tx_beacon_en ■ tx_disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	BEACON_EN	R/W	This bit is the override value for tx_beacon_en. **Value After Reset:** 0x0

Table 2-371 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
1	DISABLE	R/W	This bit is the override value for tx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	NYQUIST_DATA	R/W	This bit overrides incoming data to nyquist. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.368 LANEN_DIG_ASIC_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)04
- **Exists:** Always

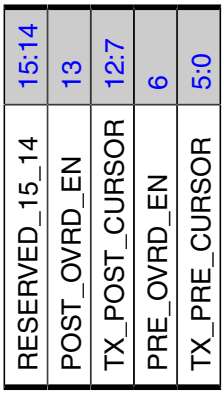


Table 2-372 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	POST_OVRD_EN	R/W	This bit enables override values for TX EQ post input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:7	TX_POST_CURSOR	R/W	This field is the override value for tx_eq_post. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
6	PRE_OVRD_EN	R/W	This bit enables override values for TX EQ pre input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:0	TX_PRE_CURSOR	R/W	This field is the override value for tx_eq_pre. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.369 LANEN_DIG_ASIC_TX_OVRD_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)05
- **Exists:** Always

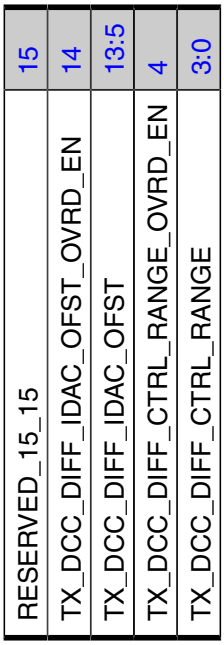


Table 2-373 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_DCC_DIFF_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for tx_dcc_diff_idac_ofst[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:5	TX_DCC_DIFF_IDAC_OFST	R/W	This field is the override value for tx_dcc_diff_idac_ofst[8:0]. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff
4	TX_DCC_DIFF_CTRL_RANGE_OVRD_EN	R/W	This bit enables override value for tx_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0x0

Table 2-373 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3:0	TX_DCC_DIFF_CTRL_RANGE	R/W	This field is the override value for tx_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0xd **Exists:** Always **Reset Mask:** 0xf

2.1.370 LANEN_DIG_ASIC_TX_OVRD_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)06
- **Exists:** Always

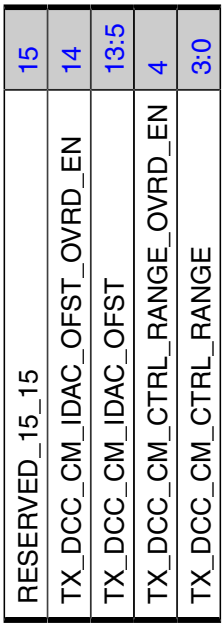


Table 2-374 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_DCC_CM_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for tx_dcc_cm_idac_ofst[8:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:5	TX_DCC_CM_IDAC_OFST	R/W	This field is the override value for tx_dcc_cm_idac_ofst[8:0]. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff
4	TX_DCC_CM_CTRL_RANGE_OVRD_EN	R/W	This bit enables override value for tx_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-374 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_IN_5 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3:0	TX_DCC_CM_CTRL_RANGE	R/W	This field is the override value for tx_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0xd **Exists:** Always **Reset Mask:** 0xf

2.1.371 LANEN_DIG_ASIC_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing TX Controls to ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)07
- **Exists:** Always

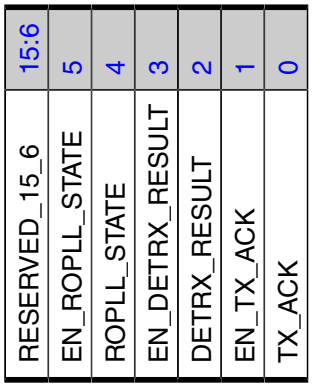


Table 2-375 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	EN_ROPLL_STATE	R/W	This bit enables for override value for tx_ropll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ROPLL_STATE	R/W	This bit enables for override value for tx_ropll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	EN_DETRX_RESULT	R/W	This bit enables for override value for tx_detrx_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	DETRX_RESULT	R/W	This bit is the override value for tx_detrx_result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-375 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	EN_TX_ACK	R/W	This bit enables for override value for tx_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ACK	R/W	This bit is the override value for tx_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.372 LANEN_DIG_ASIC_ROPLL_OVRD_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)08
- **Exists:** Always

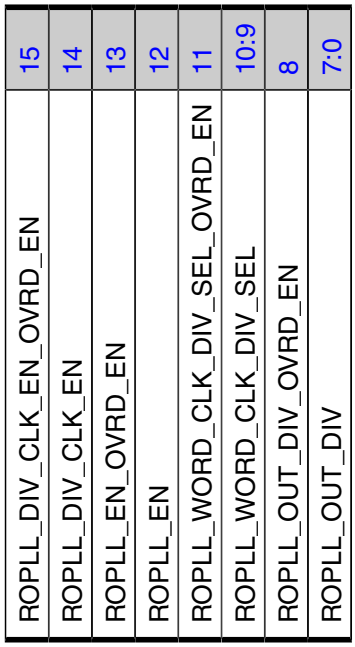


Table 2-376 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_DIV_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	ROPLL_DIV_CLK_EN	R/W	This bit is the override value for tx_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	ROPLL_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	ROPLL_EN	R/W	This bit is the override value for tx_ropll_en. **Value After Reset:** 0x0

Table 2-376 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11	ROPLL_WORD_CLK_DIV_SEL_OVRD_EN	R/W	This bit is the override enable for tx_ropll_word_clk_div_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:9	ROPLL_WORD_CLK_DIV_SEL	R/W	This field is the override value for tx_ropll_word_clk_div_sel[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
8	ROPLL_OUT_DIV_OVRD_EN	R/W	This bit is the override enable for tx_ropll_out_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	ROPLL_OUT_DIV	R/W	This field is the override value for tx_ropll_out_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.373 LANEN_DIG_ASIC_ROPLL_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)09
- **Exists:** Always

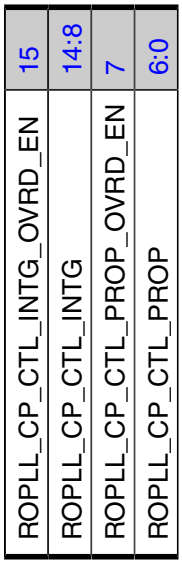


Table 2-377 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_CP_CTL_INTG_OVRD_EN	R/W	This bit is the override enable for tx_ropll_cp_ctl_intg. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:8	ROPLL_CP_CTL_INTG	R/W	This field is the override value for tx_ropll_cp_ctl_intg. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
7	ROPLL_CP_CTL_PROP_OVRD_EN	R/W	This bit is the override enable for tx_ropll_cp_ctl_prop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	ROPLL_CP_CTL_PROP	R/W	This field is the override value for tx_ropll_cp_ctl_prop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.374 LANEN_DIG_ASIC_ROPLL_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0a
- **Exists:** Always

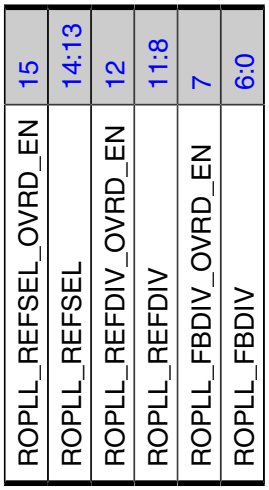


Table 2-378 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_REFSEL_OVRD_EN	R/W	This bit is the override enable for tx_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:13	ROPLL_REFSEL	R/W	This field is the override value for tx_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
12	ROPLL_REFDIV_OVRD_EN	R/W	This bit is the override enable for tx_ropll_refdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:8	ROPLL_REFDIV	R/W	This field is the override value for tx_ropll_refdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7	ROPLL_FBDIV_OVRD_EN	R/W	This bit is the override enable for tx_ropll_fbdiv. **Value After Reset:** 0x0

Table 2-378 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
6:0	ROPLL_FBDIV	R/W	This field is the override value for tx_ropll_fbdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.375 LANEN_DIG_ASIC_ROPLL_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0b
- **Exists:** Always

Table 2-379 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_BYPASS_OVRD_EN	R/W	This bit is the override enable for tx_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	ROPLL_BYPASS	R/W	This bit is the override value for tx_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	ROPLL_RC_FILTER_OVRD_EN	R/W	This bit is the override enable for tx_ropll_rc_filter. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:10	ROPLL_RC_FILTER	R/W	This field is the override value for tx_ropll_rc_filter. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

Table 2-379 Fields for Register: LANEN_DIG_ASIC_ROPLL_OVRD_IN_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	ROPLL_V2I_MODE_OVRD_EN	R/W	This bit is the override enable for tx_ropll_v2i_mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:7	ROPLL_V2I_MODE	R/W	This field is the override value for tx_ropll_v2i_mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6	ROPLL_VCO_LOW_FREQ_OVRD_EN	R/W	This bit is the override enable for tx_ropll_vco_low_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	ROPLL_VCO_LOW_FREQ	R/W	This field is the override value for tx_ropll_vco_low_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	ROPLL_POSTDIV_OVRD_EN	R/W	This bit is the override enable for tx_ropll_postdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	ROPLL_POSTDIV	R/W	This field is the override value for tx_ropll_postdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.376 LANEN_DIG_ASIC_TX_OVRD_MISC (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX MISC BUS from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0c
- **Exists:** Always

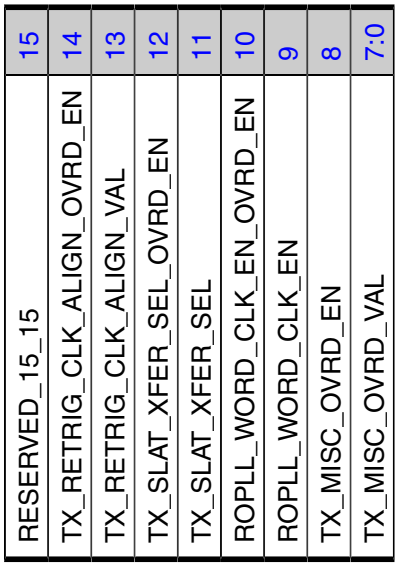


Table 2-380 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_MISC (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_RETRIG_CLK_ALIGN_OVRD_EN	R/W	This bit is the override enable for tx_retrig_clk_align. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	TX_RETRIG_CLK_ALIGN_VAL	R/W	This bit is the override enable for tx_retrig_clk_align. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	TX_SLAT_XFER_SEL_OVRD_EN	R/W	This bit is the override enable for tx_slat_xfer_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-380 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_MISC (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	TX_SLAT_XFER_SEL	R/W	This bit is the override value for tx_slat_xfer_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	ROPLL_WORD_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	ROPLL_WORD_CLK_EN	R/W	This bit is the override value for tx_ropll_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TX_MISC_OVRD_EN	R/W	This bit is the override enable for tx_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	TX_MISC_OVRD_VAL	R/W	This field is the override value for tx_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.377 LANEN_DIG_ASIC_TX_ASIC_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0d
- **Exists:** Always

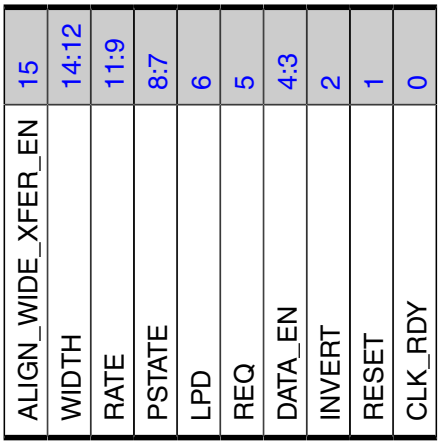


Table 2-381 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ALIGN_WIDE_XFER_EN	R	This bit is the value from ASIC for tx_align_wide_xfer_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14:12	WIDTH	R	This field is the value from ASIC for tx_width[2:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:9	RATE	R	This field is the value from ASIC for tx_rate[2:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:7	PSTATE	R	This field is the value from ASIC for tx_pstate[1:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-381 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
6	LPD	R	This bit is the value from ASIC for tx_lpd. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	REQ	R	This bit is the value from ASIC for tx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4:3	DATA_EN	R	This field is the value from ASIC for tx_data_en[1:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	INVERT	R	This bit is the value from ASIC for tx_invert. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	RESET	R	This bit is the value from ASIC for tx_reset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	CLK_RDY	R	This bit is the value from ASIC for tx_clk_rdy. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-381 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.378 LANEN_DIG_ASIC_TX_ASIC_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0e
- **Exists:** Always

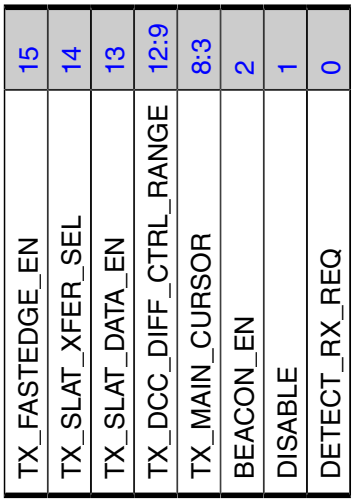


Table 2-382 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_FASTEDGE_EN	R	This bit is the value from ASIC for tx_fastedge_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	TX_SLAT_XFER_SEL	R	This field is the value from ASIC for tx_slat_xfer_sel (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	TX_SLAT_DATA_EN	R	This field is the value from ASIC for tx_slat_data_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-382 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12:9	TX_DCC_DIFF_CTRL_RANGE	R	This field is the value from ASIC for tx_dcc_ctrl_diff_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:3	TX_MAIN_CURSOR	R	This field is the value from ASIC for tx_eq_main. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	BEACON_EN	R	This bit is the value from ASIC for tx_beacon_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	DISABLE	R	This bit is the value from ASIC for tx_disable. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	DETECT_RX_REQ	R	This bit is the value from ASIC for tx_detrx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.379 LANEN_DIG_ASIC_TX_ASIC_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)0f
- **Exists:** Always

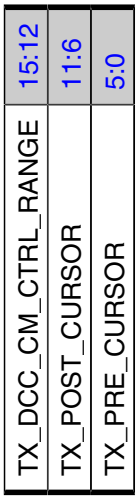


Table 2-383 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	TX_DCC_CM_CTRL_RANGE	R	This field is the value from ASIC for tx_dcc_ctrl_cm_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:6	TX_POST_CURSOR	R	This field is the value from ASIC for tx_eq_post. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	TX_PRE_CURSOR	R	This field is the value from ASIC for tx_eq_pre. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.380 LANEN_DIG_ASIC_TX_ASIC_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)10
- **Exists:** Always

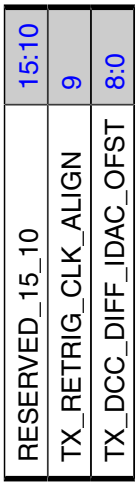


Table 2-384 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TX_RETRIG_CLK_ALIGN	R	This field is the value from ASIC for tx_retrig_clk_align. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:0	TX_DCC_DIFF_IDAC_OFST	R	This field is the value from ASIC for tx_dcc_diff_idac_ofst[8:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.381 LANEN_DIG_ASIC_TX_ASIC_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Drive Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)11
- **Exists:** Always

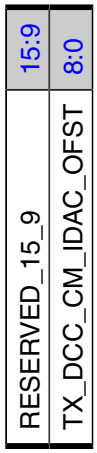


Table 2-385 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	TX_DCC_CM_IDAC_OFST	R	This field is the value from ASIC for tx_dcc_cm_idac_ofst[8:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.382 LANEN_DIG_ASIC_TX_ASIC_OUT (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Outgoing TX Status Controls from PHY.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)12
- **Exists:** Always

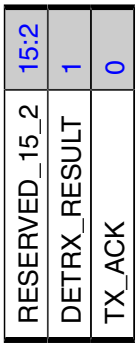


Table 2-386 Fields for Register: LANEN_DIG_ASIC_TX_ASIC_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DETRX_RESULT	R	This bit is the value from PHY for tx_detrx_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TX_ACK	R	This bit is the value from PHY for tx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.383 LANEN_DIG_ASIC_ROPLL_ASIC_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)13
- **Exists:** Always

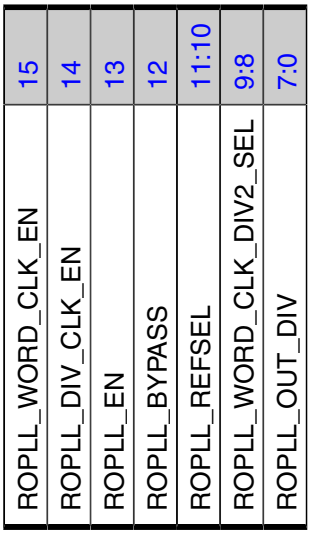


Table 2-387 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_WORD_CLK_EN	R	This bit is the value from ASIC for ropll_word_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	ROPLL_DIV_CLK_EN	R	This bit is the value from ASIC for ropll_div_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	ROPLL_EN	R	This bit is the value from ASIC for tx_ropll_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-387 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12	ROPLL_BYPASS	R	This bit is the value from ASIC for tx_ropll_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:10	ROPLL_REFSEL	R	This field is the value from ASIC for tx_ropll_refsel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:8	ROPLL_WORD_CLK_DIV2_SEL	R	This field is the value from ASIC for tx_ropll_word_clk_div_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	ROPLL_OUT_DIV	R	This field is the value from ASIC for tx_ropll_out_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.384 LANEN_DIG_ASIC_ROPLL_ASIC_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)14
- **Exists:** Always

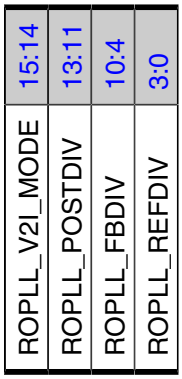


Table 2-388 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	ROPLL_V2I_MODE	R	This field is the value from ASIC for tx_ropll_v2i_mode. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:11	ROPLL_POSTDIV	R	This field is the value from ASIC for tx_ropll_postdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:4	ROPLL_FBDIV	R	This field is the value from ASIC for tx_ropll_fbdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	ROPLL_REFDIV	R	This field is the value from ASIC for tx_ropll_refdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-388 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.385 LANEN_DIG_ASIC_ROPLL_ASIC_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)15
- **Exists:** Always

Table 2-389 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	ROPLL_VCO_LOW_FREQ	R	This field is the value from ASIC for tx_ropll_vco_low_freq. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:7	ROPLL_CP_CTL_PROP	R	This field is the value from ASIC for tx_ropll_cp_ctl_prop. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	ROPLL_CP_CTL_INTG	R	This field is the value from ASIC for tx_ropll_cp_ctl_intg. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.386 LANEN_DIG_ASIC_ROPLL_ASIC_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)16
- **Exists:** Always

Table 2-390 Fields for Register: LANEN_DIG_ASIC_ROPLL_ASIC_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	ROPLL_RC_FILTER	R	This field is the value from ASIC for tx_ropll_rc_filter. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.387 LANEN_DIG_ASIC_TX_DLY_CAL_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Delay Calibration Signals from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)17
- **Exists:** Always

Table 2-391 Fields for Register: LANEN_DIG_ASIC_TX_DLY_CAL_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DLY_CAL_FLIP_OVRD_EN	R/W	This bit is the override enable for tx_dly_cal_flip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	DLY_CAL_FLIP	R/W	This bit is the override value for tx_dly_cal_flip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	DLY_CAL_GAIN_OVRD_EN	R/W	This bit is the override enable for tx_dly_cal_gain[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:11	DLY_CAL_GAIN	R/W	This field is the override value for tx_dly_cal_gain[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
10	DLY_CAL_CODE_OVRD_EN	R/W	This bit is the override enable for tx_dly_cal_code[9:0]. **Value After Reset:** 0x0

Table 2-391 Fields for Register: LANEN_DIG_ASIC_TX_DLY_CAL_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9:0	DLY_CAL_CODE	R/W	This field is the override value for tx_dly_cal_code[9:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.388 LANEN_DIG_ASIC_TX_DLY_CAL_ASIC_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Delay Calibration Controls from ASIC Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)18
- **Exists:** Always

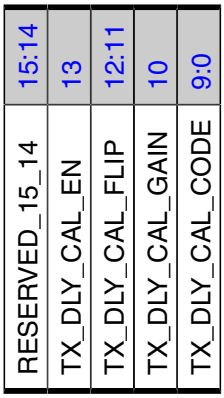


Table 2-392 Fields for Register: LANEN_DIG_ASIC_TX_DLY_CAL_ASIC_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	TX_DLY_CAL_EN	R	This bit is the value from ASIC for tx_dly_cal_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:11	TX_DLY_CAL_FLIP	R	This field is the value from ASIC for tx_dly_cal_flip. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	TX_DLY_CAL_GAIN	R	This bit is the value from ASIC for tx_dly_cal_gain[1:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-392 Fields for Register: LANEN_DIG_ASIC_TX_DLY_CAL_ASIC_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
9:0	TX_DLY_CAL_CODE	R	This field is the value from ASIC for tx_dly_cal_code[9:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.389 LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_UP (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TXUP termination code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)19
- **Exists:** Always

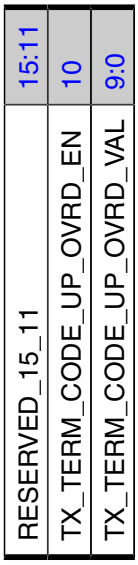


Table 2-393 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_UP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_TERM_CODE_UP_OVRD_EN	R/W	This bit enables override value for tx_term_code_up. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_TERM_CODE_UP_OVRD_VAL	R/W	This bit is the override value for tx_term_code_up. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.390 LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_DN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TXDN termination code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)1a
- **Exists:** Always

Table 2-394 Fields for Register: LANEN_DIG_ASIC_TX_OVRD_TERM_CODE_DN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_TERM_CODE_DN_OVRD_EN	R/W	This bit enables override value for tx_term_code_dn. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_TERM_CODE_DN_OVRD_ VAL	R/W	This bit is the override value for tx_term_code_dn. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.391 LANEN_DIG_ASIC_TX_TERM_CODE_UP_ASIC_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TXUP termination code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)1b
- **Exists:** Always

Table 2-395 Fields for Register: LANEN_DIG_ASIC_TX_TERM_CODE_UP_ASIC_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TX_TERM_CODE_UP	R	This field is the value from ASIC for tx_term_code_up[9:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.392 LANEN_DIG_ASIC_TX_TERM_CODE_DN_ASIC_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TXDN termination code from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)1c
- **Exists:** Always

Table 2-396 Fields for Register: LANEN_DIG_ASIC_TX_TERM_CODE_DN_ASIC_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	TX_TERM_CODE_DN	R	This field is the value from ASIC for tx_term_code_dn[9:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.393 LANEN_DIG_ASIC_RX_OVRD_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)20
- **Exists:** Always

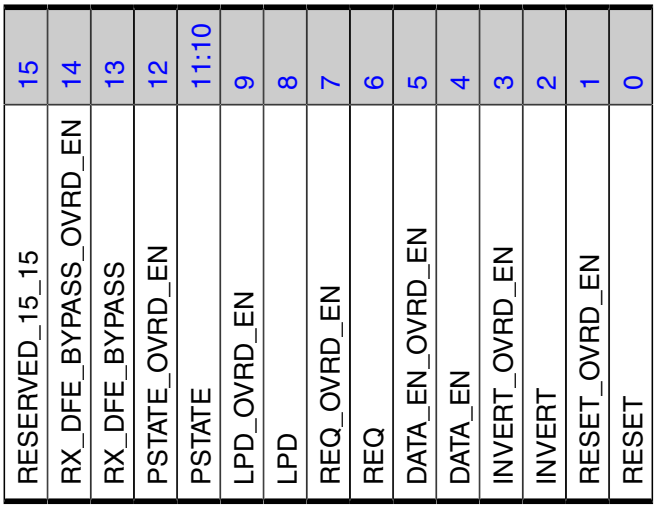


Table 2-397 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_DFE_BYPASS_OVRD_EN	R/W	This bit enables override value for rx_dfe_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_DFE_BYPASS	R/W	This bit is the override value for rx_dfe_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	PSTATE_OVRD_EN	R/W	This bit enables override value for rx_pstate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:10	PSTATE	R/W	This field is the override value for rx_pstate. **Value After Reset:** 0x0

Table 2-397 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3
9	LPD_OVRD_EN	R/W	This bit enables override value for rx_lpd. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	LPD	R/W	This bit is the override value for rx_lpd. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	REQ_OVRD_EN	R/W	This bit enables override value for rx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	REQ	R/W	This bit is the override value for rx_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	DATA_EN_OVRD_EN	R/W	This bit enables override value for rx_data_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	DATA_EN	R/W	This bit is the override value for rx_data_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	INVERT_OVRD_EN	R/W	This bit enables override value for rx_invert. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	INVERT	R/W	This bit is the override value for rx_invert. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-397 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RESET_OVRD_EN	R/W	This bit enables override value for rx_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RESET	R/W	This bit is the override value for rx_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.394 LANEN_DIG_ASIC_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)21
- **Exists:** Always

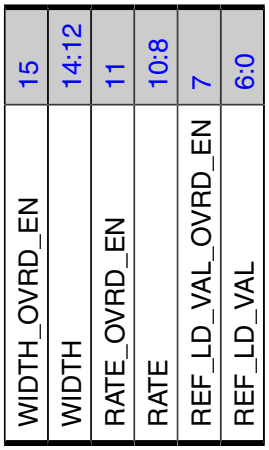


Table 2-398 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	WIDTH_OVRD_EN	R/W	This bit enables override value for rx_width. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:12	WIDTH	R/W	This field is the override value for rx_width. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
11	RATE_OVRD_EN	R/W	This bit enables override value for rx_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:8	RATE	R/W	This field is the override value for rx_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
7	REF_LD_VAL_OVRD_EN	R/W	This bit enables override value for rx_ref_ld_val. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-398 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6:0	REF_LD_VAL	R/W	This field is the override value for rx_ref_ld_val. **Value After Reset:** 0x14 **Exists:** Always **Reset Mask:** 0x7f

2.1.395 LANEN_DIG_ASIC_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)22
- **Exists:** Always

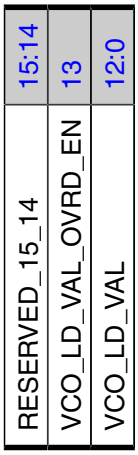


Table 2-399 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	VCO_LD_VAL_OVRD_EN	R/W	This bit enables override value for tx_vco_ld_val. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:0	VCO_LD_VAL	R/W	This field is the override value for rx_vco_ld_val. **Value After Reset:** 0x3e8 **Exists:** Always **Reset Mask:** 0x1fff

2.1.396 LANEN_DIG_ASIC_RX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)23
- **Exists:** Always

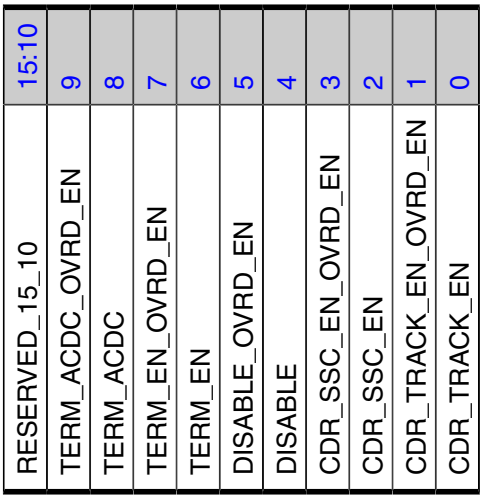


Table 2-400 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TERM_ACDC_OVRD_EN	R/W	This bit enables override value for rx_term_acdc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TERM_ACDC	R/W	This bit is the override value for rx_term_acdc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TERM_EN_OVRD_EN	R/W	This bit enables override value for rx_term_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TERM_EN	R/W	This bit is the override value for rx_term_en. **Value After Reset:** 0x0

Table 2-400 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
5	DISABLE_OVRD_EN	R/W	This bit enables override value for rx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	DISABLE	R/W	This bit is the override value for rx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	CDR_SSC_EN_OVRD_EN	R/W	This bit enables override value for rx_cdr_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	CDR_SSC_EN	R/W	This bit is the override value for rx_cdr_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	CDR_TRACK_EN_OVRD_EN	R/W	This bit enables override value for rx_cdr_track_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	CDR_TRACK_EN	R/W	This bit is the override value for rx_cdr_track_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.397 LANEN_DIG_ASIC_RX_OVRD_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Drive Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)24
- **Exists:** Always

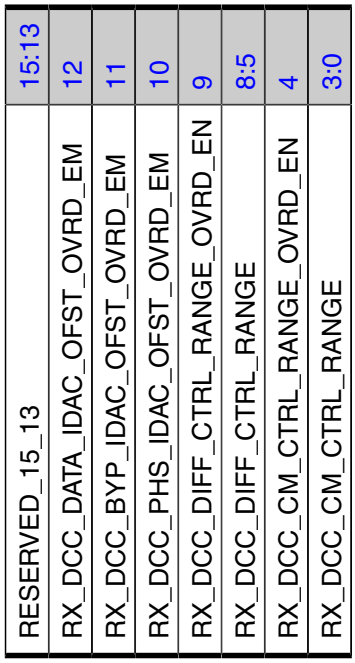


Table 2-401 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_DCC_DATA_IDAC_OFST_OVRD_EM	R/W	This bit is the override enable for RX DCC data signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	RX_DCC_BYP_IDAC_OFST_OVRD_EM	R/W	This bit is the override enable for RX DCC bypass signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX_DCC_PHS_IDAC_OFST_OVRD_EM	R/W	This bit is the override enable for RX DCC phase signals. **Value After Reset:** 0x0

Table 2-401 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9	RX_DCC_DIFF_CTRL_RANGE_OVRD_EN	R/W	This bit enables override value for rx_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:5	RX_DCC_DIFF_CTRL_RANGE	R/W	This field is the override value for rx_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf
4	RX_DCC_CM_CTRL_RANGE_OVRD_EN	R/W	This bit enables override value for rx_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	RX_DCC_CM_CTRL_RANGE	R/W	This field is the override value for rx_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf

2.1.398 LANEN_DIG_ASIC_RX_OVRD_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Drive Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)25
- **Exists:** Always

Table 2-402 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_DATA_DIFF_IDAC_OFST	R/W	This field is the override value for rx_dcc_data_diff_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	RX_DCC_DATA_CM_IDAC_OFST	R/W	This field is the override value for rx_dcc_data_cm_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.399 LANEN_DIG_ASIC_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Drive Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)26
- **Exists:** Always

Table 2-403 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_BYP_DIFF_IDAC_OFST	R/W	This field is the override value for rx_dcc_byp_diff_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	RX_DCC_BYP_CM_IDAC_OFST	R/W	This field is the override value for rx_dcc_byp_cm_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.400 LANEN_DIG_ASIC_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Drive Controls from ASIC, Register 7.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)27
- **Exists:** Always

Table 2-404 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_PHS_DIFF_IDAC_OFST	R/W	This field is the override value for rx_dcc_phs_diff_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	RX_DCC_PHS_CM_IDAC_OFST	R/W	This field is the override value for rx_dcc_phs_cm_idac_ofst[7:0]. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.401 LANEN_DIG_ASIC_RX_OVRD_SIGDET_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX SIGDET Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)28
- **Exists:** Always

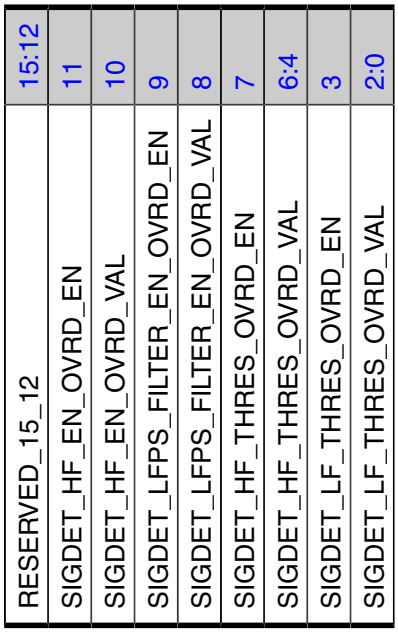


Table 2-405 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_SIGDET_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	SIGDET_HF_EN_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	SIGDET_HF_EN_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_hf_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	SIGDET_LFPS_FILTER_EN_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lfps_filter_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-405 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_SIGDET_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8	SIGDET_LFPS_FILTER_EN_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_lfps_filter_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	SIGDET_HF_THRES_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:4	SIGDET_HF_THRES_OVRD_VAL	R/W	This field is the override value for rx_sigdet_hf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3	SIGDET_LF_THRES_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	SIGDET_LF_THRES_OVRD_VAL	R/W	This field is the override value for rx_sigdet_lf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.402 LANEN_DIG_ASIC_RX_OVRD_VCO_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX VCO Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)29
- **Exists:** Always

Table 2-406 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_VCO_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	CDR_VCO_CONFIG_OVRD_EN	R/W	This bit enables override value for rx_cdr_vco_config. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	CDR_VCO_CONFIG	R/W	This field is the override value for rx_cdr_vco_config. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.403 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2a
- **Exists:** Always

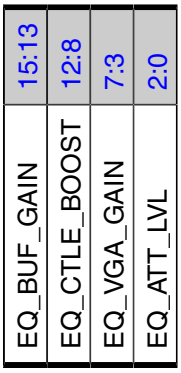


Table 2-407 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	EQ_BUF_GAIN	R/W	This field is the override value for rx_eq_buf_gain. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7
12:8	EQ_CTLE_BOOST	R/W	This field is the override value for rx_eq_ctle_boost. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x1f
7:3	EQ_VGA_GAIN	R/W	This field is the override value for rx_eq_vga_gain. **Value After Reset:** 0xe **Exists:** Always **Reset Mask:** 0x1f
2:0	EQ_ATT_LVL	R/W	This field is the override value for rx_eq_att_lvl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.404 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2b
- **Exists:** Always

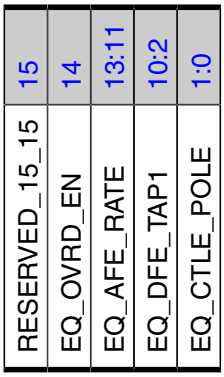


Table 2-408 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	EQ_OVRD_EN	R/W	This bit enables override value for rx_eq_* inputs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:11	EQ_AFE_RATE	R/W	This field is the override value for rx_eq_afe_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
10:2	EQ_DFE_TAP1	R/W	This field is the override value for rx_eq_dfe_tap1. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff
1:0	EQ_CTLE_POLE	R/W	This field is the override value for rx_eq_ctle_pole. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.405 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2c
- **Exists:** Always

Table 2-409 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	EQ_IND_BYPASS	R/W	This bit enables override value for rx_eq_ind_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:8	EQ_DFE_TAP3	R/W	This field is the override value for rx_eq_dfe_tap3. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f
7:0	EQ_DFE_TAP2	R/W	This field is the override value for rx_eq_dfe_tap2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.406 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2d
- **Exists:** Always

Table 2-410 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_TAP5	R/W	This field is the override value for rx_eq_dfe_tap5. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f
6:0	EQ_DFE_TAP4	R/W	This field is the override value for rx_eq_dfe_tap4. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f

2.1.407 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2e
- **Exists:** Always

Table 2-411 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	EQ_IQ_OVRD_EN	R/W	This bit enables override value for rx_eq_iq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	EQ_IQ	R/W	This field is the override value for rx_eq_iq. **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff

2.1.408 LANEN_DIG_ASIC_RX_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing RX Controls to ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)2f
- **Exists:** Always

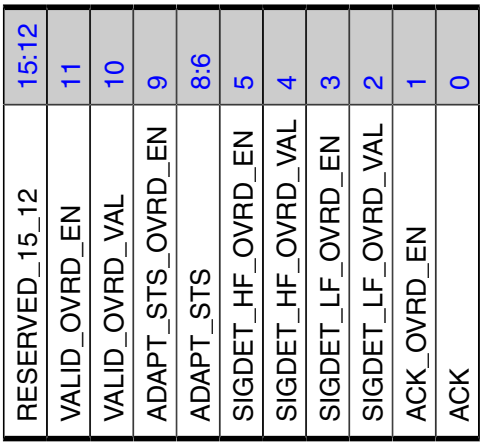


Table 2-412 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	VALID_OVRD_EN	R/W	This bit is the override enable for rx_valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	VALID_OVRD_VAL	R/W	This bit is the override value for rx_valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	ADAPT_STS_OVRD_EN	R/W	This bit is the override enable for rx_adapt_sts. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:6	ADAPT_STS	R/W	This field is the override value for rx_adapt_sts. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

Table 2-412 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5	SIGDET_HF_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SIGDET_HF_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_hf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SIGDET_LF_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SIGDET_LF_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_lf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ACK_OVRD_EN	R/W	This bit is the override enable for rx_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ACK	R/W	This bit is the override value for rx_ack. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.409 LANEN_DIG_ASIC_LANE_ASIC_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming LANE Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)30
- **Exists:** Always

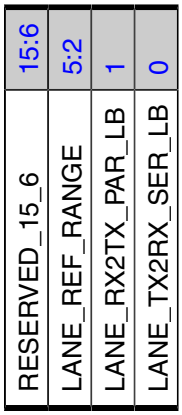


Table 2-413 Fields for Register: LANEN_DIG_ASIC_LANE_ASIC_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:2	LANE_REF_RANGE	R	This field is the value from ASIC for lane_ref_range[3:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	LANE_RX2TX_PAR_LB	R	This bit is the value from ASIC for lane_rx2tx_par_lb_en_r. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	LANE_TX2RX_SER_LB	R	This bit is the value from ASIC for lane_tx2rx_ser_lb_en_r. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-413 Fields for Register: LANEN_DIG_ASIC_LANE_ASIC_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.410 LANEN_DIG_ASIC_RX_ASIC_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)31
- **Exists:** Always

15	14	13	12:10	9:7	6:5	4	3	2	1	0
RESERVED_15_15	CDR_TRACK_EN	RX_DFE_BYPASS	WIDTH	RATE	PSTATE	LPD	REQ	DATA_EN	INVERT	RESET

Table 2-414 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	CDR_TRACK_EN	R	This bit is the value from ASIC for rx_cdr_track_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	RX_DFE_BYPASS	R	This bit is the value from ASIC for rx_dfe_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:10	WIDTH	R	This field is the value from ASIC for rx_width[2:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-414 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9:7	RATE	R	This field is the value from ASIC for rx_rate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:5	PSTATE	R	This field is the value from ASIC for rx_pstate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	LPD	R	This bit is the value from ASIC for rx_lpd. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	REQ	R	This bit is the value from ASIC for rx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	DATA_EN	R	This bit is the value from ASIC for rx_data_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	INVERT	R	This bit is the value from ASIC for rx_invert. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-414 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	RESET	R	This bit is the value from ASIC for rx_reset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.411 LANEN_DIG_ASIC_RX_ASIC_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)32
- **Exists:** Always

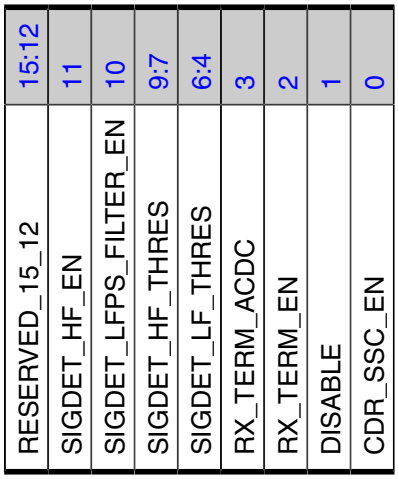


Table 2-415 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	SIGDET_HF_EN	R	This bit is the value from ASIC for rx_sigdet_hf_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	SIGDET_LFPS_FILTER_EN	R	This bit is the value from ASIC for rx_sigdet_lfps_filter_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:7	SIGDET_HF_THRES	R	This field is the value from ASIC for rx_sigdet_hf_threshold. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-415 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
6:4	SIGDET_LF_THRES	R	This field is the value from ASIC for rx_sigdet_lf_threshold. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	RX_TERM_ACDC	R	This bit is the value from ASIC for rx_term_acdc. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	RX_TERM_EN	R	This bit is the value from ASIC for rx_term_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	DISABLE	R	This bit is the value from ASIC for rx_disable. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	CDR_SSC_EN	R	This bit is the value from ASIC for rx_cdr_ssc_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.412 LANEN_DIG_ASIC_RX_ASIC_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)33
- **Exists:** Always

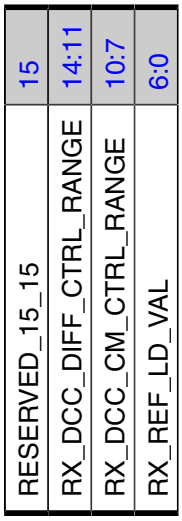


Table 2-416 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:11	RX_DCC_DIFF_CTRL_RANGE	R	This field is the value from ASIC for rx_dcc_ctrl_diff_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:7	RX_DCC_CM_CTRL_RANGE	R	This field is the value from ASIC for rx_dcc_ctrl_cm_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	RX_REF_LD_VAL	R	This field is the value from ASIC for rx_ref_ld_val. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-416 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true

2.1.413 LANEN_DIG_ASIC_RX_ASIC_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)34
- **Exists:** Always

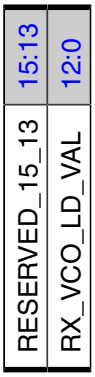


Table 2-417 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	RX_VCO_LD_VAL	R	This field is the value from ASIC for rx_vco_ld_val. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.414 LANEN_DIG_ASIC_RX_ASIC_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)35
- **Exists:** Always

Table 2-418 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_DATA_DIFF_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_data_diff_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_DATA_CM_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_data_cm_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.415 LANEN_DIG_ASIC_RX_ASIC_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)36
- **Exists:** Always

Table 2-419 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_BYP_DIFF_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_byp_diff_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_BYP_CM_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_byp_cm_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.416 LANEN_DIG_ASIC_RX_ASIC_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)37
- **Exists:** Always

Table 2-420 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_PHS_DIFF_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_phs_diff_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_PHS_CM_IDAC_OFST	R	This field is the value from ASIC for rx_dcc_phs_cm_idac_ofst[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.417 LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX CDR VCO Controls from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)38
- **Exists:** Always

Table 2-421 Fields for Register: LANEN_DIG_ASIC_RX_CDR_VCO_ASIC_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	RX_CDR_VCO_CONFIG	R	This field is the value from ASIC for rx_cdr_vco_config. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.418 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX EQ Controls from ASIC, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)39
- **Exists:** Always

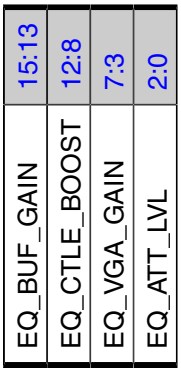


Table 2-422 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	EQ_BUF_GAIN	R	This field is the value from ASIC for rx_eq_buf_gain. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:8	EQ_CTLE_BOOST	R	This field is the value from ASIC for rx_eq_ctle_boost. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:3	EQ_VGA_GAIN	R	This field is the value from ASIC for rx_eq_vga_gain. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2:0	EQ_ATT_LVL	R	This field is the value from ASIC for rx_eq_att_lvl. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-422 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.419 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX EQ Controls from ASIC, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3a
- **Exists:** Always

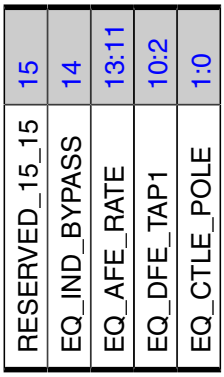


Table 2-423 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	EQ_IND_BYPASS	R	This field is the value from ASIC for rx_eq_ind_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:11	EQ_AFE_RATE	R	This field is the value from ASIC for rx_eq_afe_rate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:2	EQ_DFE_TAP1	R	This field is the value from ASIC for rx_eq_dfe_tap1. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-423 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1:0	EQ_CTLE_POLE	R	This field is the value from ASIC for rx_eq_ctle_pole. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.420 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX EQ Controls from ASIC, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3b
- **Exists:** Always

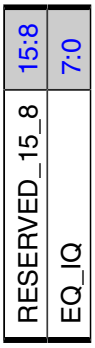


Table 2-424 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	EQ_IQ	R	This field is the value from ASIC for rx_eq_iq. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.421 LANEN_DIG_ASIC_RX_ASIC_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Outgoing RX Status Controls from PHY, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3c
- **Exists:** Always

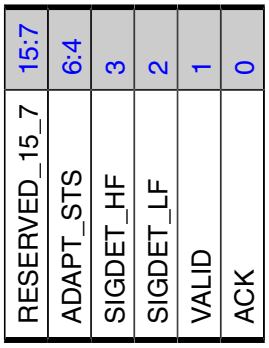


Table 2-425 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:4	ADAPT_STS	R	This field is the value from PHY for rx_adapt_sts. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	SIGDET_HF	R	This bit is the value from PHY for rx_sigdet_hf. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	SIGDET_LF	R	This bit is the value from PHY for rx_sigdet_lf. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-425 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	VALID	R	This bit is the value from PHY for rx_valid. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	ACK	R	This bit is the value from PHY for rx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.422 LANEN_DIG_ASIC_RX_OVRD_MISC (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX MISC BUS from ASIC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3d
- **Exists:** Always

Table 2-426 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_MISC (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_TERM_CODE_OVRD_EN	R/W	This bit enables override value for rx_term_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:9	RX_TERM_CODE_OVRD_VAL	R/W	This bit is the override value for rx_term_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
8	RX_MISC_OVRD_EN	R/W	This bit is the override enable for rx_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	RX_MISC_OVRD_VAL	R/W	This field is the override value for rx_misc[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.423 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3e
- **Exists:** Always

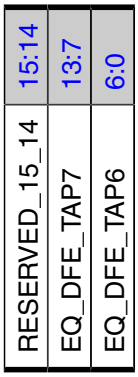


Table 2-427 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_TAP7	R/W	This field is the override value for rx_eq_dfe_tap7. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f
6:0	EQ_DFE_TAP6	R/W	This field is the override value for rx_eq_dfe_tap6. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f

2.1.424 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)3f
- **Exists:** Always

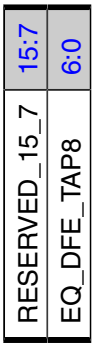


Table 2-428 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	EQ_DFE_TAP8	R/W	This field is the override value for rx_eq_dfe_tap8. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7f

2.1.425 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 7.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)40
- **Exists:** Always

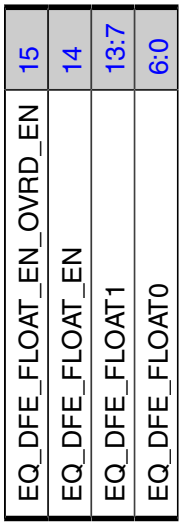


Table 2-429 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	EQ_DFE_FLOAT_EN_OVRD_EN	R/W	This bit is the override value for rx_eq_dfe_float_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	EQ_DFE_FLOAT_EN	R/W	This bit is the override value for rx_eq_dfe_float_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:7	EQ_DFE_FLOAT1	R/W	This field is the override value for rx_eq_dfe_float1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
6:0	EQ_DFE_FLOAT0	R/W	This field is the override value for rx_eq_dfe_float0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.426 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 8.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)41
- **Exists:** Always

Table 2-430 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_FLOAT3	R/W	This field is the override value for rx_eq_dfe_float3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
6:0	EQ_DFE_FLOAT2	R/W	This field is the override value for rx_eq_dfe_float2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.427 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 9.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)42
- **Exists:** Always

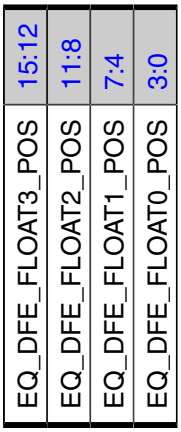


Table 2-431 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	EQ_DFE_FLOAT3_POS	R/W	This field is the override value for rx_eq_dfe_float3_pos. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:8	EQ_DFE_FLOAT2_POS	R/W	This field is the override value for rx_eq_dfe_float2_pos. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:4	EQ_DFE_FLOAT1_POS	R/W	This field is the override value for rx_eq_dfe_float1_pos. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	EQ_DFE_FLOAT0_POS	R/W	This field is the override value for rx_eq_dfe_float0_pos. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.428 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_10 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 10.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)43
- **Exists:** Always

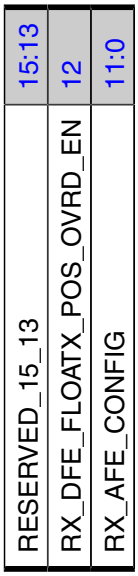


Table 2-432 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_DFE_FLOATX_POS_OVRD_EN	R/W	This bit is the override enable for rx_eq_dfe_floatX_pos. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	RX_AFE_CONFIG	R/W	This field is the override value for rx_eq_afe_config. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.429 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_11 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 11.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)44
- **Exists:** Always

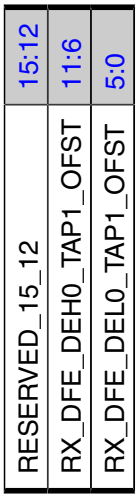


Table 2-433 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DEH0_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_deh0_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_DFE_DEL0_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_del0_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.430 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_12 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 12.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)45
- **Exists:** Always

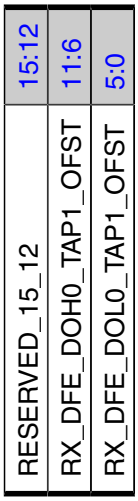


Table 2-434 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DOH0_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_doh0_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_DFE_DOL0_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_dol0_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.431 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_13 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 13.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)46
- **Exists:** Always

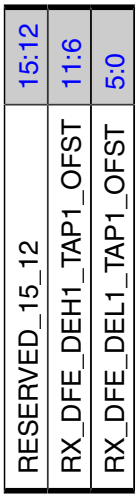


Table 2-435 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DEH1_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_deh1_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_DFE_DEL1_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_del1_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.432 LANEN_DIG_ASIC_RX_OVRD_EQ_IN_14 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from ASIC, Register 14.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)47
- **Exists:** Always

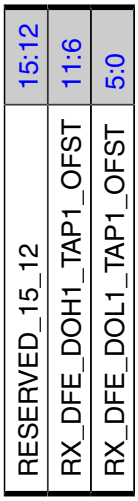


Table 2-436 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_EQ_IN_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DOH1_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_doh1_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_DFE_DOL1_TAP1_OFST	R/W	This field is the override value for rx_eq_dfe_dol1_tap1_ofst. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.433 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)48
- **Exists:** Always

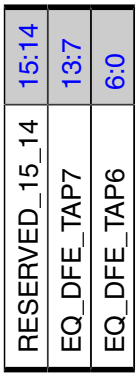


Table 2-437 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_TAP7	R	This field is the value from ASIC for rx_eq_dfe_tap7. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	EQ_DFE_TAP6	R	This field is the value from ASIC for rx_eq_dfe_tap6. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.434 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)49
- **Exists:** Always

Table 2-438 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:1	EQ_DFE_TAP8	R	This field is the value from ASIC for rx_eq_dfe_tap8. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	EQ_DFE_FLOAT_EN	R	This bit is the value from ASIC for rx_eq_dfe_float_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.435 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4a
- **Exists:** Always

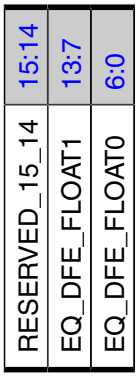


Table 2-439 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_FLOAT1	R	This field is the value from ASIC for rx_eq_dfe_float1. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	EQ_DFE_FLOAT0	R	This field is the value from ASIC for rx_eq_dfe_float0. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.436 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4b
- **Exists:** Always

Table 2-440 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	EQ_DFE_FLOAT3	R	This field is the value from ASIC for rx_eq_dfe_float3. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	EQ_DFE_FLOAT2	R	This field is the value from ASIC for rx_eq_dfe_float2. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.437 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 7.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4c
- **Exists:** Always

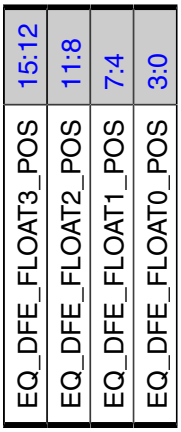


Table 2-441 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	EQ_DFE_FLOAT3_POS	R	This field is the value from ASIC for rx_eq_dfe_float3_pos. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:8	EQ_DFE_FLOAT2_POS	R	This field is the value from ASIC for rx_eq_dfe_float2_pos. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:4	EQ_DFE_FLOAT1_POS	R	This field is the value from ASIC for rx_eq_dfe_float1_pos. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	EQ_DFE_FLOAT0_POS	R	This field is the value from ASIC for rx_eq_dfe_float0_pos. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-441 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_7 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true

2.1.438 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 8.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4d
- **Exists:** Always

Table 2-442 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	RX_AFE_CONFIG	R	This field is the value from ASIC for rx_eq_afe_config. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.439 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 9.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4e
- **Exists:** Always

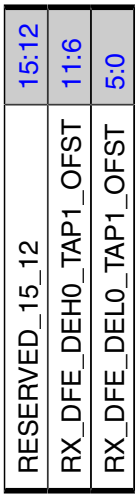


Table 2-443 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DEH0_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_deh_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	RX_DFE_DEL0_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_del_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.440 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_10 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 10.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)4f
- **Exists:** Always

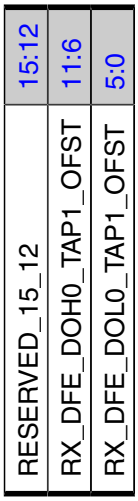


Table 2-444 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DOH0_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_doh_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	RX_DFE_DOL0_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_dol_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.441 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_11 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 11.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)50
- **Exists:** Always

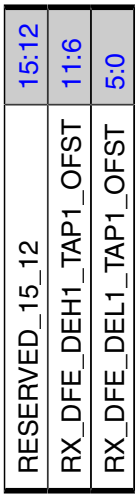


Table 2-445 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DEH1_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_deh_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	RX_DFE_DEL1_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_del_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.442 LANEN_DIG_ASIC_RX_EQ_ASIC_IN_12 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 12.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)51
- **Exists:** Always

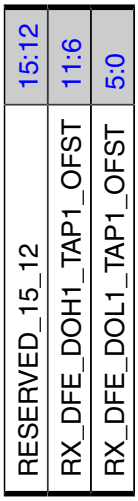


Table 2-446 Fields for Register: LANEN_DIG_ASIC_RX_EQ_ASIC_IN_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_DFE_DOH1_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_doh_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	RX_DFE_DOL1_TAP1_OFST	R	This field is the value from ASIC for rx_eq_dfe_dol_tap1_ofst. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.443 LANEN_DIG_ASIC_RX_OVRD_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from ASIC, Register 8
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)52
- **Exists:** Always

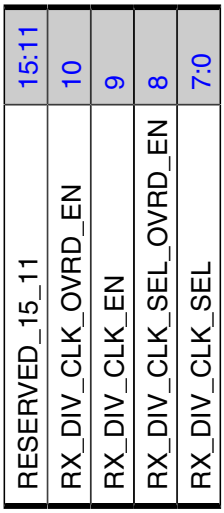


Table 2-447 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RX_DIV_CLK_OVRD_EN	R/W	This bit is the override enable for tx_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RX_DIV_CLK_EN	R/W	This bit is the override value for tx_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_DIV_CLK_SEL_OVRD_EN	R/W	This bit is the override enable for tx_div_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	RX_DIV_CLK_SEL	R/W	This field is the override value for tx_div_clk_sel. **Value After Reset:** 0x0

Table 2-447 Fields for Register: LANEN_DIG_ASIC_RX_OVRD_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0xff

2.1.444 LANEN_DIG_ASIC_RX_ASIC_IN_13 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 13.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)53
- **Exists:** Always

Table 2-448 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:6	TX_MISC	R	This field is the value from ASIC for rx_misc[7:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:0	RX_MISC	R	This field is the value from ASIC for rx_term_code[5:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.445 LANEN_DIG_ASIC_RX_ASIC_IN_14 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from ASIC, Register 14.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)54
- **Exists:** Always

Table 2-449 Fields for Register: LANEN_DIG_ASIC_RX_ASIC_IN_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_TERM_CODE	R	This field is the value from ASIC for rx_term_code[5:0]. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.446 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 (for N = 0; N <= NLANES-1)

- **Description:** TX Power State Control Register for P0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)55
- **Exists:** Always

15:12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_12	TX_P0_ANA_DLY_CAL_EN	TX_P0_ANA_DCC_EN	TX_P0_ROPLL_EN	TX_P0_ALLOW_RXDET	TX_P0_DATA_EN	TX_P0_DIG_CLK_EN	TX_P0_ANA_SERIAL_EN	TX_P0_ANA_RESET	TX_P0_ANA_WORD_CLK_EN	TX_P0_ANA_CLK_EN	TX_P0_ANA_VCM_HOLD	TX_P0_ANA_REFGEN_EN

Table 2-450 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_P0_ANA_DLY_CAL_EN	R/W	This bit is the value of TX ana DCC enable in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	TX_P0_ANA_DCC_EN	R/W	This bit is the value of TX ana DCC enable in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	TX_P0_ROPLL_EN	R/W	This bit is the value of TX POPLL enable in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	TX_P0_ALLOW_RXDET	R/W	If this bit is asserted, then rxdet request is allowed in P0. **Value After Reset:** 0x0

Table 2-450 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	TX_P0_DATA_EN	R/W	This bit is ANDed with top-level tx_data_en asic input. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	TX_P0_DIG_CLK_EN	R/W	This bit enables/disables TX digital clocks in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	TX_P0_ANA_SERIAL_EN	R/W	This bit is the value of TX ana serial_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	TX_P0_ANA_RESET	R/W	This bit is the value of TX ana reset in P0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_P0_ANA_WORD_CLK_EN	R/W	This bit is the value of TX ana word_clk_en in P0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_P0_ANA_CLK_EN	R/W	This bit is the value of TX ana clk_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	TX_P0_ANA_VCM_HOLD	R/W	This bit is the value of TX ana vcm_hold in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TX_P0_ANA_REFGEN_EN	R/W	This bit is the value of TX ana refgen_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.447 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S (for N = 0; N <= NLANES-1)

- **Description:** TX Power State Control Register for P0S.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)56
- **Exists:** Always

15:12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_12	TX_P0S_ANA_DLY_CAL_EN	TX_P0S_ANA_DCC_EN	TX_P0S_ROPLL_EN	TX_P0S_ALLOW_RXDET	TX_P0S_DATA_EN	TX_P0S_DIG_CLK_EN	TX_P0S_ANA_SERIAL_EN	TX_P0S_ANA_RESET	TX_P0S_ANA_WORD_CLK_EN	TX_P0S_ANA_CLK_EN	TX_P0S_ANA_VCM_HOLD	TX_P0S_ANA_REFGEN_EN

Table 2-451 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S (for N = 0; N <= NLANES-1)

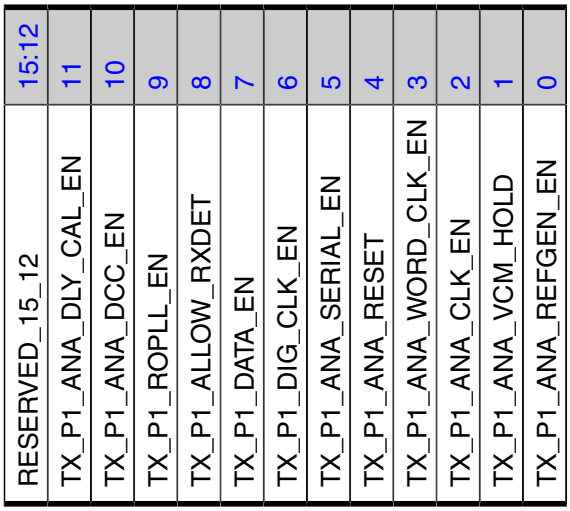
Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_P0S_ANA_DLY_CAL_EN	R/W	This bit is the value of TX ana DCC enable in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	TX_P0S_ANA_DCC_EN	R/W	This bit is the value of TX ana DCC enable in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	TX_P0S_ROPLL_EN	R/W	This bit is the value of TX POPLL enable in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	TX_P0S_ALLOW_RXDET	R/W	If this bit is asserted, then rxdet request is allowed in P0S. **Value After Reset:** 0x0

Table 2-451 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P0S (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	TX_P0S_DATA_EN	R/W	This bit is ANDed with top-level tx_data_en asic input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_P0S_DIG_CLK_EN	R/W	Tis bit enables/disables TX digital clocks in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	TX_P0S_ANA_SERIAL_EN	R/W	This bit is the value of TX ana serial_en in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_P0S_ANA_RESET	R/W	This bit is the value of TX ana reset in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_P0S_ANA_WORD_CLK_EN	R/W	This bit is the value of TX ana word_clk_en in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_P0S_ANA_CLK_EN	R/W	This bit is the value of TX ana clk_en in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	TX_P0S_ANA_VCM_HOLD	R/W	This bit is the value of TX ana vcm_hold in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TX_P0S_ANA_REFGEN_EN	R/W	This bit is the value of TX ana refgen_en in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.448 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 (for N = 0; N <= NLANES-1)

- **Description:** TX Power State Control Register for P1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)57
- **Exists:** Always



15:12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_12	TX_P1_ANA_DLY_CAL_EN	TX_P1_ANA_DCC_EN	TX_P1_ROPLL_EN	TX_P1_ALLOW_RXDET	TX_P1_DATA_EN	TX_P1_DIG_CLK_EN	TX_P1_ANA_SERIAL_EN	TX_P1_ANA_RESET	TX_P1_ANA_WORD_CLK_EN	TX_P1_ANA_CLK_EN	TX_P1_ANA_VCM_HOLD	TX_P1_ANA_REFGEN_EN

Table 2-452 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_P1_ANA_DLY_CAL_EN	R/W	This bit is the value of TX ana DCC enable in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	TX_P1_ANA_DCC_EN	R/W	This bit is the value of TX ana DCC enable in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_P1_ROPLL_EN	R/W	This bit is the value of TX POPLL enable in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	TX_P1_ALLOW_RXDET	R/W	If this bit is asserted, then rxdet request is allowed in P1. **Value After Reset:** 0x1

Table 2-452 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	TX_P1_DATA_EN	R/W	This bit is ANDed with top-level tx_data_en asic input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_P1_DIG_CLK_EN	R/W	This bit enables/disables TX digital clocks in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_P1_ANA_SERIAL_EN	R/W	This bit is the value of TX ana serial_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_P1_ANA_RESET	R/W	This bit is the value of TX ana reset in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	TX_P1_ANA_WORD_CLK_EN	R/W	This bit is the value of TX ana word_clk_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_P1_ANA_CLK_EN	R/W	This bit is the value of TX ana clk_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_P1_ANA_VCM_HOLD	R/W	This bit is the value of TX ana vcm_hold in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TX_P1_ANA_REFGEN_EN	R/W	This bit is the value of TX ana refgen_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.449 LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 (for N = 0; N <= NLANES-1)

- **Description:** TX Power State Control Register for P2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)58
- **Exists:** Always

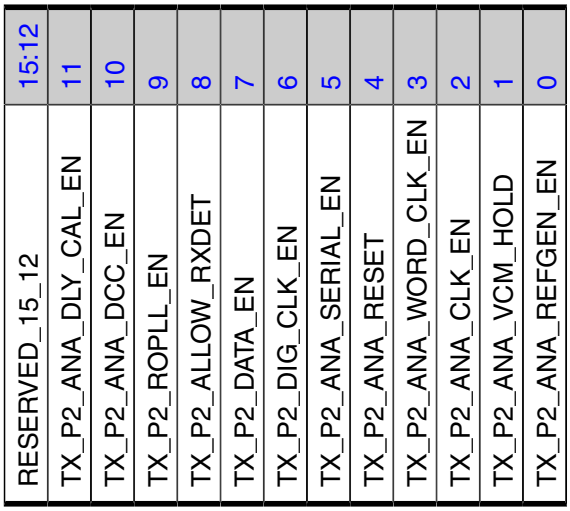


Table 2-453 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_P2_ANA_DLY_CAL_EN	R/W	This bit is the value of TX ana DCC enable in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	TX_P2_ANA_DCC_EN	R/W	This bit is the value of TX ana DCC enable in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_P2_ROPLL_EN	R/W	This bit is the value of TX POPLL enable in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	TX_P2_ALLOW_RXDET	R/W	If this bit is asserted, then rxdet request is allowed. **Value After Reset:** 0x1

Table 2-453 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PSTATE_P2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	TX_P2_DATA_EN	R/W	This bit is ANDed with top-level tx_data_en asic input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_P2_DIG_CLK_EN	R/W	This bit enables/disables TX digital clocks in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_P2_ANA_SERIAL_EN	R/W	This bit is the value of TX ana serial_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_P2_ANA_RESET	R/W	This bit is the value of TX ana reset in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	TX_P2_ANA_WORD_CLK_EN	R/W	This bit is the value of TX ana word_clk_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_P2_ANA_CLK_EN	R/W	This bit is the value of TX ana clk_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_P2_ANA_VCM_HOLD	R/W	This bit is the value of TX ana vcm_hold in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_P2_ANA_REFGEN_EN	R/W	This bit is the value of TX ana refgen_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.450 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Time Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)59
- **Exists:** Always

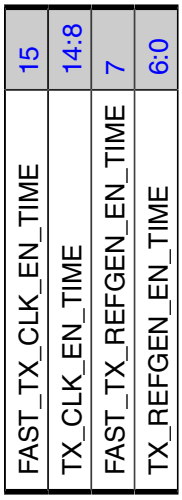


Table 2-454 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	FAST_TX_CLK_EN_TIME	R/W	This bit enables fast TX ana clock enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:8	TX_CLK_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for TX ana clock enable (spec:>=1us). **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0x7f
7	FAST_TX_REFGEN_EN_TIME	R/W	This bit enables fast TX ana refgen enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	TX_REFGEN_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for TX ana refgen enable (spec:>=1us). **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0x7f

2.1.451 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Time Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5a
- **Exists:** Always

Table 2-455 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SKIP_TX_VCM_HOLD_WAIT	R/W	This bit skips wait for TX common mode hold power up. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14:0	TX_VCM_HOLD_TIME	R/W	This field is the power up time (in ref_range cycles) for TX ana vreg enable (spec:>=802us). This value is multiplied by 16 to get the actual tx_vcm_hold_time. **Value After Reset:** 0x518 **Exists:** Always **Reset Mask:** 0x7fff

2.1.452 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Time Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5b
- **Exists:** Always

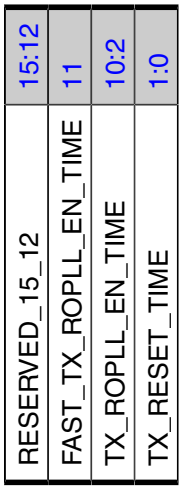


Table 2-456 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	FAST_TX_ROPLL_EN_TIME	R/W	This bit enables fast TX ropll wait time. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:2	TX_ROPLL_EN_TIME	R/W	This field is the threshold for the ROPLL WAIT time. The default timr is 2us. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0x1ff
1:0	TX_RESET_TIME	R/W	This field is TX Reset de-assertion time (in ref_range cycles) (spec: >=50ns). **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.453 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Time Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5c
- **Exists:** Always

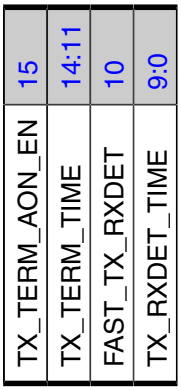


Table 2-457 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_TERM_AON_EN	R/W	This bit enables the old behavior of TX term code. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14:11	TX_TERM_TIME	R/W	This field is TX termination up or dn time (in ref_range cycles), (spec: 160ns < time). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
10	FAST_TX_RXDET	R/W	This bit enables fast TX RX-detection (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_RXDET_TIME	R/W	This field is RX detect up time (in ref_range cycles), starting from asserting rxdet_en (spec: 11us < time < 22us). **Value After Reset:** 0x125 **Exists:** Always **Reset Mask:** 0x3ff

2.1.454 LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Time Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5d
- **Exists:** Always

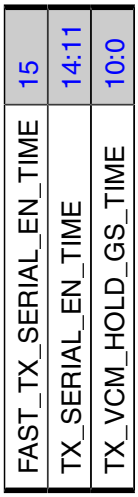


Table 2-458 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_PWRUP_TIME_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	FAST_TX_SERIAL_EN_TIME	R/W	This bit enables fast TX ana serial enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:11	TX_SERIAL_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for TX ana serial enable (spec: 130ns < time). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
10:0	TX_VCM_HOLD_GS_TIME	R/W	This field is TX common mode gear-shift time (in ref range cycles) (spec: >=202us). This value is multiplied by 16 to get the actual gear-shift time. **Value After Reset:** 0x149 **Exists:** Always **Reset Mask:** 0x7ff

2.1.455 LANEN_DIG_TX_PWRCTL_TX_CTL (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Control Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5e
- **Exists:** Always

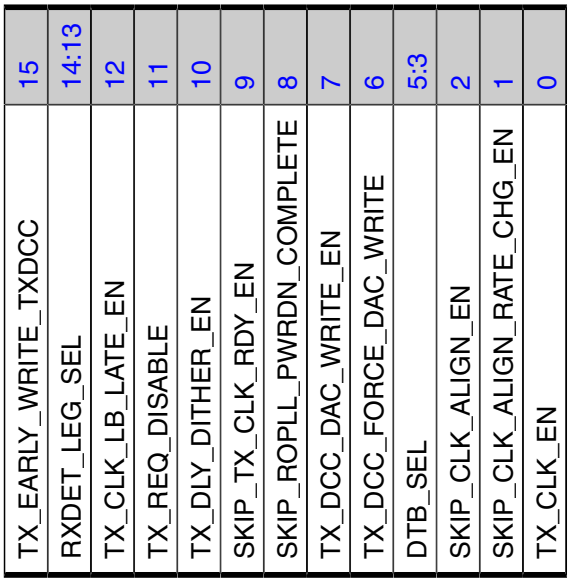


Table 2-459 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_EARLY_WRITE_TXDCC	R/W	This bit enables writing TXDCC DACs before ropll_state is ON. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14:13	RXDET_LEG_SEL	R/W	This field selects which RX leg is used for rxdet_result The possible options are as follows: ■ 0 - use P & N legs for rxdet_result ■ 1 - use P | N legs for rxdet_result ■ 2 - use P leg only for rxdet_result ■ 3 - use N leg only for rxdet_result **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
12	TX_CLK_LB_LATE_EN	R/W	This bit waits till tx_dig_clk_en is asserted to start rx2tx loopback. **Value After Reset:** 0x0

Table 2-459 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11	TX_REQ_DISABLE	R/W	This bit ignores tx_req during data transmission. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	TX_DLY_DITHER_EN	R/W	This bit enables dithering the final code of TX delay calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	SKIP_TX_CLK_RDY_EN	R/W	This bit skips clock ready signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	SKIP_ROPLL_PWRDN_COMPLETE	R/W	This bit skips ROPLL power-down complete skip_ropll_pwrdn_complete_r. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_DCC_DAC_WRITE_EN	R/W	This bit enables the TX DCC control block to write the TX DCC DAC values. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	TX_DCC_FORCE_DAC_WRITE	R/W	This bit forces a TX DCC DAC write. This will force a TX DCC DAC write even when there is no rate change. Note that this bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:3	DTB_SEL	R/W	This field selects data to drive on DTB. ■ 0 - disabled ■ 1 - tx_ack and tx_pwrsm_state[0] ■ 2 - tx_ana_rxdetp_result_i, tx_ana_rxdetm_result_i ■ 3 - tx_ana_reset_i, tx_ana_clk_en_i ■ 4 - analog/asic clocks ■ 5 - asic early signal / clock aligner shift

Table 2-459 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			■ 6 - tx_clk_state counter / lbert strobe ■ 7 - ref_dig_rst/tx_dig_rst **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
2	SKIP_CLK_ALIGN_EN	R/W	This bit skips clock alignment stage. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_CLK_ALIGN_RATE_CHG_EN	R/W	This bit skips clock alignment stage when a rate change is detected. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CLK_EN	R/W	This bit enables TX clocks. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.456 LANEN_DIG_TX_PWRCTL_TX_CTL_2 (for N = 0; N <= NLANES-1)

- **Description:** TX Power UP Control Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)5f
- **Exists:** Always

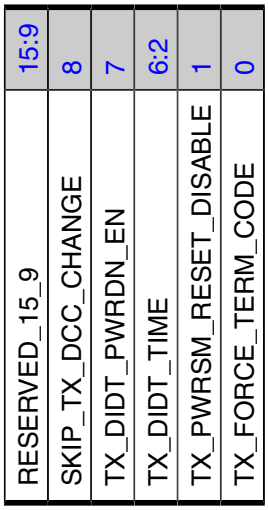


Table 2-460 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_CTL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	SKIP_TX_DCC_CHANGE	R/W	This bit enables skipping rate change when TXDCC changes along with Pstate change. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_DIDT_PWRDN_EN	R/W	This bit enables di/dt protection during power-down. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6:2	TX_DIDT_TIME	R/W	This field is TX di/dt protection time (in ref_range cycles) (spec: >=100ns). **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f

Table 2-460 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_CTL_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	TX_PWRSM_RESET_DISABLE	R/W	This bit disable the first loop when rx_reset is applied. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TX_FORCE_TERM_CODE	R/W	This bit forces a TX TERM CODE latch. This will force an TX TERM CODE write even when there is no rate change. Note that this bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.457 LANEN_DIG_TX_PWRCTL_TX_STATUS (for N = 0; N <= NLANES-1)

- **Description:** TX Power Up Control FSM StatusLane Input Status Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)60
- **Exists:** Always

Table 2-461 Fields for Register: LANEN_DIG_TX_PWRCTL_TX_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:1	TX_FSM_STATE	R	This field is the current value of the TX PWR FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TX_RATE_IRQ	R	This bit is the value of req_tx_rate_change_s01 register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.458 LANEN_DIG_TX_ROPLL_CTL_ROPLL_OVRD (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Override Controls.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)61
- **Exists:** Always

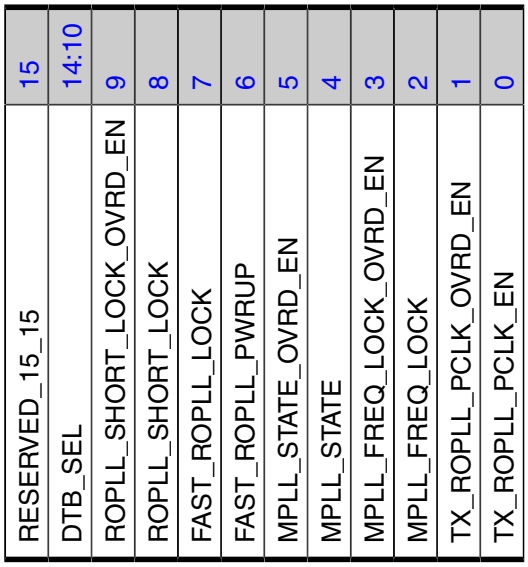


Table 2-462 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:10	DTB_SEL	R/W	This field is the DTB select for ROPLL dtb signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
9	ROPLL_SHORT_LOCK_OVRD_EN	R/W	This bit is the override enable for short ROPLL locking. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	ROPLL_SHORT_LOCK	R/W	This bit is the override value for short ROPLL locking. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-462 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7	FAST_ROPLL_LOCK	R/W	This bit enables fast ROPLL locking. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	FAST_ROPLL_PWRUP	R/W	This bit enables fast ROPLL powerup. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	MPLL_STATE_OVRD_EN	R/W	This bit is the override enable for mpll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	MPLL_STATE	R/W	This bit is the override value for mpll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	MPLL_FREQ_LOCK_OVRD_EN	R/W	This bit is the override enable for mpll_freq_lock. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	MPLL_FREQ_LOCK	R/W	This bit is the override value for mpll_freq_lock. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_ROPLL_PCLK_OVRD_EN	R/W	This bit is the override enable for tx_ropll_pclk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ROPLL_PCLK_EN	R/W	This bit is the override value for tx_ropll_pclk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.459 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_0 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)62
- **Exists:** Always

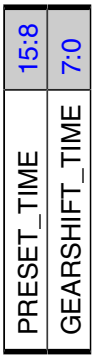


Table 2-463 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	PRESET_TIME	R/W	This field is the threshold for the ROPLL preset time (2us). It can be up to 4us. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xff
7:0	GEARSHIFT_TIME	R/W	This field is the threshold for the ROPLL gearshift time (2us). It can be up to 6us. **Value After Reset:** 0x34 **Exists:** Always **Reset Mask:** 0xff

2.1.460 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_1 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)63
- **Exists:** Always

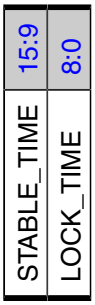


Table 2-464 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	STABLE_TIME	R/W	This field is the threshold for the ROPLL stability time. The default time is 2us. **Value After Reset:** 0x34 **Exists:** Always **Reset Mask:** 0x7f
8:0	LOCK_TIME	R/W	This field is the threshold for the ROPLL lock time. The default time is 6us. **Value After Reset:** 0x9c **Exists:** Always **Reset Mask:** 0x1ff

2.1.461 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_2 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)64
- **Exists:** Always

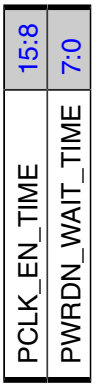


Table 2-465 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	PCLK_EN_TIME	R/W	This field is the threshold for the ROPLL PCLK. The default time is 1us. **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0xff
7:0	PWRDN_WAIT_TIME	R/W	This field is the threshold for the ROPLL power-down wait time to discharge the PLL capacitors. Minimum is set to 1us. **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0xff

2.1.462 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_3 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)65
- **Exists:** Always

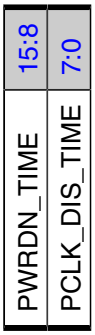


Table 2-466 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	PWRDN_TIME	R/W	This field is the threshold for the ROPLL power-down (1us). **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0xff
7:0	PCLK_DIS_TIME	R/W	This field is the threshold for the PCLK disable time (1us). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.463 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_4 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)66
- **Exists:** Always

Table 2-467 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	GEARSHIFT_SHORT_TIME	R/W	This field is the threshold for the ROPLL gearshift time (400ns). It can be up to 6us. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0xff
7:0	FDIV_RST_TIME	R/W	This field is the threshold for the FDIV reset time (1us). **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0xff

2.1.464 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_5 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)67
- **Exists:** Always

Table 2-468 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	TX_VREG_FAST_START_TIME	R/W	This field is the power up time (in ref_range cycles) for tx_ana_vreg_fast_start (spec: >=400ns). **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x7f
8:0	LOCK_SHORT_TIME	R/W	This field is the threshold for the ROPLL lock time. The default time is 1.2us. **Value After Reset:** 0x1f **Exists:** Always **Reset Mask:** 0x1ff

2.1.465 LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_6 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL Timers Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)68
- **Exists:** Always

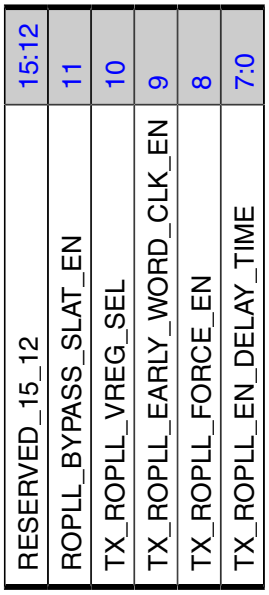


Table 2-469 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	ROPLL_BYPASS_SLAT_EN	R/W	This bit enables fast ROPLL locking in BYPASS MODE by skipping some states. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	TX_ROPLL_VREG_SEL	R/W	This bit disable ANDing the VREG signals from the repeaters and ROPLL FSM.. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_ROPLL_EARLY_WORD_CLK_EN	R/W	This bit turns ON word_clk during STABLE state, else turn it ON at the end of STABLE state. **Value After Reset:** 0x0

Table 2-469 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_ROPLL_TIMERS_6 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8	TX_ROPLL_FORCE_EN	R/W	This bit forces ROPLL enable signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	TX_ROPLL_EN_DELAY_TIME	R/W	This bit delays the ROPLL enable signal to insure all configuration signals are stable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.466 LANEN_DIG_TX_ROPLL_CTL_STAT (for N = 0; N <= NLANES-1)

- **Description:** ROPLL status register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)69
- **Exists:** Always

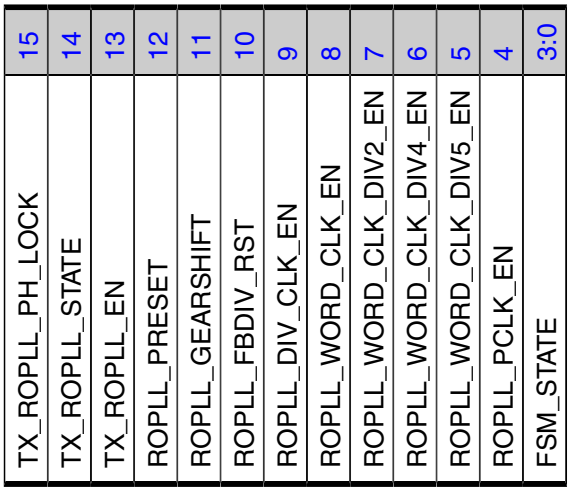


Table 2-470 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_STAT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_ROPLL_PH_LOCK	R	This bit is the current value of tx_ropll_phase_locked_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_ROPLL_STATE	R	This bit is the current value of tx_ropll_state_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	TX_ROPLL_EN	R	This bit is the current value of tx_ana_ropll_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ROPLL_PRESET	R	This bit is the current value of tx_ana_ropll_preset_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-470 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_STAT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	ROPLL_GEARSHIFT	R	This bit is the current value of tx_ana_ropll_gearshift_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	ROPLL_FBDIV_RST	R	This bit is the current value of tx_ana_ropll_fbdiv_rst_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	ROPLL_DIV_CLK_EN	R	This bit is the current value of tx_ana_ropll_div_clk_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	ROPLL_WORD_CLK_EN	R	This bit is the current value of tx_ana_ropll_word_clk_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ROPLL_WORD_CLK_DIV2_EN	R	This bit is the current value of tx_ana_ropll_word_clk_div2_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	ROPLL_WORD_CLK_DIV4_EN	R	This bit is the current value of tx_ana_ropll_word_clk_div4_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	ROPLL_WORD_CLK_DIV5_EN	R	This bit is the current value of tx_ana_ropll_word_clk_div5_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-470 Fields for Register: LANEN_DIG_TX_ROPLL_CTL_STAT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	ROPLL_PCLK_EN	R	This bit is the current value of tx_ropll_pclk_en. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	FSM_STATE	R	This field is the current value of the PWR FSM state register. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.467 LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_FR (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Differential IDAC OFFSET for Full Rate.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6a
- **Exists:** Always

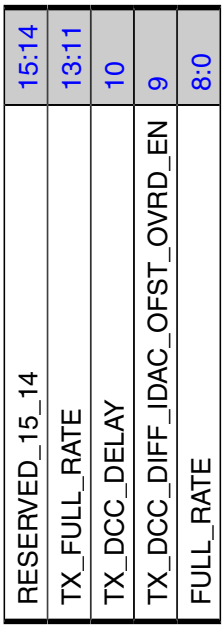


Table 2-471 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_FR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:11	TX_FULL_RATE	R/W	This field is TX full rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
10	TX_DCC_DELAY	R/W	This bit enables delaying tx_dcc_update_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_DCC_DIFF_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for TX DCC DIFF offset. **Value After Reset:** 0x0

Table 2-471 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_FR (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8:0	FULL_RATE	R/W	This field is TX differential DCC IDAC offset for full rate. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.468 LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_HR (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Differential IDAC OFFSET for Half Rate.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6b
- **Exists:** Always

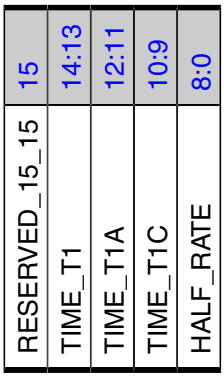


Table 2-472 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_HR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:13	TIME_T1	R/W	This field is for DAC pre-write time that should be at least one ref_range_clk cycle. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
12:11	TIME_T1A	R/W	This field is for DAC write en pulse (one ref_range_clk cycle). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
10:9	TIME_T1C	R/W	This field is for DAC post-write time that should be at least one ref_range_clk cycle. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
8:0	HALF_RATE	R/W	This field is TX differential DCC IDAC offset for half rate. **Value After Reset:** 0x100

Table 2-472 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_DIFF_IDAC_OFST_HR (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1ff

2.1.469 LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_FR (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Common-Mode IDAC OFFSET for Full Rate.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6c
- **Exists:** Always

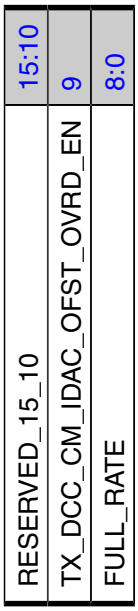


Table 2-473 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_FR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TX_DCC_CM_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for TX DCC DIFF offset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:0	FULL_RATE	R/W	This field is TX common-mode DCC IDAC offset for full rate. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.470 LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_HR (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Common-Mode IDAC OFFSET for Half Rate.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6d
- **Exists:** Always

Table 2-474 Fields for Register: LANEN_DIG_TX_DCC_CTL_TX_DCC_CM_IDAC_OFST_HR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	HALF_RATE	R/W	This field is TX common-mode DCC IDAC offset for half rate. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.471 LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0 (for N = 0; N <= NLANES-1)

- **Description:** TX Clock Alignment Control Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6e
- **Exists:** Always

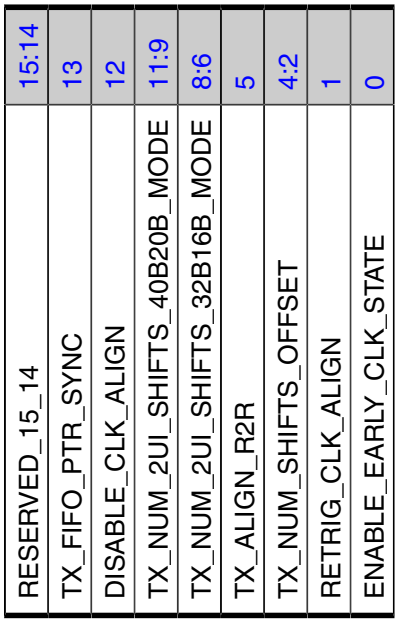


Table 2-475 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	TX_FIFO_PTR_SYNC	R/W	This bit skips sync TX FIFO pointers . **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	DISABLE_CLK_ALIGN	R/W	This bit disable running clock alignment in short latency mode.. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-475 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_TX_CTL_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11:9	TX_NUM_2UI_SHIFTS_40B20B_MODE	R/W	This field is the number of shifts in 40B/20B mode. This is the value for 20B shifts or 40B shifts when txX_align_wide_xfer_en is asserted. This value is incremented first by 3 and then shifted up by 1 (x2) for 40B mode. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7
8:6	TX_NUM_2UI_SHIFTS_32B16B_MODE	R/W	This field is the number of shifts in 32B/16B mode. This is the value for 16B shifts or 32B shifts when txX_align_wide_xfer_en is asserted. This value is incremented first by 3 and then shifted up by 1 (x2) for 32B mode. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x7
5	TX_ALIGN_R2R	R/W	This bit enables running rise to rise clock alignment. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4:2	TX_NUM_SHIFTS_OFFSET	R/W	This field is the offset to be added to the number of shifts. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
1	RETRIG_CLK_ALIGN	R/W	This is a self-clearing bit that retriggers the clock align FSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ENABLE_EARLY_CLK_STATE	R/W	This bit asserts the clock state before startup counter expires. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.472 LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1 (for N = 0; N <= NLANES-1)

- **Description:** TX Clock Alignment Control Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)6f
- **Exists:** Always

Table 2-476 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_TX_CTL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:0	STARTUP_DELAY	R/W	This field is the startup delay value. **Value After Reset:** 0x450 **Exists:** Always **Reset Mask:** 0x7ff

2.1.473 LANEN_DIG_TX_CLK_ALIGN_CLK_ALIGN_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Tx Clock Align FSM Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)70
- **Exists:** Always

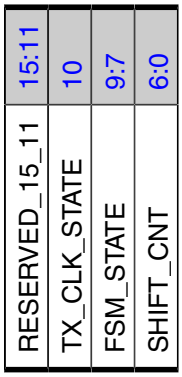


Table 2-477 Fields for Register: LANEN_DIG_TX_CLK_ALIGN_CLK_ALIGN_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_CLK_STATE	R	This bit asserts when FSM finishes the clock alignment. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:7	FSM_STATE	R	This field states the current value of the TX clk align FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	SHIFT_CNT	R	This field states the total number of clock shifts. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.474 LANEN_DIG_TX_LBERT_CTL (for N = 0; N <= NLANES-1)

- **Description:** Pattern Generator controls
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)71
- **Exists:** Always

Table 2-478 Fields for Register: LANEN_DIG_TX_LBERT_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:5	PAT0	R/W	Pattern for modes 9-11 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
4	TRIGGER_ERR	R/W	Insert a single error into a lsb. Any write of a 1 to this bit will insert an error **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	MODE	R/W	Pattern to generate. When changing modes, you must change to disabled first. ■ 0 - Disabled ■ 1 - lfsr31: X^31 + X^28 + 1 ■ 2 - lfsr23: X^23 + X^18 + 1 ■ 3 - lfsr23: x^23 + x^21 + x^16 + x^8 + x^5 + x^2 + 1 ■ 4 - lfsr16: x^16 + x^5 + x^4 + x^3 + 1 ■ 5 - lfsr15: X^15 + X^14 + 1 ■ 6 - lfsr11: X^11 + X^9 + 1 ■ 7 - lfsr9: X^9 + X^5 + 1 ■ 8 - lfsr7: X^7 + X^6 + 1 ■ 9 - Fixed word (PAT0)

Table 2-478 Fields for Register: LANEN_DIG_TX_LBERT_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			■ 10 - DC balanced word (PAT0, ~PAT0) ■ 11 - Fixed pattern: (000, PAT0, 3ff, ~PAT0) ■ 12 - Extended pattern (64 bit pattern) ■ 13 -lfsr13: x^13 + x^12 + x^11 + x^8 + 1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.475 LANEN_DIG_TX_LBERT_PAT1_0 (for N = 0; N <= NLANES-1)

- **Description:** Extended Pattern register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)72
- **Exists:** Always

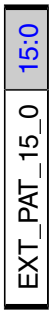


Table 2-479 Fields for Register: LANEN_DIG_TX_LBERT_PAT1_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	EXT_PAT_15_0	R/W	Extended Pattern for mode 12 **Value After Reset:** 0x3333 **Exists:** Always **Reset Mask:** 0xffff

2.1.476 LANEN_DIG_TX_LBERT_PAT1_1 (for N = 0; N <= NLANES-1)

- **Description:** Extended Pattern register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)73
- **Exists:** Always

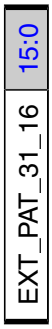


Table 2-480 Fields for Register: LANEN_DIG_TX_LBERT_PAT1_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	EXT_PAT_31_16	R/W	Extended Pattern for mode 12 **Value After Reset:** 0x5555 **Exists:** Always **Reset Mask:** 0xffff

2.1.477 LANEN_DIG_TX_LBERT_PAT1_2 (for N = 0; N <= NLANES-1)

- **Description:** Extended Pattern register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)74
- **Exists:** Always

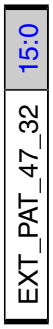


Table 2-481 Fields for Register: LANEN_DIG_TX_LBERT_PAT1_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	EXT_PAT_47_32	R/W	Extended Pattern for mode 12 **Value After Reset:** 0x7777 **Exists:** Always **Reset Mask:** 0xffff

2.1.478 LANEN_DIG_TX_LBERT_PAT1_3 (for N = 0; N <= NLANES-1)

- **Description:** Extended Pattern register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)75
- **Exists:** Always

Table 2-482 Fields for Register: LANEN_DIG_TX_LBERT_PAT1_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	EXT_PAT_63_48	R/W	Extended Pattern for mode 12 **Value After Reset:** 0xeeee **Exists:** Always **Reset Mask:** 0xffff

2.1.479 LANEN_DIG_TX_FIFO_CTL (for N = 0; N <= NLANES-1)

- **Description:** TX Digital Control Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)76
- **Exists:** Always

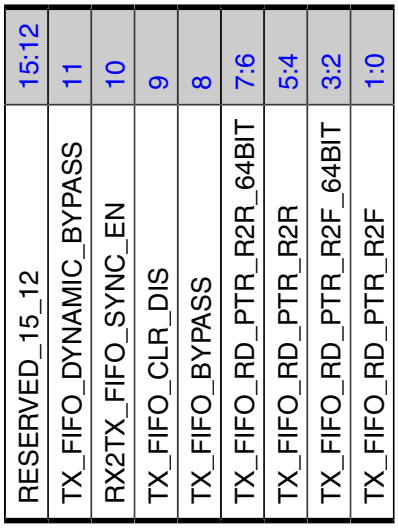


Table 2-483 Fields for Register: LANEN_DIG_TX_FIFO_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_FIFO_DYNAMIC_BYPASS	R/W	This bit enables bypassing the TX FIFO when TX clock align in in progress. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX2TX_FIFO_SYNC_EN	R/W	This bit enables using TX clock align to sync the FIFO pointers in RX2TX loopback. when set to 0, we use txX_clk_rdy input to sync the pointers **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	TX_FIFO_CLR_DIS	R/W	This bit disables clearing TX datapath FIFO. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-483 Fields for Register: LANEN_DIG_TX_FIFO_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8	TX_FIFO_BYPASS	R/W	This bit is By-pass TX datapath FIFO. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:6	TX_FIFO_RD_PTR_R2R_64BIT	R/W	These bits are TX FIFO read pointer starting location in 64 bit mode rise to rise. This should only change when the TX is out of P0. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3
5:4	TX_FIFO_RD_PTR_R2R	R/W	These bits are TX FIFO read pointer starting location in other modes rise to rise. This should only change when the TX is out of P0. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3
3:2	TX_FIFO_RD_PTR_R2F_64BIT	R/W	These bits are TX FIFO read pointer starting location in 64 bit mode rise to fall. This should only change when the TX is out of P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
1:0	TX_FIFO_RD_PTR_R2F	R/W	These bits are TX FIFO read pointer starting location in other modes rise to fall. This should only change when the TX is out of P0. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3

2.1.480 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_0 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)77
- **Exists:** Always

Table 2-484 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:3	REF_DIV_CNT	R/W	This field is ref range clock count (e.g. 5'd3 = 4 ref_range cycles). **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
2:0	PRECHRGE_CNT	R/W	This field is precharge count (e.g. 5'd0 = 2 ref_range cycles). **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x7

2.1.481 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_1 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)78
- **Exists:** Always

Table 2-485 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	DLY_LD_VAL	R/W	This field is the number of samples taken after a code update. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
12:10	DLY_N_ITER	R/W	This field is the number of iterations to perform the algorithm. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
9:0	DLY_CAL_STEP	R/W	This field is DLY step size. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x3ff

2.1.482 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_2 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)79
- **Exists:** Always

Table 2-486 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DLY_DITH_ENABLE	R/W	This bit enables dithering for the final code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	DLY_STAT_MODE	R/W	This bit uses statistical approach to decide on the cal result. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	DLY_SMPL_THRESH	R/W	This field is the threshold used to define the XOR output as 1 or 0-based. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
9:0	DLY_OFFSET_CODE	R/W	This field is DLY offset code. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x3ff

2.1.483 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_3 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)7a
- **Exists:** Always

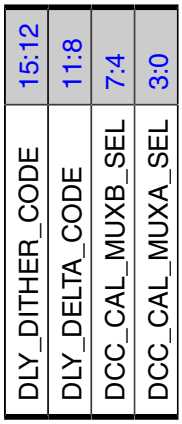


Table 2-487 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	DLY_DITHER_CODE	R/W	This field dithers the final code by the amount specified in this register. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
11:8	DLY_DELTA_CODE	R/W	This field increments/decrements the code by delta when sampling the ideal point. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
7:4	DCC_CAL_MUXB_SEL	R/W	This field is vref muxb select. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
3:0	DCC_CAL_MUXA_SEL	R/W	This field is vref muxa select. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf

2.1.484 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_4 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)7b
- **Exists:** Always

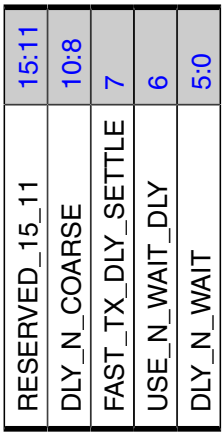


Table 2-488 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:8	DLY_N_COARSE	R/W	This field is the number of coarse stages used. **Value After Reset:** 0x5 **Exists:** Always **Reset Mask:** 0x7
7	FAST_TX_DLY_SETTLE	R/W	This bit enables fast DAC settling time (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	USE_N_WAIT_DLY	R/W	when high DLY_N_WAIT is used, else DLY_CNFG_0.REF_DIV_CNT is used. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:0	DLY_N_WAIT	R/W	This field is the number of wait cycles following an update. **Value After Reset:** 0x7

Table 2-488 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3f

2.1.485 LANEN_DIG_TX_DLY_CAL_DLY_CNFG_5 (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Configuration Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)7c
- **Exists:** Always

Table 2-489 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DLY_CAL_GAIN_OVRD_EN	R/W	This bit is the override enable for dly_cal_gain. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:13	DLY_CAL_GAIN	R/W	This field is the value for dly_cal_gain. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3
12	DLY_CAL_FLIP_OVRD_EN	R/W	This bit is the override enable for dly_cal_flip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	DLY_CAL_FLIP	R/W	This bit is the override value for dly_cal_flip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	DLY_INIT_CODE_OVRD_EN	R/W	This bit is the override enable for dly_init_code_r. **Value After Reset:** 0x0

Table 2-489 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_CNFG_5 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9:0	DLY_INIT_CODE	R/W	This field is initial DLY calibration code. **Value After Reset:** 0x160 **Exists:** Always **Reset Mask:** 0x3ff

2.1.486 LANEN_DIG_TX_DLY_CAL_DLY_STAT_0 (for N = 0; N <= NLANES-1)

- **Description:** Pre-Driver DLY Status Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)7d
- **Exists:** Always

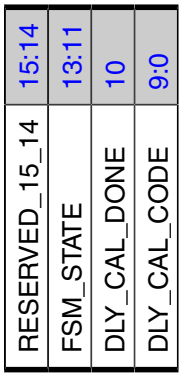


Table 2-490 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_STAT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:11	FSM_STATE	R	This field is the current value of the DLY FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	DLY_CAL_DONE	R	This bit asserts when delay calibration is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:0	DLY_CAL_CODE	R	This field is the final code of the DLY calibration. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.487 LANEN_DIG_TX_DLY_CAL_DLY_STAT_1 (for N = 0; N <= NLANES-1)

- **Description:** Pre-driver DLY Status Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)7e
- **Exists:** Always

Table 2-491 Fields for Register: LANEN_DIG_TX_DLY_CAL_DLY_STAT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	SMPL_CAL_RESULT	R	This field compiles the last 8 samples of tx_cal_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	DLY_CAL_TREE	R	This field is the parcour taken by FSM from initial code to final code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.488 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 (for N = 0; N <= NLANES-1)

- **Description:** RX Power State Control Register for P0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)80
- **Exists:** Always

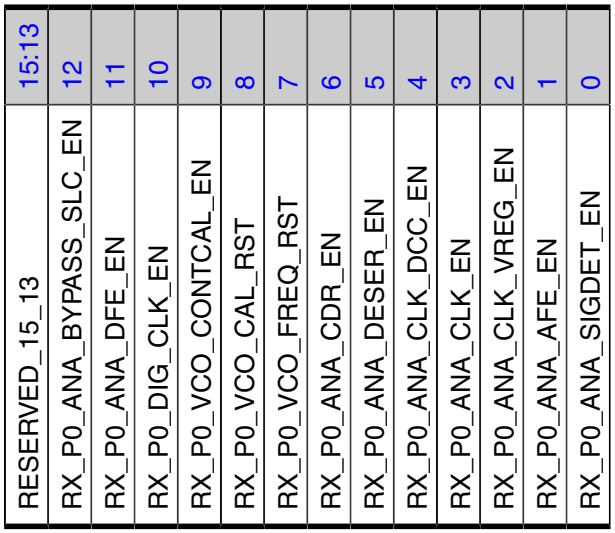


Table 2-492 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_P0_ANA_BYPASS_SLC_EN	R/W	This bit enables/disables RX bypass slicer in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
11	RX_P0_ANA_DFE_EN	R/W	This bit enables/disables RX DFE in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	RX_P0_DIG_CLK_EN	R/W	This bit enables/disables RX digital clocks in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-492 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 (for N = 0; N <= NLANES-1) (continued)

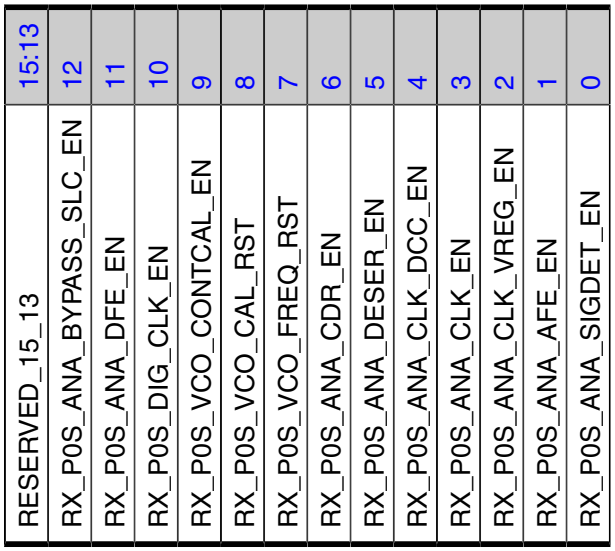
Bits	Name	Memory Access	Description
9	RX_P0_VCO_CONTCAL_EN	R/W	This bit enables/disables continuous calibration of the RX VCO in P0. If RX_P0_DIG_CLK_EN and the top-level rx_data_en are both asserted, then continuous calibration is turned off and this value is ignored. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	RX_P0_VCO_CAL_RST	R/W	This bit enables/disables resetting the RX VCO in P0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	RX_P0_VCO_FREQ_RST	R/W	This bit enables/disables resetting the RX VCO frequency in P0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_P0_ANA_CDR_EN	R/W	This bit is the value of RX ana cdr_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	RX_P0_ANA_DESER_EN	R/W	This bit is the value of RX ana deserial_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	RX_P0_ANA_CLK_DCC_EN	R/W	This bit is the value of RX ana CLK_DCC_EN in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	RX_P0_ANA_CLK_EN	R/W	This bit is the value of RX ana clk_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RX_P0_ANA_CLK_VREG_EN	R/W	This bit is the value of RX ana clk_vreg_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-492 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RX_P0_ANA_AFE_EN	R/W	This bit is the value of RX ana afe_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_P0_ANA_SIGDET_EN	R/W	This bit is the value of RX ana sigdet_en in P0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.489 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S (for N = 0; N <= NLANES-1)

- **Description:** RX Power State Control Register for P0S.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)81
- **Exists:** Always



15:13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_13	RX_P0S_ANA_BYPASS_SLC_EN	RX_P0S_ANA_DFE_EN	RX_P0S_DIG_CLK_EN	RX_P0S_VCO_CONTCAL_EN	RX_P0S_VCO_CAL_RST	RX_P0S_VCO_FREQ_RST	RX_P0S_ANA_CDR_EN	RX_P0S_ANA_DESER_EN	RX_P0S_ANA_CLK_DCC_EN	RX_P0S_ANA_CLK_EN	RX_P0S_ANA_CLK_VREG_EN	RX_P0S_ANA_AFE_EN	RX_P0S_ANA_SIGDET_EN

Table 2-493 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_P0S_ANA_BYPASS_SLC_EN	R/W	This bit enables/disables RX bypass slicer in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
11	RX_P0S_ANA_DFE_EN	R/W	This bit enables/disables RX DFE in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	RX_P0S_DIG_CLK_EN	R/W	This bit enables/disables RX digital clocks in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-493 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	RX_P0S_VCO_CONTCAL_EN	R/W	This bit enables/disables continuous calibration of the RX VCO in P0S. If RX_P0S_DIG_CLK_EN and the top-level rx_data_en are both asserted, then continuous calibration is turned off and this value is ignored. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	RX_P0S_VCO_CAL_RST	R/W	This bit enables/disables resetting the RX VCO in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	RX_P0S_VCO_FREQ_RST	R/W	This bit enables/disables resetting the RX VCO frequency in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_P0S_ANA_CDR_EN	R/W	This bit is the value of RX ana cdr_en in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_P0S_ANA_DESER_EN	R/W	This bit is the value of RX ana deserial_en in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_P0S_ANA_CLK_DCC_EN	R/W	This bit is the value of RX ana CLK_DCC_EN in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	RX_P0S_ANA_CLK_EN	R/W	This bit is the value of RX ana clk_en in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RX_P0S_ANA_CLK_VREG_EN	R/W	This bit is the value of RX ana clk_vreg_en in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-493 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P0S (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RX_P0S_ANA_AFE_EN	R/W	This bit is the value of RX ana afe_en in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_P0S_ANA_SIGDET_EN	R/W	This bit is the value of RX ana sigdet_en in P0S. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.490 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 (for N = 0; N <= NLANES-1)

- **Description:** RX Power State Control Register for P1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)82
- **Exists:** Always

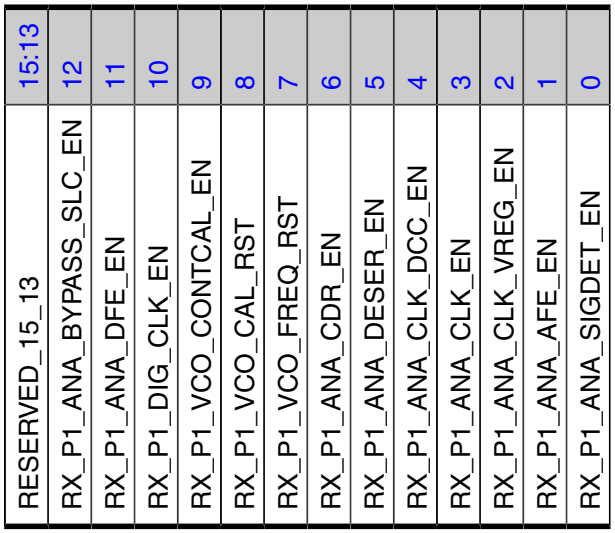


Table 2-494 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_P1_ANA_BYPASS_SLC_EN	R/W	This bit enables/disables RX bypass slicer in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	RX_P1_ANA_DFE_EN	R/W	This bit enables/disables RX DFE in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX_P1_DIG_CLK_EN	R/W	This bit enables/disables RX digital clocks in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-494 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	RX_P1_VCO_CONTCAL_EN	R/W	This bit enables/disables continuous calibration of the RX VCO in P1. If RX_P1_DIG_CLK_EN and the top-level rx_data_en are both asserted, then continuous calibration is turned off and this value is ignored. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_P1_VCO_CAL_RST	R/W	This bit enables/disables resetting the RX VCO in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	RX_P1_VCO_FREQ_RST	R/W	This bit enables/disables resetting the RX VCO frequency in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	RX_P1_ANA_CDR_EN	R/W	This bit is the value of RX ana cdr_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_P1_ANA_DESER_EN	R/W	This bit is the value of RX ana deserial_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_P1_ANA_CLK_DCC_EN	R/W	This bit is the value of RX ana CLK_DCC_EN in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_P1_ANA_CLK_EN	R/W	This bit is the value of RX ana clk_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_P1_ANA_CLK_VREG_EN	R/W	This bit is the value of RX ana clk_vreg_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-494 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RX_P1_ANA_AFE_EN	R/W	This bit is the value of RX ana afe_en in P1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_P1_ANA_SIGDET_EN	R/W	This bit is the value of RX ana sigdet_en in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.491 LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 (for N = 0; N <= NLANES-1)

- **Description:** RX Power State Control Register for P2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)83
- **Exists:** Always

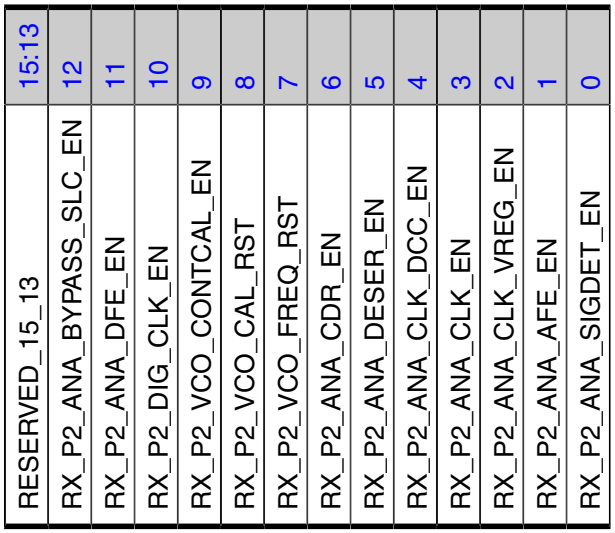


Table 2-495 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_P2_ANA_BYPASS_SLC_EN	R/W	This bit enables/disables RX bypass slicer in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	RX_P2_ANA_DFE_EN	R/W	This bit enables/disables RX DFE in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX_P2_DIG_CLK_EN	R/W	This bit enables/disables RX digital clocks in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-495 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	RX_P2_VCO_CONTCAL_EN	R/W	This bit enables/disables continuous calibration of the RX VCO in P2. If RX_P2_DIG_CLK_EN and the top-level rx_data_en are both asserted, then continuous calibration is turned off and this value is ignored. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_P2_VCO_CAL_RST	R/W	This bit enables/disables resetting the RX VCO in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	RX_P2_VCO_FREQ_RST	R/W	This bit enables/disables resetting the RX VCO frequency in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	RX_P2_ANA_CDR_EN	R/W	This bit is the value of RX ana cdr_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_P2_ANA_DESER_EN	R/W	This bit is the value of RX ana deserial_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_P2_ANA_CLK_DCC_EN	R/W	This bit is the value of RX ana CLK_DCC_EN in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_P2_ANA_CLK_EN	R/W	This bit is the value of RX ana clk_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_P2_ANA_CLK_VREG_EN	R/W	This bit is the value of RX ana clk_vreg_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-495 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PSTATE_P2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RX_P2_ANA_AFE_EN	R/W	This bit is the value of RX ana afe_en in P2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_P2_ANA_SIGDET_EN	R/W	This bit is the value of RX ana sigdet_en in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.492 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_0 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Time Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)84
- **Exists:** Always

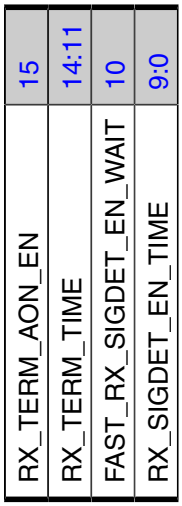


Table 2-496 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_TERM_AON_EN	R/W	This bit enables the old behavior of RX term code. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14:11	RX_TERM_TIME	R/W	This field is the RX termination time (in ref_range cycles), (spec: 160ns < time). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
10	FAST_RX_SIGDET_EN_WAIT	R/W	This bit enables fast wait for RX SIGDET enable (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	RX_SIGDET_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for RX ana sigdet enable (spec >=1.1us). **Value After Reset:** 0x1d **Exists:** Always **Reset Mask:** 0x3ff

2.1.493 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Time Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)85
- **Exists:** Always

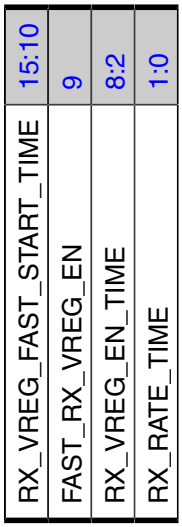


Table 2-497 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RX_VREG_FAST_START_TIME	R/W	This field is the power up time (in ref_range cycles) for rx_ana_vreg_fast_start (spec >=400ns). **Value After Reset:** 0xb **Exists:** Always **Reset Mask:** 0x3f
9	FAST_RX_VREG_EN	R/W	This bit enables fast RX VREG enable (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:2	RX_VREG_EN_TIME	R/W	This field is the power up time (in ref_range cycles divided by 2) for RX ana vreg enable (spec >2us). **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0x7f
1:0	RX_RATE_TIME	R/W	This field is the power up time (in ref_range cycles) for RX ana rate or width change (spec >=50ns). **Value After Reset:** 0x1

Table 2-497 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3

2.1.494 LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Time Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)86
- **Exists:** Always

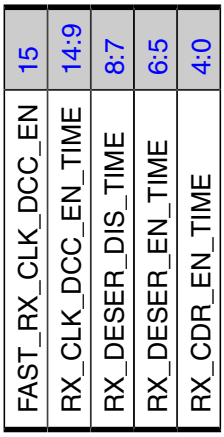


Table 2-498 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	FAST_RX_CLK_DCC_EN	R/W	This bit enables fast RX clock dcc enable (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:9	RX_CLK_DCC_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for RX ana clk dcc enable (spec 0ns). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
8:7	RX_DESER_DIS_TIME	R/W	This field is the power down time in (ref_range cycles) for RX ana deserial enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6:5	RX_DESER_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for RX ana deserial enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-498 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_PWRUP_TIME_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4:0	RX_CDR_EN_TIME	R/W	This field is the power up time (in ref_range cycles) for RX ana cdr (or dfe/dfe_taps) enable (spec 200ns). **Value After Reset:** 0x5 **Exists:** Always **Reset Mask:** 0x1f

2.1.495 LANEN_DIG_RX_PWRCTL_RX_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Control Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)87
- **Exists:** Always

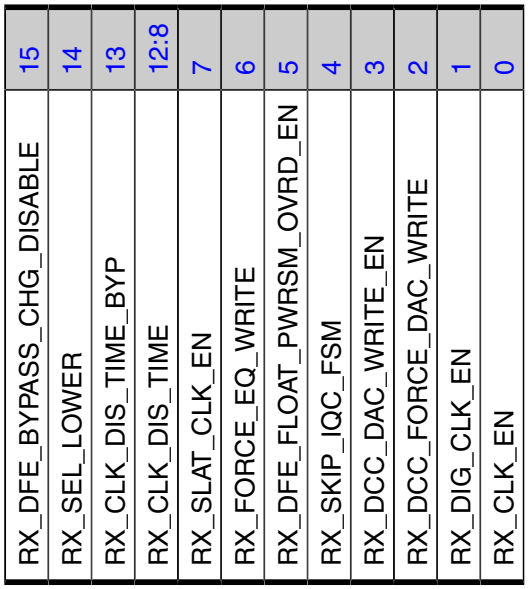


Table 2-499 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_DFE_BYPASS_CHG_DISABLE	R/W	This bit disables a change in dfe_bypass to power-down the RX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	RX_SEL_LOWER	R/W	This bit enables the lower byte to output first in 8/10 bits mode **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_CLK_DIS_TIME_BYP	R/W	This bit enables RX_SLAT_CLK clock. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12:8	RX_CLK_DIS_TIME	R/W	This field is the disable time (in ref_range cycles) for RX ana clock. **Value After Reset:** 0xa

Table 2-499 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1f
7	RX_SLAT_CLK_EN	R/W	This bit enables RX_SLAT_CLK clock. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_FORCE_EQ_WRITE	R/W	This bit enables the RX adapt block to write the EQ values. This will force an RX EQ write even when there is no rate change. Note that this bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_DFE_FLOAT_PWRSM_OVRD_EN	R/W	This bit overrides the individual DFE float tap positions and tap enables. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_SKIP_IQC_FSM	R/W	This bit skip IQCs calibration even if rx_eq_iq is changed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_DCC_DAC_WRITE_EN	R/W	This bit enables the RX adapt block to write the RX DCC DAC values. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RX_DCC_FORCE_DAC_WRITE	R/W	This bit enables the RX adapt block to write the RX DCC DAC values. This will force a RX DCC DAC write even when there is no rate change. Note that this bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_DIG_CLK_EN	R/W	This bit enables RX dig clock. This will control the rx_pwrsm_dig_clk_en signal which will reset the RX datapath and cause the RX DPLL GLCM to switch to the reference clock. **Value After Reset:** 0x1

Table 2-499 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	RX_CLK_EN	R/W	This bit enables RX clocks. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.496 LANEN_DIG_RX_PWRCTL_RX_CTL_2 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Control Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)88
- **Exists:** Always

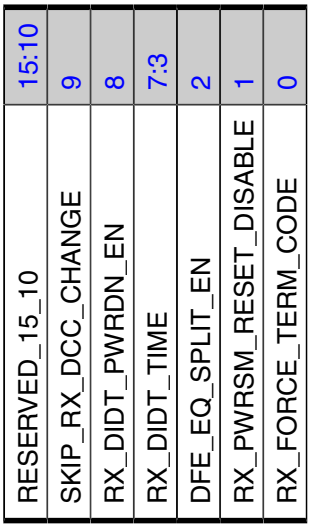


Table 2-500 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_CTL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	SKIP_RX_DCC_CHANGE	R/W	This bit enables skipping rate change when RXDCC changes along with Pstate change. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_DIDT_PWRDN_EN	R/W	This bit enables di/dt protection during power-down. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7:3	RX_DIDT_TIME	R/W	This field is RX di/dt protection time (in ref_range cycles) (spec: >=100ns). **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f

Table 2-500 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_CTL_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2	DFE_EQ_SPLIT_EN	R/W	This bit splits the DFE parameters from the AFE parameters. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_PWRSM_RESET_DISABLE	R/W	This bit disable the first loop when rx_reset is applied. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_FORCE_TERM_CODE	R/W	This bit forces a RX TERM CODE latch. This will force an RX TERM CODE write even when there is no rate change. Note that this bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.497 LANEN_DIG_RX_PWRCTL_RX_STATUS (for N = 0; N <= NLANES-1)

- **Description:** RX Power Up Control FSM StatusLane Input Status Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)89
- **Exists:** Always

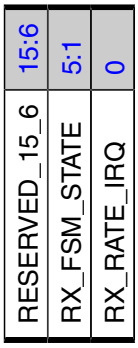


Table 2-501 Fields for Register: LANEN_DIG_RX_PWRCTL_RX_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:1	RX_FSM_STATE	R	This field is the current value of the RX PWR FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	RX_RATE_IRQ	R	This bit is the value of req_rx_rate_change_s01 register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.498 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Calibration Controls Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8a
- **Exists:** Always

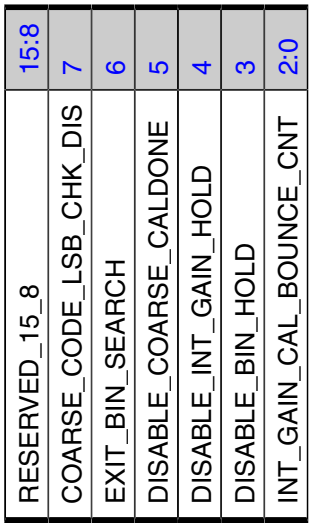


Table 2-502 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	COARSE_CODE_LSB_CHK_DIS	R/W	This bit disables checking for coarse_tune[1:0] lsb bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	EXIT_BIN_SEARCH	R/W	This bit enables the FSM to exit the binary serach when hold condition is detected **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	DISABLE_COARSE_CALDONE	R/W	This bit disables coarse value update during linear search once calibration done is asserted. This register only works in conjunction with disable_int_cal_mode_r. **Value After Reset:** 0x0

Table 2-502 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
4	DISABLE_INT_GAIN_HOLD	R/W	This bit disables hold of the integral value when vco correct is detected during bounce check. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	DISABLE_BIN_HOLD	R/W	This bit disables hold of the coarse value when VCO_CORRECT is detected during binary search. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2:0	INT_GAIN_CAL_BOUNCE_CNT	R/W	This field is the number of bounces (i.e. direction changes) on the int_gain code before indicating that the RX VCO calibration is done. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7

2.1.499 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Calibration Controls Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8b
- **Exists:** Always

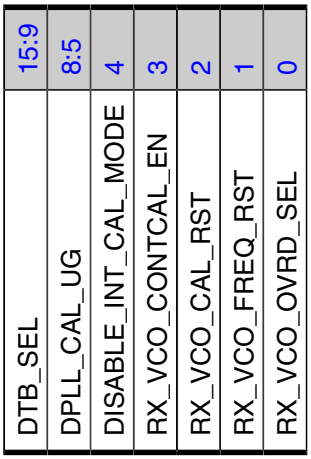


Table 2-503 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	DTB_SEL	R/W	This field is DTB select for RX VCO dtb signals. ■ 7'h01 - {chkfrq_en, ref_dig_clk} ■ 7'h02 - {rx_ana_cdr_vco_en_i, rx_ana_cdr_startup_i} ■ 7'h04 - {rx_vco_up, dpll_freq_rst} ■ 7'h08 - {rx_vco_contcal_en, rx_vco_cal_rst} ■ 7'h10 - {chkfrq_done, vcoclk_too_fast} ■ 7'h20 - {cal_dir, rx_vco_cal_done} ■ 7'h40 - {curr_state[0], rx_vco_cnt[0]} **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
8:5	DPLL_CAL_UG	R/W	This field is DPLL calibration update on int_gain code. Encoding: ■ 3'h0 - 0 ■ Else - (1/16)*2^(DPLL_CAL_UG-1) LSB/update. Maximum DPLL_CAL_UG=10, i.e. 32 LSB/update **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

Table 2-503 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	DISABLE_INT_CAL_MODE	R/W	When this bit is asserted, then the DPLL frequency register is never modified by the RX VCO calibration FSM (even if DPLL_CAL_UG is non-zero). In this case, the calibration will always be performed on the VCO freq_tune code. This allows disabling of integral calibration feature, and hence only using freq_tune calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_VCO_CONTCAL_EN	R/W	This bit is an override value for the continuous calibration enable from the RX PWRSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_VCO_CAL_RST	R/W	This bit is an override value for the calibration reset from the RX PWRSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_VCO_FREQ_RST	R/W	This bit is an override value for the frequency reset from the RX PWRSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_VCO_OVRD_SEL	R/W	This bit overrides the calibration controls from the RX PWRSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.500 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Calibration Controls Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8c
- **Exists:** Always

Table 2-504 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_CTRL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:10	FREG_TUNE_CAL_STEPS	R/W	This field is the number of cal steps of freq tune. **Value After Reset:** 0x9 **Exists:** Always **Reset Mask:** 0xf
9:0	FREG_TUNE_START_VAL	R/W	This field is the starting value of freq tune code. **Value After Reset:** 0x200 **Exists:** Always **Reset Mask:** 0x3ff

2.1.501 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_0 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Time Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8d
- **Exists:** Always

Table 2-505 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	FAST_RX_VCO_WAIT	R/W	This bit enables fast RX VCO wait times (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:4	RX_VCO_CNTR_PWRUP_TIME	R/W	This field is the power up time for RX ana vco counter (spec: >2us). **Value After Reset:** 0x34 **Exists:** Always **Reset Mask:** 0xff
3:0	RX_VCO_UPDATE_TIME	R/W	This field settles time in (ref_range cycles) for RX ana vco update (freq_tune or int_gain) (spec >350ns). **Value After Reset:** 0x9 **Exists:** Always **Reset Mask:** 0xf

2.1.502 LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1 (for N = 0; N <= NLANES-1)

- **Description:** RX Power UP Time Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8e
- **Exists:** Always

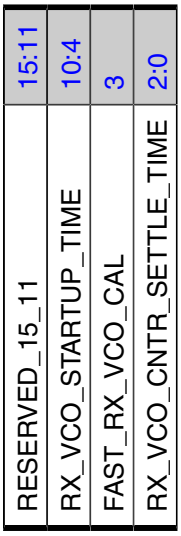


Table 2-506 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:4	RX_VCO_STARTUP_TIME	R/W	This field is the power up time in ref_range cycles for RX ana vco startup (spec >250ns). **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7f
3	FAST_RX_VCO_CAL	R/W	This bit enables fast RX VCO calibration (skips calibration) (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	RX_VCO_CNTR_SETTLE_TIME	R/W	THis field is the RX VCO counter value settling time in (ref_dig_clk cycles) (spec: 1 ref_dig_clk cycle). **Value After Reset:** 0x3

Table 2-506 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_CAL_TIME_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7

2.1.503 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Status Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)8f
- **Exists:** Always

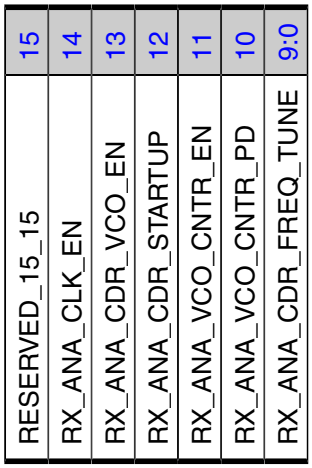


Table 2-507 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_ANA_CLK_EN	R	This bit is the current value of rx_ana_clk_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	RX_ANA_CDR_VCO_EN	R	This bit is the current value of rx_ana_cdr_vco_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_ANA_CDR_STARTUP	R	This bit is the current value of rx_ana_cdr_startup_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-507 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	RX_ANA_VCO_CNTR_EN	R	This bit is the current value of rx_ana_vco_cntr_en_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RX_ANA_VCO_CNTR_PD	R	This bit is the current value of rx_ana_vco_cntr_pd_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RX_ANA_CDR_FREQ_TUNE	R	This field is the current value of rx_ana_cdr_freq_tune_i. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.504 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Status Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)90
- **Exists:** Always

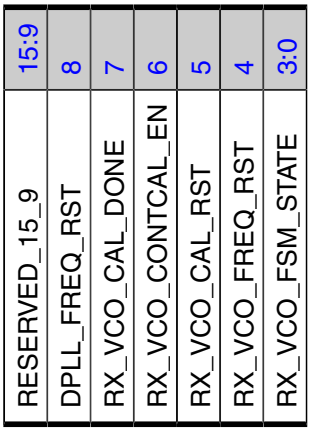


Table 2-508 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	DPLL_FREQ_RST	R	This bit indicates that the RX integral frequency is reset or not. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	RX_VCO_CAL_DONE	R	This bit indicates that the RX VCO has completed calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	RX_VCO_CONTCAL_EN	R	This bit is the value of the continuous calibration enable from the RX PWRSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

Table 2-508 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5	RX_VCO_CAL_RST	R	This bit is the value of the calibration reset from the RX PWRSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	RX_VCO_FREQ_RST	R	This bit is the value of the RX VCO frequency reset from the RX PWRSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	RX_VCO_FSM_STATE	R	This field is the value of the RX VCO CAL FSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.505 LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2 (for N = 0; N <= NLANES-1)

- **Description:** RX VCO Status Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)91
- **Exists:** Always

Table 2-509 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_VCO_STAT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_VCO_UP	R	This bit indicates that the RX VCO is ready. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_VCO_CORRECT	R	This bit indicates that the RX VCO clock has the correct frequency. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	VCOCLK_TOO_FAST	R	This bit indicates that the RX VCO clock frequency is too fast. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	VCO_CNTR_FINAL	R	This field is the value of Rx VCO counter when refclk counter expired. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.506 LANEN_DIG_RX_VCOCAL_RX_DPLL_FREQ_SUM_RST (for N = 0; N <= NLANES-1)

- **Description:** RX DPLL frequency integrator reset value.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)92
- **Exists:** Always

Table 2-510 Fields for Register: LANEN_DIG_RX_VCOCAL_RX_DPLL_FREQ_SUM_RST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	RX_DPLL_FREQ_INT_RST	R/W	This field is the DPLL frequncy integrator reset value. **Value After Reset:** 0x2000 **Exists:** Always **Reset Mask:** 0x3fff

2.1.507 LANEN_DIG_RX_LBERT_CTL (for N = 0; N <= NLANES-1)

- **Description:** Pattern Matcher controls
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)93
- **Exists:** Always

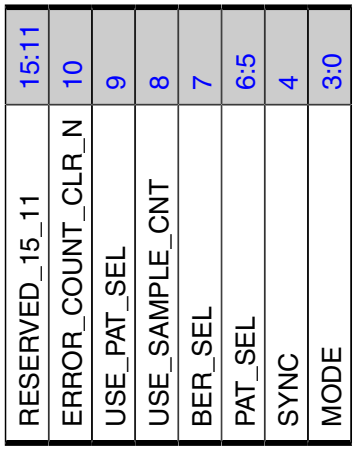


Table 2-511 Fields for Register: LANEN_DIG_RX_LBERT_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	ERROR_COUNT_CLR_N	R/W	Use 1 sync_start to clear the error count instead of two **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	USE_PAT_SEL	R/W	Use [6:5] field to select what to read into STAT0 and STAT1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	USE_SAMPLE_CNT	R/W	Enable bit to run the lbert for the duration of SMPL_CNT **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	BER_SEL	R/W	Enable bit Error instead of word errors during pattern matching **Value After Reset:** 0x0

Table 2-511 Fields for Register: LANEN_DIG_RX_LBERT_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
6:5	PAT_SEL	R/W	Select which 16 bits of the erroneous captured pattern will be read into stat0 and stat1. For a given select value, the following will be read. ■ 2'b00 - {pat_r[15:00], xor_r[15:00]} ■ 2'b01 - {pat_r[31:16], xor_r[31:16]} ■ 2'b10 - {pat_r[39:24], xor_r[39:24]} ■ 2'b11 - {pat_r[31:16], pat_r[15:00]} **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
4	SYNC	R/W	Synchronize pattern matcher LFSR with incoming data A write of a one to this bit will reset the error counter and start a synchronization of the PM. There is no need to write this back to zero to run normally. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	MODE	R/W	Pattern to match. When changing modes, you must change to disabled first ■ 0 - Disabled ■ 1 - lfsr31: X^31 + X^28 + 1 ■ 2 - lfsr23: X^23 + X^18 + 1 ■ 3 - lfsr23: x^23 + x^21 + x^16 + x^8 + x^5 + x^2 + 1 ■ 4 - lfsr16: x^16 + x^5 + x^4 + x^3 + 1 ■ 5 - lfsr15: X^15 + X^14 + 1 ■ 6 - lfsr11: X^11 + X^9 + 1 ■ 7 - lfsr9: X^9 + X^5 + 1 ■ 8 - lfsr7: X^7 + X^6 + 1 ■ 9 - d[n] = d[n-10] ■ 10 - d[n] = !d[n-10] ■ 11 - d[n] = !d[n-20] ■ 12 - d[n] = d[n] ■ 13 - lfsr13: x^13 + x^12 + x^11 + x^8 + 1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.508 LANEN_DIG_RX_LBERT_ERR (for N = 0; N <= NLANES-1)

- **Description:** Pattern Match Error Counter.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)94
- **Exists:** Always

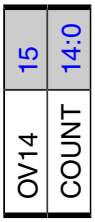


Table 2-512 Fields for Register: LANEN_DIG_RX_LBERT_ERR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	OV14	R/W	If this bit is active, multiply count by 128. If OV14=1 and COUNT=2^15-1, signals overflow of counter. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
14:0	COUNT	R/W	This field is the current error count. If OV14 field is active, then multiply count by 128. (volatile and 2 reads needed to read value) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff **Volatile:** true

2.1.509 LANEN_DIG_RX_LBERT_LBERT_PM_STAT0 (for N = 0; N <= NLANES-1)

- **Description:** LBERT PM Status Register #0
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)95
- **Exists:** Always

Table 2-513 Fields for Register: LANEN_DIG_RX_LBERT_LBERT_PM_STAT0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	PM_PAT	R	Current value of the XOR comparison (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.510 LANEN_DIG_RX_LBERT_LBERT_PM_STAT1 (for N = 0; N <= NLANES-1)

- **Description:** E32_LANE0_DIG_RX_LBERT_LBERT_PM_STAT1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)96
- **Exists:** Always

Table 2-514 Fields for Register: LANEN_DIG_RX_LBERT_LBERT_PM_STAT1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	PM_XOR	R	Current value of the received pattern (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.511 LANEN_DIG_RX_CDR_CDR_CTL_0 (for N = 0; N <= NLANES-1)

- **Description:** Control Bits for Receiver in Recovered Domain.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)97
- **Exists:** Always

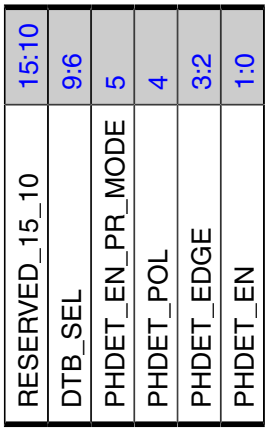


Table 2-515 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:6	DTB_SEL	R/W	This field is selected to drive various signals onto the dtb. ■ 0 - disabled ■ 1 - rx_pr_stable, rx_afe_stable from rx_ana_ctl ■ 2 - lbert_err[1:0] from lbert_pm ■ 3 - lbert_err[3:2] from lbert_pm for 4 symbol mode ■ 4 - sigdet_hf, sigdet_lf from analog ■ 5 - cdr_valid, msb of FSM state from cdr_ctl ■ 6 - 2 lsb's of FSM state from cdr_ctl ■ 7 - rx_dig_rst,rx_dig_en ■ 8 - DTB outputs from VCO cal block ■ 9 - rx_ana_word_clk_i, rx_ana_dword_clk_i **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
5	PHDET_EN_PR_MODE	R/W	This bit enables partial response phase detector mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-515 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	PHDET_POL	R/W	This bit reverses polarity of phase error. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	PHDET_EDGE	R/W	This field is the edges to use for phase detection. ■ 10 - Use both edges ■ 01 - Use rising edges only ■ 11 - Use falling edges only ■ 00 - Ignore all edges **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3
1:0	PHDET_EN	R/W	This field enables phase detector. The top bit is odd slicers, the bottom is even. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3

2.1.512 LANEN_DIG_RX_CDR_CDR_CTL_1 (for N = 0; N <= NLANES-1)

- **Description:** CDR Control Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)98
- **Exists:** Always

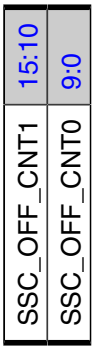


Table 2-516 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	SSC_OFF_CNT1	R/W	When SSC mode is disabled, the 12-bit word count in gain stage 1 is: ■ (SSC_OFF_CNT1 * 4) in 20b mode ■ (SSC_OFF_CNT1 * 5) in 16b mode **Value After Reset:** 0x13 **Exists:** Always **Reset Mask:** 0x3f
9:0	SSC_OFF_CNT0	R/W	When SSC mode is disabled, the 12-bit word count in gain stage 0 is: ■ (SSC_OFF_CNT0 * 4) in 20b mode ■ (SSC_OFF_CNT0 * 5) in 16b mode **Value After Reset:** 0x39 **Exists:** Always **Reset Mask:** 0x3ff

2.1.513 LANEN_DIG_RX_CDR_CDR_CTL_2 (for N = 0; N <= NLANES-1)

- **Description:** CDR Control Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)99
- **Exists:** Always

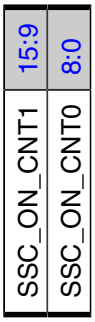


Table 2-517 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	SSC_ON_CNT1	R/W	When SSC mode is enabled, the 12-bit word count in gain stage 1 is: ■ (SSC_ON_CNT1 * 8) in 20b mode ■ (SSC_ON_CNT1 * 10) in 16b mode **Value After Reset:** 0x25 **Exists:** Always **Reset Mask:** 0x7f
8:0	SSC_ON_CNT0	R/W	When SSC mode is enabled, the 12-bit word count in gain stage 0 is: ■ (SSC_ON_CNT0 * 8) in 20b mode ■ (SSC_ON_CNT0 * 10) in 16b mode **Value After Reset:** 0xbb **Exists:** Always **Reset Mask:** 0x1ff

2.1.514 LANEN_DIG_RX_CDR_CDR_CTL_3 (for N = 0; N <= NLANES-1)

- **Description:** CDR Control Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9a
- **Exists:** Always

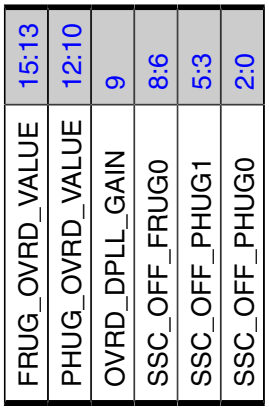


Table 2-518 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	FRUG_OVRD_VALUE	R/W	This field is the override value for FRUG (frequency update gain). ■ 3'h0 - 0 ■ 3'h1 - 1/16 LSB/update ■ 3'h2 - 1/8 LSB/update ■ 3'h3 - 1/4 LSB/update ■ 3'h4 - 1/2 LSB/update ■ 3'h5 - 1 LSB/update ■ 3'h6 - 2 LSB/update ■ 3'h7 - 4 LSB/update **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
12:10	PHUG_OVRD_VALUE	R/W	This field is the override value for PHUG (phase update gain). ■ 3'h0 - 0 ■ 3'h1 - 1000 ppm ■ 3'h2 - 2000 ppm ■ 3'h3 - 3000 ppm ■ 3'h4 - 4000 ppm ■ 3'h5 - 5000 ppm ■ 3'h6 - 6000 ppm ■ 3'h7 - 7000 ppm **Value After Reset:** 0x0

Table 2-518 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7
9	OVRD_DPLL_GAIN	R/W	This bit is the override PHUG and FRUG values. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:6	SSC_OFF_FRUG0	R/W	When SSC mode is disabled, the frug value in gain stage 0 is SSC_OFF_FRUG0. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
5:3	SSC_OFF_PHUG1	R/W	When SSC mode is disabled, the phug value in gain stage 1 is SSC_OFF_PHUG1. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7
2:0	SSC_OFF_PHUG0	R/W	When SSC mode is disabled, the phug value in gain stage 0 is SSC_OFF_PHUG0. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7

2.1.515 LANEN_DIG_RX_CDR_CDR_CTL_4 (for N = 0; N <= NLANES-1)

- **Description:** CDR Control Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9b
- **Exists:** Always

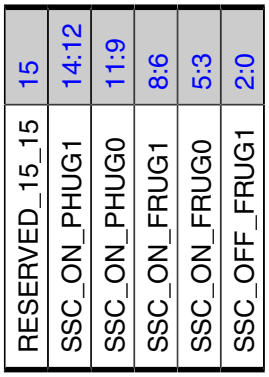


Table 2-519 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:12	SSC_ON_PHUG1	R/W	When SSC mode is enabled, the phug value in gain stage 1 is SSC_ON_PHUG1. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7
11:9	SSC_ON_PHUG0	R/W	When SSC mode is enabled, the phug value in gain stage 0 is SSC_ON_PHUG0. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7
8:6	SSC_ON_FRUG1	R/W	When SSC mode is enabled, the frug value in gain stage 1 is SSC_ON_FRUG1. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
5:3	SSC_ON_FRUG0	R/W	When SSC mode is enabled, the frug value in gain stage 0 is SSC_ON_FRUG0. **Value After Reset:** 0x3

Table 2-519 Fields for Register: LANEN_DIG_RX_CDR_CDR_CTL_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7
2:0	SSC_OFF_FRUG1	R/W	When SSC mode is disabled, the frug value in gain stage 1 is SSC_OFF_FRUG1. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7

2.1.516 LANEN_DIG_RX_CDR_STAT (for N = 0; N <= NLANES-1)

- **Description:** Current Output Values to Dpll (phug, frug).
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9c
- **Exists:** Always

Table 2-520 Fields for Register: LANEN_DIG_RX_CDR_STAT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:3	FRUG_VALUE	R	This field is the current value for dpll_frug[2:0]. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	PHUG_VALUE	R	This field is the current value for dpll_phug[2:0]. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.517 LANEN_DIG_RX_DPLL_FREQ (for N = 0; N <= NLANES-1)

- **Description:** Current Frequency Integrator Value.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9d
- **Exists:** Always

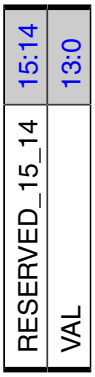


Table 2-521 Fields for Register: LANEN_DIG_RX_DPLL_FREQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	VAL	R/W	The freq is 125*VAL ppm from the reference. (volatile and 2 reads needed to read value) **Value After Reset:** 0x2000 **Exists:** Always **Reset Mask:** 0x3fff **Volatile:** true

2.1.518 LANEN_DIG_RX_DPLL_FREQ_BOUND_0 (for N = 0; N <= NLANES-1)

- **Description:** Frequency Bounds for Incoming Data Stream 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9e
- **Exists:** Always

Table 2-522 Fields for Register: LANEN_DIG_RX_DPLL_FREQ_BOUND_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:1	UPPER_FREQ_BOUND	R/W	This field is upper frequency bound in terms of LSBs of the integral control code. **Value After Reset:** 0x264 **Exists:** Always **Reset Mask:** 0x3ff
0	FREQ_BOUND_EN	R/W	This bit enables the frequency bounds feature. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.519 LANEN_DIG_RX_DPLL_FREQ_BOUND_1 (for N = 0; N <= NLANES-1)

- **Description:** Frequency Bounds for Incoming Data Stream 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)9f
- **Exists:** Always

Table 2-523 Fields for Register: LANEN_DIG_RX_DPLL_FREQ_BOUND_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	LOWER_FREQ_BOUND	R/W	This field is lower frequency bound in terms of LSBs of the integral control code. **Value After Reset:** 0x19c **Exists:** Always **Reset Mask:** 0x3ff

2.1.520 LANEN_DIG_RX_IQC_CTL_RESET_ADJUST (for N = 0; N <= NLANES-1)

- **Description:** RX IQC Reset iq_adjust Bus Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a0
- **Exists:** Always

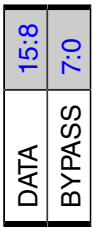


Table 2-524 Fields for Register: LANEN_DIG_RX_IQC_CTL_RESET_ADJUST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DATA	R/W	This field is the RX IQC data reset value (iq_adjust bus value). **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff
7:0	BYPASS	R/W	This field is the RX IQC bypass reset value (iq_adjust bus value). **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff

2.1.521 LANEN_DIG_RX_IQC_CTL_CONFIG (for N = 0; N <= NLANES-1)

- **Description:** RX IQC Config Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a1
- **Exists:** Always

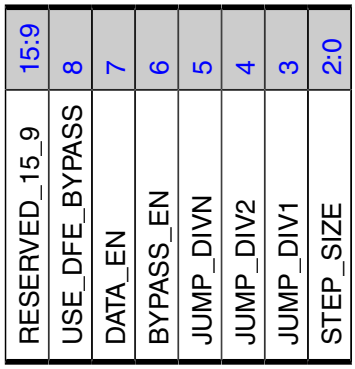


Table 2-525 Fields for Register: LANEN_DIG_RX_IQC_CTL_CONFIG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	USE_DFE_BYPASS	R/W	This bit uses the dfe_bypass input to determine whether to move the bypass or data adjust buses. When this bit is set, the DATA_EN and BYPASS_EN registers are ignored. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	DATA_EN	R/W	This bit enables walking/jumping the data IQ adjust value. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	BYPASS_EN	R/W	This bit enables walking/jumping the bypass IQ adjust value. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	JUMP_DIVN	R/W	This bit enables jumping to DIVN iq_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-525 Fields for Register: LANEN_DIG_RX_IQC_CTL_CONFIG (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	JUMP_DIV2	R/W	This bit enables jumping to DIV2 iq_code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	JUMP_DIV1	R/W	This bit enables jumping to DIV1 iq_code. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2:0	STEP_SIZE	R/W	This field is the RX IQC step size. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x7

2.1.522 LANEN_DIG_RX_IQC_CTL_STATUS (for N = 0; N <= NLANES-1)

- **Description:** RX IQC Status Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a2
- **Exists:** Always

Table 2-526 Fields for Register: LANEN_DIG_RX_IQC_CTL_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	RX_FSM_STATE	R	This field is the current value of the RX IQC FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.523 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a5
- **Exists:** Always

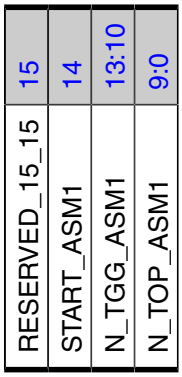


Table 2-527 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	START_ASM1	R/W	This bit starts adaptation state machine 1 (VGA, CTLE, DFE, EYEH). This register-bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	N_TGG_ASM1	R/W	This field is the number of toggle loop iterations for ASM1. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
9:0	N_TOP_ASM1	R/W	This field is the number of top level loop iterations for ASM1. **Value After Reset:** 0x10 **Exists:** Always **Reset Mask:** 0x3ff

2.1.524 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a6
- **Exists:** Always

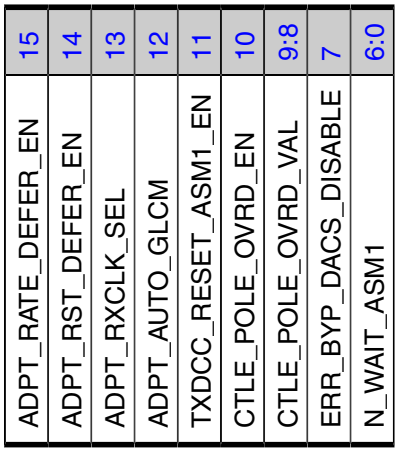


Table 2-528 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ADPT_RATE_DEFER_EN	R/W	This bit defers adaptation rate request till SSM is done when doing TX DCC. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	ADPT_RST_DEFER_EN	R/W	This bit defers adaptation reset request till SSM is done when doing TX DCC. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	ADPT_RXCLK_SEL	R/W	This bit selects ADPT clock. ■ 0x0 - ref_range_clk ■ 0x1 - rx_dig_qword_clk **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	ADPT_AUTO_GLCM	R/W	This bit enables the automatic switching of the ADPT GLCM. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-528 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	TXDCC_RESET_ASM1_EN	R/W	This bit enables the RESET_ASM1 register to reset the SSM, even when doing TX DCC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	CTLE_POLE_OVRD_EN	R/W	This bit is the override CTLE pole value (only valid if adaptation is run). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:8	CTLE_POLE_OVRD_VAL	R/W	This field is the CTLE Pole override value to load at start of adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
7	ERR_BYP_DACS_DISABLE	R/W	This bit is used to disable the.error bypass DACs **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	N_WAIT_ASM1	R/W	This field is the number of wait cycles for adaptation SM 1. **Value After Reset:** 0x9 **Exists:** Always **Reset Mask:** 0x7f

2.1.525 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a7
- **Exists:** Always

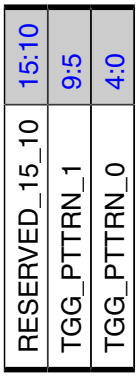


Table 2-529 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	TGG_PTTRN_1	R/W	This field is the pattern for the second toggle loop. The error slicer is moved upward by Data TAP1 if this pattern is matched. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x1f
4:0	TGG_PTTRN_0	R/W	This field is the pattern for the first toggle loop. The error slicer is moved downward by Data TAP1 if this pattern is matched. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f

2.1.526 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a8
- **Exists:** Always

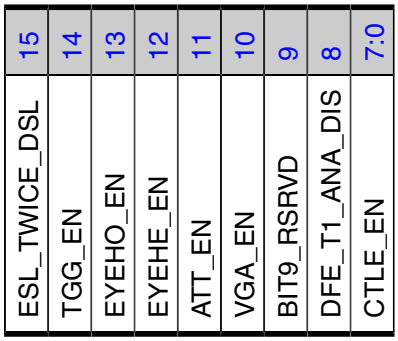


Table 2-530 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ESL_TWICE_DSL	R/W	This bit asserts if error slicer has twice the voltage range as the data slicer. (for the same 8 bits). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TGG_EN	R/W	This bit enables toggling of the error slicer. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	EYEHO_EN	R/W	This bit enables eye height measurement using odd error slicer. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	EYEHE_EN	R/W	This bit enables eye height measurement using even error slicer. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	ATT_EN	R/W	This bit enables ATT adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-530 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
10	VGA_EN	R/W	This bit enables VGA adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	BIT9_RSRVD	R/W	This bit is reserved. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	DFE_T1_ANA_DIS	R/W	This bit disables updating DFE tap1 analog values during adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	CTLE_EN	R/W	This field enables CTLE boost adaptation. These bits determine which correlators are used to adapt the CTLE. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.527 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)a9
- **Exists:** Always

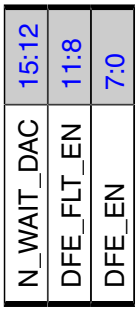


Table 2-531 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	N_WAIT_DAC	R/W	This field selects setup time and pulse width time in terms of qword_clk for RX DFE tapes and AFE update. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0xf
11:8	DFE_FLT_EN	R/W	This field is DFE float enable signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:0	DFE_EN	R/W	This bit enables DFE adaptation for taps 8-1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.528 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)aa
- **Exists:** Always

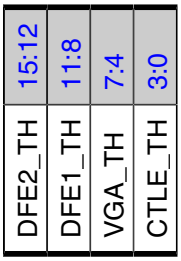


Table 2-532 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	DFE2_TH	R/W	This field is DFE TAP2 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:8	DFE1_TH	R/W	This field is DFE TAP1 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:4	VGA_TH	R/W	This field is VGA correlation decision threshold (2^N-1). During eye height measurement, the VGA_TH is repurposed for error slicer updates. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	CTLE_TH	R/W	This field is CTLE correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.529 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ab
- **Exists:** Always

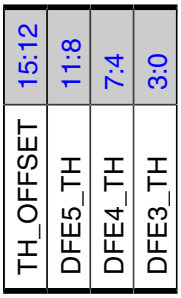


Table 2-533 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	TH_OFFSET	R/W	This field applies an offset to the decision threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:8	DFE5_TH	R/W	This field is DFE TAP5 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:4	DFE4_TH	R/W	This field is DFE TAP4 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	DFE3_TH	R/W	This field is DFE TAP3 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.530 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 7.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ac
- **Exists:** Always

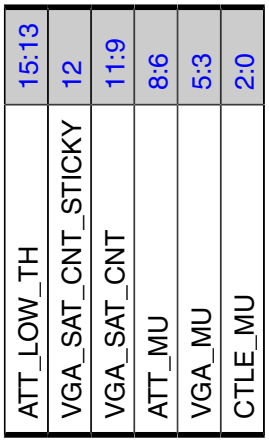


Table 2-534 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	ATT_LOW_TH	R/W	This field is the ATT low threshold. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
12	VGA_SAT_CNT_STICKY	R/W	If this bit is de-asserted, then VGA saturation counts must be consecutive to change ATT. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
11:9	VGA_SAT_CNT	R/W	This field is the VGA saturation count. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
8:6	ATT_MU	R/W	This field is the ATT gain code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
5:3	VGA_MU	R/W	This field is the VGA gain code update gain (2^N). During eye height measurement, the VGA_MU is repurposed for error slicer updates. **Value After Reset:** 0x5

Table 2-534 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_7 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7
2:0	CTLE_MU	R/W	This field is the CTLE boost code update gain (2^N). **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7

2.1.531 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 8.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ad
- **Exists:** Always

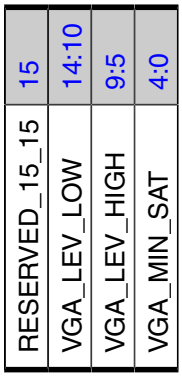


Table 2-535 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:10	VGA_LEV_LOW	R/W	This field is the VGA level low saturation limit. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0x1f
9:5	VGA_LEV_HIGH	R/W	This field is the VGA level high saturation limit. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f
4:0	VGA_MIN_SAT	R/W	This field is the VGA minimum saturation limit. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f

2.1.532 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 9.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ae
- **Exists:** Always

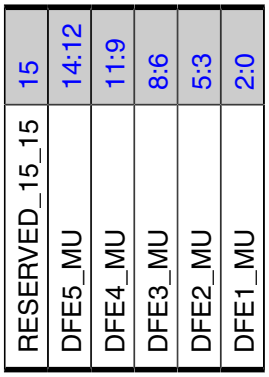


Table 2-536 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:12	DFE5_MU	R/W	This field is the DFE TAP5 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
11:9	DFE4_MU	R/W	This field is the DFE TAP4 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
8:6	DFE3_MU	R/W	This field is the DFE TAP3 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
5:3	DFE2_MU	R/W	This field is the DFE TAP2 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7

Table 2-536 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2:0	DFE1_MU	R/W	This field is the DFE TAP1 code update gain (2^N). **Value After Reset:** 0x5 **Exists:** Always **Reset Mask:** 0x7

2.1.533 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 10.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)af
- **Exists:** Always

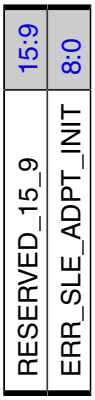


Table 2-537 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	ERR_SLE_ADPT_INIT	R/W	The error even slicer is initialized to this value at the start of a new adaptation request. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.534 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_11 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 11.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b0
- **Exists:** Always

Table 2-538 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	ERR_SLO_ADPT_INIT	R/W	The error odd slicer is initialized to this value, at the start of a new adaptation request. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.535 LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG (for N = 0; N <= NLANES-1)

- **Description:** Reset Adaptation Configuration Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b1
- **Exists:** Always

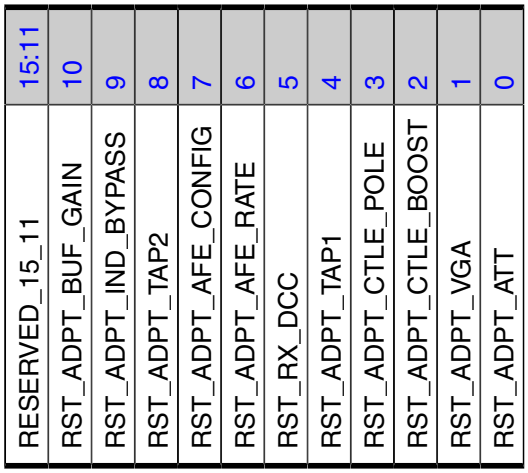


Table 2-539 Fields for Register: LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RST_ADPT_BUF_GAIN	R/W	This bit resets BUF Gain when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	RST_ADPT_IND_BYPASS	R/W	This bit resets IND BYPASS when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	RST_ADPT_TAP2	R/W	This bit resets data TAP2 when turning off DFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	RST_ADPT_AFE_CONFIG	R/W	This bit resets AFE config when turning off AFE adaptation. **Value After Reset:** 0x1

Table 2-539 Fields for Register: LANEN_DIG_RX_ADPTCTL_RST_ADPT_CFG (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
6	RST_ADPT_AFE_RATE	R/W	This bit resets AFE rate when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	RST_RX_DCC	R/W	This bit resets RX DCC DACs **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	RST_ADPT_TAP1	R/W	This bit resets data TAP1 when turning off DFE adaptation. (TAPs 3-5 are always turned off when DFE adaptation is turned off). **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	RST_ADPT_CTLE_POLE	R/W	This bit resets CTLE Pole when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RST_ADPT_CTLE_BOOST	R/W	This bit resets CTLE Boost when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	RST_ADPT_VGA	R/W	This bit resets VGA when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RST_ADPT_ATT	R/W	This bit resets ATT when turning off AFE adaptation. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.536 LANEN_DIG_RX_ADPTCTL_ATT_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of ATT Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b2
- **Exists:** Always

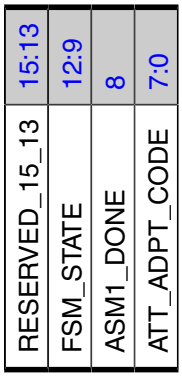


Table 2-540 Fields for Register: LANEN_DIG_RX_ADPTCTL_ATT_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:9	FSM_STATE	R	This field is the current value of the ADAPT FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	ATT_ADPT_CODE	R	This field is the value of ATT adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.537 LANEN_DIG_RX_ADPTCTL_VGA_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of VGA Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b3
- **Exists:** Always

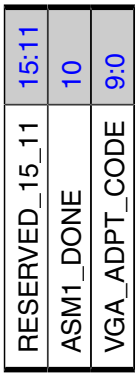


Table 2-541 Fields for Register: LANEN_DIG_RX_ADPTCTL_VGA_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:0	VGA_ADPT_CODE	R	This field is the value of VGA adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.538 LANEN_DIG_RX_ADPTCTL_CTLE_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of CTLE Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b4
- **Exists:** Always

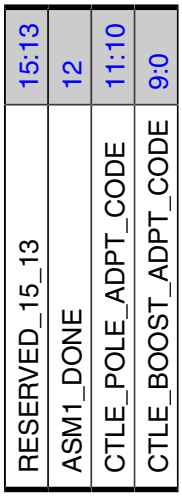


Table 2-542 Fields for Register: LANEN_DIG_RX_ADPTCTL_CTLE_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:10	CTLE_POLE_ADPT_CODE	R	This field is the value of CTLE pole adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:0	CTLE_BOOST_ADPT_CODE	R	This field is the value of CTLE boost adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-542 Fields for Register: LANEN_DIG_RX_ADPTCTL_CTLE_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true

2.1.539 LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP1 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b5
- **Exists:** Always

Table 2-543 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP1_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:0	DFE_TAP1_ADPT_CODE	R	This field is the value of DFE TAP1 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.540 LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP2 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b6
- **Exists:** Always

Table 2-544 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP2_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:0	DFE_TAP2_ADPT_CODE	R	This field is the value of DFE TAP2 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.541 LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP3 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b7
- **Exists:** Always

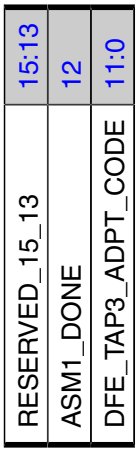


Table 2-545 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP3_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP3_ADPT_CODE	R	This field is the value of DFE TAP3 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.542 LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP4 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b8
- **Exists:** Always

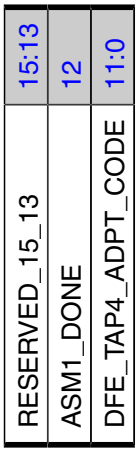


Table 2-546 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP4_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP4_ADPT_CODE	R	This field is the value of DFE TAP4 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.543 LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP5 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)b9
- **Exists:** Always

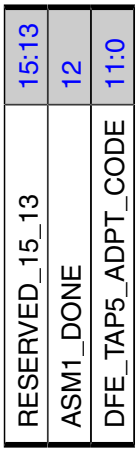


Table 2-547 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP5_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP5_ADPT_CODE	R	This field is the value of DFE TAP5 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.544 LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Even High vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ba
- **Exists:** Always

Table 2-548 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA0_EVEN_HIGH_VDAC_OFST	R/W	This field is the offset value for DFE data0 even high vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.545 LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Even Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)bb
- **Exists:** Always

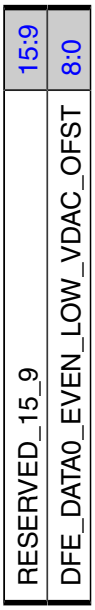


Table 2-549 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA0_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA0_EVEN_LOW_VDAC_OFST	R/W	This field is the offset value for DFE data0 even low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.546 LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Odd High vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)bc
- **Exists:** Always

Table 2-550 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA0_ODD_HIGH_VDAC_OFST	R/W	This field is the offset value for DFE data0 odd high vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.547 LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Odd Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)bd
- **Exists:** Always

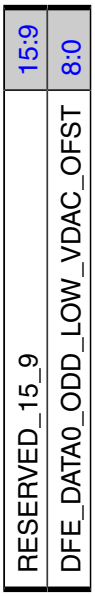


Table 2-551 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA0_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA0_ODD_LOW_VDAC_OFST	R/W	This field is the offset value for DFE data0 odd low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.548 LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX SLICER CTRL Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)be
- **Exists:** Always

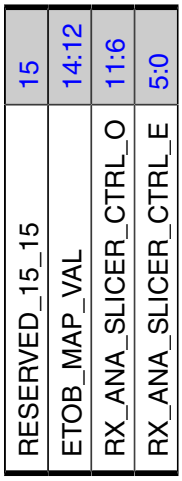


Table 2-552 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:12	ETOB_MAP_VAL	R/W	This field is the mapping value for etob_o and etob_e signals. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
11:6	RX_ANA_SLICER_CTRL_O	R/W	This field is the value for rx_ana_slicer_ctrl_o[5:0]. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_ANA_SLICER_CTRL_E	R/W	This field is the value for rx_ana_slicer_ctrl_e[5:0]. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0x3f

2.1.549 LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_MAP (for N = 0; N <= NLANES-1)

- **Description:** Sets Mapping Values for RX SLICER CTRL.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)bf
- **Exists:** Always

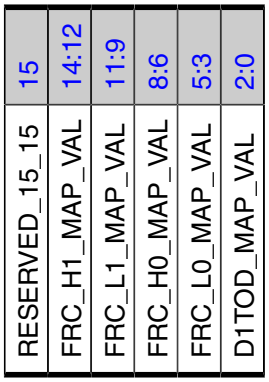


Table 2-553 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_MAP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:12	FRC_H1_MAP_VAL	R/W	This field is the mapping value for frc_h1_o and frc_h1_e signals. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x7
11:9	FRC_L1_MAP_VAL	R/W	This field is the mapping value for frc_l1_o and frc_l1_e signals. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7
8:6	FRC_H0_MAP_VAL	R/W	This field is the mapping value for frc_h0_o and frc_h0_e signals. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
5:3	FRC_L0_MAP_VAL	R/W	This field is the mapping value for frc_l0_o and frc_l0_e signals. **Value After Reset:** 0x5

Table 2-553 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_SLICER_CTRL_MAP (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7
2:0	D1TOD_MAP_VAL	R/W	This field is the mapping value for d1tod_o and d1tod_e signals. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x7

2.1.550 LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even High vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c0
- **Exists:** Always

Table 2-554 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA1_EVEN_HIGH_VDAC_OFST	R/W	This field is the offset value for DFE data1 even high vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.551 LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd High vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c1
- **Exists:** Always

Table 2-555 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA1_ODD_HIGH_VDAC_OFST	R/W	This field is the offset value for DFE data1 odd high vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.552 LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_ODD_LEVEL (for N = 0; N <= NLANES-1)

- **Description:** Value of Odd Error Slicer Level.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c2
- **Exists:** Always

Table 2-556 Fields for Register: LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_ODD_LEVEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	E_SLO_LVL	R	This field is the odd error slicer level. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.553 LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_EVEN_LEVEL (for N = 0; N <= NLANES-1)

- **Description:** Value of Even Error Slicer Level.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c3
- **Exists:** Always

Table 2-557 Fields for Register: LANEN_DIG_RX_ADPTCTL_ERROR_SLICER_EVEN_LEVEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	E_SLE_LVL	R	This field is the even error slicer level. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.554 LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE By-Pass Even vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c4
- **Exists:** Always

Table 2-558 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_BYPASS_EVEN_VDAC_OFST	R/W	This field is the offset value for DFE by-pass even vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.555 LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset values for RX DFE Bypass Odd vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c5
- **Exists:** Always

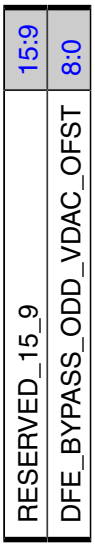


Table 2-559 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_BYPASS_ODD_VDAC_OFST	R/W	This field is the offset value for DFE bypass odd vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.556 LANEN_DIG_RX_ADPTCTL_ADPT_RESET (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Reset Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c6
- **Exists:** Always

Table 2-560 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_RESET (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RESET_ASM1	R/W	This bit resets adaptation state machine (ASM1), binary search as well as the stats capture block. This is a self-clearing bit, and requires re-start of ASM1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.557 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_12 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 12.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c7
- **Exists:** Always

Table 2-561 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	CTLE_T1_WT	R/W	When calculating CTLE correlation, TAP1 correlation will be right shifted by CTLE_T1_WT before being used. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.558 LANEN_DIG_RX_ADPTCTL_RX_DCC_IDAC_OFST_OVRD_EN (for N = 0; N <= NLANES-1)

- **Description:** Offset Override Enables for RX DCC iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c8
- **Exists:** Always

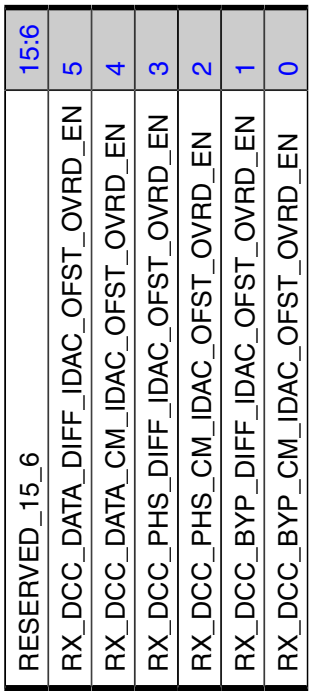


Table 2-562 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_IDAC_OFST_OVRD_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	RX_DCC_DATA_DIFF_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC data differential iDAC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_DCC_DATA_CM_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC data common-mode iDAC. **Value After Reset:** 0x0

Table 2-562 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_IDAC_OFST_OVRD_EN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	RX_DCC_PHS_DIFF_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC phase differential iDAC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_DCC_PHS_CM_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC phase common-mode iDAC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_DCC_BYP_DIFF_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC bypass differential iDAC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_DCC_BYP_CM_IDAC_OFST_OVRD_EN	R/W	This bit is the override enable for RX DCC bypass common-mode iDAC. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.559 LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Phase Differential iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)c9
- **Exists:** Always

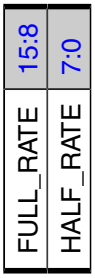


Table 2-563 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC phase differential iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC phase differential iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.560 LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Phase Common Mode iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ca
- **Exists:** Always

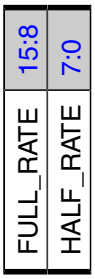


Table 2-564 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_PHASE_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC phase common mode iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC phase common mode iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.561 LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Data Differential iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)cb
- **Exists:** Always

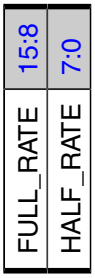


Table 2-565 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC data differential iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC data differential iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.562 LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Data Common Mode iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)cc
- **Exists:** Always

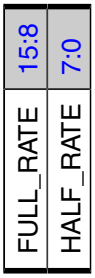


Table 2-566 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_DATA_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC data common mode iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC data common mode iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.563 LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Bypass Differential iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)cd
- **Exists:** Always

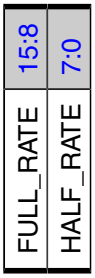


Table 2-567 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_DIFF_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC bypass differential iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC bypass differential iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.564 LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DCC Bypass Common Mode iDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ce
- **Exists:** Always

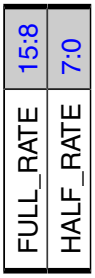


Table 2-568 Fields for Register: LANEN_DIG_RX_ADPTCTL_RX_DCC_BYPASS_CM_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	This field is the offset value for RX DCC bypass common mode iDAC in full-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	This field is the offset value for RX DCC bypass common mode iDAC in half-rate mode. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.565 LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)cf
- **Exists:** Always

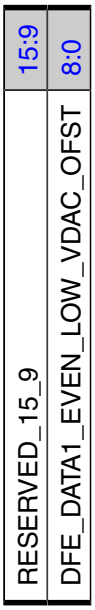


Table 2-569 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA1_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA1_EVEN_LOW_VDAC_OFST	R/W	This field is the offset value for DFE data1 even low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.566 LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd Low vDAC
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d0
- **Exists:** Always

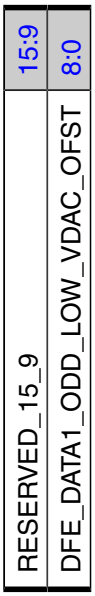


Table 2-570 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_DATA1_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_DATA1_ODD_LOW_VDAC_OFST	R/W	This field is the offset value for DFE data1 odd low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.567 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_13 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 13.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d1
- **Exists:** Always

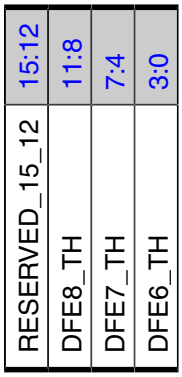


Table 2-571 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:8	DFE8_TH	R/W	This field is DFE TAP8 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:4	DFE7_TH	R/W	This field is DFE TAP7 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	DFE6_TH	R/W	This field is DFE TAP6 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.568 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_14 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 14.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d2
- **Exists:** Always

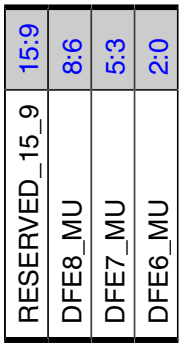


Table 2-572 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:6	DFE8_MU	R/W	This field is DFE TAP8 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
5:3	DFE7_MU	R/W	This field is DFE TAP7 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
2:0	DFE6_MU	R/W	This field is DFE TAP6 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7

2.1.569 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_15 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 15.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d3
- **Exists:** Always

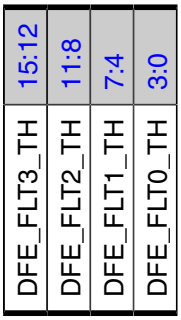


Table 2-573 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_15 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	DFE_FLT3_TH	R/W	This field is DFE float3 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:8	DFE_FLT2_TH	R/W	This field is DFE float2 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:4	DFE_FLT1_TH	R/W	This field is DFE float1 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	DFE_FLT0_TH	R/W	This field is DFE float0 correlation decision threshold (2^N-1). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.570 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_16 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 16.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d4
- **Exists:** Always

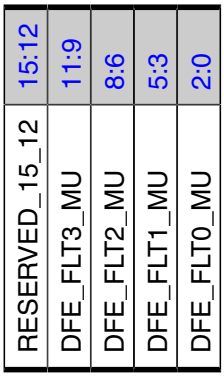


Table 2-574 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_16 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:9	DFE_FLT3_MU	R/W	This field is DFE float3 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
8:6	DFE_FLT2_MU	R/W	This field is DFE float2 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
5:3	DFE_FLT1_MU	R/W	This field is DFE float1 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7
2:0	DFE_FLT0_MU	R/W	This field is DFE float0 code update gain (2^N). **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x7

2.1.571 LANEN_DIG_RX_ADPTCTL_DFE_TAP6_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP6 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d5
- **Exists:** Always

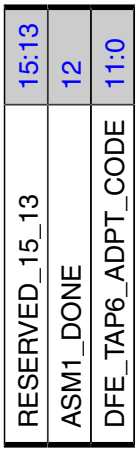


Table 2-575 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP6_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP6_ADPT_CODE	R	This field is the value of DFE TAP6 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.572 LANEN_DIG_RX_ADPTCTL_DFE_TAP7_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP7 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d6
- **Exists:** Always

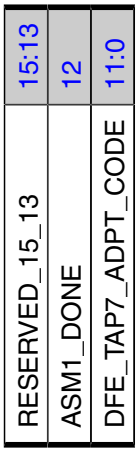


Table 2-576 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP7_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP7_ADPT_CODE	R	This field is the value of DFE TAP7 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.573 LANEN_DIG_RX_ADPTCTL_DFE_TAP8_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE TAP8 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d7
- **Exists:** Always

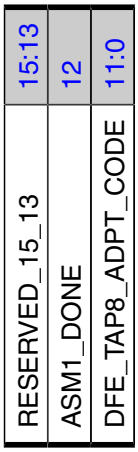


Table 2-577 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_TAP8_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_TAP8_ADPT_CODE	R	This field is the value of DFE TAP8 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.574 LANEN_DIG_RX_ADPTCTL_DFE_FLOAT0_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE Float0 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d8
- **Exists:** Always

Table 2-578 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_FLOAT0_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_FLOAT0_ADPT_CODE	R	This field is the value of DFE float0 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.575 LANEN_DIG_RX_ADPTCTL_DFE_FLOAT1_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE Float1 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)d9
- **Exists:** Always

Table 2-579 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_FLOAT1_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_FLOAT1_ADPT_CODE	R	This field is the value of DFE float1 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.576 LANEN_DIG_RX_ADPTCTL_DFE_FLOAT2_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE Float2 Adaptation Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)da
- **Exists:** Always

Table 2-580 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_FLOAT2_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_FLOAT2_ADPT_CODE	R	This field is the value of DFE float2 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.577 LANEN_DIG_RX_ADPTCTL_DFE_FLOAT3_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Value of DFE Float3 Adaptation Code
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)db
- **Exists:** Always

Table 2-581 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_FLOAT3_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	ASM1_DONE	R	This bit asserts when adaptation state machine 1 is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	DFE_FLOAT3_ADPT_CODE	R	This field is the value of DFE float3 adaptation code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.578 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG0 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)dc
- **Exists:** Always

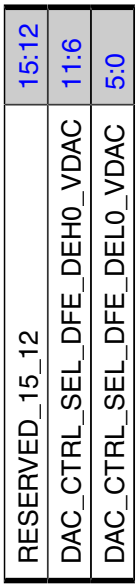


Table 2-582 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_DEH0_VDAC	R/W	This field is data0 even high vDAC address. **Value After Reset:** 0x17 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_DEL0_VDAC	R/W	This field is data0 even low vDAC address. **Value After Reset:** 0x18 **Exists:** Always **Reset Mask:** 0x3f

2.1.579 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG1 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)dd
- **Exists:** Always

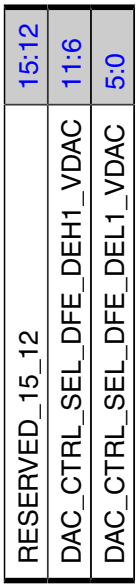


Table 2-583 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_DEH1_VDAC	R/W	This field is data1 even high DAC address. **Value After Reset:** 0x19 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_DEL1_VDAC	R/W	This field is data1 even low DAC address. **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0x3f

2.1.580 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG2 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)de
- **Exists:** Always

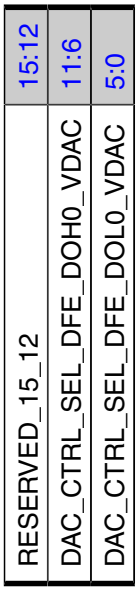


Table 2-584 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_DOH0_VDAC	R/W	This field is data0 odd high vDAC address. **Value After Reset:** 0x1b **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_DOL0_VDAC	R/W	This field is data0 odd low vDAC address. **Value After Reset:** 0x1c **Exists:** Always **Reset Mask:** 0x3f

2.1.581 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG3 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)df
- **Exists:** Always

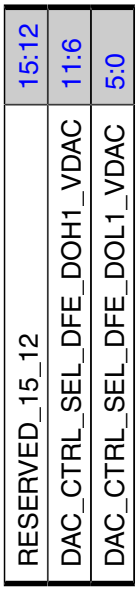


Table 2-585 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_DOH1_VDAC	R/W	This field is data1 odd high DAC address. **Value After Reset:** 0x1d **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_DOL1_VDAC	R/W	This field is data1 odd low DAC address. **Value After Reset:** 0x1e **Exists:** Always **Reset Mask:** 0x3f

2.1.582 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG4 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e0
- **Exists:** Always

Table 2-586 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_T3_IDAC	R/W	This field is T3 DFE DAC address. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_T2_IDAC	R/W	This field is T2 DFE DAC address. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x3f

2.1.583 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG5 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e1
- **Exists:** Always

Table 2-587 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_T5_IDAC	R/W	This field is T5 DFE DAC address. **Value After Reset:** 0x9 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_T4_IDAC	R/W	This field is T4 DFE DAC address. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0x3f

2.1.584 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG6 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 6 .
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e2
- **Exists:** Always

Table 2-588 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_T7_IDAC	R/W	This field is T7 DFE DAC address. **Value After Reset:** 0xb **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_T6_IDAC	R/W	This field is T6 DFE DAC address. **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x3f

2.1.585 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG7 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 7.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e3
- **Exists:** Always

Table 2-589 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DAC_CTRL_SEL_DFE_T8_IDAC	R/W	This field is T8 DFE DAC address. **Value After Reset:** 0xc **Exists:** Always **Reset Mask:** 0x3f

2.1.586 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG8 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 8.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e4
- **Exists:** Always

Table 2-590 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_F1_IDAC	R/W	This field is float 1 DFE DAC address. **Value After Reset:** 0xe **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_F0_IDAC	R/W	This field is float 0 DFE DAC address. **Value After Reset:** 0xd **Exists:** Always **Reset Mask:** 0x3f

2.1.587 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG9 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 9.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e5
- **Exists:** Always

Table 2-591 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_DFE_F3_IDAC	R/W	This field is float 3 DFE DAC address. **Value After Reset:** 0x10 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_DFE_F2_IDAC	R/W	This field is float 2 DFE DAC address. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0x3f

2.1.588 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG10 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 10.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e6
- **Exists:** Always

Table 2-592 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_BYP_EVEN_VDAC	R/W	This field is DEB DFE DAC address. **Value After Reset:** 0x21 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_BYP_ODD_VDAC	R/W	This field is DOB DFE DAC address. **Value After Reset:** 0x22 **Exists:** Always **Reset Mask:** 0x3f

2.1.589 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG11 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 11.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e7
- **Exists:** Always

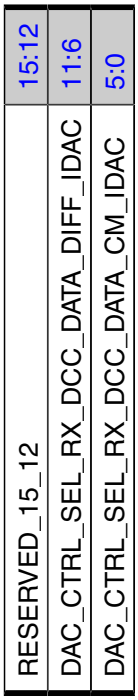


Table 2-593 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_RX_DCC_DATA_DIFF_IDAC	R/W	This field is DCC data clock diff DAC address. **Value After Reset:** 0x16 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_RX_DCC_DATA_CM_IDAC	R/W	This field is DCC data CM DAC address. **Value After Reset:** 0x15 **Exists:** Always **Reset Mask:** 0x3f

2.1.590 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG12 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 12.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e8
- **Exists:** Always

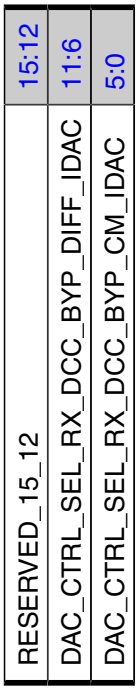


Table 2-594 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_RX_DCC_BYP_DIFF_IDAC	R/W	This field is DCC bypass clock diff DAC address. **Value After Reset:** 0x12 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_RX_DCC_BYP_CM_IDAC	R/W	This field is DCC bypass CM DAC address. **Value After Reset:** 0x11 **Exists:** Always **Reset Mask:** 0x3f

2.1.591 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG13 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 13.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)e9
- **Exists:** Always

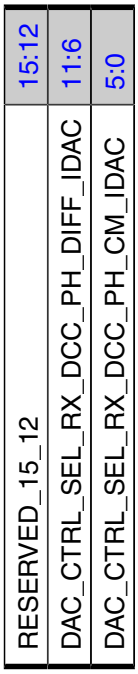


Table 2-595 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_RX_DCC_PH_DIFF_IDAC	R/W	This field is DCC phase clock diff DAC address. **Value After Reset:** 0x14 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_RX_DCC_PH_CM_IDAC	R/W	This field is DCC phase CM DAC address. **Value After Reset:** 0x13 **Exists:** Always **Reset Mask:** 0x3f

2.1.592 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_17 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 17.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ea
- **Exists:** Always

Table 2-596 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_17 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	CTLE_MAX_SAT	R/W	This field is CTLE upper saturation level. **Value After Reset:** 0x1f **Exists:** Always **Reset Mask:** 0x1f
4:0	CTLE_MIN_SAT	R/W	This field is CTLE lower saturation level. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.593 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_18 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Configuration Register 18.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)eb
- **Exists:** Always

Table 2-597 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_18 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	DFE1_MAX_SAT	R/W	This field is DFE TAP1 upper saturation level. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.594 LANEN_DIG_RX_ADPTCTL_ADPT_CFG_19 (for N = 0; N <= NLANES-1)

- **Description:** ADPT and SSM Fast Flags
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ec
- **Exists:** Always

Table 2-598 Fields for Register: LANEN_DIG_RX_ADPTCTL_ADPT_CFG_19 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_AFE_DFE_SETTLE	R/W	This bit enables fast AFE and DFE settling time (simulation only). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.595 LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG14 (for N = 0; N <= NLANES-1)

- **Description:** DAC Address Config 14.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ed
- **Exists:** Always

Table 2-599 Fields for Register: LANEN_DIG_RX_ADPTCTL_DAC_CTRL_SEL_CFG14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	DAC_CTRL_SEL_ERR_BYP_EVEN_VDAC	R/W	This field is DEB DFE DAC address. **Value After Reset:** 0x26 **Exists:** Always **Reset Mask:** 0x3f
5:0	DAC_CTRL_SEL_ERR_BYP_ODD_VDAC	R/W	This field is DOB DFE DAC address. **Value After Reset:** 0x27 **Exists:** Always **Reset Mask:** 0x3f

2.1.596 LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Error Bypass Even vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ee
- **Exists:** Always

Table 2-600 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_ERR_BYPASS_EVEN_VDAC_OFST	R/W	This field is the offset value for DFE error bypass even vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.597 LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset values for RX DFE Error Bypass Odd vDAC.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)ef
- **Exists:** Always

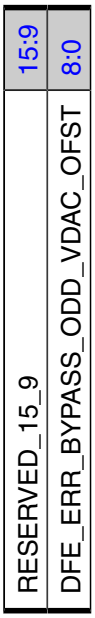


Table 2-601 Fields for Register: LANEN_DIG_RX_ADPTCTL_DFE_ERR_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_ERR_BYPASS_ODD_VDAC_OFST	R/W	This field is the offset value for DFE error bypass odd vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.598 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_0 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f1
- **Exists:** Always

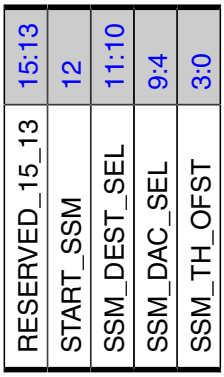


Table 2-602 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	START_SSM	R/W	This bit starts search state machine. This register-bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:10	SSM_DEST_SEL	R/W	This field selects update destination. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:4	SSM_DAC_SEL	R/W	This field is the calibration DAC address. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
3:0	SSM_TH_OFST	R/W	This field is the correlation threshold offset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.599 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_1 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f2
- **Exists:** Always

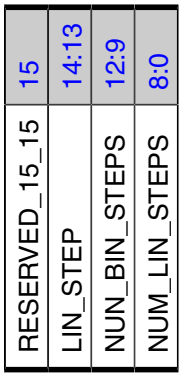


Table 2-603 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:13	LIN_STEP	R/W	This field is linear step size (2^N). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
12:9	NUN_BIN_STEPS	R/W	This field is the number of binary search steps. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
8:0	NUM_LIN_STEPS	R/W	This field is the number of linear search steps. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x1ff

2.1.600 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_2 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f3
- **Exists:** Always

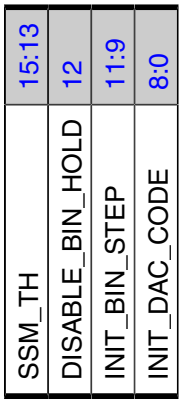


Table 2-604 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	SSM_TH	R/W	This field is the correlation threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
12	DISABLE_BIN_HOLD	R/W	This bit disables DAC hold during binary seach. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:9	INIT_BIN_STEP	R/W	This field is initial binary search step size (2^N). **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x7
8:0	INIT_DAC_CODE	R/W	This field is initial code (center of a 9-bit DAC). **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.601 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_3 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f4
- **Exists:** Always

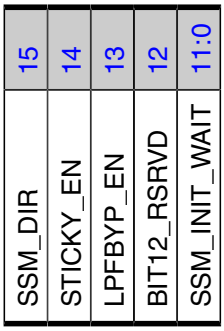


Table 2-605 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SSM_DIR	R/W	This bit selects update direction. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	STICKY_EN	R/W	This bit enables sticky linear search **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	LPFBYP_EN	R/W	This bit enables LPF bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	BIT12_RSRVD	R/W	This bit is reserved. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	SSM_INIT_WAIT	R/W	This field is the number of initial wait cycles following DAC write. **Value After Reset:** 0x1f4 **Exists:** Always **Reset Mask:** 0xfff

2.1.602 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_4 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f5
- **Exists:** Always

Table 2-606 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	SSM_N_WAIT_LPFBYP	R/W	This field is the number of wait cycles while LPF bypass is enabled. **Value After Reset:** 0x5 **Exists:** Always **Reset Mask:** 0xff
7:0	SSM_N_WAIT	R/W	This field is the number of wait cycles following an update. **Value After Reset:** 0x32 **Exists:** Always **Reset Mask:** 0xff

2.1.603 LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_5 (for N = 0; N <= NLANES-1)

- **Description:** Search State Machine Configuration Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f6
- **Exists:** Always

Table 2-607 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_SSM_CFG_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RESET_SSM	R/W	This bit resets the search state machine. This register-bit is self-clearing. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.604 LANEN_DIG_RX_ADPTCTL_SSM_FINAL_CODE (for N = 0; N <= NLANES-1)

- **Description:** Value of Final DAC Code.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+0)f7
- **Exists:** Always

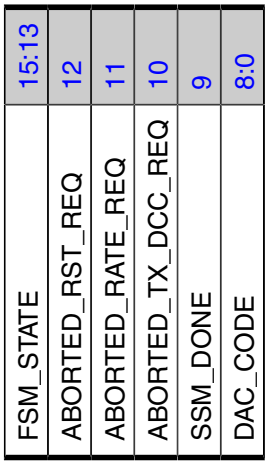


Table 2-608 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_FINAL_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	FSM_STATE	R	This field is the current value of the SSM FSM state register. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	ABORTED_RST_REQ	R	This bit is SSM aborted when receiving reset request. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	ABORTED_RATE_REQ	R	This bit is SSM aborted when receiving rate change request. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-608 Fields for Register: LANEN_DIG_RX_ADPTCTL_SSM_FINAL_CODE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
10	ABORTED_TX_DCC_REQ	R	This bit is SSM aborted when receiving TX DCC update request. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	SSM_DONE	R	This bit asserts when binary/linear state machine is done. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:0	DAC_CODE	R	This field is the result code of the SSM search. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.605 LANEN_DIG_RX_STAT_LD_VAL_1 (for N = 0; N <= NLANES-1)

- **Description:** Stat Load Value for the Sample Counter 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)00
- **Exists:** Always

Table 2-609 Fields for Register: LANEN_DIG_RX_STAT_LD_VAL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SC1_START	R/W	This bit starts sample counter 1. This is a self-clearing bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:0	SC1_LD_VAL	R/W	This field is sample counter 1 load value. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7fff

2.1.606 LANEN_DIG_RX_STAT_DATA_MSK (for N = 0; N <= NLANES-1)

- **Description:** Stat Data Mask Bits [15:0]
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)01
- **Exists:** Always

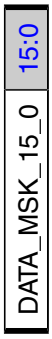


Table 2-610 Fields for Register: LANEN_DIG_RX_STAT_DATA_MSK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	DATA_MSK_15_0	R/W	This field is the value of data_msk_r[15:0]. **Value After Reset:** 0xffff **Exists:** Always **Reset Mask:** 0xffff

2.1.607 LANEN_DIG_RX_STAT_MATCH_CTL0 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)02
- **Exists:** Always

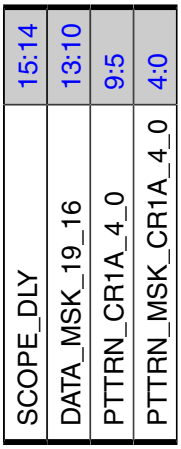


Table 2-611 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	SCOPE_DLY	R/W	This field is the number of clock cycle delays on scope_data_rxclk. An additional MSB is added in SCOPE_DLY_2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
13:10	DATA_MSK_19_16	R/W	This field is the value of data_msk_r[19:16]. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf
9:5	PTTRN_CR1A_4_0	R/W	This field is the value of pattern A for 1st correlator (bits 4:0). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
4:0	PTTRN_MSK_CR1A_4_0	R/W	This field is the value of pattern A mask for 1st correlator (bits 4:0). **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0x1f

2.1.608 LANEN_DIG_RX_STAT_MATCH_CTL1 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)03
- **Exists:** Always

Table 2-612 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	PTTRN_CR1A_ADPT_EN	R/W	This bit enables ORing of adaptation pattern with pattern CR1A. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10:6	PTTRN_CR1B_4_0	R/W	This field is the value of pattern B for 1st correlator (bits 4:0). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
5:1	PTTRN_MSK_CR1B_4_0	R/W	This field is the value of pattern B mask for 1st correlator (bits 4:0). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
0	PTTRN_CR1B_EN	R/W	This bit enables pattern B matching for 1st correlator. **Value After Reset:** 0x0

Table 2-612 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.609 LANEN_DIG_RX_STAT_STAT_CTL0 (for N = 0; N <= NLANES-1)

- **Description:** Stat Controls Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)04
- **Exists:** Always

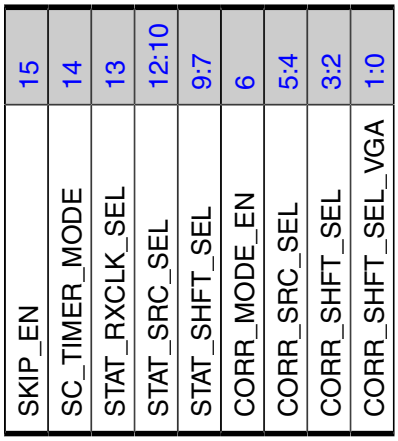


Table 2-613 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SKIP_EN	R/W	This bit is the value of skip_en_r. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	SC_TIMER_MODE	R/W	This bit is the sample counter operation mode. ■ 0x0 - counts number of matched samples. ■ 0x1 - counts clock cycles (i.e. a timer). **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	STAT_RXCLK_SEL	R/W	This bit selects stat clock. ■ 0x0 - ref_range_clk ■ 0x1 - rx_dig_qword_clk (i.e. rx dword clk) Before changing stat_rxclk_sel_r from 1->0, the rx_dig_qword_clk must be active (i.e. enabled). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-613 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12:10	STAT_SRC_SEL	R/W	This field selects stat source input. ■ 0x0 - {40{rx_cal_result}} ■ 0x1 - {{20{scope_data_rxclk_dly_s01}},{20{scope_data_rxclk_dly}}} ■ 0x2 - rx_phase[39:0] ■ 0x3 - rx_error[39:0] ■ 0x4 - rx_data[39:0] ■ 0x5 - rx_phdir[39:0] ■ 0x6 - 40'hFF_FFFF_FFFF ■ 0x7 - {40{tx_dcc_cal_result}} **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
9:7	STAT_SHFT_SEL	R/W	This field selects stat source shift value. ■ 0x0 - Correlate N-1 -> N+6 (use N for offset calibration) ■ 0x1 - Correlate N+1 -> N+8 (for taps1-8) ■ 0x2 - Correlate N+9 -> N+16 ■ 0x3 - Correlate N+17 -> N+24 ■ 0x4 - Correlate N+25 -> N+32 ■ 0x5 - Correlate N+33 -> N+39 Setting 0x5 is only used in 20b mode (for checking corr on bits 33-39). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
6	CORR_MODE_EN	R/W	This bit enables correlation mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	CORR_SRC_SEL	R/W	This field selects correlation input source. ■ 0x0 - rx_error[39:0] ■ 0x1 - rx_phase[39:0] ■ 0x2 - {{20{scope_data_rxclk_dly_s01}},{20{scope_data_rxclk_dly}}} ■ 0x3 - No correlation **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-613 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3:2	CORR_SHFT_SEL	R/W	This field selects shift for phase. ■ 0x0 - corr_src[40:1] ■ 0x1 - corr_src[41:2] ■ 0x2 - corr_src[42:3] ■ 0x3 - corr_src[43:4] **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1:0	CORR_SHFT_SEL_VGA	R/W	This field selects shift for error going to VGA. ■ 0x0 - corr_src_vga[40:1] ■ 0x1 - corr_src_vga[41:2] ■ 0x2 - corr_src_vga[42:3] ■ 0x3 - corr_src_vga[43:4] **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.610 LANEN_DIG_RX_STAT_STAT_CTL1 (for N = 0; N <= NLANES-1)

- **Description:** Stat Controls Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)05
- **Exists:** Always

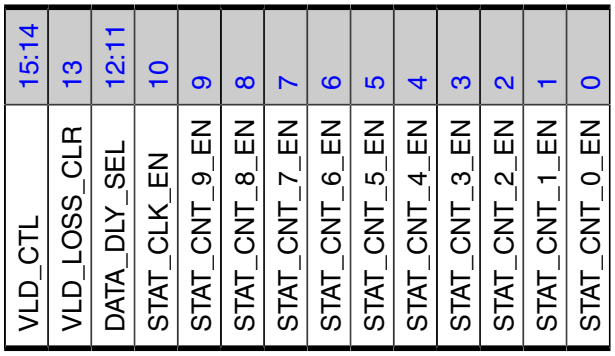


Table 2-614 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	VLD_CTL	R/W	This field is gating configuration of stats collection. ■ 0x0 - ignore both cdr_valid and rx_valid ■ 0x1 - gate stats collection with cdr_valid ■ 0x2 - gate stats collection with rx_valid ■ 0x3 - ignore both cdr_valid and rx_valid **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
13	VLD_LOSS_CLR	R/W	This bit is the clearing of stats collection upon loss of valid. ■ 0x0 - hold sample and stat counters ■ 0x1 - clear sample and stat counters **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:11	DATA_DLY_SEL	R/W	This field is the number of clock cycle delays on rx_data[19:0]. An additional MSB is added in DATA_DLY_SEL_2 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
10	STAT_CLK_EN	R/W	This bit is the clock gate enable for stat clock. **Value After Reset:** 0x0

Table 2-614 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9	STAT_CNT_9_EN	R/W	This bit enables for stat counter 9. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	STAT_CNT_8_EN	R/W	This bit enables for stat counter 8. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	STAT_CNT_7_EN	R/W	This bit enables for stat counter 7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	STAT_CNT_6_EN	R/W	This bit enables for stat counter 6. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	STAT_CNT_5_EN	R/W	This bit enables for stat counter 5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	STAT_CNT_4_EN	R/W	This bit enables for stat counter 4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	STAT_CNT_3_EN	R/W	This bit enables for stat counter 3. Only counter to be enabled by default, since used for offset calibration. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	STAT_CNT_2_EN	R/W	This bit enables for stat counter 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-614 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	STAT_CNT_1_EN	R/W	This bit enables for stat counter 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	STAT_CNT_0_EN	R/W	This bit enables for stat counter 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.611 LANEN_DIG_RX_STAT_SMPL_CNT1 (for N = 0; N <= NLANES-1)

- **Description:** Bits[14:0] of Sample Counter 1 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)06
- **Exists:** Always

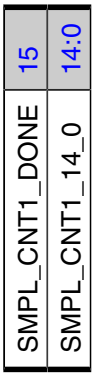


Table 2-615 Fields for Register: LANEN_DIG_RX_STAT_SMPL_CNT1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	SMPL_CNT1_14_0	R	This field is the current value of sample counter 1 bits[14:0]. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.612 LANEN_DIG_RX_STAT_SMPL_CNT1_29DN15 (for N = 0; N <= NLANES-1)

- **Description:** Bits[29:15] of Sample Counter 1 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)07
- **Exists:** Always

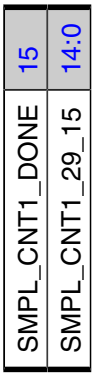


Table 2-616 Fields for Register: LANEN_DIG_RX_STAT_SMPL_CNT1_29DN15 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	SMPL_CNT1_29_15	R	This field is the current value of sample counter 1 bits[29:15]. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.613 LANEN_DIG_RX_STAT_SMPL_CNT1_44DN30 (for N = 0; N <= NLANES-1)

- **Description:** Bits[44:30] of Sample Counter 1 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)08
- **Exists:** Always

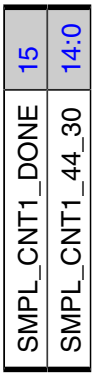


Table 2-617 Fields for Register: LANEN_DIG_RX_STAT_SMPL_CNT1_44DN30 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1 bits[44:30]. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	SMPL_CNT1_44_30	R	This field is the current value of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.614 LANEN_DIG_RX_STAT_STAT_CNT_0 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 0 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)09
- **Exists:** Always

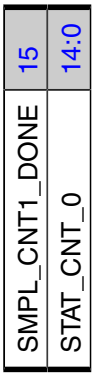


Table 2-618 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_0	R	This field is the current value of stat counter 0. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.615 LANEN_DIG_RX_STAT_STAT_CNT_1 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 1 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0a
- **Exists:** Always

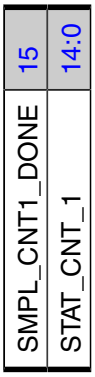


Table 2-619 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_1	R	This field is the current value of stat counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.616 LANEN_DIG_RX_STAT_STAT_CNT_2 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 2 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0b
- **Exists:** Always

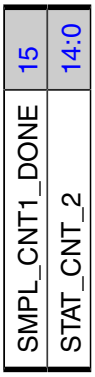


Table 2-620 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_2	R	This field is the current value of stat counter 2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.617 LANEN_DIG_RX_STAT_STAT_CNT_3 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 3 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0c
- **Exists:** Always

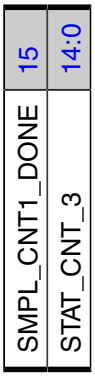


Table 2-621 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_3	R	This field is the current value of stat counter 3. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.618 LANEN_DIG_RX_STAT_STAT_CNT_4 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 4 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0d
- **Exists:** Always

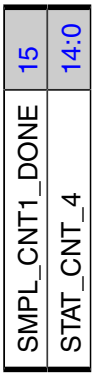


Table 2-622 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_4	R	This field is the current value of stat counter 4. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.619 LANEN_DIG_RX_STAT_STAT_CNT_5 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 5 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0e
- **Exists:** Always

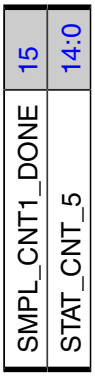


Table 2-623 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_5	R	This field is the current value of stat counter 5. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.620 LANEN_DIG_RX_STAT_STAT_CNT_6 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 6 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)0f
- **Exists:** Always

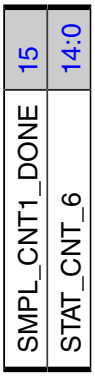


Table 2-624 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_6	R	This field is the current value of stat counter 6. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.621 LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL (for N = 0; N <= NLANES-1)

- **Description:** Calibration Comparator Control.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)10
- **Exists:** Always

Table 2-625 Fields for Register: LANEN_DIG_RX_STAT_CAL_COMP_CLK_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:3	REF_DIV_CNT	R/W	This field is ref range clock count (e.g. 5'd3 = 4 ref_range cycles). **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
2:0	PRECHRGE_CNT	R/W	This field is the precharge count (e.g. 5'd1 = 2 ref_range cycles). **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x7

2.1.622 LANEN_DIG_RX_STAT_MATCH_CTL2 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)11
- **Exists:** Always

Table 2-626 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	PTTRN_CR1A_19_5	R/W	This field is the value of pattern A for 1st correlator (bits 19:5). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff

2.1.623 LANEN_DIG_RX_STAT_MATCH_CTL3 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)12
- **Exists:** Always

Table 2-627 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	PTTRN_MSK_CR1A_19_5	R/W	This field is the value of pattern A mask for 1st correlator (bits 19:5). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff

2.1.624 LANEN_DIG_RX_STAT_MATCH_CTL4 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)13
- **Exists:** Always

Table 2-628 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	PTTRN_CR1B_19_5	R/W	This field is the value of pattern B for 1st correlator (bits 19:5). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff

2.1.625 LANEN_DIG_RX_STAT_MATCH_CTL5 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 5.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)14
- **Exists:** Always

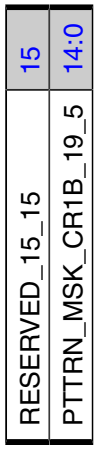


Table 2-629 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	PTTRN_MSK_CR1B_19_5	R/W	This field is the value of pattern B mask for 1st correlator (bits 19:5). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7fff

2.1.626 LANEN_DIG_RX_STAT_STAT_CTL2 (for N = 0; N <= NLANES-1)

- **Description:** Stat Controls Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)15
- **Exists:** Always

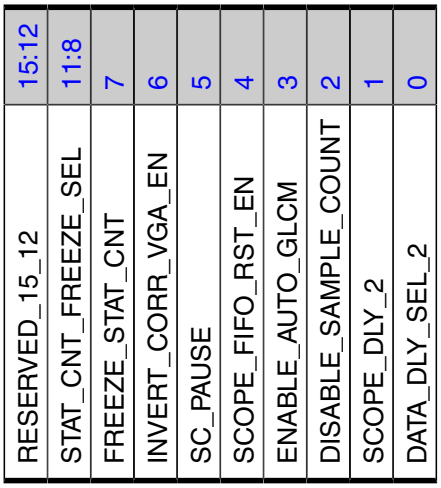


Table 2-630 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:8	STAT_CNT_FREEZE_SEL	R/W	This field selects which statcnt to read as shadow register. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
7	FREEZE_STAT_CNT	R/W	When this bit is enabled, it freezes the value of stat_cnt_n. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	INVERT_CORR_VGA_EN	R/W	This bit controls the inversion of corr_src_vga in stat_cnt_1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	SC_PAUSE	R/W	This bit pauses the sample counter and stat counters. **Value After Reset:** 0x0

Table 2-630 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
4	SCOPE_FIFO_RST_EN	R/W	This bit enables resetting the scope fifo when the rx_ana_scope_en signal is de-asserted. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	ENABLE_AUTO_GLCM	R/W	This bit enables the automatic switching of the stats GLCM. This will automatically switch the stats GLCM to the ref_range_clk, when the rx_dig_qword_clk is disabled. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	DISABLE_SAMPLE_COUNT	R/W	This bit disables the sample counter. This will keep the sample counter at the current value. It is used to determine the error rate over long periods of time. This must be set back to zero to allow the sample count to complete its countdown and finish the stats call. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SCOPE_DLY_2	R/W	This is an additional MSB bit for SCOPE_DLY to extend the delay range to 0->7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	DATA_DLY_SEL_2	R/W	This is an additional MSB bit for DATA_DLY_SEL to extend the delay range to 0->7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.627 LANEN_DIG_RX_STAT_STAT_STOP (for N = 0; N <= NLANES-1)

- **Description:** Stat Stop Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)16
- **Exists:** Always

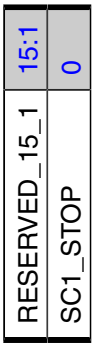


Table 2-631 Fields for Register: LANEN_DIG_RX_STAT_STAT_STOP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SC1_STOP	R/W	This bit stops sample counters 1 and associated stat counters. This is a self-clearing bit, and requires re-start of sample counter 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.628 LANEN_DIG_RX_STAT_STAT_CNT_7 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 7 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)17
- **Exists:** Always

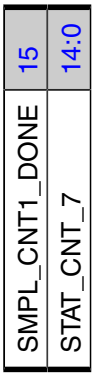


Table 2-632 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_7	R	This field is the current value of stat counter 7. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.629 LANEN_DIG_RX_STAT_STAT_CNT_8 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 8 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)18
- **Exists:** Always

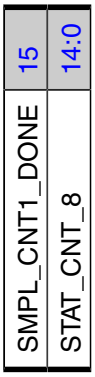


Table 2-633 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_8	R	This field is the current value of stat counter 8. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.630 LANEN_DIG_RX_STAT_STAT_CNT_9 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 9 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)19
- **Exists:** Always

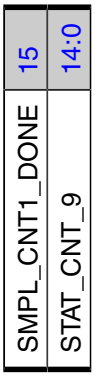


Table 2-634 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_9	R	This field is the current value of stat counter 9. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.631 LANEN_DIG_RX_STAT_STAT_CNT_10 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 10 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1a
- **Exists:** Always

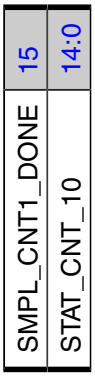


Table 2-635 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_10	R	This field is the current value of stat counter 10. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.632 LANEN_DIG_RX_STAT_STAT_CNT_11 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 11 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1b
- **Exists:** Always

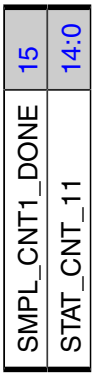


Table 2-636 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_11	R	This field is the current value of stat counter 11. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.633 LANEN_DIG_RX_STAT_STAT_CNT_12 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 12 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1c
- **Exists:** Always

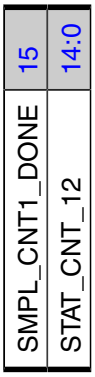


Table 2-637 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_12	R	This field is the current value of stat counter 12. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.634 LANEN_DIG_RX_STAT_STAT_CNT_13 (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter 13 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1d
- **Exists:** Always

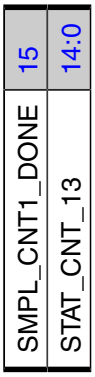


Table 2-638 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_13	R	This field is the current value of stat counter 13. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.635 LANEN_DIG_RX_STAT_STAT_CTL3 (for N = 0; N <= NLANES-1)

- **Description:** Stat Controls Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1e
- **Exists:** Always

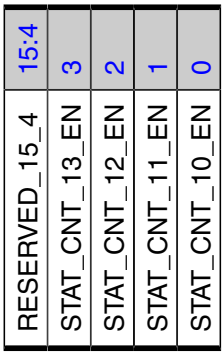


Table 2-639 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	STAT_CNT_13_EN	R/W	This bit enables for stat counter 13. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	STAT_CNT_12_EN	R/W	This bit enables for stat counter 12. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	STAT_CNT_11_EN	R/W	This bit enables for stat counter 11. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	STAT_CNT_10_EN	R/W	This bit enables for stat counter 10. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.636 LANEN_DIG_RX_STAT_STAT_CTL4 (for N = 0; N <= NLANES-1)

- **Description:** Stat Controls Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)1f
- **Exists:** Always

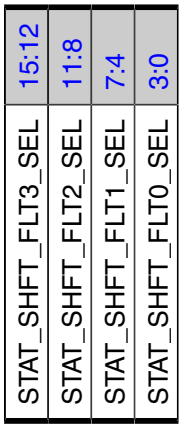


Table 2-640 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	STAT_SHFT_FLT3_SEL	R/W	This field selects stat source shift value for float TAP3. The same encoding as STAT_SHFT_FLT0_SEL. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
11:8	STAT_SHFT_FLT2_SEL	R/W	This field selects stat source shift value for float TAP2. The same encoding as STAT_SHFT_FLT0_SEL. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
7:4	STAT_SHFT_FLT1_SEL	R/W	This field selects stat source shift value for float TAP1. The same encoding as STAT_SHFT_FLT0_SEL. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
3:0	STAT_SHFT_FLT0_SEL	R/W	This field selects stat source shift value for float TAP0. ■ 0x0 - Not used ■ 0x1 - Correlate N+9 ■ 0x2 - Correlate N+10 ■ 0x3 - Correlate N+11 ■ 0x4 - Correlate N+12 ■ 0x5 - Correlate N+13 ■ 0x6 - Correlate N+14

Table 2-640 Fields for Register: LANEN_DIG_RX_STAT_STAT_CTL4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			■ 0x7 - Correlate N+15 ■ 0x8 - Correlate N+16 ■ 0x9 - Correlate N+17 ■ 0x10 - Correlate N+18 ■ 0x11 - Correlate N+19 ■ 0x12 - Correlate N+20 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf

2.1.637 LANEN_DIG_RX_STAT_MATCH_CTL6 (for N = 0; N <= NLANES-1)

- **Description:** Stat Match Controls Register 6.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)20
- **Exists:** Always

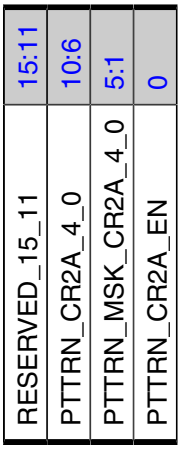


Table 2-641 Fields for Register: LANEN_DIG_RX_STAT_MATCH_CTL6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:6	PTTRN_CR2A_4_0	R/W	This field is the value of pattern 2A for 1st correlator (bits 4:0). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
5:1	PTTRN_MSK_CR2A_4_0	R/W	This field is the value of pattern 2A mask for 1st correlator (bits 4:0). **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f
0	PTTRN_CR2A_EN	R/W	This bit enables pattern 2A matching for 1st correlator. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.638 LANEN_DIG_RX_STAT_STAT_CNT_N_SHD (for N = 0; N <= NLANES-1)

- **Description:** Stat Counter N Shadow Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)21
- **Exists:** Always

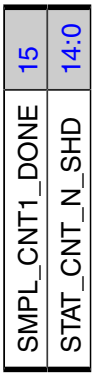


Table 2-642 Fields for Register: LANEN_DIG_RX_STAT_STAT_CNT_N_SHD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT1_DONE	R	This bit is the status of sample counter 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	STAT_CNT_N_SHD	R	This field is the current value of stat counter N with the freeze bit enabled. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.639 LANEN_DIG_RX_STAT_SMPL_CNT2 (for N = 0; N <= NLANES-1)

- **Description:** Sample Counter 2 Status.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)22
- **Exists:** Always

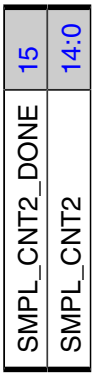


Table 2-643 Fields for Register: LANEN_DIG_RX_STAT_SMPL_CNT2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SMPL_CNT2_DONE	R	This bit is the status of sample counter 2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	SMPL_CNT2	R	This field is the current value of sample counter 2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.640 LANEN_DIG_RX_STAT_LD_VAL_EXT_1 (for N = 0; N <= NLANES-1)

- **Description:** Bits[14:0] of Stat Extended Load Value for the Sample Counter 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)23
- **Exists:** Always

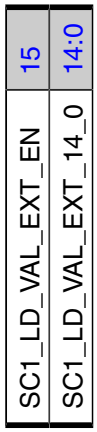


Table 2-644 Fields for Register: LANEN_DIG_RX_STAT_LD_VAL_EXT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SC1_LD_VAL_EXT_EN	R/W	This bit enables for extended load value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:0	SC1_LD_VAL_EXT_14_0	R/W	This field is the sample counter 1 load value. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7fff

2.1.641 LANEN_DIG_RX_STAT_LD_VAL_EXT_2 (for N = 0; N <= NLANES-1)

- **Description:** Bits[29:15] of Stat Extended Load Value for the Sample Counter 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)24
- **Exists:** Always

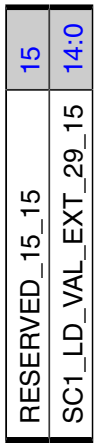


Table 2-645 Fields for Register: LANEN_DIG_RX_STAT_LD_VAL_EXT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:0	SC1_LD_VAL_EXT_29_15	R/W	This field is the sample counter 1 load value. **Value After Reset:** 0x40 **Exists:** Always **Reset Mask:** 0x7fff

2.1.642 LANEN_DIG_ANA_TX_PWR_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX PWR UP/DN Signals Going to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)40
- **Exists:** Always

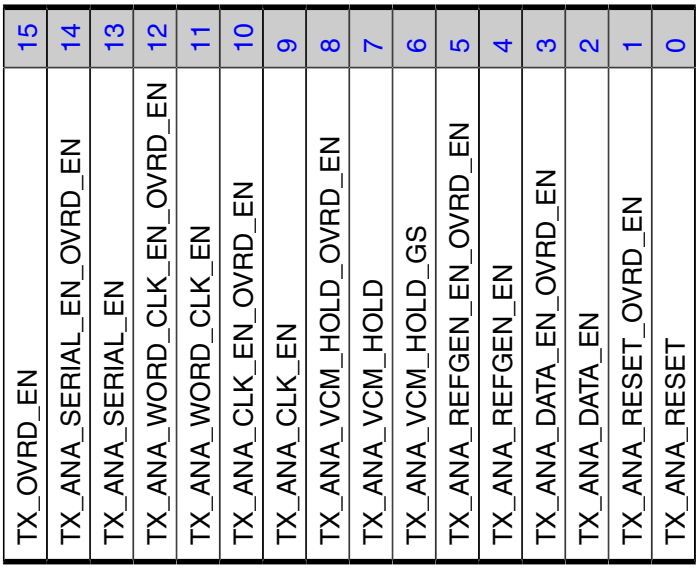


Table 2-646 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_OVRD_EN	R/W	This bit is the global override enable for all outputs controlled by TX_PWR_OVRD_OUT_0,1,2 registers. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_ANA_SERIAL_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_serial_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	TX_ANA_SERIAL_EN	R/W	This bit is the override value for tx_ana_serial_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	TX_ANA_WORD_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-646 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	TX_ANA_WORD_CLK_EN	R/W	This bit is the override value for tx_ana_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	TX_ANA_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_ANA_CLK_EN	R/W	This bit is the override value for tx_ana_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TX_ANA_VCM_HOLD_OVRD_EN	R/W	This bit is the override enable for tx_ana_vcm_hold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_ANA_VCM_HOLD	R/W	This bit is the override value for tx_ana_vcm_hold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_ANA_VCM_HOLD_GS	R/W	This bit is the override value for tx_ana_vcm_hold_gs. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_ANA_REFGEN_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_refgen_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_ANA_REFGEN_EN	R/W	This bit is the override value for tx_ana_refgen_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_ANA_DATA_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_data_en. **Value After Reset:** 0x0

Table 2-646 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	TX_ANA_DATA_EN	R/W	This bit is the override value for tx_ana_data_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_ANA_RESET_OVRD_EN	R/W	This bit is the override enable for tx_ana_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ANA_RESET	R/W	This bit is the override value for tx_ana_reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.643 LANEN_DIG_ANA_TX_PWR_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX PWR UP/DN Signals Going to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)41
- **Exists:** Always

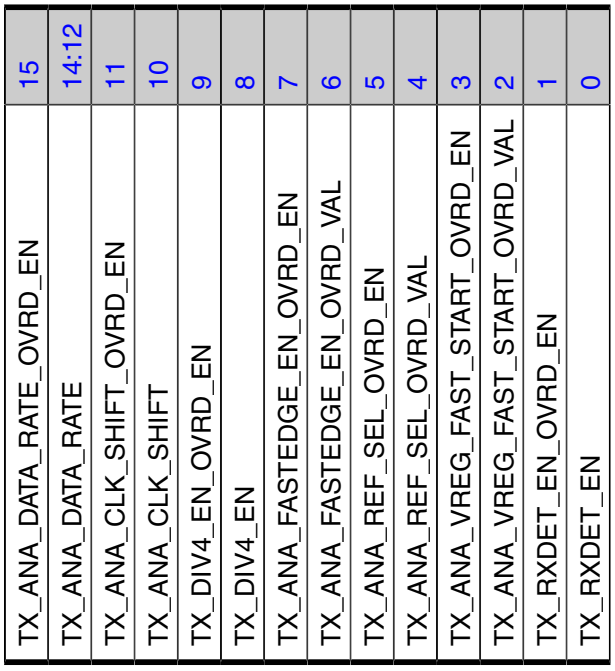


Table 2-647 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_ANA_DATA_RATE_OVRD_EN	R/W	This bit is the override enable for tx_ana_data_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:12	TX_ANA_DATA_RATE	R/W	This field is the override value for tx_ana_data_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
11	TX_ANA_CLK_SHIFT_OVRD_EN	R/W	This bit is the override enable for tx_ana_clk_shift. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	TX_ANA_CLK_SHIFT	R/W	This bit is the override value for tx_ana_clk_shift. **Value After Reset:** 0x0

Table 2-647 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9	TX_DIV4_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TX_DIV4_EN	R/W	This bit is the override value for tx_ana_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_ANA_FASTEDGE_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_fastedge_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_ANA_FASTEDGE_EN_OVRD_VAL	R/W	This bit is the override value for tx_ana_fastedge_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_ANA_REF_SEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_ref_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_ANA_REF_SEL_OVRD_VAL	R/W	This bit is the override value for tx_ana_ref_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_ANA_VREG_FAST_START_OVRD_EN	R/W	This bit is the override enable for tx_ana_vreg_fast_start. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_ANA_VREG_FAST_START_OVRD_VAL	R/W	This bit is the override value for tx_ana_vreg_fast_start. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-647 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	TX_RXDET_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_rxdet_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_RXDET_EN	R/W	This bit is the override value for tx_ana_rxdet_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.644 LANEN_DIG_ANA_TX_PWR_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX PWR UP/DN Signals Going to ANA, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)42
- **Exists:** Always

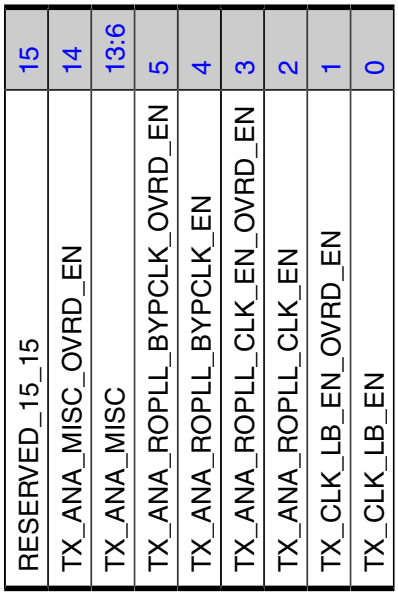


Table 2-648 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_ANA_MISC_OVRD_EN	R/W	This bit is the override enable for tx_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:6	TX_ANA_MISC	R/W	This field is the override value for tx_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
5	TX_ANA_ROPLL_BYPCLK_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_bypclk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-648 Fields for Register: LANEN_DIG_ANA_TX_PWR_OVRD_OUT_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	TX_ANA_ROPLL_BYPCLK_EN	R/W	This bit is the override value for tx_ana_ropll_bypclk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_ANA_ROPLL_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_ANA_ROPLL_CLK_EN	R/W	This bit is the override value for tx_ana_ropll_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_CLK_LB_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_clk_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CLK_LB_EN	R/W	This bit is the override value for tx_ana_clk_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.645 LANEN_DIG_ANA_TX_TERM_CODE_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Value for TX Termination Code Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)43
- **Exists:** Always

Table 2-649 Fields for Register: LANEN_DIG_ANA_TX_TERM_CODE_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_TERM_OVRD_EN	R/W	This bit is the override enable for the tx_ana_term_code[9:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_TERM_CODE	R/W	This field overrides the tx_ana_term_code[9:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.646 LANEN_DIG_ANA_TX_TERM_CODE_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Value for TX Termination Code Clocks Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)44
- **Exists:** Always

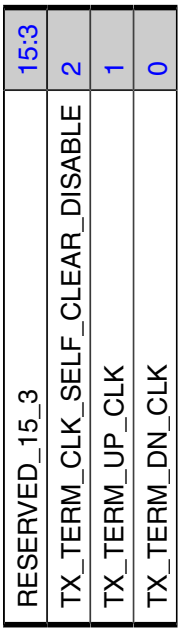


Table 2-650 Fields for Register: LANEN_DIG_ANA_TX_TERM_CODE_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	TX_TERM_CLK_SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the tx_ana_term_up/dn_clk register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_TERM_UP_CLK	R/W	This bit is the override for TX term UP clock. This bit is self-clearing (4 cr_clks later). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-650 Fields for Register: LANEN_DIG_ANA_TX_TERM_CODE_CLK_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	TX_TERM_DN_CLK	R/W	This bit is the override for TX term DN clock. This bit is self_clearing (4 cr_clks later). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.647 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX EQ Signals Going to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)45
- **Exists:** Always

Table 2-651 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_EQ_OVRD_EN	R/W	This bit is the global override enable for tx eq signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_ANA_CTRL_ATTEN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ctrl_atten. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:0	TX_ANA_CTRL_ATTEN_13_0	R/W	This field is the override value for tx_ana_ctrl_atten[13:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.648 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX EQ Signals Going to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)46
- **Exists:** Always

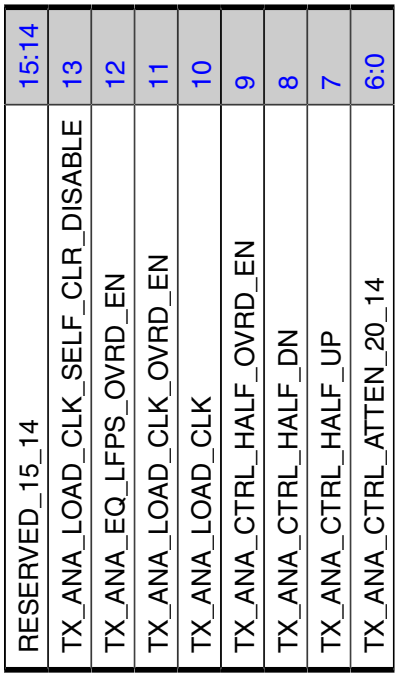


Table 2-652 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	TX_ANA_LOAD_CLK_SELF_CLR_DISABLE	R/W	Disable self-clearing for the tx_ana_load_clk register **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	TX_ANA_EQ_LFPS_OVRD_EN	R/W	Override Tx EQ signals when LFPS is enabled This will override the Tx EQ signals to the analog when txX_beacon_en is asserted **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-652 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	TX_ANA_LOAD_CLK_OVRD_EN	R/W	This bit is the override enable for tx_ana_load_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	TX_ANA_LOAD_CLK	R/W	This bit is the override value for tx_ana_load_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_ANA_CTRL_HALF_OVRD_EN	R/W	This bit is the override enable for tx_ana_ctrl_half_up and tx_ana_ctrl_half_dn. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TX_ANA_CTRL_HALF_DN	R/W	This bit is the override value for tx_ana_ctrl_half_dn. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_ANA_CTRL_HALF_UP	R/W	This bit is the override value for tx_ana_ctrl_half_up. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	TX_ANA_CTRL_ATTEN_20_14	R/W	This field is the override value for tx_ana_ctrl_atten[20:14]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.649 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX EQ Signals Going to ANA, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)47
- **Exists:** Always

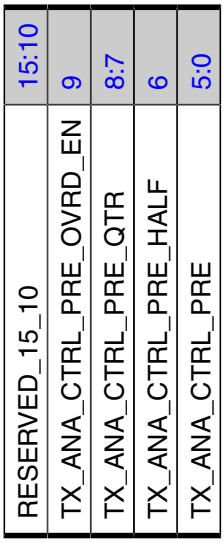


Table 2-653 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	TX_ANA_CTRL_PRE_OVRD_EN	R/W	This bit is the override enable for tx_ana_ctrl_pre, tx_ana_ctrl_pre_qtr and tx_ana_ctrl_pre_half. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:7	TX_ANA_CTRL_PRE_QTR	R/W	This field is the override value for tx_ana_ctrl_pre_qtr[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6	TX_ANA_CTRL_PRE_HALF	R/W	This bit is the override value for tx_ana_ctrl_pre_half. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-653 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5:0	TX_ANA_CTRL_PRE	R/W	This field is the override value for tx_ana_ctrl_pre[5:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.650 LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for TX EQ Signals Going to ANA, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)48
- **Exists:** Always

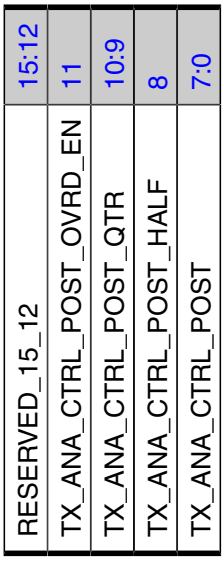


Table 2-654 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	TX_ANA_CTRL_POST_OVRD_EN	R/W	This bit is the override enable for tx_ana_ctrl_post, tx_ana_ctrl_post_qtr and tx_ana_ctrl_post_half. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:9	TX_ANA_CTRL_POST_QTR	R/W	This field is the override value for tx_ana_ctrl_post_qtr[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
8	TX_ANA_CTRL_POST_HALF	R/W	This bit is the override value for tx_ana_ctrl_post_half. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-654 Fields for Register: LANEN_DIG_ANA_TX_EQ_OVRD_OUT_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7:0	TX_ANA_CTRL_POST	R/W	This field is the override value for tx_ana_ctrl_post[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.651 LANEN_DIG_ANA_TX_DCC_EN (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_en.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)49
- **Exists:** Always

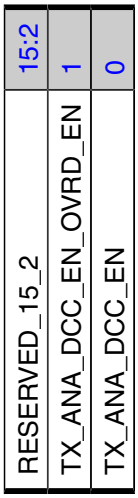


Table 2-655 Fields for Register: LANEN_DIG_ANA_TX_DCC_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	TX_ANA_DCC_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_dcc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ANA_DCC_EN	R/W	This bit is the value for tx_ana_dcc_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.652 LANEN_DIG_ANA_TX_DCC_DSQ_EN (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_dsq_en.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4a
- **Exists:** Always

Table 2-656 Fields for Register: LANEN_DIG_ANA_TX_DCC_DSQ_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_ANA_DCC_DSQ_EN	R/W	This bit is the value for tx_ana_dcc_dsq_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.653 LANEN_DIG_ANA_TX_DCC_CAL_OVRD_EN (for N = 0; N <= NLANES-1)

- **Description:** Overrides TX DCC Control BUS (en/val/sel) Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4b
- **Exists:** Always

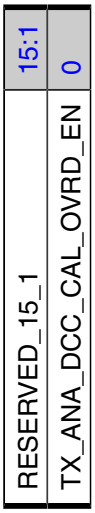


Table 2-657 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_OVRD_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_ANA_DCC_CAL_OVRD_EN	R/W	This bit is the override enable for TX DCC control bus. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.654 LANEN_DIG_ANA_TX_DCC_CAL_COMP_EN (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_cal_comp_en.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4c
- **Exists:** Always

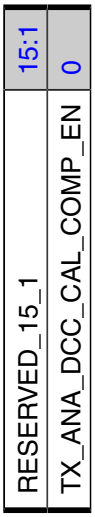


Table 2-658 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_COMP_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_ANA_DCC_CAL_COMP_EN	R/W	This bit is the value for tx_ana_dcc_cal_comp_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.655 LANEN_DIG_ANA_TX_DCC_CAL_CTRL_EN (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_cal_ctrl_en.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4d
- **Exists:** Always

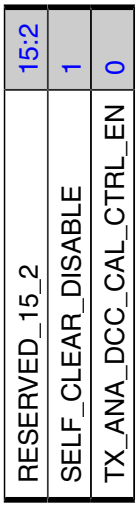


Table 2-659 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_CTRL_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the tx_ana_dcc_dac_ctrl_en register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ANA_DCC_CAL_CTRL_EN	R/W	This bit is the value for tx_ana_dcc_cal_ctrl_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.656 LANEN_DIG_ANA_TX_DCC_CAL_CTRL_SEL (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_cal_ctrl_sel.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4e
- **Exists:** Always

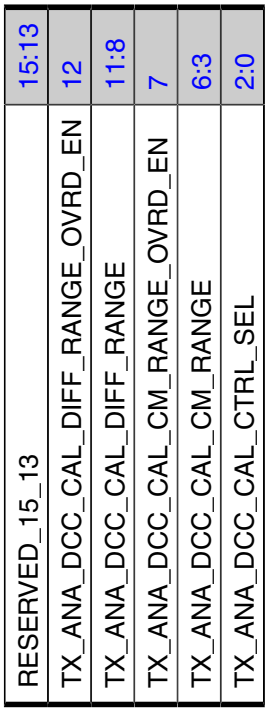


Table 2-660 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_CTRL_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	TX_ANA_DCC_CAL_DIFF_RANGE_OVRD_EN	R/W	This bit enables override value for tx_ana_dcc_cal_diff_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:8	TX_ANA_DCC_CAL_DIFF_RANGE	R/W	This field is the override value for tx_ana_dcc_cal_diff_range[3:0]. **Value After Reset:** 0xd **Exists:** Always **Reset Mask:** 0xf

Table 2-660 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_CTRL_SEL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7	TX_ANA_DCC_CAL_CM_RANGE_OVRD_EN	R/W	This bit enables override value for tx_ana_dcc_cal_cm_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:3	TX_ANA_DCC_CAL_CM_RANGE	R/W	This field is the override value for tx_ana_dcc_cal_cm_range[3:0]. **Value After Reset:** 0xd **Exists:** Always **Reset Mask:** 0xf
2:0	TX_ANA_DCC_CAL_CTRL_SEL	R/W	This bit is the value for tx_ana_dcc_cal_ctrl_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.657 LANEN_DIG_ANA_TX_DCC_CAL_DATA (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Control for tx_ana_dcc_cal_data.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)4f
- **Exists:** Always

Table 2-661 Fields for Register: LANEN_DIG_ANA_TX_DCC_CAL_DATA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	TX_ANA_DCC_CAL_DATA	R/W	This field is the value for tx_ana_dcc_cal_data. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.658 LANEN_DIG_ANA_ROPLL_ANA_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing ROPLL Controls to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)50
- **Exists:** Always

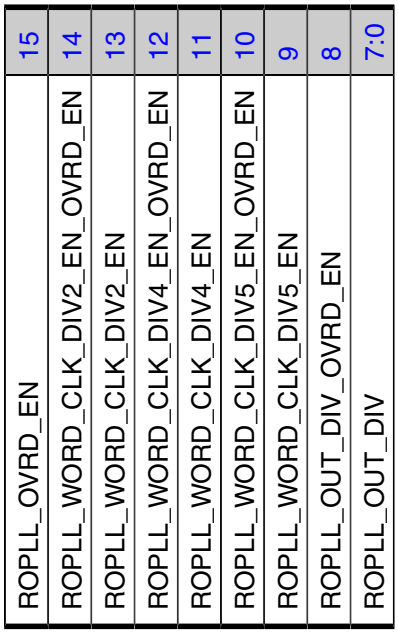


Table 2-662 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_OVRD_EN	R/W	This bit is the global override enable for all outputs controlled by ROPLL_ANA_OUT_0,1,2,3,4 registers. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	ROPLL_WORD_CLK_DIV2_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_word_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	ROPLL_WORD_CLK_DIV2_EN	R/W	This bit is the override value for tx_ana_ropll_word_clk_div2_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-662 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12	ROPLL_WORD_CLK_DIV4_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_word_clk_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	ROPLL_WORD_CLK_DIV4_EN	R/W	This bit is the override value for tx_ana_ropll_word_clk_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	ROPLL_WORD_CLK_DIV5_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_word_clk_div5_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	ROPLL_WORD_CLK_DIV5_EN	R/W	This bit is the override value for tx_ana_ropll_word_clk_div5_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	ROPLL_OUT_DIV_OVRD_EN	R/W	This bit is the override enable for ropll_ana_out_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	ROPLL_OUT_DIV	R/W	This field is the override value for ropll_ana_out_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.659 LANEN_DIG_ANA_ROPLL_ANA_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing ROPLL Controls to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)51
- **Exists:** Always

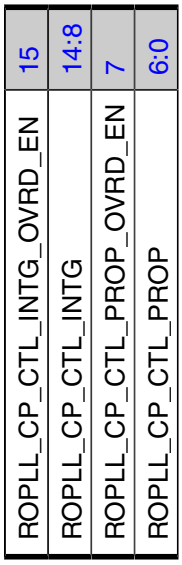


Table 2-663 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_CP_CTL_INTG_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_cp_ctl_intg. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:8	ROPLL_CP_CTL_INTG	R/W	This field is the override value for tx_ana_ropll_cp_ctl_intg. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
7	ROPLL_CP_CTL_PROP_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_cp_ctl_prop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	ROPLL_CP_CTL_PROP	R/W	This field is the override value for tx_ana_ropll_cp_ctl_prop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.660 LANEN_DIG_ANA_ROPLL_ANA_OUT_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing ROPLL Controls to ANA, Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)52
- **Exists:** Always

Table 2-664 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_REFDIV_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_refdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:11	ROPLL_REFDIV	R/W	This field is the override value for tx_ana_ropll_refdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
10	ROPLL_REFSEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:8	ROPLL_REFSEL	R/W	This field is the override value for tx_ana_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
7	ROPLL_FBDIV_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_fbdiv. **Value After Reset:** 0x0

Table 2-664 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
6:0	ROPLL_FBDIV	R/W	This field is an override value for tx_ana_ropll_fbdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.661 LANEN_DIG_ANA_ROPLL_ANA_OUT_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing ROPLL Controls to ANA, Register 3.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)53
- **Exists:** Always

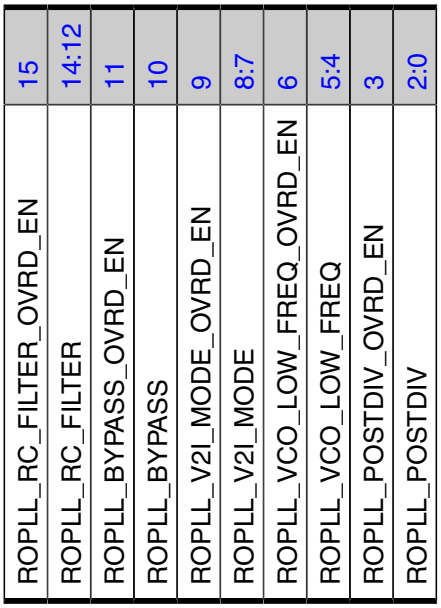


Table 2-665 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	ROPLL_RC_FILTER_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_rc_filter. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:12	ROPLL_RC_FILTER	R/W	This field is the override value for tx_ana_ropll_rc_filter. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
11	ROPLL_BYPASS_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	ROPLL_BYPASS	R/W	This bit is the override value for tx_ana_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-665 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	ROPLL_V2I_MODE_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_v2i_mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:7	ROPLL_V2I_MODE	R/W	This field is an override value for tx_ana_ropll_v2i_mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6	ROPLL_VCO_LOW_FREQ_OVRD_EN	R/W	This bit is an override enable for tx_ana_ropll_vco_low_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	ROPLL_VCO_LOW_FREQ	R/W	This field is an override value for tx_ana_ropll_vco_low_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	ROPLL_POSTDIV_OVRD_EN	R/W	This bit is an override enable for tx_ana_ropll_postdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	ROPLL_POSTDIV	R/W	This field is an override value for tx_ana_ropll_postdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.662 LANEN_DIG_ANA_ROPLL_ANA_OUT_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing ROPLL Controls to ANA, Register 4.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)54
- **Exists:** Always

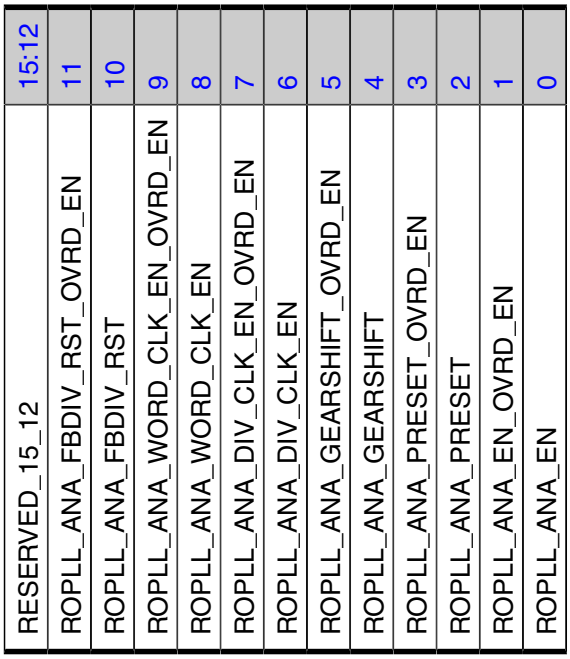


Table 2-666 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	ROPLL_ANA_FBDIV_RST_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_fbdiv_rst_i. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	ROPLL_ANA_FBDIV_RST	R/W	This bit is the override value for tx_ana_ropll_fbdiv_rst_ir **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	ROPLL_ANA_WORD_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_word_clk_en. **Value After Reset:** 0x0

Table 2-666 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8	ROPLL_ANA_WORD_CLK_EN	R/W	This bit is the override value for tx_ana_ropll_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	ROPLL_ANA_DIV_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ROPLL_ANA_DIV_CLK_EN	R/W	This bit is the override value for tx_ana_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	ROPLL_ANA_GEARSHIFT_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_gearshift. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ROPLL_ANA_GEARSHIFT	R/W	This bit is the override value for tx_ana_ropll_gearshift. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	ROPLL_ANA_PRESET_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_preset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ROPLL_ANA_PRESET	R/W	This bit is the override value for tx_ana_ropll_preset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ROPLL_ANA_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ropll_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-666 Fields for Register: LANEN_DIG_ANA_ROPLL_ANA_OUT_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	ROPLL_ANA_EN	R/W	This bit is the override value for tx_ana_ropll_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.663 LANEN_DIG_ANA_RX_CTL_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX Control Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)55
- **Exists:** Always

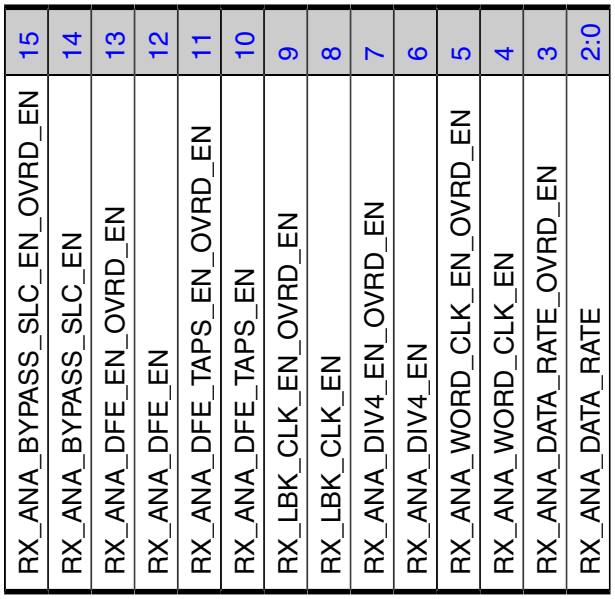


Table 2-667 Fields for Register: LANEN_DIG_ANA_RX_CTL_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_ANA_BYPASS_SLC_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_bypass_slc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	RX_ANA_BYPASS_SLC_EN	R/W	This bit is the override value for rx_ana_bypass_slc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_ANA_DFE_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_dfe_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_ANA_DFE_EN	R/W	This bit is the override value for rx_ana_dfe_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-667 Fields for Register: LANEN_DIG_ANA_RX_CTL_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	RX_ANA_DFE_TAPS_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_dfe_taps_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX_ANA_DFE_TAPS_EN	R/W	This bit is the override value for rx_ana_dfe_taps_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RX_LBK_CLK_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_loopback_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_LBK_CLK_EN	R/W	This bit is the override value for rx_ana_loopback_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	RX_ANA_DIV4_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_ANA_DIV4_EN	R/W	This bit is the override value for rx_ana_div4_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_ANA_WORD_CLK_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_ANA_WORD_CLK_EN	R/W	This bit is the override value for rx_ana_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_ANA_DATA_RATE_OVRD_EN	R/W	This bit is the override enable for rx_ana_data_rate. **Value After Reset:** 0x0

Table 2-667 Fields for Register: LANEN_DIG_ANA_RX_CTL_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2:0	RX_ANA_DATA_RATE	R/W	This field is the override value for rx_ana_data_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.664 LANEN_DIG_ANA_RX_PWR_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX PWR UP/DN Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)56
- **Exists:** Always

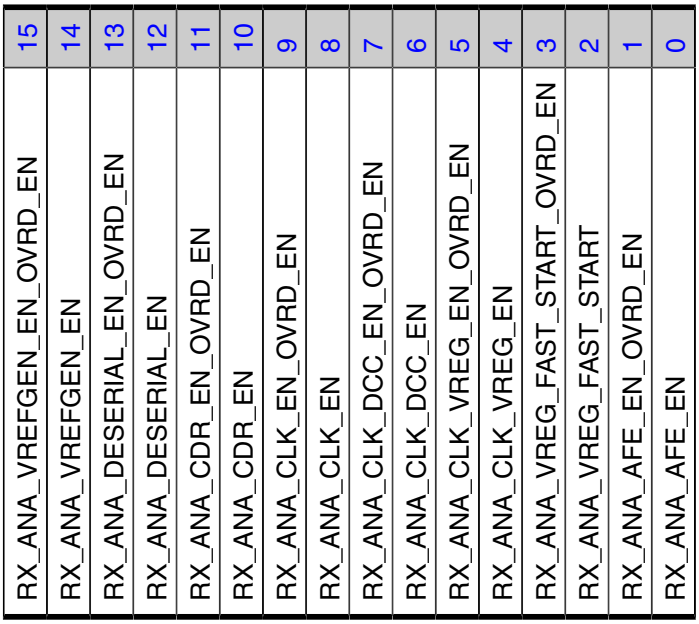


Table 2-668 Fields for Register: LANEN_DIG_ANA_RX_PWR_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_ANA_VREFGEN_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_vrefgen_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	RX_ANA_VREFGEN_EN	R/W	This bit is the override value for rx_ana_vrefgen_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_ANA_DESERIAL_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_deserial_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_ANA_DESERIAL_EN	R/W	This bit is the override value for rx_ana_deserial_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-668 Fields for Register: LANEN_DIG_ANA_RX_PWR_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	RX_ANA_CDR_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_cdr_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	RX_ANA_CDR_EN	R/W	This bit is the override value for rx_ana_cdr_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RX_ANA_CLK_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_ANA_CLK_EN	R/W	This bit is the override value for rx_ana_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	RX_ANA_CLK_DCC_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_clk_dcc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_ANA_CLK_DCC_EN	R/W	This bit is the override value for rx_ana_clk_dcc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_ANA_CLK_VREG_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_clk_vreg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_ANA_CLK_VREG_EN	R/W	This bit is the override value for rx_ana_clk_vreg_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_ANA_VREG_FAST_START_OVRD_EN	R/W	This bit is the override enable for rx_ana_vreg_fast_start. **Value After Reset:** 0x0

Table 2-668 Fields for Register: LANEN_DIG_ANA_RX_PWR_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	RX_ANA_VREG_FAST_START	R/W	This bit is the override value for rx_ana_vreg_fast_start. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_ANA_AFE_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_AFE_EN	R/W	This bit is the override value for rx_ana_afe_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.665 LANEN_DIG_ANA_RX_SIGDET_CAL_EN (for N = 0; N <= NLANES-1)

- **Description:** Value for RX Sigdet Calibration Enable Signal Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)57
- **Exists:** Always

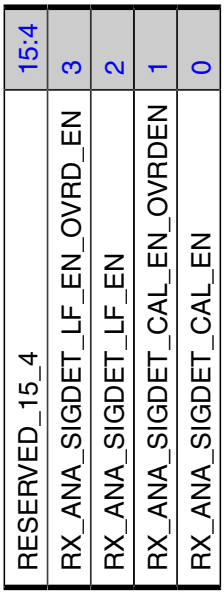


Table 2-669 Fields for Register: LANEN_DIG_ANA_RX_SIGDET_CAL_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	RX_ANA_SIGDET_LF_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_sigdet_lf_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_ANA_SIGDET_LF_EN	R/W	This bit is the override value for rx_ana_sigdet_lf_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_ANA_SIGDET_CAL_EN_OVRDEN	R/W	This bit is the override enable for rx_ana_sigdet_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-669 Fields for Register: LANEN_DIG_ANA_RX_SIGDET_CAL_EN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	RX_ANA_SIGDET_CAL_EN	R/W	This bit is the override value for rx_ana_sigdet_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.666 LANEN_DIG_ANA_RX_SIGDET_HF_CAL (for N = 0; N <= NLANES-1)

- **Description:** Value for RX Sigdet High Frequency Calibration Signal Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)58
- **Exists:** Always

Table 2-670 Fields for Register: LANEN_DIG_ANA_RX_SIGDET_HF_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_ANA_SIGDET_HF_CAL_TUNE	R/W	This field is the override value for rx_ana_sigdet_hf_cal_tune. **Value After Reset:** 0x20 **Exists:** Always **Reset Mask:** 0x3f

2.1.667 LANEN_DIG_ANA_RX_SIGDET_LF_CAL (for N = 0; N <= NLANES-1)

- **Description:** Value for RX Sigdet Low Frequency Calibration Signal Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)59
- **Exists:** Always

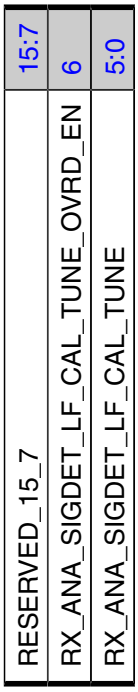


Table 2-671 Fields for Register: LANEN_DIG_ANA_RX_SIGDET_LF_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	RX_ANA_SIGDET_LF_CAL_TUNE_OVRD_EN	R/W	This bit is the override enable for rx_ana_sigdet_lf_cal_tune. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_ANA_SIGDET_LF_CAL_TUNE	R/W	This field is the override value for rx_ana_sigdet_lf_cal_tune. **Value After Reset:** 0x20 **Exists:** Always **Reset Mask:** 0x3f

2.1.668 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX VCO Signals Going to ANA 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5a
- **Exists:** Always

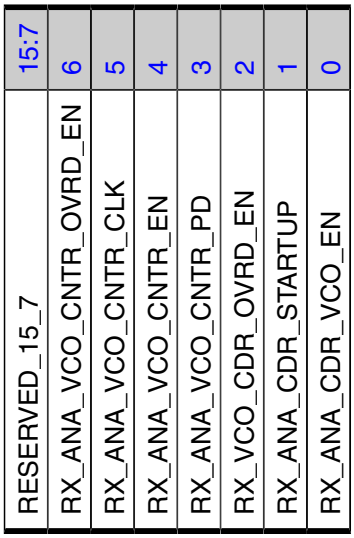


Table 2-672 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	RX_ANA_VCO_CNTR_OVRD_EN	R/W	This bit is the override enable for rx_ana_vco_cntr_clk, rx_ana_vco_cntr_en, and rx_ana_vco_cntr_pd. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_ANA_VCO_CNTR_CLK	R/W	This bit is the override value for rx_ana_vco_cntr_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_ANA_VCO_CNTR_EN	R/W	This bit is the override value for rx_ana_vco_cntr_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-672 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	RX_ANA_VCO_CNTR_PD	R/W	This bit is the override value for rx_ana_vco_cntr_pd. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RX_VCO_CDR_OVRD_EN	R/W	This bit enables override values for rx_ana_cdr_vco_en and rx_ana_cdr_startup. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_ANA_CDR_STARTUP	R/W	This bit is the override value for rx_ana_cdr_startup. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_CDR_VCO_EN	R/W	This bit is the override value for rx_ana_cdr_vco_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.669 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX VCO Signals Going to ANA 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5b
- **Exists:** Always

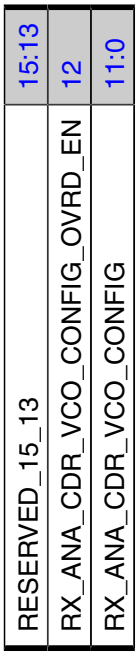


Table 2-673 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_ANA_CDR_VCO_CONFIG_OVRD_EN	R/W	This bit enables override value for rx_ana_cdr_vco_config. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	RX_ANA_CDR_VCO_CONFIG	R/W	This field is the override value for rx_ana_cdr_vco_config. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.670 LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX VCO Signals Going to ANA 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5c
- **Exists:** Always

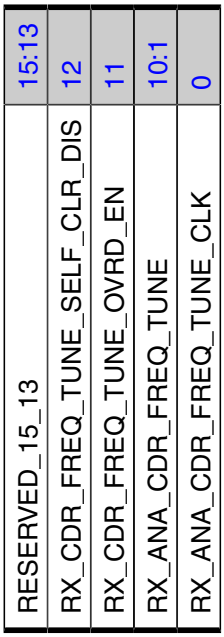


Table 2-674 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_CDR_FREQ_TUNE_SELF_CLR_DIS	R/W	This bit disables the self clear for rx_ana_cdr_freq_tune_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	RX_CDR_FREQ_TUNE_OVRD_EN	R/W	This bit enables override value for rx_ana_cdr_freq_tune and rx_ana_cdr_freq_tune_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:1	RX_ANA_CDR_FREQ_TUNE	R/W	This field is the override value for rx_ana_cdr_freq_tune. **Value After Reset:** 0x0

Table 2-674 Fields for Register: LANEN_DIG_ANA_RX_VCO_OVRD_OUT_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3ff
0	RX_ANA_CDR_FREQ_TUNE_CLK	R/W	This bit is the override value for rx_ana_cdr_freq_tune_clk. - This bit self-cleares to generate a pulse 1 cr_clk wide. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.671 LANEN_DIG_ANA_RX_CAL_0 (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX CAL Signals Going to ANA, Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5d
- **Exists:** Always

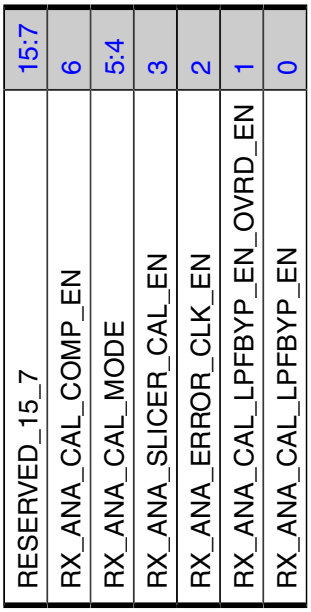


Table 2-675 Fields for Register: LANEN_DIG_ANA_RX_CAL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	RX_ANA_CAL_COMP_EN	R/W	This bit is the value for rx_ana_cal_comp_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	RX_ANA_CAL_MODE	R/W	This field is the value for rx_ana_cal_mode[1:0]. ■ 00 - Differential comparison ■ 01 - Single-ended, positive node to positive node ■ 10 - Single-ended, negative node to negative node ■ 11 - Common mode comparison **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-675 Fields for Register: LANEN_DIG_ANA_RX_CAL_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	RX_ANA_SLICER_CAL_EN	R/W	This bit is the value for rx_ana_slicer_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_ANA_ERROR_CLK_EN	R/W	This bit is the value for rx_ana_error_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_ANA_CAL_LPFBYP_EN_OVRD_EN	R/W	This bit is the override enable for rx_ana_cal_lpfbyp_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_CAL_LPFBYP_EN	R/W	This bit is the override value for rx_ana_cal_lpfbyp_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.672 LANEN_DIG_ANA_RX_CAL_1 (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX CAL Signals Going to ANA, Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5e
- **Exists:** Always

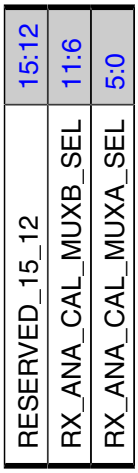


Table 2-676 Fields for Register: LANEN_DIG_ANA_RX_CAL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	RX_ANA_CAL_MUXB_SEL	R/W	This field is the value for rx_ana_cal_muxb_sel[5:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_ANA_CAL_MUXA_SEL	R/W	This field is the value for rx_ana_cal_muxa_sel[5:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.673 LANEN_DIG_ANA_RX_VDAC_RANGE_SEL (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX DAC CTRL Value Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)5f
- **Exists:** Always

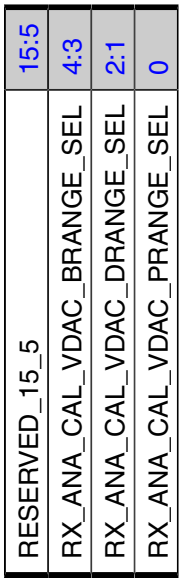


Table 2-677 Fields for Register: LANEN_DIG_ANA_RX_VDAC_RANGE_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4:3	RX_ANA_CAL_VDAC_BRANGE_SEL	R/W	This field is the value for rx_ana_cal_vdac_brange. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
2:1	RX_ANA_CAL_VDAC_DRANGE_SEL	R/W	This field is the value for rx_ana_cal_vdac_drange. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3
0	RX_ANA_CAL_VDAC_PRANGE_SEL	R/W	This bit is the value for rx_ana_cal_vdac_prange. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.674 LANEN_DIG_ANA_RX_DAC_CTRL (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX DAC CTRL Value Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)60
- **Exists:** Always

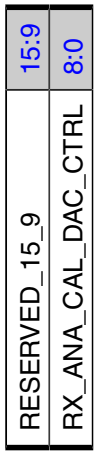


Table 2-678 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	RX_ANA_CAL_DAC_CTRL	R/W	This field is the value for rx_ana_cal_dac_ctrl[8:0]. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.675 LANEN_DIG_ANA_RX_ANA_RTRIM (for N = 0; N <= NLANES-1)

- **Description:** Set Value for RX RTRIM Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)61
- **Exists:** Always

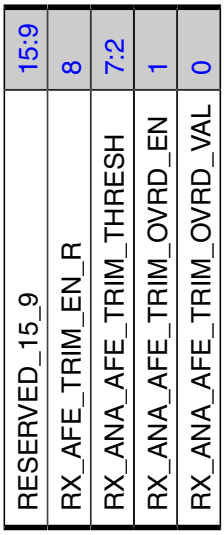


Table 2-679 Fields for Register: LANEN_DIG_ANA_RX_ANA_RTRIM (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	RX_AFE_TRIM_EN_R	R/W	This bit is the enable for signal rx_afe_trim_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:2	RX_ANA_AFE_TRIM_THRESH	R/W	This field is the threshold value for rx_ana_afe_trim. **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0x3f
1	RX_ANA_AFE_TRIM_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_trim_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_AFE_TRIM_OVRD_VAL	R/W	This bit is the override value for rx_ana_afe_trim_ovrd_val. **Value After Reset:** 0x0

Table 2-679 Fields for Register: LANEN_DIG_ANA_RX_ANA_RTRIM (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.676 LANEN_DIG_ANA_RX_DAC_CTRL_OVRD (for N = 0; N <= NLANES-1)

- **Description:** Overrides RX DAC CTRL Bus (en/val/sel) Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)62
- **Exists:** Always

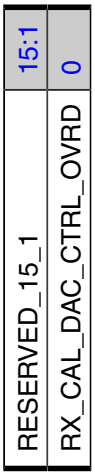


Table 2-680 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_CAL_DAC_CTRL_OVRD	R/W	This bit is the override enable for cal DAC control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.677 LANEN_DIG_ANA_RX_DAC_CTRL_SEL (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX DAC CTRL Select Signal Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)63
- **Exists:** Always

Table 2-681 Fields for Register: LANEN_DIG_ANA_RX_DAC_CTRL_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_ANA_CAL_DAC_CTRL_SEL	R/W	This field is the value for rx_ana_cal_dac_ctrl_sel[5:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.678 LANEN_DIG_ANA_RX_AFE_ATT_VGA (for N = 0; N <= NLANES-1)

- **Description:** Value for RX AFE ATT & VGA Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)64
- **Exists:** Always

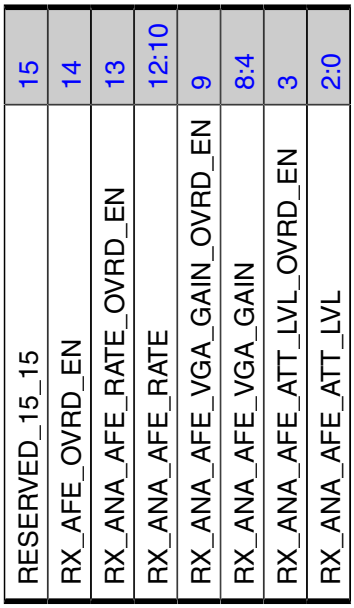


Table 2-682 Fields for Register: LANEN_DIG_ANA_RX_AFE_ATT_VGA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_AFE_OVRD_EN	R/W	This bit is the global override enable for all AFE signals. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_ANA_AFE_RATE_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_rate[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:10	RX_ANA_AFE_RATE	R/W	This field is the value for rx_ana_afe_rate[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

Table 2-682 Fields for Register: LANEN_DIG_ANA_RX_AFE_ATT_VGA (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	RX_ANA_AFE_VGA_GAIN_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_vga_gain[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:4	RX_ANA_AFE_VGA_GAIN	R/W	This field is the value for rx_ana_afe_vga_gain[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
3	RX_ANA_AFE_ATT_LVL_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_att_lvl[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	RX_ANA_AFE_ATT_LVL	R/W	This field is the value for rx_ana_afe_att_lvl[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.679 LANEN_DIG_ANA_RX_AFE_CTLE (for N = 0; N <= NLANES-1)

- **Description:** Values for RX AFE CTLE Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)65
- **Exists:** Always

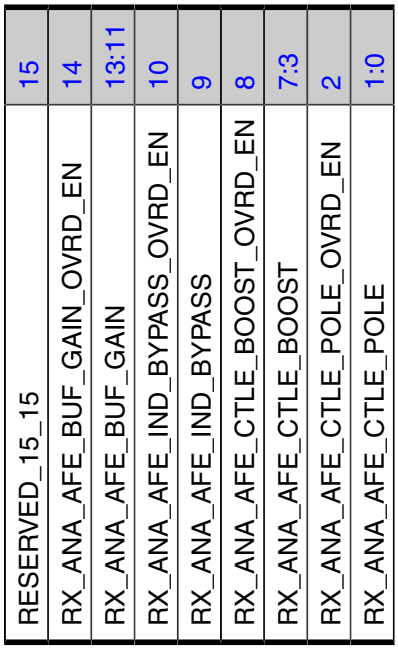


Table 2-683 Fields for Register: LANEN_DIG_ANA_RX_AFE_CTLE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_ANA_AFE_BUF_GAIN_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_buf_gain[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:11	RX_ANA_AFE_BUF_GAIN	R/W	This bit is the override enable for rx_ana_afe_buf_gain[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
10	RX_ANA_AFE_IND_BYPASS_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_ind_bypass. **Value After Reset:** 0x0

Table 2-683 Fields for Register: LANEN_DIG_ANA_RX_AFE_CTLE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
9	RX_ANA_AFE_IND_BYPASS	R/W	This bit is the override enable for rx_ana_afe_ind_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_ANA_AFE_CTLE_BOOST_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_ctle_boost[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:3	RX_ANA_AFE_CTLE_BOOST	R/W	This field is the value for rx_ana_afe_ctle_boost[4:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
2	RX_ANA_AFE_CTLE_POLE_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_ctle_pole[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	RX_ANA_AFE_CTLE_POLE	R/W	This field is the value for rx_ana_afe_ctle_pole[1:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.680 LANEN_DIG_ANA_RX_ANA_AFE_CONFIG (for N = 0; N <= NLANES-1)

- **Description:** RX AFE Config Control.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)66
- **Exists:** Always

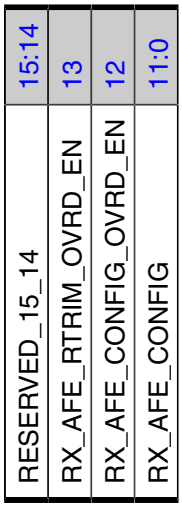


Table 2-684 Fields for Register: LANEN_DIG_ANA_RX_ANA_AFE_CONFIG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	RX_AFE_RTRIM_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_config[9:8]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_AFE_CONFIG_OVRD_EN	R/W	This bit is the override enable for all rx_ana_afe_config bits, except bits [9:8]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	RX_AFE_CONFIG	R/W	This field is the override value for rx_ana_afe_config[11:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.681 LANEN_DIG_ANA_RX_SCOPE (for N = 0; N <= NLANES-1)

- **Description:** Values for RX SCOPE Signals Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)67
- **Exists:** Always

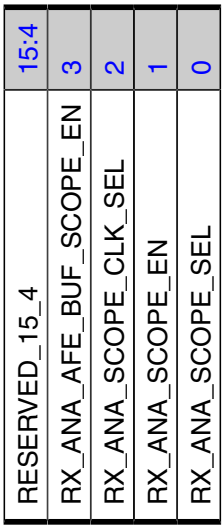


Table 2-685 Fields for Register: LANEN_DIG_ANA_RX_SCOPE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	RX_ANA_AFE_BUF_SCOPE_EN	R/W	This bit sets the value for rx_ana_afe_buf_scope_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_ANA_SCOPE_CLK_SEL	R/W	This bit sets the value for rx_ana_scope_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_ANA_SCOPE_EN	R/W	This bit sets the value for rx_ana_scope_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_SCOPE_SEL	R/W	This bit sets the value for rx_ana_scope_sel. **Value After Reset:** 0x0

Table 2-685 Fields for Register: LANEN_DIG_ANA_RX_SCOPE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.682 LANEN_DIG_ANA_RX_SLICER_CTRL (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX Slicer Ctrl Signals Going to ANA Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)68
- **Exists:** Always

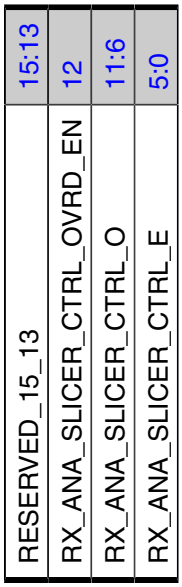


Table 2-686 Fields for Register: LANEN_DIG_ANA_RX_SLICER_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_ANA_SLICER_CTRL_OVRD_EN	R/W	This bit is the override enable for RX ANA slicer ctrl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:6	RX_ANA_SLICER_CTRL_O	R/W	This field is the value for rx_ana_slicer_ctrl_o[5:0]. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0x3f
5:0	RX_ANA_SLICER_CTRL_E	R/W	This field is the value for rx_ana_slicer_ctrl_e[5:0]. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0x3f

2.1.683 LANEN_DIG_ANA_RX_ANA_IQ (for N = 0; N <= NLANES-1)

- **Description:** Sets Values for RX ANA IQ Signals.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)69
- **Exists:** Always

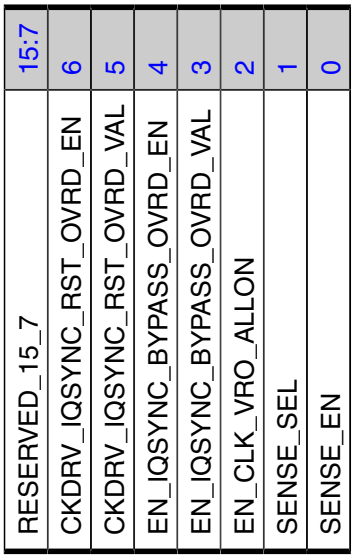


Table 2-687 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	CKDRV_IQSYNC_RST_OVRD_EN	R/W	This bit is the override enable for rx_ana_en_iqsync_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	CKDRV_IQSYNC_RST_OVRD_VAL	R/W	This bit is the override value for rx_ana_en_iqsync_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	EN_IQSYNC_BYPASS_OVRD_EN	R/W	This bit is the override enable for rx_ana_en_iqsync_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-687 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQ (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	EN_IQSYNC_BYPASS_OVRD_VAL	R/W	This bit is the override value for rx_ana_en_iqsync_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	EN_CLK_VRO_ALLON	R/W	This bit is the value for rx_ana_en_clk_vro_allon. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SENSE_SEL	R/W	This bit is the value for rx_ana_iq_sense_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SENSE_EN	R/W	This bit is the value for rx_ana_iq_sense_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.684 LANEN_DIG_ANA_RX_ANA_IQC_BYP_OVRD (for N = 0; N <= NLANES-1)

- **Description:** Sets Value for RX ANA IQC BYPASS Adjust Signal Going to ANA Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6a
- **Exists:** Always

Table 2-688 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQC_BYP_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	EN	R/W	This bit enables override of rx_ana_iqc_bypass_adjust[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	VAL	R/W	This field is the value for rx_ana_iqc_bypass_adjust[7:0]. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xff

2.1.685 LANEN_DIG_ANA_RX_ANA_IQC_BYPASS_ADJUST_CLK (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Bypass Adjust Clock Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6b
- **Exists:** Always

Table 2-689 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQC_BYPASS_ADJUST_CLK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the rx_ana_iqc_bypass_adjust_clk register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit is the value for rx_ana_iqc_bypass_adjust_clk. If SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.686 LANEN_DIG_ANA_RX_ANA_IQC_DATA_OVRD (for N = 0; N <= NLANES-1)

- **Description:** Sets Value for RX ANA IQC DATA Adjust Signal Going to ANA Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6c
- **Exists:** Always

Table 2-690 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQC_DATA_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	EN	R/W	This bit enables override of rx_ana_iqc_data_adjust[7:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	VAL	R/W	This bit is the value for rx_ana_iqc_data_adjust[7:0]. **Value After Reset:** 0x6 **Exists:** Always **Reset Mask:** 0xff

2.1.687 LANEN_DIG_ANA_RX_ANA_IQC_DATA_ADJUST_CLK (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Data Adjust Clock Register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6d
- **Exists:** Always

Table 2-691 Fields for Register: LANEN_DIG_ANA_RX_ANA_IQC_DATA_ADJUST_CLK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the rx_ana_iqc_data_adjust_clk register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit is the value for rx_ana_iqc_data_adjust_clk. If SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.688 LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN (for N = 0; N <= NLANES-1)

- **Description:** DAC CTRL Enable Signal.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6e
- **Exists:** Always

Table 2-692 Fields for Register: LANEN_DIG_ANA_RX_ANA_CAL_DAC_CTRL_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DAC_CTRL_SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the rx_ana_cal_dac_ctrl_en register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_CAL_DAC_CTRL_EN	R/W	This bit is the value for rx_ana_cal_dac_ctrl_en. If DAC_CTRL_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.689 LANEN_DIG_ANA_RX_ANA_AFE_UPDATE_EN (for N = 0; N <= NLANES-1)

- **Description:** AFE Update Enable Clock.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)6f
- **Exists:** Always

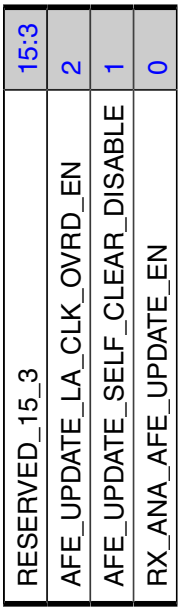


Table 2-693 Fields for Register: LANEN_DIG_ANA_RX_ANA_AFE_UPDATE_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	AFE_UPDATE_LA_CLK_OVRD_EN	R/W	This bit is the override enable for rx_ana_afe_update_la_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	AFE_UPDATE_SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the rx_ana_afe_update_en register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ANA_AFE_UPDATE_EN	R/W	This bit is the value for rx_ana_afe_update_en. If AFE_UPDATE_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle). **Value After Reset:** 0x0

Table 2-693 Fields for Register: LANEN_DIG_ANA_RX_ANA_AFE_UPDATE_EN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.690 LANEN_DIG_ANA_RX_ANA_DFE_SAMP_SEL (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Control for rx_ana_dfe_samp_sel.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)70
- **Exists:** Always

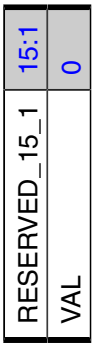


Table 2-694 Fields for Register: LANEN_DIG_ANA_RX_ANA_DFE_SAMP_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	This bit is the value for rx_ana_dfe_samp_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.691 LANEN_DIG_ANA_RX_ANA_BYP_SAMP_SEL (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Control for rx_ana_byp_samp_sel.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)71
- **Exists:** Always

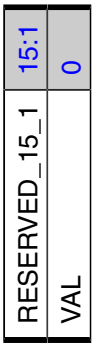


Table 2-695 Fields for Register: LANEN_DIG_ANA_RX_ANA_BYP_SAMP_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	This bit is the value for rx_ana_byp_samp_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.692 LANEN_DIG_ANA_RX_ANA_PHS_SAMP_SEL (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Control for rx_ana_phs_samp_sel.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)72
- **Exists:** Always

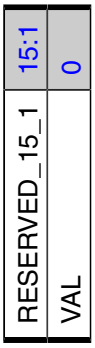


Table 2-696 Fields for Register: LANEN_DIG_ANA_RX_ANA_PHS_SAMP_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	This bit the value for rx_ana_phs_samp_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.693 LANEN_DIG_ANA_RX_TERM_CODE_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Value for RX Termination Code Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)73
- **Exists:** Always

Table 2-697 Fields for Register: LANEN_DIG_ANA_RX_TERM_CODE_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	RX_TERM_OVRD_EN	R/W	This bit is the override enable for the rx_ana_term_code[5:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:0	RX_TERM_CODE	R/W	This field overrides the rx_ana_term_code[5:0] signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.694 LANEN_DIG_ANA_RX_TERM_CODE_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Value for RX Termination Code Clock Going to ANA.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)74
- **Exists:** Always

Table 2-698 Fields for Register: LANEN_DIG_ANA_RX_TERM_CODE_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_TERM_CLK_SELF_CLEAR_DISABLE	R/W	This bit disables the self-clearing of rx_ana_term_clk register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_TERM_CLK	R/W	This bit is the override for TX term DN clock. This bit is self_clearing (4 cr_clks later). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.695 LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_ENABLE (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Control for Floating TAP Enables.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)75
- **Exists:** Always

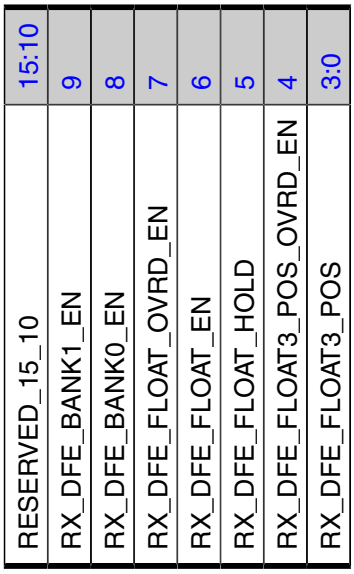


Table 2-699 Fields for Register: LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_ENABLE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	RX_DFE_BANK1_EN	R/W	This bit is the override value for RX DFE Bank1 enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_DFE_BANK0_EN	R/W	This bit is the override value for RX DFE Bank0 enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	RX_DFE_FLOAT_OVRD_EN	R/W	This bit is the override enable for RX DFE Float enables and positions. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-699 Fields for Register: LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_ENABLE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6	RX_DFE_FLOAT_EN	R/W	This bit is the override for RX DFE Floating taps enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_DFE_FLOAT_HOLD	R/W	This bit is the override for RX DFE Float hold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_DFE_FLOAT3_POS_OVRD_EN	R/W	This bit is the override enable for RX DFE Float3 position. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	RX_DFE_FLOAT3_POS	R/W	This field is the override value for RX DFE Float3 position. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf

2.1.696 LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_POS (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Control for Floating Positions.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)76
- **Exists:** Always

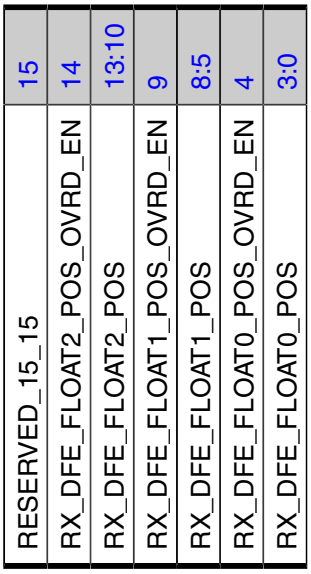


Table 2-700 Fields for Register: LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_POS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_DFE_FLOAT2_POS_OVRD_EN	R/W	This bit is the override enable for RX DFE Float2 position. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:10	RX_DFE_FLOAT2_POS	R/W	This field is the override value for RX DFE Float2 position. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
9	RX_DFE_FLOAT1_POS_OVRD_EN	R/W	This bit is the override enable for RX DFE Float1 position. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-700 Fields for Register: LANEN_DIG_ANA_RX_ANA_DFE_FLOAT_POS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8:5	RX_DFE_FLOAT1_POS	R/W	This field is the override value for RX DFE Float1 position. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
4	RX_DFE_FLOAT0_POS_OVRD_EN	R/W	This bit is the override enable for RX DFE Float0 position. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	RX_DFE_FLOAT0_POS	R/W	This field is the override value for RX DFE Float0 position. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf

2.1.697 LANEN_DIG_ANA_RX_DCC_CTRL_RANGE (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Control for .rx_ana_dcc_cal_diff_range and rx_ana_dcc_cal_cm_range.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)77
- **Exists:** Always

Table 2-701 Fields for Register: LANEN_DIG_ANA_RX_DCC_CTRL_RANGE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	RX_ANA_DCC_CTRL_DIFF_RANGE_OVRD_EN	R/W	This bit enables override value for rx_ana_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:5	RX_ANA_DCC_CTRL_DIFF_RANGE	R/W	This field is the override value for rx_ana_dcc_ctrl_diff_range[3:0]. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf

Table 2-701 Fields for Register: LANEN_DIG_ANA_RX_DCC_CTRL_RANGE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	RX_ANA_DCC_CTRL_CM_RANGE_OVRD_EN	R/W	This bit enables override value for rx_ana_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	RX_ANA_DCC_CTRL_CM_RANGE	R/W	This field is the override value for rx_ana_dcc_ctrl_cm_range[3:0]. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xf

2.1.698 LANEN_DIG_ANA_STATUS_0 (for N = 0; N <= NLANES-1)

- **Description:** Lane Input Status Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)78
- **Exists:** Always

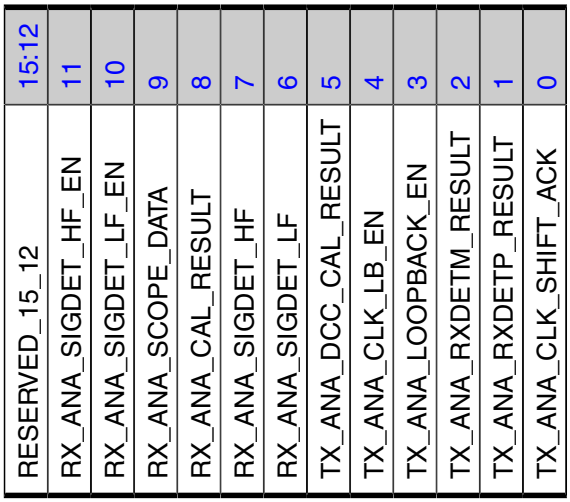


Table 2-702 Fields for Register: LANEN_DIG_ANA_STATUS_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	RX_ANA_SIGDET_HF_EN	R	This bit is the value of rx_ana_sigdet_hf_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	RX_ANA_SIGDET_LF_EN	R	This bit is the value of rx_ana_sigdet_lf_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	RX_ANA_SCOPE_DATA	R	This bit is the value from ANA for rx_ana_scope_data. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-702 Fields for Register: LANEN_DIG_ANA_STATUS_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
8	RX_ANA_CAL_RESULT	R	This bit is the value from ANA for rx_ana_cal_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	RX_ANA_SIGDET_HF	R	This bit is the value from ANA for rx_ana_sigdet_hf. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	RX_ANA_SIGDET_LF	R	This bit is the value from ANA for rx_ana_sigdet_lf. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	TX_ANA_DCC_CAL_RESULT	R	This bit is the value from ANA for tx_ana_dcc_cal_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	TX_ANA_CLK_LB_EN	R	This bit is the value of tx_ana_clk_lb_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	TX_ANA_LOOPBACK_EN	R	This bit is the value of tx_ana_loopback_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-702 Fields for Register: LANEN_DIG_ANA_STATUS_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
2	TX_ANA_RXDETM_RESULT	R	This bit is the value from ANA for tx_ana_rxdetm_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	TX_ANA_RXDETP_RESULT	R	This bit is the value from ANA for tx_ana_rxdetp_result. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TX_ANA_CLK_SHIFT_ACK	R	This bit is the value from ANA for tx_ana_clk_shift_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.699 LANEN_DIG_ANA_STATUS_1 (for N = 0; N <= NLANES-1)

- **Description:** Lane Input Status Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)79
- **Exists:** Always

Table 2-703 Fields for Register: LANEN_DIG_ANA_STATUS_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	RX_ANA_VCO_CNTR	R	This field is the value from ANA for rx_ana_vco_cntr. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.700 LANEN_DIG_ANA_RX_MISC_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX MISC Signals Going to ANA register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)7a
- **Exists:** Always

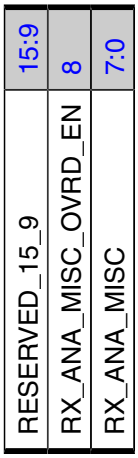


Table 2-704 Fields for Register: LANEN_DIG_ANA_RX_MISC_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	RX_ANA_MISC_OVRD_EN	R/W	This bit is the override enable for tx_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	RX_ANA_MISC	R/W	This field is the override value for tx_ana_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.701 LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_0 (for N = 0; N <= NLANES-1)

- **Description:** TX Data Delay Enable Clock Register 0.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)7b
- **Exists:** Always

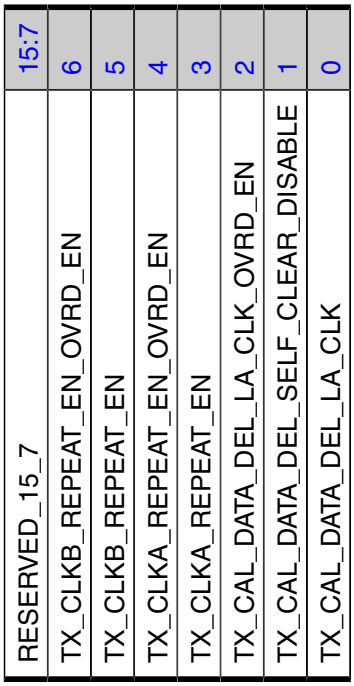


Table 2-705 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	TX_CLKB_REPEAT_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_clkb_repeat_en_i. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_CLKB_REPEAT_EN	R/W	This bit is the override value for tx_ana_clkb_repeat_en_i. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_CLKA_REPEAT_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_clka_repeat_en_i. **Value After Reset:** 0x0

Table 2-705 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	TX_CLKA_REPEAT_EN	R/W	This bit is the override value for tx_ana_clka_repeat_en_i. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_CAL_DATA_DEL_LA_CLK_OVRD_EN	R/W	This bit is the override enable for tx_ana_cal_data_del_la_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_CAL_DATA_DEL_SELF_CLEAR_DISABLE	R/W	This bit disables self-clearing for the tx_ana_cal_data_del_ovrd_en register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CAL_DATA_DEL_LA_CLK	R/W	This bit is the value for tx_ana_cal_data_del_la_clk_r. If TX_CAL_DATA_DEL_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (i.e. only asserts for one cr_clk cycle). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.702 LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_1 (for N = 0; N <= NLANES-1)

- **Description:** TX Data Delay Enable Clock Register 1.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)7c
- **Exists:** Always

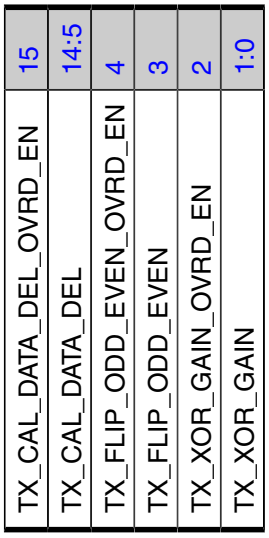


Table 2-706 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_CAL_DATA_DEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_cal_data_del[9:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:5	TX_CAL_DATA_DEL	R/W	This field is the override value for tx_ana_cal_data_del[9:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff
4	TX_FLIP_ODD_EVEN_OVRD_EN	R/W	This bit is the override enable for tx_ana_flip_odd_even. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_FLIP_ODD_EVEN	R/W	This bit is the override value for tx_ana_flip_odd_even. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-706 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2	TX_XOR_GAIN_OVRD_EN	R/W	This bit is the override enable for tx_ana_xor_gain. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	TX_XOR_GAIN	R/W	This field is the override value for tx_ana_xor_gain. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.703 LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_2 (for N = 0; N <= NLANES-1)

- **Description:** TX Data Delay Enable Clock Register 2.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)7d
- **Exists:** Always

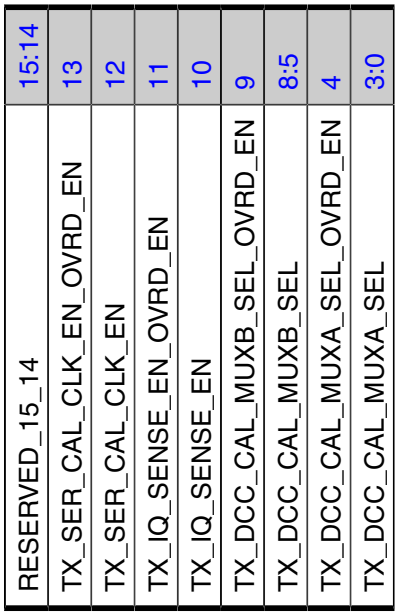


Table 2-707 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	TX_SER_CAL_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_ser_cal_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	TX_SER_CAL_CLK_EN	R/W	This bit is the override value for tx_ana_ser_cal_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	TX_IQ_SENSE_EN_OVRD_EN	R/W	This bit is the override enable for tx_ana_iq_sense_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-707 Fields for Register: LANEN_DIG_ANA_TX_ANA_CAL_DATA_DEL_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
10	TX_IQ_SENSE_EN	R/W	This bit is the override value for tx_ana_iq_sense_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_DCC_CAL_MUXB_SEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_dcc_cal_muxb_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:5	TX_DCC_CAL_MUXB_SEL	R/W	This field is the override value for tx_ana_dcc_cal_muxb_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
4	TX_DCC_CAL_MUXA_SEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_dcc_cal_muxa_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	TX_DCC_CAL_MUXA_SEL	R/W	This field is the override value for tx_ana_dcc_cal_muxa_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.704 LANEN_DIG_ANA_RX_DIV_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for RX DIV CLK Signals Going to ANA register.
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)7e
- **Exists:** Always

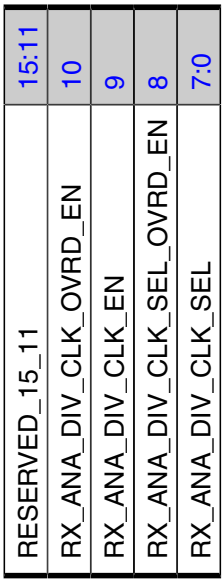


Table 2-708 Fields for Register: LANEN_DIG_ANA_RX_DIV_CLK_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RX_ANA_DIV_CLK_OVRD_EN	R/W	This bit is the override enable for tx_ana_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RX_ANA_DIV_CLK_EN	R/W	This bit is the override value for tx_ana_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_ANA_DIV_CLK_SEL_OVRD_EN	R/W	This bit is the override enable for tx_ana_div_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-708 Fields for Register: LANEN_DIG_ANA_RX_DIV_CLK_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7:0	RX_ANA_DIV_CLK_SEL	R/W	This field is the override value for tx_ana_div_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.705 LANEN_ANA_TX_OVRD_MEAS (for N = 0; N <= NLANES-1)

- **Description:** TX_OVRD_MEAS
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c0
- **Exists:** Always

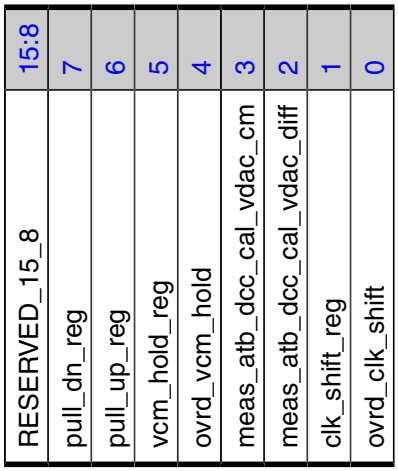


Table 2-709 Fields for Register: LANEN_ANA_TX_OVRD_MEAS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	pull_dn_reg	R/W	Pull down TX output if asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	pull_up_reg	R/W	Pull up TX output if asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	vcm_hold_reg	R/W	Set Tx in common mode if asserted together with bit 4 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_vcm_hold	R/W	If asserted, bit 5 takes effect on control Tx common mode **Value After Reset:** 0x0

Table 2-709 Fields for Register: LANEN_ANA_TX_OVRD_MEAS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	meas_atb_dcc_cal_vdac_cm	R/W	If asserted, atb_s_p/m reads voltage on idcc_cm_p/m nodes, respectively. This represents Common-Mode DCC DAC output voltage. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_atb_dcc_cal_vdac_diff	R/W	If asserted, atb_s_p/m reads voltage on idcc_diff_p/m nodes, respectively. This represents Differential DCC DAC output voltage. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	clk_shift_reg	R/W	Controls clock shift if asserted with bit 0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_clk_shift	R/W	If asserted, allow analog register to control clock shift function **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.706 LANEN_ANA_TX_PWR_OVRD (for N = 0; N <= NLANES-1)

- **Description:** TX_PWR_OVRD
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c1
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	ovrd_en	serial_en_reg	clk_en_reg	data_en_reg_int	clk_div_en_reg	refgen_en_reg	loopback_en_reg	ovrd_tx_loopback

Table 2-710 Fields for Register: LANEN_ANA_TX_PWR_OVRD (for N = 0; N <= NLANES-1)

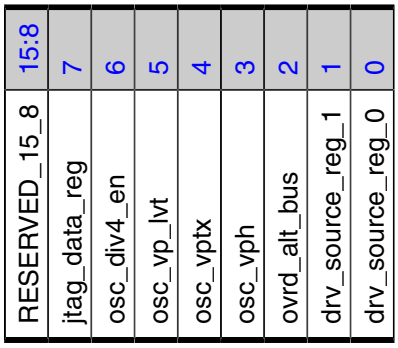
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ovrd_en	R/W	Enable analog register to take control TX power state if asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	serial_en_reg	R/W	Enable TX serializer if assered with bit 7 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	clk_en_reg	R/W	Enable TX clock if asserted with bit 7 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	data_en_reg_int	R/W	Enable TX driver data path if asserted with bit 7 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-710 Fields for Register: LANEN_ANA_TX_PWR_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	clk_div_en_reg	R/W	Enable TX divider if asserted with bit 7, overrides !tx_reset **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	refgen_en_reg	R/W	Enable TX biasing if asserted with bit 7 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	loopback_en_reg	R/W	Enable TX loopback path to RX if asserted along with bit 0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_tx_loopback	R/W	Enable Tx loopback mode over ridden by analog register if asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.707 LANEN_ANA_TX_ALT_BUS (for N = 0; N <= NLANES-1)

- **Description:** TX_ALT_BUS
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c2
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	jtag_data_reg	osc_div4_en	osc_vp_lvt	osc_vptx	osc_vph	ovrd_alt_bus	drv_source_reg_1	drv_source_reg_0

Table 2-711 Fields for Register: LANEN_ANA_TX_ALT_BUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	jtag_data_reg	R/W	When bit 2 is asserted, It replace jtag data **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	osc_div4_en	R/W	ring oscillator selected outputs divide-by-4 from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	osc_vp_lvt	R/W	Vp powered lvt device ring oscillator outputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	osc_vptx	R/W	Vptx powered svt device ring oscillator outputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-711 Fields for Register: LANEN_ANA_TX_ALT_BUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	osc_vph	R/W	Vph powered ring oscillator outputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_alt_bus	R/W	Jtag data and tx data source selection function is controlled by bit _1:0 and 7 when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	drv_source_reg_1	R/W	When bit 2 is asserted, drv_source_reg_1:0 takes control of TX function, overrides tx_data_source_1:0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	drv_source_reg_0	R/W	When bit 2 is asserted, drv_source_reg_1:0 takes control of TX function, overrides tx_data_source_1:0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.708 LANEN_ANA_TX_ATB1 (for N = 0; N <= NLANES-1)

- **Description:** TX_ATB1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c3
- **Exists:** Always

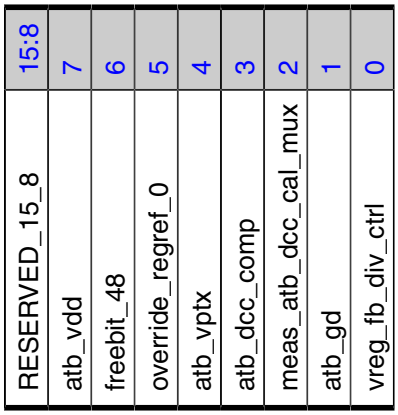


Table 2-712 Fields for Register: LANEN_ANA_TX_ATB1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	atb_vdd	R/W	If asserted, atb_s_p reads vdd_h/3 (charge pump output voltage) for tx_vreg **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	freebit_48	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	override_regref_0	R/W	use atb_s_m as TX regulator reference voltage when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	atb_vptx	R/W	If asserted, atb_s_p reads vptx **Value After Reset:** 0x0

Table 2-712 Fields for Register: LANEN_ANA_TX_ATB1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	atb_dcc_comp	R/W	If asserted, atb_s_p/m reads p/m outputs of opamp which amplifies signal entering DCC comparator. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_atb_dcc_cal_mux	R/W	If asserted, atb_s_p/m reads p/m inputs entering AC coupling mux inside tx_cal_top. This would generally be the filtered clock used by the DCC loop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	atb_gd	R/W	if asserted, atb_s_p reads gd for tx_vreg **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	vreg_fb_div_ctrl	R/W	Adjust gain of vreg_cp; 0: vr/vref=2, 1:vr/vref=2.1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.709 LANEN_ANA_TX_ATB2 (for N = 0; N <= NLANES-1)

- **Description:** TX_ATB2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c4
- **Exists:** Always

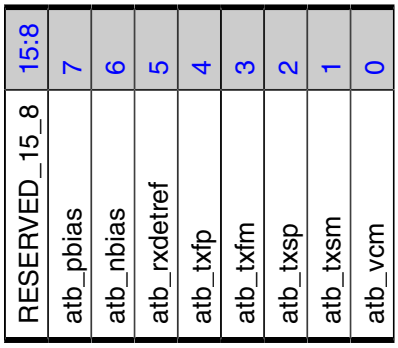


Table 2-713 Fields for Register: LANEN_ANA_TX_ATB2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	atb_pbias	R/W	Put TX pbias on atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	atb_nbias	R/W	Put TXnbias on atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	atb_rxdetref	R/W	Put TX RX detection reference voltage on atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	atb_txfp	R/W	Overdrive tx_p through atb_f_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-713 Fields for Register: LANEN_ANA_TX_ATB2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	atb_txfm	R/W	Overdrive tx_m through atb_f_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	atb_txsp	R/W	Measure tx_p through atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	atb_txsm	R/W	Measure tx_m through atb_s_m when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	atb_vcm	R/W	Measure Tx vcm through atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.710 LANEN_ANA_TX_ATB3 (for N = 0; N <= NLANES-1)

- **Description:** TX_ATB3
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c5
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	freebit_49	freebit_50	freebit_51	freebit_52	freebit_53	freebit_54	atb_s_enable	meas_atb_vph_half

Table 2-714 Fields for Register: LANEN_ANA_TX_ATB3 (for N = 0; N <= NLANES-1)

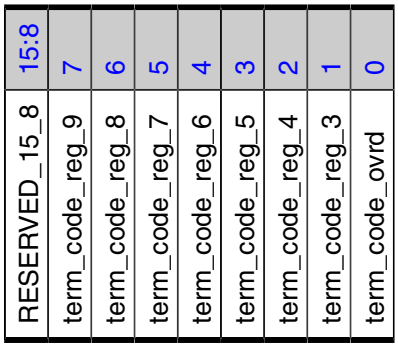
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	freebit_49	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	freebit_50	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	freebit_51	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	freebit_52	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	freebit_53	R/W	Not used **Value After Reset:** 0x0

Table 2-714 Fields for Register: LANEN_ANA_TX_ATB3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	freebit_54	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	atb_s_enable	R/W	Enables TX atb function if asserted This bit has to be set to 1 in order to make TX atb_s_p/m visible **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_atb_vph_half	R/W	Measure vph/3 for tx_vreg on atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.711 LANEN_ANA_TX_TERM_CODE_DN (for N = 0; N <= NLANES-1)

- **Description:** TX_TERM_CODE_DN
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c6
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	term_code_reg_9	term_code_reg_8	term_code_reg_7	term_code_reg_6	term_code_reg_5	term_code_reg_4	term_code_reg_3	term_code_ovrd

Table 2-715 Fields for Register: LANEN_ANA_TX_TERM_CODE_DN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	term_code_reg_9	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	term_code_reg_8	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	term_code_reg_7	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	term_code_reg_6	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	term_code_reg_5	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0

Table 2-715 Fields for Register: LANEN_ANA_TX_TERM_CODE_DN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	term_code_reg_4	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	term_code_reg_3	R/W	TX nmos leg biasing register (7 MSBs) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	term_code_ovrd	R/W	Enable analog register to overdrive TX nmos leg biasing **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.712 LANEN_ANA_TX_TERM_CODE_UP (for N = 0; N <= NLANES-1)

- **Description:** TX_TERM_CODE_UP
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c7
- **Exists:** Always

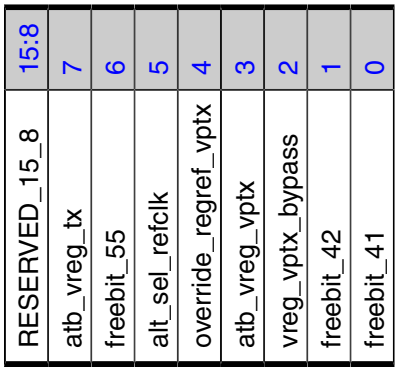


Table 2-716 Fields for Register: LANEN_ANA_TX_TERM_CODE_UP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	atb_vreg_tx	R/W	Measure vreg_tx through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	freebit_55	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	alt_sel_refclk	R/W	Turning on transmitting refclk to TX when drv_source_1:0=00 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	override_regref_vptx	R/W	if asserted, TX regulator vreg_vptx input reference voltage is forced through atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-716 Fields for Register: LANEN_ANA_TX_TERM_CODE_UP (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	atb_vreg_vptx	R/W	Measure vreg_vptx/2 through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	vreg_vptx_bypass	R/W	if asserted, the vreg_vptx regulator is disabled and its output shorted to VPTX **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	freebit_42	R/W	reserved bits **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	freebit_41	R/W	reserved bits **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.713 LANEN_ANA_TX_IBOOST_CODE (for N = 0; N <= NLANES-1)

- **Description:** TX_IBOOST_CODE
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c8
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	freebit_43	freebit_44	freebit_45	freebit_46	freebit_47	term_code_reg_2	lfps_high_priority	freebit_40

Table 2-717 Fields for Register: LANEN_ANA_TX_IBOOST_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	freebit_43	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	freebit_44	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	freebit_45	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	freebit_46	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	freebit_47	R/W	Not used **Value After Reset:** 0x0

Table 2-717 Fields for Register: LANEN_ANA_TX_IBOOST_CODE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	term_code_reg_2	R/W	TX pmos leg biasing register bit 2 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	lfps_high_priority	R/W	If asserted, lfps/beacon enable has higher priority than data enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	freebit_40	R/W	reserved bits **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.714 LANEN_ANA_TX_OVRD_CLK (for N = 0; N <= NLANES-1)

- **Description:** TX_OVRD_CLK
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)c9
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	ovrd_word_clk_en	word_clk_en_reg	vreg_mplla_boost_1	vreg_mplla_boost_0	tx_dsq_acfb_enb	ovrd_lb_en	clk_lb_en_reg	osc_vp_ulvt

Table 2-718 Fields for Register: LANEN_ANA_TX_OVRD_CLK (for N = 0; N <= NLANES-1)

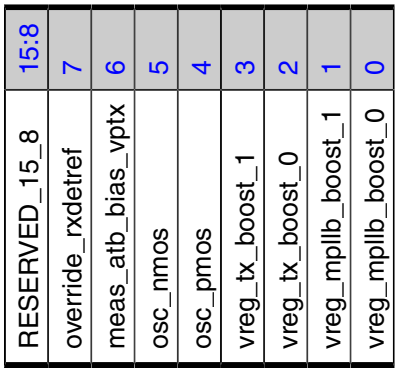
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ovrd_word_clk_en	R/W	If asserted, it allows bit 6 to override tx word clock enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	word_clk_en_reg	R/W	Tx word clock enable/disable when bit 7 is asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	vreg_mplla_boost_1	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	vreg_mplla_boost_0	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-718 Fields for Register: LANEN_ANA_TX_OVRD_CLK (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	tx_dsq_acfb_enb	R/W	If asserted this disables the High Frequency feedback path for the desqew cells along the clock path **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_lb_en	R/W	If asserted, it allows bit 1 takes control of RX clock loopback to TX **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	clk_lb_en_reg	R/W	When asserted with bit 2, selects RX clock for TX data output clock **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	osc_vp_ulvt	R/W	Vp powered ulvt device ring oscillator outputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.715 LANEN_ANA_TX_MISC (for N = 0; N <= NLANES-1)

- **Description:** TX_MISC
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ca
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	override_rxdetref	meas_atb_bias_vptx	osc_nmos	osc_pmos	vreg_tx_boost_1	vreg_tx_boost_0	vreg_mpllb_boost_1	vreg_mpllb_boost_0

Table 2-719 Fields for Register: LANEN_ANA_TX_MISC (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	override_rxdetref	R/W	If asserted, atb_s_m is used to override RX detection reference voltage **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_atb_bias_vptx	R/W	Measure TX bias local vptx through atb_s_p when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	osc_nmos	R/W	nmos device ring oscillator ouputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	osc_pmos	R/W	pmos device ring oscillator ouputs from TX when asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-719 Fields for Register: LANEN_ANA_TX_MISC (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	vreg_tx_boost_1	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	vreg_tx_boost_0	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	vreg_mpllb_boost_1	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	vreg_mpllb_boost_0	R/W	Adjust gain of regulator for tx clock path. 00: gain=2, 01: gain =2.333, 10: gain=2.5, 11: gain=2.6 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.716 LANEN_ANA_RX_ATB_IQSKEW (for N = 0; N <= NLANES-1)

- **Description:** RX_ATB_IQSKEW
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)cb
- **Exists:** Always

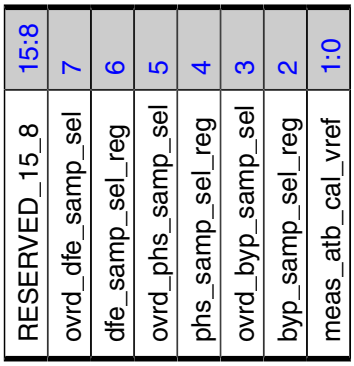


Table 2-720 Fields for Register: LANEN_ANA_RX_ATB_IQSKEW (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ovrd_dfe_samp_sel	R/W	If asserted, enable override for dfe_samp_sel **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	dfe_samp_sel_reg	R/W	when asserted with bit 7, enables DFE clock sample to rx_cal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	ovrd_phs_samp_sel	R/W	If asserted, enable override for phs_samp_sel **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	phs_samp_sel_reg	R/W	when asserted with bit 5, enables PHS clock sample to rx_cal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	ovrd_byp_samp_sel	R/W	If asserted, enable override for byp_samp_sel **Value After Reset:** 0x0

Table 2-720 Fields for Register: LANEN_ANA_RX_ATB_IQSKEW (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	byp_samp_sel_reg	R/W	when asserted with bit 5, enables PHS clock sample to rx_cal **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	meas_atb_cal_vref	R/W	When bit 0 is asserted, enable ATB measure for vref_pre When bit 1 is asserted, enable ATB measure for vref_lpf **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.717 LANEN_ANA_RX_DCC_OVRD (for N = 0; N <= NLANES-1)

- **Description:** RX_DCC_OVRD
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)cc
- **Exists:** Always

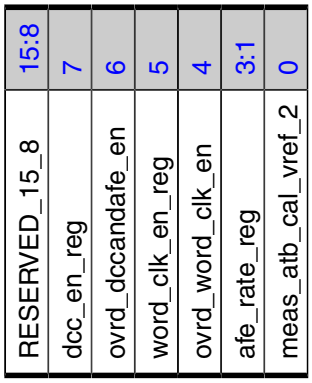


Table 2-721 Fields for Register: LANEN_ANA_RX_DCC_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	dcc_en_reg	R/W	When bit 6 is asserted, this bit takes control of RX DCC enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ovrd_dccandafe_en	R/W	When asserted, override RX DCC enable and afe data rate by analog register **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	word_clk_en_reg	R/W	If asserted, it allows bit 4 to override rx word clock enable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_word_clk_en	R/W	rx word clock enable/disable when bit 5 is asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-721 Fields for Register: LANEN_ANA_RX_DCC_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3:1	afe_rate_reg	R/W	When bit 6 is asserted, these bits takes control of afe_rate **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
0	meas_atb_cal_vref_2	R/W	When bit 0 is asserted, enable ATB measure for vref_sigdet **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.718 LANEN_ANA_RX_PWR_CTRL1 (for N = 0; N <= NLANES-1)

- **Description:** RX_PWR_CTRL1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)cd
- **Exists:** Always

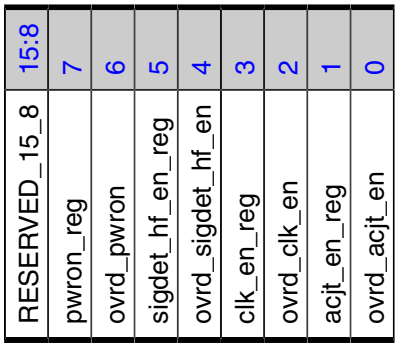


Table 2-722 Fields for Register: LANEN_ANA_RX_PWR_CTRL1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	pwron_reg	R/W	When asserted with bit 6, enables AFE **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ovrd_pwron	R/W	If asserted, bit 7 takes control of AFE enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	sigdet_hf_en_reg	R/W	When asserted with bit 4, emables rx_sigdet_hf **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_sigdet_hf_en	R/W	If asserted, bit 5 take control of sigdet_hf enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	clk_en_reg	R/W	When asserted with bit 2, emables RX clock **Value After Reset:** 0x0

Table 2-722 Fields for Register: LANEN_ANA_RX_PWR_CTRL1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	ovrd_clk_en	R/W	If asserted, bit 3 take control of RX clock enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	acjt_en_reg	R/W	When asserted with bit 0, emables ACJTAG **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_acjt_en	R/W	If asserted, bit 1 take control of ACJTAG enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.719 LANEN_ANA_RX_ATB_REGREF (for N = 0; N <= NLANES-1)

- **Description:** RX_ATB_REGREF
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ce
- **Exists:** Always

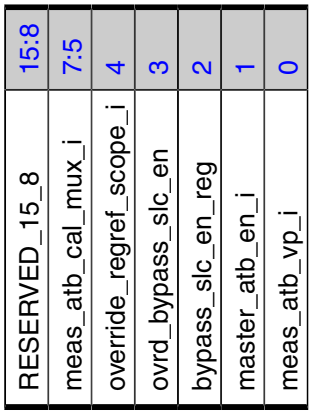


Table 2-723 Fields for Register: LANEN_ANA_RX_ATB_REGREF (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:5	meas_atb_cal_mux_i	R/W	if bit 7 is asserted, cal comparator outputs are sensed through ATB if bit 6 is asserted, cal lpf outputs (1) are sensed through ATB if bit 5 is asserted, cal lpf outputs (2) are sensed through ATB **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
4	override_regref_scope_i	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	ovrd_bypass_slc_en	R/W	If asserted, bit 2 take control of bypass slicer path **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	bypass_slc_en_reg	R/W	When asserted with bit 3, emables bypass slicer path **Value After Reset:** 0x0

Table 2-723 Fields for Register: LANEN_ANA_RX_ATB_REGREF (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
1	master_atb_en_i	R/W	if asserted, master ATB is enabled **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_atb_vp_i	R/W	if asserted, vp is sensed through ATB **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.720 LANEN_ANA_RX_CDR_AFE (for N = 0; N <= NLANES-1)

- **Description:** RX_CDR_AFE
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)cf
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	phdet_even_reg	phdet_odd_reg	meas_atb_sum_e	meas_atb_sum_o	Meas_atb_rx_9	Meas_atb_rx_8	Meas_atb_rx_7	Meas_atb_rx_6

Table 2-724 Fields for Register: LANEN_ANA_RX_CDR_AFE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	phdet_even_reg	R/W	If asserted, CDR phase detector only uses even data path **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	phdet_odd_reg	R/W	If asserted, CDR phase detector only uses odd data path **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	meas_atb_sum_e	R/W	if asserted, even DFE summer output is sensed through atb **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_atb_sum_o	R/W	if asserted, odd DFE summer output is sensed through atb **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	Meas_atb_rx_9	R/W	If asserted, AFE biasing vbp is measured through atb_s_p **Value After Reset:** 0x0

Table 2-724 Fields for Register: LANEN_ANA_RX_CDR_AFE (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	Meas_atb_rx_8	R/W	If asserted, AFE biasing vbn is measured through atb_s_m **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	Meas_atb_rx_7	R/W	If asseted, rx_p is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	Meas_atb_rx_6	R/W	If asserted, rx_p is driven by atb_f_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.721 LANEN_ANA_RX_PWR_CTRL2 (for N = 0; N <= NLANES-1)

- **Description:** RX_PWR_CTRL2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d0
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	loopback_en_reg	ovrd_loopback_en	deserial_en_reg	ovrd_des_en	dfe_en_reg	ovrd_dfe_en	sigdet_lf_en_reg	ovrd_sigdet_lf_en

Table 2-725 Fields for Register: LANEN_ANA_RX_PWR_CTRL2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	loopback_en_reg	R/W	If asserted with bit 6, enables Tx to Rx data loopback **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	ovrd_loopback_en	R/W	If asserted, bit 7 takes control of TX to Rx data loopback enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	deserial_en_reg	R/W	If asserted with bit 4, enables Rx deserializer **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_des_en	R/W	If asserted, bit 5 takes control of Rx deserializer enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-725 Fields for Register: LANEN_ANA_RX_PWR_CTRL2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	dfe_en_reg	R/W	If asserted with bit 2, enables RX DFE **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ovrd_dfe_en	R/W	If asserted, bit 3 takes control of RX DFE enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	sigdet_lf_en_reg	R/W	When asserted with bit 0, emables rx_sigdet_lf **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_sigdet_lf_en	R/W	If asserted, bit 1 take control of sigdet_lf enable/disable **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.722 LANEN_ANA_RX_MISC_OVRD (for N = 0; N <= NLANES-1)

- **Description:** RX_MISC_OVRD
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d1
- **Exists:** Always

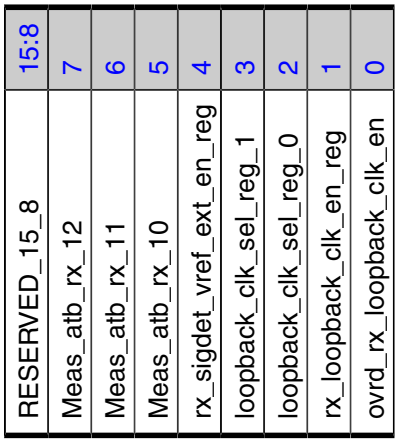


Table 2-726 Fields for Register: LANEN_ANA_RX_MISC_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	Meas_atb_rx_12	R/W	if asserted, enables sigdet HF sig to atb_s_p_int and HF ref to atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	Meas_atb_rx_11	R/W	if asserted, enables sigdet LF sig_lfps to atb_s_p_int and HF ref_lfps to atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	Meas_atb_rx_10	R/W	if asserted, enables sigdet bias test point **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	rx_sigdet_vref_ext_en_reg	R/W	if asserted, enables external vref for rx_sigdet **Value After Reset:** 0x0

Table 2-726 Fields for Register: LANEN_ANA_RX_MISC_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	loopback_clk_sel_reg_1	R/W	if asserted with bit 1, enable/disables loop_clk_sel_1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	loopback_clk_sel_reg_0	R/W	if asserted with bit 1, enable/disables loop_clk_sel_0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	rx_loopback_clk_en_reg	R/W	loopback clk en .valid only when ovrd_rx_loopback_clk_en is asserted **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ovrd_rx_loopback_clk_en	R/W	overides loopback en from digital and selects loopback_clk_en from regs **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.723 LANEN_ANA_RX_CAL_MUXA (for N = 0; N <= NLANES-1)

- **Description:** RX_CAL_MUXA
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d2
- **Exists:** Always

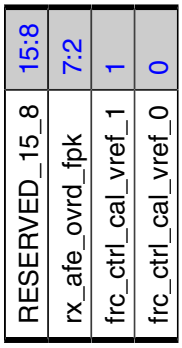
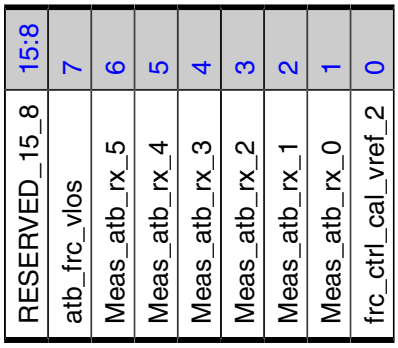


Table 2-727 Fields for Register: LANEN_ANA_RX_CAL_MUXA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:2	rx_afe_ovrd_fpk	R/W	AFE control overide bits, bits 5-4 for CTLE, bits 3-2 for VGA2, bits 1-0 for VGA1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
1	frc_ctrl_cal_vref_1	R/W	if bit_1 is asserted, vref_lpf is forced from atb_f_p_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	frc_ctrl_cal_vref_0	R/W	if bit_0 is asserted, vref_pre is forced from atb_f_p_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.724 LANEN_ANA_RX_ATB_MEAS1 (for N = 0; N <= NLANES-1)

- **Description:** RX_ATB_MEAS1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d3
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	atb_frc_vlos	Meas_atb_rx_5	Meas_atb_rx_4	Meas_atb_rx_3	Meas_atb_rx_2	Meas_atb_rx_1	Meas_atb_rx_0	frc_ctrl_cal_vref_2

Table 2-728 Fields for Register: LANEN_ANA_RX_ATB_MEAS1 (for N = 0; N <= NLANES-1)

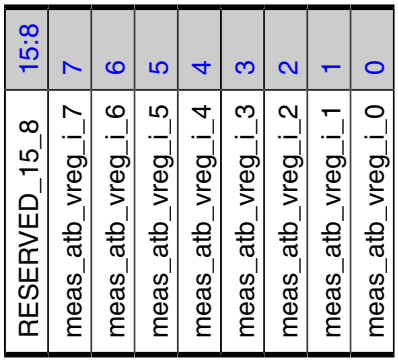
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	atb_frc_vlos	R/W	If asserted, force los detection reference voltage through atb_s_m (not used) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	Meas_atb_rx_5	R/W	If asseted, rx_m is sensed through atb_s_m **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	Meas_atb_rx_4	R/W	If asserted, rx_m is driven by atb_f_m **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	Meas_atb_rx_3	R/W	If asserted, measure RX LOS detection threshold voltage (not used) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-728 Fields for Register: LANEN_ANA_RX_ATB_MEAS1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	Meas_atb_rx_2	R/W	If asserted, measure RX bias reference voltage **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	Meas_atb_rx_1	R/W	If asserted, measure RX bias pmos biasing voltage **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	Meas_atb_rx_0	R/W	If asserted, measure RX bias cascaded pmos biasing voltage **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	frc_ctrl_cal_vref_2	R/W	if asserted, force sigdet reference voltage through atb_s_p_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.725 LANEN_ANA_RX_ATB_MEAS2 (for N = 0; N <= NLANES-1)

- **Description:** RX_ATB_MEAS2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d4
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_atb_vreg_i_7	meas_atb_vreg_i_6	meas_atb_vreg_i_5	meas_atb_vreg_i_4	meas_atb_vreg_i_3	meas_atb_vreg_i_2	meas_atb_vreg_i_1	meas_atb_vreg_i_0

Table 2-729 Fields for Register: LANEN_ANA_RX_ATB_MEAS2 (for N = 0; N <= NLANES-1)

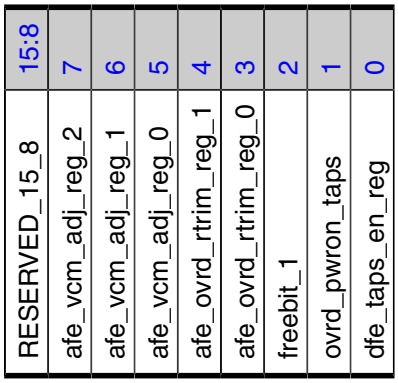
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_atb_vreg_i_7	R/W	Measure vreg_byp through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_atb_vreg_i_6	R/W	Measure RX regulator gd through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_atb_vreg_i_5	R/W	Measure RX regulator vdd_h (charge pump output) through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_atb_vreg_i_4	R/W	unused **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-729 Fields for Register: LANEN_ANA_RX_ATB_MEAS2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	meas_atb_vreg_i_3	R/W	Measure vreg_clk through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_atb_vreg_i_2	R/W	Measure vreg_dfe through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_atb_vreg_i_1	R/W	Measure RX regulator vph through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	meas_atb_vreg_i_0	R/W	Measure vreg_clk_vro through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.726 LANEN_ANA_RX_CAL_MUXB (for N = 0; N <= NLANES-1)

- **Description:** RX_CAL_MUXB
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d5
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	afe_vcm_adj_reg_2	afe_vcm_adj_reg_1	afe_vcm_adj_reg_0	afe_ovrd_rtrim_reg_1	afe_ovrd_rtrim_reg_0	freebit_1	ovrd_pwron_taps	dfe_taps_en_reg

Table 2-730 Fields for Register: LANEN_ANA_RX_CAL_MUXB (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	afe_vcm_adj_reg_2	R/W	sets bit 2 of AFE ATT output common-mode level **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	afe_vcm_adj_reg_1	R/W	sets bit 1 of AFE ATT output common-mode level **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	afe_vcm_adj_reg_0	R/W	sets bit 0 of AFE ATT output common-mode level **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	afe_ovrd_rtrim_reg_1	R/W	sets bit 1 of AFE rtrim override **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-730 Fields for Register: LANEN_ANA_RX_CAL_MUXB (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	afe_ovrd_rtrim_reg_0	R/W	sets bit 0 of AFE rtrim override **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	freebit_1	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ovrd_pwron_taps	R/W	If 1, allows for bit 0 to control DFE tap2 to tap5 currents **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	dfe_taps_en_reg	R/W	If bit 1 is asserted, controls DFE tap2 to tap5 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.727 LANEN_ANA_RX_TERM (for N = 0; N <= NLANES-1)

- **Description:** RX_TERM
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d6
- **Exists:** Always

15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	vreg_ring_ctrl_reg_1	vreg_ring_ctrl_reg_0	ovrd_rx_term_dc_en	rx_term_dc_en_reg	ovrd_rx_term_gd_en	rx_term_gd_en_reg	frc_ctrl_cal_vref_4	frc_ctrl_cal_vref_3

Table 2-731 Fields for Register: LANEN_ANA_RX_TERM (for N = 0; N <= NLANES-1)

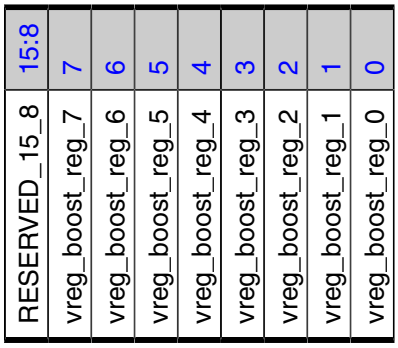
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	vreg_ring_ctrl_reg_1	R/W	if asserted, enable/disables rx_vreg ring_ctrl_1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	vreg_ring_ctrl_reg_0	R/W	if asserted, enable/disables rx_vreg ring_ctrl_0 **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	ovrd_rx_term_dc_en	R/W	If asserted, bit 4 takes control of rx_term_dc_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	rx_term_dc_en_reg	R/W	if asserted with bit 5, enable/disabes rx_term_dc_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-731 Fields for Register: LANEN_ANA_RX_TERM (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	ovrd_rx_term_gd_en	R/W	If asserted, bit 2 takes control of rx_term_gd_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	rx_term_gd_en_reg	R/W	if asserted with bit 3, enable/disabes rx_term_gd_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	frc_ctrl_cal_vref_4	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	frc_ctrl_cal_vref_3	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.728 LANEN_ANA_RX_SLC_CTRL (for N = 0; N <= NLANES-1)

- **Description:** RX_SLC_CTRL
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d7
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	vreg_boost_reg_7	vreg_boost_reg_6	vreg_boost_reg_5	vreg_boost_reg_4	vreg_boost_reg_3	vreg_boost_reg_2	vreg_boost_reg_1	vreg_boost_reg_0

Table 2-732 Fields for Register: LANEN_ANA_RX_SLC_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	vreg_boost_reg_7	R/W	bit_7:6 controls vreg_clk boost setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	vreg_boost_reg_6	R/W	bit_7:6 controls vreg_clk boost setting **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	vreg_boost_reg_5	R/W	bit_5:4 controls vreg_dfe boost setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	vreg_boost_reg_4	R/W	bit_5:4 controls vreg_dfe boost setting **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	vreg_boost_reg_3	R/W	bit_3:2. controls vreg_byp boost setting **Value After Reset:** 0x0

Table 2-732 Fields for Register: LANEN_ANA_RX_SLC_CTRL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	vreg_boost_reg_2	R/W	bit_3:2. controls vreg_byp boost setting **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	vreg_boost_reg_1	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	vreg_boost_reg_0	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.729 LANEN_ANA_RX_ATB_VREG (for N = 0; N <= NLANES-1)

- **Description:** RX_ATB_VREG
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d8
- **Exists:** Always

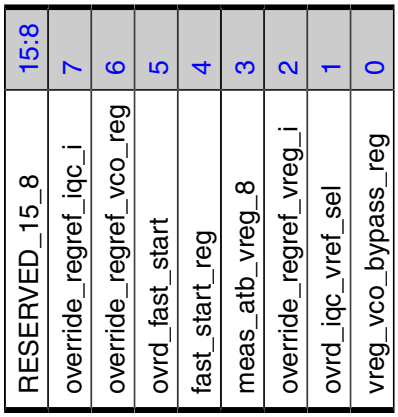


Table 2-733 Fields for Register: LANEN_ANA_RX_ATB_VREG (for N = 0; N <= NLANES-1)

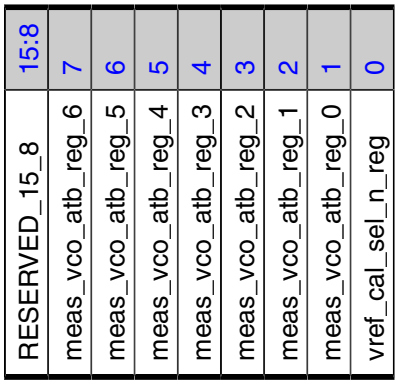
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	override_regref_iqc_i	R/W	if asserted, RX regulator reg_clk_vro input reference voltage is forced through atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	override_regref_vco_reg	R/W	if asserted, RX regulator vreg_vco input reference voltage is forced through atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	ovrd_fast_start	R/W	If asserted, bit 4 takes control of RX_VREG fast start **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	fast_start_reg	R/W	if asserted with bit 5, enable/disabes RX_VREG fast start **Value After Reset:** 0x0

Table 2-733 Fields for Register: LANEN_ANA_RX_ATB_VREG (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	meas_atb_vreg_8	R/W	Measure vreg_vco through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	override_regref_vreg_i	R/W	if asserted, the input reference for RX regulators charge pump, vreg_clk, vreg_dfe and vreg_byp is forced through atb_s_m_int **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ovrd_iqc_vref_sel	R/W	If asserted the reference for the reg_clk_vro regulator is controlled via registers **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	vreg_vco_bypass_reg	R/W	if asserted, vreg_vco regulator output is pulled to VPH (TBD) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.730 LANEN_ANA_RX_CDR_VCO_CTRL (for N = 0; N <= NLANES-1)

- **Description:** RX_CDR_VCO_CTRL
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)d9
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	meas_vco_atb_reg_6	meas_vco_atb_reg_5	meas_vco_atb_reg_4	meas_vco_atb_reg_3	meas_vco_atb_reg_2	meas_vco_atb_reg_1	meas_vco_atb_reg_0	vref_cal_sel_n_reg

Table 2-734 Fields for Register: LANEN_ANA_RX_CDR_VCO_CTRL (for N = 0; N <= NLANES-1)

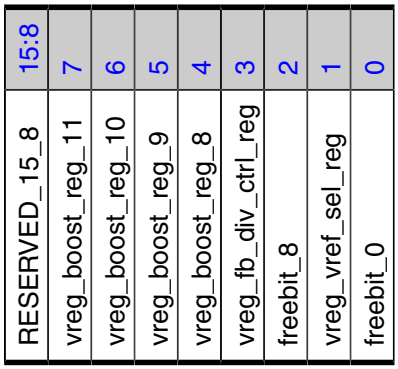
Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	meas_vco_atb_reg_6	R/W	if asserted, VCO vregbg is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	meas_vco_atb_reg_5	R/W	if asserted, VCO rx_vref_ext is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	meas_vco_atb_reg_4	R/W	if asserted, VCO rx_vref is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	meas_vco_atb_reg_3	R/W	if asserted, VCO Vosc is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-734 Fields for Register: LANEN_ANA_RX_CDR_VCO_CTRL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	meas_vco_atb_reg_2	R/W	if asserted, VCO gd is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	meas_vco_atb_reg_1	R/W	if asserted, VCO Inref_1 is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	meas_vco_atb_reg_0	R/W	if asserted, VCO gd is sensed through atb_s_p **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	vref_cal_sel_n_reg	R/W	if de-asserted, vref_cal is set to be 0.45V in RX_CAL vgen **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.731 LANEN_ANA_RX_VREG_CTRL2 (for N = 0; N <= NLANES-1)

- **Description:** RX_VREG_CTRL2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)da
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	vreg_boost_reg_11	vreg_boost_reg_10	vreg_boost_reg_9	vreg_boost_reg_8	vreg_fb_div_ctrl_reg	freebit_8	vreg_vref_sel_reg	freebit_0

Table 2-735 Fields for Register: LANEN_ANA_RX_VREG_CTRL2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	vreg_boost_reg_11	R/W	bit_11:10 controls vreg_vco boost setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	vreg_boost_reg_10	R/W	bit_11:10 controls vreg_vco boost setting **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	vreg_boost_reg_9	R/W	bit_9:8 controls vreg_clk_vro boost setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	vreg_boost_reg_8	R/W	bit_9:8 controls vreg_clk_vro boost setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-735 Fields for Register: LANEN_ANA_RX_VREG_CTRL2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	vreg_fb_div_ctrl_reg	R/W	if asserted, enable/disables rx_vreg CP reference voltage gain setting **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	freebit_8	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	vreg_vref_sel_reg	R/W	if asserted, enable vp/2-reference going to vregs **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	freebit_0	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.732 LANEN_ANA_RX_MISC_OVRD_2 (for N = 0; N <= NLANES-1)

- **Description:** RX_MISC_OVRD_2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)db
- **Exists:** Always

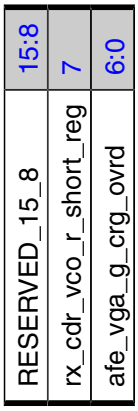
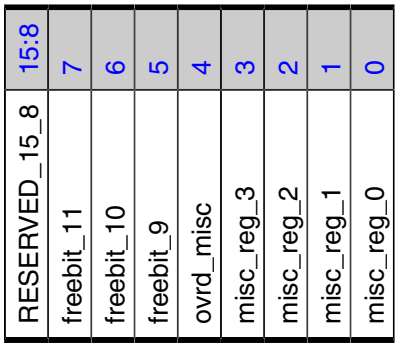


Table 2-736 Fields for Register: LANEN_ANA_RX_MISC_OVRD_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	rx_cdr_vco_r_short_reg	R/W	Resistor is shorted with a switch in VCO refgen block and if set to b'0 switch is open **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	afe_vga_g_crg_ovrd	R/W	AFE vga override bits **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.733 LANEN_ANA_RX_MISC_OVRD_3 (for N = 0; N <= NLANES-1)

- **Description:** RX_MISC_OVRD_3
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)dc
- **Exists:** Always



15:8	7	6	5	4	3	2	1	0
RESERVED_15_8	freebit_11	freebit_10	freebit_9	ovrd_misc	misc_reg_3	misc_reg_2	misc_reg_1	misc_reg_0

Table 2-737 Fields for Register: LANEN_ANA_RX_MISC_OVRD_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	freebit_11	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	freebit_10	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	freebit_9	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ovrd_misc	R/W	if asserted, enables override of rx_misc<3:0> with misc_reg<3:0> **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-737 Fields for Register: LANEN_ANA_RX_MISC_OVRD_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	misc_reg_3	R/W	override values for rx_misc<3>, override occurs when ovrd_misc is asserted. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	misc_reg_2	R/W	override values for rx_misc<2>, override occurs when ovrd_misc is asserted. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	misc_reg_1	R/W	override values for rx_misc<1>, override occurs when ovrd_misc is asserted. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	misc_reg_0	R/W	override values for rx_misc<0>, override occurs when ovrd_misc is asserted. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.734 LANEN_ANA_RX_AFE_OVRD_1 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)dd
- **Exists:** Always

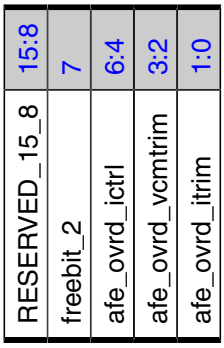
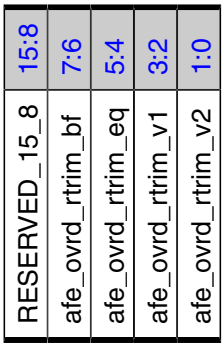


Table 2-738 Fields for Register: LANEN_ANA_RX_AFE_OVRD_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	freebit_2	R/W	not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:4	afe_ovrd_ictrl	R/W	override negC biasing currrent **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3:2	afe_ovrd_vcmtrim	R/W	override afe common mode voltage **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1:0	afe_ovrd_itrim	R/W	override afe constant biasing current **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.735 LANEN_ANA_RX_AFE_OVRD_2 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)de
- **Exists:** Always



15:8	7:6	5:4	3:2	1:0
RESERVED_15_8	afe_ovrd_rtrim_bf	afe_ovrd_rtrim_eq	afe_ovrd_rtrim_v1	afe_ovrd_rtrim_v2

Table 2-739 Fields for Register: LANEN_ANA_RX_AFE_OVRD_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:6	afe_ovrd_rtrim_bf	R/W	override buf stage rtrim code, reserved but not actually used in X321 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5:4	afe_ovrd_rtrim_eq	R/W	override eq stage rtrim code, reserved but not actually used in X321 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3:2	afe_ovrd_rtrim_v1	R/W	override vga1 stage rtrim code, reserved but not actually used in X321 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1:0	afe_ovrd_rtrim_v2	R/W	override vga2 stage rtrim code, reserved but not actually used in X321 **Value After Reset:** 0x0

Table 2-739 Fields for Register: LANEN_ANA_RX_AFE_OVRD_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3

2.1.736 LANEN_ANA_RX_AFE_OVRD_3 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_3
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)df
- **Exists:** Always

Table 2-740 Fields for Register: LANEN_ANA_RX_AFE_OVRD_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	afe_ovrd_en_lms	R/W	if asserted, enables all AFE overrides except eq, vga1 and vga2 fpeak bits; this ensures LMS can still function properly **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:4	afe_ovrd_gain_bf	R/W	override afe buf stage gain **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3:0	afe_ovrd_fpk_bf	R/W	override afe buf stage fpk **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.737 LANEN_ANA_ROPLL_0 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_0
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e0
- **Exists:** Always

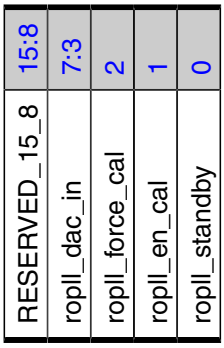
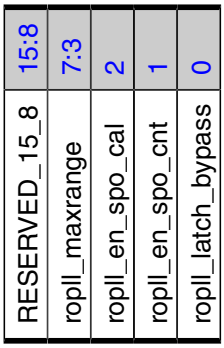


Table 2-741 Fields for Register: LANEN_ANA_ROPLL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:3	ropll_dac_in	R/W	Sets the initial open loop ICO current to set the open loop frequency when ropll_preset=1 **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0x1f
2	ropll_force_cal	R/W	Forces the DAC to use ropll_dac_in **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	ropll_en_cal	R/W	Enables calibration mode which isn't used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ropll_standby	R/W	Enables standby mode which isn't used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.738 LANEN_ANA_ROPLL_1 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_1
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e1
- **Exists:** Always



15:8	7:3	2	1	0
RESERVED_15_8	ropll_maxrange	ropll_en_spo_cal	ropll_en_spo_cnt	ropll_latch_bypass

Table 2-742 Fields for Register: LANEN_ANA_ROPLL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:3	ropll_maxrange	R/W	Sets max range for the cycle counter on the SPO calibration **Value After Reset:** 0x1f **Exists:** Always **Reset Mask:** 0x1f
2	ropll_en_spo_cal	R/W	Enables SPO calibration **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	ropll_en_spo_cnt	R/W	Sets the counter mode for SPO calibration **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	ropll_latch_bypass	R/W	sets the latch mode for SPO calibration **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.739 LANEN_ANA_ROPLL_2 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_2
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e2
- **Exists:** Always

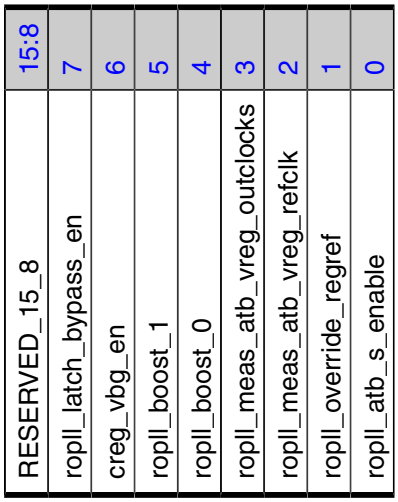


Table 2-743 Fields for Register: LANEN_ANA_ROPLL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ropll_latch_bypass_en	R/W	bypasses the latch in the SPO calibration engine **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	creg_vbg_en	R/W	Enables bandgap as primary reference for uPLL regulator **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	ropll_boost_1	R/W	Boost setting for ropll wrapper regulators **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ropll_boost_0	R/W	Boost setting for ropll wrapper regulators **Value After Reset:** 0x1

Table 2-743 Fields for Register: LANEN_ANA_ROPLL_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
3	ropll_meas_atb_vreg_outclocks	R/W	if asserted allows observation of outclocks regulator output in the ropll wrapper **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	ropll_meas_atb_vreg_refclk	R/W	if asserted allows observation of refclock regulator output in the ropll wrapper **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ropll_override_regref	R/W	if asserted allows regulator reference to be forced through ATB **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ropll_atb_s_enable	R/W	if asserted enables ATB observation on the ropll **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.740 LANEN_ANA_ROPLL_3 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_3
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e3
- **Exists:** Always

Table 2-744 Fields for Register: LANEN_ANA_ROPLL_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ropll_testmode	R/W	ropll testmode bits 7-0 **Value After Reset:** 0x9 **Exists:** Always **Reset Mask:** 0xff

2.1.741 LANEN_ANA_ROPLL_4 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_4
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e4
- **Exists:** Always

Table 2-745 Fields for Register: LANEN_ANA_ROPLL_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ropll_testmode	R/W	ropll testmode bits 15-8 **Value After Reset:** 0x65 **Exists:** Always **Reset Mask:** 0xff

2.1.742 LANEN_ANA_ROPLL_5 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_5
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e5
- **Exists:** Always

Table 2-746 Fields for Register: LANEN_ANA_ROPLL_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ropll_testmode	R/W	ropll testmode bits 23-16 **Value After Reset:** 0x81 **Exists:** Always **Reset Mask:** 0xff

2.1.743 LANEN_ANA_ROPLL_6 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_6
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e6
- **Exists:** Always

Table 2-747 Fields for Register: LANEN_ANA_ROPLL_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ropll_testmode	R/W	ropll testmode bits 31-24 **Value After Reset:** 0x18 **Exists:** Always **Reset Mask:** 0xff

2.1.744 LANEN_ANA_ROPLL_7 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_7
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e7
- **Exists:** Always

Table 2-748 Fields for Register: LANEN_ANA_ROPLL_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ropll_testanamode	R/W	ropll analog testmode bits 7-0 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.745 LANEN_ANA_ROPLL_8 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_8
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e8
- **Exists:** Always

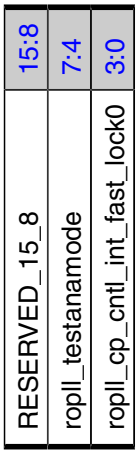


Table 2-749 Fields for Register: LANEN_ANA_ROPLL_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	ropll_testanamode	R/W	ropll analog testmode bits 11-8 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	ropll_cp_cntl_int_fast_lock0	R/W	fast lock integral CP setting for high bandwidth application bits 3-0 **Value After Reset:** 0xc **Exists:** Always **Reset Mask:** 0xf

2.1.746 LANEN_ANA_ROPLL_9 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_9
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)e9
- **Exists:** Always

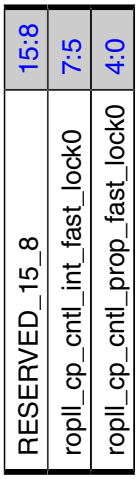


Table 2-750 Fields for Register: LANEN_ANA_ROPLL_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:5	ropll_cp_cntl_int_fast_lock0	R/W	fast lock integral CP setting for high bandwidth application bits 6-4 **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
4:0	ropll_cp_cntl_prop_fast_lock0	R/W	fast lock proportional CP setting for high bandwidth application bits 4-0 **Value After Reset:** 0x10 **Exists:** Always **Reset Mask:** 0x1f

2.1.747 LANEN_ANA_ROPLL_10 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_10
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ea
- **Exists:** Always

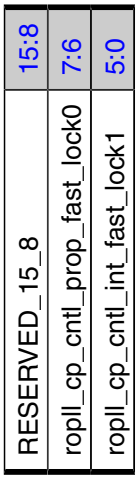


Table 2-751 Fields for Register: LANEN_ANA_ROPLL_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:6	ropll_cp_cntl_prop_fast_lock0	R/W	fast lock proportional CP setting for high bandwidth application bits 6-5 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5:0	ropll_cp_cntl_int_fast_lock1	R/W	fast lock integral CP setting for low bandwidth application bits 5-0 **Value After Reset:** 0x3f **Exists:** Always **Reset Mask:** 0x3f

2.1.748 LANEN_ANA_ROPLL_11 (for N = 0; N <= NLANES-1)

- **Description:** ROPLL_11
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)eb
- **Exists:** Always

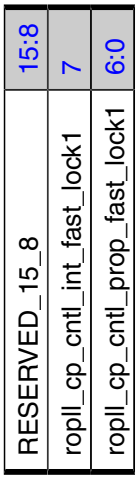


Table 2-752 Fields for Register: LANEN_ANA_ROPLL_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ropll_cp_cntl_int_fast_lock1	R/W	fast lock integral CP setting for low bandwidth application bit 6 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6:0	ropll_cp_cntl_prop_fast_lock1	R/W	fast lock proportional CP setting for low bandwidth application bits 6-0 **Value After Reset:** 0x7f **Exists:** Always **Reset Mask:** 0x7f

2.1.749 LANEN_ANA_RX_AFE_OVRD_4 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_4
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ec
- **Exists:** Always

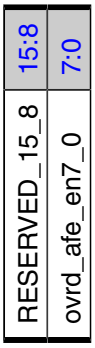


Table 2-753 Fields for Register: LANEN_ANA_RX_AFE_OVRD_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ovrd_afe_en7_0	R/W	AFE Inductor bypass configuration **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.750 LANEN_ANA_RX_AFE_OVRD_5 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_5
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ed
- **Exists:** Always

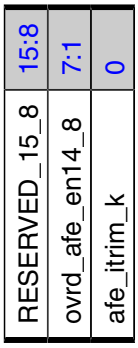


Table 2-754 Fields for Register: LANEN_ANA_RX_AFE_OVRD_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:1	ovrd_afe_en14_8	R/W	AFE Inductor bypass configuration **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
0	afe_itrim_k	R/W	Negc Current control **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.751 LANEN_ANA_RX_AFE_OVRD_6 (for N = 0; N <= NLANES-1)

- **Description:** RX_AFE_OVRD_6
- **Size:** 16 bits
- **Offset:** 0x1(N*2+1)ee
- **Exists:** Always

Table 2-755 Fields for Register: LANEN_ANA_RX_AFE_OVRD_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	freebit19_12	R/W	Not used **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.752 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)00
- **Exists:** Always

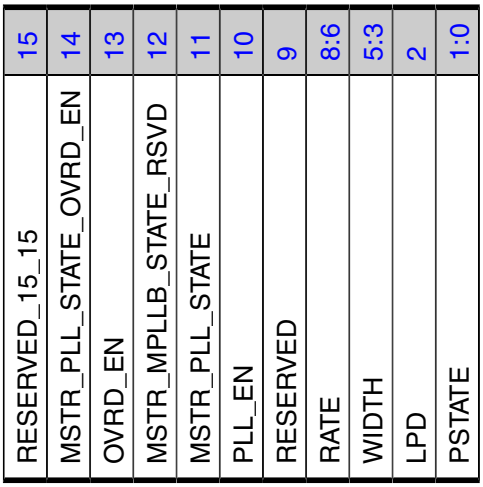


Table 2-756 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	MSTR_PLL_STATE_OVRD_EN	R/W	This bit is the override enable for tx_master_pll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	OVRD_EN	R/W	This bit is the override enable for the following input signals. This bit overrides the following input signals: ■ tx_pll_en ■ tx_ropll_refsel ■ lane_ref_sel ■ tx_rate ■ tx_width ■ tx_lpd ■ tx_pstate **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-756 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12	MSTR_MPLLB_STATE_RSVD	R/W	This bit is reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	MSTR_PLL_STATE	R/W	This bit is the override value for tx_master_pll_state. Its override enable is TX_OVRD_IN.MSTR_PLL_STATE_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	PLL_EN	R/W	This bit is the override value for tx_pll_en. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RESERVED	R/W	This bit is reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:6	RATE	R/W	This bit is the override value for tx_rate. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7
5:3	WIDTH	R/W	This bit is the override value for tx_width. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
2	LPD	R/W	This bit is the override value for tx_lpd. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	PSTATE	R/W	This bit is the override value for tx_pstate. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x3

Table 2-756 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x3

2.1.753 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)01
- **Exists:** Always

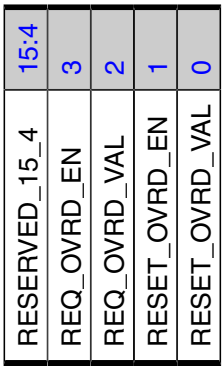


Table 2-757 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	REQ_OVRD_EN	R/W	This bit is the override enable for tx_req. This bit overrides the tx_req input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	REQ_OVRD_VAL	R/W	This bit is the override value for tx_req. Its override enable is TX_OVRD_IN_1.REQ_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	RESET_OVRD_EN	R/W	This bit is the override enable for tx_reset. This bit overrides the tx_reset input signal. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RESET_OVRD_VAL	R/W	This bit is the override value for tx_reset. Its override enable is TX_OVRD_IN_1.RESET_OVRD_EN. **Value After Reset:** 0x1

Table 2-757 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.754 RAWLANEN_DIG_PCS_XF_TX_PCS_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX ontrols from PCS (Pre-Override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)02
- **Exists:** Always

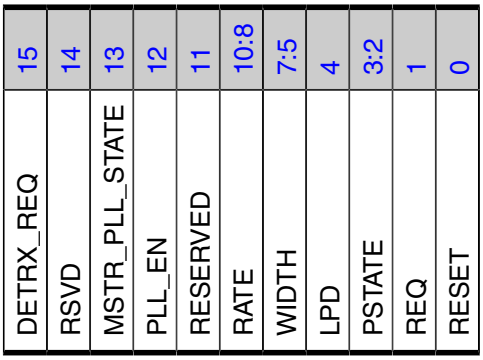


Table 2-758 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DETRX_REQ	R	This bit is the value from PCS for tx_detrx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	RSVD	R	This bit is reserved for future use. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	MSTR_PLL_STATE	R	This bit is the value from PCS for tx_master_pll_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	PLL_EN	R	This bit is the value from PCS for tx_pll_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-758 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11	RESERVED	R	This bit is reserved for future use. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:8	RATE	R	These bits are the value from PCS for tx_rate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:5	WIDTH	R	These bits are the value from PCS for tx_width. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	LPD	R	This bit is the value from PCS for tx_lpd. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:2	PSTATE	R	These bits are the value from PCS for tx_pstate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	REQ	R	This bit is the value from PCS for tx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-758 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
0	RESET	R	This bit is the value from PCS for tx_reset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.755 RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing TX Controls to PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)03
- **Exists:** Always

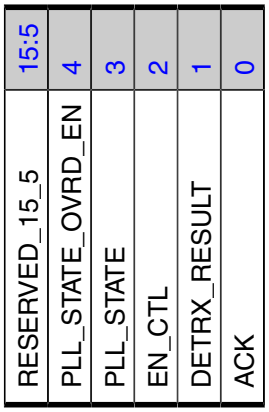


Table 2-759 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	PLL_STATE_OVRD_EN	R/W	This bit is the override enable for tx_pll_state. This bit overrides the tx_pll_state output. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	PLL_STATE	R/W	This bit is the override the value for tx_pll_state. Its override enable is TX_OVRD_OUT.PLL_STATE_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	EN_CTL	R/W	This bit enables override values for all control outputs of this register. This bit overrides the following output signals: ■ tx_detrx_result ■ tx_ack **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-759 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	DETRX_RESULT	R/W	This bit is the override value for tx_detrx_result. Its override enable is TX_OVRD_OUT.EN_CTL. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ACK	R/W	This bit is the override value for tx_ack. Its override enable is TX_OVRD_OUT.EN_CTL. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.756 RAWLANEN_DIG_PCS_XF_TX_PCS_OUT (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Outgoing TX Status Controls from Raw PCS (Pre-Override Output Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)04
- **Exists:** Always

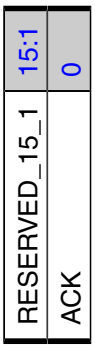


Table 2-760 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ACK	R	This bit is the value from Raw PCS for tx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.757 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from PCS, Register 0.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)05
- **Exists:** Always

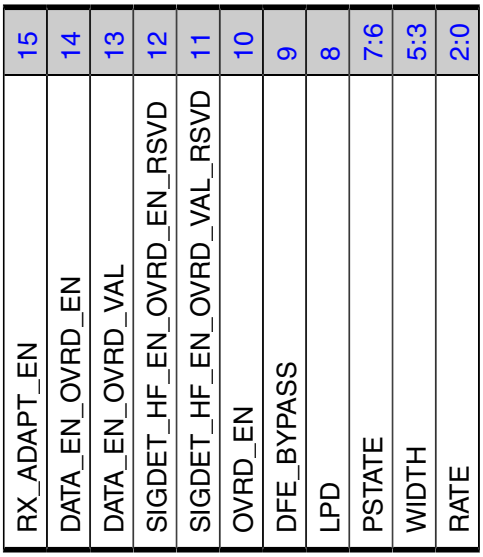


Table 2-761 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_ADAPT_EN	R/W	This bit is the override enable for rx_adapt_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	DATA_EN_OVRD_EN	R/W	This bit is the override enable for rx_data_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	DATA_EN_OVRD_VAL	R/W	This bit is the override value for rx_data_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	SIGDET_HF_EN_OVRD_EN_RSVD	R/W	Reserved bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-761 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	SIGDET_HF_EN_OVRD_VAL_RSVD	R/W	Reserved bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	OVRD_EN	R/W	This bit enables the override values for RX inputs. This bit overrides the following input signals: ■ rx_adapt_en ■ rx_dfe_bypass ■ rx_lpd ■ rx_pstate ■ rx_width ■ rx_rate **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	DFE_BYPASS	R/W	This bit is the override value for rx_dfe_bypass. Its override enable is RX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	LPD	R/W	This bit is the override value for rx_lpd. Its override enable is RX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:6	PSTATE	R/W	This bit is the override value for rx_pstate. Its override enable is RX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5:3	WIDTH	R/W	This bit is the override value for rx_width. Its override enable is RX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
2:0	RATE	R/W	This bit is the override value for rx_rate. Its override enable is RX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0

Table 2-761 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x7

2.1.758 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from PCS, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)06
- **Exists:** Always

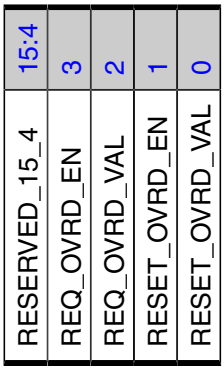


Table 2-762 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	REQ_OVRD_EN	R/W	This bit is the override enable for rx_req. This bit overrides the rx_req input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	REQ_OVRD_VAL	R/W	This bit is the override value for rx_req. Its override enable is RX_OVRD_IN_1.REQ_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	RESET_OVRD_EN	R/W	This bit is the override enable for rx_reset. This bit overrides the rx_reset input. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RESET_OVRD_VAL	R/W	This bit is the override value for rx_reset. Its override enable is RX_OVRD_IN_1.RESET_OVRD_EN. **Value After Reset:** 0x1

Table 2-762 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.759 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from PCS, Register 2.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)07
- **Exists:** Always

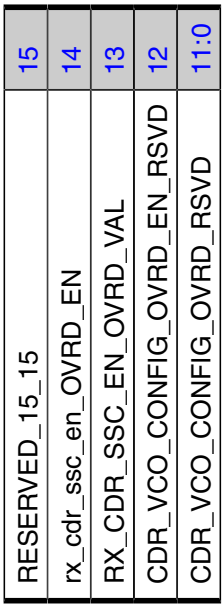


Table 2-763 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	rx_cdr_ssc_en_OVRD_EN	R/W	This bit is the override enable for rx_cdr_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_CDR_SSC_EN_OVRD_VAL	R/W	This bit is the override value for rx_cdr_ssc_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	CDR_VCO_CONFIG_OVRD_EN_RSVD	R/W	This bit is the override enable for rx_cdr_vco_config. This bit overrides the rx_cdr_vco_config input. **Value After Reset:** 0x0

Table 2-763 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11:0	CDR_VCO_CONFIG_OVRD_RSVD	R/W	These bits are the override value for rx_cdr_vco_config. Its override enable is RX_OVRD_IN_2.CDR_VCO_CONFIG_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.760 RAWLANEN_DIG_PCS_XF_CNTX_RES_IN_PROG (for N = 0; N <= NLANES-1)

- **Description:** TX/RX cntx res in prog .
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)08
- **Exists:** Always

Table 2-764 Fields for Register: RAWLANEN_DIG_PCS_XF_CNTX_RES_IN_PROG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_CNTX_RES_IN_PROG	R/W	RX cntx restore in prog **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CNTX_RES_IN_PROG	R/W	TX cntx restore in prog **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.761 RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_1 (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANE0_DIG_PCS_XF_RX_OVRD_FOR_FW_1
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)09
- **Exists:** Always

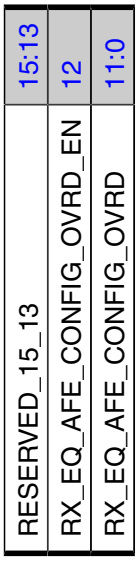


Table 2-765 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_EQ_AFE_CONFIG_OVRD_EN	R/W	Enable Override the rx_eq_afe_config setting Input signals overridden by it: rx_eq_afe_config **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:0	RX_EQ_AFE_CONFIG_OVRD	R/W	Override the rx_eq_afe_config setting. Its override enable is RX_OVRD_IN_11.RX_EQ_AFE_CONFIG_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.762 RAWLANEN_DIG_PCS_XF_ADAPT_CONT_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming Adaptation Controls from PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0a
- **Exists:** Always

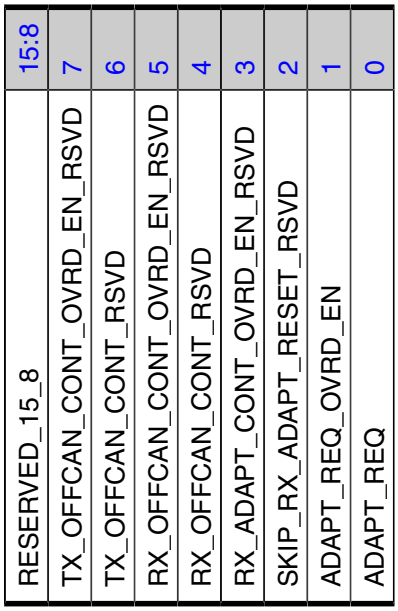


Table 2-766 Fields for Register: RAWLANEN_DIG_PCS_XF_ADAPT_CONT_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	TX_OFFCAN_CONT_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_OFFCAN_CONT_RSVD	R/W	This bit is the override value for tx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_OFFCAN_CONT_OVRD_EN_RSVD	R/W	This bit is the override enable for rx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-766 Fields for Register: RAWLANEN_DIG_PCS_XF_ADAPT_CONT_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	RX_OFFCAN_CONT_RSVD	R/W	This bit is the override value for rx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_ADAPT_CONT_OVRD_EN_RSVD	R/W	This bit is the override enable for rx_adapt_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SKIP_RX_ADAPT_RESET_RSVD	R/W	This bit is to skip rx adapt reset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	ADAPT_REQ_OVRD_EN	R/W	This bit is the override enable values for rx_adapt_req. This bit overrides the rx_adapt_req input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	ADAPT_REQ	R/W	This bit is the override value for rx_adapt_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.763 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Mode Override Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0b
- **Exists:** Always

15	14	13	12	11:7	6	5:3	2	1:0
RESERVED_15_15	RX_ADAPT_IN_PROG_OVRD_EN	RX_ADAPT_IN_PROG_OVRD	RX_CDR_PPM_MAX_OVRD_EN_RSVD	RX_CDR_PPM_MAX_OVRD_RSVD	RX_ADAPT_MODE_OVRD_EN_RSVD	RX_ADAPT_MODE_RSVD	RX_ADAPT_SEL_OVRD_EN	RX_ADAPT_SEL_OVRD

Table 2-767 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_ADAPT_IN_PROG_OVRD_EN	R/W	This bit is the override enable for RX adaptation in progress. This bit overrides the rx_adapt_in_prog input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_ADAPT_IN_PROG_OVRD	R/W	This bit is the overridden value for the RX adaptation in progress. Its override enable is RX_OVRD_IN_6.RX_ADAPT_IN_PROG_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-767 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_6 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
12	RX_CDR_PPM_MAX_OVRD_EN_RSVD	R/W	This bit is the enable override for the RX CDR PPM Max. This bit overrides the rx_cdr_ppm_max input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:7	RX_CDR_PPM_MAX_OVRD_RSVD	R/W	This bit is the overridden value for RX CDR PPM Max. Its override enable is RX_OVRD_IN_5.RX_CDR_PPM_MAX_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
6	RX_ADAPT_MODE_OVRD_EN_RSVD	R/W	This bit is the override enable for the RX adaptation mode. This bit overrides the rx_adapt_mode input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:3	RX_ADAPT_MODE_RSVD	R/W	This bit is the overridden value for RX adaptation mode. Its override enable is RX_OVRD_IN_5.RX_ADAPT_MODE_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
2	RX_ADAPT_SEL_OVRD_EN	R/W	This bit is the override enable the RX adaptation setting. This bit overrides the rx_adapt_sel input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	RX_ADAPT_SEL_OVRD	R/W	This bit is the overridden value for the RX adaptation setting. Its override enable is RX_OVRD_IN_5.RX_ADAPT_SEL_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.764 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX IQ Setting.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0c
- **Exists:** Always

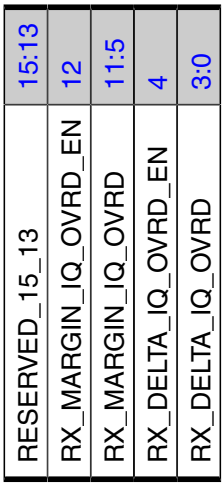


Table 2-768 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	RX_MARGIN_IQ_OVRD_EN	R/W	This bit is the override enable the RX margin IQ setting. This bit overrides the rx_margin_iq input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:5	RX_MARGIN_IQ_OVRD	R/W	These bits override the RX margin IQ setting. Their override enable is RX_OVRD_IN_7.RX_MARGIN_IQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
4	RX_DELTA_IQ_OVRD_EN	R/W	This bit is the override enable the RX Delta IQ setting. This bit overrides the rx_delta_iq input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-768 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_7 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3:0	RX_DELTA_IQ_OVRD	R/W	These bits override the RX Delta IQ seeting. Their override enable is RX_OVRD_IN_6.RX_DELTA_IQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.765 RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from PCS, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0d
- **Exists:** Always

15:14	13	12	11:9	8	7	6	5	4:0
RESERVED_15_14	rx_eq_dfe_tap1_ovrd_en	RX_EQ_OVRD_EN	RX_EQ_ATT_LVL_OVRD_VAL	RX_EQ_ATT_LVL_OVRD_EN	rx_eq_ctle_boost_ovrd_en	rx_eq_ctle_pole_ovrd_en	RX_EQ_VGA_GAIN_OVRD_EN	RX_EQ_VGA_GAIN_OVRD_VAL

Table 2-769 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	rx_eq_dfe_tap1_ovrd_en	R/W	This bit are the override value for rx_eq_dfe_tap1_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_EQ_OVRD_EN	R/W	This bit enables override values for all RX EQ settings. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:9	RX_EQ_ATT_LVL_OVRD_VAL	R/W	These bits are the override value for rx_eq_att_lvl[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

Table 2-769 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8	RX_EQ_ATT_LVL_OVRD_EN	R/W	This bit are the override value for rx_eq_att_lvl_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	rx_eq_ctle_boost_ovrd_en	R/W	This bit are the override value for rx_eq_ctle_boost_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	rx_eq_ctle_pole_ovrd_en	R/W	This bit are the override value for rx_eq_ctle_pole_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_EQ_VGA_GAIN_OVRD_EN	R/W	These bits are reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4:0	RX_EQ_VGA_GAIN_OVRD_VAL	R/W	These bits are the override value for rx_eq_vga_gain[2:0]. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.766 RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override values for incoming RX EQ controls from PCS, register #2
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0e
- **Exists:** Always

Table 2-770 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_EQ_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RX_EQ_CTLE_POLE_OVRD_VAL	R/W	Override value for rx_eq_ctle_pole[2]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
13:9	RX_EQ_CTLE_BOOST_OVRD_VAL	R/W	Override value for rx_eq_ctle_boost[4:0]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
8:0	RX_EQ_DFE_TAP1_OVRD_VAL	R/W	Override value for rx_eq_dfe_tap1[7:0] Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.767 RAWLANEN_DIG_PCS_XF_RX_PCS_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS (Monitor on Post ATE Override).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)0f
- **Exists:** Always

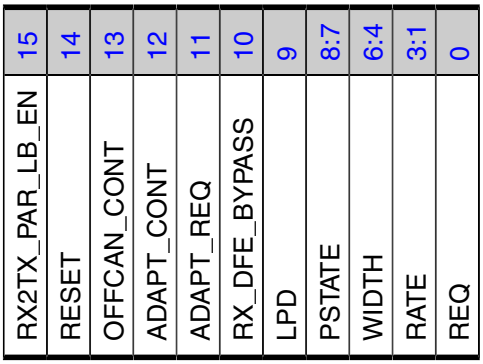


Table 2-771 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX2TX_PAR_LB_EN	R	This bit is the value from PCS for lane_rx2tx_par_lb_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	RESET	R	This bit is the value from PCS for rx_reset. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	OFFCAN_CONT	R	This bit is the value from PCS for rx_offcan_cont. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	ADAPT_CONT	R	This bit is the value from PCS for rx_adapt_cont. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-771 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
11	ADAPT_REQ	R	This bit is the value from PCS for rx_adapt_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	RX_DFE_BYPASS	R	This bit is the value from PCS for rx_dfe_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	LPD	R	This bit is the value from PCS for rx_lpd. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:7	PSTATE	R	This bit is the value from PCS for rx_pstate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:4	WIDTH	R	This bit is the value from PCS for rx_width. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:1	RATE	R	This bit is the value from PCS for rx_rate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-771 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
0	REQ	R	This bit is the value from PCS for rx_req. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.768 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 1 (Monitor on Pre-Override Signals).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)10
- **Exists:** Always

Table 2-772 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_ADAPT_EN	R	This bit is the value from PCS for rx_adapt_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14:7	RX_DIV_CLK_SEL	R	read value for rx_div_clk_sel_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	REF_LD_VAL	R	These bits are the value from PCS for rx_ref_ld_val. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.769 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 2 (Monitor on Pre-Override Signals).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)11
- **Exists:** Always

Table 2-773 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	VCO_LD_VAL	R	These bits are the value from PCS for rx_vco_ld_val. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.770 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 3 (Monitor on Pre-Override Signals).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)12
- **Exists:** Always

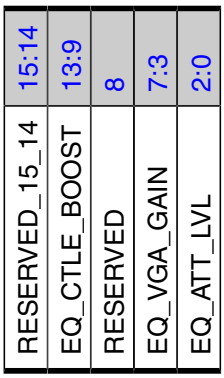


Table 2-774 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:9	EQ_CTLE_BOOST	R	These bits are the value from ASIC for rx_eq_ctle_boost. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	RESERVED	R	These bits are reserved for future use. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:3	EQ_VGA_GAIN	R	These bits are the value from ASIC for rx_eq_vga_gain. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-774 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
2:0	EQ_ATT_LVL	R	These bits are the value from ASIC for rx_eq_att_lvl. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.771 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 4 (Monitor on Pre-Override Signals).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)13
- **Exists:** Always

Table 2-775 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10:2	EQ_DFE_TAP1	R	These bits are the value from ASIC for rx_eq_dfe_tap1. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1:0	EQ_CTLE_POLE	R	These bits are the value from ASIC for rx_eq_ctle_pole. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.772 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 5.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)14
- **Exists:** Always

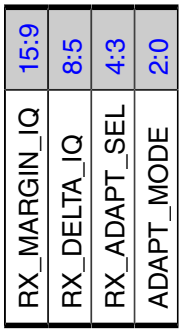


Table 2-776 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RX_MARGIN_IQ	R	These bits are the value from ASIC for rx_margin_iq. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8:5	RX_DELTA_IQ	R	These bits are the value from ASIC for rx_delta_iq. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4:3	RX_ADAPT_SEL	R	These bits are the value from ASIC for rx_adapt_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2:0	ADAPT_MODE	R	These bits are the value from ASIC for rx_adapt_mode. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-776 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_5 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.773 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 6.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)15
- **Exists:** Always

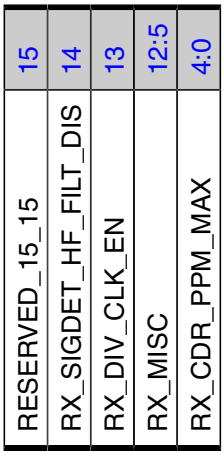


Table 2-777 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_SIGDET_HF_FILT_DIS	R	read vale for rx_sigdet_hf_filt_dis_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	RX_DIV_CLK_EN	R	read value for rx_div_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12:5	RX_MISC	R	These bits are the value from ASIC for rx_misc. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-777 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_6 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
4:0	RX_CDR_PPM_MAX	R	These bits are the value from ASIC for rx_cdr_ppm_max. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.774 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 7.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)16
- **Exists:** Always

Table 2-778 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RX_DIG_DIV_CLK_SEL	R	read value for rx_dig_div_clk_sel_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:0	CDR_VCO_CONFIG	R	These bits are the value from ASIC for rx_cdr_vco_config. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.775 RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing RX Controls to PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)17
- **Exists:** Always

Table 2-779 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	EN_CTL	R/W	This bit enables override values for all control outputs of this register. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	ACK	R/W	This bit is the override value for rx_ack. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.776 RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing RX Controls to PCS, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)18
- **Exists:** Always

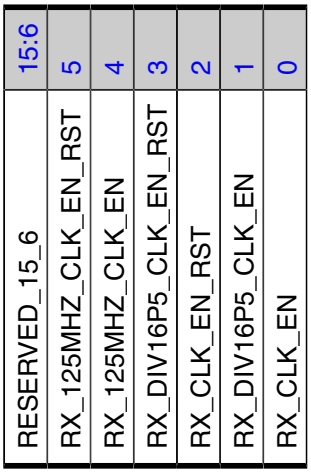


Table 2-780 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	RX_125MHZ_CLK_EN_RST	R/W	This bit resets the rx_125mhz_clk_en synchronizer. This bit is used to reset the synchronizer for the rx_125mhz_clk_en_r register because if the rx_125mhz_clk is already off (such as, P1), it takes up to four clocks to turn off the clock gate when the rx_125mhz_clk is subsequently enabled. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_125MHZ_CLK_EN	R/W	This bit enables the outgoing rx_125mhz_clk. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-780 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_OUT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	RX_DIV16P5_CLK_EN_RST	R/W	This bit resets the rx_div16p5_clk_en synchronizer. This bit is used to reset the synchronizer for the rx_div16p5_clk_en_r register because if the rx_div16p5_clk is already off (such as, P1) it takes up to four clocks to turn off the clock gate when the rx_div16p5_clk is subsequently enabled. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_CLK_EN_RST	R/W	This bit resets the rx_clk_en synchronizer. This bit is used to reset the synchronizer for the rx_clk_en_r register because if the rx_clk is already off (such as, P1) it takes up to four clocks to turn off the clock gate when the rx_clk is subsequently enabled. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_DIV16P5_CLK_EN	R/W	This bit enables the outgoing rx_div16p5_clk. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_CLK_EN	R/W	This bit enables the outgoing rx_clk. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.777 RAWLANEN_DIG_PCS_XF_RX_PCS_OUT (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Outgoing RX Status Controls from Raw PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)19
- **Exists:** Always

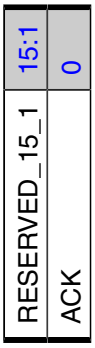


Table 2-781 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ACK	R	This bit is the value from Raw PCS for rx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.778 RAWLANEN_DIG_PCS_XF_RX_ADAPT_ACK (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Acknowledge.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1a
- **Exists:** Always

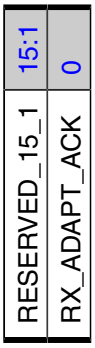


Table 2-782 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_ADAPT_ACK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_ACK	R/W	This bit is the RX adaptation acknowledge. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.779 RAWLANEN_DIG_PCS_XF_RX_ADAPT_FOM (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Figure of Merit.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1b
- **Exists:** Always

Table 2-783 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_ADAPT_FOM (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	RX_ADAPT_FOM	R/W	These bits are the RX adaptation figure of merit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.780 RAWLANEN_DIG_PCS_XF_RX_TXPRE_DIR (for N = 0; N <= NLANES-1)

- **Description:** RX Calculated Direction for TX-pre.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1c
- **Exists:** Always

Table 2-784 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXPRE_DIR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	RX_TXPRE_DIR	R/W	These bits are the RX calculated direction for TX-pre. ■ 00: No change ■ 01: Increment ■ 10: Decrement ■ 11: Reserved **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.781 RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DIR (for N = 0; N <= NLANES-1)

- **Description:** RX Calculated Direction for TX-Main.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1d
- **Exists:** Always

Table 2-785 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXMAIN_DIR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	RX_TXMAIN_DIR	R/W	These bits are the RX calculated direction for TX-Main. ■ 00: No change ■ 01: Increment ■ 10: Decrement ■ 11: Reserved **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.782 RAWLANEN_DIG_PCS_XF_RX_TXPOST_DIR (for N = 0; N <= NLANES-1)

- **Description:** RX Calculated Direction for TX-Post.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1e
- **Exists:** Always

Table 2-786 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_TXPOST_DIR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	RX_TXPOST_DIR	R/W	These bits are the RX calculated direction for TX-Post. ■ 00: No change ■ 01: Increment ■ 10: Decrement ■ 11: Reserved **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.783 RAWLANEN_DIG_PCS_XF_LANE_NUMBER (for N = 0; N <= NLANES-1)

- **Description:** Current Lane Number.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)1f
- **Exists:** Always

Table 2-787 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_NUMBER (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	LANE_NUMBER	R	These bits are the current lane number. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.784 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** ATE Override Input to Control Top-Level Inputs.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)20
- **Exists:** Always

15	14	13	12:10	9	8	7	6	5	4	3	2	1	0
RX_DATA_EN_ATE_OVRD_EN	RX_DATA_EN_ATE_OVRD_VAL	RX_RATE_ATE_OVRD_EN_RSVD	RX_RATE_ATE_OVRD_VAL_RSVD	RX_DFE_BYPASS_OVRD_EN_RSVD	RX_DFE_BYPASS_OVRD_VAL_RSVD	TX_REQ_ATE_OVRD_EN	TX_REQ_ATE_OVRD_VAL	RX_REQ_ATE_OVRD_EN	RX_REQ_ATE_OVRD_VAL	TX_RESET_ATE_OVRD_EN	TX_RESET_ATE_OVRD_VAL	RX_RESET_ATE_OVRD_EN	RX_RESET_ATE_OVRD_VAL

Table 2-788 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_DATA_EN_ATE_OVRD_EN	R/W	This bit enables the ATE override value for rx_data_en input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	RX_DATA_EN_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for rx_data_en input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_RATE_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for rx_rate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:10	RX_RATE_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for rx_rate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

Table 2-788 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
9	RX_DFE_BYPASS_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for rx_dfe_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_DFE_BYPASS_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for rx_dfe_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_REQ_ATE_OVRD_EN	R/W	This bit enables the ATE override vale for tx_req input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	TX_REQ_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level tx_req input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_REQ_ATE_OVRD_EN	R/W	This bit enables the ATE override value for rx_req input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_REQ_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level rx_req input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_RESET_ATE_OVRD_EN	R/W	This bit enables the ATE override value for tx_reset input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_RESET_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level tx_reset input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_RESET_ATE_OVRD_EN	R/W	This bit enables the ATE override value for rx_reset input. **Value After Reset:** 0x0

Table 2-788 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	RX_RESET_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level rx_reset input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.785 RAWLANEN_DIG_PCS_XF_RX_DELTA_IQ (for N = 0; N <= NLANES-1)

- **Description:** RX Delta IQ after Override. This CREG is in RX_PCS_IN category
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)21
- **Exists:** Always

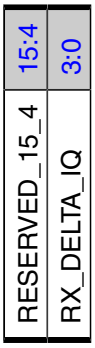


Table 2-789 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_DELTA_IQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3:0	RX_DELTA_IQ	R	RX Delta IQ after Override (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.786 RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RX_RO_OUT (for N = 0; N <= NLANES-1)

- **Description:** read only registers for TX/RX related output signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)22
- **Exists:** Always

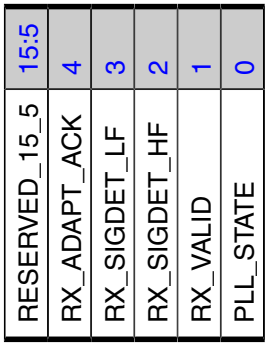


Table 2-790 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RX_RO_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	RX_ADAPT_ACK	R	read value for rx_adapt_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	RX_SIGDET_LF	R	read value for rx_sigdet_lf (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2	RX_SIGDET_HF	R	read value for rx_sigdet_hf. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-790 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RX_RO_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
1	RX_VALID	R	read value for rx_valid (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	PLL_STATE	R	read value for tx_pll_state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.787 RAWLANEN_DIG_PCS_XF_TXRX_TERM_CTRL_IN (for N = 0; N <= NLANES-1)

- **Description:** tx/rx_term_ctrl status. This CREG is in RX_PCS_IN category
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)23
- **Exists:** Always

Table 2-791 Fields for Register: RAWLANEN_DIG_PCS_XF_TXRX_TERM_CTRL_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	TX_TERM_CTRL	R	tx_term_ctrl value (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	RX_TERM_CTRL	R	tx_term_ctrl value (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.788 RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing TX Controls to PCS, Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)24
- **Exists:** Always

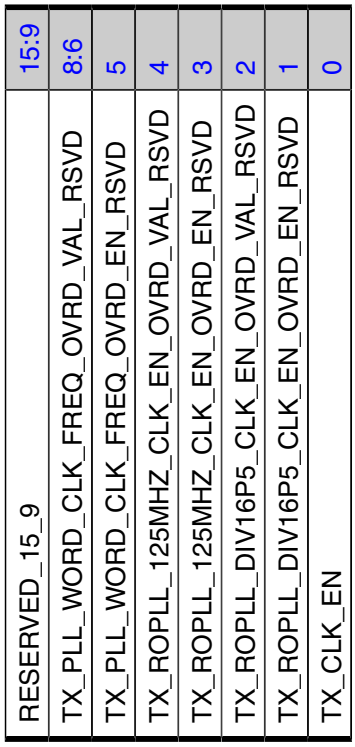


Table 2-792 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:6	TX_PLL_WORD_CLK_FREQ_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_pll_word_clk_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
5	TX_PLL_WORD_CLK_FREQ_OVRD_EN_RSVD	R/W	This bit enables the override value for tx_pll_word_clk_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-792 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_OUT_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	TX_ROPLL_125MHZ_CLK_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_ropll_125mhz_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_ROPLL_125MHZ_CLK_EN_OVRD_EN_RSVD	R/W	This bit enables the override value for tx_ropll_125mhz_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_ROPLL_DIV16P5_CLK_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_ropll_div16p5_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_ROPLL_DIV16P5_CLK_EN_OVRD_EN_RSVD	R/W	This bit enables the override value for tx_ropll_div16p5_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CLK_EN	R/W	This bit enables the incoming tx_clk. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.789 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** ATE Override input to control top-level inputs, Register 2.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)25
- **Exists:** Always

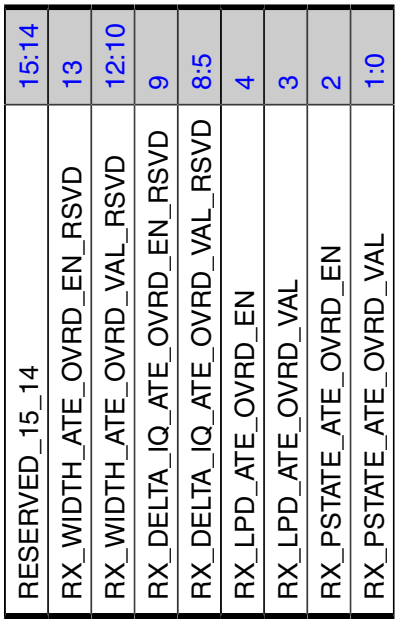


Table 2-793 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	RX_WIDTH_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for rx_width input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:10	RX_WIDTH_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for rx_width input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
9	RX_DELTA_IQ_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for rx_delta_iq input. **Value After Reset:** 0x0

Table 2-793 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8:5	RX_DELTA_IQ_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for top-level rx_delta_iq input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
4	RX_LPD_ATE_OVRD_EN	R/W	This bit enables the ATE override value for rx_lpd input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_LPD_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level rx_lpd input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_PSTATE_ATE_OVRD_EN	R/W	This bit enables the ATE override value for rx_pstate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	RX_PSTATE_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level rx_pstate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.790 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** ATE Override Input to Control Top-Level Inputs, Register 3.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)26
- **Exists:** Always

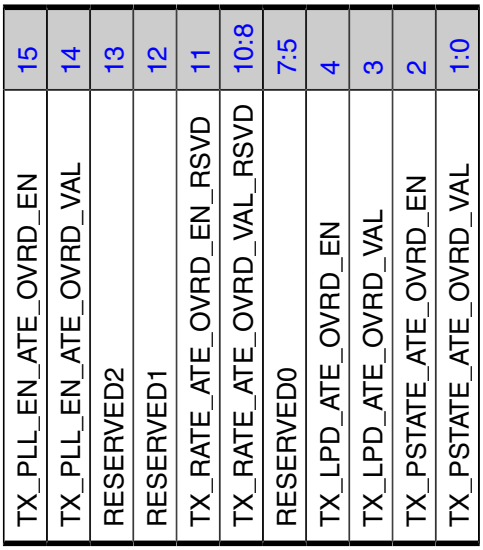


Table 2-794 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_PLL_EN_ATE_OVRD_EN	R/W	This bit enables the ATE override value for tx_pll_en input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_PLL_EN_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for tx_pll_en input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RESERVED2	R/W	This bit is reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RESERVED1	R/W	This bit is reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-794 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	TX_RATE_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for tx_rate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:8	TX_RATE_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for tx_rate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
7:5	RESERVED0	R/W	This bit is reserved for future use. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
4	TX_LPD_ATE_OVRD_EN	R/W	This bit enables the ATE override value for tx_lpd input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_LPD_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level tx_lpd input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_PSTATE_ATE_OVRD_EN	R/W	This bit enables the ATE override value for tx_pstate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	TX_PSTATE_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for top-level tx_pstate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.791 RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** ATE Override Input to Control Top-Level Inputs, Register 3.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)27
- **Exists:** Always

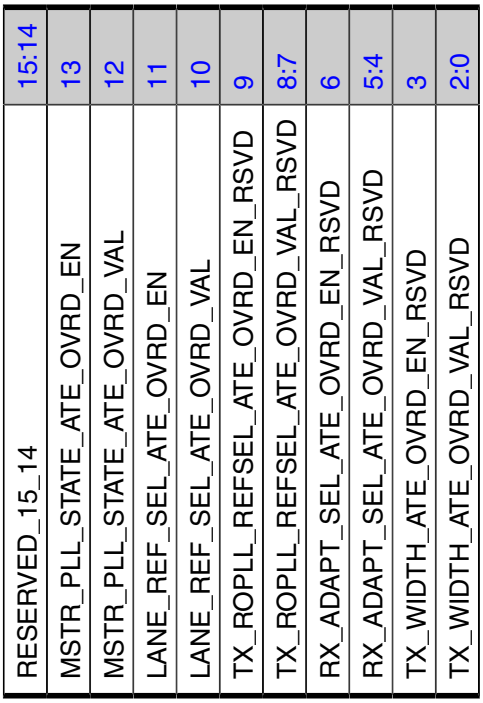


Table 2-795 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	MSTR_PLL_STATE_ATE_OVRD_EN	R/W	This bit enables the ATE override value for tx_master_pll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	MSTR_PLL_STATE_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for tx_master_pll_state. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-795 Fields for Register: RAWLANEN_DIG_PCS_XF_ATE_OVRD_IN_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	LANE_REF_SEL_ATE_OVRD_EN	R/W	This bit enables the ATE override value for lane_ref_sel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	LANE_REF_SEL_ATE_OVRD_VAL	R/W	This bit is the ATE Override value for lane_ref_sel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	TX_ROPLL_REFSEL_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for tx_ropll_refsel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8:7	TX_ROPLL_REFSEL_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for tx_ropll_refsel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6	RX_ADAPT_SEL_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for rx_adapt_sel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5:4	RX_ADAPT_SEL_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for top-level rx_adapt_sel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
3	TX_WIDTH_ATE_OVRD_EN_RSVD	R/W	This bit enables the ATE override value for tx_width input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	TX_WIDTH_ATE_OVRD_VAL_RSVD	R/W	This bit is the ATE Override value for tx_width input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.792 RAWLANEN_DIG_PCS_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming Lane Controls from PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)28
- **Exists:** Always

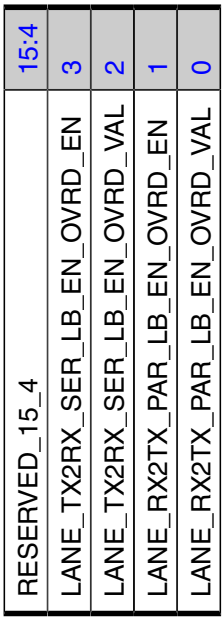


Table 2-796 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	LANE_TX2RX_SER_LB_EN_OVRD_EN	R/W	This bit is the override enable for lane_tx2rx_ser_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	LANE_TX2RX_SER_LB_EN_OVRD_VAL	R/W	This bit is the override value for lane_tx2rx_ser_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	LANE_RX2TX_PAR_LB_EN_OVRD_EN	R/W	This bit is the verride enable for lane_rx2tx_par_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-796 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	LANE_RX2TX_PAR_LB_EN_OVRD_VAL	R/W	This bit is the verride value for lane_rx2tx_par_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.793 RAWLANEN_DIG_PCS_XF_SIGDET_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing SIGDET Signals from PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)29
- **Exists:** Always

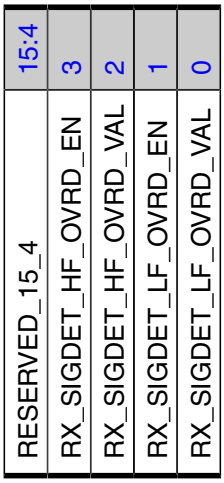


Table 2-797 Fields for Register: RAWLANEN_DIG_PCS_XF_SIGDET_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	RX_SIGDET_HF_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	RX_SIGDET_HF_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_hf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_SIGDET_LF_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lf. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_SIGDET_LF_OVRD_VAL	R/W	This bit is the override value for rx_sigdet_lf. **Value After Reset:** 0x0

Table 2-797 Fields for Register: RAWLANEN_DIG_PCS_XF_SIGDET_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.794 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2a
- **Exists:** Always

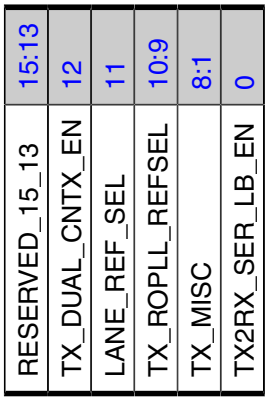


Table 2-798 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	TX_DUAL_CNTX_EN	R	This bit is the value from the output of the tx_packed for tx_dual_cntx_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	LANE_REF_SEL	R	This bit is the value from PCS for lane_ref_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:9	TX_ROPLL_REFSEL	R	This bit is the value from PCS for tx_ropll_refsel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-798 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8:1	TX_MISC	R	This bit is the value from PCS for tx_misc. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TX2RX_SER_LB_EN	R	This bit is the value from PCS for lane_tx2rx_ser_lb_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.795 RAWLANEN_DIG_PCS_XF_RX_VALID_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Value for RX VALID Signal from PCS.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2b
- **Exists:** Always

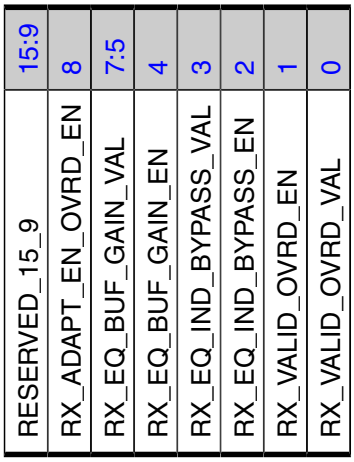


Table 2-799 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_VALID_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	RX_ADAPT_EN_OVRD_EN	R/W	This bit is the override enable for rx_adapt_en_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:5	RX_EQ_BUF_GAIN_VAL	R/W	This bit is the override enable for rx_eq_buf_gain_ovrd_val. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
4	RX_EQ_BUF_GAIN_EN	R/W	This bit is the override enable for rx_eq_buf_gain_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_EQ_IND_BYPASS_VAL	R/W	This bit is the override enable for rx_eq_ind_bypass_ovrd_val. **Value After Reset:** 0x0

Table 2-799 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_VALID_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	RX_EQ_IND_BYPASS_EN	R/W	This bit is the override enable for rx_eq_ind_bypass_ovrd_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_VALID_OVRD_EN	R/W	This bit is the override enable for rx_valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_VALID_OVRD_VAL	R/W	This bit is the override value for rx_valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.796 RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_2 (for N = 0; N <= NLANES-1)

- **Description:** read only registers for context restore related signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2c
- **Exists:** Always

Table 2-800 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:8	LANE_LINK_NUM_RO	R	read value for LANE_LINK_NUM (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_CNTX_SEL_RO	R	read value for RX_CNTX_SEL (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.797 RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RO_1 (for N = 0; N <= NLANES-1)

- **Description:** read only registers for TX related signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2d
- **Exists:** Always

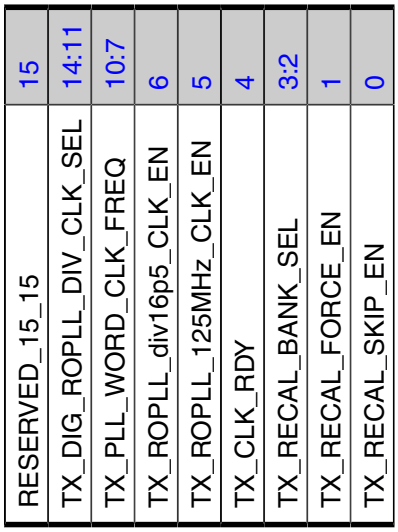


Table 2-801 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RO_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:11	TX_DIG_ROPLL_DIV_CLK_SEL	R	read value for tx_dig_ropll_div_clk_sel_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:7	TX_PLL_WORD_CLK_FREQ	R	read value for tx_pll_word_clk_freq_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-801 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_TX_RO_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6	TX_ROPLL_div16p5_CLK_EN	R	read value for tx_ropll_div16p5_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	TX_ROPLL_125MHz_CLK_EN	R	read value for tx_ropll_125mhz_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	TX_CLK_RDY	R	read value for tx_clk_rdy_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:2	TX_RECAL_BANK_SEL	R	read value for tx_recal_bank_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	TX_RECAL_FORCE_EN	R	read value for tx_recal_force_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	TX_RECAL_SKIP_EN	R	read value for tx_recal_skip_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.798 RAWLANEN_DIG_PCS_XF_LANE_PCS_RX_RO_1 (for N = 0; N <= NLANES-1)

- **Description:** read only registers for RX related signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2e
- **Exists:** Always

15	14	13:11	10	9	8	7	6	5	4	3:2	1	0
RX_SIGDET_HF_EN	RX_SIGDET_LFPS_FILTER_EN	RX_SIGDET_LF_THRESHOLD	RX_125MHZ_CLK_EN	RX_DIV16P5_CLK_EN	RX_TERM_ACDC	RX_ADAPT_IN_PROG	RX_COARSE_ADPT_FORCE_EN-	RX_COARSE_ADPT_SKIP_EN	RX_DATA_EN	RX_RECAL_BANK_SEL	RX_RECAL_FORCE_EN	RX_RECAL_SKIP_EN

Table 2-802 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_RX_RO_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_SIGDET_HF_EN	R	read value for rx_sigdet_hf_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
14	RX_SIGDET_LFPS_FILTER_EN	R	read value for rx_sigdet_lfps_filter_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:11	RX_SIGDET_LF_THRESHOLD	R	read value for rx_sigdet_lf_threshold_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-802 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_RX_RO_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
10	RX_125MHZ_CLK_EN	R	read value for rx_125mhz_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	RX_DIV16P5_CLK_EN	R	read value for rx_div16p5_clk_en_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	RX_TERM_ACDC	R	read value for rx_term_acdc_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	RX_ADAPT_IN_PROG	R	read value for rx_adapt_in_prog (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	RX_COARSE_ADPT_FORCE_EN-	R	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	RX_COARSE_ADPT_SKIP_EN	R	read value for rx_coarse_adpt_skip_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-802 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_PCS_RX_RO_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	RX_DATA_EN	R	read value for rx_data_en_ate (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:2	RX_RECAL_BANK_SEL	R	read value for rx_recal_bank_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1	RX_RECAL_FORCE_EN	R	read value for rx_recal_force_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	RX_RECAL_SKIP_EN	R	read value for rx_recal_skip_en (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.799 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 3.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)2f
- **Exists:** Always

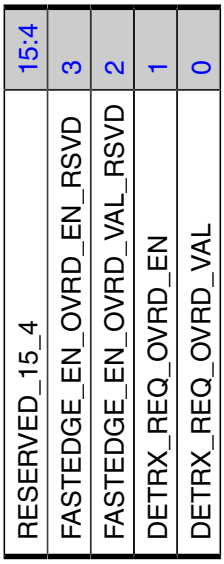


Table 2-803 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	FASTEDGE_EN_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_fastedge_en. This bit overrides the tx_fastedge_en input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	FASTEDGE_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_fastedge_en. Its override enable is TX_OVRD_IN_3.FASTEDGE_E_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	DETRX_REQ_OVRD_EN	R/W	This bit is the override enable for tx_detrx_req. This bit overrides the tx_detrx_req input signal. **Value After Reset:** 0x0

Table 2-803 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_3 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
0	DETRX_REQ_OVRD_VAL	R/W	This bit is the override value for tx_detrx_req. Its override enable is TX_OVRD_IN_3.DETRX_REQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.800 RAWLANEN_DIG_PCS_XF_RX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1)

- **Description:** Override the Incoming rx_recal_inputs.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)30
- **Exists:** Always

15:11	10	9	8	7	6	5	4	3	2	1:0
RESERVED_15_11	RX_COARSE_ADPT_SKIP_EN_OVRD_EN	RX_COARSE_ADPT_SKIP_EN_OVRD_VAL	RX_COARSE_ADPT_FORCE_EN_OVRD_EN	RX_COARSE_ADPT_FORCE_EN_OVRD_VAL	RX_RECAL_SKIP_EN_OVRD_EN	RX_RECAL_SKIP_EN_OVRD_VAL	RX_RECAL_FORCE_EN_OVRD_EN	RX_RECAL_FORCE_EN_OVRD_VAL	RX_RECAL_BANK_SEL_OVRD_EN	RX_RECAL_BANK_SEL_OVRD_VAL

Table 2-804 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RX_COARSE_ADPT_SKIP_EN_OVRD_EN	R/W	This bit enables the override for rx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	RX_COARSE_ADPT_SKIP_EN_OVRD_VAL	R/W	This bit is the override value for rx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	RX_COARSE_ADPT_FORCE_EN_OVRD_EN	R/W	This bit enables the override for rx_recal_force_en. **Value After Reset:** 0x0

Table 2-804 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
7	RX_COARSE_ADPT_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for rx_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_RECAL_SKIP_EN_OVRD_EN	R/W	This bit enables the override for rx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_RECAL_SKIP_EN_OVRD_VAL	R/W	This bit is the override value for rx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	RX_RECAL_FORCE_EN_OVRD_EN	R/W	This bit enables the override for rx_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_RECAL_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for rx_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_RECAL_BANK_SEL_OVRD_EN	R/W	This bit enables the override for rx_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	RX_RECAL_BANK_SEL_OVRD_VAL	R/W	This bit is the override value for rx_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.801 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)31
- **Exists:** Always

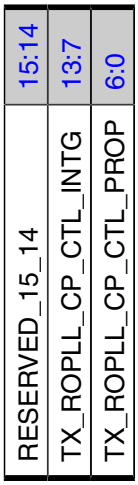


Table 2-805 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:7	TX_ROPLL_CP_CTL_INTG	R	This bit is the value from PCS for tx_ropll_cp_ctl_intg. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6:0	TX_ROPLL_CP_CTL_PROP	R	This bit is the value from PCS for tx_ropll_cp_ctl_prop. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.802 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)32
- **Exists:** Always

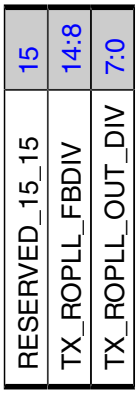


Table 2-806 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:8	TX_ROPLL_FBDIV	R	This bit is the value from PCS for tx_ropll_fbdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	TX_ROPLL_OUT_DIV	R	This bit is the value from PCS for tx_ropll_out_div. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.803 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)33
- **Exists:** Always

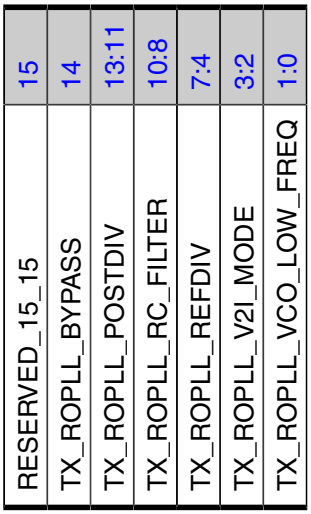


Table 2-807 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	TX_ROPLL_BYPASS	R	This bit is the value from PCS for tx_ropll_bypass. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:11	TX_ROPLL_POSTDIV	R	This bit is the value from PCS for tx_ropll_postdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10:8	TX_ROPLL_RC_FILTER	R	This bit is the value from PCS for tx_ropll_rc_filter. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-807 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
7:4	TX_ROPLL_REFDIV	R	This bit is the value from PCS for tx_ropll_refdiv. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:2	TX_ROPLL_V2I_MODE	R	This bit is the value from PCS for tx_ropll_v2i_mode. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1:0	TX_ROPLL_VCO_LOW_FREQ	R	This bit is the value from PCS for tx_ropll_vco_low_freq. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.804 RAWLANEN_DIG_PCS_XF_CUST_TX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

- **Description:** Current Values for TX_CNTX_RSTR_DONE.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)34
- **Exists:** Always

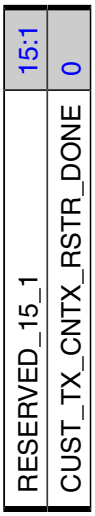


Table 2-808 Fields for Register: RAWLANEN_DIG_PCS_XF_CUST_TX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	CUST_TX_CNTX_RSTR_DONE	R	This bit is the value for tx_cntx_rstr_done_r for cusotmer to poll (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.805 RAWLANEN_DIG_PCS_XF_CUST_RX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

- **Description:** read values for rx_eq_dfe_tap2 and RX_CNTX_RSTR_DONE.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)35
- **Exists:** Always

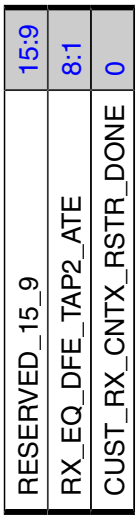


Table 2-809 Fields for Register: RAWLANEN_DIG_PCS_XF_CUST_RX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:1	RX_EQ_DFE_TAP2_ATE	R	read value for rx_eq_dfe_tap2 (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	CUST_RX_CNTX_RSTR_DONE	R	This bit is the value for rx_cntx_rstr_done_r for cusotmer to poll (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.806 RAWLANEN_DIG_PCS_XF_PAUSE_AFTER_CNTX_RSTR (for N = 0; N <= NLANES-1)

- **Description:** A register for customer to pause the FW from proceeding after the context restore is done
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)36
- **Exists:** Always

Table 2-810 Fields for Register: RAWLANEN_DIG_PCS_XF_PAUSE_AFTER_CNTX_RSTR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	PAUSE_AFTER_RX_CNTX_VAL	R/W	Override value for pause_after_rx_cntx_res **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	PAUSE_AFTER_TX_CNTX_VAL	R/W	Override value for pause_after_tx_cntx_res **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.807 RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_1 (for N = 0; N <= NLANES-1)

- **Description:** read only registers for context restore related signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)37
- **Exists:** Always

Table 2-811 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_RO_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	TX_CNTX_SEL_RO	R	read value for TX_CNTX_SEL (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	TX_CMN_CNTX_SEL_RO	R	read value for TX_CMN_CNTX_SEL (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.808 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 8.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)38
- **Exists:** Always

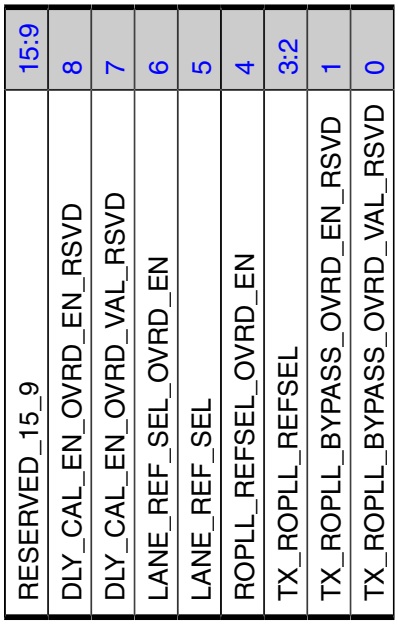


Table 2-812 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	DLY_CAL_EN_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_dly_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	DLY_CAL_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_dly_cal_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	LANE_REF_SEL_OVRD_EN	R/W	This bit is the override enable for lane_ref_sel. **Value After Reset:** 0x0

Table 2-812 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
5	LANE_REF_SEL	R/W	This bit is the override value for lane_ref_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	ROPLL_REFSEL_OVRD_EN	R/W	This bit is the override enable for tx_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:2	TX_ROPLL_REFSEL	R/W	This bit is the override value for tx_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
1	TX_ROPLL_BYPASS_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_ROPLL_BYPASS_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.809 RAWLANEN_DIG_PCS_XF_TX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1)

- **Description:** Override the Incoming tx_recal_inputs.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)39
- **Exists:** Always

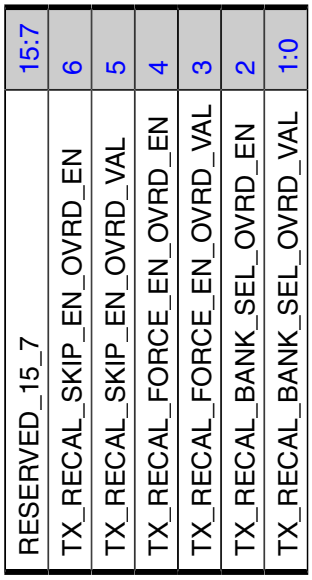


Table 2-813 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	TX_RECAL_SKIP_EN_OVRD_EN	R/W	This bit is the override enable for tx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_RECAL_SKIP_EN_OVRD_ VAL	R/W	This bit is the override value for tx_recal_skip_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_RECAL_FORCE_EN_OVRD _EN	R/W	This bit is the override enable for tx_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-813 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_RECAL_BANK_OVRD (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	TX_RECAL_FORCE_EN_OVRD_VAL	R/W	This bit is the override value for tx_recal_force_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_RECAL_BANK_SEL_OVRD_EN	R/W	This bit is the override enable for tx_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	TX_RECAL_BANK_SEL_OVRD_VAL	R/W	This bit is the override value for tx_recal_bank_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.810 RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX IQ Setting.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3a
- **Exists:** Always

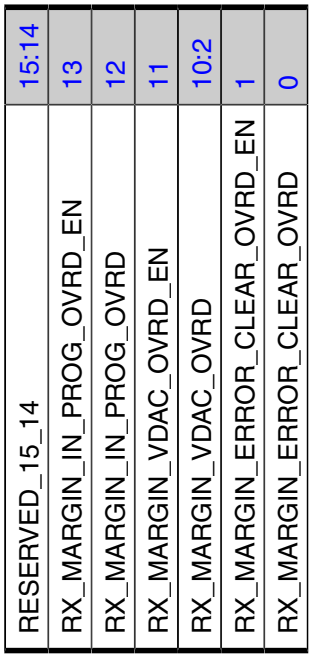


Table 2-814 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	RX_MARGIN_IN_PROG_OVRD_EN	R/W	This bit is the override enable the RX margin in prog. This bit overrides the rx_margin_in_prog input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_MARGIN_IN_PROG_OVRD	R/W	This bit is the override the RX margin in prog. Its override enable is RX_OVRD_IN_9.RX_MARGIN_IN_PROG_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-814 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_IN_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	RX_MARGIN_VDAC_OVRD_EN	R/W	This bit is the override enable the RX margin VDAC setting. This bit overrides the rx_margin_vdac input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10:2	RX_MARGIN_VDAC_OVRD	R/W	This bit overrides the RX margin VDAC setting. Its override enable is RX_OVRD_IN_9.RX_MARGIN_VDAC_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff
1	RX_MARGIN_ERROR_CLEAR_OVRD_EN	R/W	This bit is the override enable the RX margin error clear. This bit overrides the rx_margin_error_clear input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_MARGIN_ERROR_CLEAR_OVRD	R/W	This bit overrides the RX margin error clear. Its override enable is RX_OVRD_IN_9.RX_MARGIN_ERROR_CLEAR_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.811 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 8.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3b
- **Exists:** Always

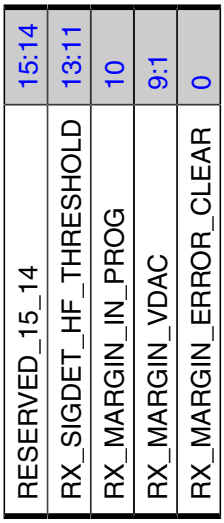


Table 2-815 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:11	RX_SIGDET_HF_THRESHOLD	R	read value for rx_sigdet_hf_threshold_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
10	RX_MARGIN_IN_PROG	R	This bit is the value from the ASIC for rx_margin_in_prog. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:1	RX_MARGIN_VDAC	R	This bit is the value from the ASIC for rx_margin_vdac. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-815 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
0	RX_MARGIN_ERROR_CLEAR	R	This bit is the value from the ASIC for rx_margin_error_clear. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.812 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3c
- **Exists:** Always

Table 2-816 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	TX_DCC_CTRL_CM_RANGE	R	This bit is the value from PCS for tx_dcc_ctrl_cm_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	TX_DCC_CTRL_DIFF_RANGE	R	This bit is the value from PCS for tx_dcc_ctrl_diff_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.813 RAWLANEN_DIG_PCS_XF_TX_PCS_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming TX Controls from PCS (Pre-override Input Monitor).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3d
- **Exists:** Always

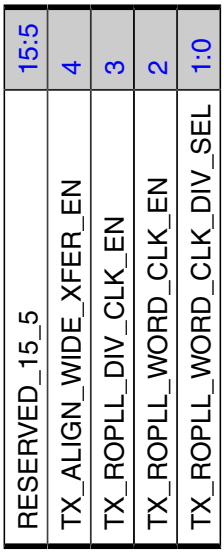


Table 2-817 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	TX_ALIGN_WIDE_XFER_EN	R	This bit is the value from PCS for tx_align_wide_xfer_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3	TX_ROPLL_DIV_CLK_EN	R	This bit is the value from PCS for tx_ropll_div_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-817 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PCS_IN_6 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2	TX_ROPLL_WORD_CLK_EN	R	This bit is the value from PCS for tx_ropll_word_clk_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
1:0	TX_ROPLL_WORD_CLK_DIV_SEL	R	This bit is the value from PCS for tx_ropll_word_clk_div_sel. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.814 RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_2 (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANE0_DIG_PCS_XF_RX_OVRD_FOR_FW_2
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3e
- **Exists:** Always

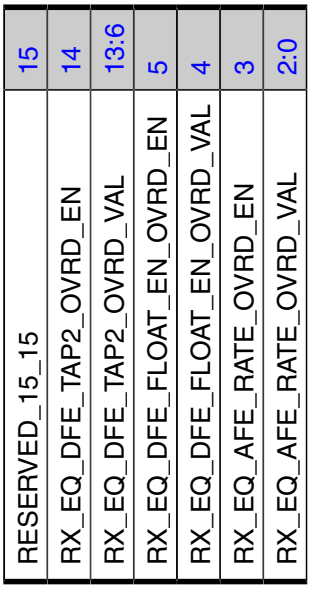


Table 2-818 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_EQ_DFE_TAP2_OVRD_EN	R/W	Override value for rx_eq_dfe_tap2_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:6	RX_EQ_DFE_TAP2_OVRD_VAL	R/W	Override value for rx_eq_dfe_tap2[7:0] **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
5	RX_EQ_DFE_FLOAT_EN_OVRD_EN	R/W	Override enable for rx_eq_dfe_float_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-818 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_OVRD_FOR_FW_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	RX_EQ_DFE_FLOAT_EN_OVRD_VAL	R/W	Override value for rx_eq_dfe_float_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_EQ_AFE_RATE_OVRD_EN	R/W	Override enable for rx_eq_afe_rate **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	RX_EQ_AFE_RATE_OVRD_VAL	R/W	Override value for rx_eq_afe_rate **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.815 RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_10 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 10.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)3f
- **Exists:** Always

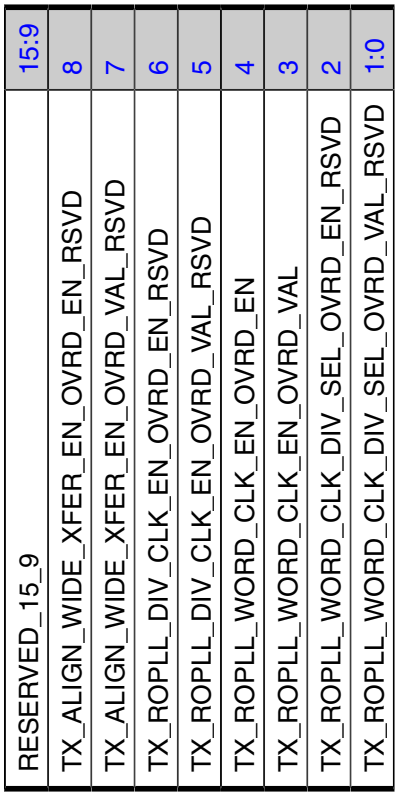


Table 2-819 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	TX_ALIGN_WIDE_XFER_EN_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_align_wide_xfer_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	TX_ALIGN_WIDE_XFER_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_align_wide_xfer_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-819 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_OVRD_IN_10 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6	TX_ROPLL_DIV_CLK_EN_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_ROPLL_DIV_CLK_EN_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_ROPLL_WORD_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	TX_ROPLL_WORD_CLK_EN_OVRD_VAL	R/W	This bit is the override value for tx_ropll_word_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_ROPLL_WORD_CLK_DIV_SEL_OVRD_EN_RSVD	R/W	This bit is the override enable for tx_ropll_word_clk_div_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	TX_ROPLL_WORD_CLK_DIV_SEL_OVRD_VAL_RSVD	R/W	This bit is the override value for tx_ropll_word_clk_div_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.816 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 9.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)40
- **Exists:** Always

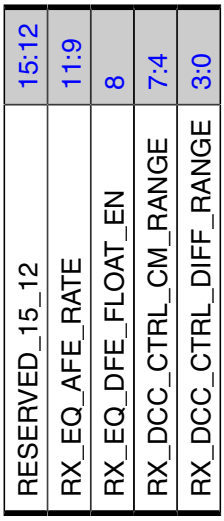


Table 2-820 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:9	RX_EQ_AFE_RATE	R	This bit is the value from PCS for rx_eq_afe_rate. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	RX_EQ_DFE_FLOAT_EN	R	This bit is the value from PCS for rx_eq_dfe_float_en. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:4	RX_DCC_CTRL_CM_RANGE	R	This bit is the value from PCS for rx_dcc_ctrl_cm_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-820 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0 **Volatile:** true
3:0	RX_DCC_CTRL_DIFF_RANGE	R	This bit is the value from PCS for rx_dcc_ctrl_diff_range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.817 RAWLANEN_DIG_PCS_XF_RX_PCS_IN_10 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Incoming RX Controls from PCS, Register 10.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)41
- **Exists:** Always

Table 2-821 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PCS_IN_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	RX_EQ_AFE_CONFIG	R	This bit is the value from the ASIC for rx_eq_afe_config. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.818 RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_1 (for N = 0; N <= NLANES-1)

- **Description:** Override register for CNTX_RSTR primary input signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)42
- **Exists:** Always

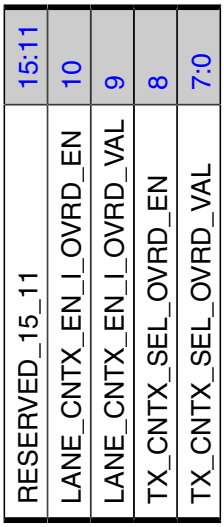


Table 2-822 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	LANE_CNTX_EN_I_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	LANE_CNTX_EN_I_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	TX_CNTX_SEL_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	TX_CNTX_SEL_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.819 RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_2 (for N = 0; N <= NLANES-1)

- **Description:** Override register for CNTX_RSTR primary input signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)43
- **Exists:** Always

Table 2-823 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	TX_CMN_CNTX_SEL_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	TX_CMN_CNTX_SEL_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.820 RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_3 (for N = 0; N <= NLANES-1)

- **Description:** Override register for CNTX_RSTR primary input signals
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)44
- **Exists:** Always

Table 2-824 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_OVRD_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_CNTX_SEL_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:7	RX_CNTX_SEL_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
6	LANE_CNTX_EN_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	LANE_CNTX_EN_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	LANE_LINK_NUM_OVRD_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	LANE_LINK_NUM_OVRD_VAL	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.821 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** ATE Override Input to Control Top-Level Inputs, Register 1. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)45
- **Exists:** Always

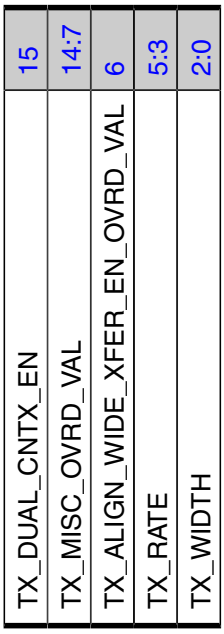


Table 2-825 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_DUAL_CNTX_EN	R/W	The bit that determines whether the dual_cntx feature is used in the current protocol **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:7	TX_MISC_OVRD_VAL	R/W	This field is the override value for tx_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
6	TX_ALIGN_WIDE_XFER_EN_OVRD_VAL	R/W	This field is the override value for tx_align_wide_xfer_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-825 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5:3	TX_RATE	R/W	This field is the override value for tx_rate. Its override enable is TX_OVRD_IN.OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
2:0	TX_WIDTH	R/W	This field is the override value for tx_width. Its override enable is TX_OVRD_IN_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.822 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 2. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)46
- **Exists:** Always

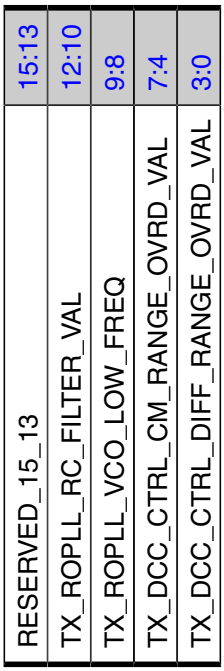


Table 2-826 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:10	TX_ROPLL_RC_FILTER_VAL	R/W	This field is the value from PCS for tx_ropll_rc_filter. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
9:8	TX_ROPLL_VCO_LOW_FREQ	R/W	This field is the value from PCS for tx_ropll_vco_low_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-826 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7:4	TX_DCC_CTRL_CM_RANGE_OVRD_VAL	R/W	This field is the override value for tx_dcc_ctrl_cm_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
3:0	TX_DCC_CTRL_DIFF_RANGE_OVRD_VAL	R/W	This field is the override value for tx_dcc_ctrl_diff_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.823 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 3. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)47
- **Exists:** Always

Table 2-827 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	TX_ROPLL_V2I_MODE_OVRD_VAL	R/W	This field is the value from PCS for tx_ropll_v2i_mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
13:7	TX_ROPLL_CP_CTL_INTG_OVRD_VAL	R/W	This field is the override value for tx_ropll_cp_ctl_intg. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
6:0	TX_ROPLL_CP_CTL_PROP_OVRD_VAL	R/W	This field is the override value for tx_ropll_cp_ctl_prop. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.824 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 4. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)48
- **Exists:** Always

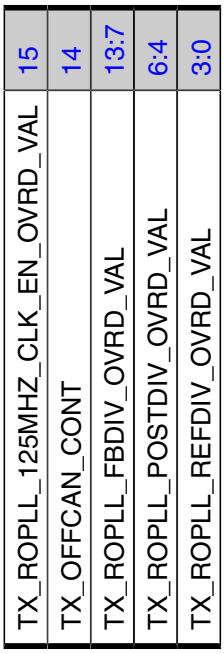


Table 2-828 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_ROPLL_125MHZ_CLK_EN_OVRD_VAL	R/W	This bit is the override value for tx_ropll_125mhz_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_OFFCAN_CONT	R/W	This bit is the override value for tx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:7	TX_ROPLL_FBDIV_OVRD_VAL	R/W	These bits are the override value for tx_ropll_fbdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

Table 2-828 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_4 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6:4	TX_ROPLL_POSTDIV_OVRD_VAL	R/W	These bits are the override value for tx_ropll_postdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3:0	TX_ROPLL_REFDIV_OVRD_VAL	R/W	These bits are the override value for tx_ropll_refdiv. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.825 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 5. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)49
- **Exists:** Always

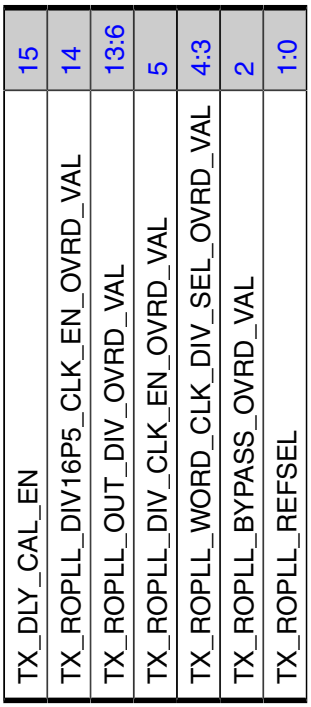


Table 2-829 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_DLY_CAL_EN	R/W	This bit is the override value for tx_dly_cal_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_ROPLL_DIV16P5_CLK_EN_OVRD_VAL	R/W	This bit is the override value for tx_ropll_div16p5_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:6	TX_ROPLL_OUT_DIV_OVRD_VAL	R/W	These bits are the override value for tx_ropll_out_div. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

Table 2-829 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5	TX_ROPLL_DIV_CLK_EN_OVRD_VAL	R/W	This bit is the override value for tx_ropll_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4:3	TX_ROPLL_WORD_CLK_DIV_SEL_OVRD_VAL	R/W	These bits are the override value for tx_ropll_word_clk_div_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
2	TX_ROPLL_BYPASS_OVRD_VAL	R/W	This bit is the override value for tx_ropll_bypass. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1:0	TX_ROPLL_REFSEL	R/W	This bit is the override value for tx_ropll_refsel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.826 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Override Incoming Values for tx/rx_term_ctrl, Register 6. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4a
- **Exists:** Always

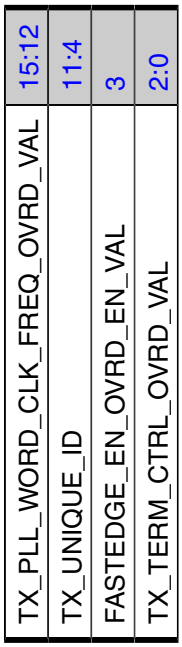


Table 2-830 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	TX_PLL_WORD_CLK_FREQ_OVRD_VAL	R/W	This bit is the override value for tx_pll_word_clk_freq. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:4	TX_UNIQUE_ID	R/W	The lane columnn ID in the config master that the current context is taken from. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
3	FASTEDGE_EN_OVRD_EN_VAL	R/W	This bit is the override enable for tx_fastedge_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-830 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_6 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2:0	TX_TERM_CTRL_OVRD_VAL	R/W	These bits are the override value for tx_term_ctrl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.827 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming TX Controls from PCS, Register 7. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4b
- **Exists:** Always

Table 2-831 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	TX_TERM_CTRL_MSB	R/W	This field is the override value for tx_term_ctrl_msb_r MSB bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:0	TX_DIG_ROPLL_DIV_CLK_SEL_VAL	R/W	This field is the override value for tx_dig_ropll_div_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf

2.1.828 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input, Register 8. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4c
- **Exists:** Always

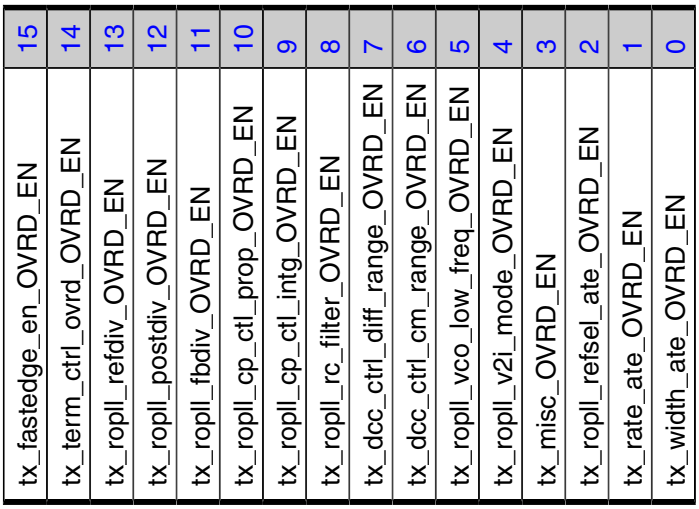


Table 2-832 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	tx_fastedge_en_OVRD_EN	R/W	This bit is the override enable for tx_fastedge_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	tx_term_ctrl_ovrd_OVRD_EN	R/W	This bit is the override enable for tx_term_ctrl_in. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	tx_ropll_refdiv_OVRD_EN	R/W	This bit is the override enable for tx_ropll_refdiv. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	tx_ropll_postdiv_OVRD_EN	R/W	This bit is the override enable for tx_ropll_postdiv. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-832 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	tx_ropll_fbdiv_OVRD_EN	R/W	This bit is the override enable for tx_ropll_fbdiv. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	tx_ropll_cp_ctl_prop_OVRD_EN	R/W	This bit is the override enable for tx_ropll_cp_ctl_prop. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	tx_ropll_cp_ctl_intg_OVRD_EN	R/W	This bit is the override enable for tx_ropll_cp_ctl_intg. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	tx_ropll_rc_filter_OVRD_EN	R/W	This bit is the override enable for tx_ropll_rc_filter. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	tx_dcc_ctrl_diff_range_OVRD_EN	R/W	This bit is the override enable for tx_dcc_ctrl_diff_range. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	tx_dcc_ctrl_cm_range_OVRD_EN	R/W	This bit is the override enable for tx_dcc_ctrl_cm_range. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	tx_ropll_vco_low_freq_OVRD_EN	R/W	This bit is the override enable for tx_ropll_vco_low_freq. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	tx_ropll_v2i_mode_OVRD_EN	R/W	This bit is the override enable for tx_ropll_v2i_mode. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	tx_misc_OVRD_EN	R/W	This bit is the override enable for tx_misc. **Value After Reset:** 0x1

Table 2-832 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	tx_ropll_refsel_ate_OVRD_EN	R/W	This bit is the override enable for tx_ropll_refsel_ate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	tx_rate_ate_OVRD_EN	R/W	This bit is the override enable for tx_rate_ate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	tx_width_ate_OVRD_EN	R/W	This bit is the override enable for tx_width_ate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.829 RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input, Register 9. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4d
- **Exists:** Always

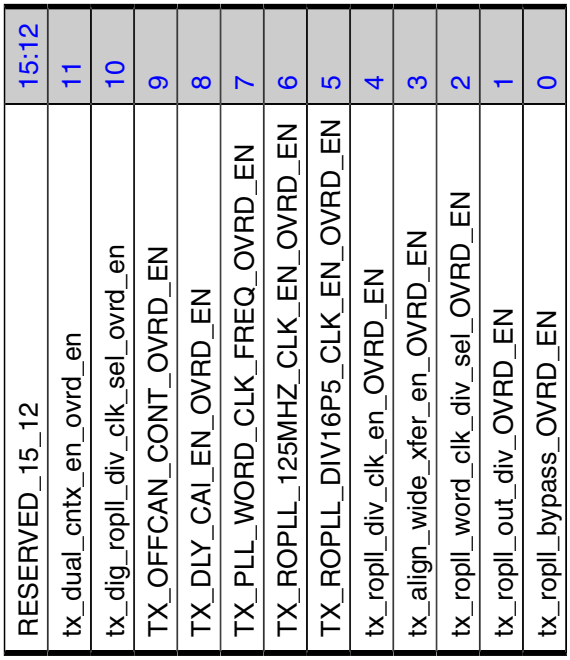


Table 2-833 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	tx_dual_cntx_en_ovrd_en	R/W	This bit is the override enable for tx_dual_cntx_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	tx_dig_ropll_div_clk_sel_ovrd_en	R/W	This bit is the override enable for tx_dig_ropll_div_clk_sel. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	TX_OFFCAN_CONT_OVRD_EN	R/W	This bit is the override enable for tx_offcan_cont. **Value After Reset:** 0x1

Table 2-833 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8	TX_DLY_CAl_EN_OVRD_EN	R/W	This bit is the override enable for tx_dly_cal_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	TX_PLL_WORD_CLK_FREQ_OVRD_EN	R/W	This bit is the override enable for tx_pll_word_clk_freq. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	TX_ROPLL_125MHZ_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_125mhz_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	TX_ROPLL_DIV16P5_CLK_EN_OVRD_EN	R/W	This bit is the override enable for tx_ropll_div16p5_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	tx_ropll_div_clk_en_OVRD_EN	R/W	This bit is the override enable for tx_ropll_div_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	tx_align_wide_xfer_en_OVRD_EN	R/W	This bit is the override enable for tx_align_wide_xfer_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	tx_ropll_word_clk_div_sel_OVRD_EN	R/W	This bit is the override enable for tx_ropll_word_clk_div_sel. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	tx_ropll_out_div_OVRD_EN	R/W	This bit is the override enable for tx_ropll_out_div. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-833 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_PACKED_CNTX_TX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	tx_ropll_bypass_OVRD_EN	R/W	This bit is the override enable for tx_ropll_bypass. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.830 RAWLANEN_DIG_PCS_XF_PWR_EN_LANE (for N = 0; N <= NLANES-1)

- **Description:** power_en new register to sequence isolation
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4e
- **Exists:** Always

Table 2-834 Fields for Register: RAWLANEN_DIG_PCS_XF_PWR_EN_LANE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	PWR_EN	R/W	**Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.831 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX EQ Controls from PCS, Register 1. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)4f
- **Exists:** Always

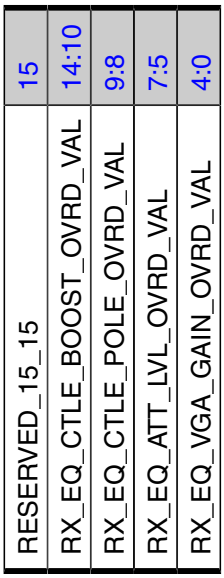


Table 2-835 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:10	RX_EQ_CTLE_BOOST_OVRD_VAL	R/W	This field is the override value for rx_eq_ctle_boost[4:0]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
9:8	RX_EQ_CTLE_POLE_OVRD_VAL	R/W	This field is the override value for rx_eq_ctle_pole[2]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

Table 2-835 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7:5	RX_EQ_ATT_LVL_OVRD_VAL	R/W	This field is the override value for rx_eq_att_lvl[2:0]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
4:0	RX_EQ_VGA_GAIN_OVRD_VAL	R/W	This field is the override value for rx_eq_vga_gain[2:0]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.832 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_2 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)50
- **Exists:** Always

Table 2-836 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14:12	RX_EQ_AFE_RATE_OVRD_VAL	R/W	This field is the override value for rx_eq_afe_rate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
11:0	RX_EQ_AFE_CONFIG_OVRD	R/W	This field overrides the rx_eq_afe_config setting. Its override enable is RX_OVRD_IN_11.RX_EQ_AFE_CONFIG_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.833 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_3 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)51
- **Exists:** Always

Table 2-837 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RX_DELTA_IQ_ATE_OVRD	R/W	This field overrides the RX Delta IQ seeting. Its override enable is RX_OVRD_IN_6.RX_DELTA_IQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:0	RX_CDR_VCO_CONFIG_OVRD	R/W	This field is the override value for rx_cdr_vco_config. Its override enable is RX_OVRD_IN_2.CDR_VCO_CONFIG_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.834 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_4 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)52
- **Exists:** Always

Table 2-838 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RX_RATE_ATE_OVRD_VAL	R/W	This field is the ATE Override value for the rx_rate input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
12:0	RX_VCO_LD_VAL_OVRD	R/W	This field is the override value for rx_vco_ld_val. Its override enable is RX_OVRD_IN_2.VCO_LD_VAL_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1fff

2.1.835 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)53
- **Exists:** Always

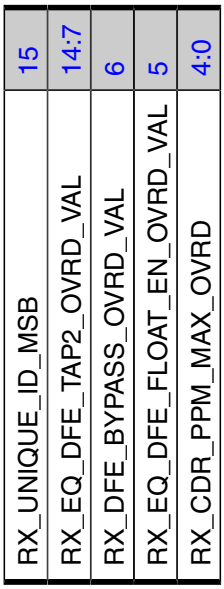


Table 2-839 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_UNIQUE_ID_MSB	R/W	The MSB of the sup columnn ID in the config master that the current context is taken from. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:7	RX_EQ_DFE_TAP2_OVRD_VAL	R/W	This field is the override value for rx_eq_dfe_tap2[7:0]. Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
6	RX_DFE_BYPASS_OVRD_VAL	R/W	This field is the ATE override value for rx_dfe_bypass input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	RX_EQ_DFE_FLOAT_EN_OVRD_VAL	R/W	This field is the override value for rx_eq_dfe_float_en. **Value After Reset:** 0x0

Table 2-839 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_5 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
4:0	RX_CDR_PPM_MAX_OVRD	R/W	This field is the overridden value for RX CDR PPM Max. Its override enable is RX_OVRD_IN_5.RX_CDR_PPM_MAX_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.836 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_6 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)54
- **Exists:** Always

Table 2-840 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	REF_LD_VAL_OVRD	R/W	This field is the override value for rx_ref_ld_val. Its override enable is RX_OVRD_IN_4.REF_LD_VAL_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f
8:0	RX_EQ_DFE_TAP1_OVRD_VAL	R/W	This field is the override value for rx_eq_dfe_tap1[7:0] Its override enable is RX_EQ_OVRD_IN_1.RX_EQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.837 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_7 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)55
- **Exists:** Always

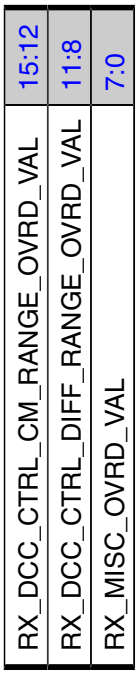


Table 2-841 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RX_DCC_CTRL_CM_RANGE_OVRD_VAL	R/W	This field is the override value for rx_dcc_ctrl_cm_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
11:8	RX_DCC_CTRL_DIFF_RANGE_OVRD_VAL	R/W	This field is the override value for rx_dcc_ctrl_diff_range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xf
7:0	RX_MISC_OVRD_VAL	R/W	This field is the override value for rx_misc. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.838 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)56
- **Exists:** Always

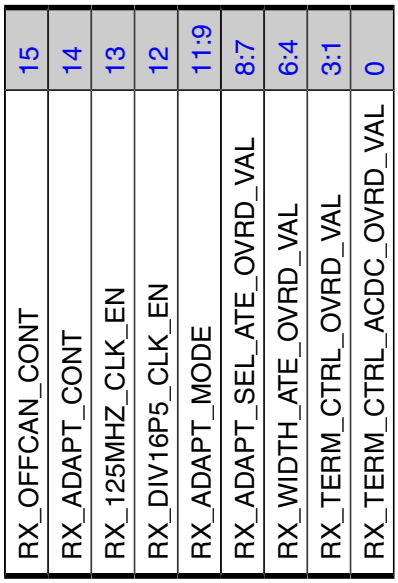


Table 2-842 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RX_OFFCAN_CONT	R/W	This bit is the override value for rx_offcan_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	RX_ADAPT_CONT	R/W	This bit is the override value for rx_adapt_cont. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_125MHZ_CLK_EN	R/W	This bit enables the outgoing rx_125mhz_clk. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	RX_DIV16P5_CLK_EN	R/W	This bit enables the outgoing rx_div16p5_clk. **Value After Reset:** 0x1

Table 2-842 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_8 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11:9	RX_ADAPT_MODE	R/W	This field is the overridden value for RX adaptation mode. Its override enable is RX_OVRD_IN_5.RX_ADAPT_MODE_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
8:7	RX_ADAPT_SEL_ATE_OVRD_VAL	R/W	This field is the ATE override value for the top-level rx_adapt_sel input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
6:4	RX_WIDTH_ATE_OVRD_VAL	R/W	This field is the ATE Override value for rx_width input. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
3:1	RX_TERM_CTRL_OVRD_VAL	R/W	This field is the Override value for rx_term_ctrl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
0	RX_TERM_CTRL_ACDC_OVRD_VAL	R/W	This field is the Override value for rx_term_ctrl. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.839 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)57
- **Exists:** Always

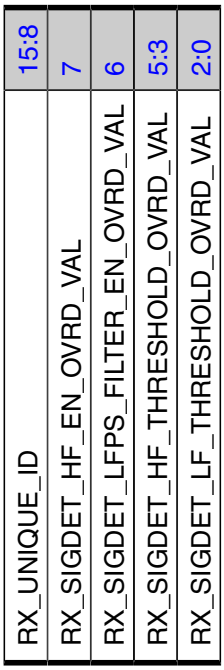


Table 2-843 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_UNIQUE_ID	R/W	The sup columnn ID in the config master that the current context is taken from. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
7	RX_SIGDET_HF_EN_OVRD_VAL	R/W	Override value for rx_sigdet_hf_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	RX_SIGDET_LFPS_FILTER_EN_OVRD_VAL	R/W	This field is the override value for rx_sigdet_lfps_filter_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-843 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_9 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5:3	RX_SIGDET_HF_THRESHOLD_OVRD_VAL	R/W	This field is the override value for rx_sigdet_hf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
2:0	RX_SIGDET_LF_THRESHOLD_OVRD_VAL	R/W	This field is the override value for rx_sigdet_lf_threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.840 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_10 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)58
- **Exists:** Always

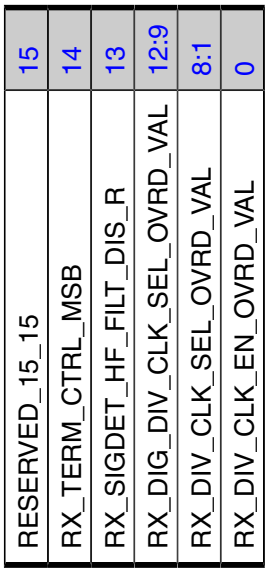


Table 2-844 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_TERM_CTRL_MSB	R/W	This field is the override value for rx_term_ctrl_msb_r MSB bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	RX_SIGDET_HF_FILT_DIS_R	R/W	This bit enables/disables SIGDET_HF circuit when it is present in analog **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12:9	RX_DIG_DIV_CLK_SEL_OVRD_VAL	R/W	This field is the override value for rx_dig_div_clk_sel. **Value After Reset:** 0x0

Table 2-844 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_10 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0xf
8:1	RX_DIV_CLK_SEL_OVRD_VAL	R/W	This field is the override value for rx_div_clk_sel. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
0	RX_DIV_CLK_EN_OVRD_VAL	R/W	This field is the override value for rx_div_clk_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.841 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_11 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)59
- **Exists:** Always

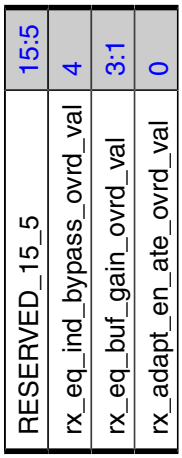


Table 2-845 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	rx_eq_ind_bypass_ovrd_val	R/W	This bit is the override value for rx_eq_ind_bypass **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3:1	rx_eq_buf_gain_ovrd_val	R/W	This bit is the override value for rx_eq_buf_gain. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
0	rx_adapt_en_ate_ovrd_val	R/W	This bit is the override value for rx_adapt_en_ate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.842 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_12 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5a
- **Exists:** Always

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
rx_offcan_cont_OVRD_EN	rx_ref_ld_val_OVRD_EN	rx_eq_dfe_tap1_OVRD_EN	rx_eq_dfe_tap2_OVRD_EN	rx_dfe_bypass_ate_OVRD_EN	rx_cdr_ppm_max_OVRD_EN	rx_rate_ate_OVRD_EN	rx_vco_ld_val_OVRD_EN	rx_delta_iq_ate_OVRD_EN	rx_cdr_vco_config_OVRD_EN	rx_eq_afe_rate_OVRD_EN	rx_eq_afe_config_OVRD_EN	rx_eq_ctle_boost_OVRD_EN	rx_eq_ctle_pole_OVRD_EN	rx_eq_att_lvl_OVRD_EN	rx_eq_vga_gain_OVRD_EN

Table 2-846 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	rx_offcan_cont_OVRD_EN	R/W	This bit the override enable for rx_offcan_cont. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	rx_ref_ld_val_OVRD_EN	R/W	This bit is the override enable for rx_ref_ld_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	rx_eq_dfe_tap1_OVRD_EN	R/W	This bit is the override enable for rx_eq_dfe_tap1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	rx_eq_dfe_tap2_OVRD_EN	R/W	This bit is the override enable for rx_eq_dfe_tap2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
11	rx_dfe_bypass_ate_OVRD_EN	R/W	This bit is the override enable for rx_dfe_bypass_ate. **Value After Reset:** 0x1

Table 2-846 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_12 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
10	rx_cdr_ppm_max_OVRD_EN	R/W	This bit is the override enable for rx_cdr_ppm_max. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	rx_rate_ate_OVRD_EN	R/W	This bit is the override enable for rx_rate_ate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	rx_vco_ld_val_OVRD_EN	R/W	This bit is the override enable for rx_vco_ld_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	rx_delta_iq_ate_OVRD_EN	R/W	This bit is the override enable for rx_delta_iq_ate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	rx_cdr_vco_config_OVRD_EN	R/W	This bit is the override enable for rx_cdr_vco_config. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	rx_eq_afe_rate_OVRD_EN	R/W	This bit is the override enable for rx_eq_afe_rate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	rx_eq_afe_config_OVRD_EN	R/W	This bit is the override enable for rx_eq_afe_config. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	rx_eq_ctle_boost_OVRD_EN	R/W	This bit is the override enable for rx_eq_ctle_boost. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-846 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_12 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
2	rx_eq_ctle_pole_OVRD_EN	R/W	This bit is the override enable for rx_eq_ctle_pole. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	rx_eq_att_lvl_OVRD_EN	R/W	This bit is the override enable for rx_eq_att_lvl. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	rx_eq_vga_gain_OVRD_EN	R/W	This bit is the override enable for rx_eq_vga_gain. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.843 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_13 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5b
- **Exists:** Always

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
rx_adapt_cont_ovrd_en_OVRD_EN	rx_125mhz_clk_en_ovrd_en_OVRD_EN	rx_div16p5_clk_en_ovrd_en_OVRD_EN	rx_eq_dfe_float_en_OVRD_EN	rx_sigdet_hf_en_OVRD_EN	rx_sigdet_lfps_filter_en_OVRD_EN	rx_sigdet_lf_threshold_OVRD_EN	rx_sigdet_hf_threshold_OVRD_EN	rx_width_ate_OVRD_EN	rx_term_ctrl_OVRD_EN	rx_adapt_mode_OVRD_EN	rx_adapt_sel_ate_OVRD_EN	rx_term_ctrl_acdc_OVRD_EN	rx_dcc_ctrl_diff_range_OVRD_EN	rx_dcc_ctrl_cm_range_OVRD_EN	rx_misc_OVRD_EN

Table 2-847 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	rx_adapt_cont_ovrd_en_OVRD_EN	R/W	This bit is the override enable for rx_adapt_cont. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
14	rx_125mhz_clk_en_ovrd_en_OVRD_EN	R/W	This bit is the override enable for rx_125mhz_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	rx_div16p5_clk_en_ovrd_en_OVRD_EN	R/W	This bit is the override enable for rx_div16p5_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	rx_eq_dfe_float_en_OVRD_EN	R/W	This bit is the override enable for rx_eq_dfe_float_en. **Value After Reset:** 0x1

Table 2-847 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_13 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
11	rx_sigdet_hf_en_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	rx_sigdet_lfps_filter_en_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lfps_filter_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	rx_sigdet_lf_threshold_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_lf_threshold. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	rx_sigdet_hf_threshold_OVRD_EN	R/W	This bit is the override enable for rx_sigdet_hf_threshold. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	rx_width_ate_OVRD_EN	R/W	This bit is the override enable for rx_width_ate. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	rx_term_ctrl_OVRD_EN	R/W	This bit is the override enable for rx_term_ctrl. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	rx_adapt_mode_OVRD_EN	R/W	This bit is the override enable for rx_adapt_mode. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	rx_adapt_sel_ate_OVRD_EN	R/W	This bit is the override enable for rx_adapt_sel_ate. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-847 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_13 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	rx_term_ctrl_acdc_OVRD_EN	R/W	This bit is the override enable for rx_term_ctrl_acdc. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	rx_dcc_ctrl_diff_range_OVRD_EN	R/W	This bit is the override enable for rx_dcc_ctrl_diff_range. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	rx_dcc_ctrl_cm_range_OVRD_EN	R/W	This bit is the override enable for rx_dcc_ctrl_cm_range. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	rx_misc_OVRD_EN	R/W	This bit is the override enable for rx_misc. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.844 RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_14 (for N = 0; N <= NLANES-1)

- **Description:** Override Register for RX PCS Input. RO: stands for re-ordered
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5c
- **Exists:** Always

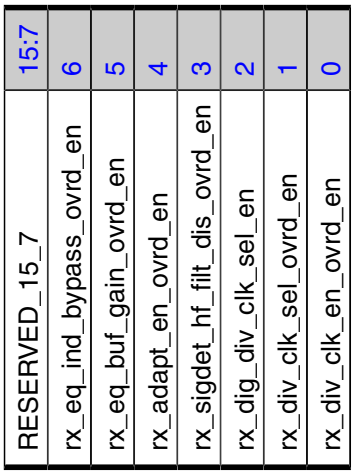


Table 2-848 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	rx_eq_ind_bypass_ovrd_en	R/W	This bit is the override enable for rx_eq_ind_bypass. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	rx_eq_buf_gain_ovrd_en	R/W	This bit is the override enable for rx_eq_buf_gain_ovrd_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	rx_adapt_en_ovrd_en	R/W	This bit is the override enable for rx_adapt_en_ovrd_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	rx_sigdet_hf_filt_dis_ovrd_en	R/W	This bit is the override enable for rx_div_clk_sel. **Value After Reset:** 0x1

Table 2-848 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_PACKED_CNTX_RX_RO_OVRD_IN_14 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	rx_dig_div_clk_sel_en	R/W	This bit is the override enable for rx_dig_div_clk_sel. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	rx_div_clk_sel_ovrd_en	R/W	This bit is the override enable for rx_div_clk_sel. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	rx_div_clk_en_ovrd_en	R/W	This bit is the override enable for rx_div_clk_en. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.845 RAWLANEN_DIG_PCS_XF_TX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

- **Description:** TX Context Restore done.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5d
- **Exists:** Always

Table 2-849 Fields for Register: RAWLANEN_DIG_PCS_XF_TX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_CNTX_RSTR_DONE	R	TX Context Restore done (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.846 RAWLANEN_DIG_PCS_XF_RX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

- **Description:** RX Context Restore done.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5e
- **Exists:** Always

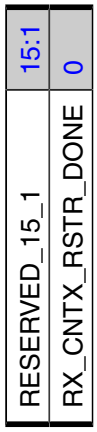


Table 2-850 Fields for Register: RAWLANEN_DIG_PCS_XF_RX_CNTX_RSTR_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_CNTX_RSTR_DONE	R	RX Context Restore done (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.847 RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_EN (for N = 0; N <= NLANES-1)

- **Description:** Lane context restore enable
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)5f
- **Exists:** Always

Table 2-851 Fields for Register: RAWLANEN_DIG_PCS_XF_LANE_CNTX_RSTR_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	LANE_CNTX_EN	R	Lane context restore enable after two internal overrides in xface (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.848 RAWLANEN_DIG_PCS_XF_DEBUG_MODE (for N = 0; N <= NLANES-1)

- **Description:** A register added for FW to know if we are in the debug mode or not
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)60
- **Exists:** Always

Table 2-852 Fields for Register: RAWLANEN_DIG_PCS_XF_DEBUG_MODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	DEBUG_MODE_EN	R/W	**Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14:0	DEBUG_MODE_SRAM_ADDR	R/W	**Value After Reset:** 0x7fff **Exists:** Always **Reset Mask:** 0x7fff

2.1.849 RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming LANE Controls.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b0
- **Exists:** Always

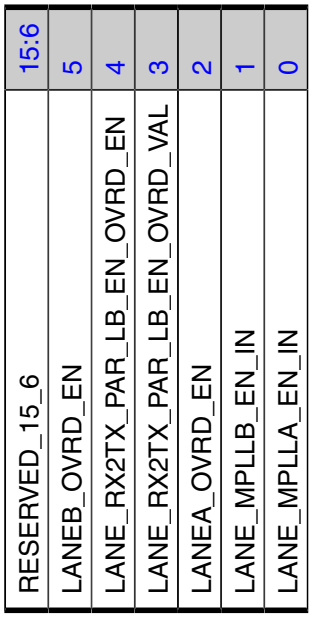


Table 2-853 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	LANEB_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the lane_mpllb_en_in input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	LANE_RX2TX_PAR_LB_EN_OVRD_EN	R/W	This bit is the override enable for lane_rx2tx_par_lb_en. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	LANE_RX2TX_PAR_LB_EN_OVRD_VAL	R/W	This bit is the override value for lane_rx2tx_par_lb_en. **Value After Reset:** 0x0

Table 2-853 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	LANEA_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the lane_mplla_en_in input signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	LANE_MPLLB_EN_IN	R/W	This bit is the override value for lane_mpllb_en_in. Its override enable is LANE_OVRD_IN.LANEB_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	LANE_MPLLA_EN_IN	R/W	This bit is the override value for lane_mplla_en_in. Its override enable is LANE_OVRD_IN.LANEA_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.850 RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing LANE Controls.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b1
- **Exists:** Always

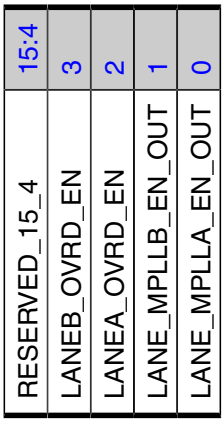


Table 2-854 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	LANEB_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the lane_mpllb_en_out output signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	LANEA_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the lane_mplla_en_out output signal. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	LANE_MPLLB_EN_OUT	R/W	This bit is the override value for lane_mpllb_en_out. Its override enable is LANE_OVRD_OUT.LANEB_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-854 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	LANE_MPLLA_EN_OUT	R/W	This bit is the override value for lane_mplla_en_out. Its override enable is LANE_OVRD_OUT.LANEA_OVRD_EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.851 RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming SUP Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b2
- **Exists:** Always

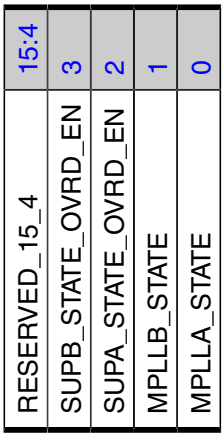


Table 2-855 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	SUPB_STATE_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the mpllb_state_crclk input signal, which is the synchronized version of mpllb_state, coming from CMN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SUPA_STATE_OVRD_EN	R/W	This bit is the override enable for signals in this register. This bit overrides the mplla_state_crclk input signal, which is the synchronized version of mplla_state, coming from CMN . **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	MPLLB_STATE	R/W	This bit is the override value for mpllb_state. Its override enable is SUP_OVRD_IN.SUPB_STATE_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-855 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_OVRD_IN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	MPLLA_STATE	R/W	This bit is the override value for mplla_state. Its override enable is SUP_OVRD_IN.SUPA_STATE_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.852 RAWLANEN_DIG_PMA_XF_SUP_PMA_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Coming SUP Status Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b3
- **Exists:** Always

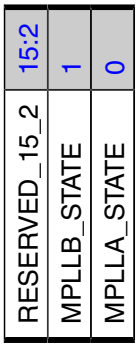


Table 2-856 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_PMA_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	MPLLB_STATE	R	This it is the value from PMA for mpllb_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	MPLLA_STATE	R	This it is the value from PMA for mplla_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.853 RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing TX Controls to PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b4
- **Exists:** Always

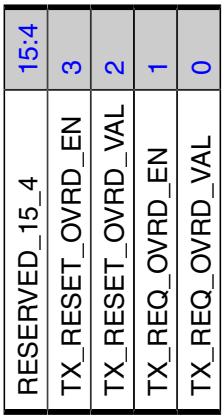


Table 2-857 Fields for Register: RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	TX_RESET_OVRD_EN	R/W	This bit is the override enable for tx_reset. This bit overrides the tx_reset output signal to the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_RESET_OVRD_VAL	R/W	This bit is the override value for tx_reset. Its override enable is TX_OVRD_OUT.TX_RESET_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_REQ_OVRD_EN	R/W	This bit is the override enable for tx_req. This bit overrides the tx_req output signal to the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-857 Fields for Register: RAWLANEN_DIG_PMA_XF_TX_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	TX_REQ_OVRD_VAL	R/W	This bit is the override value for tx_req. Its override enable is TX_OVRD_OUT.TX_REQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.854 RAWLANEN_DIG_PMA_XF_TX_PMA_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Coming TX Status Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b5
- **Exists:** Always

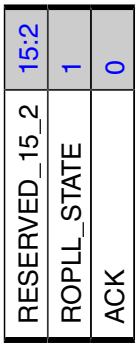


Table 2-858 Fields for Register: RAWLANEN_DIG_PMA_XF_TX_PMA_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	ROPLL_STATE	R	This bit is the value from PMA for tx_ropll_state. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	ACK	R	This bit is the value from PMA for tx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.855 RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Outgoing RX Controls to PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b6
- **Exists:** Always

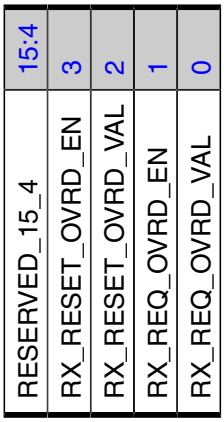


Table 2-859 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	RX_RESET_OVRD_EN	R/W	This bit is the override enable for rx_reset. This bit overrides the rx_reset output signal to PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_RESET_OVRD_VAL	R/W	This bit is the override value for rx_reset. Its override enable is RX_OVRD_OUT.RX_RESET_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_REQ_OVRD_EN	R/W	This bit is the override enable for rx_req. This bit overrides the rx_req output signal to the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-859 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_OVRD_OUT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	RX_REQ_OVRD_VAL	R/W	This bit is the override value for rx_req. Its override enable is RX_OVRD_OUT.RX_REQ_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.856 RAWLANEN_DIG_PMA_XF_RX_PMA_IN (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Coming RX Status Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b7
- **Exists:** Always

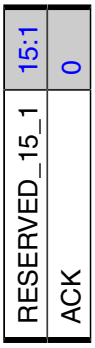


Table 2-860 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_PMA_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ACK	R	This bit is the value from the PMA for rx_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.857 RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL (for N = 0; N <= NLANES-1)

- **Description:** Lane Rtune Controls.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b8
- **Exists:** Always

Table 2-861 Fields for Register: RAWLANEN_DIG_PMA_XF_LANE_RTUNE_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	LANE_RTUNE_REQ	R/W	This bit is the lane value for rtune_req. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.858 RAWLANEN_DIG_PMA_XF_SUP_PMA_IN_1 (for N = 0; N <= NLANES-1)

- **Description:** Current Values for Coming SUP Status Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)b9
- **Exists:** Always

Table 2-862 Fields for Register: RAWLANEN_DIG_PMA_XF_SUP_PMA_IN_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RTUNE_ACK	R	This bit is the value from the PMA for rtune_ack. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.859 RAWLANEN_DIG_PMA_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming RX Controls from PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ba
- **Exists:** Always

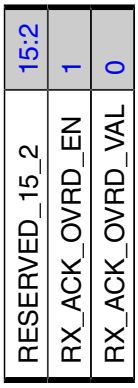


Table 2-863 Fields for Register: RAWLANEN_DIG_PMA_XF_RX_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_ACK_OVRD_EN	R/W	This bit is the override enable for rx_ack. This bit overrides the rx_ack input signal from the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ACK_OVRD_VAL	R/W	This bit is the override value for rx_ack. Its override enable is RX_OVRD_IN.RX_ACK_OVRD_EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.860 RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ (for N = 0; N <= NLANES-1)

- **Description:** Reset Routine Request.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c0
- **Exists:** Always

Table 2-864 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RESET_RTN_REQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RESET_RTN_REQ	R/W	This bit is the reset routine request. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.861 RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX Reset Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c1
- **Exists:** Always

Table 2-865 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_RESET	R	This bit is the RX reset interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.862 RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX Request Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c2
- **Exists:** Always

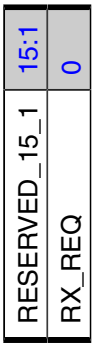


Table 2-866 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_REQ	R	This bit is the RX request interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.863 RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX Rate Change Interrupt Request.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c3
- **Exists:** Always

Table 2-867 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_RATE_IRQ	R	This bit is the RX rate change interrupt request. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.864 RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX PSTATE Change Interrupt Request.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c4
- **Exists:** Always

Table 2-868 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_PSTATE_IRQ	R	This bit is the RX pstate change interrupt request. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.865 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX adaptation request interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c5
- **Exists:** Always

Table 2-869 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_REQ_IRQ	R	This bit is the RX adaptation request interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.866 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX adaptation disable interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c6
- **Exists:** Always

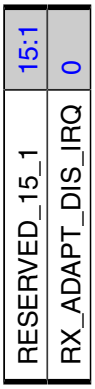


Table 2-870 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_DIS_IRQ	R	This bit is the RX adaptation disable interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.867 RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX reset interrupt clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c7
- **Exists:** Always

Table 2-871 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RESET_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_RESET_IRQ_CLR	R/W	This bit is the RX reset interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.868 RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX request interrupt clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c8
- **Exists:** Always

Table 2-872 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_REQ_IRQ_CLR	R/W	This bit is the RX request interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.869 RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX rate change interrupt clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)c9
- **Exists:** Always

Table 2-873 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_RATE_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_RATE_IRQ_CLR	R/W	This bit is the RX rate change interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.870 RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX pstate change interrupt clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ca
- **Exists:** Always

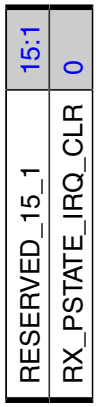


Table 2-874 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_PSTATE_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_PSTATE_IRQ_CLR	R/W	This bit is the RX pstate change interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.871 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Request Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)cb
- **Exists:** Always

Table 2-875 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_REQ_IRQ_CLR	R/W	This bit is the RX adaptation request interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.872 RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Disable Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)cc
- **Exists:** Always

Table 2-876 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_ADAPT_DIS_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_DIS_IRQ_CLR	R/W	This bit is the RX adaptation disable interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.873 RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (for N = 0; N <= NLANES-1)

- **Description:** Interrupt Mask.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)cd
- **Exists:** Always

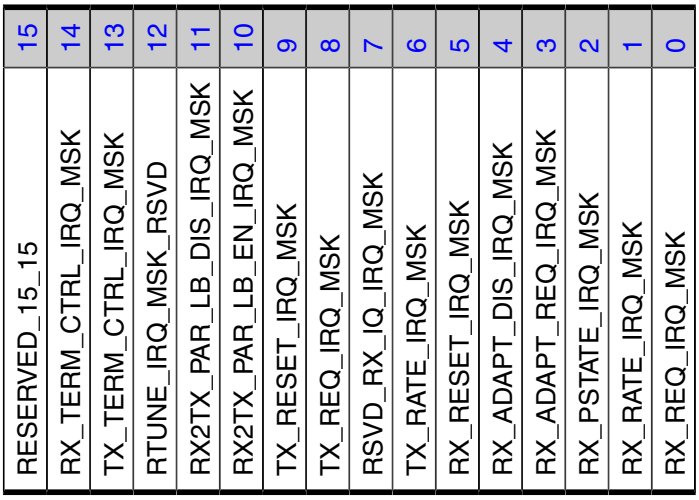


Table 2-877 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_TERM_CTRL_IRQ_MSK	R/W	This bit is the mask for the TX termination control interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
13	TX_TERM_CTRL_IRQ_MSK	R/W	This bit is the mask for the TX termination control interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
12	RTUNE_IRQ_MSK_RSVD	R/W	This bit is reserved; it is the mask for RTUNE interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-877 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
11	RX2TX_PAR_LB_DIS_IRQ_MSK	R/W	This bit is the mask for the RX2TX parallel loopback disable interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
10	RX2TX_PAR_LB_EN_IRQ_MSK	R/W	This bit is the mask for the RX2TX parallel loopback enable interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
9	TX_RESET_IRQ_MSK	R/W	This bit is the mask for the TX reset interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
8	TX_REQ_IRQ_MSK	R/W	This bit is the mask for the TX req interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
7	RSVD_RX_IQ_IRQ_MSK	R/W	This bit is reserved; it is the mask for the RX IQ interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
6	TX_RATE_IRQ_MSK	R/W	This bit is the mask for the TX rate interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-877 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
5	RX_RESET_IRQ_MSK	R/W	This bit is the mask for the RX reset interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4	RX_ADAPT_DIS_IRQ_MSK	R/W	This bit is the mask for the RX adaptation disable interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
3	RX_ADAPT_REQ_IRQ_MSK	R/W	This bit is the mask for the RX adaptation request interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	RX_PSTATE_IRQ_MSK	R/W	This bit is the mask for the RX pstate change interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
1	RX_RATE_IRQ_MSK	R/W	This bit is the mask for the RX rate change interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	RX_REQ_IRQ_MSK	R/W	This bit is the mask for the RX request interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-877 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true

2.1.874 RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ (for N = 0; N <= NLANES-1)

- **Description:** TX Rate Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ce
- **Exists:** Always

Table 2-878 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_RATE_IRQ	R	This bit is the TX rate interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.875 RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** TX Rate Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)cf
- **Exists:** Always

Table 2-879 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_RATE_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_RATE_IRQ_CLR	R/W	This bit is the TX rate interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.876 RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ (for N = 0; N <= NLANES-1)

- **Description:** TX Reset Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d2
- **Exists:** Always

Table 2-880 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_RESET_IRQ	R	This bit is the TX reset interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.877 RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** TX Reset Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d3
- **Exists:** Always

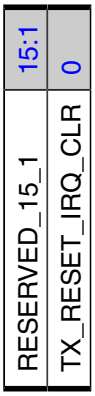


Table 2-881 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_RESET_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_RESET_IRQ_CLR	R/W	This bit is the TX reset interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.878 RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ (for N = 0; N <= NLANES-1)

- **Description:** TX REQ Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d4
- **Exists:** Always

Table 2-882 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_REQ_IRQ	R	This bit is the TX req interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.879 RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** TX req Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d5
- **Exists:** Always

Table 2-883 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_REQ_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_REQ_IRQ_CLR	R/W	This bit is the TX req interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.880 RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX2TX Parallel Loopback Enable Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d6
- **Exists:** Always

Table 2-884 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX2TX_PAR_LB_EN_IRQ	R	This bit is the RX2TX parallel loopback enable interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.881 RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX2TX Parallel Loopback Enable Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d7
- **Exists:** Always

Table 2-885 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_EN_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX2TX_PAR_LB_EN_IRQ_CLR	R/W	This bit is the RX2TX parallel loopback enable interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.882 RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX2TX Parallel Loopback Disable Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d8
- **Exists:** Always

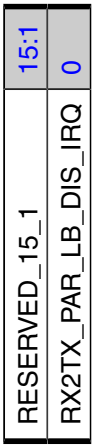


Table 2-886 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX2TX_PAR_LB_DIS_IRQ	R	This bit is the RX2TX parallel loopback disable interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.883 RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX2TX Parallel Loopback Disable Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)d9
- **Exists:** Always

Table 2-887 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX2TX_PAR_LB_DIS_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX2TX_PAR_LB_DIS_IRQ_CLR	R/W	This bit is the RX2TX parallel loopback disable interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.884 RAWLANEN_DIG_IRQ_CTL_IRQ_EN_FLAGS (for N = 0; N <= NLANES-1)

- **Description:** Interrupt Enable Flags.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)da
- **Exists:** Always

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESERVED_15_15	RX_MARGIN_FINISH_IRQ_EN	RX_MARGIN_INIT_IRQ_EN	RX_MARGIN_ERROR_CLEAR_IRQ_EN	RX_MARGIN_VDAC_START_IRQ_EN	RX_MARGIN_IQ_START_IRQ_EN	RX_MARGIN_GLB_IRQ_EN	RX_TERM_CTRL_IRQ_EN	TX_TERM_CTRL_IRQ_EN	RTUNE_IRQ_EN_RSVD	RXTX_REQ_IRQ_BLOCK_ACK_EN	TX_RATE_IRQ_EN	RX_RATE_IRQ_EN	RSVD_RX_IQ_IRQ_EN	RXTX_REQ_IRQ_EN	RX2TX_PAR_LB_IRQ_EN

Table 2-888 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_EN_FLAGS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_MARGIN_FINISH_IRQ_EN	R/W	This bit enables the RX margin finish interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
13	RX_MARGIN_INIT_IRQ_EN	R/W	This bit enables RX margin init interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
12	RX_MARGIN_ERROR_CLEAR_IRQ_EN	R/W	This bit enables RX margin error clear interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-888 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_EN_FLAGS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	RX_MARGIN_VDAC_START_IRQ_EN	R/W	This bit enables RX margin vdac start interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
10	RX_MARGIN_IQ_START_IRQ_EN	R/W	This bit enables RX margin iq start interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9	RX_MARGIN_GLB_IRQ_EN	R/W	This bit enables RX margin glb interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
8	RX_TERM_CTRL_IRQ_EN	R/W	This bit enables RX termination control interrupt. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	TX_TERM_CTRL_IRQ_EN	R/W	This bit enables TX termination control interrupt. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	RTUNE_IRQ_EN_RSVD	R/W	This bit is reserved for Enable RTUNE interrupt. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	RXTX_REQ_IRQ_BLOCK_ACK_EN	R/W	This bit enables the TX/RX request interrupts to block the ACK signal coming from the RAW_PCS. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	TX_RATE_IRQ_EN	R/W	This bit enables TX rate interrupt. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	RX_RATE_IRQ_EN	R/W	This bit enables RX rate interrupt. **Value After Reset:** 0x1

Table 2-888 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_EN_FLAGS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	RSVD_RX_IQ_IRQ_EN	R/W	This bit is reserved; it enables RX IQ interrupts. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RXTX_REQ_IRQ_EN	R/W	This bit enables RX/TX request interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX2TX_PAR_LB_IRQ_EN	R/W	This bit enables RX2TX parallel loopback interrupts. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.885 RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ (for N = 0; N <= NLANES-1)

- **Description:** TX Termination Control Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)dd
- **Exists:** Always

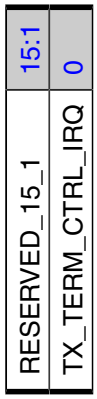


Table 2-889 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_TERM_CTRL_IRQ	R	This bit is the TX termination control interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.886 RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** TX Termination Control Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)de
- **Exists:** Always

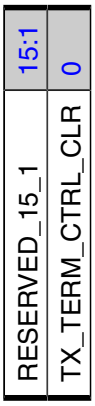


Table 2-890 Fields for Register: RAWLANEN_DIG_IRQ_CTL_TX_TERM_CTRL_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_TERM_CTRL_CLR	R/W	This bit is the TX termination control interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.887 RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX Termination Control Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)df
- **Exists:** Always

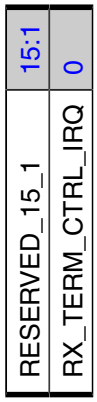


Table 2-891 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_TERM_CTRL_IRQ	R	This bit is the RX termination control interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.888 RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX Termination Control Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e0
- **Exists:** Always

Table 2-892 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_TERM_CTRL_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_TERM_CTRL_CLR	R/W	This bit is the RX termination control interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.889 RAWLANEN_DIG_IRQ_CTL_IRQ_MASK_2 (for N = 0; N <= NLANES-1)

- **Description:** Interrupt Mask 2.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e1
- **Exists:** Always

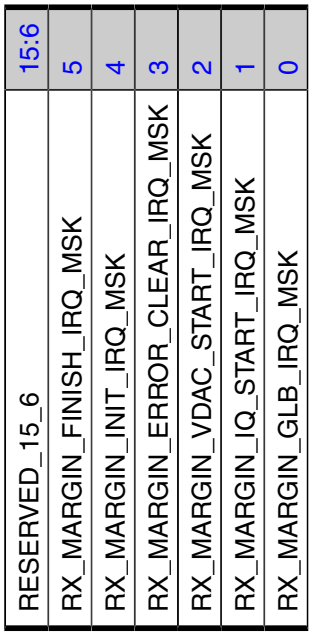


Table 2-893 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	RX_MARGIN_FINISH_IRQ_MSK	R/W	This bit is the mask for the RX margin finish interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4	RX_MARGIN_INIT_IRQ_MSK	R/W	This bit is the mask for the RX margin init interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

Table 2-893 Fields for Register: RAWLANEN_DIG_IRQ_CTL_IRQ_MASK_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3	RX_MARGIN_ERROR_CLEAR_IRQ_MSK	R/W	This bit is the mask for the RX margin error clear interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	RX_MARGIN_VDAC_START_IRQ_MSK	R/W	This bit is the mask for the RX margin vdac start interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
1	RX_MARGIN_IQ_START_IRQ_MSK	R/W	This bit is the mask for the RX margin iq start interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	RX_MARGIN_GLB_IRQ_MSK	R/W	This bit is the mask for the RX margin glb interrupt (0 = cannot interrupt). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.890 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN iq_start Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e2
- **Exists:** Always

Table 2-894 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_IQ_START_IRQ	R	This bit is the RX MARGIN iq_start interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.891 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN iq_start Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e3
- **Exists:** Always

Table 2-895 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_IQ_START_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_IQ_START_IRQ_CLR	R/W	This bit is the RX MARGIN iq_start interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.892 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN vdac_start Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e4
- **Exists:** Always

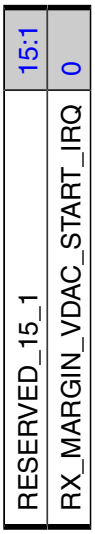


Table 2-896 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_VDAC_START_IRQ	R	This is the RX MARGIN vdac_start interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.893 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN vdac_start Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e5
- **Exists:** Always

Table 2-897 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_VDAC_START_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_VDAC_START_IRQ_CLR	R/W	This bit is the RX MARGIN vdac_start interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.894 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN Error Clear Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e6
- **Exists:** Always

Table 2-898 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_ERROR_CLEAR_IRQ	R	This bit is the RX MARGIN error clear interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.895 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN Error Clear Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e7
- **Exists:** Always

Table 2-899 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_ERROR_CLEAR_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_ERROR_CLEAR_IRQ_CLR	R/W	This bit is the RX MARGIN error clear interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.896 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN init Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e8
- **Exists:** Always

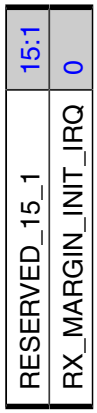


Table 2-900 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_INIT_IRQ	R	This bit is the RX MARGIN init interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.897 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN init Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)e9
- **Exists:** Always

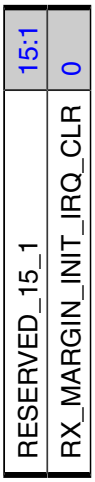


Table 2-901 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_INIT_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_INIT_IRQ_CLR	R/W	This bit is the RX MARGIN init interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.898 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN Finish Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ea
- **Exists:** Always

Table 2-902 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_FINISH_IRQ	R	This bit is the RX MARGIN finish interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.899 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN Finish Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)eb
- **Exists:** Always

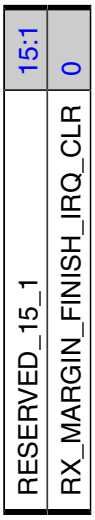


Table 2-903 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_FINISH_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_FINISH_IRQ_CLR	R/W	This bit is the RX MARGIN finish interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.900 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN glb Interrupt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ec
- **Exists:** Always

Table 2-904 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_MARGIN_GLB_IRQ	R	This bit is the RX MARGIN glb interrupt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.901 RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ_CLR (for N = 0; N <= NLANES-1)

- **Description:** RX MARGIN glb Interrupt Clear.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ed
- **Exists:** Always

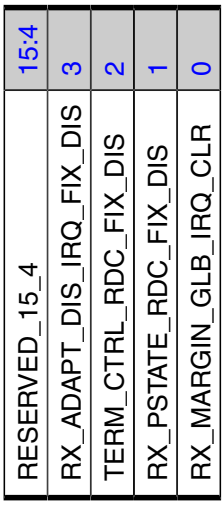


Table 2-905 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ_CLR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	RX_ADAPT_DIS_IRQ_FIX_DIS	R/W	This bit is the checken bit to disable rx_adapt_dis_irq fix (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	TERM_CTRL_RDC_FIX_DIS	R/W	This bit is the checken bit to disable term_ctrl rdc fix (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
1	RX_PSTATE_RDC_FIX_DIS	R/W	This bit is the checken bit to disable rx_pstate rdc fix (volatile) **Value After Reset:** 0x0

Table 2-905 Fields for Register: RAWLANEN_DIG_IRQ_CTL_RX_MARGIN_GLB_IRQ_CLR (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1 **Volatile:** true
0	RX_MARGIN_GLB_IRQ_CLR	R/W	This bit is the RX MARGIN glb interrupt clear (self-clearing). (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.902 RAWLANEN_DIG_IRQ_CTL_LANE_PERST_CTL (for N = 0; N <= NLANES-1)

- **Description:** PERST related control.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ee
- **Exists:** Always

15:7	6	5	4	3	2	1	0
LANE_PERST_CTL_RSVD	LANE_PERST_SCAN_MODE	PERST_TX_RESET_DIS	PERST_RX_RESET_DIS	MPLLB_CMNCAL_SELF	MPLLA_CMNCAL_SELF	RTUNE_CMNCAL_SELF	LANE_PERST_CTL_DIS

Table 2-906 Fields for Register: RAWLANEN_DIG_IRQ_CTL_LANE_PERST_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	LANE_PERST_CTL_RSVD	R/W	This bit is reserved for PERST control (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff **Volatile:** true
6	LANE_PERST_SCAN_MODE	R/W	This bit is used to set lane_perst in scan_mode (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
5	PERST_TX_RESET_DIS	R/W	This bit disables PERST trigger tx_reset (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
4	PERST_RX_RESET_DIS	R/W	This bit disables PERST trigger rx_reset (volatile) **Value After Reset:** 0x0 **Exists:** Always

Table 2-906 Fields for Register: RAWLANEN_DIG_IRQ_CTL_LANE_PERST_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x1 **Volatile:** true
3	MPLLB_CMNCAL_SELF	R/W	This bit indicates MPLLB calib is initiated by the current lane (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
2	MPLLA_CMNCAL_SELF	R/W	This bit indicates MPLLA calib is initiated by the current lane (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
1	RTUNE_CMNCAL_SELF	R/W	This bit indicates RTUNE is initiated by the current lane (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true
0	LANE_PERST_CTL_DIS	R/W	This chicken bit disables PERST control (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.903 RAWLANEN_DIG_TX_CTL_TX_FSM_CTL (for N = 0; N <= NLANES-1)

- **Description:** TX FSM Control.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)f8
- **Exists:** Always

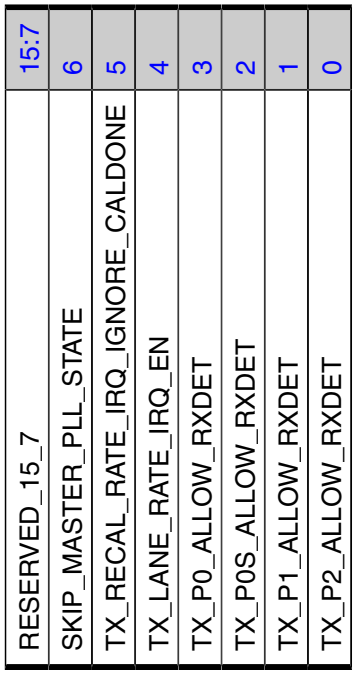


Table 2-907 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_FSM_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	SKIP_MASTER_PLL_STATE	R/W	This bit skips the master PLL state check in tx_ctl FSM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	TX_RECAL_RATE_IRQ_IGNORE_CALDONE	R/W	This bit ignores a trigger of a rate-change IRQ when tx_recal signal is updated. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-907 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_FSM_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
4	TX_LANE_RATE_IRQ_EN	R/W	This bit enables a rate-change IRQ when only the tx_rate signal is updated. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	TX_P0_ALLOW_RXDET	R/W	If asserted, then a rxdet request is allowed in P0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_P0S_ALLOW_RXDET	R/W	If asserted, then a rxdet request is allowed in P0S. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TX_P1_ALLOW_RXDET	R/W	If asserted, then a rxdet request is allowed in P1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	TX_P2_ALLOW_RXDET	R/W	If asserted, then a rxdet request is allowed in P2. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.904 RAWLANEN_DIG_TX_CTL_TX_CLK_CTL (for N = 0; N <= NLANES-1)

- **Description:** Select Clock to Act as TX Input Clock.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)f9
- **Exists:** Always

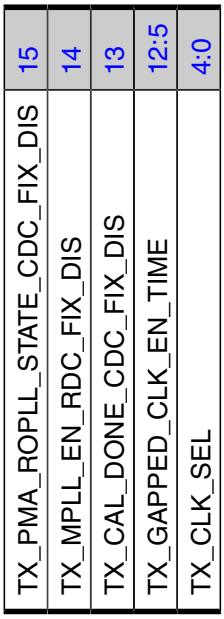


Table 2-908 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CLK_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	TX_PMA_ROPLL_STATE_CDC_FIX_DIS	R/W	This field is used as chicken bit to disable CDC fix of tx_pma_ropll_state **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	TX_MPLL_EN_RDC_FIX_DIS	R/W	This field is used as chicken bit to disable RDC fix of tx_mpll_en **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	TX_CAL_DONE_CDC_FIX_DIS	R/W	This field is used as chicken bit to disable CDC fix of tx_cal_done **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12:5	TX_GAPPED_CLK_EN_TIME	R/W	This field is the Eth TX gapped clock enable timer. **Value After Reset:** 0xf

Table 2-908 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CLK_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0xff
4:0	TX_CLK_SEL	R/W	This field is the select clock source for tx_pma_clk. ■ 0 - tx_pcs_clk (input clock from pcs) ■ 1 - mplla_word_clk ■ 2 - mplla_dword_clk ■ 3 - mplla_qword_clk ■ 4 - mplla_oword_clk ■ 5 - tx_ropll_125mhz_clk ■ 6 - tx_ropll_div33_clk ■ 7 - tx_ropll_div66_clk ■ 8 - tx_ropll_div_clk ■ 9 - mpllb_word_clk ■ 10 - mpllb_dword_clk ■ 11 - mpllb_qword_clk ■ 12 - mpllb_oword_clk ■ 13 - na ■ 14 - na ■ 15 - na ■ 16 - na ■ 17 - rx_clk (from pma) ■ 18 - tx_ropll_word_clk ■ 19 - tx_ropll_dword_clk ■ 20 - tx_ropll_qword_clk ■ 21 - tx_ropll_oword_clk ■ 22 - rx_div_clk (from pma) ■ 23 - tx_dig_ropll_div_clk ■ 24 - rx_dig_div_clk **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.905 RAWLANEN_DIG_TX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Continuous Calibration Status.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)fa
- **Exists:** Always

Table 2-909 Fields for Register: RAWLANEN_DIG_TX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ENABLE	R	This bit is the enable status for TX DCC continuous cancellation. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.906 RAWLANEN_DIG_TX_CTL_TX_RATE_IRQ_ACK (for N = 0; N <= NLANES-1)

- **Description:** TX FSM Rate IRQ ACK Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)fb
- **Exists:** Always

Table 2-910 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_RATE_IRQ_ACK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	TX_RATE_IRQ_ACK	R/W	This bit acknowledges the rate IRQ. This bit is self-clearing. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.907 RAWLANEN_DIG_TX_CTL_REQ_TX_PLL_EN_CTL (for N = 0; N <= NLANES-1)

- **Description:** TX PLL Enable override register to be used by FW
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)fc
- **Exists:** Always

Table 2-911 Fields for Register: RAWLANEN_DIG_TX_CTL_REQ_TX_PLL_EN_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	REQ_TX_PLL_EN_OVRD_VAL	R/W	The override enable value (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
0	REQ_TX_PLL_EN_OVRD_EN	R/W	The override enable signal (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.908 RAWLANEN_DIG_TX_CTL_TX_CTL_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Tx_ctl status
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)fd
- **Exists:** Always

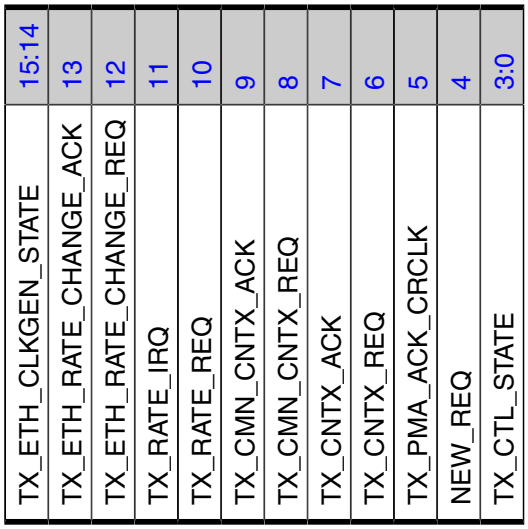


Table 2-912 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CTL_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	TX_ETH_CLKGEN_STATE	R	tx_eth_clkgen FSM state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13	TX_ETH_RATE_CHANGE_ACK	R	tx_eth_rate_change_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	TX_ETH_RATE_CHANGE_REQ	R	tx_eth_rate_change_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	TX_RATE_IRQ	R	tx_rate_irq (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-912 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CTL_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
10	TX_RATE_REQ	R	tx_rate_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	TX_CMN_CNTX_ACK	R	tx_cmn_cntx_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	TX_CMN_CNTX_REQ	R	tx_cmn_cntx_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	TX_CNTX_ACK	R	tx_cntx_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	TX_CNTX_REQ	R	tx_cntx_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	TX_PMA_ACK_CRCLK	R	tx_pma_ack_crclk (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	NEW_REQ	R	new_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-912 Fields for Register: RAWLANEN_DIG_TX_CTL_TX_CTL_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
3:0	TX_CTL_STATE	R	tx_ctl FSM state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.909 RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_LEFT (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Left Code.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+0)ff
- **Exists:** Always

Table 2-913 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_LEFT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	RX_ADPT_IQ_LEFT	R/W	This field is RX adapted value of left code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.910 RAWLANEN_DIG_RX_CTL_RX_SIGDET_MASK_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX SIGDET Mask Control.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)00
- **Exists:** Always

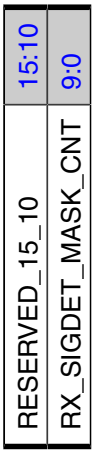


Table 2-914 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_SIGDET_MASK_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	RX_SIGDET_MASK_CNT	R/W	This field is the number of cycles (ref_range_clk) to mask out the rx_sigdet/lfpsdet outputs, from the time the sigdet is powered on. The default is set for a minimum of 12us (after scaling). **Value After Reset:** 0x138 **Exists:** Always **Reset Mask:** 0x3ff

2.1.911 RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_RIGHT (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Right Code.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)01
- **Exists:** Always

Table 2-915 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_IQ_RIGHT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	RX_ADPT_IQ_RIGHT	R/W	This field is RX adapted value of right code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.912 RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)

- **Description:** RX continuous offset cancellation status.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)02
- **Exists:** Always

Table 2-916 Fields for Register: RAWLANEN_DIG_RX_CTL_OFFCAN_CONT_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ENABLE	R	This bit is the enable status for the RX continuous offset cancellation. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.913 RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS (for N = 0; N <= NLANES-1)

- **Description:** RX Continuous Adaptation Status.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)03
- **Exists:** Always

Table 2-917 Fields for Register: RAWLANEN_DIG_RX_CTL_ADAPT_CONT_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	ENABLE	R	This bit is the enable status for RX continuous adaptation. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.914 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Mode.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)04
- **Exists:** Always

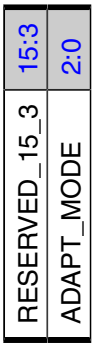


Table 2-918 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	ADAPT_MODE	R	This field is the RX adaptation mode. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.915 RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL (for N = 0; N <= NLANES-1)

- **Description:** Select Between Two Banks Adaptation Settings for RX Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)05
- **Exists:** Always

Table 2-919 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	RX_ADAPT_SEL	R	This field selects between two banks adaptation settings for RX adaptation. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.916 RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT (for N = 0; N <= NLANES-1)

- **Description:** RX CDR PPM Drift on RX Clock.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)06
- **Exists:** Always

Table 2-920 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PPM_DRIFT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	RX_PPM_DRIFT_VLD	R	This bit indicates if the PPM Drift is valid. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
13:0	RX_PPM_DRIFT	R	This field indicates the amount of drift on RX clock when RX_PPM_DRIFT_VLD is asserted. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.917 RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS (for N = 0; N <= NLANES-1)

- **Description:** RX Misc Status Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)07
- **Exists:** Always

Table 2-921 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:6	ETH_CLKGEN_STATE	R	This field indicates the current state for the state machine in ETH Clk Gen. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5:3	COARSE_ADPT_STATE	R	This field indicates the current state for the state machine in Coarse Adapt Ctl. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2:0	CDR_DET_STATE	R	This field indicates the current state for the state machine in CDR Detector. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-921 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CDR_DET_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true

2.1.918 RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL (for N = 0; N <= NLANES-1)

- **Description:** Values for Outgoing RX Controls to PMA.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)08
- **Exists:** Always

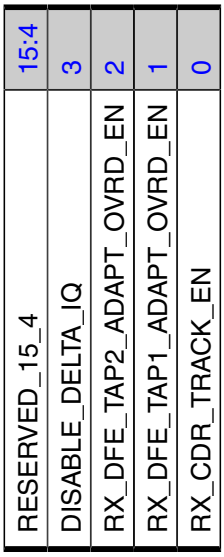


Table 2-922 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	DISABLE_DELTA_IQ	R/W	This bit disables adding the delta_iq value to the PMA. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_DFE_TAP2_ADAPT_OVRD_EN	R/W	This bit enables the override of DFE TAP2 to zero during RX Adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_DFE_TAP1_ADAPT_OVRD_EN	R/W	This bit enables the override of DFE TAP1 to zero during RX Adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-922 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PMA_MISC_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	RX_CDR_TRACK_EN	R/W	This bit is the value for rx_cdr_track_en to PMA. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.919 RAWLANEN_DIG_RX_CTL_RX_CTL_STATUS (for N = 0; N <= NLANES-1)

- **Description:** Rx_ctl status
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)09
- **Exists:** Always

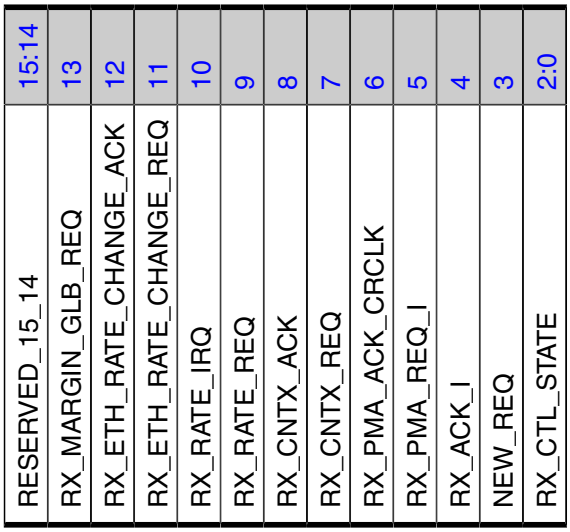


Table 2-923 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CTL_STATUS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13	RX_MARGIN_GLB_REQ	R	rx_margin_glb_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
12	RX_ETH_RATE_CHANGE_ACK	R	rx_eth_rate_change_ack (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11	RX_ETH_RATE_CHANGE_REQ	R	rx_eth_rate_change_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-923 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CTL_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
10	RX_RATE_IRQ	R	rx_rate_irq (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9	RX_RATE_REQ	R	rx_rate_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
8	RX_CNTX_ACK	R	rx_cntx_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7	RX_CNTX_REQ	R	rx_cntx_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
6	RX_PMA_ACK_CRCLK	R	rx_pma_ack_crclk (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
5	RX_PMA_REQ_I	R	rx_pma_req_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
4	RX_ACK_I	R	rx_ack_i (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always

Table 2-923 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_CTL_STATUS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Reset Mask: 0x0 **Volatile:** true
3	NEW_REQ	R	new_req (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
2:0	RX_CTL_STATE	R	current FSM state (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.920 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN (for N = 0; N <= NLANES-1)

- **Description:** Enable RX Adapt Mode Override (using Custom Setting for RX Adaptation).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0a
- **Exists:** Always

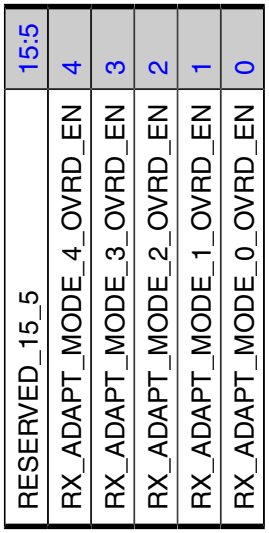


Table 2-924 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4	RX_ADAPT_MODE_4_OVRD_EN	R/W	This bit enables the RX adapt mode override when adapt_mode is 3'b100. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	RX_ADAPT_MODE_3_OVRD_EN	R/W	This bit enables the RX adapt mode override when adapt_mode is 3'b011. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	RX_ADAPT_MODE_2_OVRD_EN	R/W	This bit enables the RX adapt mode override when adapt_mode is 3'b010. **Value After Reset:** 0x0

Table 2-924 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_OVRD_EN (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
1	RX_ADAPT_MODE_1_OVRD_EN	R/W	This bit enables the RX adapt mode override when adapt_mode is 3'b001. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ADAPT_MODE_0_OVRD_EN	R/W	This bit enables the RX adapt mode override when adapt_mode is 3'b000. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.921 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN (for N = 0; N <= NLANES-1)

- **Description:** Enable Using Four Different RX Adaptation Modes.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0b
- **Exists:** Always

Table 2-925 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MODE_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_MODE_EN	R	This bit enables using four different RX adaptation modes. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.922 RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Mission-Mode Figure of Merit.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0c
- **Exists:** Always

Table 2-926 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_MM_FOM (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	RX_ADAPT_MM_FOM	R/W	This field is the RX adaptation mission-mode figure of merit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.923 RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Start-Up Figure of Merit (Base Mode or Extended Mode).
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0d
- **Exists:** Always

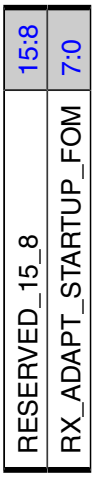


Table 2-927 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADAPT_STARTUP_FOM (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	RX_ADAPT_STARTUP_FOM	R/W	This bit is the RX adaptation start-up figure of merit (base mode or extended mode). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.924 RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Even Slicer.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0e
- **Exists:** Always

Table 2-928 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_EVEN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	RX_ADPT_REF_ERR_EVEN	R/W	This field is the RX adapted value of reference level for error even slicer for Bank 0. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.925 RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Odd Slicer.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)0f
- **Exists:** Always

Table 2-929 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_ADPT_REF_ERR_ODD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	RX_ADPT_REF_ERR_ODD	R/W	This field is the RX adapted value of reference level for the error odd slicer. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.926 RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN (for N = 0; N <= NLANES-1)

- **Description:** RX Phase Adjust Linear Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)10
- **Exists:** Always

Table 2-930 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	RX_PHSADJ_LIN	R	This field is the linear value for the RX phase adjust. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.927 RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP (for N = 0; N <= NLANES-1)

- **Description:** RX Phase Adjust Mapped Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)11
- **Exists:** Always

Table 2-931 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	RX_PHSADJ_MAP	R	This field is the mapped value for the RX phase adjust. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.928 RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN_UPDATE (for N = 0; N <= NLANES-1)

- **Description:** RX Phase Adjust Linear Update Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)12
- **Exists:** Always

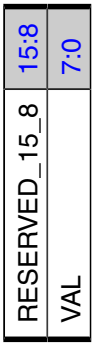


Table 2-932 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PHSADJ_LIN_UPDATE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	This field is the RX phase adjust linear update value. Firmware can use this register to translate between linear IQ codes and adjust bus IQ codes. **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff

2.1.929 RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP_UPDATE (for N = 0; N <= NLANES-1)

- **Description:** RX Phase Adjust Mapped Update Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)13
- **Exists:** Always

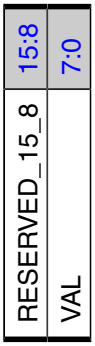


Table 2-933 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_PHSADJ_MAP_UPDATE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R	This field is the mapped value for the RX phase adjust. Firmware can use this register to translate between linear IQ codes and adjust bus IQ codes. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.930 RAWLANEN_DIG_RX_CTL_RX_MARGIN_IQ_DELTA (for N = 0; N <= NLANES-1)

- **Description:** RX Margin IQ Delta Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)14
- **Exists:** Always

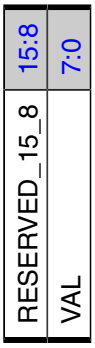


Table 2-934 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_MARGIN_IQ_DELTA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R	This field is the RX margin IQ delta value. This register is currently reserved. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.931 RAWLANEN_DIG_RX_CTL_RX_FSM_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX FSM Control.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)15
- **Exists:** Always

Table 2-935 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_FSM_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_RECAL_RATE_IRQ_IGNORE_CALDONE	R/W	This bit ignores a trigger of a rate-change IRQ when the rx recal signal is updated. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_LANE_RATE_IRQ_EN	R/W	This bit enables a rate-change IRQ when only the rx_rate signal is updated. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.932 RAWLANEN_DIG_RX_CTL_RX_RATE_IRQ_ACK (for N = 0; N <= NLANES-1)

- **Description:** RX FSM Rate IRQ ACK Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)16
- **Exists:** Always

Table 2-936 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_RATE_IRQ_ACK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_RATE_IRQ_ACK	R/W	This bit acknowledges the rate IRQ. This bit is self-clearing. (volatile) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1 **Volatile:** true

2.1.933 RAWLANEN_DIG_RX_CTL_DFE_EEH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for DFE Error Even High Slicer in Mission Mode.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)17
- **Exists:** Always

Table 2-937 Fields for Register: RAWLANEN_DIG_RX_CTL_DFE_EEH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_EEH_VDAC_OFST	R/W	This field is the value for the DFE error even high slicer in mission mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.934 RAWLANEN_DIG_RX_CTL_DFE_EOH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for DFE Error Odd High Slicer in Mission Mode.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)18
- **Exists:** Always

Table 2-938 Fields for Register: RAWLANEN_DIG_RX_CTL_DFE_EOH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_EOH_VDAC_OFST	R/W	This field is the value for the DFE error odd high slicer in mission mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.935 RAWLANEN_DIG_RX_CTL_DFE_EEL_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for DFE Error Even Low Slicer in Mission Mode.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)19
- **Exists:** Always

Table 2-939 Fields for Register: RAWLANEN_DIG_RX_CTL_DFE_EEL_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_EEL_VDAC_OFST	R/W	This field is the value for the DFE error even low slicer in mission mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.936 RAWLANEN_DIG_RX_CTL_DFE_EOL_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for DFE Error Odd Low Slicer in Mission Mode.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)1a
- **Exists:** Always

Table 2-940 Fields for Register: RAWLANEN_DIG_RX_CTL_DFE_EOL_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DFE_EOL_VDAC_OFST	R/W	This field is the value for the DFE error odd low slicer in mission mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1ff

2.1.937 RAWLANEN_DIG_RX_CTL_SETUP_REF_SCO_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for AFE SCO IDAC During Reference-Level Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)1b
- **Exists:** Always

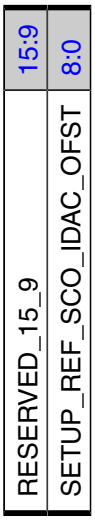


Table 2-941 Fields for Register: RAWLANEN_DIG_RX_CTL_SETUP_REF_SCO_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	SETUP_REF_SCO_IDAC_OFST	R/W	This field is the offset value for AFE CTLE SCO during reference-level calibration. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.938 RAWLANEN_DIG_RX_CTL_SETUP_REF_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for AFE VGA2 IDAC During Reference-Level Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)1c
- **Exists:** Always

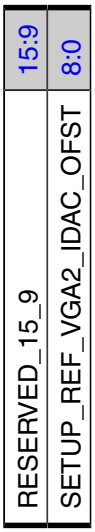


Table 2-942 Fields for Register: RAWLANEN_DIG_RX_CTL_SETUP_REF_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	SETUP_REF_VGA2_IDAC_OFST	R/W	This field is the offset value for AFE CTLE VGA2 during reference-level calibration. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.939 RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA1_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for AFE VGA1 for Slicer Set-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)1d
- **Exists:** Always

Table 2-943 Fields for Register: RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA1_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	SETUP_SLC_VGA1_IDAC_OFST	R/W	This field is the offset value for AFE VGA1 for slicer set-up calibration. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.940 RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for AFE VGA2 for Slicer Set-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)1e
- **Exists:** Always

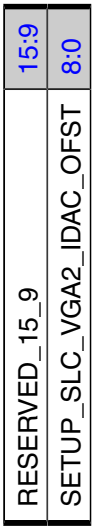


Table 2-944 Fields for Register: RAWLANEN_DIG_RX_CTL_SETUP_SLC_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	SETUP_SLC_VGA2_IDAC_OFST	R/W	This field is the offset value for AFE VGA2 for slicer set-up calibration. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.941 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP0_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 0.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)20
- **Exists:** Always

Table 2-945 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP0_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap0_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.942 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP1_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)21
- **Exists:** Always

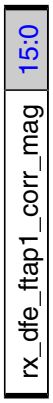


Table 2-946 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP1_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap1_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.943 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP2_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 2.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)22
- **Exists:** Always

Table 2-947 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP2_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap2_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.944 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP3_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 3.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)23
- **Exists:** Always

Table 2-948 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP3_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap3_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.945 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP4_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 4.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)24
- **Exists:** Always

Table 2-949 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP4_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap4_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.946 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP5_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 5.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)25
- **Exists:** Always

Table 2-950 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP5_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap5_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.947 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP6_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 6.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)26
- **Exists:** Always

Table 2-951 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP6_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap6_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 6. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.948 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP7_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 7.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)27
- **Exists:** Always

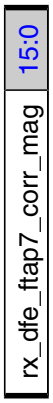


Table 2-952 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP7_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap7_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 7. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.949 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP8_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 8.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)28
- **Exists:** Always

Table 2-953 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP8_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap8_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 8. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.950 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP9_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 9.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)29
- **Exists:** Always

Table 2-954 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP9_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap9_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 9. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.951 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP10_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 10.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2a
- **Exists:** Always

Table 2-955 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP10_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap10_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 10. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.952 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP11_CORR_MAG (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Correlation Magnitude Sorting Input 11.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2b
- **Exists:** Always

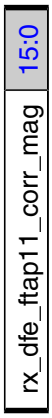


Table 2-956 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP11_CORR_MAG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	rx_dfe_ftap11_corr_mag	R/W	This field is the RX DFE floating TAP correlation magnitude sorting input 11. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.953 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Sorting Control.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2c
- **Exists:** Always

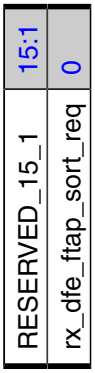


Table 2-957 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	rx_dfe_ftap_sort_req	R/W	This bit is the RX DFE floating TAP sorting request. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.954 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_STATE (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Sorting ACK.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2d
- **Exists:** Always

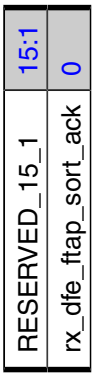


Table 2-958 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_STATE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	rx_dfe_ftap_sort_ack	R	This bit is the RX DFE floating TAP sorting ACK . (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.955 RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_RESULT (for N = 0; N <= NLANES-1)

- **Description:** RX DFE Floating TAP Sorting Result.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2e
- **Exists:** Always

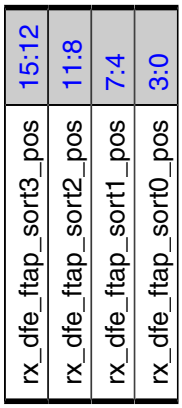


Table 2-959 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_RESULT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	rx_dfe_ftap_sort3_pos	R	This field is the RX DFE floating TAP sorting position 3. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
11:8	rx_dfe_ftap_sort2_pos	R	This field is the RX DFE floating TAP sorting position 2. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:4	rx_dfe_ftap_sort1_pos	R	This field is the RX DFE floating TAP sorting position 1. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

Table 2-959 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_DFE_FTAP_SORT_RESULT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
3:0	rx_dfe_ftap_sort0_pos	R	This field is the RX DFE floating TAP sorting position 0. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.956 RAWLANEN_DIG_RX_CTL_RX_MARGIN_VDAC_DELTA (for N = 0; N <= NLANES-1)

- **Description:** RX Margin VDAC Delta Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)2f
- **Exists:** Always

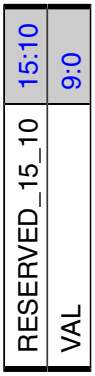


Table 2-960 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_MARGIN_VDAC_DELTA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VAL	R	This field is the RX margin VDAC delta value. This register is currently reserved. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.957 RAWLANEN_DIG_RX_CTL_RX_MARGIN_ERROR (for N = 0; N <= NLANES-1)

- **Description:** RX Margin Error.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)30
- **Exists:** Always

Table 2-961 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_MARGIN_ERROR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	RX_MARGIN_ERROR	R/W	This field is the RX margin error. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.958 RAWLANEN_DIG_RX_CTL_RX_VALID_COARSE_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** RX Valid Coarse Adapt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)31
- **Exists:** Always

Table 2-962 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_VALID_COARSE_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_VALID_COARSE_ADAPT	R/W	This bit is the RX valid coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.959 RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_WR (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Linear Code Input Register for Conversion.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)32
- **Exists:** Always

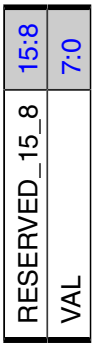


Table 2-963 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_WR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	This field is the input of RX IQ linear code conversion. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.960 RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_RD (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Step Code Readback Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)33
- **Exists:** Always

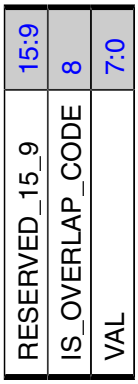


Table 2-964 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_RD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	IS_OVERLAP_CODE	R	This bit signals whether the rx_iq_lin_code_wr code is an overlap code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	VAL	R	This field is the readback value for RX IQ step code conversion. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.961 RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_WR (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Step Code Input Register for Conversion.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)34
- **Exists:** Always

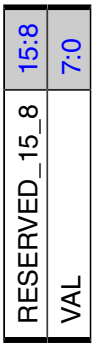


Table 2-965 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_IQ_STEP_CODE_WR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	This field is the input of the RX IQ step code conversion. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.962 RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_RD (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Linear Code Readback Value.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)35
- **Exists:** Always

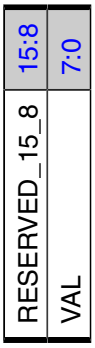


Table 2-966 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_IQ_LIN_CODE_RD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R	This field is the readback value for the RX IQ linear code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.963 RAWLANEN_DIG_RX_CTL_FOM_CTL (for N = 0; N <= NLANES-1)

- **Description:** FOM CTL Setting.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)36
- **Exists:** Always

Table 2-967 Fields for Register: RAWLANEN_DIG_RX_CTL_FOM_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FOM_AFE_EYEH	R/W	This field is the FOM AFE eye height. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
7:0	FOM_STARTUP_EXTEND	R/W	This field is the FOM start-up extended. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.964 RAWLANEN_DIG_RX_CTL_RX_MARGIN_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX Margin CTL.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)37
- **Exists:** Always

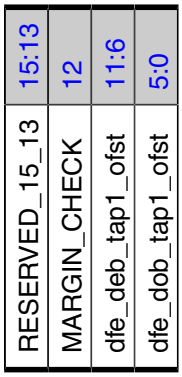


Table 2-968 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_MARGIN_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	MARGIN_CHECK	R/W	This bit enables and disables the error check. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11:6	dfe_deb_tap1_ofst	R/W	This field is the signed DEB TAP1 offset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f
5:0	dfe_dob_tap1_ofst	R/W	This field is the signed DOB TAP1 offset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.965 RAWLANEN_DIG_RX_CTL_PRE_EE_DAC_CODE (for N = 0; N <= NLANES-1)

- **Description:** EE Precursor Measurement Code.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)38
- **Exists:** Always

Table 2-969 Fields for Register: RAWLANEN_DIG_RX_CTL_PRE_EE_DAC_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	PRE_EE_DAC_CODE	R/W	This field is the EE precursor measurement code. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.966 RAWLANEN_DIG_RX_CTL_PRE_EO_DAC_CODE (for N = 0; N <= NLANES-1)

- **Description:** EO Precursor Measurement Code.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)39
- **Exists:** Always

Table 2-970 Fields for Register: RAWLANEN_DIG_RX_CTL_PRE_EO_DAC_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	PRE_EO_DAC_CODE	R/W	This field is the EO precursor measurement code. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.967 RAWLANEN_DIG_RX_CTL_SETUP_SLC_CTLE_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for AFE CTLE for Slicer Set-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)3a
- **Exists:** Always

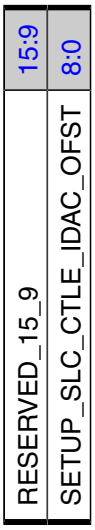


Table 2-971 Fields for Register: RAWLANEN_DIG_RX_CTL_SETUP_SLC_CTLE_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	SETUP_SLC_CTLE_IDAC_OFST	R/W	This field is the offset value for AFE CTLE for slicer set-up calibration. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.968 RAWLANEN_DIG_RX_CTL_RX_GAPPED_CLK_EN_TIME (for N = 0; N <= NLANES-1)

- **Description:** Eth RX Gapped Clock Enable Timer.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)3b
- **Exists:** Always

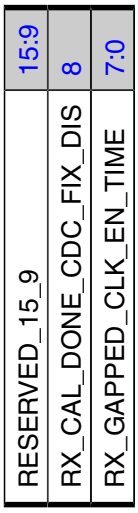


Table 2-972 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_GAPPED_CLK_EN_TIME (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	RX_CAL_DONE_CDC_FIX_DIS	R/W	This field is used as chicken bit to disable CDC fix of rx_cal_done **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7:0	RX_GAPPED_CLK_EN_TIME	R/W	This field is the Eth RX gapped clock enable timer. **Value After Reset:** 0xf **Exists:** Always **Reset Mask:** 0xff

2.1.969 RAWLANEN_DIG_RX_CTL_RX_SIGDET_FILT_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX SIGDET LF/HF Filtering Control
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)3c
- **Exists:** Always

Table 2-973 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_SIGDET_FILT_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RX_SIGDET_LF_HOLD	R/W	When this signal is set to 0, the sigdet LF output is gated to 1'b0 when sigdet HF after filtering is 1'b1 and sigdet HF filter is enabled **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
9:5	RX_SIGDET_HF_FILT_CNT	R/W	The number of consecutive of sigdet HF output symbols to consider it a new value **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0x1f
4:0	RX_SIGDET_LF_FILT_CNT	R/W	The number of consecutive of sigdet LF output symbols to consider it a new value. The filtered version is part of the sigdet HF filtering mechanism **Value After Reset:** 0x7

Table 2-973 Fields for Register: RAWLANEN_DIG_RX_CTL_RX_SIGDET_FILT_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1f

2.1.970 RAWLANEN_DIG_FSM_FSM_OVRD_CTL (for N = 0; N <= NLANES-1)

- **Description:** FSM Override Control Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)80
- **Exists:** Always

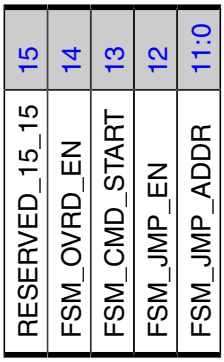


Table 2-974 Fields for Register: RAWLANEN_DIG_FSM_FSM_OVRD_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	FSM_OVRD_EN	R/W	This bit enables overriding the FSM execution of commands. This bit must be asserted to use FSM_CMD_START and FSM_JMP_EN features. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	FSM_CMD_START	R/W	This bit executes the current command or resume from breakpoint. This is a self-clearing bit. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	FSM_JMP_EN	R/W	This bit forces the FSM to jump to FSM_JMP_ADDR in the program memory. This bit is applied when FSM_CMD_START is pulsed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-974 Fields for Register: RAWLANEN_DIG_FSM_FSM_OVRD_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11:0	FSM_JMP_ADDR	R/W	These bits are the jump address used when FSM_JUMP_EN=1. The address is encoded as follows: ■ [11:8]: mem_lane ■ [7:5]: bank ■ [4:0]: register **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.971 RAWLANEN_DIG_FSM_FSM_JMP_BANK (for N = 0; N <= NLANES-1)

- **Description:** FSM Jump Bank Register.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)81
- **Exists:** Always

Table 2-975 Fields for Register: RAWLANEN_DIG_FSM_FSM_JMP_BANK (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	ENABLE_CREG_MASK_AUTO_RST	R/W	Enable creg rd/wr mask HW aoto reset function **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2:0	BANK	R/W	These bits are the jump bank used when FSM_JUMP_EN=1. This bank needs to be updated accordingly along with FSM_OVRD_CTL.FSM_JMP_ADDR. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.972 RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0 (for N = 0; N <= NLANES-1)

- **Description:** FSM Breakpoint 0 on SRAM Address for Debugging.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)82
- **Exists:** Always

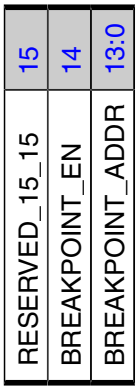


Table 2-976 Fields for Register: RAWLANEN_DIG_FSM_MEM_BREAKPOINT_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	BREAKPOINT_EN	R/W	This bit enables the Breakpoint 0. Clearing this bit automatically resumes the FSM if it is stopped at MEM_BREAKPOINT_0.BREAKPOINT_ADDR. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:0	BREAKPOINT_ADDR	R/W	These bits are the Memory Address for Breakpoint 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.973 RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1 (for N = 0; N <= NLANES-1)

- **Description:** FSM Breakpoint 1 on SRAM Address for Debugging.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)83
- **Exists:** Always

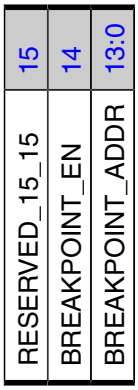


Table 2-977 Fields for Register: RAWLANEN_DIG_FSM_MEM_BREAKPOINT_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	BREAKPOINT_EN	R/W	This bit enables the Breakpoint 1. Clearing this bit automatically resumes the FSM if it is stopped at MEM_BREAKPOINT_1.BREAKPOINT_ADDR **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:0	BREAKPOINT_ADDR	R/W	These bits are the Memory Address for Breakpoint 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.974 RAWLANEN_DIG_FSM_SKIP_RX_MARGINING (for N = 0; N <= NLANES-1)

- **Description:** Status of Skip RX Margining.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)84
- **Exists:** Always

Table 2-978 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_MARGINING (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_MARGINING	R	This bit is the status of skip RX margining. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.975 RAWLANEN_DIG_FSM_MEM_ADDR_MON (for N = 0; N <= NLANES-1)

- **Description:** Memory Address Monitor.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)85
- **Exists:** Always

Table 2-979 Fields for Register: RAWLANEN_DIG_FSM_MEM_ADDR_MON (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	MEM_ADDR	R	These bits are the current value of memory address used in Lane FSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.976 RAWLANEN_DIG_FSM_STATUS_MON (for N = 0; N <= NLANES-1)

- **Description:** FSM Status Monitor.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)86
- **Exists:** Always

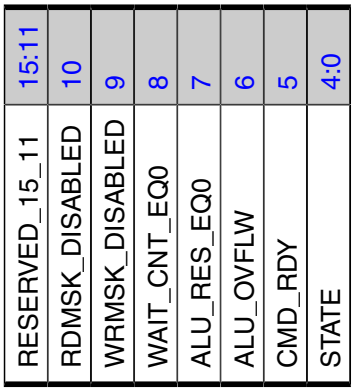


Table 2-980 Fields for Register: RAWLANEN_DIG_FSM_STATUS_MON (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	RDMSK_DISABLED	R	This bit checks if the read mask is currently disabled (for instance, mask is all ones). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	WRMSK_DISABLED	R	This bit checks if the write mask is currently disabled (for example, mask is all ones). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	WAIT_CNT_EQ0	R	This bit checks if the wait counter currently equals zero. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7	ALU_RES_EQ0	R	This bit checks if the ALU result register currently equals zero. (actual reset value is 0xX) **Value After Reset:** 0x0

Table 2-980 Fields for Register: RAWLANEN_DIG_FSM_STATUS_MON (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x0
6	ALU_OVFLW	R	This bit is the current value of ALU overflow bit. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5	CMD_RDY	R	The new command is ready for execution (applicable when FSM_OVRD_EN=1). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4:0	STATE	R	These bits are the current state of Lane FSM. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.977 RAWLANEN_DIG_FSM_FAST_RX_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)87
- **Exists:** Always

Table 2-981 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_STARTUP_CAL	R	This bit is the status of fast the RX start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.978 RAWLANEN_DIG_FSM_FAST_RX_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)88
- **Exists:** Always

Table 2-982 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_ADAPT	R	This bit is the status of fast RX adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.979 RAWLANEN_DIG_FSM_FAST_SUP (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast Support Block.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)89
- **Exists:** Always

Table 2-983 Fields for Register: RAWLANEN_DIG_FSM_FAST_SUP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_SUP	R	This bit is the status of fast support block (MPLL and Rtune). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.980 RAWLANEN_DIG_FSM_FAST_TX_CMN_MODE (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast TX Common-Mode Charge Up.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8a
- **Exists:** Always

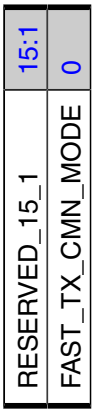


Table 2-984 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_CMN_MODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_TX_CMN_MODE	R	This bit is the status of fast TX common-mode charge up. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.981 RAWLANEN_DIG_FSM_FAST_TX_RXDET (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast TX Detect RX.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8b
- **Exists:** Always

Table 2-985 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_RXDET (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_TX_RXDET	R	This bit is the status of fast TX detect RX. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.982 RAWLANEN_DIG_FSM_FAST_RX_PWRUP (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Power Up.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8c
- **Exists:** Always

Table 2-986 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_PWRUP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_PWRUP	R	This bit is the status of fast RX power up (LOS, VREG/AFE and DCC). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.983 RAWLANEN_DIG_FSM_FAST_RX_VCO_WAIT (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX VCO Wait Times.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8d
- **Exists:** Always

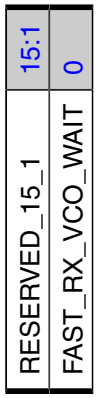


Table 2-987 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_VCO_WAIT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_VCO_WAIT	R	This bit is the status of fast RX VCO wait times. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.984 RAWLANEN_DIG_FSM_FAST_RX_VCO_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX VCO Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8e
- **Exists:** Always

Table 2-988 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_VCO_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_VCO_CAL	R	This bit is the status of fast RX VCO calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.985 RAWLANEN_DIG_FSM_FAST_RX_CONT_CAL_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Continuous Calibration/Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)8f
- **Exists:** Always

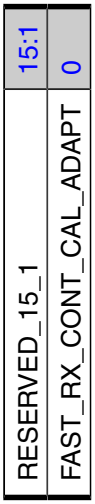


Table 2-989 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_CONT_CAL_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_CONT_CAL_ADAPT	R	This bit is the status of fast RX continuous calibration/adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.986 RAWLANEN_DIG_FSM_FAST_RX_CONT_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Continuous Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)90
- **Exists:** Always

Table 2-990 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_CONT_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_CONT_ADAPT	R	This bit is the status of fast RX continuous adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.987 RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX AFE Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)91
- **Exists:** Always

Table 2-991 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_AFE_STARTUP_CAL	R	This bit is the status of RX AFE start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.988 RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DFE Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)92
- **Exists:** Always

Table 2-992 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DFE_STARTUP_CAL	R	This bit is the status of RX DFE start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.989 RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of DFE Extended Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)93
- **Exists:** Always

Table 2-993 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_EXT_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DFE_EXT_STARTUP_CAL	R	This bit is the status of DFE extended start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.990 RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELTA_STARTUP (for N = 0; N <= NLANES-1)

- **Description:** Status of RX IQ Fixed Offset.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)94
- **Exists:** Always

Table 2-994 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_DELTA_STARTUP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_IQ_DELTA_STARTUP	R	This bit is the status of RX IQ fixed offset. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.991 RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX IQ Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)95
- **Exists:** Always

Table 2-995 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_IQ_STARTUP_CAL	R	This bit is the status of RX IQ start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.992 RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX AFE Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)96
- **Exists:** Always

Table 2-996 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_AFE_STARTUP_ADAPT	R	This bit is the status of RX AFE DAC start-up adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.993 RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DFE Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)97
- **Exists:** Always

Table 2-997 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DFE_STARTUP_ADAPT	R	This bit is the status of RX DFE DAC start-up adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.994 RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX IO Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)98
- **Exists:** Always

Table 2-998 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_IQ_STARTUP_ADAPT	R	This bit is the status of RX IQ start-up adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.995 RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Continuous Phase Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)99
- **Exists:** Always

Table 2-999 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_CONT_PHASE_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_CONT_PHASE_CAL	R	This bit is the status of RX continuous phase calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.996 RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX AFE Continuous Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9a
- **Exists:** Always

Table 2-1000 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_AFE_CONT_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_AFE_CONT_CAL	R	This bit is the status of RX AFE continuous calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.997 RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Reference Level Continuous Adaptation .
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9b
- **Exists:** Always

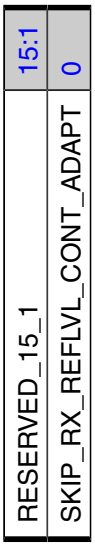


Table 2-1001 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_REFLVL_CONT_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_REFLVL_CONT_ADAPT	R	This bit is the status of RX reference level continuous adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.998 RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX VGA Continuous Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9c
- **Exists:** Always

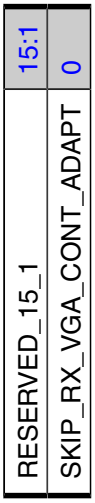


Table 2-1002 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_VGA_CONT_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_VGA_CONT_ADAPT	R	This bit is the status of RX VGA continuous adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.999 RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Phase Start-up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9d
- **Exists:** Always

Table 2-1003 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_PHS_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_PHS_STARTUP_CAL	R	This bit is the status of RX Phase start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1000 RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Phase Extended Start-up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9e
- **Exists:** Always

Table 2-1004 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_PHS_EXT_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_PHS_EXT_STARTUP_CAL	R	This bit is the status of RX Phase extended start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1001 RAWLANEN_DIG_FSM_SKIP_RX_DCC_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)9f
- **Exists:** Always

Table 2-1005 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_STARTUP_CAL	R	This bit is the status of RX DCC start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1002 RAWLANEN_DIG_FSM_SKIP_RX_DCC_CONT_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC Continuous Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a0
- **Exists:** Always

Table 2-1006 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_CONT_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_CONT_CAL	R	This bit is the status of RX DCC continuous calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1003 RAWLANEN_DIG_FSM_SKIP_TX_DCC_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of TX DCC Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a1
- **Exists:** Always

Table 2-1007 Fields for Register: RAWLANEN_DIG_FSM_SKIP_TX_DCC_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_TX_DCC_STARTUP_CAL	R	This bit is the status of TX DCC start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1004 RAWLANEN_DIG_FSM_SKIP_TX_DCC_CONT_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of TX DCC Continuous Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a2
- **Exists:** Always

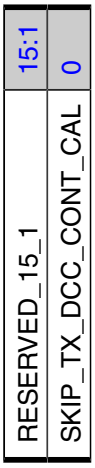


Table 2-1008 Fields for Register: RAWLANEN_DIG_FSM_SKIP_TX_DCC_CONT_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_TX_DCC_CONT_CAL	R	This bit is the status of TX DCC continuous calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1005 RAWLANEN_DIG_FSM_SKIP_RX_DCC_RATE_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC Rate-Change Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a3
- **Exists:** Always

Table 2-1009 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_RATE_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_RATE_CAL	R	This bit is the status of RX_DCC rate-change calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1006 RAWLANEN_DIG_FSM_SKIP_TX_DCC_RATE_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of TX DCC Rate-Change Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a4
- **Exists:** Always

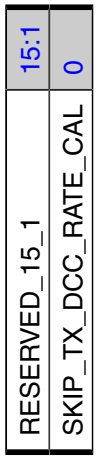


Table 2-1010 Fields for Register: RAWLANEN_DIG_FSM_SKIP_TX_DCC_RATE_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_TX_DCC_RATE_CAL	R	This bit is the status of TX_DCC rate-change calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1007 RAWLANEN_DIG_FSM_SKIP_SIGDET_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of SIGDET Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a5
- **Exists:** Always

Table 2-1011 Fields for Register: RAWLANEN_DIG_FSM_SKIP_SIGDET_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_SIGDET_STARTUP_CAL	R	This bit is the status of SIGDET start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1008 RAWLANEN_DIG_FSM_SKIP_VGEN_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of VGEN Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a6
- **Exists:** Always

Table 2-1012 Fields for Register: RAWLANEN_DIG_FSM_SKIP_VGEN_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_VGEN_STARTUP_CAL	R	This bit is the status of VGEN start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1009 RAWLANEN_DIG_FSM_SKIP_RX_IQC_RATE_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX IQC Rate-hange Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a7
- **Exists:** Always

Table 2-1013 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQC_RATE_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_IQC_RATE_CAL	R	This bit is the status of RX IQC rate-change calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1010 RAWLANEN_DIG_FSM_SKIP_RX_DCC_DATA_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC DATA Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a8
- **Exists:** Always

Table 2-1014 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_DATA_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_DATA_STARTUP_CAL	R	This bit is the status of RX DCC DATA start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1011 RAWLANEN_DIG_FSM_SKIP_RX_DCC_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC BYPASS Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)a9
- **Exists:** Always

Table 2-1015 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_BYPASS_START UP_CAL	R	This bit is the status of RX DCC BYPASS start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1012 RAWLANEN_DIG_FSM_SKIP_RX_DCC_PHASE_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DCC PHASE Start-up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)aa
- **Exists:** Always

Table 2-1016 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DCC_PHASE_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DCC_PHASE_STARTUP_CAL	R	This bit is the status of RX DCC PHASE start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1013 RAWLANEN_DIG_FSM_SKIP_RX_HALF_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Half-Rate Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ab
- **Exists:** Always

Table 2-1017 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_HALF_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_HALF_RATE_STARTUP_CAL	R	This bit is the status of the RX half-rate start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1014 RAWLANEN_DIG_FSM_SKIP_RX_FULL_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Full-Rate Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ac
- **Exists:** Always

Table 2-1018 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_FULL_RATE_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_FULL_RATE_STARTUP_CAL	R	This bit is the status of RX full-rate start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1015 RAWLANEN_DIG_FSM_SKIP_RX_ERROR_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Error Slicer Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ad
- **Exists:** Always

Table 2-1019 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_ERROR_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_ERROR_STARTUP_CAL	R	This bit is the status of RX Error slicer start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1016 RAWLANEN_DIG_FSM_SKIP_RX_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Bypass Slicer Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ae
- **Exists:** Always

Table 2-1020 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_BYPASS_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_ERROR_STARTUP_CAL	R	This bit is the status of RX Bypass slicer start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1017 RAWLANEN_DIG_FSM_SKIP_RX_VGA_SLICER_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX VGA Slicer Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)af
- **Exists:** Always

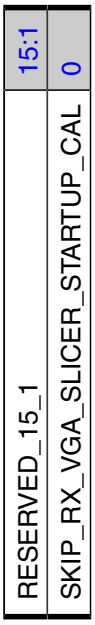


Table 2-1021 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_VGA_SLICER_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_VGA_SLICER_START UP_CAL	R	This bit is the status of RX VGA slicer start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1018 RAWLANEN_DIG_FSM_FW_SCRATCH_0 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 0.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b0
- **Exists:** Always

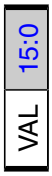


Table 2-1022 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1019 RAWLANEN_DIG_FSM_FW_SCRATCH_1 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 1.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b1
- **Exists:** Always

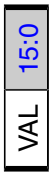


Table 2-1023 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1020 RAWLANEN_DIG_FSM_FW_SCRATCH_2 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 2.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b2
- **Exists:** Always

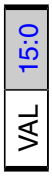


Table 2-1024 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1021 RAWLANEN_DIG_FSM_FW_SCRATCH_3 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 3.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b3
- **Exists:** Always

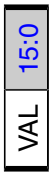


Table 2-1025 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1022 RAWLANEN_DIG_FSM_FAST_TX_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast TX Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b4
- **Exists:** Always

Table 2-1026 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_TX_STARTUP_CAL	R	This bit is the status of fast TX start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1023 RAWLANEN_DIG_FSM_SKIP_RX_BUF_STARTUP_CAL (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Buffer Start-Up Calibration.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b5
- **Exists:** Always

Table 2-1027 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_BUF_STARTUP_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_BUF_STARTUP_CAL	R	This bit is the status of RX Buffer start-up calibration. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1024 RAWLANEN_DIG_FSM_FAST_TX_PWRUP (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast TX Power Up.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b6
- **Exists:** Always

Table 2-1028 Fields for Register: RAWLANEN_DIG_FSM_FAST_TX_PWRUP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_TX_PWRUP	R	This bit is the status of fast TX power up (LOS, VREG/AFE and DCC). (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1025 RAWLANEN_DIG_FSM_SKIP_RX_ADAPT_RELOAD (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Adaptation Initialization.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b7
- **Exists:** Always

Table 2-1029 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_ADAPT_RELOAD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_ADAPT_RELOAD	R	This bit is the status of RX adaptation initialization. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1026 RAWLANEN_DIG_FSM_FW_SCRATCH_4 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 4.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b8
- **Exists:** Always

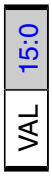


Table 2-1030 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1027 RAWLANEN_DIG_FSM_FW_SCRATCH_5 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 5.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)b9
- **Exists:** Always

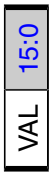


Table 2-1031 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1028 RAWLANEN_DIG_FSM_FW_SCRATCH_6 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 6.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ba
- **Exists:** Always

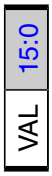


Table 2-1032 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1029 RAWLANEN_DIG_FSM_FW_SCRATCH_7 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 7.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)bb
- **Exists:** Always

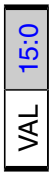


Table 2-1033 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1030 RAWLANEN_DIG_FSM_FW_SCRATCH_8 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 8.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)bc
- **Exists:** Always

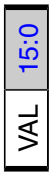


Table 2-1034 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1031 RAWLANEN_DIG_FSM_FW_SCRATCH_9 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 9.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)bd
- **Exists:** Always

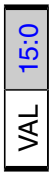


Table 2-1035 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1032 RAWLANEN_DIG_FSM_FW_SCRATCH_10 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 10.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)be
- **Exists:** Always

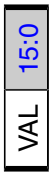


Table 2-1036 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1033 RAWLANEN_DIG_FSM_FW_SCRATCH_11 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 11.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)bf
- **Exists:** Always

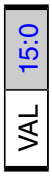


Table 2-1037 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1034 RAWLANEN_DIG_FSM_FW_SCRATCH_12 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 12.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c0
- **Exists:** Always

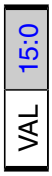


Table 2-1038 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1035 RAWLANEN_DIG_FSM_FW_SCRATCH_13 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 13.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c1
- **Exists:** Always

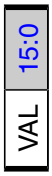


Table 2-1039 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1036 RAWLANEN_DIG_FSM_FW_SCRATCH_14 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 14.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c2
- **Exists:** Always

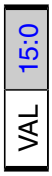


Table 2-1040 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1037 RAWLANEN_DIG_FSM_FW_SCRATCH_15 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 15.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c3
- **Exists:** Always

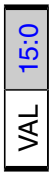


Table 2-1041 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_15 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1038 RAWLANEN_DIG_FSM_SKIP_RX_ATT_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX ATT Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c4
- **Exists:** Always

Table 2-1042 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_ATT_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_ATT_COARSE_START UP_ADAPT	R	This bit is the status of RX ATT coarse start-up adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1039 RAWLANEN_DIG_FSM_SKIP_RX_VGA_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX VGA Ccoarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c5
- **Exists:** Always

Table 2-1043 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_VGA_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_VGA_COARSE_STARTUP_ADAPT	R	This bit is the status of RX VGA coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1040 RAWLANEN_DIG_FSM_SKIP_RX_CTLE_POLE_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX CTLE_POLE Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c6
- **Exists:** Always

Table 2-1044 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_CTLE_POLE_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_CTLE_POLE_COARSE_STARTUP_ADAPT	R	This bit is the status of RX CTLE_POLE coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1041 RAWLANEN_DIG_FSM_SKIP_RX_CTLE_BOOST_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX CTLE_BOOST Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c7
- **Exists:** Always

Table 2-1045 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_CTLE_BOOST_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_CTLE_BOOST_COARSE_STARTUP_ADAPT	R	This bit is the status of RX CTLE_BOOST coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1042 RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP1_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DFE_TAP1 Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c8
- **Exists:** Always

Table 2-1046 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP1_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DFE_TAP1_COARSE_STARTUP_ADAPT	R	This bit is the status of RX DFE_TAP1 coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1043 RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP2_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX DFE_TAP2 Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)c9
- **Exists:** Always

Table 2-1047 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_DFE_TAP2_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_DFE_TAP2_COARSE_STARTUP_ADAPT	R	This bit is the status of RX DFE_TAP2 coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1044 RAWLANEN_DIG_FSM_SKIP_RX_IQ_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of RX IQ Coarse Start-Up Adaptation.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ca
- **Exists:** Always

Table 2-1048 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RX_IQ_COARSE_STARTUP_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RX_IQ_COARSE_STARTUP_ADAPT	R	This bit is the status of RX IQ coarse adaptation. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1045 RAWLANEN_DIG_FSM_FAST_RX_COARSE_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Status of Fast RX Coarse Adapt.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)cb
- **Exists:** Always

Table 2-1049 Fields for Register: RAWLANEN_DIG_FSM_FAST_RX_COARSE_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	FAST_RX_COARSE_ADAPT	R	This bit is the status of fast RX coarse adapt. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1046 RAWLANEN_DIG_FSM_SKIP_INIT_PWRUP_END_RX_RESET_TOGGLE (for N = 0; N <= NLANES-1)

- **Description:** Status of RX Init Power-Up End rx_reset Toggle.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)cc
- **Exists:** Always

Table 2-1050 Fields for Register: RAWLANEN_DIG_FSM_SKIP_INIT_PWRUP_END_RX_RESET_TOGGLE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_INIT_PWRUP_END_RX_RESET_TOGGLE	R	This bit is the status of RX init power-up end rx_reset toggle. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1047 RAWLANEN_DIG_FSM_SKIP_RATE_CHANGE_SETTING (for N = 0; N <= NLANES-1)

- **Description:** Status of Skip Rate Change Setting.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)cd
- **Exists:** Always

Table 2-1051 Fields for Register: RAWLANEN_DIG_FSM_SKIP_RATE_CHANGE_SETTING (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	SKIP_RATE_CHANGE_SETTING	R	This bit is the status of skip rate change setting. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1048 RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_STARTUP (for N = 0; N <= NLANES-1)

- **Description:** Status of Skip DFE FLT Adapt Start Up.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)ce
- **Exists:** Always

Table 2-1052 Fields for Register: RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_STARTUP (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_SKIP_DFE_FLT_ADAPT_STARTUP	R	This bit is the status of skip dfe flt adapt start up. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1049 RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_CONT (for N = 0; N <= NLANES-1)

- **Description:** Status of Skip DFE FLT Adapt Continuous.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)cf
- **Exists:** Always

Table 2-1053 Fields for Register: RAWLANEN_DIG_FSM_RX_SKIP_DFE_FLT_ADAPT_CONT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_SKIP_DFE_FLT_ADAPT_CONT	R	This bit is the status of skip dfe flt adapt continuous. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0

2.1.1050 RAWLANEN_DIG_FSM_FW_SCRATCH_16 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 16.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d0
- **Exists:** Always

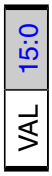


Table 2-1054 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_16 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1051 RAWLANEN_DIG_FSM_FW_SCRATCH_17 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 17.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d1
- **Exists:** Always

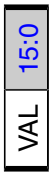


Table 2-1055 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_17 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1052 RAWLANEN_DIG_FSM_FW_SCRATCH_18 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 18.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d2
- **Exists:** Always

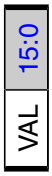


Table 2-1056 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_18 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1053 RAWLANEN_DIG_FSM_FW_SCRATCH_19 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 19.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d3
- **Exists:** Always

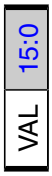


Table 2-1057 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_19 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1054 RAWLANEN_DIG_FSM_FW_SCRATCH_20 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 20.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d4
- **Exists:** Always

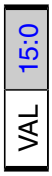


Table 2-1058 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_20 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1055 RAWLANEN_DIG_FSM_FW_SCRATCH_21 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 21.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d5
- **Exists:** Always

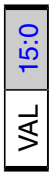


Table 2-1059 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_21 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1056 RAWLANEN_DIG_FSM_FW_SCRATCH_22 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 22.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d6
- **Exists:** Always

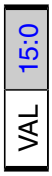


Table 2-1060 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_22 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1057 RAWLANEN_DIG_FSM_FW_SCRATCH_23 (for N = 0; N <= NLANES-1)

- **Description:** Firmware Scratch Register 23.
- **Size:** 16 bits
- **Offset:** 0x2(N*2+1)d7
- **Exists:** Always

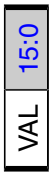


Table 2-1061 Fields for Register: RAWLANEN_DIG_FSM_FW_SCRATCH_23 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	Firmware scratch register. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1058 RAWLANEAONN_DIG_AFE_ATT_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for RX AFE ATT iDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)00
- **Exists:** Always

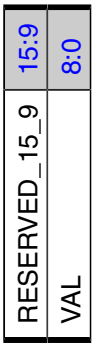


Table 2-1062 Fields for Register: RAWLANEAONN_DIG_AFE_ATT_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for AFE ATT iDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1059 RAWLANEAONN_DIG_AFE_CTLE_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for RX AFE CTLE iDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)01
- **Exists:** Always

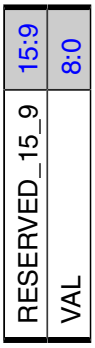


Table 2-1063 Fields for Register: RAWLANEAONN_DIG_AFE_CTLE_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for AFE CTLE iDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1060 RAWLANEAONN_DIG_AFE_VGA1_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX AFE VGA1 iDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)02
- **Exists:** Always

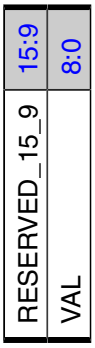


Table 2-1064 Fields for Register: RAWLANEAONN_DIG_AFE_VGA1_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for AFE VGA1 iDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1061 RAWLANEAONN_DIG_DFE_PHASE_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Phase Even vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)03
- **Exists:** Always

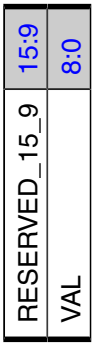


Table 2-1065 Fields for Register: RAWLANEAONN_DIG_DFE_PHASE_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Phase Even vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1062 RAWLANEAONN_DIG_DFE_PHASE_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Phase Odd vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)04
- **Exists:** Always

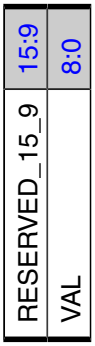


Table 2-1066 Fields for Register: RAWLANEAONN_DIG_DFE_PHASE_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Phase Odd vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1063 RAWLANEAONN_DIG_DFE_PHASE_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Phase Even Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)05
- **Exists:** Always

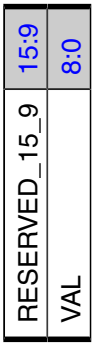


Table 2-1067 Fields for Register: RAWLANEAONN_DIG_DFE_PHASE_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Phase Even Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1064 RAWLANEAONN_DIG_DFE_PHASE_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Phase Odd Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)06
- **Exists:** Always

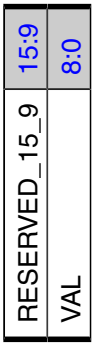


Table 2-1068 Fields for Register: RAWLANEAONN_DIG_DFE_PHASE_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Phase Odd Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1065 RAWLANEAONN_DIG_AFE_BUF_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Value for RX AFE Buffer iDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)07
- **Exists:** Always

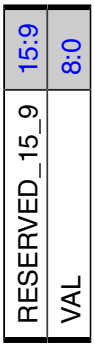


Table 2-1069 Fields for Register: RAWLANEAONN_DIG_AFE_BUF_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for AFE Buffer iDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1066 RAWLANEAONN_DIG_DFE_DATA0_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Even High vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)09
- **Exists:** Always

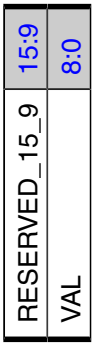


Table 2-1070 Fields for Register: RAWLANEAONN_DIG_DFE_DATA0_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Data0 Even High vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1067 RAWLANEAONN_DIG_DFE_DATA0_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Even Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)0a
- **Exists:** Always

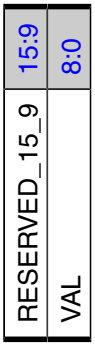


Table 2-1071 Fields for Register: RAWLANEAONN_DIG_DFE_DATA0_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Data0 Even Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1068 RAWLANEAONN_DIG_DFE_DATA0_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Odd High vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)0b
- **Exists:** Always

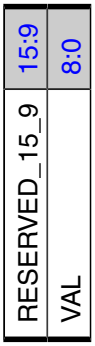


Table 2-1072 Fields for Register: RAWLANEAONN_DIG_DFE_DATA0_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Data0 Odd High vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1069 RAWLANEAONN_DIG_DFE_DATA0_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data0 Odd Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)0c
- **Exists:** Always

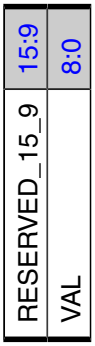


Table 2-1073 Fields for Register: RAWLANEAONN_DIG_DFE_DATA0_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE Data0 Odd Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1070 RAWLANEAONN_DIG_DFE_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE By-Pass Even vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)0d
- **Exists:** Always

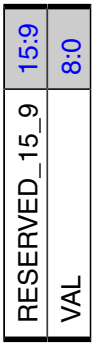


Table 2-1074 Fields for Register: RAWLANEAONN_DIG_DFE_BYPASS_EVEN_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE By-Pass Even vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1071 RAWLANEAONN_DIG_DFE_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE By-Pass Odd vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)0e
- **Exists:** Always

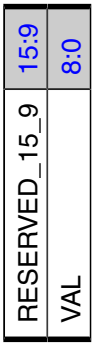


Table 2-1075 Fields for Register: RAWLANEAONN_DIG_DFE_BYPASS_ODD_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE By-Pass Odd vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1072 RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL (for N = 0; N <= NLANES-1)

- **Description:** Set to Skip Firmware Start-up Algorithms.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)12
- **Exists:** Always

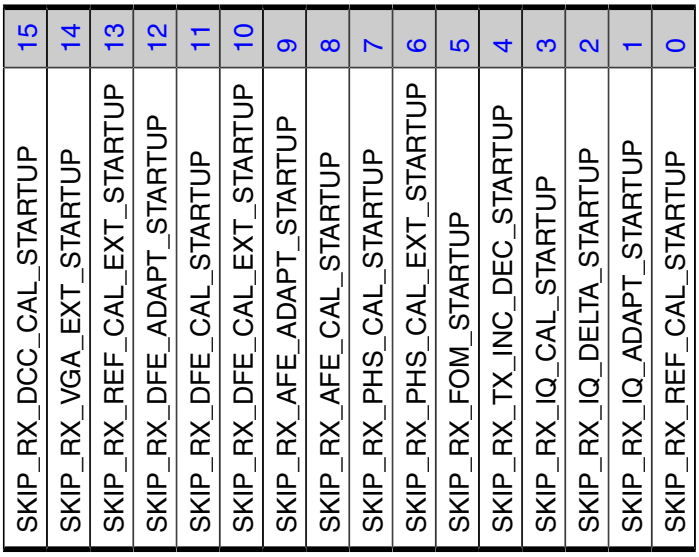


Table 2-1076 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SKIP_RX_DCC_CAL_STARTUP	R/W	This bit skips RX DCC start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	SKIP_RX_VGA_EXT_STARTUP	R/W	This bit skips RX start-up VGA calibration in extended mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	SKIP_RX_REF_CAL_EXT_STARTUP	R/W	This bit skips RX start-up reference level calibration in extended mode. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	SKIP_RX_DFE_ADAPT_STARTUP	R/W	This bit skips RX DFE DAC start-up adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1076 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	SKIP_RX_DFE_CAL_STARTUP	R/W	This bit skips RX DFE slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	SKIP_RX_DFE_CAL_EXT_STARTUP	R/W	This bit skips RX DFE slicer extended start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	SKIP_RX_AFE_ADAPT_STARTUP	R/W	This bit skips RX AFE DAC start-up adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	SKIP_RX_AFE_CAL_STARTUP	R/W	This bit skips RX AFE DAC start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	SKIP_RX_PHS_CAL_STARTUP	R/W	This bit skips RX phase slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	SKIP_RX_PHS_CAL_EXT_STARTUP	R/W	This bit skips RX extended phase slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	SKIP_RX_FOM_STARTUP	R/W	This bit skips RX start-up FOM measurement. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SKIP_RX_TX_INC_DEC_STARTUP	R/W	This bit skips RX continuous phase calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SKIP_RX_IQ_CAL_STARTUP	R/W	This bit skips RX IQ start-up calibration. **Value After Reset:** 0x0

Table 2-1076 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	SKIP_RX_IQ_DELTA_STARTUP	R/W	This bit skips RX IQ fixed offset. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_RX_IQ_ADAPT_STARTUP	R/W	This bit skips RX IQ start-up adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_RX_REF_CAL_STARTUP	R/W	This bit skips RX start-up reference level calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1073 RAWLANEAONN_DIG_RX_CONT_ALGO_CTL (for N = 0; N <= NLANES-1)

- **Description:** Set to Skip Firmware Mission-mode Algorithms.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)13
- **Exists:** Always

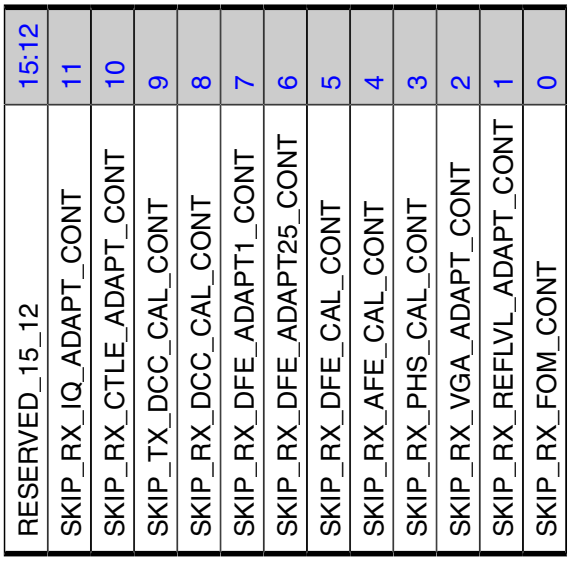


Table 2-1077 Fields for Register: RAWLANEAONN_DIG_RX_CONT_ALGO_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11	SKIP_RX_IQ_ADAPT_CONT	R/W	This bit skips RX IQ continuous adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	SKIP_RX_CTLE_ADAPT_CONT	R/W	This bit skips RX CTLE continuous adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	SKIP_TX_DCC_CAL_CONT	R/W	This bit skips TX DCC continuous calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1077 Fields for Register: RAWLANEAONN_DIG_RX_CONT_ALGO_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
8	SKIP_RX_DCC_CAL_CONT	R/W	This bit skips RX DCC continuous calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	SKIP_RX_DFE_ADAPT1_CONT	R/W	This bit skips RX DFE DAC continuous adaptation for TAP1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
6	SKIP_RX_DFE_ADAPT25_CONT	R/W	This bit skips RX DFE DAC continuous adaptation for TAP2 to TAP5. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	SKIP_RX_DFE_CAL_CONT	R/W	This bit skips RX DFE slicer continuous calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SKIP_RX_AFE_CAL_CONT	R/W	This bit skips RX AFE DAC continuous calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SKIP_RX_PHS_CAL_CONT	R/W	This bit skips RX phase slicer offset calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SKIP_RX_VGA_ADAPT_CONT	R/W	This bit skips RX VGA continuous adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_RX_REFLVL_ADAPT_CONT	R/W	This bit skips RX REF continuous adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_RX_FOM_CONT	R/W	This bit skips RX continuous FOM measurement. **Value After Reset:** 0x0

Table 2-1077 Fields for Register: RAWLANEAONN_DIG_RX_CONT_ALGO_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.1074 RAWLANEAONN_DIG_INIT_PWRUP_DONE (for N = 0; N <= NLANES-1)

- **Description:** Initial Power-Up Done Status.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)14
- **Exists:** Always

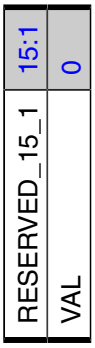


Table 2-1078 Fields for Register: RAWLANEAONN_DIG_INIT_PWRUP_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	This bit indicates whether initial power-up has completed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1075 RAWLANEAONN_DIG_RX_ADPT_ATT (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of ATT.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)15
- **Exists:** Always

Table 2-1079 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_ATT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_ADPT_VAL	R/W	These bits store the RX adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1076 RAWLANEAONN_DIG_RX_ADPT_VGA (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of VGA.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)16
- **Exists:** Always

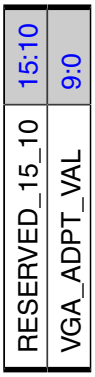


Table 2-1080 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_VGA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_ADPT_VAL	R/W	These bits store the RX adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1077 RAWLANEAONN_DIG_RX_ADPT_CTLE (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of CTLE.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)17
- **Exists:** Always

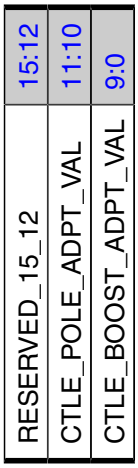


Table 2-1081 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_CTLE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_ADPT_VAL	R/W	These bits are the RX adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_ADPT_VAL	R/W	These bits store the RX adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1078 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)18
- **Exists:** Always

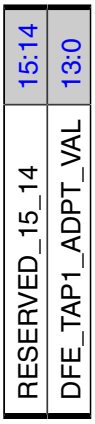


Table 2-1082 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_ADPT_VAL	R/W	These bit store the RX adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1079 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)19
- **Exists:** Always

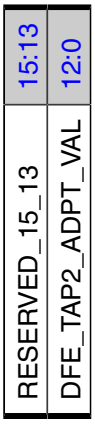


Table 2-1083 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1080 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1a
- **Exists:** Always

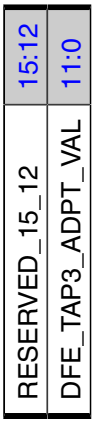


Table 2-1084 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP3_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP3 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1081 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP4.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1b
- **Exists:** Always

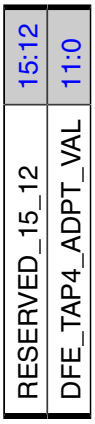


Table 2-1085 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP4_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP4 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1082 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP5.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1c
- **Exists:** Always

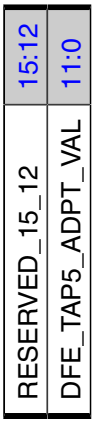


Table 2-1086 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP5_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP5 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1083 RAWLANEAONN_DIG_RX_ADPT_IQ (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of IQ for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1d
- **Exists:** Always

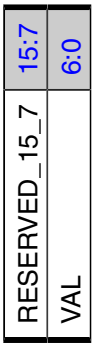


Table 2-1087 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_IQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	These bits are the RX adapted value of IQ for Bank 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1084 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Even Slicer for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1e
- **Exists:** Always

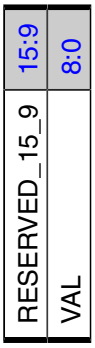


Table 2-1088 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the RX Adapted value of reference level for error even slicer for Bank 0. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1085 RAWLANEAONN_DIG_RX_ADAPT_DONE (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Done Status.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)1f
- **Exists:** Always

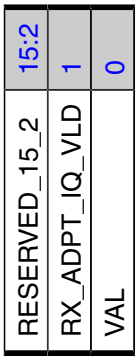


Table 2-1089 Fields for Register: RAWLANEAONN_DIG_RX_ADAPT_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_ADPT_IQ_VLD	R/W	This bit indicates if the RX adapted value of IQ for Bank 0 is valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit indicates whether RX adaptation has completed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1086 RAWLANEAONN_DIG_RX_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of ATT for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)20
- **Exists:** Always

Table 2-1090 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_ADPT_VAL	R/W	These bits store the RX adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1087 RAWLANEAONN_DIG_RX_ADPT_VGA_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of VGA for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)21
- **Exists:** Always

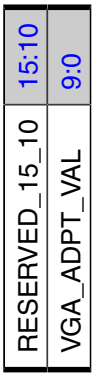


Table 2-1091 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_VGA_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_ADPT_VAL	R/W	These bits store the RX adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1088 RAWLANEAONN_DIG_RX_ADPT_CTLE_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of CTLE for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)22
- **Exists:** Always

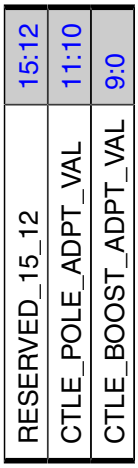


Table 2-1092 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_CTLE_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_ADPT_VAL	R/W	These bits store the RX adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_ADPT_VAL	R/W	These bits store the RX adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1089 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)23
- **Exists:** Always

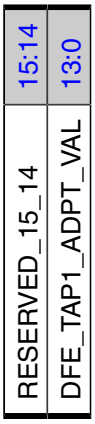


Table 2-1093 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1090 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP2 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)24
- **Exists:** Always

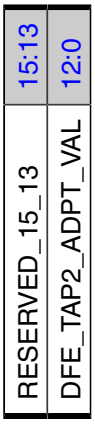


Table 2-1094 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1091 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted value of DFE TAP3 for bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)25
- **Exists:** Always

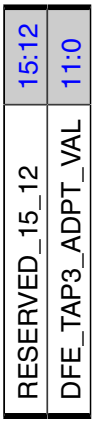


Table 2-1095 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP3_ADPT_VAL	R/W	Stored RX adapted DFE TAP3 value. Bits 4:0 are dither bits **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1092 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP4 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)26
- **Exists:** Always

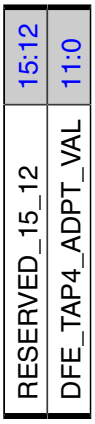


Table 2-1096 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP4_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP4 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1093 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP5 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)27
- **Exists:** Always

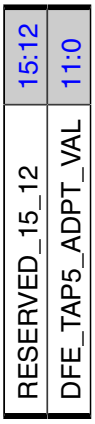


Table 2-1097 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP5_ADPT_VAL	R/W	These bits store the RX adapted DFE TAP5 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1094 RAWLANEAONN_DIG_RX_ADPT_IQ_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of IQ for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)28
- **Exists:** Always

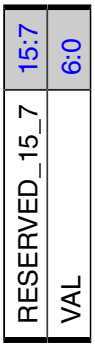


Table 2-1098 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_IQ_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	These bits are the RX adapted values of IQ for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1095 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Even Slicer for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)29
- **Exists:** Always

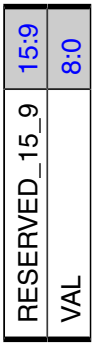


Table 2-1099 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	The bits are the RX adapted values of reference level for error even slicer for Bank 1. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1096 RAWLANEAONN_DIG_RX_ADAPT_DONE_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Done Status for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2a
- **Exists:** Always

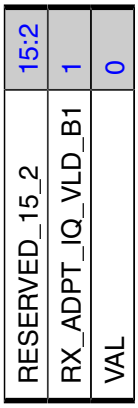


Table 2-1100 Fields for Register: RAWLANEAONN_DIG_RX_ADAPT_DONE_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_ADPT_IQ_VLD_B1	R/W	This bit indicates if the RX adapted value of IQ for Bank 1 is valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit indicates whether Bank 1 RX adaptation has completed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1097 RAWLANEAONN_DIG_FAST_FLAGS (for N = 0; N <= NLANES-1)

- **Description:** Fast Flags for Simulation Only.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2b
- **Exists:** Always

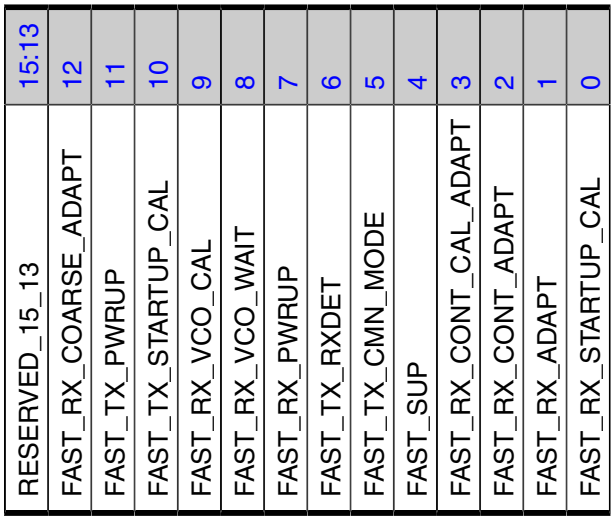


Table 2-1101 Fields for Register: RAWLANEAONN_DIG_FAST_FLAGS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12	FAST_RX_COARSE_ADAPT	R/W	This bit enables fast RX coarse adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
11	FAST_TX_PWRUP	R/W	This bit enables fast TX power up (refgen, serial_en, clk_en). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	FAST_TX_STARTUP_CAL	R/W	This bit enables fast TX start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	FAST_RX_VCO_CAL	R/W	This bit enables fast RX VCO calibration. **Value After Reset:** 0x0

Table 2-1101 Fields for Register: RAWLANEAONN_DIG_FAST_FLAGS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
8	FAST_RX_VCO_WAIT	R/W	This bit enables fast RX VCO wait times. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	FAST_RX_PWRUP	R/W	This bit enables fast RX power up (LOS, VREG/AFE and DCC). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	FAST_TX_RXDET	R/W	This bit enables fast TX Detect RX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	FAST_TX_CMN_MODE	R/W	This bit enables fast TX Common Mode charge up. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	FAST_SUP	R/W	This bit enables fast Support block (MPLL and Rtune). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	FAST_RX_CONT_CAL_ADAPT	R/W	This bit enables fast RX continuous calibration/adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	FAST_RX_CONT_ADAPT	R/W	This bit enables fast RX continuous adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	FAST_RX_ADAPT	R/W	This bit enables fast RX adaptation. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1101 Fields for Register: RAWLANEAONN_DIG_FAST_FLAGS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	FAST_RX_STARTUP_CAL	R/W	This bit enables fast RX start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1098 RAWLANEAONN_DIG_FW_STATES_0 (for N = 0; N <= NLANES-1)

- **Description:** Stores Various States for Firmware - State 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2c
- **Exists:** Always

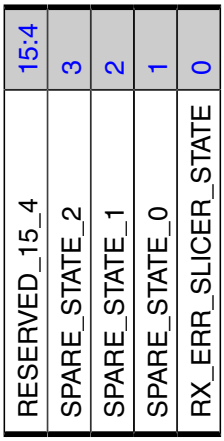


Table 2-1102 Fields for Register: RAWLANEAONN_DIG_FW_STATES_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	SPARE_STATE_2	R/W	This bit is spare field 3 to be used by Firmware. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SPARE_STATE_1	R/W	This bit is spare field 2 to be used by Firmware. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	SPARE_STATE_0	R/W	This bit is spare field 1 to be used by Firmware. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_ERR_SLICER_STATE	R/W	This bit is the value for the current RX error slicer state. This register stores the railing out direction of both the even and odd error slicer, which is flipped for each mission mode run. **Value After Reset:** 0x1

Table 2-1102 Fields for Register: RAWLANEAONN_DIG_FW_STATES_0 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.1099 RAWLANEAONN_DIG_FW_STATES_1 (for N = 0; N <= NLANES-1)

- **Description:** Stores Various States for Firmware - State 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2d
- **Exists:** Always

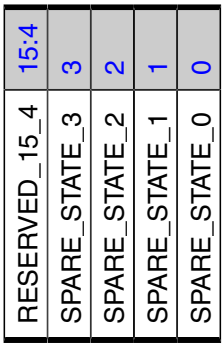


Table 2-1103 Fields for Register: RAWLANEAONN_DIG_FW_STATES_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	SPARE_STATE_3	R/W	This bit is spare field 4 to be used by Firmware. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SPARE_STATE_2	R/W	This bit is spare field 3 to be used by Firmware. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	SPARE_STATE_1	R/W	This bit is spare field 2 to be used by Firmware. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	SPARE_STATE_0	R/W	This bit is spare field 1 to be used by Firmware. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.1100 RAWLANEAONN_DIG_TXRX_OVRD_IN (for N = 0; N <= NLANES-1)

- **Description:** Override Values for Incoming AON TX/RX Controls from PCS.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2e
- **Exists:** Always

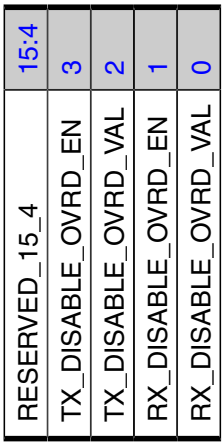


Table 2-1104 Fields for Register: RAWLANEAONN_DIG_TXRX_OVRD_IN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	TX_DISABLE_OVRD_EN	R/W	This bit the override enable for tx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	TX_DISABLE_OVRD_VAL	R/W	This bit is the override value for tx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_DISABLE_OVRD_EN	R/W	This bit is the override enable for rx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_DISABLE_OVRD_VAL	R/W	This bit is the override value for rx_disable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1101 RAWLANEAONN_DIG_TX_EQ_DIR_POLARITY_CTL (for N = 0; N <= NLANES-1)

- **Description:** Controls Equalization Direction Polarity Setting in PCS RAW.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)2f
- **Exists:** Always

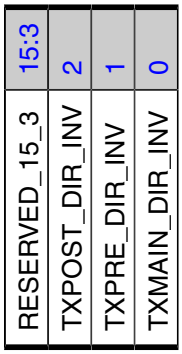


Table 2-1105 Fields for Register: RAWLANEAONN_DIG_TX_EQ_DIR_POLARITY_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	TXPOST_DIR_INV	R/W	This bit inverts the txpost_dir polarity. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	TXPRE_DIR_INV	R/W	This bit inverts the txpre_dir polarity. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TXMAIN_DIR_INV	R/W	This bit inverts teh txmain_dir polarity. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1102 RAWLANEAONN_DIG_TX_PRE_DIV (for N = 0; N <= NLANES-1)

- **Description:** TX Pre-Threshold Div.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)30
- **Exists:** Always

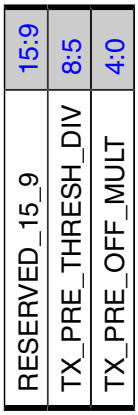


Table 2-1106 Fields for Register: RAWLANEAONN_DIG_TX_PRE_DIV (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:5	TX_PRE_THRESH_DIV	R/W	These bits are the TX pre-threshold div. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
4:0	TX_PRE_OFF_MULT	R/W	These bits are the TX pre-offset mult. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0x1f

2.1.1103 RAWLANEAONN_DIG_TX_MAIN_ATT_THRESHOLD (for N = 0; N <= NLANES-1)

- **Description:** TX Main ATT High and Low Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)31
- **Exists:** Always

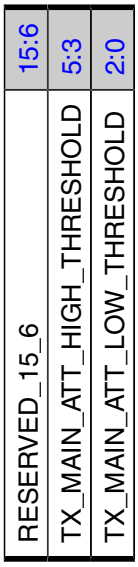


Table 2-1107 Fields for Register: RAWLANEAONN_DIG_TX_MAIN_ATT_THRESHOLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:3	TX_MAIN_ATT_HIGH_THRESHOLD	R/W	These bits are the TX main ATT high threshold. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x7
2:0	TX_MAIN_ATT_LOW_THRESHOLD	R/W	These bits are the TX main ATT low threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1104 RAWLANEAONN_DIG_TX_MAIN_VGA_THRESHOLD (for N = 0; N <= NLANES-1)

- **Description:** TX Main VGA High and Low Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)32
- **Exists:** Always

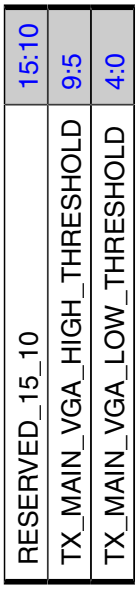


Table 2-1108 Fields for Register: RAWLANEAONN_DIG_TX_MAIN_VGA_THRESHOLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	TX_MAIN_VGA_HIGH_THRESHOLD	R/W	These bits are the TX main vga high threshold. **Value After Reset:** 0x1f **Exists:** Always **Reset Mask:** 0x1f
4:0	TX_MAIN_VGA_LOW_THRESHOLD	R/W	These bits are the TX main vga low threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.1105 RAWLANEAONN_DIG_TX_POST_BOOST_THRESHOLD (for N = 0; N <= NLANES-1)

- **Description:** TX Post Boost High and Low Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)33
- **Exists:** Always

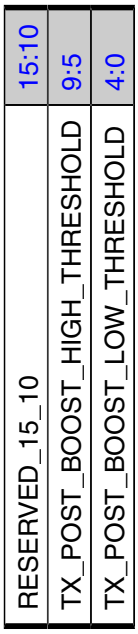


Table 2-1109 Fields for Register: RAWLANEAONN_DIG_TX_POST_BOOST_THRESHOLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	TX_POST_BOOST_HIGH_THRESHOLD	R/W	These bits are the TX post boost high threshold. **Value After Reset:** 0x14 **Exists:** Always **Reset Mask:** 0x1f
4:0	TX_POST_BOOST_LOW_THRESHOLD	R/W	These bits are the TX post boost low threshold. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1f

2.1.1106 RAWLANEAONN_DIG_TX_POST_TAP1_LOW_THRESHOLD (for N = 0; N <= NLANES-1)

- **Description:** TX Post TAP1 Low Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)34
- **Exists:** Always

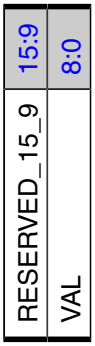


Table 2-1110 Fields for Register: RAWLANEAONN_DIG_TX_POST_TAP1_LOW_THRESHOLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the TX post TAP1 low threshold. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1107 RAWLANEAONN_DIG_RX_CAL_REF_EE_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for Reference Level Error Even Slicer.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)35
- **Exists:** Always

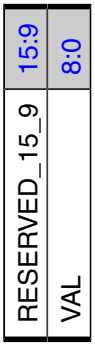


Table 2-1111 Fields for Register: RAWLANEAONN_DIG_RX_CAL_REF_EE_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the values for reference level error even slicer. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1108 RAWLANEAONN_DIG_RX_CAL_REF_EO_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Value for Reference Level Error Odd Slicer.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)36
- **Exists:** Always

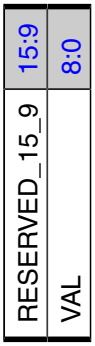


Table 2-1112 Fields for Register: RAWLANEAONN_DIG_RX_CAL_REF_EO_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the values for reference level error odd slicer. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1109 RAWLANEAONN_DIG_RX_ADPT_ERR_SLC_MODE (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Error Slicer Mode for Reference-Level Calibration.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)39
- **Exists:** Always

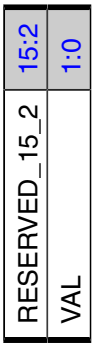


Table 2-1113 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_ERR_SLC_MODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R/W	These bits are the RX Adaptation error slicer mode for reference-level calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.1110 RAWLANEAONN_DIG_ADPT_CTL_0 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)40
- **Exists:** Always

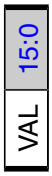


Table 2-1114 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1111 RAWLANEAONN_DIG_ADPT_CTL_1 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)41
- **Exists:** Always

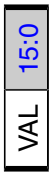


Table 2-1115 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1112 RAWLANEAONN_DIG_ADPT_CTL_2 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)42
- **Exists:** Always

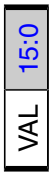


Table 2-1116 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1113 RAWLANEAONN_DIG_ADPT_CTL_3 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)43
- **Exists:** Always

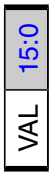


Table 2-1117 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1114 RAWLANEAONN_DIG_ADPT_CTL_4 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 4.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)44
- **Exists:** Always

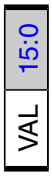


Table 2-1118 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_4 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1115 RAWLANEAONN_DIG_ADPT_CTL_5 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 5.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)45
- **Exists:** Always

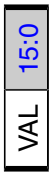


Table 2-1119 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_5 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1116 RAWLANEAONN_DIG_ADPT_CTL_6 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 6.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)46
- **Exists:** Always

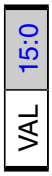


Table 2-1120 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1117 RAWLANEAONN_DIG_ADPT_CTL_7 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 7.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)47
- **Exists:** Always

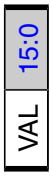


Table 2-1121 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1118 RAWLANEAONN_DIG_ADPT_CTL_8 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 8.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)48
- **Exists:** Always

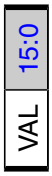


Table 2-1122 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1119 RAWLANEAONN_DIG_ADPT_CTL_9 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 9.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)49
- **Exists:** Always

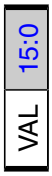


Table 2-1123 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_9 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1120 RAWLANEAONN_DIG_ADPT_CTL_10 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 10.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4a
- **Exists:** Always

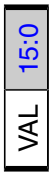


Table 2-1124 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_10 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1121 RAWLANEAONN_DIG_ADPT_CTL_11 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 11.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4b
- **Exists:** Always

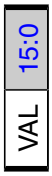


Table 2-1125 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_11 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1122 RAWLANEAONN_DIG_ADPT_CTL_12 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 12.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4c
- **Exists:** Always

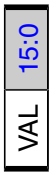


Table 2-1126 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_12 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1123 RAWLANEAONN_DIG_ADPT_CTL_13 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 13.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4d
- **Exists:** Always

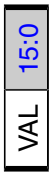


Table 2-1127 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_13 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the alue of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1124 RAWLANEAONN_DIG_ADPT_CTL_14 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 14.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4e
- **Exists:** Always

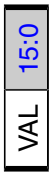


Table 2-1128 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_14 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1125 RAWLANEAONN_DIG_ADPT_CTL_15 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 15.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)4f
- **Exists:** Always

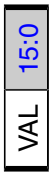


Table 2-1129 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_15 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1126 RAWLANEAONN_DIG_ADPT_CTL_16 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 16.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)50
- **Exists:** Always

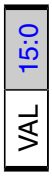


Table 2-1130 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_16 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1127 RAWLANEAONN_DIG_ADPT_CTL_17 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 17.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)51
- **Exists:** Always

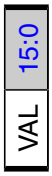


Table 2-1131 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_17 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1128 RAWLANEAONN_DIG_ADPT_CTL_18 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 18.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)52
- **Exists:** Always

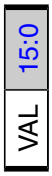


Table 2-1132 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_18 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1129 RAWLANEAONN_DIG_ADPT_CTL_19 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 19.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)53
- **Exists:** Always

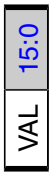


Table 2-1133 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_19 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1130 RAWLANEAONN_DIG_ADPT_CTL_20 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 20.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)54
- **Exists:** Always

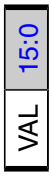


Table 2-1134 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_20 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1131 RAWLANEAONN_DIG_ADPT_CTL_21 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation ControlRregister 21.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)55
- **Exists:** Always

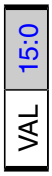


Table 2-1135 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_21 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1132 RAWLANEAONN_DIG_ADPT_CTL_22 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 22.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)56
- **Exists:** Always

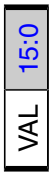


Table 2-1136 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_22 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1133 RAWLANEAONN_DIG_ADPT_CTL_23 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 23.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)57
- **Exists:** Always

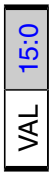


Table 2-1137 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_23 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the value of adaptation control.s **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1134 RAWLANEAONN_DIG_SRAM_REC_EN (for N = 0; N <= NLANES-1)

- **Description:** SRAM Record Enable Status.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)58
- **Exists:** Always

Table 2-1138 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_EN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	REC_EN	R	This bit enables the SRAM Recording. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1135 RAWLANEAONN_DIG_SRAM_REC_CTRL (for N = 0; N <= NLANES-1)

- **Description:** SRAM Record Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)59
- **Exists:** Always

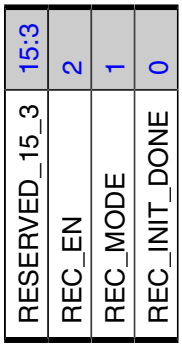


Table 2-1139 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	REC_EN	R/W	This bit enables the SRAM Recording. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	REC_MODE	R/W	SRAM Recording Mode. ■ If REC_MODE = 1'b0, only the first 16 records are stored in SRAM. ■ If REC_MODE = 1'b1, the most recent 16 records (if there are) are stored in SRAM. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	REC_INIT_DONE	R/W	This bit indicates if the SRAM_REC_ADDR and SRAM_REC_ITER have been initialized. Write 1'b0 to this field if they need be re-initialized. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1136 RAWLANEAONN_DIG_SRAM_REC_ADDR (for N = 0; N <= NLANES-1)

- **Description:** Current SRAM Recording Address.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5a
- **Exists:** Always

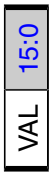


Table 2-1140 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_ADDR (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the current SRAM recording address. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1137 RAWLANEAONN_DIG_SRAM_REC_ITER (for N = 0; N <= NLANES-1)

- **Description:** Current Iteration Count for SRAM Recording.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5b
- **Exists:** Always

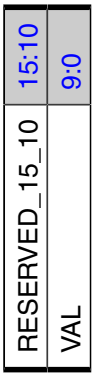


Table 2-1141 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_ITER (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VAL	R/W	These bits are the current iteration count for SRAM recording. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1138 RAWLANEAONN_DIG_IQ_MARGIN_RANGE (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Margin Range.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5c
- **Exists:** Always

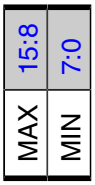


Table 2-1142 Fields for Register: RAWLANEAONN_DIG_IQ_MARGIN_RANGE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	MAX	R/W	These bits are the maximum value for the IQ margin result. **Value After Reset:** 0x46 **Exists:** Always **Reset Mask:** 0xff
7:0	MIN	R/W	These bits are the minimum value for the IQ margin result **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1139 RAWLANEAONN_DIG_RX_CAL_IQ_MAX (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Max Limit.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5d
- **Exists:** Always

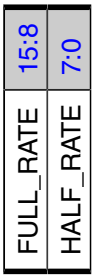


Table 2-1143 Fields for Register: RAWLANEAONN_DIG_RX_CAL_IQ_MAX (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	These bits are the maximum limit for full-rate calibration. **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	These bits are the maximum limit for half-rate calibration. **Value After Reset:** 0x42 **Exists:** Always **Reset Mask:** 0xff

2.1.1140 RAWLANEAONN_DIG_RX_CAL_IQ_MIN (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Min Limit.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5e
- **Exists:** Always

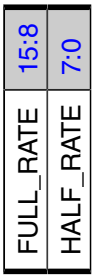


Table 2-1144 Fields for Register: RAWLANEAONN_DIG_RX_CAL_IQ_MIN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R/W	These bits are the minimum limit for full-rate calibration. **Value After Reset:** 0x1a **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE	R/W	These bits are the minimum limit for half-rate calibration. **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff

2.1.1141 RAWLANEAONN_DIG_RX_CAL_IQ_RESET (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Reset Code.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)5f
- **Exists:** Always

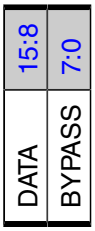


Table 2-1145 Fields for Register: RAWLANEAONN_DIG_RX_CAL_IQ_RESET (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DATA	R/W	These bits are the IQ calibration reset code for data slicer (adjust bus value). **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff
7:0	BYPASS	R/W	These bits are the IQ calibration reset code for bypass slicer (adjust bus value). **Value After Reset:** 0x2e **Exists:** Always **Reset Mask:** 0xff

2.1.1142 RAWLANEAONN_DIG_RX_CAL_IQ_ADJUST (for N = 0; N <= NLANES-1)

- **Description:** IQ Calibration Adjust Values.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)60
- **Exists:** Always

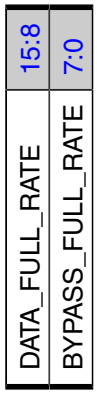


Table 2-1146 Fields for Register: RAWLANEAONN_DIG_RX_CAL_IQ_ADJUST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DATA_FULL_RATE	R/W	These bits are the IQ calibration iq_code for data slicer full-rate alignment (90 degrees). **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff
7:0	BYPASS_FULL_RATE	R/W	These bits are the IQ calibration iq_code for bypass slicer full-rate alignment (90 degrees). **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff

2.1.1143 RAWLANEAONN_DIG_RX_SIGDET_CAL (for N = 0; N <= NLANES-1)

- **Description:** SigDet Calibration Values.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)61
- **Exists:** Always

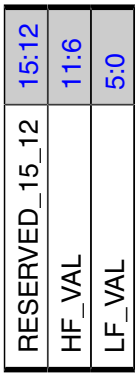


Table 2-1147 Fields for Register: RAWLANEAONN_DIG_RX_SIGDET_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:6	HF_VAL	R/W	These bits are the SigDet high-frequency calibration value. **Value After Reset:** 0x20 **Exists:** Always **Reset Mask:** 0x3f
5:0	LF_VAL	R/W	These bits are the SigDet low-frequency calibration value. **Value After Reset:** 0x20 **Exists:** Always **Reset Mask:** 0x3f

2.1.1144 RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_1 (for N = 0; N <= NLANES-1)

- **Description:** Set to Skip Firmware Start-up/Rate-change Algorithms.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)62
- **Exists:** Always

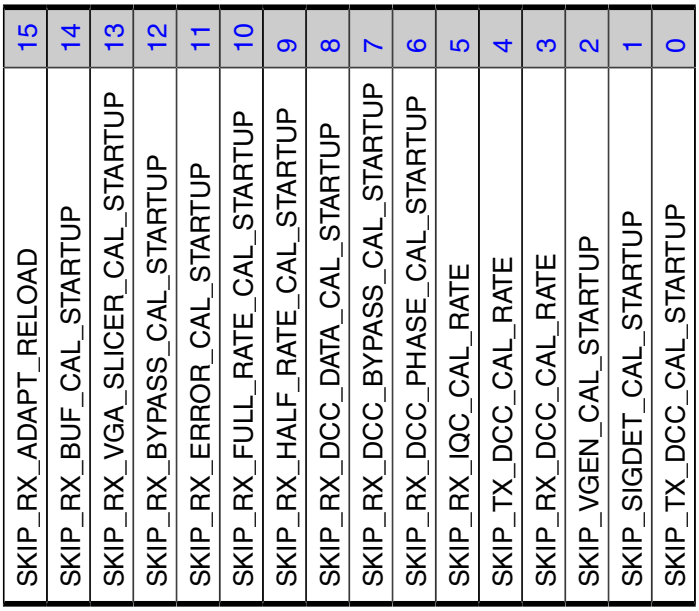


Table 2-1148 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	SKIP_RX_ADAPT_RELOAD	R/W	This bit skips RX adaptation initialization. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
14	SKIP_RX_BUF_CAL_STARTUP	R/W	This bit skips RX buffer calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13	SKIP_RX_VGA_SLICER_CAL_STARTUP	R/W	This bit skips RX VGA slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
12	SKIP_RX_BYPASS_CAL_STARTUP	R/W	This bit skips RX bypass slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1148 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
11	SKIP_RX_ERROR_CAL_STARTUP	R/W	This bit skips RX error slicer start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
10	SKIP_RX_FULL_RATE_CAL_STARTUP	R/W	This bit skips RX full-rate start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9	SKIP_RX_HALF_RATE_CAL_STARTUP	R/W	This bit skips RX half-rate start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	SKIP_RX_DCC_DATA_CAL_STARTUP	R/W	This bit skips RX DCC DATA start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	SKIP_RX_DCC_BYPASS_CAL_STARTUP	R/W	This bit skips RX DCC BYPASS start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	SKIP_RX_DCC_PHASE_CAL_STARTUP	R/W	This bit skips RX DCC PHASE start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	SKIP_RX_IQC_CAL_RATE	R/W	This bit skips RX IQC rate-change calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SKIP_TX_DCC_CAL_RATE	R/W	This bit skips TX DCC rate-change calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SKIP_RX_DCC_CAL_RATE	R/W	This bit skips RX DCC rate-change calibration. **Value After Reset:** 0x0

Table 2-1148 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_1 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1
2	SKIP_VGEN_CAL_STARTUP	R/W	This bit skips VGEN start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_SIGDET_CAL_STARTUP	R/W	This bit skips SIGDET start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_TX_DCC_CAL_STARTUP	R/W	This bit skips TX DCC start-up calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1145 RAWLANEAONN_DIG_CCA_START_LOOP_CNT (for N = 0; N <= NLANES-1)

- **Description:** Number of Back-to-Back Iterations of CCA to Run when Starting CCA.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)63
- **Exists:** Always

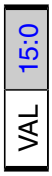


Table 2-1149 Fields for Register: RAWLANEAONN_DIG_CCA_START_LOOP_CNT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the number of back-to-back iterations of CCA to run when starting CCA. **Value After Reset:** 0x64 **Exists:** Always **Reset Mask:** 0xffff

2.1.1146 RAWLANEAONN_DIG_CCA_WAIT_CNT (for N = 0; N <= NLANES-1)

- **Description:** Wait Time Between CCA Iterations.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)64
- **Exists:** Always

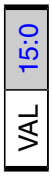


Table 2-1150 Fields for Register: RAWLANEAONN_DIG_CCA_WAIT_CNT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the wait time between CCA iterations in 20us increments. **Value After Reset:** 0xc350 **Exists:** Always **Reset Mask:** 0xffff

2.1.1147 RAWLANEAONN_DIG_VGEN_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** VGEN VDAC Offset Register.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)65
- **Exists:** Always

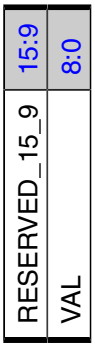


Table 2-1151 Fields for Register: RAWLANEAONN_DIG_VGEN_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the fffset value for VGEN VDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1148 RAWLANEAONN_DIG_RX_VDAC_RANGE_SEL (for N = 0; N <= NLANES-1)

- **Description:** DAC Calibration Extension Mode.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)66
- **Exists:** Always

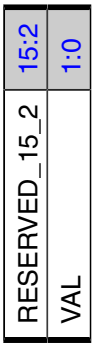


Table 2-1152 Fields for Register: RAWLANEAONN_DIG_RX_VDAC_RANGE_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R/W	These bits are the DAC calibration extension mode value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3

2.1.1149 RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX CDR Detector Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)67
- **Exists:** Always

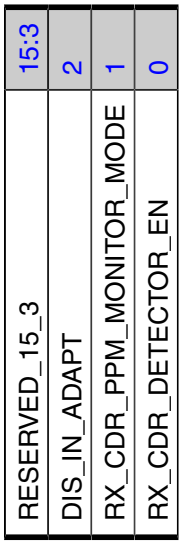


Table 2-1153 Fields for Register: RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	DIS_IN_ADAPT	R/W	This bit disables the CDR Detector state machine for during adaptation. In CDR Reference Tracking Mode (equivalently rx_data_en low), the CDR Detector state machine is not disabled. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	RX_CDR_PPM_MONITOR_MODE	R/W	This bit enables the CDR Detector in Monitor Mode. To enable monitor mode only, disable detector mode and set this field **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1153 Fields for Register: RAWLANEAONN_DIG_RX_CDR_DETECTOR_CTL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
0	RX_CDR_DETECTOR_EN	R/W	This bit enables the CDR Detector in Detector Mode. In detector mode, monitor mode is also enabled. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.1150 RAWLANEAONN_DIG_RX_CDR_RECOVERY_TIME (for N = 0; N <= NLANES-1)

- **Description:** RX CDR Recovery Time in Reference Clock for Intended PPM.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)68
- **Exists:** Always

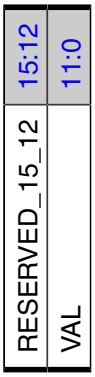


Table 2-1154 Fields for Register: RAWLANEAONN_DIG_RX_CDR_RECOVERY_TIME (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	VAL	R/W	These bits are the RX CDR Recovery Time in the reference clock for intended PPM. This default CDR Recovery Time is determined based on 15000 ppm drift on serial data. This value corresponds to the rxX_cdr_ppm_max input, which is also calculated to detect 15000 ppm on serial data. **Value After Reset:** 0xb0a **Exists:** Always **Reset Mask:** 0xfff

2.1.1151 RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate Code Value Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)69
- **Exists:** Always

Table 2-1155 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored full-rate common-mode TX DCC code for Bank 0. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1152 RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate CM Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6a
- **Exists:** Always

Table 2-1156 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored half-rate common-mode TX DCC code for Bank 0. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1153 RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate CM Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6b
- **Exists:** Always

Table 2-1157 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored full-rate common-mode TX DCC code for Bank 1. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1154 RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate CM Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6c
- **Exists:** Always

Table 2-1158 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored half-rate common-mode TX DCC code for Bank 1. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1155 RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate CM Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6d
- **Exists:** Always

Table 2-1159 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored full-rate common-mode TX DCC code for Bank 2. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1156 RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate CM Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6e
- **Exists:** Always

Table 2-1160 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored half-rate common-mode TX DCC code for Bank 2. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1157 RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate CM Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)6f
- **Exists:** Always

Table 2-1161 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_CM_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored full-rate common-mode TX DCC code for Bank 3. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1158 RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate CM Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)70
- **Exists:** Always

Table 2-1162 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_CM_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	CM_VAL	R/W	These bits are the stored half-rate common-mode TX DCC code for Bank 3. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1159 RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate Differential Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)71
- **Exists:** Always

Table 2-1163 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored the full-rate differential TX DCC code for Bank 0. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1160 RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate Differential Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)72
- **Exists:** Always

Table 2-1164 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored half-rate differential TX DCC code for Bank 0. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1161 RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate Differential Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)73
- **Exists:** Always

Table 2-1165 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored full-rate differential TX DCC code for Bank 1. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1162 RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate Differential Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)74
- **Exists:** Always

Table 2-1166 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored half-rate differential TX DCC code for Bank 1. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1163 RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate Differential Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)75
- **Exists:** Always

Table 2-1167 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored full-rate differential TX DCC code for Bank 2. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1164 RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate Differential Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)76
- **Exists:** Always

Table 2-1168 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored half-rate differential TX DCC code for Bank 2. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1165 RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Full-Rate Differential Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)77
- **Exists:** Always

Table 2-1169 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored full-rate differential TX DCC code for Bank 3. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1166 RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Half-Rate Differential Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)78
- **Exists:** Always

Table 2-1170 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	DIFF_VAL	R/W	These bits are the stored half-rate differential TX DCC code for Bank 3. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1167 RAWLANEAONN_DIG_RX_DCC_FULL_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Data/Bypass Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)79
- **Exists:** Always

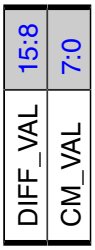


Table 2-1171 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC data/bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC data/bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1168 RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Phase Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7a
- **Exists:** Always

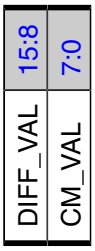


Table 2-1172 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC phase code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC phase code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1169 RAWLANEAONN_DIG_RX_DCC_HALF_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Data/Bypass Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7b
- **Exists:** Always

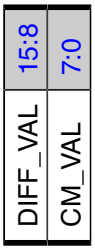


Table 2-1173 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC data/bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC data/bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1170 RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Phase Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7c
- **Exists:** Always

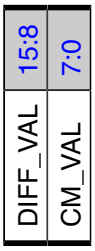


Table 2-1174 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC phase code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC phase code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1171 RAWLANEAONN_DIG_RX_IQ_CAL_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Calibration Codes Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7d
- **Exists:** Always

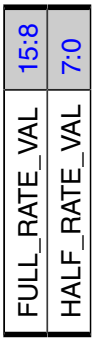


Table 2-1175 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE_VAL	R/W	These bits are the stored full-rate RX IQ calibration code for Bank 0. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE_VAL	R/W	These bits are the stored half-rate RX IQ calibration code for Bank 0. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0xff

2.1.1172 RAWLANEAONN_DIG_RX_CAL_DONE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX Calibration Done Flag for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7e
- **Exists:** Always

Table 2-1176 Fields for Register: RAWLANEAONN_DIG_RX_CAL_DONE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1173 RAWLANEAONN_DIG_RX_DCC_FULL_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Data/Bypass Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)7f
- **Exists:** Always

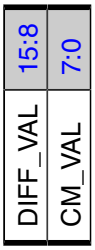


Table 2-1177 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the full-rate differential RX DCC data/bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the full-rate common-mode RX DCC data/bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1174 RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Phase Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)80
- **Exists:** Always

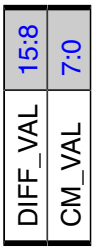


Table 2-1178 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the full-rate differential RX DCC phase code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the full-rate common-mode RX DCC phase code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1175 RAWLANEAONN_DIG_RX_DCC_HALF_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Data/Bypass Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)81
- **Exists:** Always

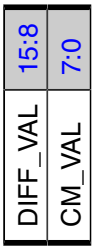


Table 2-1179 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the half-rate differential RX DCC data/bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the half-rate common-mode RX DCC data/bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1176 RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Phase Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)82
- **Exists:** Always

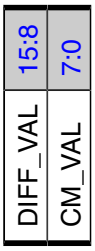


Table 2-1180 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the half-rate differential RX DCC phase code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the half-rate common-mode RX DCC phase code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1177 RAWLANEAONN_DIG_RX_IQ_CAL_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Calibration Codes Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)83
- **Exists:** Always

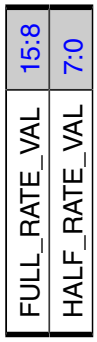


Table 2-1181 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE_VAL	R/W	These bits are the half-rate RX IQ calibration code for Bank 1. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE_VAL	R/W	These bits are the half-rate RX IQ calibration code for Bank 1. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0xff

2.1.1178 RAWLANEAONN_DIG_RX_CAL_DONE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX Calibration Done Flag for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)84
- **Exists:** Always

Table 2-1182 Fields for Register: RAWLANEAONN_DIG_RX_CAL_DONE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1179 RAWLANEAONN_DIG_RX_DCC_FULL_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Data/Bypass Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)85
- **Exists:** Always

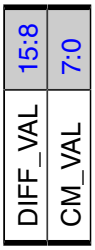


Table 2-1183 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC data/bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC data/bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1180 RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Phase Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)86
- **Exists:** Always

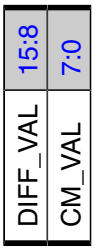


Table 2-1184 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC phase code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC phase code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1181 RAWLANEAONN_DIG_RX_DCC_HALF_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Data/Bypass Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)87
- **Exists:** Always

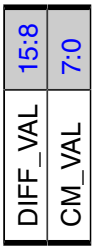


Table 2-1185 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC data/bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC data/bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1182 RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Phase Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)88
- **Exists:** Always

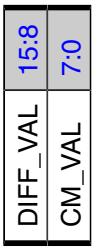


Table 2-1186 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC phase code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC phase code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1183 RAWLANEAONN_DIG_RX_IQ_CAL_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Calibration Codes Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)89
- **Exists:** Always

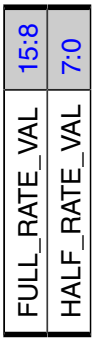


Table 2-1187 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE_VAL	R/W	These bits are the stored half-rate RX IQ calibration code for Bank 2. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE_VAL	R/W	These bits are the stored half-rate RX IQ calibration code for Bank 2. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0xff

2.1.1184 RAWLANEAONN_DIG_RX_CAL_DONE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX Calibration Done Flag for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8a
- **Exists:** Always

Table 2-1188 Fields for Register: RAWLANEAONN_DIG_RX_CAL_DONE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1185 RAWLANEAONN_DIG_RX_DCC_FULL_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Data/Bypass Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8b
- **Exists:** Always

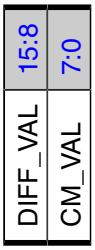


Table 2-1189 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC data/bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC data/bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1186 RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Phase Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8c
- **Exists:** Always

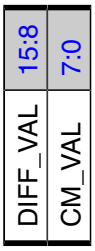


Table 2-1190 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_PHASE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC phase code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC phase code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1187 RAWLANEAONN_DIG_RX_DCC_HALF_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Data/Bypass Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8d
- **Exists:** Always

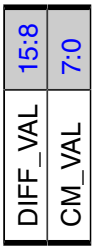


Table 2-1191 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC data/bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC data/bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1188 RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Phase Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8e
- **Exists:** Always

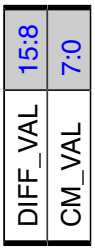


Table 2-1192 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_PHASE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC phase code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC phase code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1189 RAWLANEAONN_DIG_RX_IQ_CAL_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Calibration Codes Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)8f
- **Exists:** Always

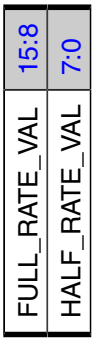


Table 2-1193 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE_VAL	R/W	These bits are the stored half-rate RX IQ calibration code for Bank 3. **Value After Reset:** 0x24 **Exists:** Always **Reset Mask:** 0xff
7:0	HALF_RATE_VAL	R/W	These bits are the stored half-rate RX IQ calibration code for Bank 3. **Value After Reset:** 0x38 **Exists:** Always **Reset Mask:** 0xff

2.1.1190 RAWLANEAONN_DIG_RX_CAL_DONE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX Calibration Done Flag for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)90
- **Exists:** Always

Table 2-1194 Fields for Register: RAWLANEAONN_DIG_RX_CAL_DONE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1191 RAWLANEAONN_DIG_RX_CAL_BANK_SEL (for N = 0; N <= NLANES-1)

- **Description:** Value for RX Bank Select.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)92
- **Exists:** Always

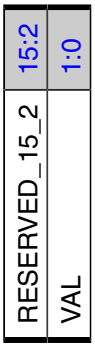


Table 2-1195 Fields for Register: RAWLANEAONN_DIG_RX_CAL_BANK_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R	These bits are the value for the RX Bank Select. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1192 RAWLANEAONN_DIG_TX_DCC_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)93
- **Exists:** Always

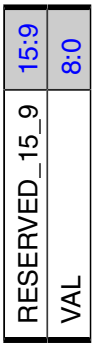


Table 2-1196 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R	These bits are the selected TX DCC common-mode code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1193 RAWLANEAONN_DIG_RX_DCC_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)94
- **Exists:** Always

Table 2-1197 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_DIFF_CODE	R	These bits are the value for Selected RX DCC differential code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_CM_CODE	R	These bits are the value for Selected RX DCC common-mode code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1194 RAWLANEAONN_DIG_RX_DCC_PHASE_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Phase Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)95
- **Exists:** Always

Table 2-1198 Fields for Register: RAWLANEAONN_DIG_RX_DCC_PHASE_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_PHASE_DIFF_CODE	R	These bits are the value for the selected RX DCC Phase differential code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_PHASE_CM_CODE	R	These bits are the value for the selected RX DCC Phase common-mode code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1195 RAWLANEAONN_DIG_RX_IQ_CAL (for N = 0; N <= NLANES-1)

- **Description:** Selected Value for RX IQ Calibration Code.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)96
- **Exists:** Always

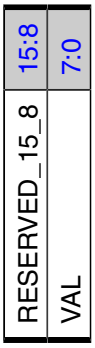


Table 2-1199 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R	Value for Selected RX IQ Calibration Code (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1196 RAWLANEAONN_DIG_RX_CAL_DONE (for N = 0; N <= NLANES-1)

- **Description:** Selected RX Calibration Bank Valid Calibrated Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)97
- **Exists:** Always

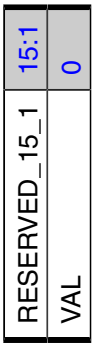


Table 2-1200 Fields for Register: RAWLANEAONN_DIG_RX_CAL_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R	This bit indicates if the selected RX Calibration bank has a valid calibrated value. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1197 RAWLANEAONN_DIG_TX_CAL_DONE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Calibration Done Flag for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)98
- **Exists:** Always

Table 2-1201 Fields for Register: RAWLANEAONN_DIG_TX_CAL_DONE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	TX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1198 RAWLANEAONN_DIG_TX_CAL_DONE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Calibration Done Flag for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)99
- **Exists:** Always

Table 2-1202 Fields for Register: RAWLANEAONN_DIG_TX_CAL_DONE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	TX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1199 RAWLANEAONN_DIG_TX_CAL_DONE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Calibration Done Flag for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)9a
- **Exists:** Always

Table 2-1203 Fields for Register: RAWLANEAONN_DIG_TX_CAL_DONE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	TX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1200 RAWLANEAONN_DIG_TX_CAL_DONE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC Calibration Done Flag for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)9b
- **Exists:** Always

Table 2-1204 Fields for Register: RAWLANEAONN_DIG_TX_CAL_DONE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	TX_CAL_HALF_DONE	R/W	This bit indicates if half rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	TX_CAL_FULL_DONE	R/W	This bit indicates if full rate calibration is complete for this bank **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1201 RAWLANEAONN_DIG_TX_CAL_DONE (for N = 0; N <= NLANES-1)

- **Description:** Selected TX DCC Calibration Bank Valid Calibrated Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a0
- **Exists:** Always

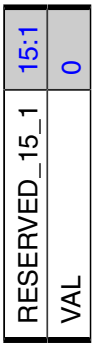


Table 2-1205 Fields for Register: RAWLANEAONN_DIG_TX_CAL_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R	This bit indicates if the selected TX DCC Calibration bank has a valid calibrated value. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1202 RAWLANEAONN_DIG_RX_IQ_CAL_DIVN (for N = 0; N <= NLANES-1)

- **Description:** Value for RX IQ Calibrated Divide-by N Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a2
- **Exists:** Always

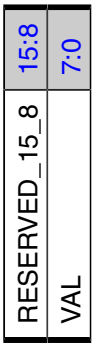


Table 2-1206 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CAL_DIVN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	These bits are the value for the RX IQ calibrated divide-by N value. **Value After Reset:** 0x10 **Exists:** Always **Reset Mask:** 0xff

2.1.1203 RAWLANEAONN_DIG_MEM_BREAKPOINT_2 (for N = 0; N <= NLANES-1)

- **Description:** FSM Breakpoint 2 on SRAM Address for Debugging.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a3
- **Exists:** Always

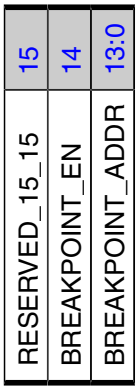


Table 2-1207 Fields for Register: RAWLANEAONN_DIG_MEM_BREAKPOINT_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15	RESERVED_15_15	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
14	BREAKPOINT_EN	R/W	This bit enables the Breakpoint 2. Clearing this bit automatically resumes the FSM if it is stopped at MEM_BREAKPOINT_2.BREAKPOINT_ADDR. This control register resides in the AON domain, which can be used to stop/resume the power-gating routine performed by firmware. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
13:0	BREAKPOINT_ADDR	R/W	These bits are the memory address for Breakpoint 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1204 RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Even High TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a4
- **Exists:** Always

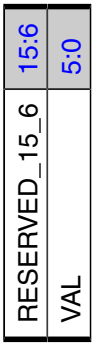


Table 2-1208 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even High TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1205 RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Even Low TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a5
- **Exists:** Always

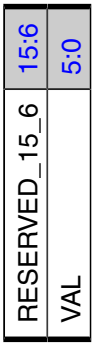


Table 2-1209 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even Low TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1206 RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd High TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a6
- **Exists:** Always

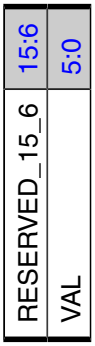


Table 2-1210 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Odd High TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1207 RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a7
- **Exists:** Always

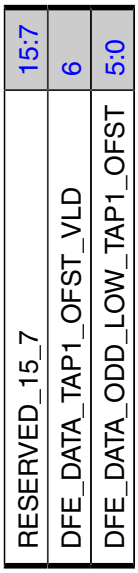


Table 2-1211 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA_TAP1_OFST_VLD	R/W	This bit is the TAP1 Offset Valid. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA_ODD_LOW_TAP1_OFST	R/W	These bits are the offset value for DFE Data Odd Low TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1208 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted value of DFE TAP6.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a8
- **Exists:** Always

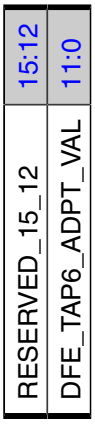


Table 2-1212 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP6_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP6 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1209 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP7.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)a9
- **Exists:** Always

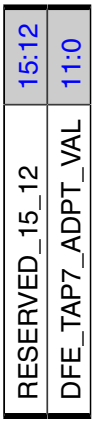


Table 2-1213 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP7_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP7 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1210 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP8.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)aa
- **Exists:** Always

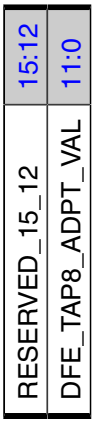


Table 2-1214 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP8_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP8 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1211 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted value of DFE TAP6 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ab
- **Exists:** Always

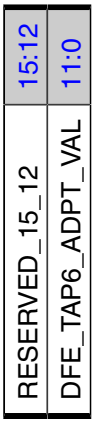


Table 2-1215 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP6_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP6 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1212 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP7 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ac
- **Exists:** Always

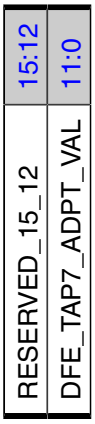


Table 2-1216 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP7_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP7 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1213 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP8 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ad
- **Exists:** Always

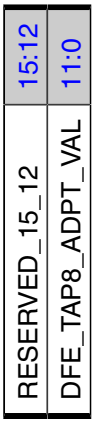


Table 2-1217 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP8_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP8 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1214 RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS (for N = 0; N <= NLANES-1)

- **Description:** Positions for RX DFE Floating TAPs 0-3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ae
- **Exists:** Always

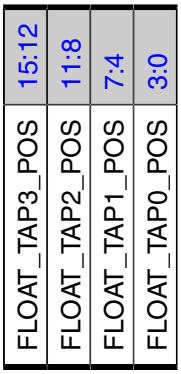


Table 2-1218 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	FLOAT_TAP3_POS	R/W	These bits are the positions for RX DFE Floating TAP3. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
11:8	FLOAT_TAP2_POS	R/W	These bits are the positions for RX DFE Floating TAP2. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
7:4	FLOAT_TAP1_POS	R/W	These bits are the positions for RX DFE Floating TAP1. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
3:0	FLOAT_TAP0_POS	R/W	These bits are the positions for RX DFE Floating TAP0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf

2.1.1215 RAWLANEAONN_DIG_DFE_FLOAT_TAP0 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP1 and Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)af
- **Exists:** Always

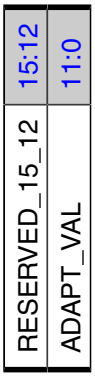


Table 2-1219 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1216 RAWLANEAONN_DIG_DFE_FLOAT_TAP1 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP2 and Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b0
- **Exists:** Always

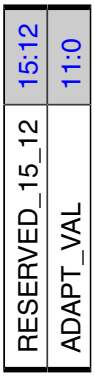


Table 2-1220 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1217 RAWLANEAONN_DIG_DFE_FLOAT_TAP2 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP3 and Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b1
- **Exists:** Always

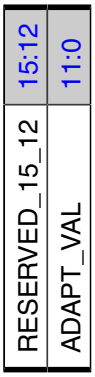


Table 2-1221 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1218 RAWLANEAONN_DIG_DFE_FLOAT_TAP3 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP4 and Control.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b2
- **Exists:** Always

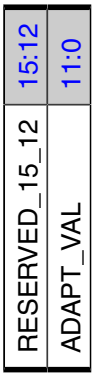


Table 2-1222 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1219 RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B1 (for N = 0; N <= NLANES-1)

- **Description:** Positions for RX DFE Floating TAPs 0-3 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b4
- **Exists:** Always

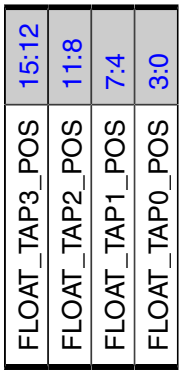


Table 2-1223 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	FLOAT_TAP3_POS	R/W	These bits are the positions for RX DFE Floating TAP 3. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
11:8	FLOAT_TAP2_POS	R/W	These bits are the positions for RX DFE Floating TAP 2. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
7:4	FLOAT_TAP1_POS	R/W	These bits are the positions for RX DFE Floating TAP 1. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
3:0	FLOAT_TAP0_POS	R/W	These bits are the positions for RX DFE Floating TAP 0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf

2.1.1220 RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B1 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP1 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b5
- **Exists:** Always

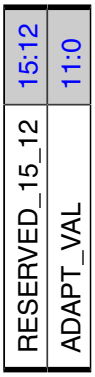


Table 2-1224 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted values for RX DFE Floating TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1221 RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B1 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP2 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b6
- **Exists:** Always

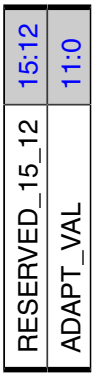


Table 2-1225 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1222 RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B1 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP3 and its Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b7
- **Exists:** Always

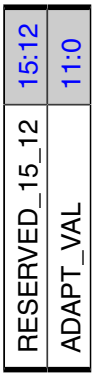


Table 2-1226 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted Value for RX DFE Floating TAP3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1223 RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B1 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP4 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)b8
- **Exists:** Always

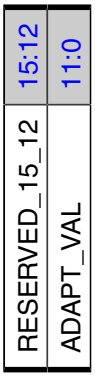


Table 2-1227 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1224 RAWLANEAONN_DIG_RX_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of ATT for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ba
- **Exists:** Always

Table 2-1228 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_ADPT_VAL	R/W	These bits are the stored RX adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1225 RAWLANEAONN_DIG_RX_ADPT_VGA_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of VGA for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)bb
- **Exists:** Always

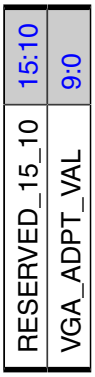


Table 2-1229 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_VGA_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_ADPT_VAL	R/W	These bits are the stored RX adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1226 RAWLANEAONN_DIG_RX_ADPT_CTLE_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of CTLE for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)bc
- **Exists:** Always

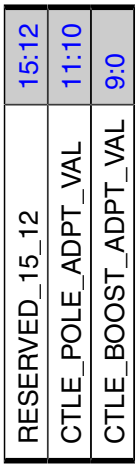


Table 2-1230 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_CTLE_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_ADPT_VAL	R/W	These bits are the stored RX adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_ADPT_VAL	R/W	These bits are the stored RX adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1227 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)bd
- **Exists:** Always

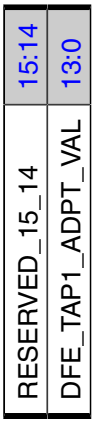


Table 2-1231 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP1_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1228 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP2 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)be
- **Exists:** Always

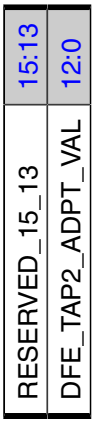


Table 2-1232 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP2_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1229 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP3 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)bf
- **Exists:** Always

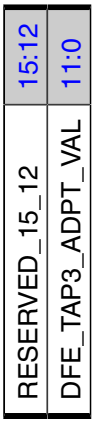


Table 2-1233 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP3_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP3_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP3 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1230 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP4 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c0
- **Exists:** Always

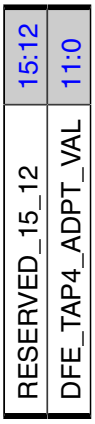


Table 2-1234 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP4_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP4_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP4 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1231 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP5 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c1
- **Exists:** Always

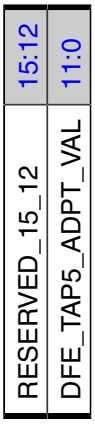


Table 2-1235 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP5_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP5_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP5 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1232 RAWLANEAONN_DIG_RX_ADPT_IQ_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of IQ for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c2
- **Exists:** Always

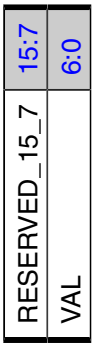


Table 2-1236 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_IQ_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	These bits are the RX adapted value of IQ for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1233 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Even Slicer for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c3
- **Exists:** Always

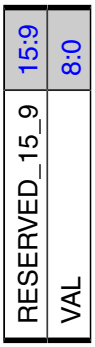


Table 2-1237 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_EVEN_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the RX adapted value of reference level for error even slicer for Bank 2. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1234 RAWLANEAONN_DIG_RX_ADAPT_DONE_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adaptation Done Status for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c4
- **Exists:** Always

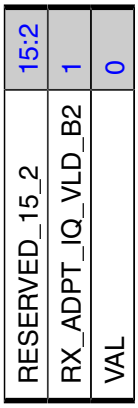


Table 2-1238 Fields for Register: RAWLANEAONN_DIG_RX_ADAPT_DONE_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_ADPT_IQ_VLD_B2	R/W	This bit indicates if the RX adapted value of IQ for bank 2 is valid. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	VAL	R/W	This bit indicates whether RX adaptation for Bank 2 has completed. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1235 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP6 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c5
- **Exists:** Always

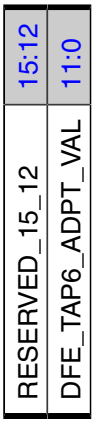


Table 2-1239 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP6_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP6_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP6 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1236 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP7 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c6
- **Exists:** Always

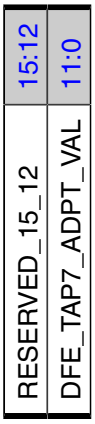


Table 2-1240 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP7_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP7_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP7 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1237 RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of DFE TAP8 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c7
- **Exists:** Always

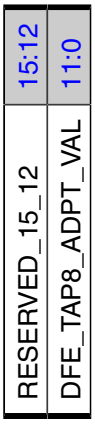


Table 2-1241 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_TAP8_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	DFE_TAP8_ADPT_VAL	R/W	These bits are the stored RX adapted DFE TAP8 value. Bits 4:0 are dither bits. **Value After Reset:** 0x800 **Exists:** Always **Reset Mask:** 0xfff

2.1.1238 RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B2 (for N = 0; N <= NLANES-1)

- **Description:** Positions for RX DFE Floating TAPs 0-3 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c8
- **Exists:** Always

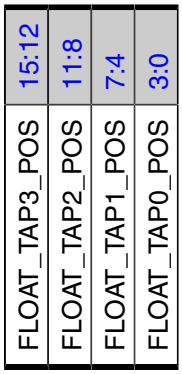


Table 2-1242 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP_POS_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	FLOAT_TAP3_POS	R/W	The bits are the positions for RX DFE Floating TAP3. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf
11:8	FLOAT_TAP2_POS	R/W	The bits are the positions for RX DFE Floating TAP2. **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0xf
7:4	FLOAT_TAP1_POS	R/W	The bits are the positions for RX DFE Floating TAP1. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0xf
3:0	FLOAT_TAP0_POS	R/W	The bits are the positions for RX DFE Floating TAP0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0xf

2.1.1239 RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B2 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP1 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)c9
- **Exists:** Always

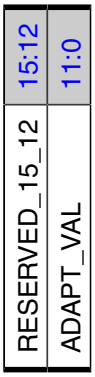


Table 2-1243 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP0_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1240 RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B2 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP2 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ca
- **Exists:** Always

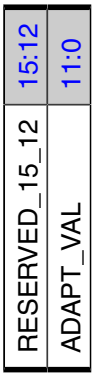


Table 2-1244 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP1_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1241 RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B2 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP3 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)cb
- **Exists:** Always

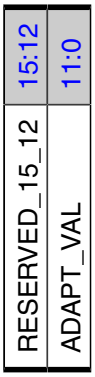


Table 2-1245 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP2_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1242 RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B2 (for N = 0; N <= NLANES-1)

- **Description:** Adapted Value for RX DFE Floating TAP4 and Control for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)cc
- **Exists:** Always

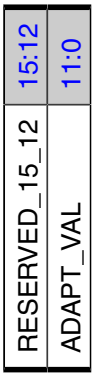


Table 2-1246 Fields for Register: RAWLANEAONN_DIG_DFE_FLOAT_TAP3_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	ADAPT_VAL	R/W	These bits are the adapted value for RX DFE Floating TAP4. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1243 RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset values for RX DFE Data1 Even High vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ce
- **Exists:** Always

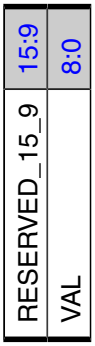


Table 2-1247 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for DFE Data1 Even High vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1244 RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)cf
- **Exists:** Always

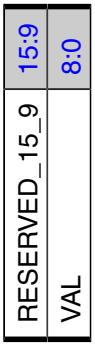


Table 2-1248 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for DFE Data1 Even Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1245 RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd High vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d0
- **Exists:** Always

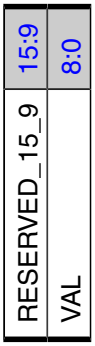


Table 2-1249 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for DFE Data1 Odd High vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1246 RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset values for RX DFE Data1 Odd Low vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d1
- **Exists:** Always

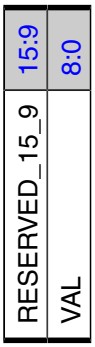


Table 2-1250 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for DFE Data1 Odd Low vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1247 RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Even High TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d2
- **Exists:** Always

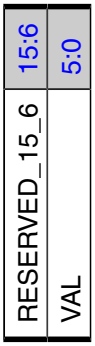


Table 2-1251 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even High TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1248 RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset values for RX DFE Data Even Low TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d3
- **Exists:** Always

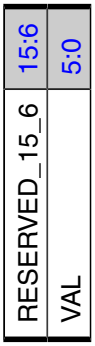


Table 2-1252 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even Low TAP1 for Bank 1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1249 RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd High TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d4
- **Exists:** Always

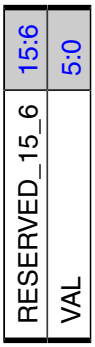


Table 2-1253 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Odd High TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1250 RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d5
- **Exists:** Always

Table 2-1254 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA_TAP1_OFST_VLD_B1	R/W	TAP1 Offset Valid for Bank 1 (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA_ODD_LOW_TAP1_OFST_B1	R/W	These bits are the offset value for DFE Data Odd Low TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1251 RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Even High TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d6
- **Exists:** Always

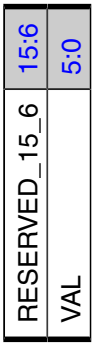


Table 2-1255 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even High TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1252 RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Even Low TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d7
- **Exists:** Always

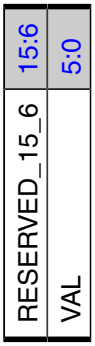


Table 2-1256 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_EVEN_LOW_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Even Low TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1253 RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd High TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d8
- **Exists:** Always

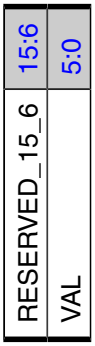


Table 2-1257 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data Odd High TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1254 RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data Odd Low TAP1 and Offset Valid for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)d9
- **Exists:** Always

Table 2-1258 Fields for Register: RAWLANEAONN_DIG_DFE_DATA_ODD_LOW_TAP1_OFST_W_VLD_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA_TAP1_OFST_VLD_B2	R/W	This bit is the TAP1 offset valid for Bank 2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA_ODD_LOW_TAP1_OFST_B2	R/W	These bits are the offset value for DFE Data Odd Low TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1255 RAWLANEAONN_DIG_AFE_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX AFE VGA2 iDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)da
- **Exists:** Always

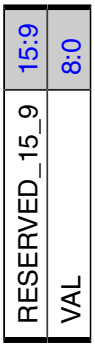


Table 2-1259 Fields for Register: RAWLANEAONN_DIG_AFE_VGA2_IDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset value for AFE VGA2 iDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1256 RAWLANEAONN_DIG_ADPT_CTL_24 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 24.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)db
- **Exists:** Always

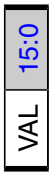


Table 2-1260 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_24 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1257 RAWLANEAONN_DIG_ADPT_CTL_25 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 25.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)dc
- **Exists:** Always

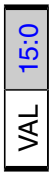


Table 2-1261 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_25 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1258 RAWLANEAONN_DIG_ADPT_CTL_26 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 26.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)dd
- **Exists:** Always

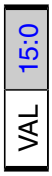


Table 2-1262 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_26 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1259 RAWLANEAONN_DIG_ADPT_CTL_27 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 27.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)de
- **Exists:** Always

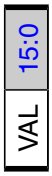


Table 2-1263 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_27 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1260 RAWLANEAONN_DIG_ADPT_CTL_28 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 28.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)df
- **Exists:** Always

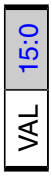


Table 2-1264 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_28 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1261 RAWLANEAONN_DIG_RX_IQ_CTL (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Adaptation Control Register 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e0
- **Exists:** Always

Table 2-1265 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CTL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:4	IQ_ADPT_SWEEP_LVL	R/W	These bits are the vertical sweep level for IQ adaptation. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x7
3:0	IQ_ADPT_OFST	R/W	These bits are the initial IQ offset for IQ adaptation. This is the additional IQ offset that is added before doing the IQ adaptation. **Value After Reset:** 0x4 **Exists:** Always **Reset Mask:** 0xf

2.1.1262 RAWLANEAONN_DIG_RX_IQ_CTL_1 (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Adaptation Control Register 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e1
- **Exists:** Always

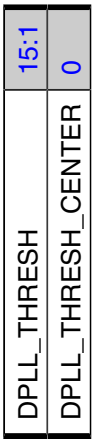


Table 2-1266 Fields for Register: RAWLANEAONN_DIG_RX_IQ_CTL_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	DPLL_THRESH	R/W	These bits are the DPLL frequency threshold. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x7fff
0	DPLL_THRESH_CENTER	R/W	This bit indicates the center of the DPLL threshold value. ■ 0: Absolute center (for example, 'h8192) ■ 1: Center is relative to the current value of DPLL frequency. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.1263 RAWLANEAONN_DIG_RX_ADPT_IQ_LIMIT (for N = 0; N <= NLANES-1)

- **Description:** RX IQ Adaptation Offset Limit.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e2
- **Exists:** Always

Table 2-1267 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_IQ_LIMIT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	RIGHT_LIMIT	R/W	These bits are the maximum IQ offset from the calibrated IQ for right-side IQ sweep. This is a signed number in sign and magnitude format; bit 4 is the sign bit, and bits [3:0] are the magnitude. **Value After Reset:** 0x7 **Exists:** Always **Reset Mask:** 0x1f
4:0	LEFT_LIMIT	R/W	These bits are the maximum IQ offset from the calibrated IQ for left-side IQ sweep. This is a signed number in sign and magnitude format; bit 4 is the sign bit, and bits [3:0] are the magnitude. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.1264 RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC Half Differential Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e3
- **Exists:** Always

Table 2-1268 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	HALF_RATE	R	These bits are the selected TX DCC differential half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1265 RAWLANEAONN_DIG_TX_DCC_HALF_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC HALF Common-Mode Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e4
- **Exists:** Always

Table 2-1269 Fields for Register: RAWLANEAONN_DIG_TX_DCC_HALF_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	HALF_RATE	R	These bits are the selected TX DCC common-mode half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1266 RAWLANEAONN_DIG_RX_DCC_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Differential Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e5
- **Exists:** Always

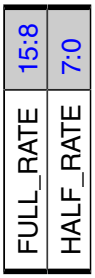


Table 2-1270 Fields for Register: RAWLANEAONN_DIG_RX_DCC_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC differential full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC differential half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1267 RAWLANEAONN_DIG_RX_DCC_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Common-Mode Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e6
- **Exists:** Always

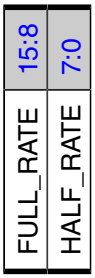


Table 2-1271 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC common-mode full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC common-mode half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1268 RAWLANEAONN_DIG_RX_DCC_PHASE_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Phase Differential Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e7
- **Exists:** Always

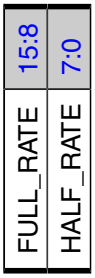


Table 2-1272 Fields for Register: RAWLANEAONN_DIG_RX_DCC_PHASE_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC Phase differential full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC Phase differential half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1269 RAWLANEAONN_DIG_RX_DCC_PHASE_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Phase Common-Mode Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e8
- **Exists:** Always

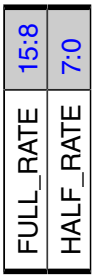


Table 2-1273 Fields for Register: RAWLANEAONN_DIG_RX_DCC_PHASE_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC Phase common-mode full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC Phase common-mode half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1270 RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even High TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)e9
- **Exists:** Always

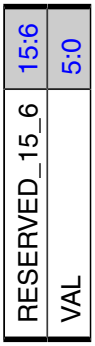


Table 2-1274 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even High TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1271 RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even Low TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ea
- **Exists:** Always

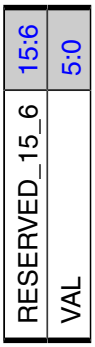


Table 2-1275 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even Low TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1272 RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd High TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)eb
- **Exists:** Always

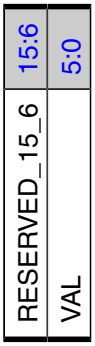


Table 2-1276 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Odd High TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1273 RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ec
- **Exists:** Always

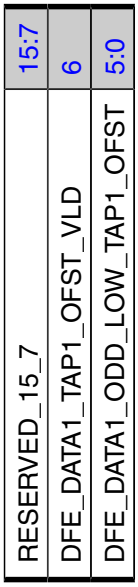


Table 2-1277 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA1_TAP1_OFST_VLD	R/W	This bit is the TAP1 Offset Valid. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA1_ODD_LOW_TAP1_OFST	R/W	These bits are the offset value for DFE Data1 Odd Low TAP1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1274 RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even High TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ed
- **Exists:** Always

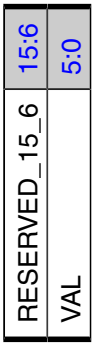


Table 2-1278 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even High TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1275 RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even Low TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ee
- **Exists:** Always

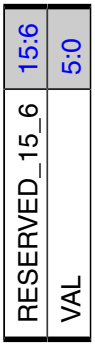


Table 2-1279 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even Low TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1276 RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd High TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ef
- **Exists:** Always

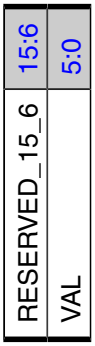


Table 2-1280 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Odd High TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1277 RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f0
- **Exists:** Always

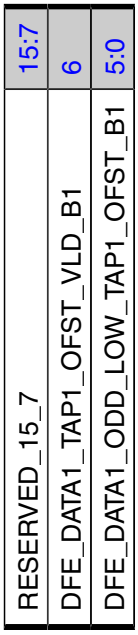


Table 2-1281 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA1_TAP1_OFST_VLD_B1	R/W	This bit is the TAP1 Offset Valid for Bank 1. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA1_ODD_LOW_TAP1_OFST_B1	R/W	These bits are the offset value for DFE Data1 Odd Low TAP1 for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1278 RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even High TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f1
- **Exists:** Always

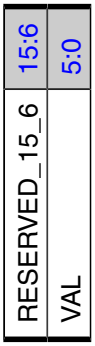


Table 2-1282 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even High TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1279 RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Even Low TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f2
- **Exists:** Always

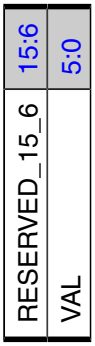


Table 2-1283 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_EVEN_LOW_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Even Low TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1280 RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd High TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f3
- **Exists:** Always

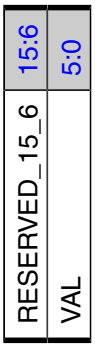


Table 2-1284 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_HIGH_TAP1_OFST_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the offset value for DFE Data1 Odd High TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1281 RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD_B2 (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE Data1 Odd Low TAP1 and Offset Valid for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f4
- **Exists:** Always

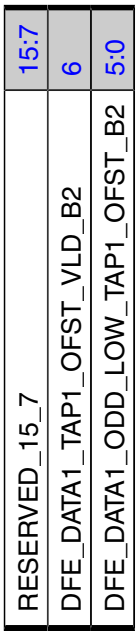


Table 2-1285 Fields for Register: RAWLANEAONN_DIG_DFE_DATA1_ODD_LOW_TAP1_OFST_W_VLD_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6	DFE_DATA1_TAP1_OFST_VLD_B2	R/W	This bit is the TAP1 Offset Valid for Bank 2. (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	DFE_DATA1_ODD_LOW_TAP1_OFST_B2	R/W	These bits are the offset value for DFE Data1 Odd Low TAP1 for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3f

2.1.1282 RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Bypass Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f6
- **Exists:** Always

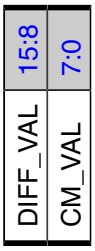


Table 2-1286 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1283 RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Bypass Code Value for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f7
- **Exists:** Always

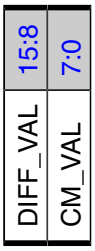


Table 2-1287 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC bypass code for Bank 0. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1284 RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Bypass Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f8
- **Exists:** Always

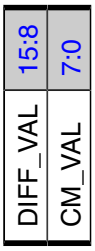


Table 2-1288 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1285 RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Bypass Code Value for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)f9
- **Exists:** Always

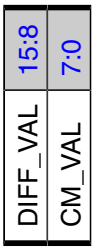


Table 2-1289 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC bypass code for Bank 1. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1286 RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Bypass Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)fa
- **Exists:** Always

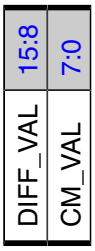


Table 2-1290 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1287 RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Bypass Code Value for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)fb
- **Exists:** Always

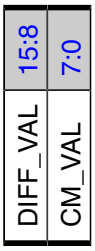


Table 2-1291 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC bypass code for Bank 2. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1288 RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Full-Rate Bypass Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)fc
- **Exists:** Always

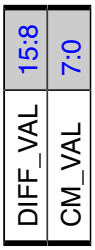


Table 2-1292 Fields for Register: RAWLANEAONN_DIG_RX_DCC_FULL_BYPASS_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored full-rate differential RX DCC bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored full-rate common-mode RX DCC bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1289 RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC Half-Rate Bypass Code Value for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)fd
- **Exists:** Always

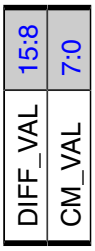


Table 2-1293 Fields for Register: RAWLANEAONN_DIG_RX_DCC_HALF_BYPASS_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	DIFF_VAL	R/W	These bits are the stored half-rate differential RX DCC bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff
7:0	CM_VAL	R/W	These bits are the stored half-rate common-mode RX DCC bypass code for Bank 3. **Value After Reset:** 0x80 **Exists:** Always **Reset Mask:** 0xff

2.1.1290 RAWLANEAONN_DIG_RX_DCC_BYPASS_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Bypass Differential Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)fe
- **Exists:** Always

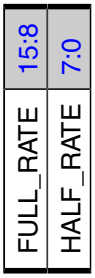


Table 2-1294 Fields for Register: RAWLANEAONN_DIG_RX_DCC_BYPASS_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC Bypass Differential Full-Rate Code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC bypass differential half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1291 RAWLANEAONN_DIG_RX_DCC_BYPASS_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Bypass Common-Mode Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+0)ff
- **Exists:** Always

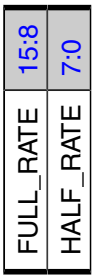


Table 2-1295 Fields for Register: RAWLANEAONN_DIG_RX_DCC_BYPASS_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	FULL_RATE	R	These bits are the selected RX DCC bypass common-mode full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	HALF_RATE	R	These bits are the selected RX DCC bypass common-mode half-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1292 RAWLANEAONN_DIG_RX_DCC_BYPASS_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for RX DCC Bypass Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)00
- **Exists:** Always

Table 2-1296 Fields for Register: RAWLANEAONN_DIG_RX_DCC_BYPASS_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_DCC_BYPASS_DIFF_CODE	R	These bits are the value for the selected RX DCC Bypass differential code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
7:0	RX_DCC_BYPASS_CM_CODE	R	These bits are the value for selected RX DCC Bypass common-mode code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1293 RAWLANEAONN_DIG_TX_CAL_BANK_SEL (for N = 0; N <= NLANES-1)

- **Description:** Value for TX Bank Select.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)01
- **Exists:** Always

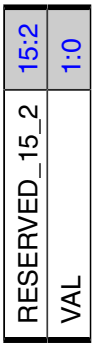


Table 2-1297 Fields for Register: RAWLANEAONN_DIG_TX_CAL_BANK_SEL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R	These bits are the value for TX Bank Select. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1294 RAWLANEAONN_DIG_TX_DCC_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)02
- **Exists:** Always

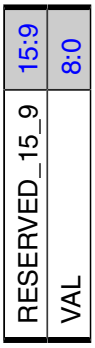


Table 2-1298 Fields for Register: RAWLANEAONN_DIG_TX_DCC_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R	These bits are the selected TX DCC differential code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1295 RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC Full Differential Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)03
- **Exists:** Always

Table 2-1299 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_DIFF_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	FULL_RATE	R	These bits are the selected TX DCC differential full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1296 RAWLANEAONN_DIG_TX_DCC_FULL_CM_CODE (for N = 0; N <= NLANES-1)

- **Description:** Selected Values for TX DCC Full Common-Mode Codes.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)04
- **Exists:** Always

Table 2-1300 Fields for Register: RAWLANEAONN_DIG_TX_DCC_FULL_CM_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	FULL_RATE	R	These bits are the selected TX DCC common-mode full-rate code. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1297 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Odd Slicer for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)05
- **Exists:** Always

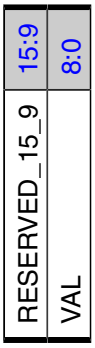


Table 2-1301 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the RX adapted value of reference level for error odd slicer for Bank 0. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1298 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Odd Slicer for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)06
- **Exists:** Always

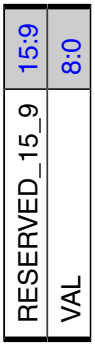


Table 2-1302 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the RX Adapted value of reference level for error odd slicer for Bank 1. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1299 RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted Value of Reference Level for Error Odd Slicer for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)07
- **Exists:** Always

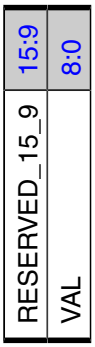


Table 2-1303 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_REF_ERR_ODD_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the RX Adapted value of reference level for error odd slicer for Bank 2. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0x1ff

2.1.1300 RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapt Done Flag for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)08
- **Exists:** Always

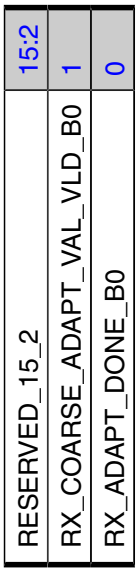


Table 2-1304 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_COARSE_ADAPT_VAL_VLD_B0	R/W	This bit indicates if RX coarse adapt value is valid for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ADAPT_DONE_B0	R/W	This bit indicates if RX coarse adapt is complete for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1301 RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapt Done Flag for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)09
- **Exists:** Always

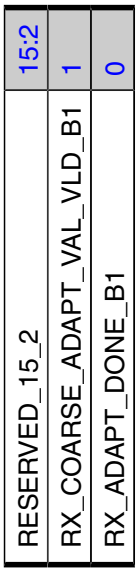


Table 2-1305 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_COARSE_ADAPT_VAL_VLD_B1	R/W	This bit indicates if RX coarse adapt value is valid for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ADAPT_DONE_B1	R/W	This bit indicates if RX coarse adapt is complete for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1302 RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapt Done Flag for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0a
- **Exists:** Always

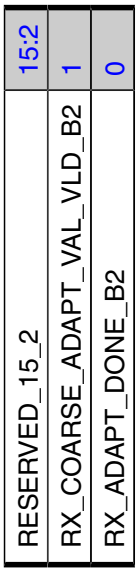


Table 2-1306 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	RX_COARSE_ADAPT_VAL_VLD_B2	R/W	This bit indicates if RX coarse adapt value is valid for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	RX_ADAPT_DONE_B2	R/W	This bit indicates if RX coarse adapt is complete for this bank. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1303 RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE (for N = 0; N <= NLANES-1)

- **Description:** Selected RX Coarse Adapt Bank Valid Sdapt Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0b
- **Exists:** Always

Table 2-1307 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DONE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX_ADAPT_DONE	R	This bit indicates if the selected RX coarse adapt bank has a valid adapt value. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1304 RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of ATT.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0c
- **Exists:** Always

Table 2-1308 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1305 RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of VGA.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0d
- **Exists:** Always

Table 2-1309 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1306 RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of CTLE.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0e
- **Exists:** Always

Table 2-1310 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1307 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)0f
- **Exists:** Always

Table 2-1311 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1308 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)10
- **Exists:** Always

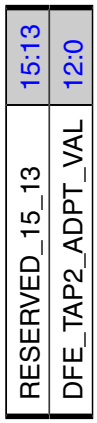


Table 2-1312 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1309 RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of IQ.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)11
- **Exists:** Always

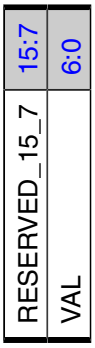


Table 2-1313 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	These bits are the RX coarse adapted value of IQ. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1310 RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of ATT for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)12
- **Exists:** Always

Table 2-1314 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1311 RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of VGA for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)13
- **Exists:** Always

Table 2-1315 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1312 RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of CTLE for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)14
- **Exists:** Always

Table 2-1316 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1313 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP1 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)15
- **Exists:** Always

Table 2-1317 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1314 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP2 for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)16
- **Exists:** Always

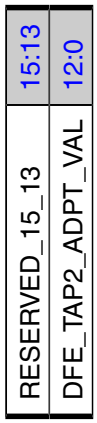


Table 2-1318 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1315 RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of IQ for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)17
- **Exists:** Always

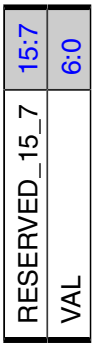


Table 2-1319 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	These bits are the RX coarse adapted value of IQ for Bank 1 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1316 RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of ATT for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)18
- **Exists:** Always

Table 2-1320 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_ATT_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	ATT_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted ATT value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1317 RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of VGA for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)19
- **Exists:** Always

Table 2-1321 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_VGA_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:0	VGA_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted VGA value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1318 RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of CTLE for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1a
- **Exists:** Always

Table 2-1322 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_CTLE_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:10	CTLE_POLE_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE pole value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
9:0	CTLE_BOOST_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted CTLE boost value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1319 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP1 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1b
- **Exists:** Always

Table 2-1323 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP1_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:14	RESERVED_15_14	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
13:0	DFE_TAP1_COARSE_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP1 value. Bits 4:0 are dither bits. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3fff

2.1.1320 RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of DFE TAP2 for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1c
- **Exists:** Always

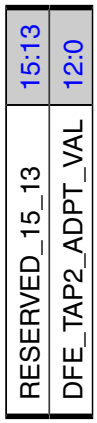


Table 2-1324 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_DFE_TAP2_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:13	RESERVED_15_13	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
12:0	DFE_TAP2_ADPT_VAL	R/W	These bits are the stored RX coarse adapted DFE TAP2 value. Bits 4:0 are dither bits. **Value After Reset:** 0x1000 **Exists:** Always **Reset Mask:** 0x1fff

2.1.1321 RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapted Value of IQ for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1d
- **Exists:** Always

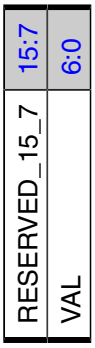


Table 2-1325 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_IQ_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:7	RESERVED_15_7	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
6:0	VAL	R/W	RX coarse adapted value of IQ for Bank 2 **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7f

2.1.1322 RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_2 (for N = 0; N <= NLANES-1)

- **Description:** Set to Skip Firmware Start-up Coarse Adapt Algorithms.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1e
- **Exists:** Always

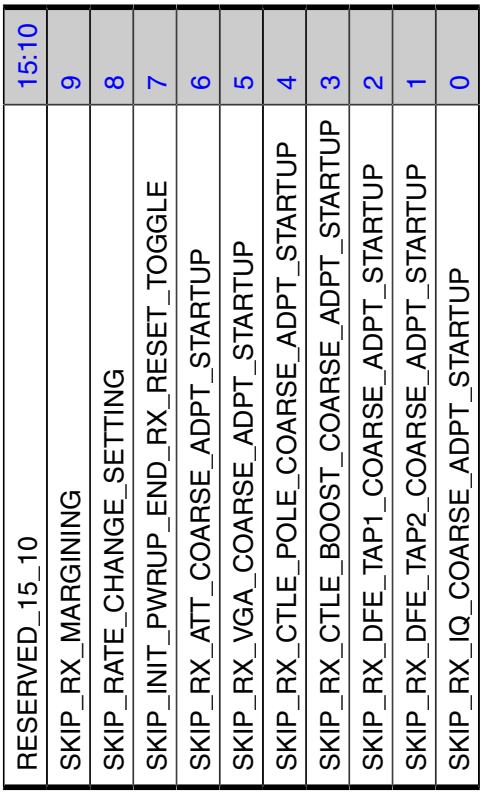


Table 2-1326 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9	SKIP_RX_MARGINING	R/W	This bit skips RX margining. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
8	SKIP_RATE_CHANGE_SETTING	R/W	This bit skips rate change setting. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1326 Fields for Register: RAWLANEAONN_DIG_RX_STARTUP_ALGO_CTL_2 (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
7	SKIP_INIT_PWRUP_END_RX_RESET_TOGGLE	R/W	This bit skips init pwrup end rx_reset toggle. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	SKIP_RX_ATT_COARSE_ADPT_STARTUP	R/W	This bit skips RX ATT coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
5	SKIP_RX_VGA_COARSE_ADPT_STARTUP	R/W	This bit skips RX VGA coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SKIP_RX_CTLE_POLE_COARSE_ADPT_STARTUP	R/W	This bit skips RX CTLE_POLE coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SKIP_RX_CTLE_BOOST_COARSE_ADPT_STARTUP	R/W	This bit skips RX CTLE_BOOST coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SKIP_RX_DFE_TAP1_COARSE_ADPT_STARTUP	R/W	This bit skips RX DFE_TAP1 coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_RX_DFE_TAP2_COARSE_ADPT_STARTUP	R/W	This bit skips RX DFE_TAP2 coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_RX_IQ_COARSE_ADPT_STARTUP	R/W	This bit skips RX IQ coarse adapt. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1323 RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC ctl Range Value Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)1f
- **Exists:** Always

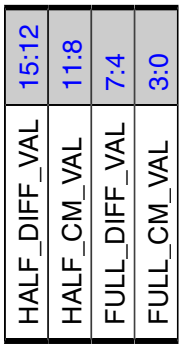


Table 2-1327 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode TX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode TX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode TX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode TX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1324 RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC ctl Range Value Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)20
- **Exists:** Always

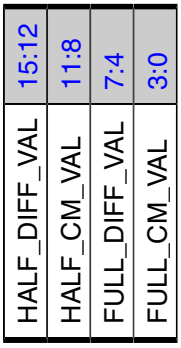


Table 2-1328 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode TX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode TX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode TX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode TX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1325 RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC ctl Range Value Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)21
- **Exists:** Always

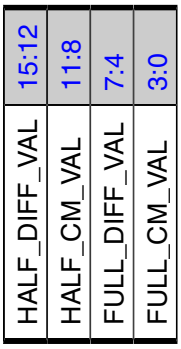


Table 2-1329 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode TX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode TX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode TX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode TX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1326 RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** TX DCC ctl Range Value Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)22
- **Exists:** Always

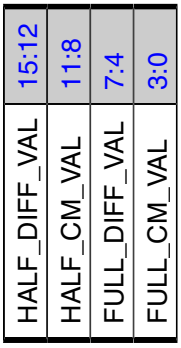


Table 2-1330 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode TX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode TX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode TX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode TX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1327 RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC ctl range value Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)23
- **Exists:** Always

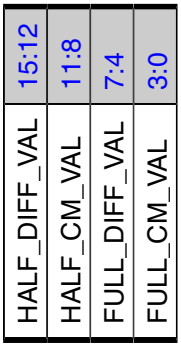


Table 2-1331 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode RX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode RX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode RX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode RX DCC ctl range for Bank 0. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1328 RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC ctl range value Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)24
- **Exists:** Always

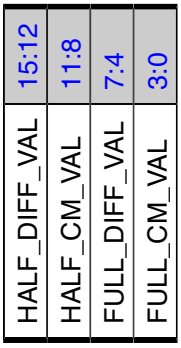


Table 2-1332 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode RX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode RX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode RX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode RX DCC ctl range for Bank 1. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1329 RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC ctl range value Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)25
- **Exists:** Always

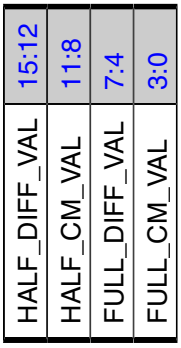


Table 2-1333 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode RX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode RX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode RX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode RX DCC ctl range for Bank 2. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1330 RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** RX DCC ctl Range Value Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)26
- **Exists:** Always

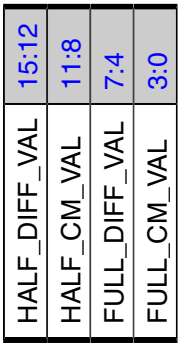


Table 2-1334 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	HALF_DIFF_VAL	R/W	These bits are the stored half rate diff-mode RX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
11:8	HALF_CM_VAL	R/W	These bits are the stored half rate common-mode RX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
7:4	FULL_DIFF_VAL	R/W	These bits are the stored full rate diff-mode RX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf
3:0	FULL_CM_VAL	R/W	These bits are the stored full rate common-mode RX DCC ctl range for Bank 3. **Value After Reset:** 0x8 **Exists:** Always **Reset Mask:** 0xf

2.1.1331 RAWLANEAONN_DIG_SRAM_REC_BASE (for N = 0; N <= NLANES-1)

- **Description:** Current SRAM Recording Base.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)27
- **Exists:** Always

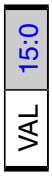


Table 2-1335 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_BASE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the current SRAM Recording Base. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1332 RAWLANEAONN_DIG_CAL_INIT_WAIT (for N = 0; N <= NLANES-1)

- **Description:** Wait Cycles During Initialization.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)28
- **Exists:** Always

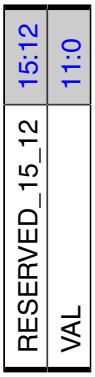


Table 2-1336 Fields for Register: RAWLANEAONN_DIG_CAL_INIT_WAIT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	VAL	R/W	These bits are the wait cycles during initialization. **Value After Reset:** 0x1f4 **Exists:** Always **Reset Mask:** 0xfff

2.1.1333 RAWLANEAONN_DIG_CAL_WAIT (for N = 0; N <= NLANES-1)

- **Description:** Wait Cycles After DAC Write.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)29
- **Exists:** Always

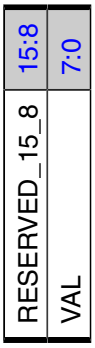


Table 2-1337 Fields for Register: RAWLANEAONN_DIG_CAL_WAIT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	These bits are the wait cycles after a DAC write. **Value After Reset:** 0x32 **Exists:** Always **Reset Mask:** 0xff

2.1.1334 RAWLANEAONN_DIG_AFE_RTRIM (for N = 0; N <= NLANES-1)

- **Description:** AFE Resistor Trim Code.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2a
- **Exists:** Always

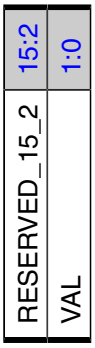


Table 2-1338 Fields for Register: RAWLANEAONN_DIG_AFE_RTRIM (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R/W	These bits are the AFE resistor trim code. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x3

2.1.1335 RAWLANEAONN_DIG_PRE_PTTRN (for N = 0; N <= NLANES-1)

- **Description:** Precursor Pattern.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2b
- **Exists:** Always

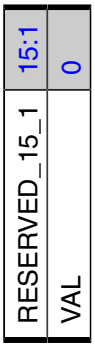


Table 2-1339 Fields for Register: RAWLANEAONN_DIG_PRE_PTTRN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	VAL	R/W	This bit is the precursor pattern. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1336 RAWLANEAONN_DIG_ADAPT_ATT_IQ_TH (for N = 0; N <= NLANES-1)

- **Description:** Adapted IQ Offset Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2c
- **Exists:** Always

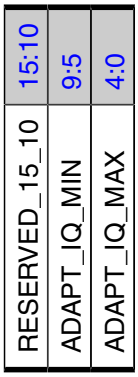


Table 2-1340 Fields for Register: RAWLANEAONN_DIG_ADAPT_ATT_IQ_TH (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:10	RESERVED_15_10	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
9:5	ADAPT_IQ_MIN	R/W	These bits are the mimimum adapted IQ offset (range 15 to -16). **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f
4:0	ADAPT_IQ_MAX	R/W	These bits are themaximum adapted IQ offset (range 15 to -16). **Value After Reset:** 0xa **Exists:** Always **Reset Mask:** 0x1f

2.1.1337 RAWLANEAONN_DIG_TX_MAIN_DFE2_TH (for N = 0; N <= NLANES-1)

- **Description:** TX Main DFE2 threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2d
- **Exists:** Always

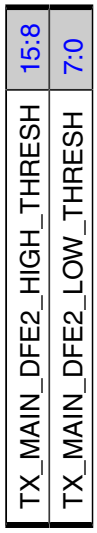


Table 2-1341 Fields for Register: RAWLANEAONN_DIG_TX_MAIN_DFE2_TH (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	TX_MAIN_DFE2_HIGH_THRESH	R/W	These bits are the TX post/DFE TAP2 high threshold. **Value After Reset:** 0xff **Exists:** Always **Reset Mask:** 0xff
7:0	TX_MAIN_DFE2_LOW_THRESH	R/W	These bits are the TX post/DFE TAP2 low threshold. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1338 RAWLANEAONN_DIG_SKIP_RX_DFE_FLT_ADAPT (for N = 0; N <= NLANES-1)

- **Description:** Skip RX DFE FLT Adapt.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2e
- **Exists:** Always

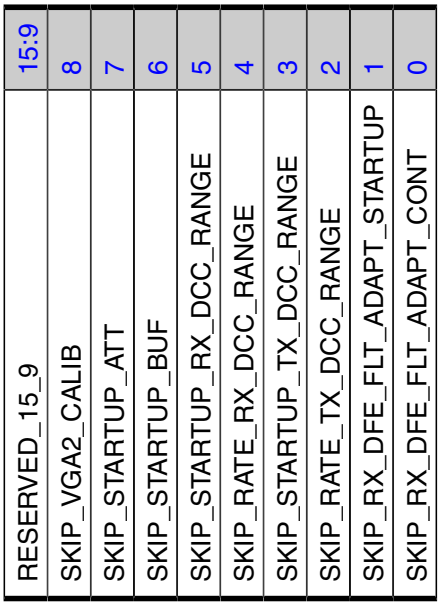


Table 2-1342 Fields for Register: RAWLANEAONN_DIG_SKIP_RX_DFE_FLT_ADAPT (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	SKIP_VGA2_CALIB	R/W	This bit skips vga2 calib. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
7	SKIP_STARTUP_ATT	R/W	This bit skips startup att. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
6	SKIP_STARTUP_BUF	R/W	This bit skips startup buf. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

Table 2-1342 Fields for Register: RAWLANEAONN_DIG_SKIP_RX_DFE_FLT_ADAPT (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
5	SKIP_STARTUP_RX_DCC_RANGE	R/W	This bit skips startup rx dcc range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
4	SKIP_RATE_RX_DCC_RANGE	R/W	This bit skips rate rx dcc range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
3	SKIP_STARTUP_TX_DCC_RANGE	R/W	This bit skips startup tx dcc range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SKIP_RATE_TX_DCC_RANGE	R/W	This bit skips rate tx dcc range. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_RX_DFE_FLT_ADAPT_STARTUP	R/W	This bit the start-up DFE floating tap adaptation state skip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_RX_DFE_FLT_ADAPT_CONT	R/W	This bit is the mission-mode DFE floating tap adaptation state skip. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1339 RAWLANEAONN_DIG_VGA_ADAPT_TH (for N = 0; N <= NLANES-1)

- **Description:** VGA Adapt Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)2f
- **Exists:** Always

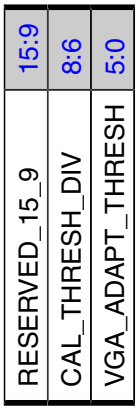


Table 2-1343 Fields for Register: RAWLANEAONN_DIG_VGA_ADAPT_TH (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:6	CAL_THRESH_DIV	R/W	These bits are the calibration threshold divider (2N). **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x7
5:0	VGA_ADAPT_THRESH	R/W	These bits update the threshold for VGA adaptation (2N). **Value After Reset:** 0x3 **Exists:** Always **Reset Mask:** 0x3f

2.1.1340 RAWLANEAONN_DIG_TX_POST_TAP1_HIGH_THRESHOLD (for N = 0; N <= NLANES-1)

- **Description:** TX Post TAP1 High Threshold.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)30
- **Exists:** Always

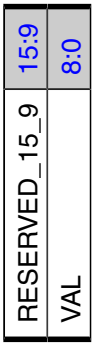


Table 2-1344 Fields for Register: RAWLANEAONN_DIG_TX_POST_TAP1_HIGH_THRESHOLD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the TX post TAP1 high threshold. **Value After Reset:** 0x14 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1341 RAWLANEAONN_DIG_RX_CAL_REF_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Reference Level Offset Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)31
- **Exists:** Always

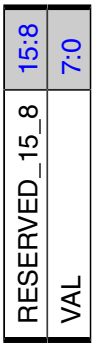


Table 2-1345 Fields for Register: RAWLANEAONN_DIG_RX_CAL_REF_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:0	VAL	R/W	These bits are the value of the reference level offset. **Value After Reset:** 0x3a **Exists:** Always **Reset Mask:** 0xff

2.1.1342 RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_MUXED (for N = 0; N <= NLANES-1)

- **Description:** TX DCC CTL Range Multiplexed Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)32
- **Exists:** Always

Table 2-1346 Fields for Register: RAWLANEAONN_DIG_TX_DCC_CTRL_RANGE_MUXED (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	DIFF_MUXED_VAL	R	These bits are the stored multiplexed diff-mode TX DCC ctl range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	CM_MUXED_VAL	R	These bits are the stored muxed common-mode TX DCC ctl range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1343 RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_MUXED (for N = 0; N <= NLANES-1)

- **Description:** RX DCC CTL Range Multiplexed Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)33
- **Exists:** Always

Table 2-1347 Fields for Register: RAWLANEAONN_DIG_RX_DCC_CTRL_RANGE_MUXED (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:4	DIFF_MUXED_VAL	R	These bits are the stored multiplexed diff-mode RX DCC ctl range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
3:0	CM_MUXED_VAL	R	These bits are the stored multiplexed common-mode RX DCC ctl range. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1344 RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** Full Rate TX Delay Calibration for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)34
- **Exists:** Always

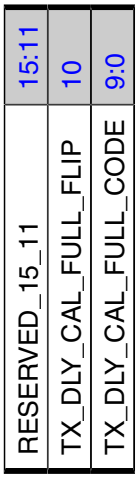


Table 2-1348 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_FULL_FLIP	R/W	This bit is the TX delay calibration flip at full rate for Bank 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_FULL_CODE	R/W	These bits are the TX delay calibration code at full rate for Bank 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1345 RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** Full Rate TX Delay Calibration for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)35
- **Exists:** Always

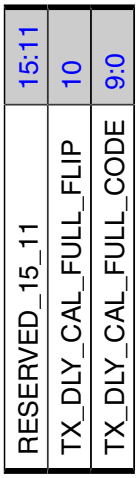


Table 2-1349 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_FULL_FLIP	R/W	This bit is the TX delay calibration flip at full rate for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_FULL_CODE	R/W	These bits are the TX delay calibration code at full rate for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1346 RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** Full Rate TX Delay Calibration for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)36
- **Exists:** Always

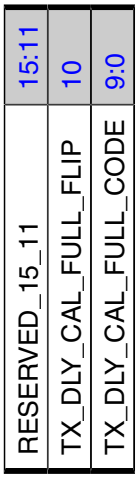


Table 2-1350 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_FULL_FLIP	R/W	This bit is the TX delay calibration flip at full rate for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_FULL_CODE	R/W	These bits are the TX delay calibration code at full rate for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1347 RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** Full Rate TX Delay Calibration for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)37
- **Exists:** Always

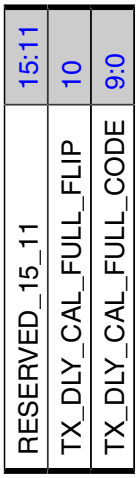


Table 2-1351 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_FULL_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_FULL_FLIP	R/W	This bit is the TX delay calibration flip at full rate for Bank 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_FULL_CODE	R/W	These bits are the TX delay calibration code at full rate for Bank 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1348 RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_0 (for N = 0; N <= NLANES-1)

- **Description:** Half Rate TX Delay Calibration for Bank 0.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)38
- **Exists:** Always

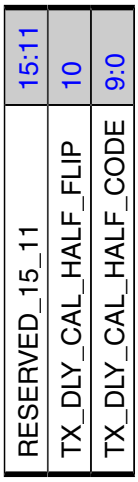


Table 2-1352 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_0 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_HALF_FLIP	R/W	This bit is the TX delay calibration flip at half rate for Bank 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_HALF_CODE	R/W	These bits are the TX delay calibration code at half rate for Bank 0. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1349 RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_1 (for N = 0; N <= NLANES-1)

- **Description:** Half Rate TX Delay Calibration for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)39
- **Exists:** Always

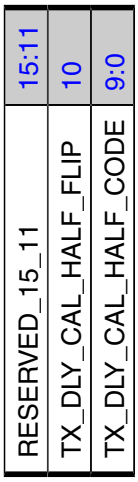


Table 2-1353 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_HALF_FLIP	R/W	This bit is the TX delay calibration flip at half rate for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_HALF_CODE	R/W	These bits are the TX delay calibration code at half rate for Bank 1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1350 RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_2 (for N = 0; N <= NLANES-1)

- **Description:** Half Rate TX Delay Calibration for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3a
- **Exists:** Always

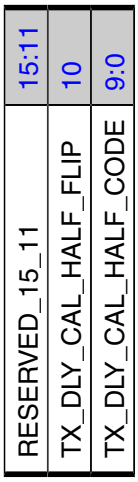


Table 2-1354 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_HALF_FLIP	R/W	This bit is the TX delay calibration flip at half rate for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_HALF_CODE	R/W	These bits are the TX delay calibration code at half rate for Bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1351 RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_3 (for N = 0; N <= NLANES-1)

- **Description:** Half Rate TX Delay Calibration for Bank 3.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3b
- **Exists:** Always

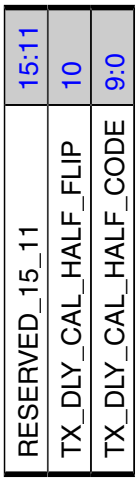


Table 2-1355 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_HALF_BANK_3 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_HALF_FLIP	R/W	This bit is the TX delay calibration flip at half rate for Bank 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
9:0	TX_DLY_CAL_HALF_CODE	R/W	These bits are the TX delay calibration code at half rate for Bank 3. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3ff

2.1.1352 RAWLANEAONN_DIG_TX_DLY_CAL_MUX (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Calibration Multiplexer Value.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3c
- **Exists:** Always

Table 2-1356 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_MUX (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:11	RESERVED_15_11	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
10	TX_DLY_CAL_FLIP_MUX	R	These bits are the TX delay calibration flip multiplexer. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true
9:0	TX_DLY_CAL_CODE_MUX	R	These bits are the TX delay calibration code multiplexer. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1353 RAWLANEAONN_DIG_TX_DLY_CAL_GAIN (for N = 0; N <= NLANES-1)

- **Description:** TX Delay Calibration Gain.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3d
- **Exists:** Always

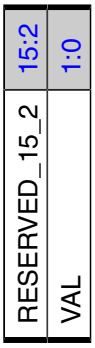


Table 2-1357 Fields for Register: RAWLANEAONN_DIG_TX_DLY_CAL_GAIN (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1:0	VAL	R/W	these bits are the TX delay calibration gain. **Value After Reset:** 0x2 **Exists:** Always **Reset Mask:** 0x3

2.1.1354 RAWLANEAONN_DIG_RX_ADPT_AFE_RATE (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE RATE.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3e
- **Exists:** Always

Table 2-1358 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_RATE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	AFE_RATE_ADPT_VAL	R/W	These bits store the RX adapted AFE RATE. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1355 RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE CONFIG.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)3f
- **Exists:** Always

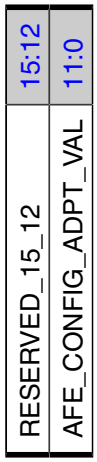


Table 2-1359 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	AFE_CONFIG_ADPT_VAL	R/W	These bits store the RX adapted AFE CONFIG. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1356 RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted DFE Enables.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)40
- **Exists:** Always

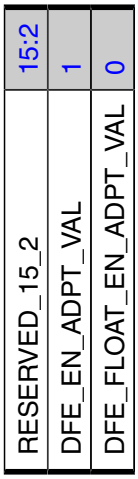


Table 2-1360 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DFE_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	DFE_FLOAT_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE FLOAT EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1357 RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE RATE for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)41
- **Exists:** Always

Table 2-1361 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	AFE_RATE_ADPT_VAL	R/W	These bits store the RX adapted AFE RATE. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1358 RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE CONFIG for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)42
- **Exists:** Always

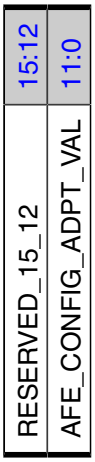


Table 2-1362 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	AFE_CONFIG_ADPT_VAL	R/W	These bits store the RX adapted AFE CONFIG. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1359 RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted DFE Enables for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)43
- **Exists:** Always

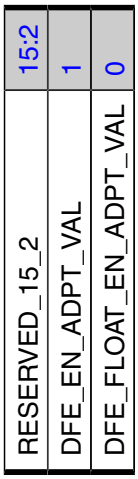


Table 2-1363 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DFE_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	DFE_FLOAT_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE FLOAT EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1360 RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE RATE for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)44
- **Exists:** Always

Table 2-1364 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_RATE_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2:0	AFE_RATE_ADPT_VAL	R/W	These bits store the RX adapted AFE RATE. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1361 RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted AFE CONFIG for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)45
- **Exists:** Always

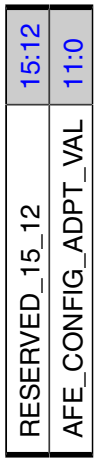


Table 2-1365 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_AFE_CONFIG_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
11:0	AFE_CONFIG_ADPT_VAL	R/W	These bits store the RX adapted AFE CONFIG. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xfff

2.1.1362 RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted DFE Enables for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)46
- **Exists:** Always

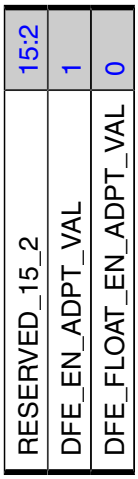


Table 2-1366 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_DFE_ENABLES_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	DFE_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE FLOAT EN. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	DFE_FLOAT_EN_ADPT_VAL	R/W	This bit stores the RX adapted DFE FLOAT EN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1363 RAWLANEAONN_DIG_RX_ADAPT_IQ_CTRL (for N = 0; N <= NLANES-1)

- **Description:** Rx Adaption IQ Controls.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)47
- **Exists:** Always

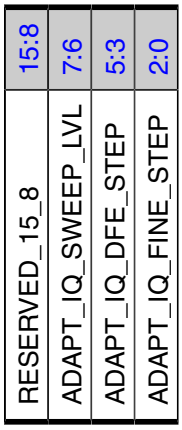


Table 2-1367 Fields for Register: RAWLANEAONN_DIG_RX_ADAPT_IQ_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
7:6	ADAPT_IQ_SWEEP_LVL	R/W	These bits are the horizontal sweep level. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x3
5:3	ADAPT_IQ_DFE_STEP	R/W	These bits are the number of DFE tap 1 steps. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7
2:0	ADAPT_IQ_FINE_STEP	R/W	These bits are the number of fine steps. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1364 RAWLANEAONN_DIG_RX_CTLE_ADAPT_CTRL (for N = 0; N <= NLANES-1)

- **Description:** Rx CTEL Adapt Controls.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)48
- **Exists:** Always

Table 2-1368 Fields for Register: RAWLANEAONN_DIG_RX_CTLE_ADAPT_CTRL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4:0	CTLE_ADAPT_MIN	R/W	These bits store the minimum allowed CTLE adaptation code. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1f

2.1.1365 RAWLANEAONN_DIG_SRAM_REC_MAX_ITER (for N = 0; N <= NLANES-1)

- **Description:** The maximum number of Iterations for SRAM Recording.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)49
- **Exists:** Always

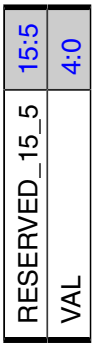


Table 2-1369 Fields for Register: RAWLANEAONN_DIG_SRAM_REC_MAX_ITER (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:5	RESERVED_15_5	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
4:0	VAL	R/W	Represents the maximum number of Iterations for SRAM Recording. **Value After Reset:** 0x10 **Exists:** Always **Reset Mask:** 0x1f

2.1.1366 RAWLANEAONN_DIG_USE_RX_COARSE_ADPT_VAL (for N = 0; N <= NLANES-1)

- **Description:** Use RX Coarse Adapt Value .
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4a
- **Exists:** Always

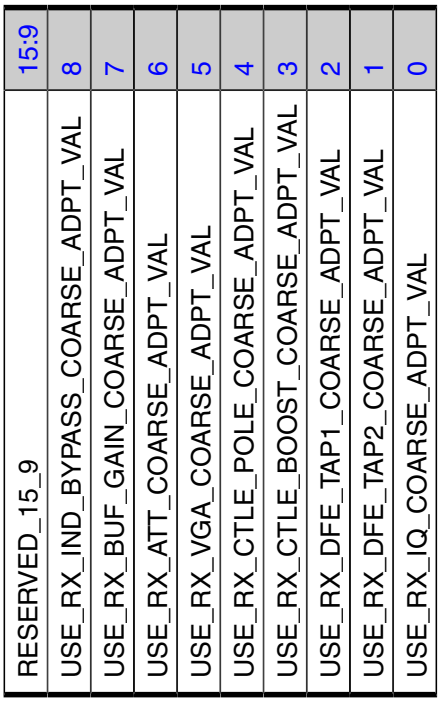


Table 2-1370 Fields for Register: RAWLANEAONN_DIG_USE_RX_COARSE_ADPT_VAL (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8	USE_RX_IND_BYPASS_COARSE_ADPT_VAL	R/W	This bit uses rx_ind_bypass_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
7	USE_RX_BUF_GAIN_COARSE_ADPT_VAL	R/W	This bit uses rx_buf_gain_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

Table 2-1370 Fields for Register: RAWLANEAONN_DIG_USE_RX_COARSE_ADPT_VAL (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
6	USE_RX_ATT_COARSE_ADPT_VAL	R/W	This bit uses rx_att_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
5	USE_RX_VGA_COARSE_ADPT_VAL	R/W	This bit uses rx_vga_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
4	USE_RX_CTLE_POLE_COARSE_ADPT_VAL	R/W	This bit uses rx_ctle_pole_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
3	USE_RX_CTLE_BOOST_COARSE_ADPT_VAL	R/W	This bit uses rx_ctle_boost_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
2	USE_RX_DFE_TAP1_COARSE_ADPT_VAL	R/W	This bit uses rx_dfe_tap1_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
1	USE_RX_DFE_TAP2_COARSE_ADPT_VAL	R/W	This bit uses rx_dfe_tap2_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	USE_RX_IQ_COARSE_ADPT_VAL	R/W	This bit uses rx_iq_coarse_adpt_val. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.1367 RAWLANEAONN_DIG_RX_RO_AONLANE_1 (for N = 0; N <= NLANES-1)

- **Description:** Value for rx_disable.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4b
- **Exists:** Always

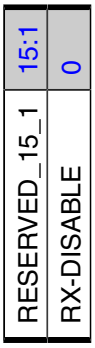


Table 2-1371 Fields for Register: RAWLANEAONN_DIG_RX_RO_AONLANE_1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:1	RESERVED_15_1	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
0	RX-DISABLE	R	Read value for rx_disable. (volatile) (actual reset value is 0xX) **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0 **Volatile:** true

2.1.1368 RAWLANEAONN_DIG_RX_DCC_IQ_SKIPS (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANEAON0_DIG_RX_DCC_IQ_SKIPS
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4c
- **Exists:** Always

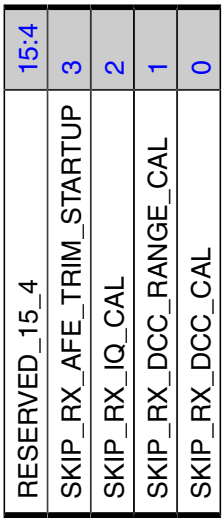


Table 2-1372 Fields for Register: RAWLANEAONN_DIG_RX_DCC_IQ_SKIPS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	SKIP_RX_AFE_TRIM_STARTUP	R/W	This bit skips the FW update to the AFE resistor trim setting. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2	SKIP_RX_IQ_CAL	R/W	This bit skips start-up/rate change RX IQ calibration **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	SKIP_RX_DCC_RANGE_CAL	R/W	This bit skips start-up/rate change RX DCC range calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_RX_DCC_CAL	R/W	This bit skips start-up/rate change RX DCC calibration. **Value After Reset:** 0x0

Table 2-1372 Fields for Register: RAWLANEAONN_DIG_RX_DCC_IQ_SKIPS (for N = 0; N <= NLANES-1) (continued)

Bits	Name	Memory Access	Description
			Exists: Always **Reset Mask:** 0x1

2.1.1369 RAWLANEAONN_DIG_TX_DCC_SKIPS (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANEAON0_DIG_TX_DCC_SKIPS
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4d
- **Exists:** Always

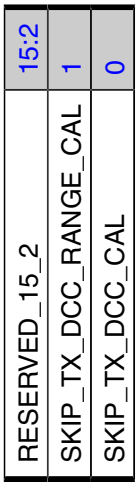


Table 2-1373 Fields for Register: RAWLANEAONN_DIG_TX_DCC_SKIPS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:2	RESERVED_15_2	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
1	SKIP_TX_DCC_RANGE_CAL	R/W	This bit skips start-up/rate change TX DCC range calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
0	SKIP_TX_DCC_CAL	R/W	This bit skips start-up/rate change TX DCC calibration. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1

2.1.1370 RAWLANEAONN_DIG_CAL_VGEN_DAC_CODE (for N = 0; N <= NLANES-1)

- **Description:** VGEN calibration code
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4e
- **Exists:** Always

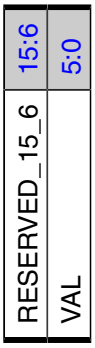


Table 2-1374 Fields for Register: RAWLANEAONN_DIG_CAL_VGEN_DAC_CODE (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
5:0	VAL	R/W	These bits are the VGEN calibration code. **Value After Reset:** 0x20 **Exists:** Always **Reset Mask:** 0x3f

2.1.1371 RAWLANEAONN_DIG_RX_COARSE_ADPT_SKIP_BITS (for N = 0; N <= NLANES-1)

- **Description:** RX Coarse Adapt Skip bits for all three banks
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)4f
- **Exists:** Always

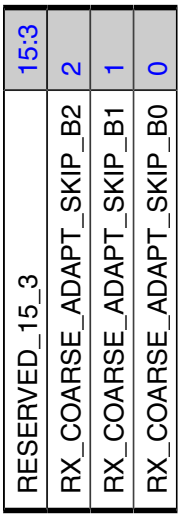


Table 2-1375 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_SKIP_BITS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:3	RESERVED_15_3	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
2	RX_COARSE_ADAPT_SKIP_B2	R/W	This bit indicates if RX coarse adapt is skipped for bank 2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
1	RX_COARSE_ADAPT_SKIP_B1	R/W	This bit indicates if RX coarse adapt is skipped for bank 1. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1
0	RX_COARSE_ADAPT_SKIP_B0	R/W	This bit indicates if RX coarse adapt is skipped for bank 0. **Value After Reset:** 0x1 **Exists:** Always **Reset Mask:** 0x1

2.1.1372 RAWLANEAONN_DIG_DFE_BYPASS_EVEN_ERROR_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE By-Pass Even Error vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)50
- **Exists:** Always

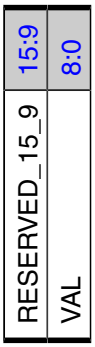


Table 2-1376 Fields for Register: RAWLANEAONN_DIG_DFE_BYPASS_EVEN_ERROR_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE By-Pass Even Error vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1373 RAWLANEAONN_DIG_DFE_BYPASS_ODD_ERROR_VDAC_OFST (for N = 0; N <= NLANES-1)

- **Description:** Offset Values for RX DFE By-Pass Odd Error vDAC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)51
- **Exists:** Always

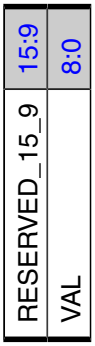


Table 2-1377 Fields for Register: RAWLANEAONN_DIG_DFE_BYPASS_ODD_ERROR_VDAC_OFST (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
8:0	VAL	R/W	These bits are the offset values for DFE By-Pass Odd Error vDAC. **Value After Reset:** 0x100 **Exists:** Always **Reset Mask:** 0x1ff

2.1.1374 RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANEAON0_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)52
- **Exists:** Always

Table 2-1378 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	EQ_IND_BYPASS_ADAPT_VAL	R/W	These bits store the RX EQ_IND_BYPASS. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	EQ_BUF_GAIN_ADAPT_VAL	R/W	These bits store the RX EQ_BUF_GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1375 RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted BUF GAIN for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)53
- **Exists:** Always

Table 2-1379 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	EQ_IND_BYPASS_ADAPT_VAL_B1	R/W	These bits store the RX EQ_IND_BYPASS. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	BUF_GAIN_ADPT_VAL_B1	R/W	These bits store the RX adapted BUF GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1376 RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted BUF GAIN for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)54
- **Exists:** Always

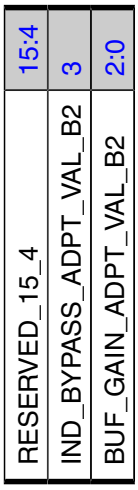


Table 2-1380 Fields for Register: RAWLANEAONN_DIG_RX_ADPT_BUF_GAIN_IND_BYPASS_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	IND_BYPASS_ADPT_VAL_B2	R/W	These bits store the RX adapted IND BYPASS. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	BUF_GAIN_ADPT_VAL_B2	R/W	These bits store the RX adapted BUF GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1377 RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS (for N = 0; N <= NLANES-1)

- **Description:** E32_RAWLANEAON0_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)55
- **Exists:** Always

Table 2-1381 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	EQ_IND_BYPASS_ADAPT_VAL	R/W	These bits store the RX EQ_IND_BYPASS. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	EQ_BUF_GAIN_COARSE_ADPT_VAL	R/W	These bits store the RX EQ_BUF_GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1378 RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B1 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted BUF GAIN for Bank 1.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)56
- **Exists:** Always

Table 2-1382 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B1 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	IND_BYPASS_COARSE_ADPT_VAL_B1	R/W	These bits store the RX adapted IND BYPASS B1. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	BUF_GAIN_COARSE_ADPT_VAL	R/W	These bits store the RX adapted BUF GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1379 RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B2 (for N = 0; N <= NLANES-1)

- **Description:** RX Adapted BUF GAIN for Bank 2.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)57
- **Exists:** Always

Table 2-1383 Fields for Register: RAWLANEAONN_DIG_RX_COARSE_ADPT_BUF_GAIN_IND_BYPASS_B2 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved for future use. Reset value is 0xX. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x0
3	IND_BYPASS_COARSE_ADPT_VAL_B2	R/W	These bits store the RX adapted IND BYPASS B2. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x1
2:0	BUF_GAIN_COARSE_ADPT_VAL	R/W	These bits store the RX adapted BUF GAIN. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0x7

2.1.1380 RAWLANEAONN_DIG_ADPT_CTL_29 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 29.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)58
- **Exists:** Always

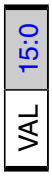


Table 2-1384 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_29 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1381 RAWLANEAONN_DIG_ADPT_CTL_30 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 30.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)59
- **Exists:** Always

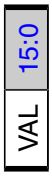


Table 2-1385 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_30 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1382 RAWLANEAONN_DIG_ADPT_CTL_31 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 31.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5a
- **Exists:** Always

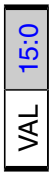


Table 2-1386 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_31 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1383 RAWLANEAONN_DIG_ADPT_CTL_32 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 32.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5b
- **Exists:** Always

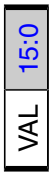


Table 2-1387 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_32 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1384 RAWLANEAONN_DIG_ADPT_CTL_33 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 33.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5c
- **Exists:** Always

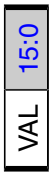


Table 2-1388 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_33 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1385 RAWLANEAONN_DIG_ADPT_CTL_34 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 34.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5d
- **Exists:** Always

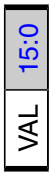


Table 2-1389 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_34 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1386 RAWLANEAONN_DIG_ADPT_CTL_35 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 35.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5e
- **Exists:** Always

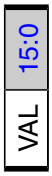


Table 2-1390 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_35 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1387 RAWLANEAONN_DIG_ADPT_CTL_36 (for N = 0; N <= NLANES-1)

- **Description:** Adaptation Control Register 36.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)5f
- **Exists:** Always

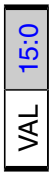


Table 2-1391 Fields for Register: RAWLANEAONN_DIG_ADPT_CTL_36 (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:0	VAL	R/W	These bits are the adaptation control value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1388 RAWLANEAONN_DIG_RX_MISC (for N = 0; N <= NLANES-1)

- **Description:** OVRD register for RX_MISC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)60
- **Exists:** Always

Table 2-1392 Fields for Register: RAWLANEAONN_DIG_RX_MISC (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	RX_MISC_EN	R/W	These bits are the rx_misc ovrd enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
7:0	RX_MISC_VAL	R/W	These bits are the rx_misc ovrd value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1389 RAWLANEAONN_DIG_TX_MISC_OVRD (for N = 0; N <= NLANES-1)

- **Description:** OVRD register for TX_MISC.
- **Size:** 16 bits
- **Offset:** 0x3(N*2+1)61
- **Exists:** Always

Table 2-1393 Fields for Register: RAWLANEAONN_DIG_TX_MISC_OVRD (for N = 0; N <= NLANES-1)

Bits	Name	Memory Access	Description
15:8	TX_MISC_EN	R/W	These bits are the tx_misc ovrd enable. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff
7:0	TX_MISC_VAL	R/W	These bits are the tx_misc ovrd value. **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xff

2.1.1390 RAWMEM_DIG_ROM_CMN0_B0_R0

- **Description:** First ROM address.
- **Size:** 16 bits
- **Offset:** 0x8000
- **Exists:** Always

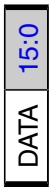


Table 2-1394 Fields for Register: RAWMEM_DIG_ROM_CMN0_B0_R0

Bits	Name	Memory Access	Description
15:0	DATA	R	Memory data **Value After Reset:** 0x81d **Exists:** Always **Reset Mask:** 0xffff

2.1.1391 RAWMEM_DIG_ROM_CMN31_B15_R31

- **Description:** Last ROM address.
- **Size:** 16 bits
- **Offset:** 0xbfff
- **Exists:** Always

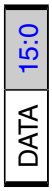


Table 2-1395 Fields for Register: RAWMEM_DIG_ROM_CMN31_B15_R31

Bits	Name	Memory Access	Description
15:0	DATA	R	Memory data **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff

2.1.1392 RAWMEM_DIG_RAM_CMN0_B0_R0

- **Description:** First RAM address.
- **Size:** 16 bits
- **Offset:** 0xc000
- **Exists:** Always

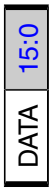


Table 2-1396 Fields for Register: RAWMEM_DIG_RAM_CMN0_B0_R0

Bits	Name	Memory Access	Description
15:0	DATA	R	Memory data **Value After Reset:** 0x81d **Exists:** Always **Reset Mask:** 0xffff

2.1.1393 RAWMEM_DIG_RAM_CMN31_B15_R31

- **Description:** Last RAM address.
- **Size:** 16 bits
- **Offset:** 0xffff
- **Exists:** Always

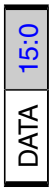


Table 2-1397 Fields for Register: RAWMEM_DIG_RAM_CMN31_B15_R31

Bits	Name	Memory Access	Description
15:0	DATA	R	Memory data **Value After Reset:** 0x0 **Exists:** Always **Reset Mask:** 0xffff